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(54) **SEMICONDUCTOR DEVICE**

Publication Classification

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South (AU)

(73) Assignee: **Silanna UV Technologies Pte Ltd,**
Singapore (SG)

(51) **Int. Cl.**
H10D 8/60 (2025.01)
H10D 8/01 (2025.01)
H10D 62/40 (2025.01)
H10D 62/832 (2025.01)

(52) **U.S. Cl.**
CPC **H10D 8/60** (2025.01); **H10D 8/051**
(2025.01); **H10D 62/40** (2025.01); **H10D**
62/8325 (2025.01)

(21) Appl. No.: **19/213,869**

(22) Filed: **May 20, 2025**

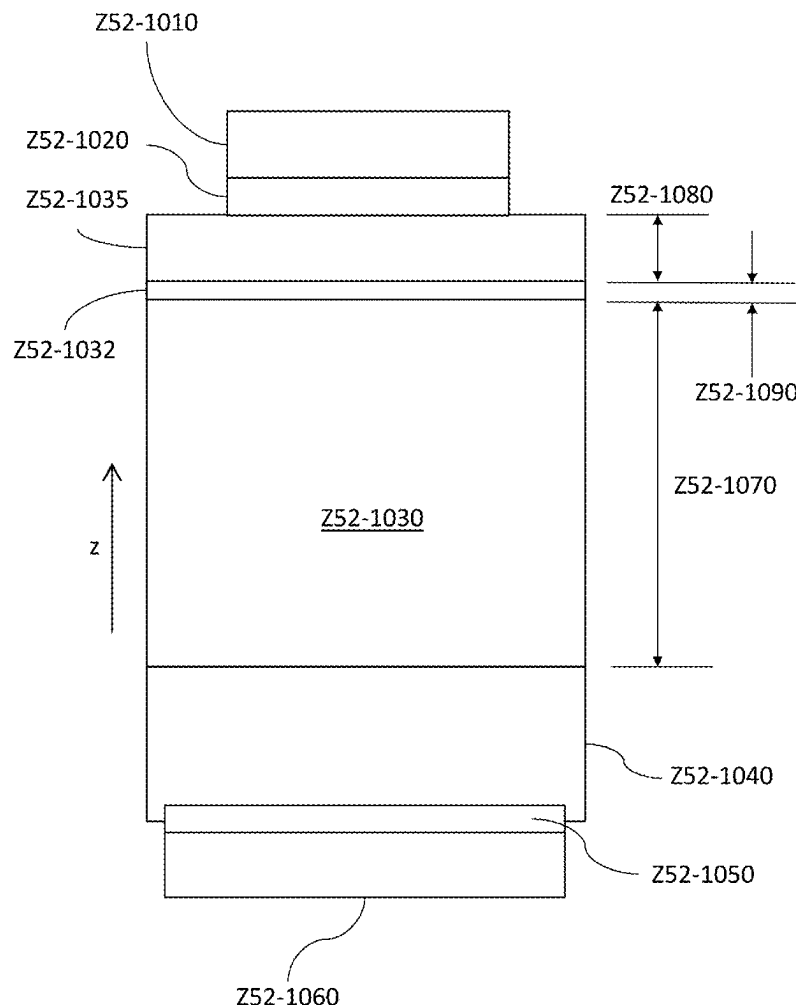
(57) **ABSTRACT**

Related U.S. Application Data

(63) Continuation of application No. 18/830,089, filed on Sep. 10, 2024, which is a continuation-in-part of application No. 18/628,178, filed on Apr. 5, 2024, which is a continuation-in-part of application No. 18/394,688, filed on Dec. 22, 2023, now Pat. No. 12,074,195.

(60) Provisional application No. 63/584,661, filed on Sep. 22, 2023.

A multilayered semiconductor diode device can include a substrate including silicon carbide (SiC) with an epitaxial drift layer including a first semiconductor oxide material above the SiC substrate with respect to a growth direction. The multilayered semiconductor diode device can further include a polar nitride layer including a polar semiconductor nitride material above the epitaxial drift layer with respect to the growth direction, and a metal layer above the polar nitride layer with respect to the growth direction.



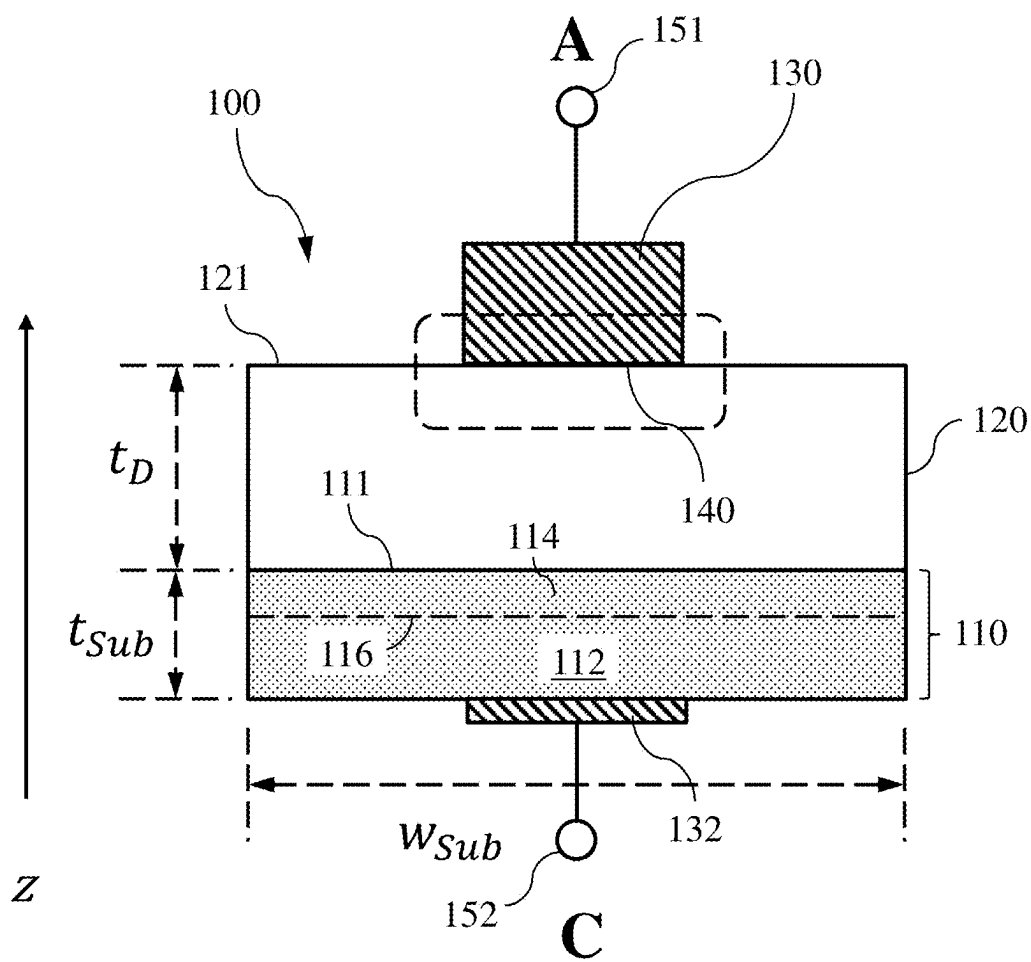


FIG. 1

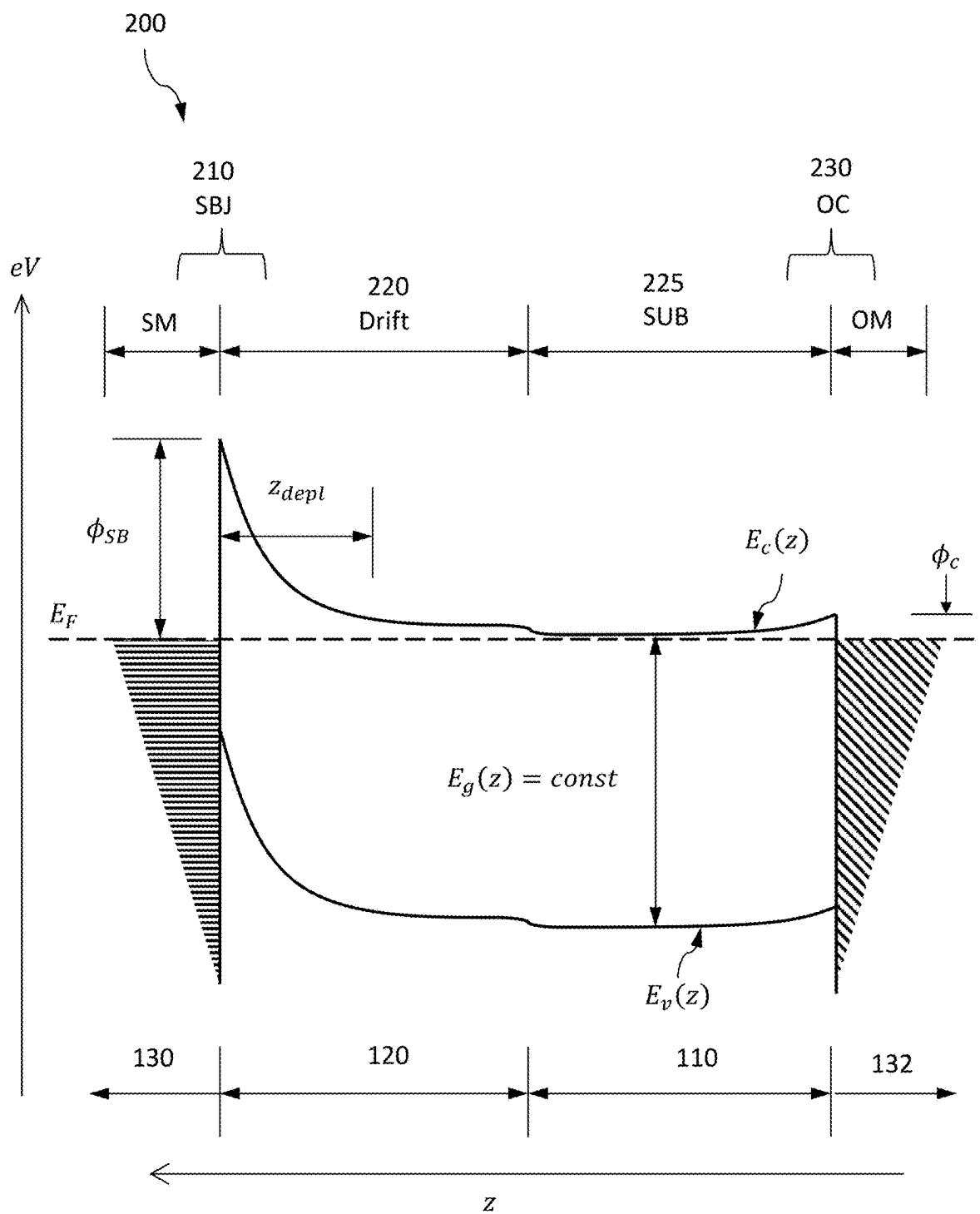


FIG. 2A

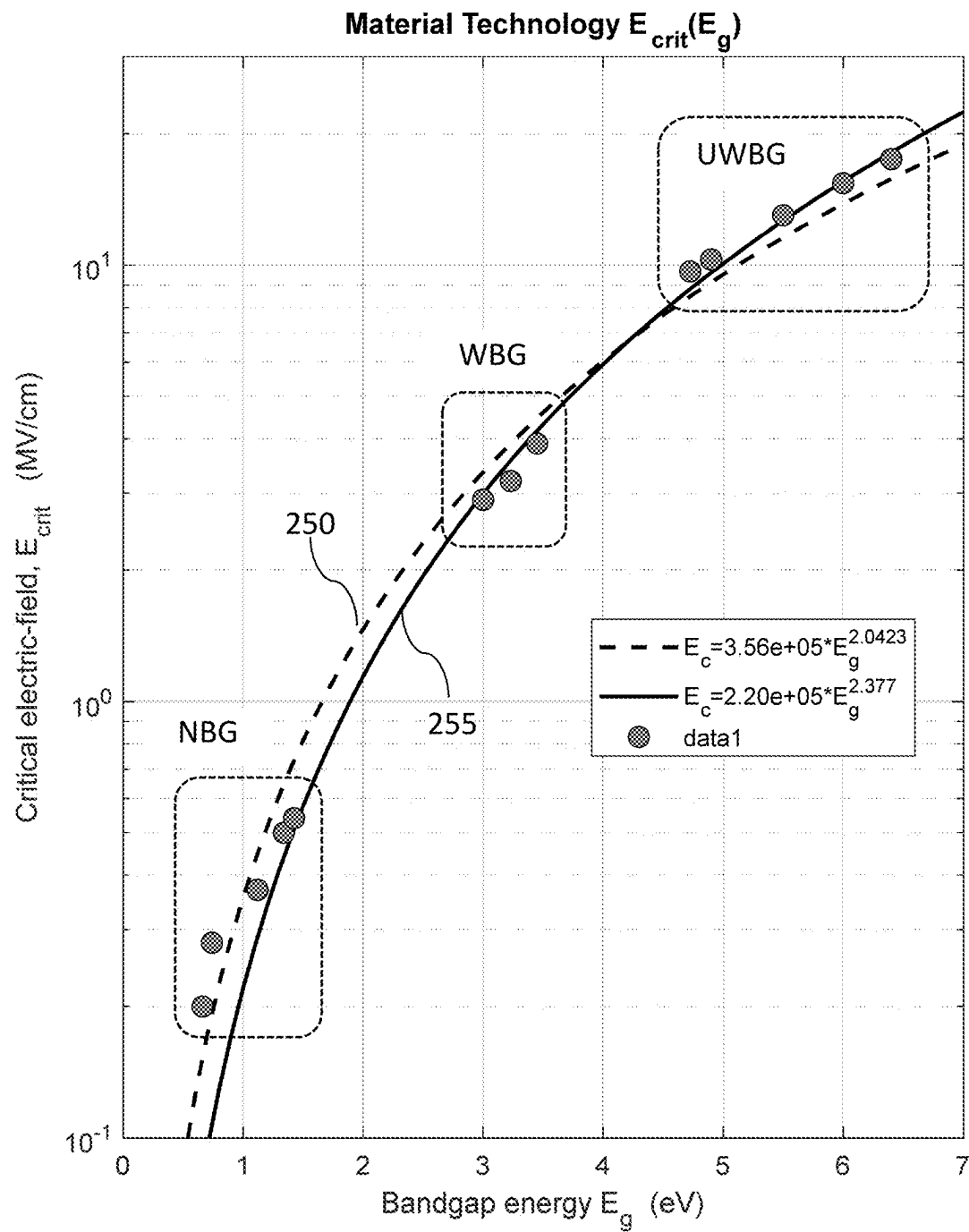


FIG. 2B

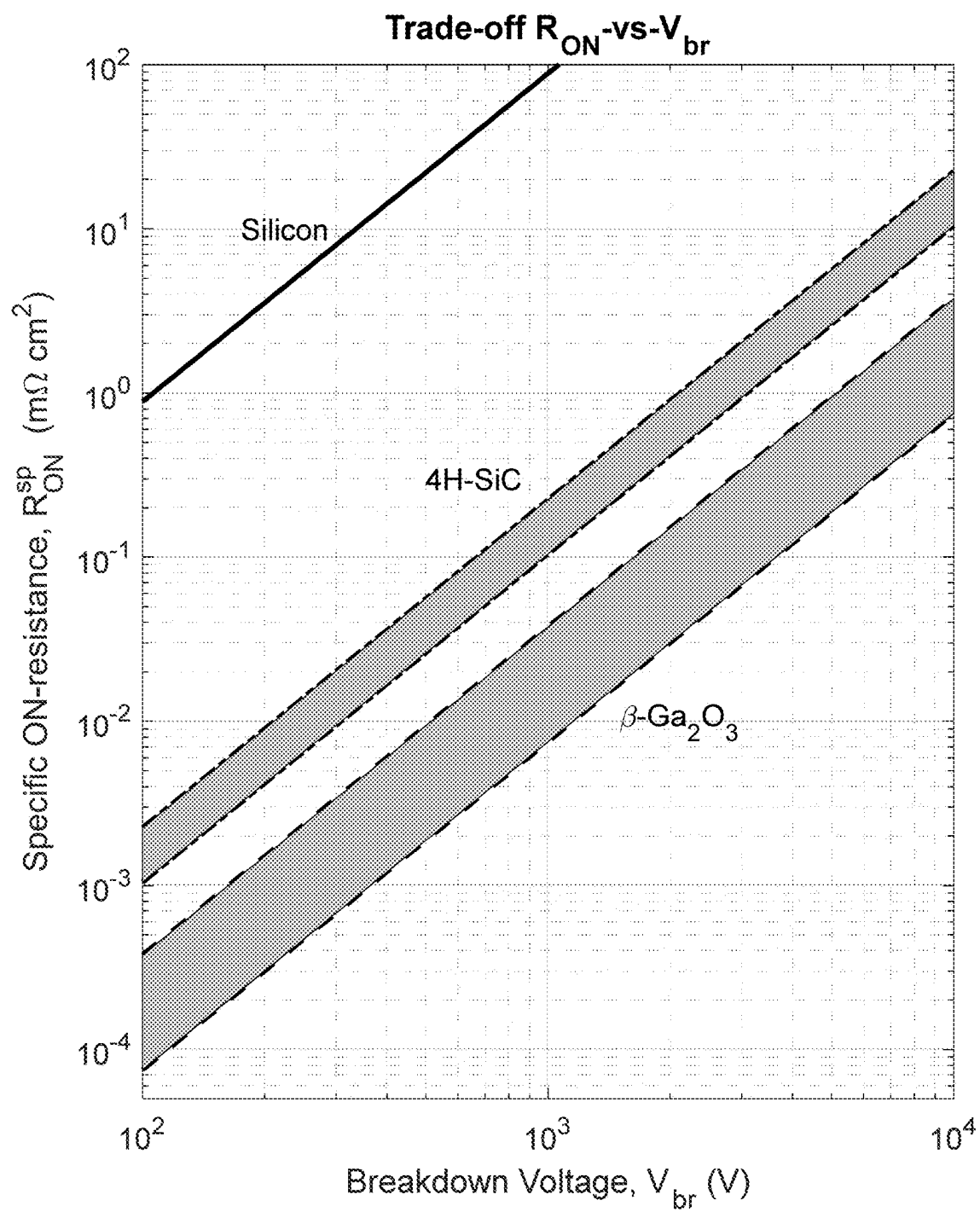


FIG. 2C

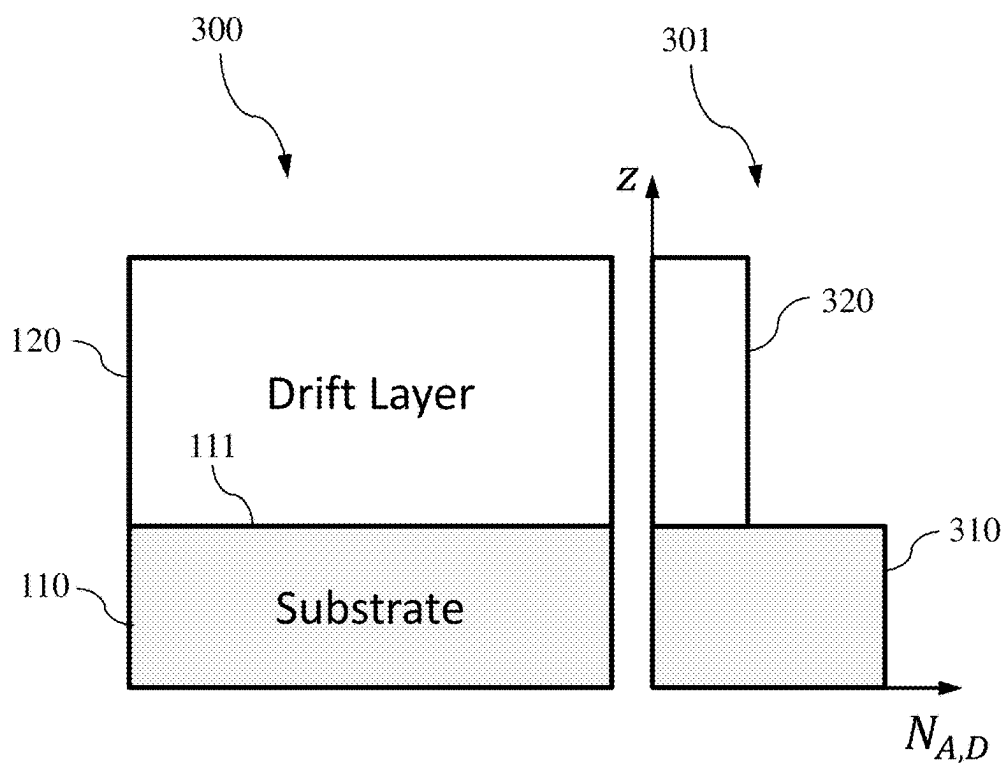


FIG. 3A

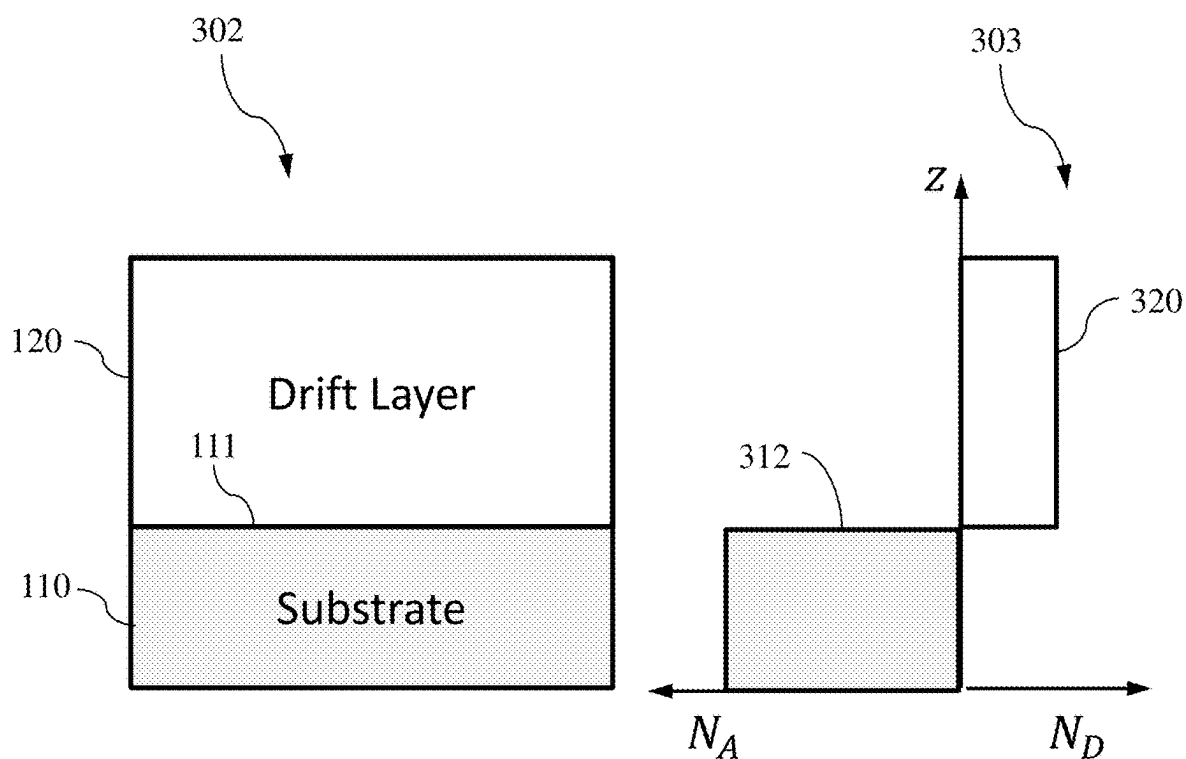


FIG. 3B

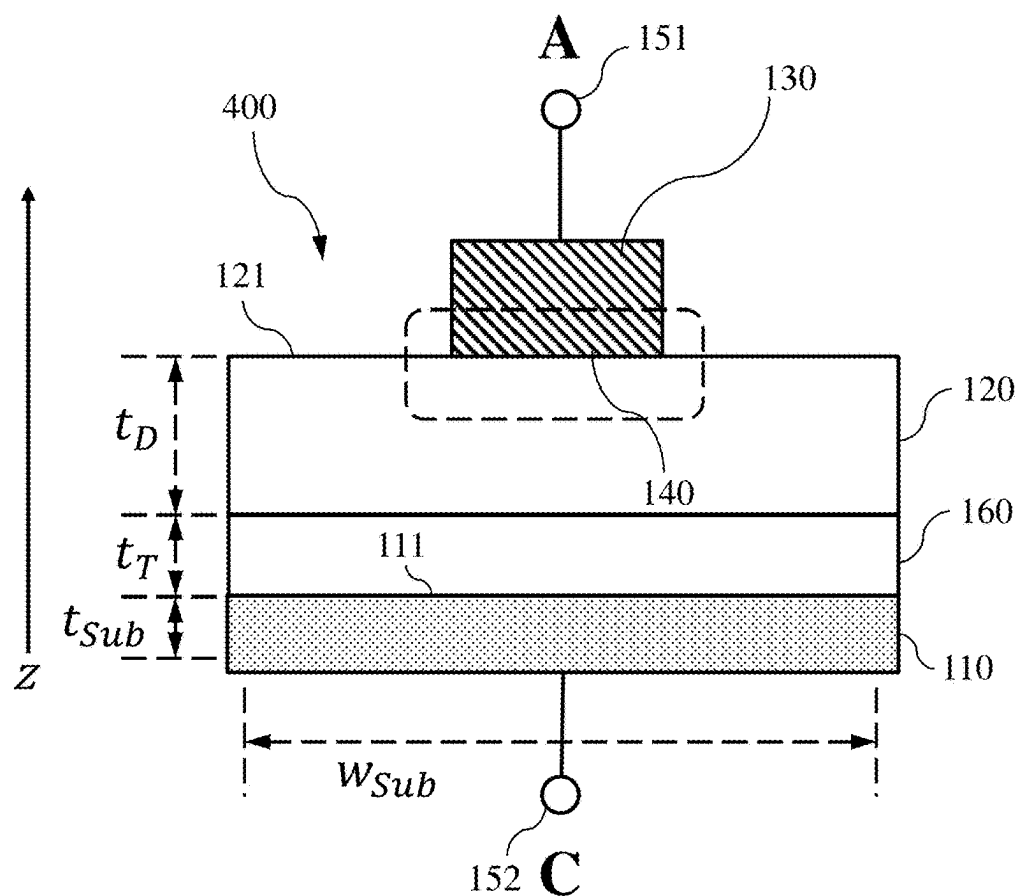
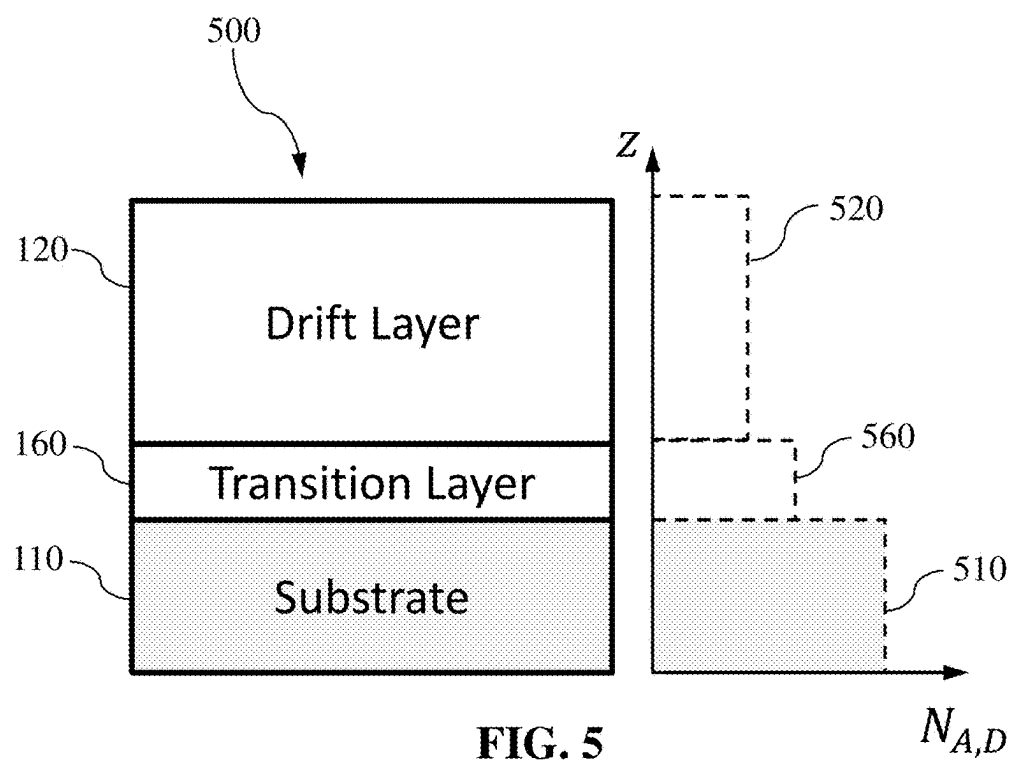


FIG. 4



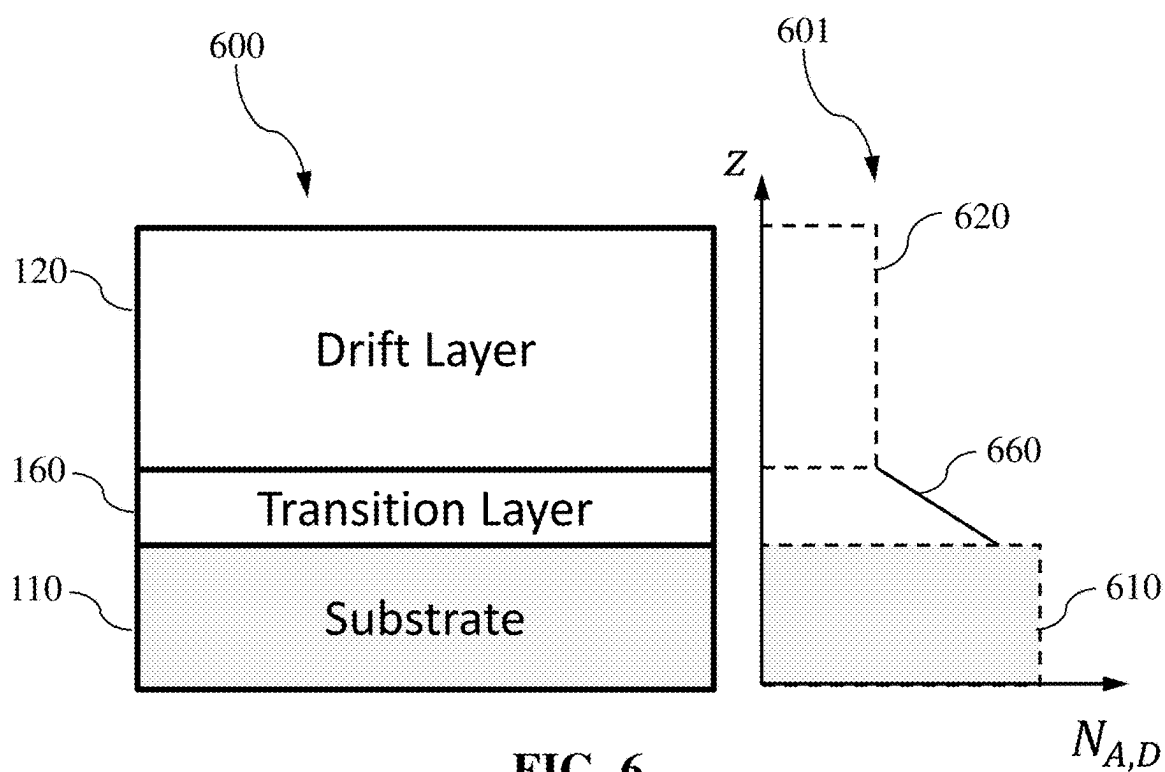


FIG. 6

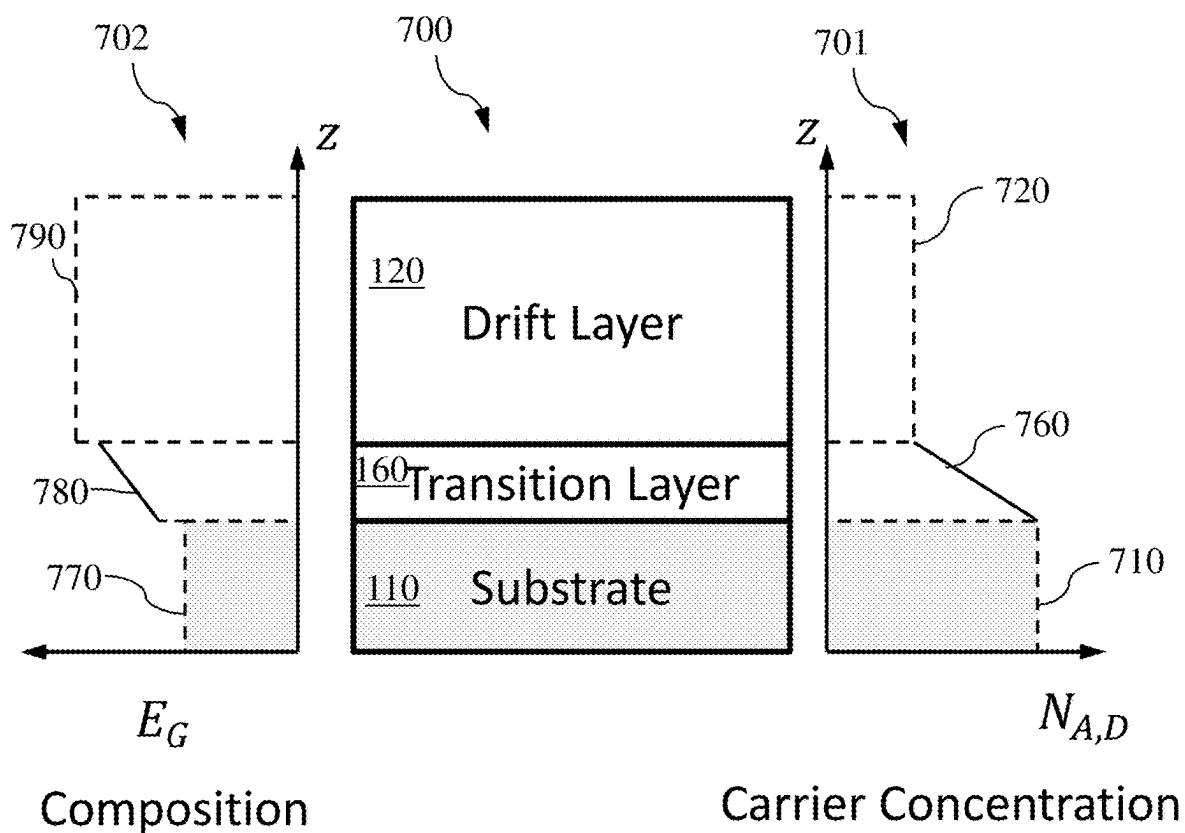


FIG. 7

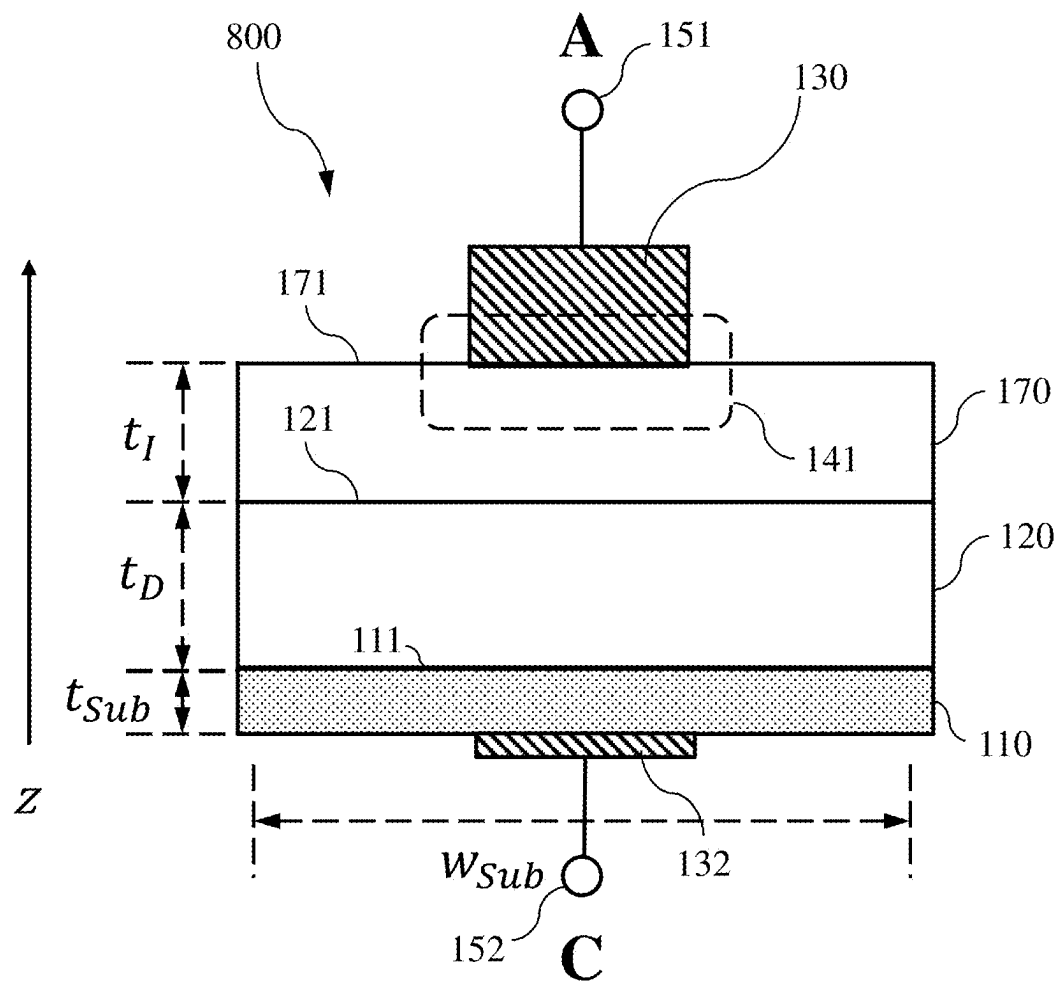


FIG. 8A

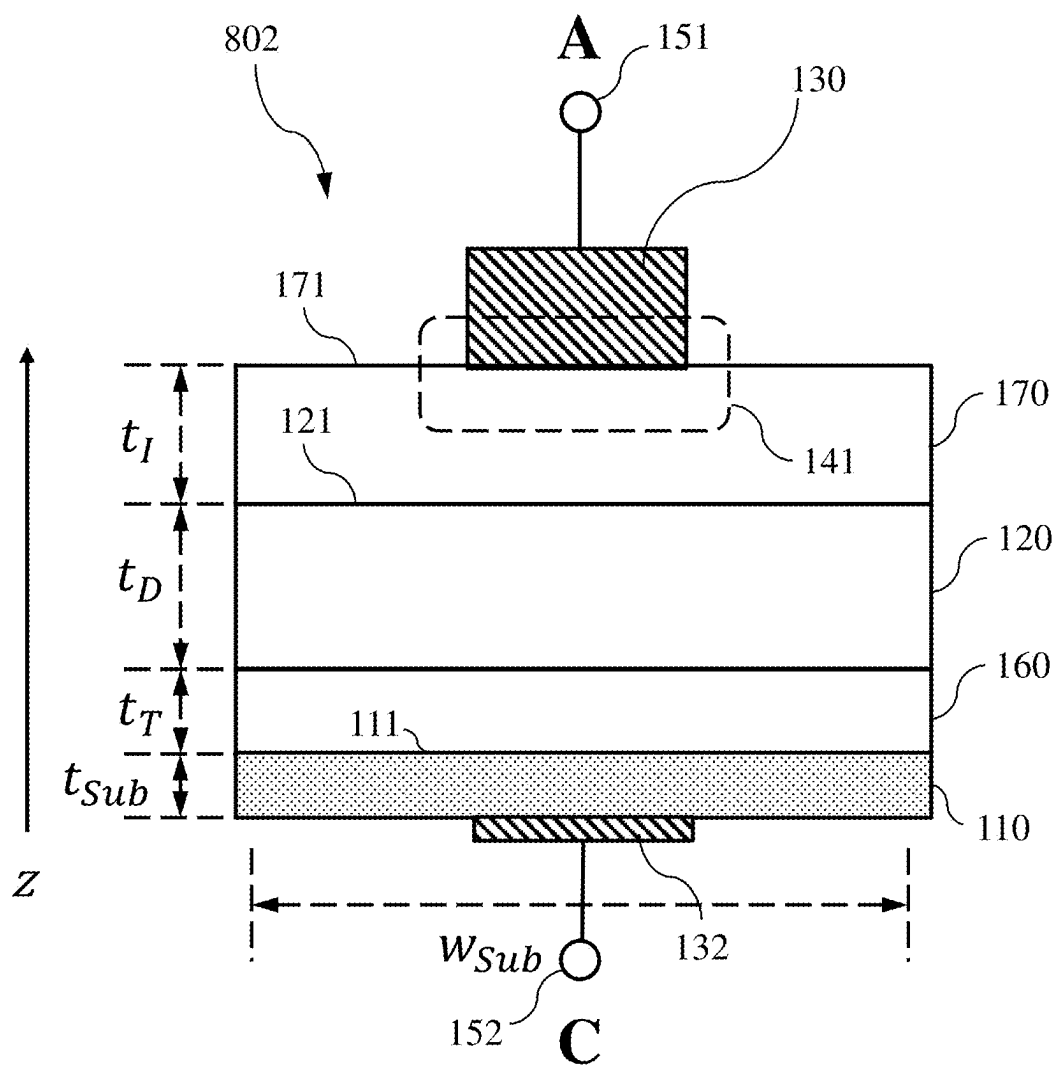


FIG. 8B

Metal E	Ohmic or Schottky			
Oxide	n	p	n	p
Substrate	n	n	p	p
Metal S	Ohmic			

FIG. 8C

Metal E	Ohmic or Schottky							
Oxide	n	p	n	p	n	p	n	p
TL	n	n	n	n	p	p	p	p
Substrate	n	n	p	p	n	n	p	p
Metal S	Ohmic							

FIG. 8D

Metal E	Ohmic or Schottky							
IL	n	n	n	n	p	p	p	p
Oxide	n	p	n	p	n	p	n	p
Substrate	n	n	p	p	n	n	p	p
Metal S	Ohmic							

FIG. 8E

Metal E	Ohmic or Schottky													
IL	p	n	p	n	p	p	n	n	p	n	p	n	p	p
Oxide	n	p	p	n	n	p	n	p	p	n	n	p	p	p
TL	n	n	n	n	n	n	p	p	p	p	p	p	p	p
Substrate	n	n	n	p	p	p	n	n	n	p	p	p	p	p
Metal S	Ohmic													

FIG. 8F

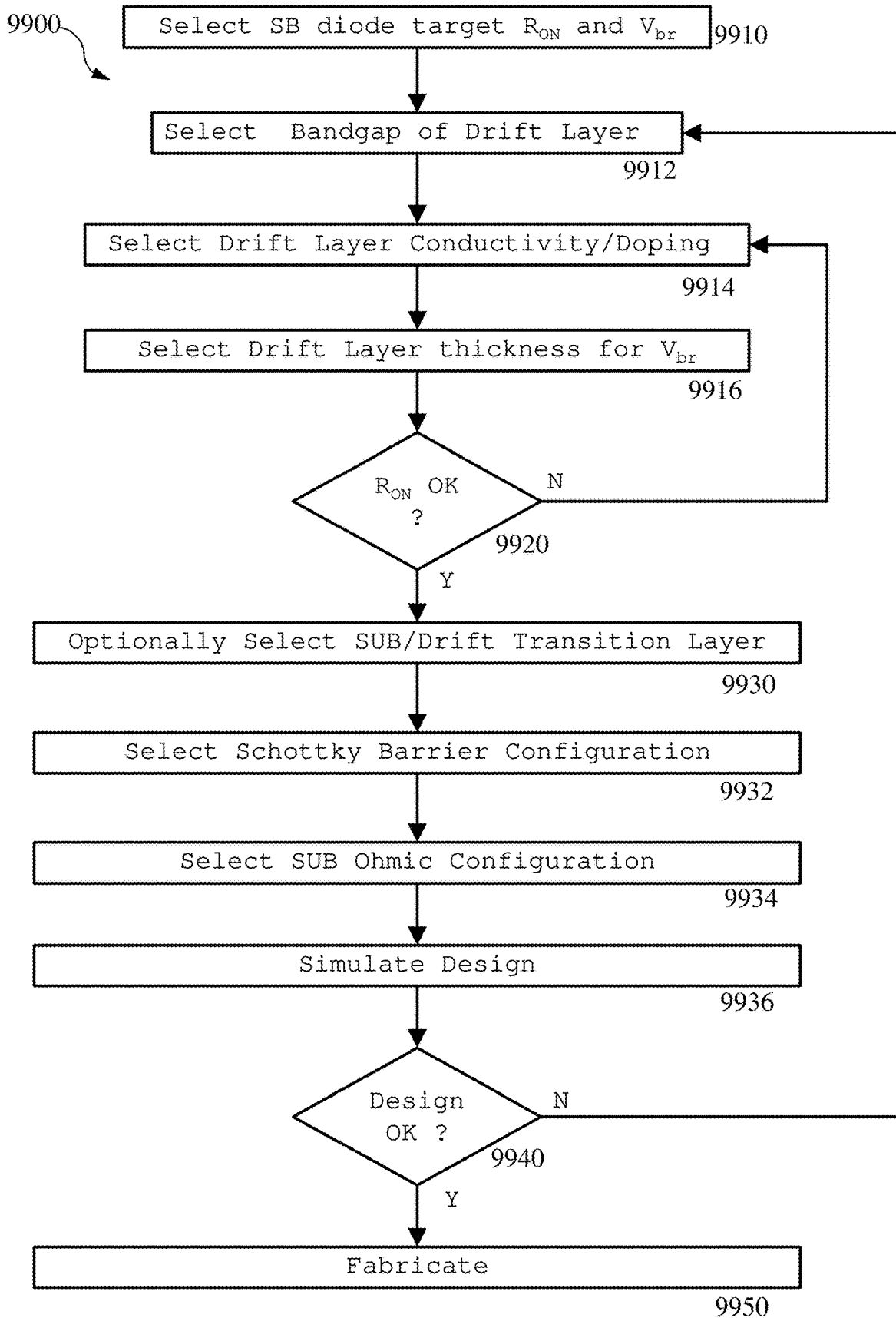
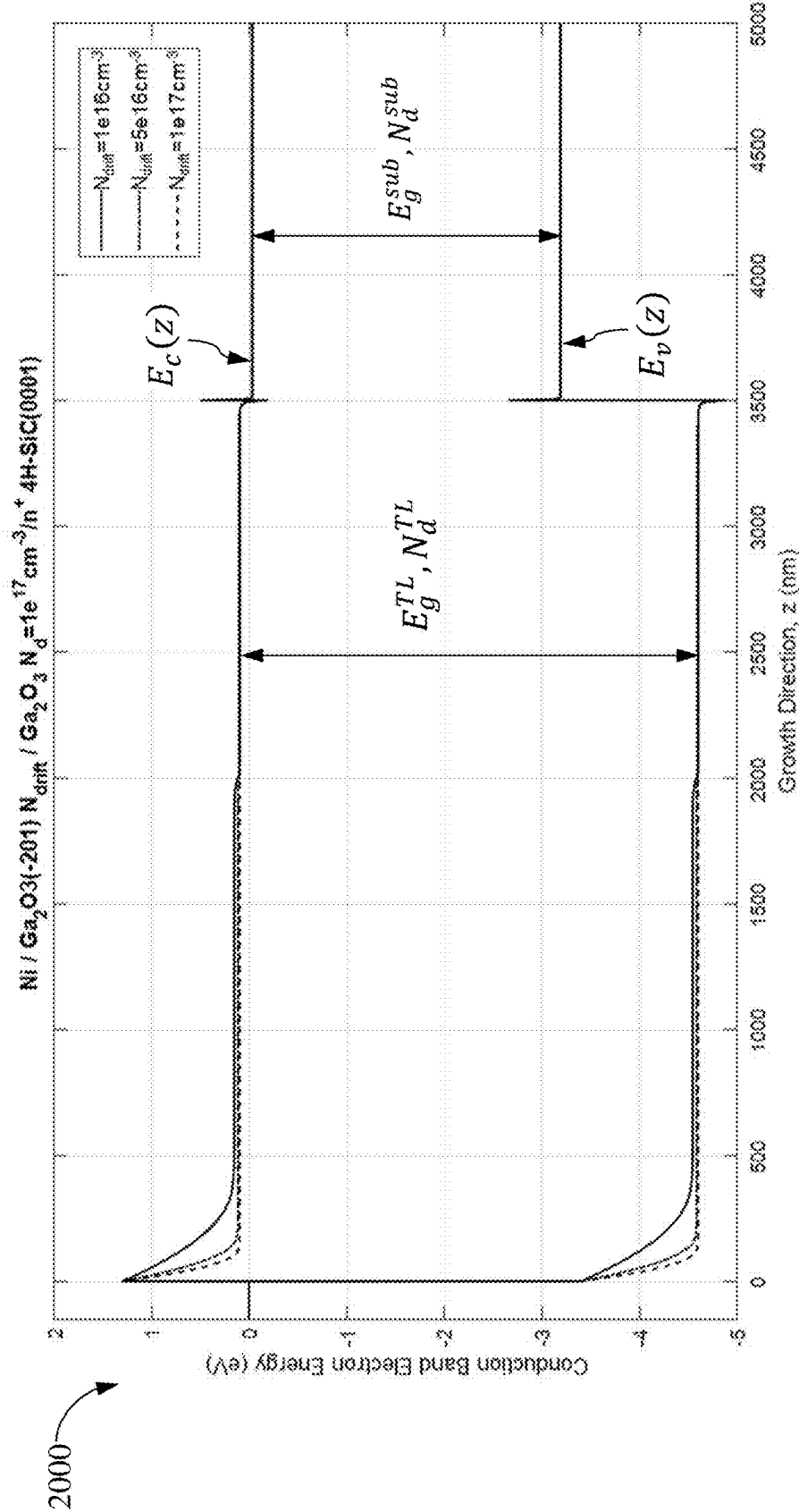


FIG. 9A



Ni	$E_g^{\text{drift}}, N_d^{\text{drift}}$	$E_g^{\text{TL}}, N_d^{\text{TL}}$	$E_g^{\text{sub}}, N_d^{\text{sub}}$
2012	2014	2016	

2010

FIG. 9B

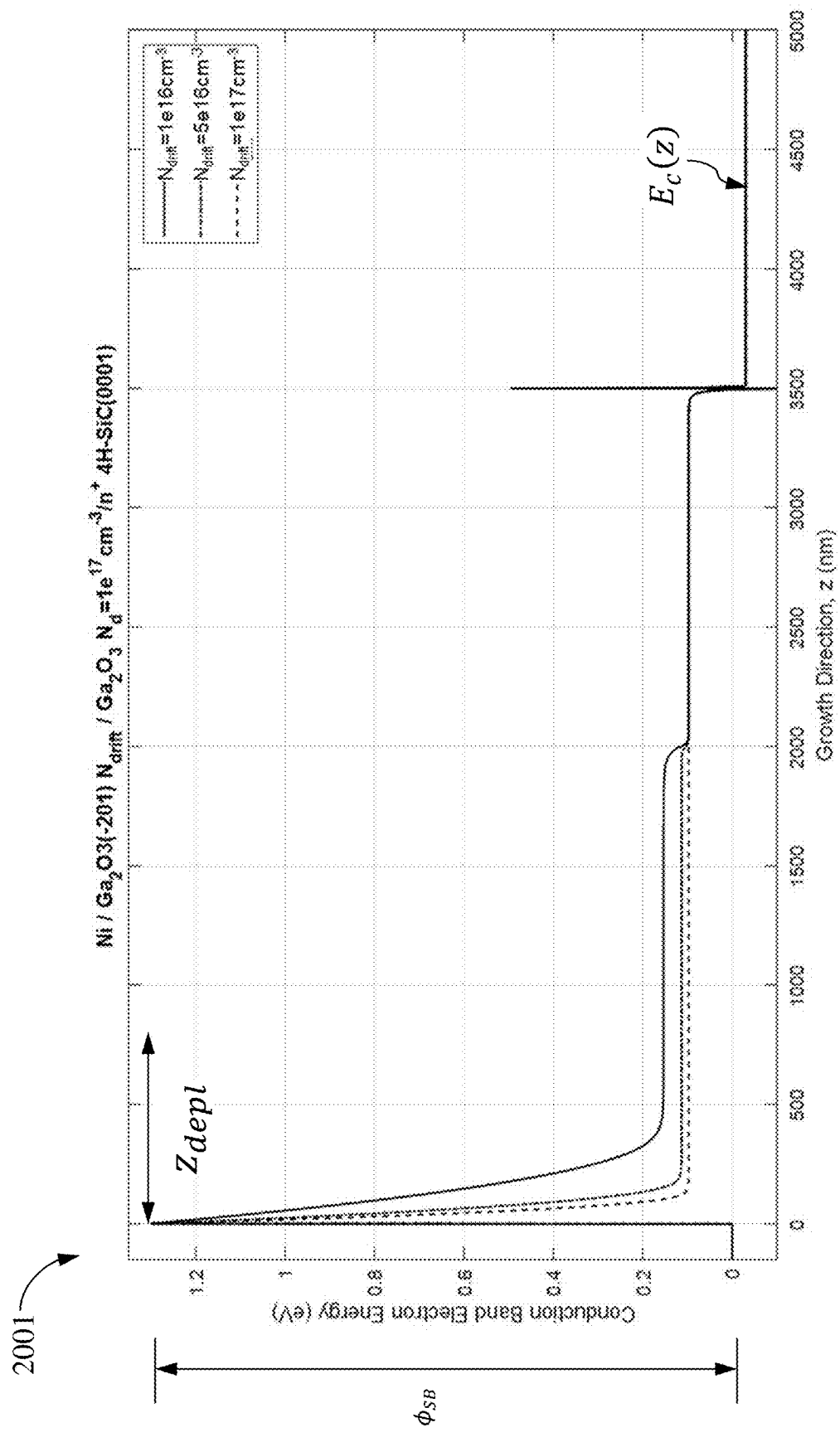


FIG. 9C

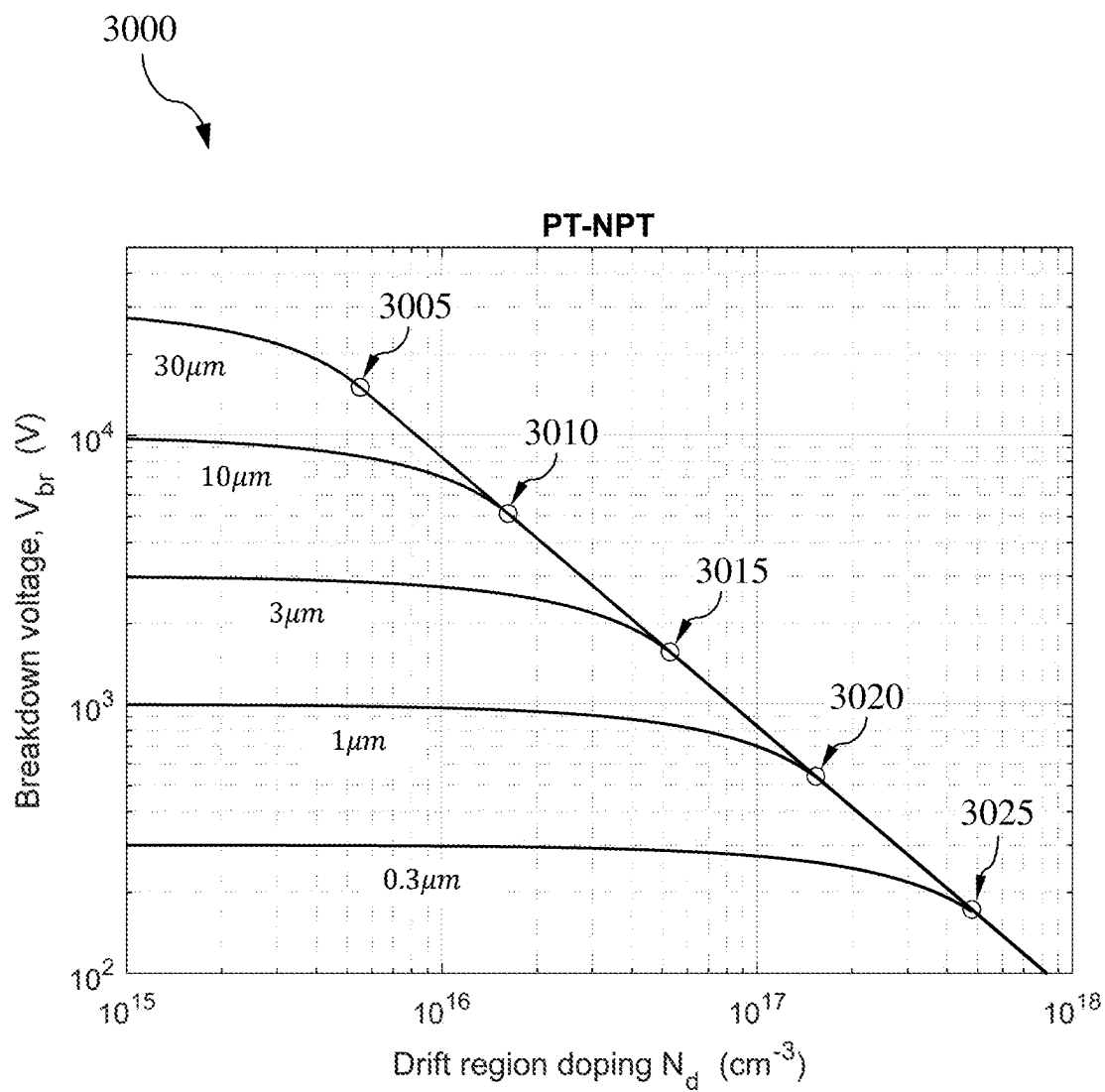


FIG. 10

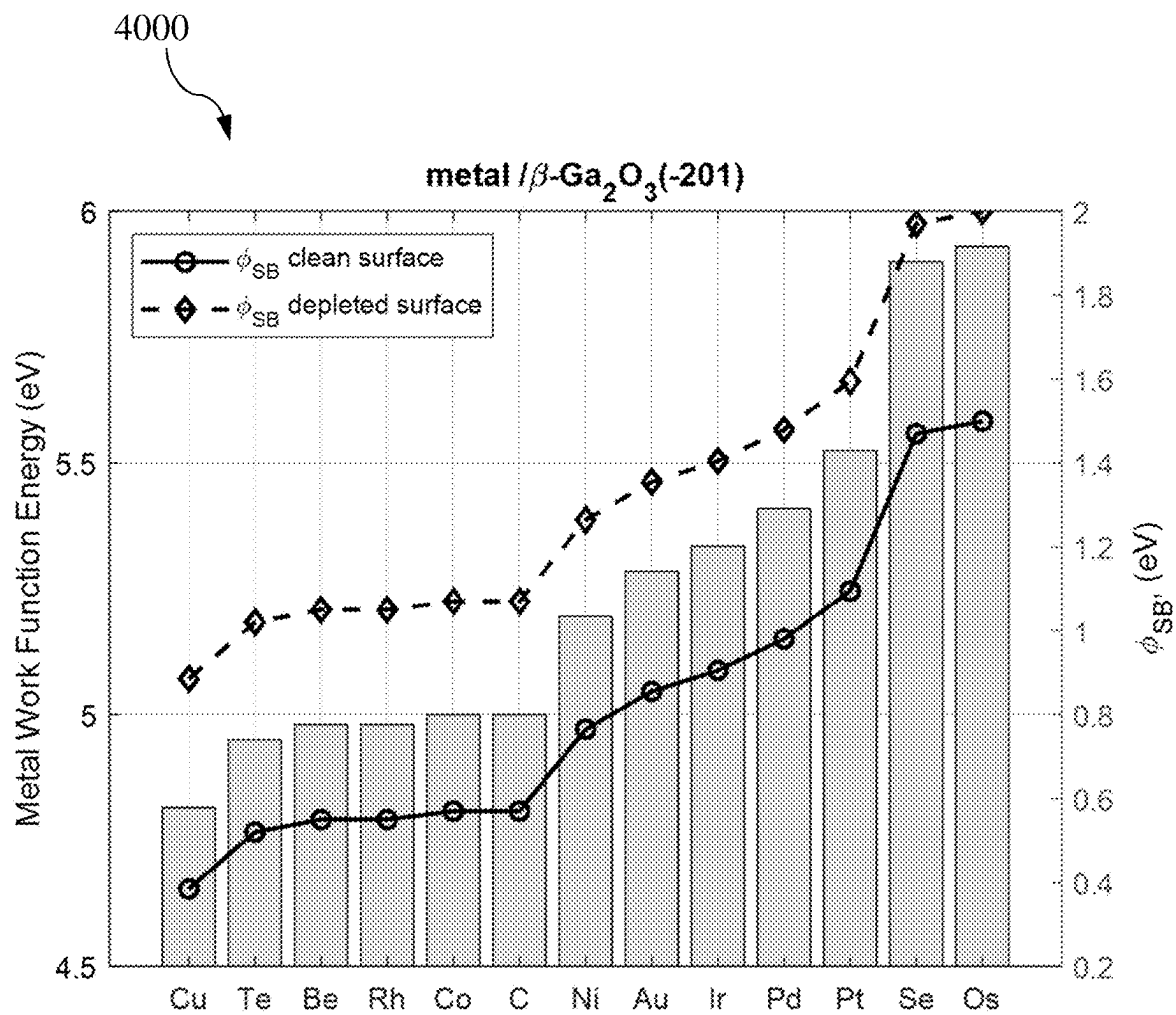


FIG. 11A

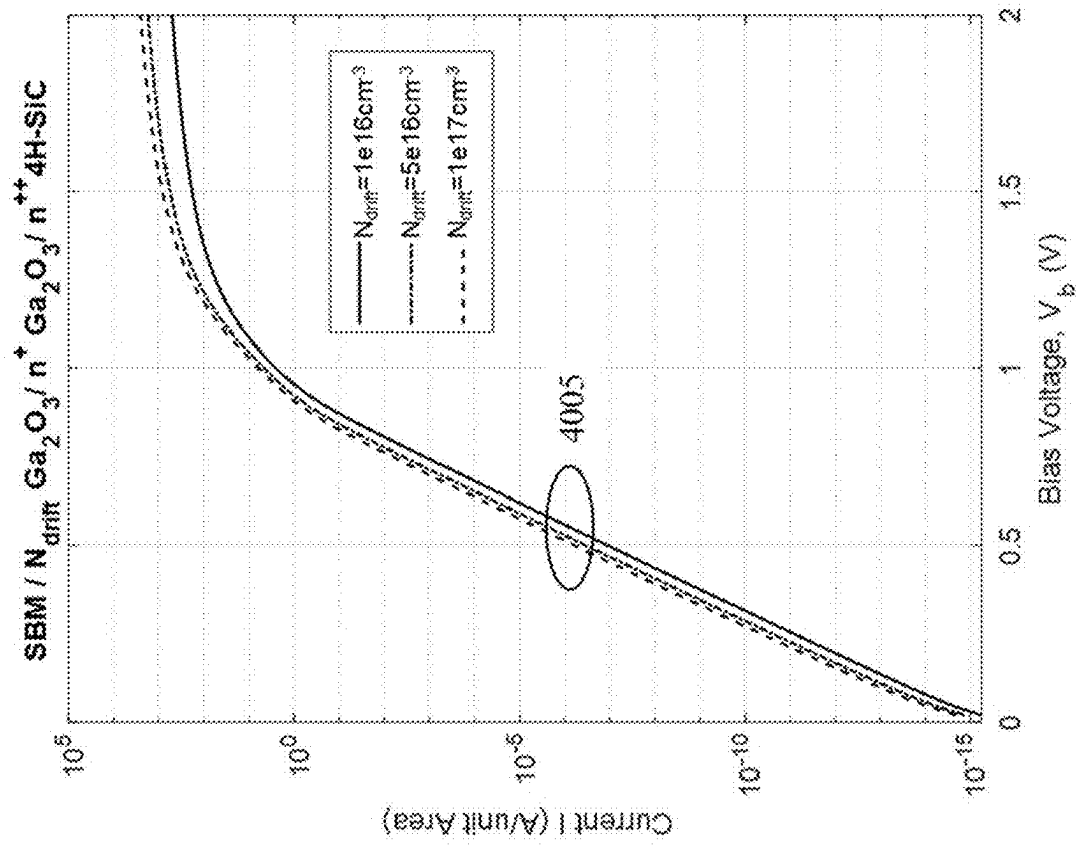


FIG. 11B

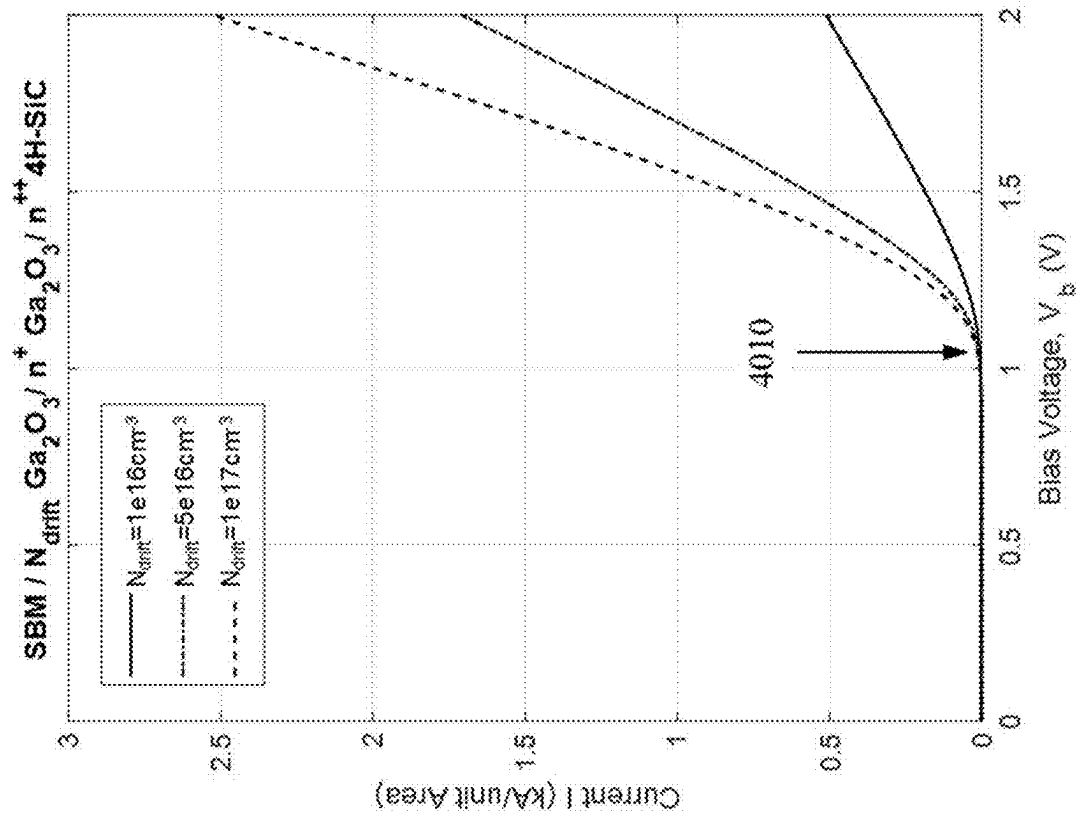
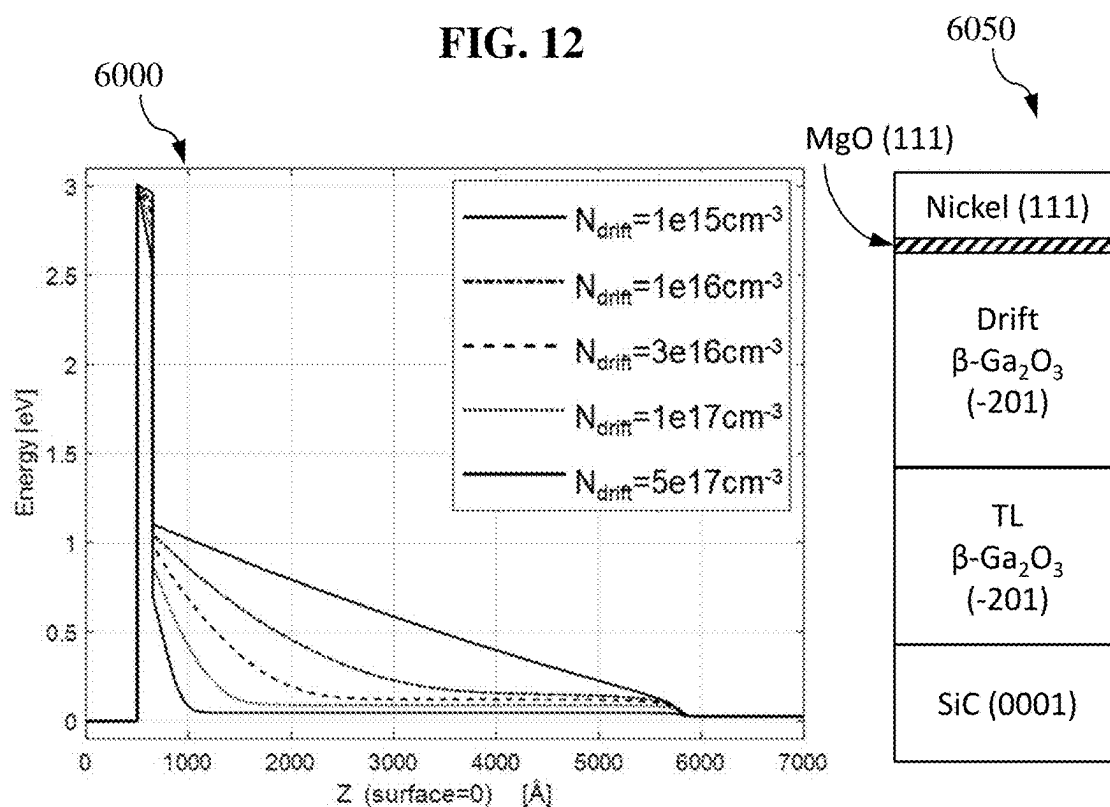
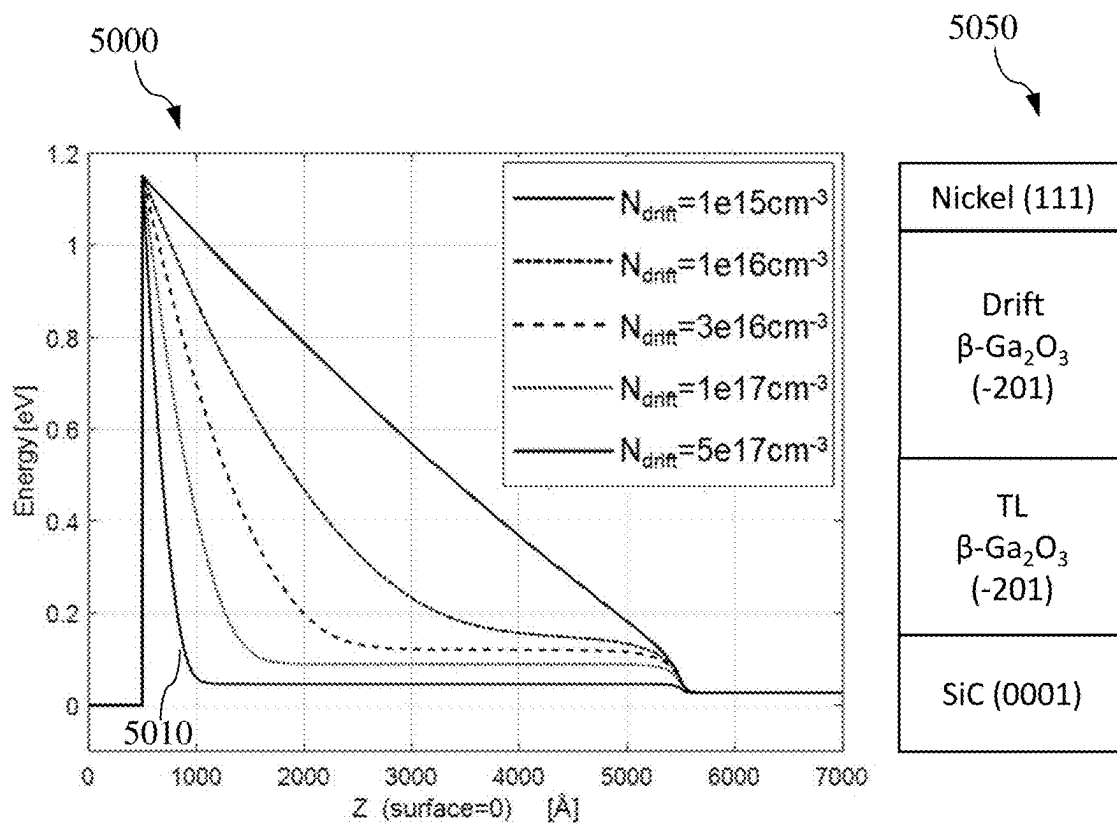


FIG. 11C



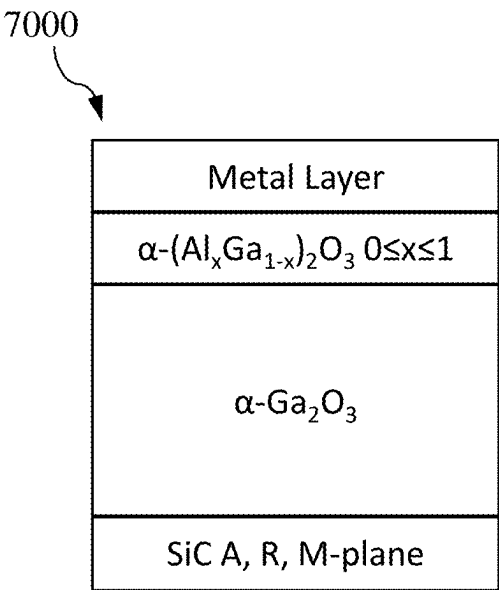


FIG. 14

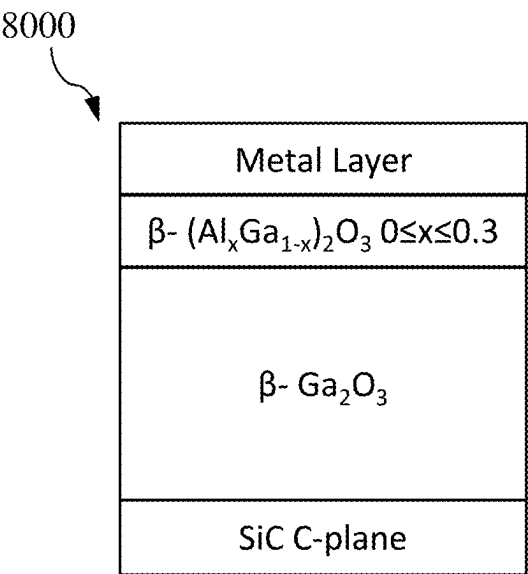


FIG. 15

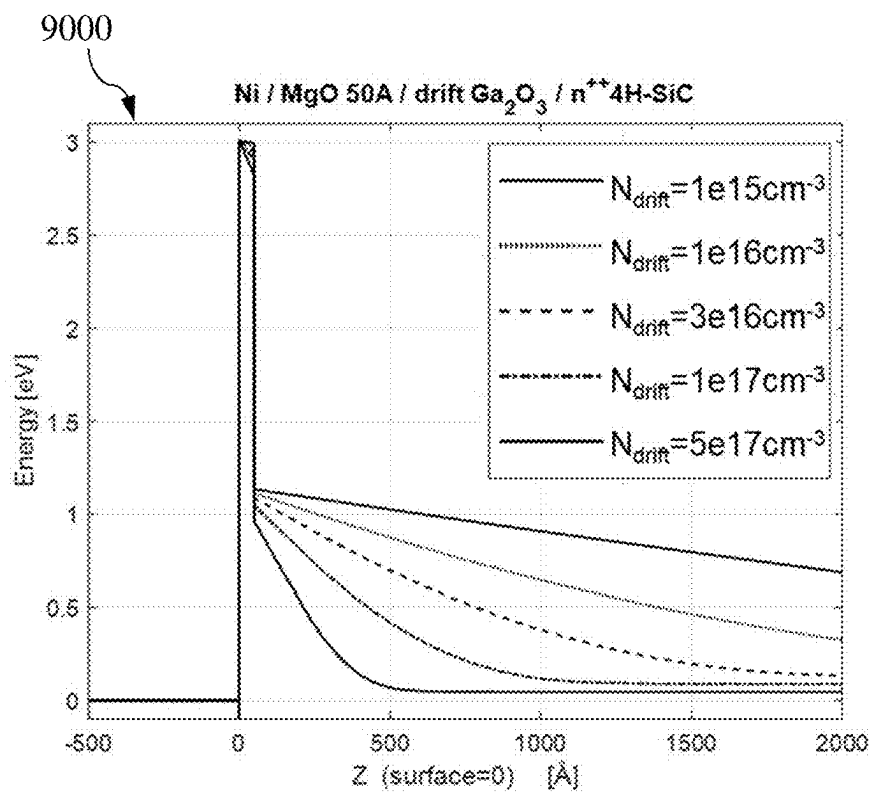


FIG. 16

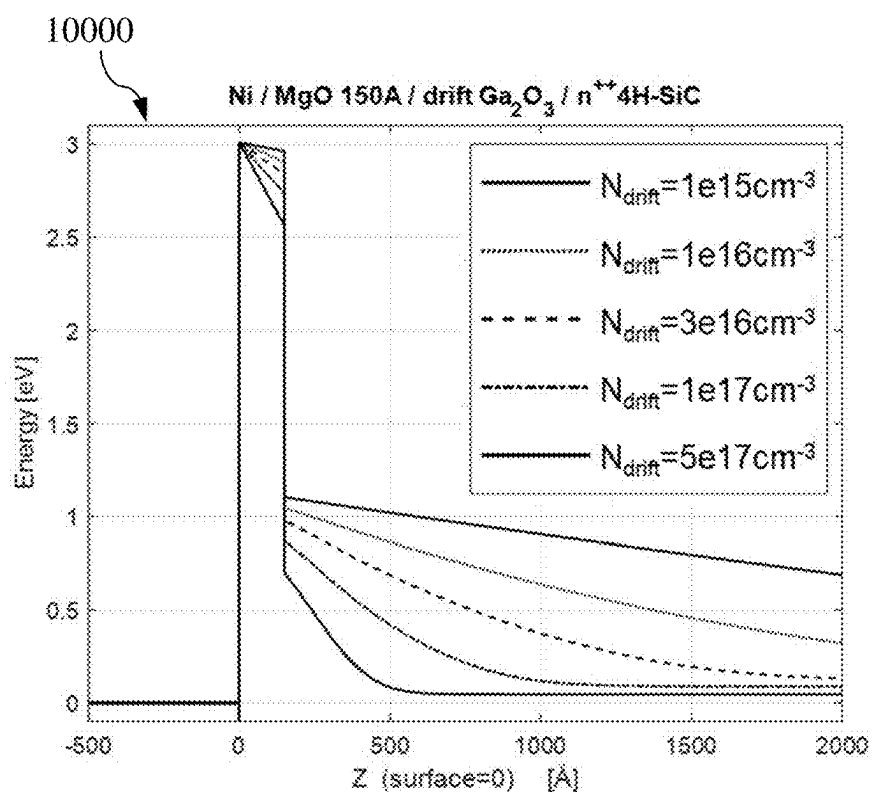


FIG. 17

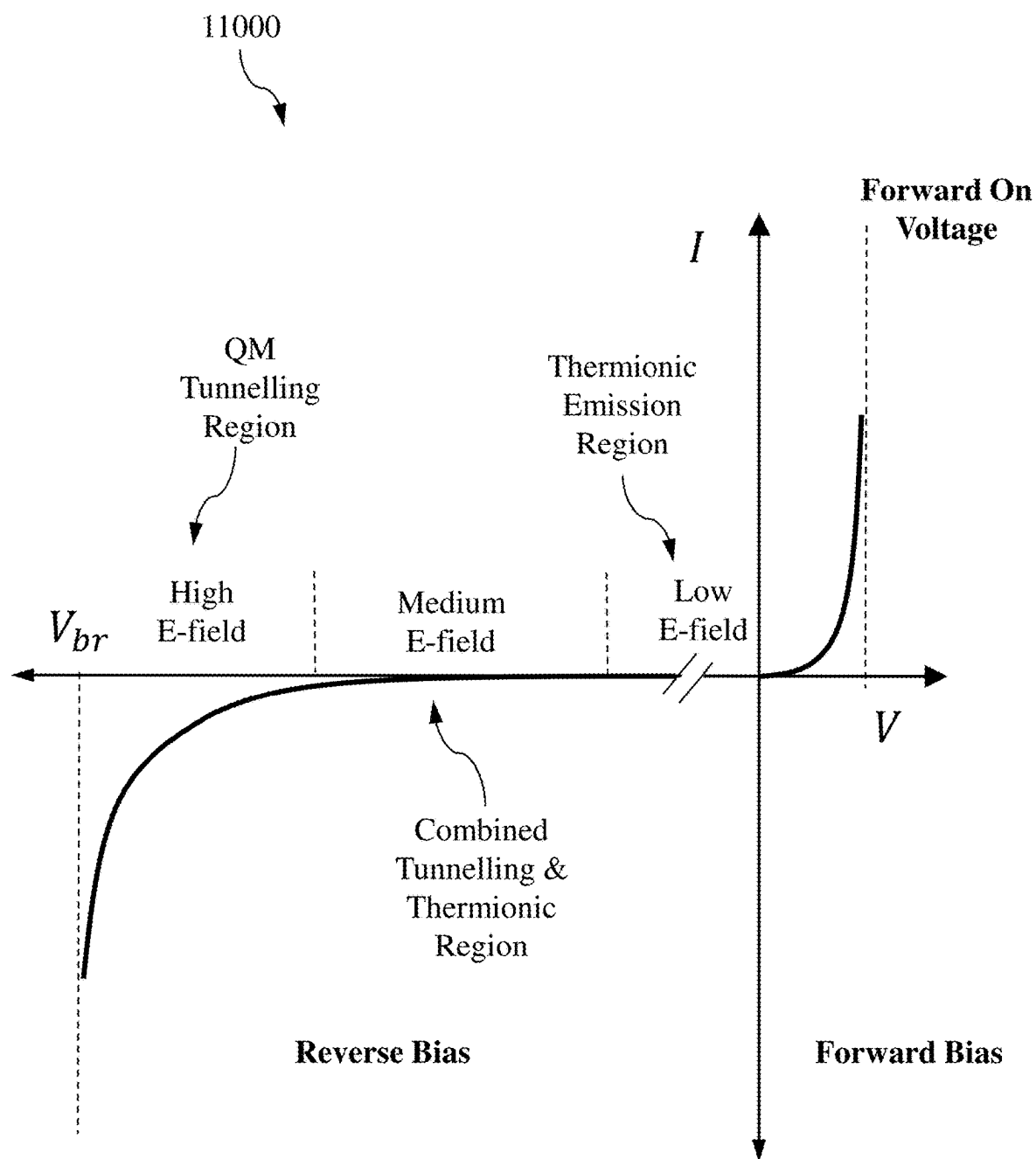


FIG. 18

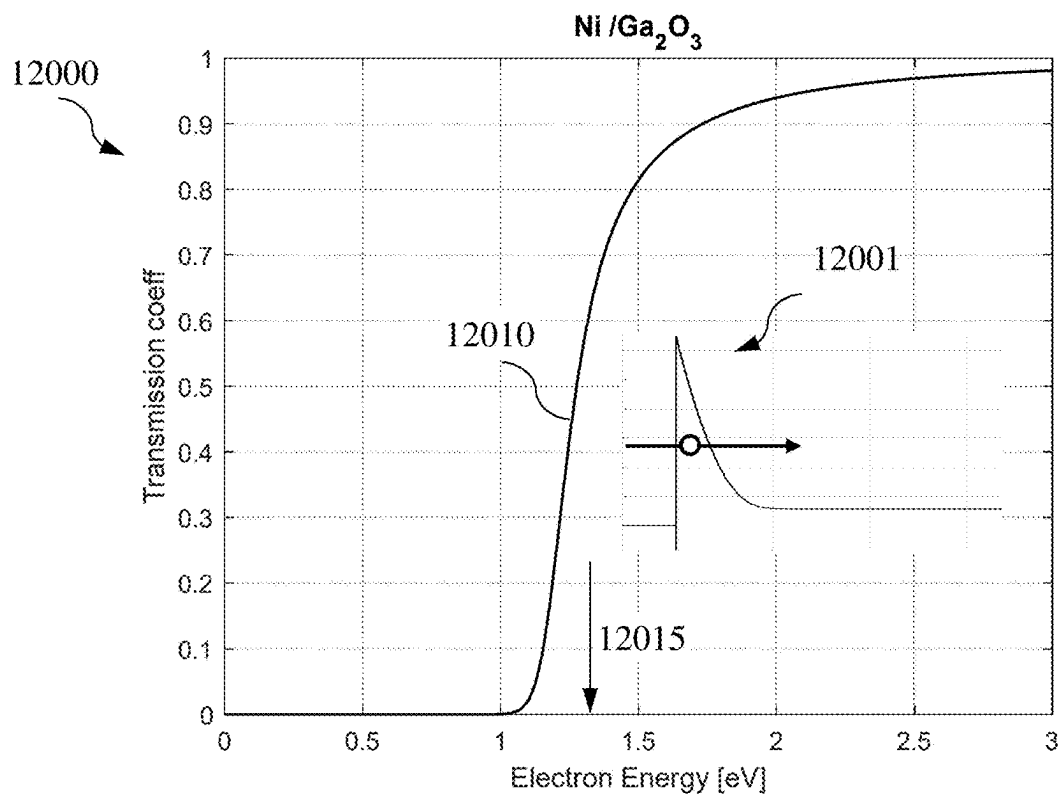


FIG. 19A

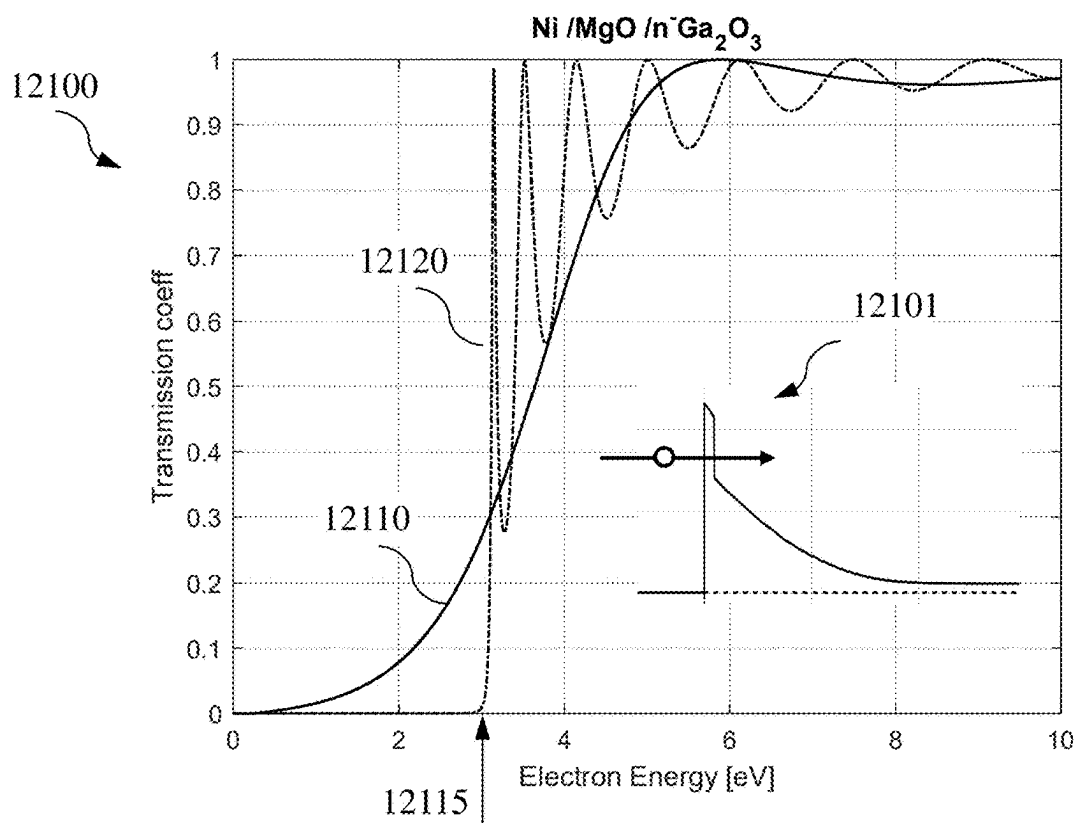


FIG. 19B

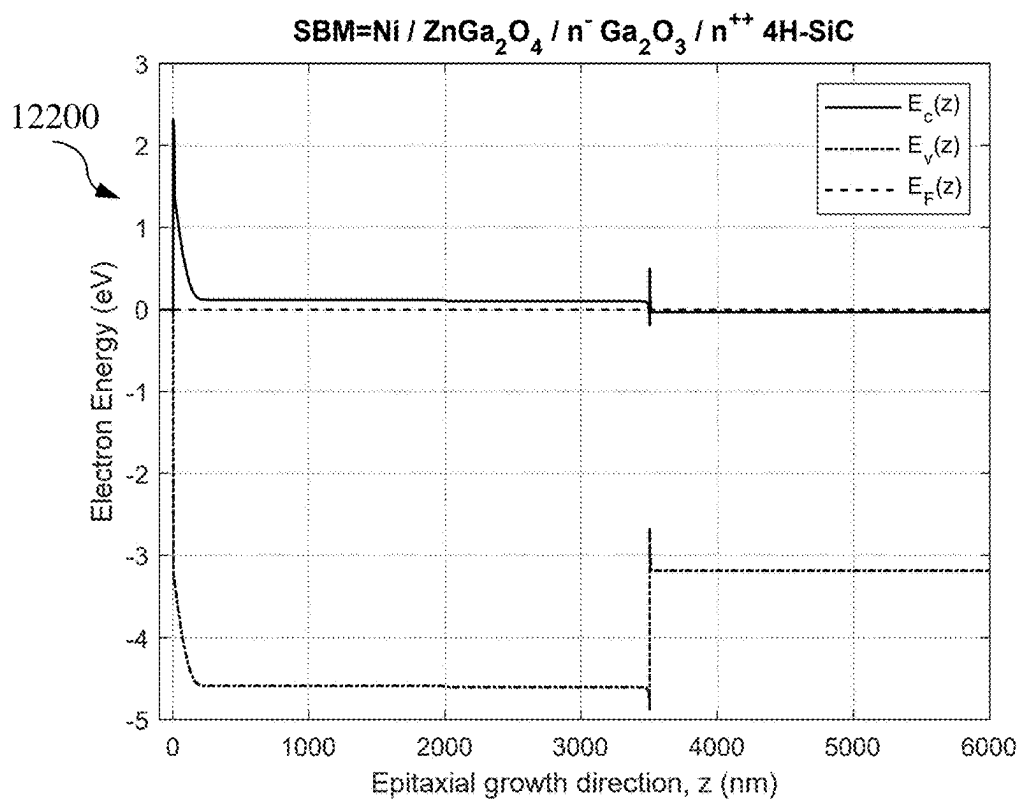


FIG. 19C

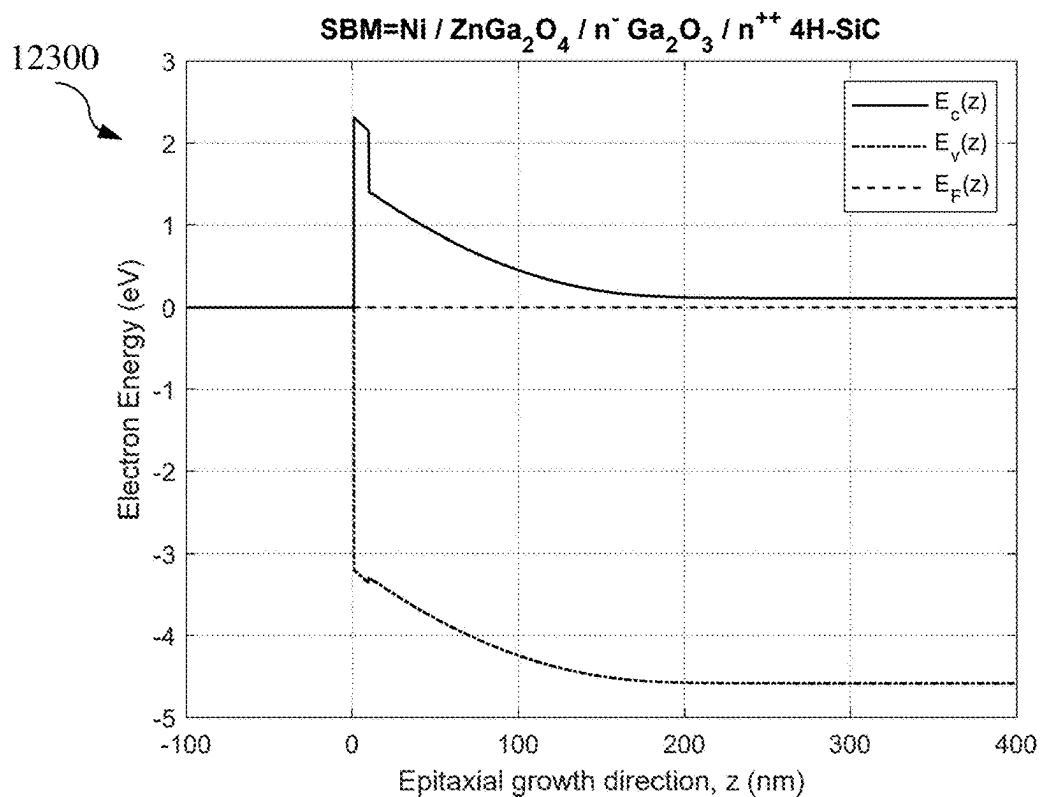


FIG. 19D

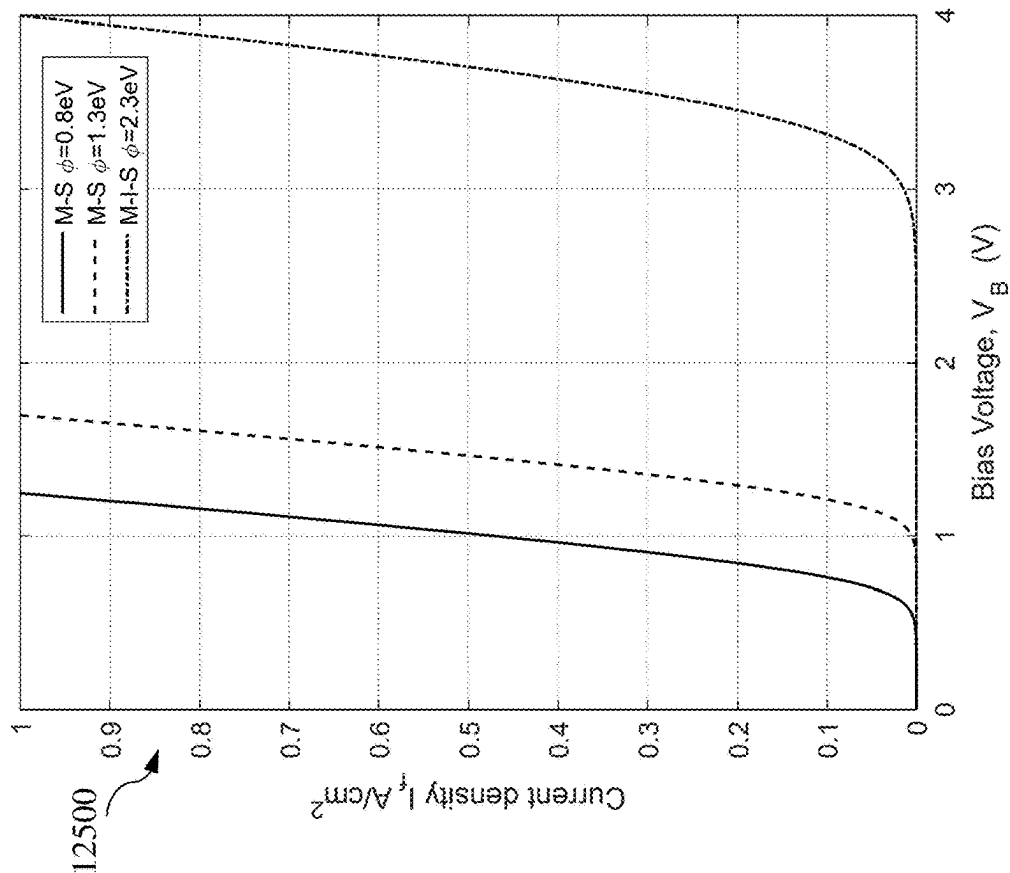


FIG. 19F

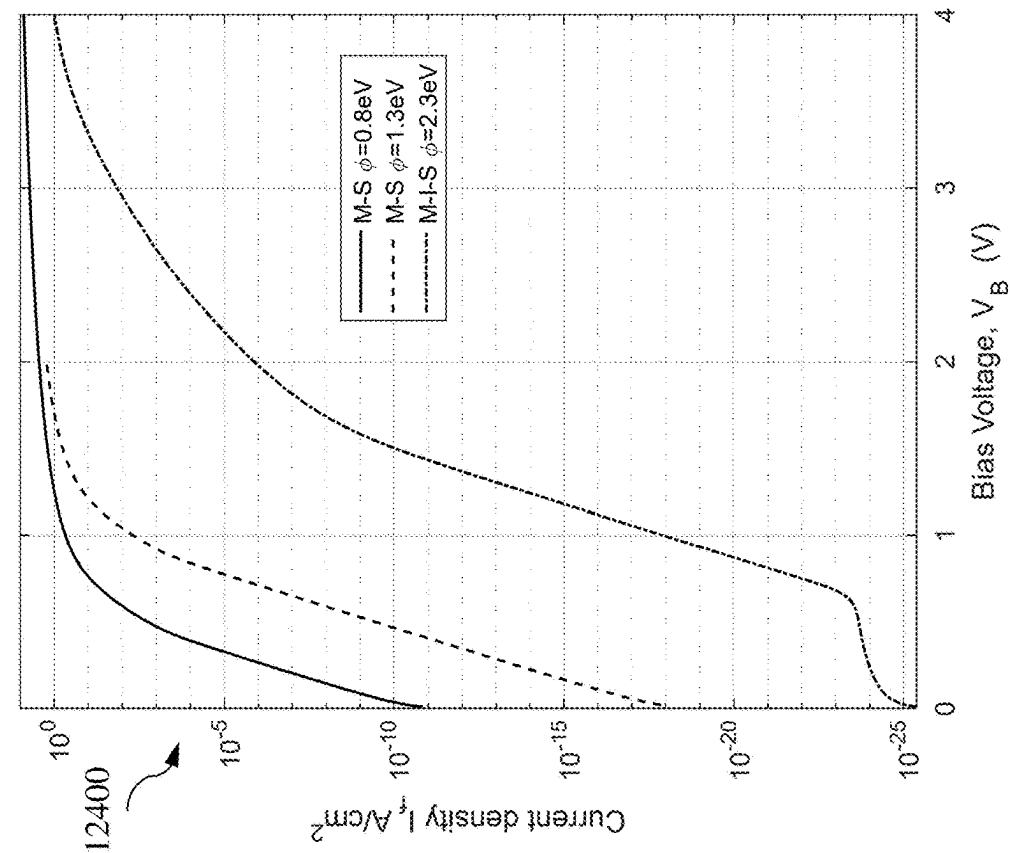


FIG. 19E

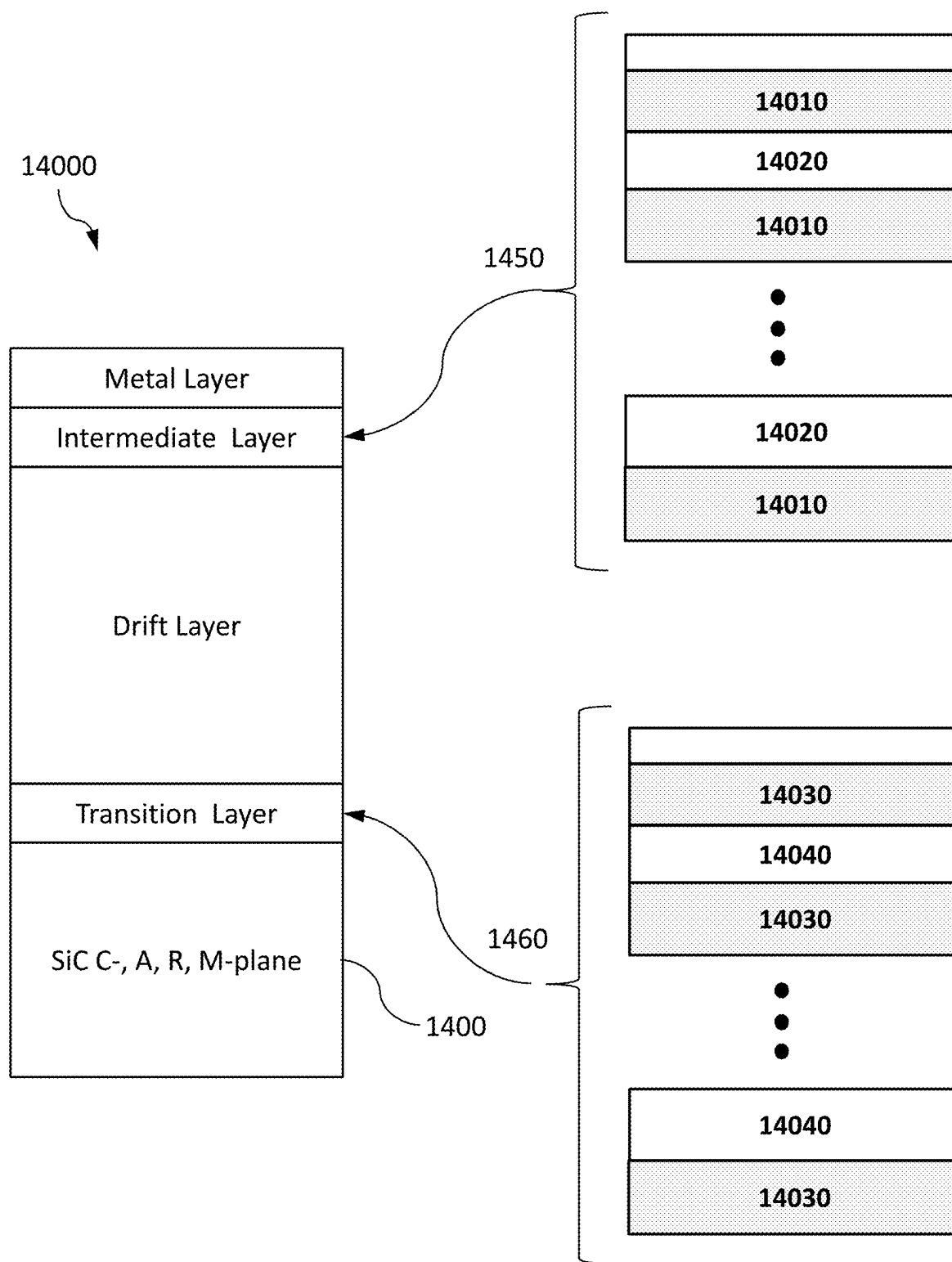


FIG. 20

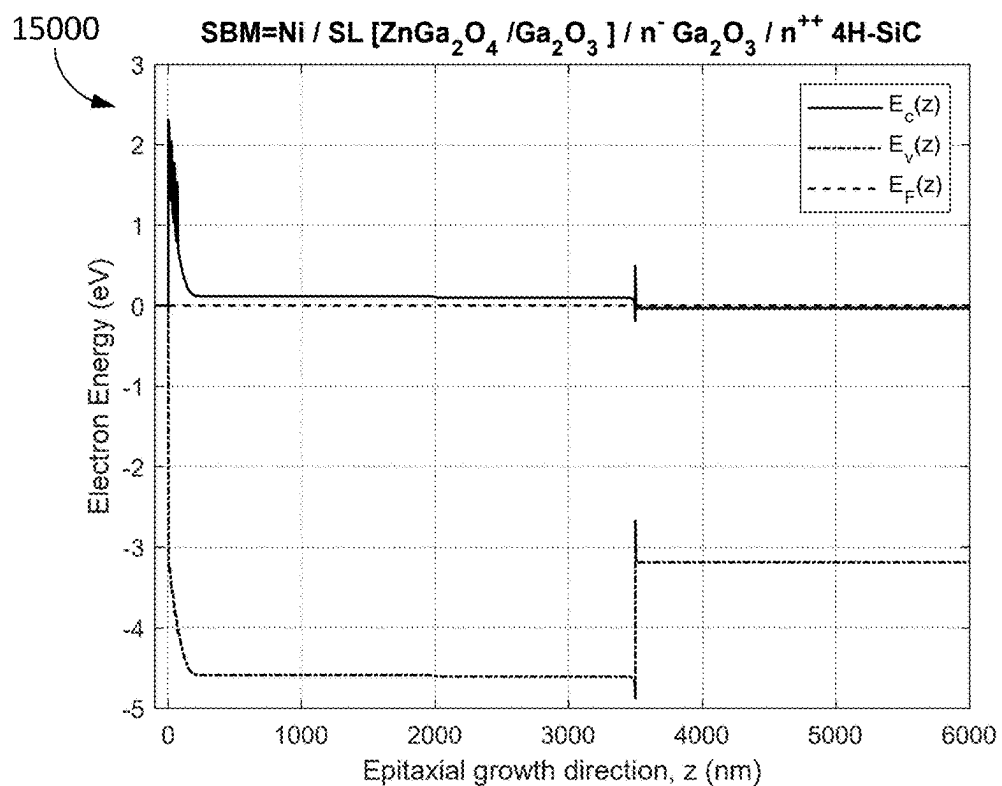


FIG. 21A

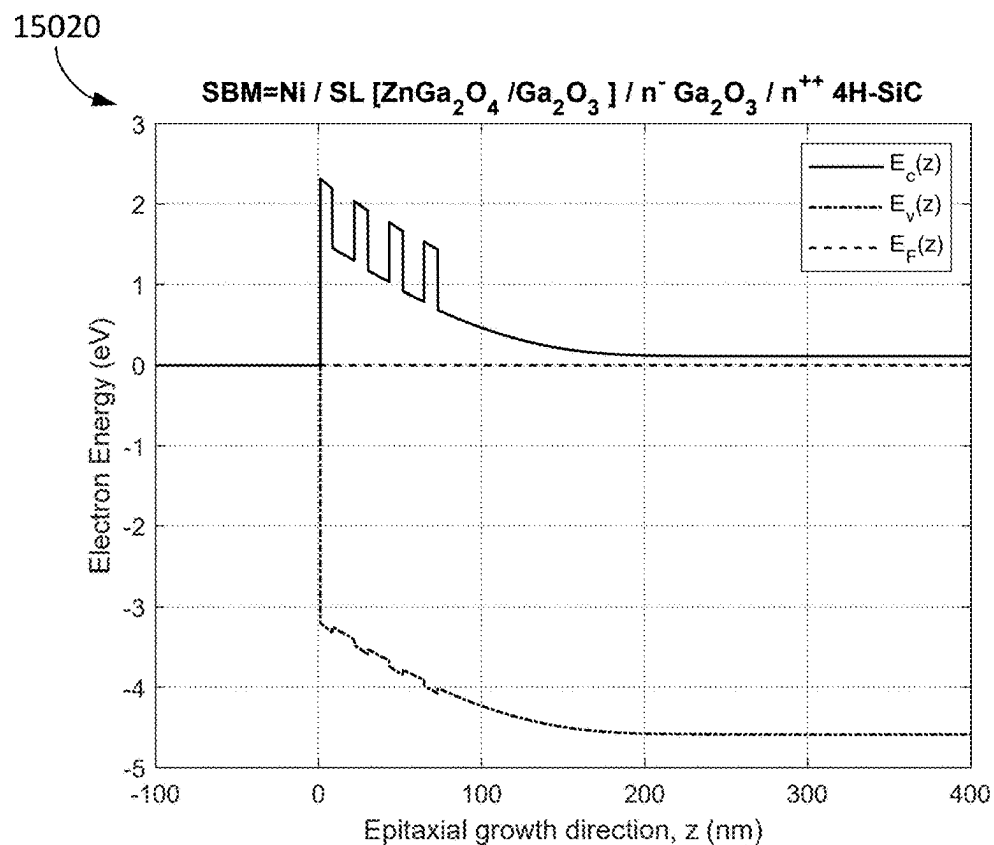


FIG. 21B

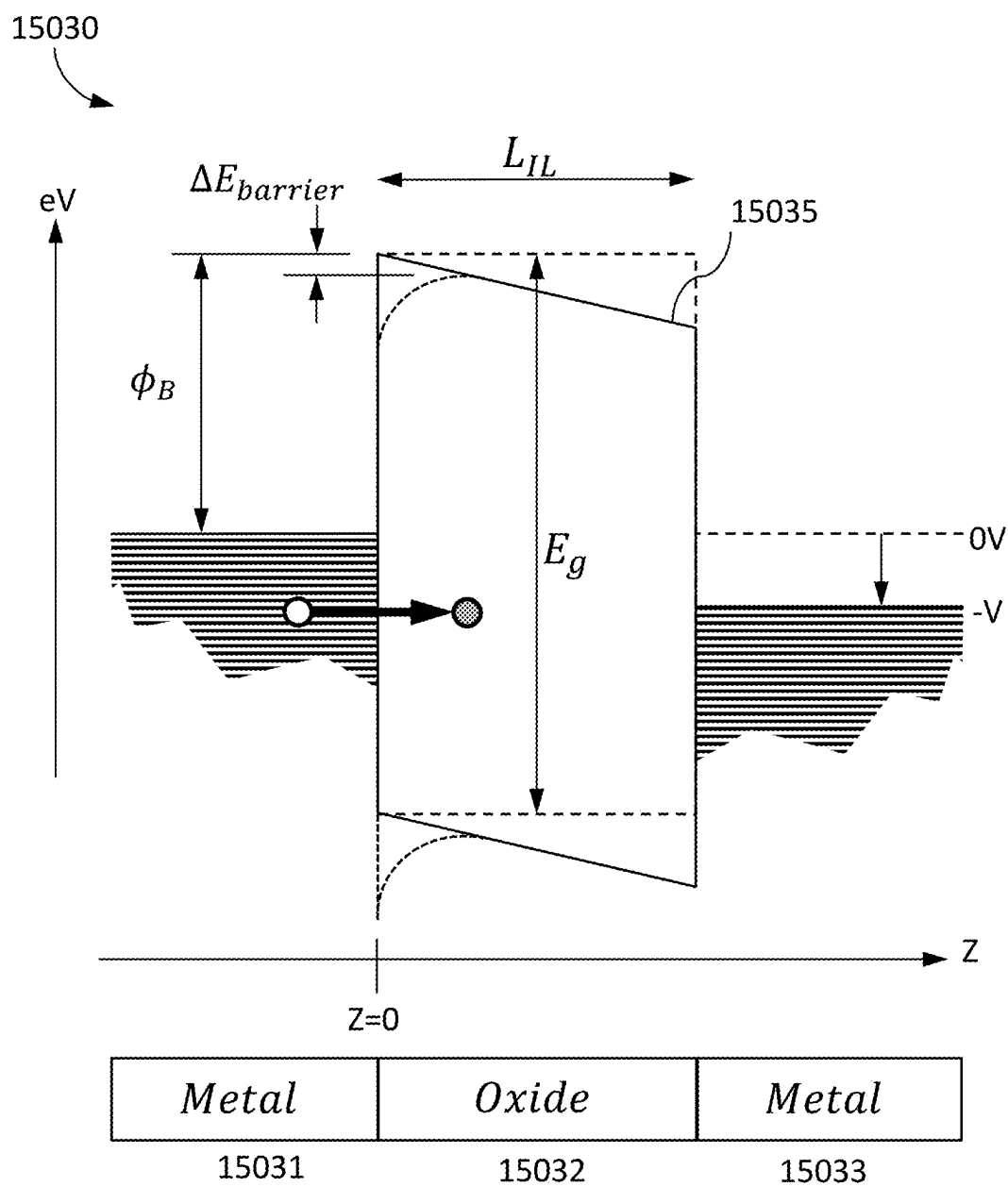


FIG. 21C

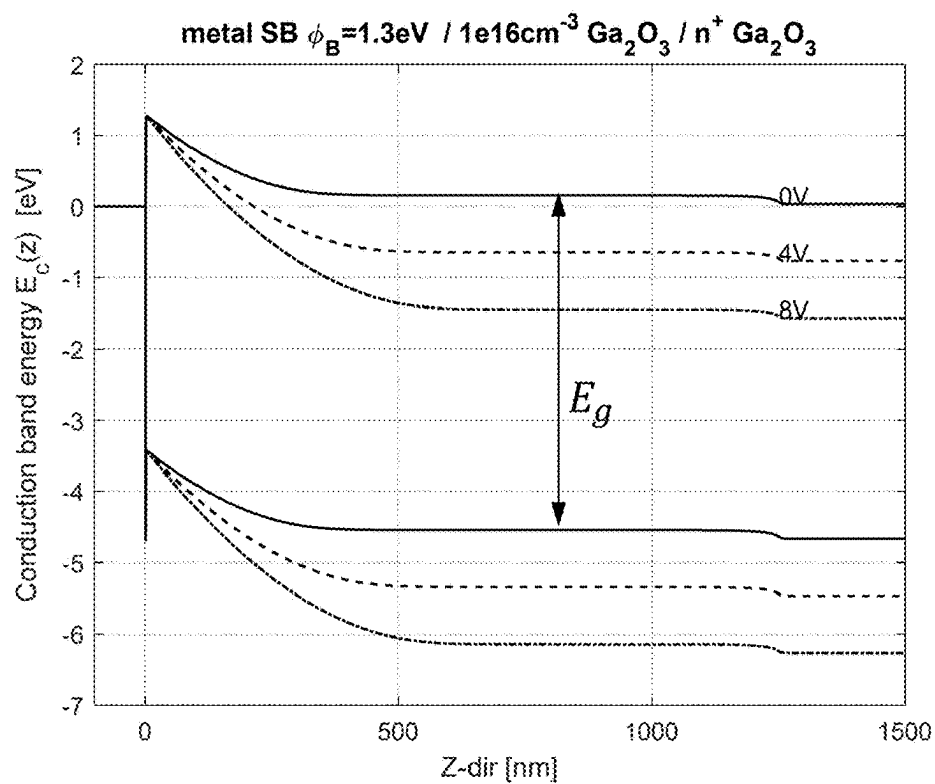


FIG. 22

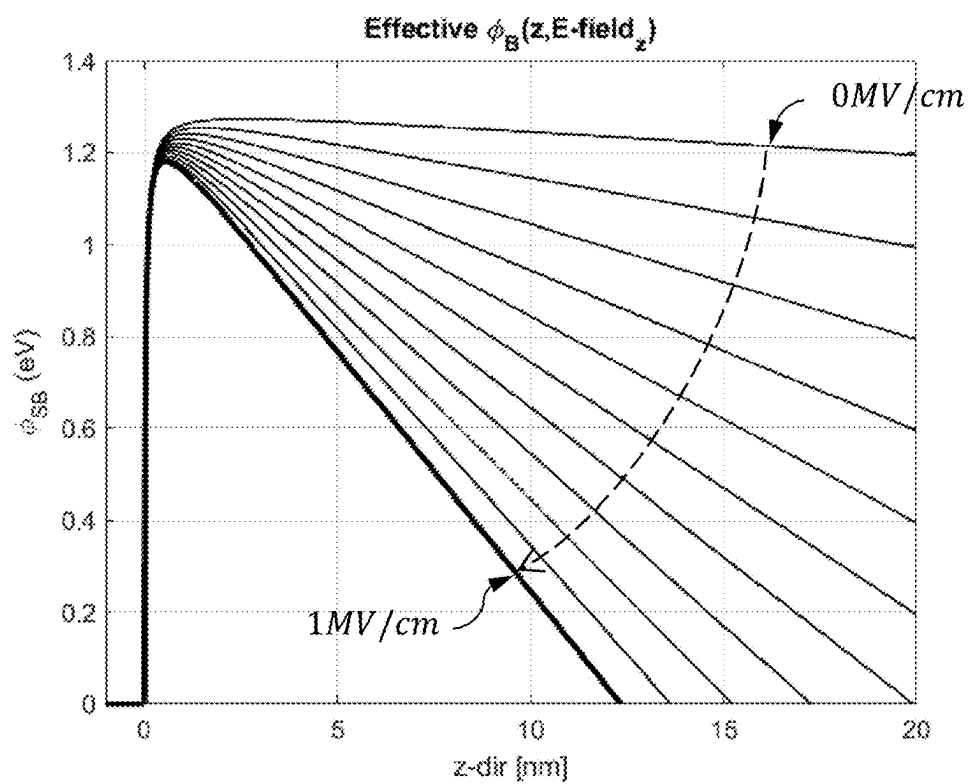


FIG. 23

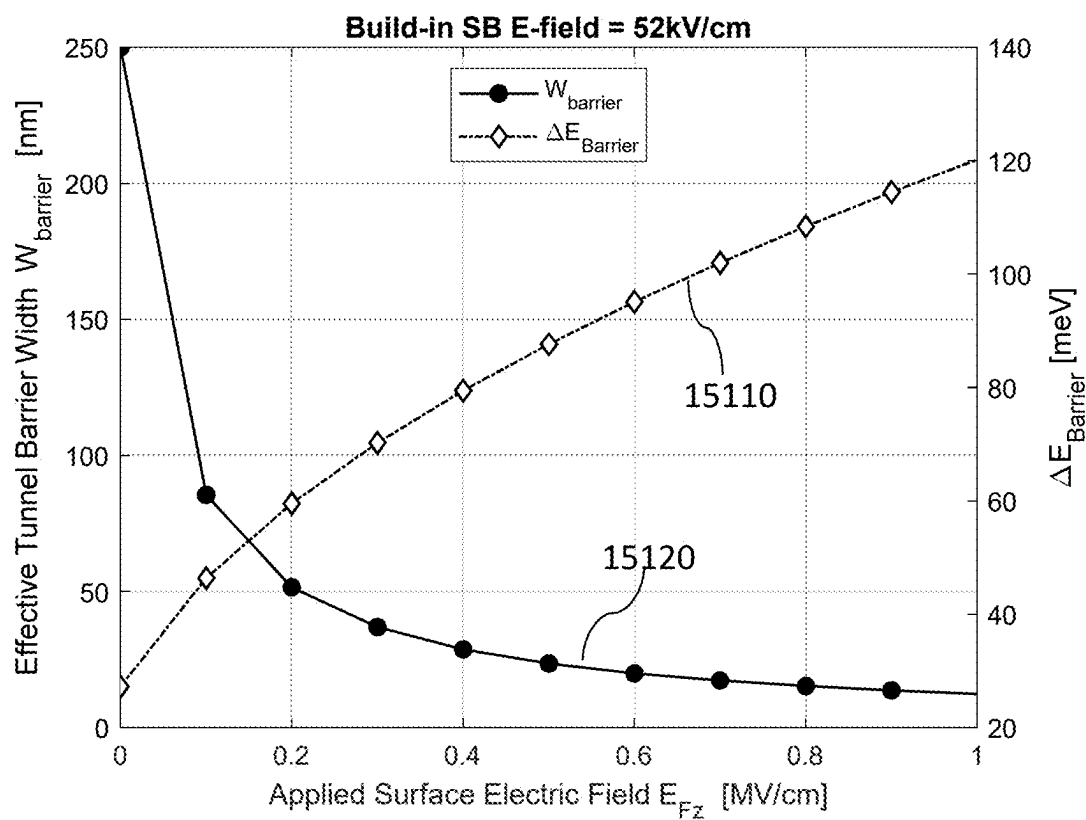


FIG. 24

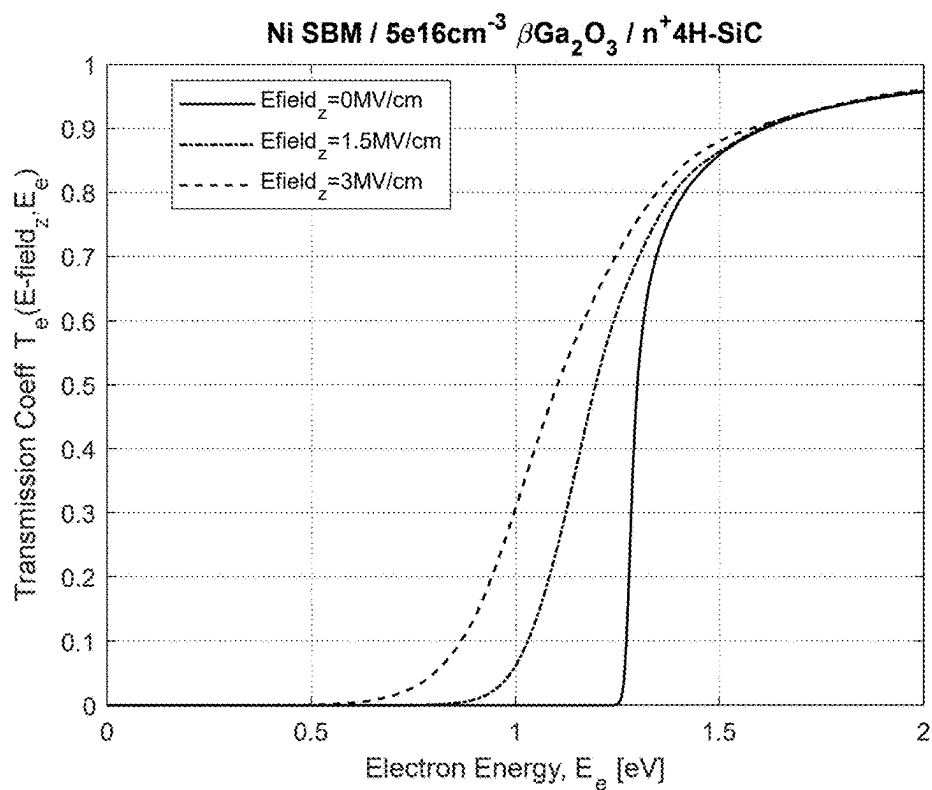


FIG. 25

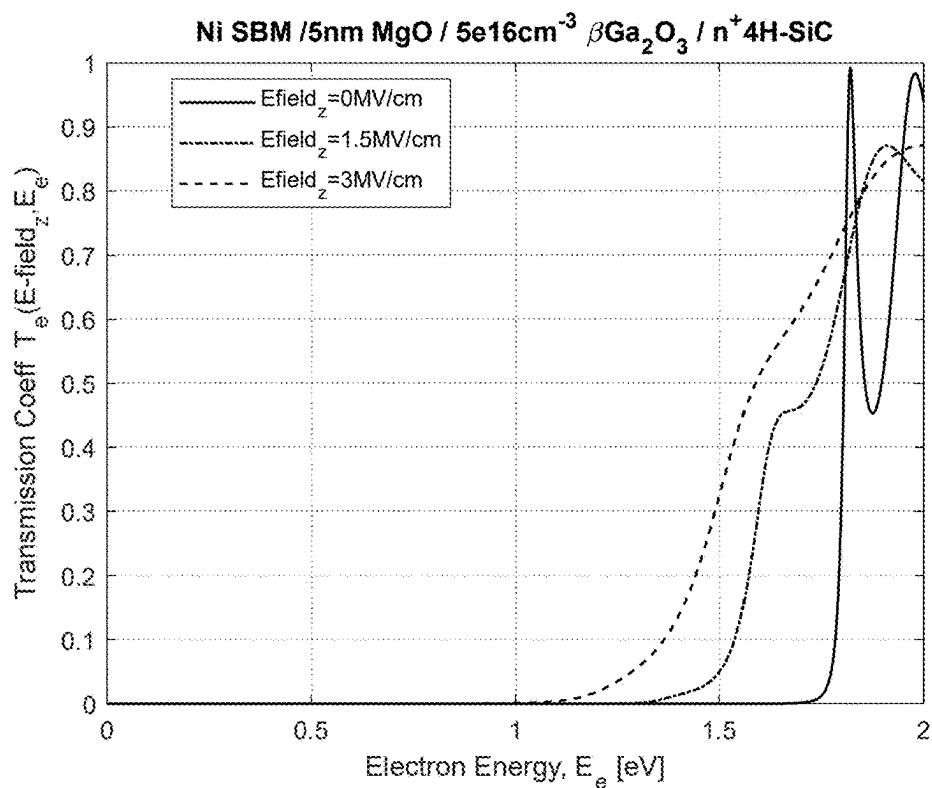


FIG. 26

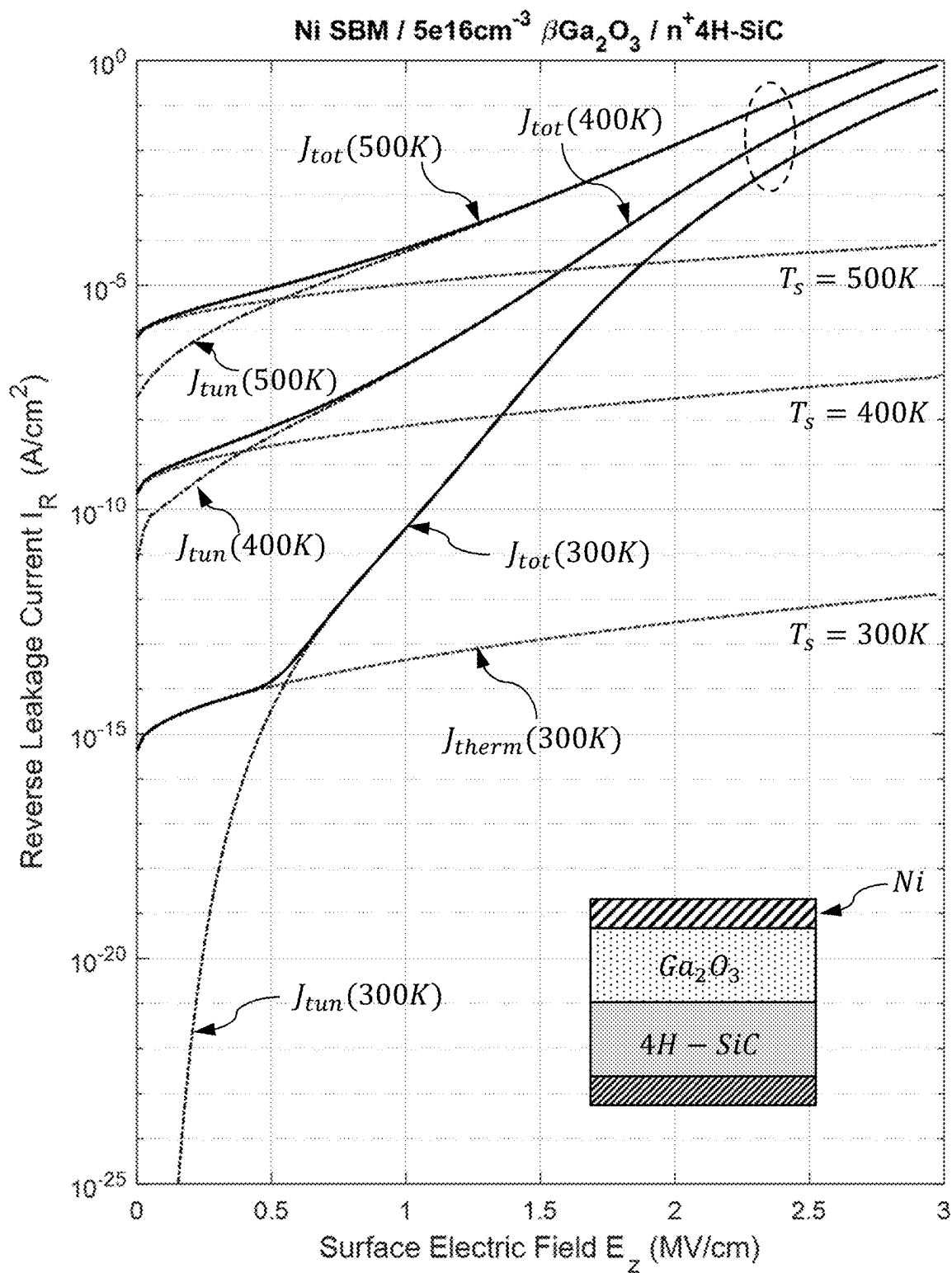


FIG. 27

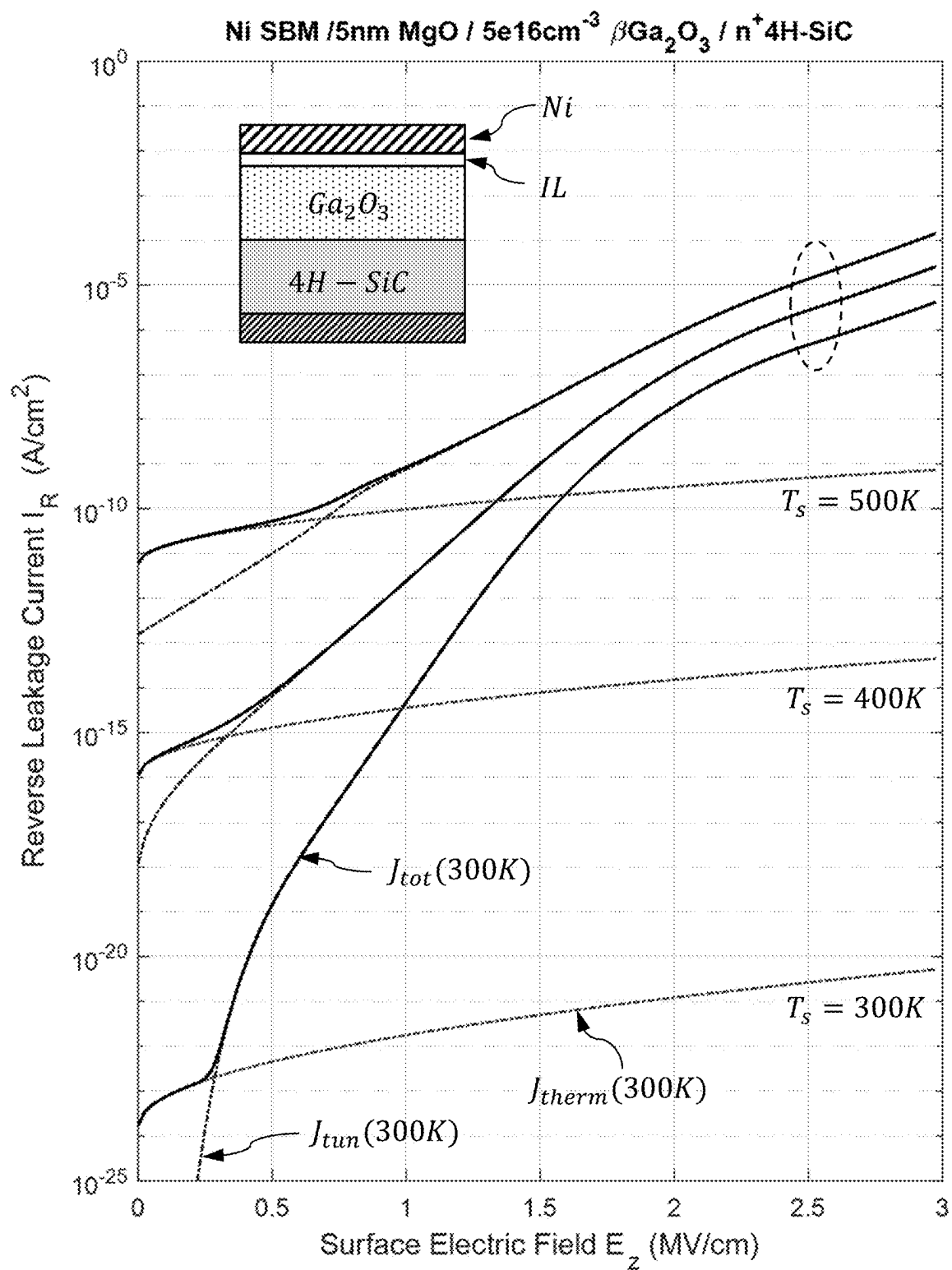


FIG. 28

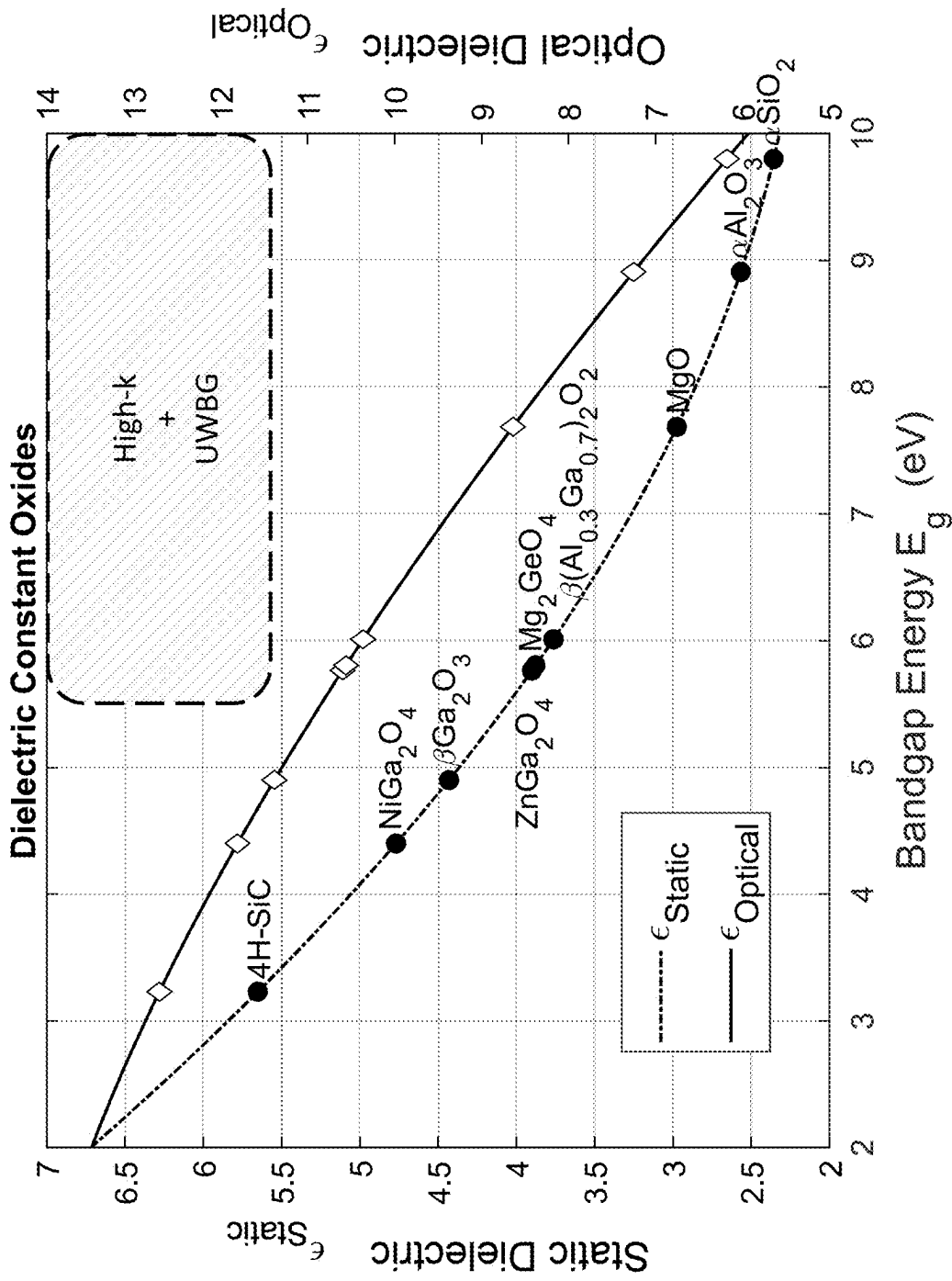


FIG. 29A

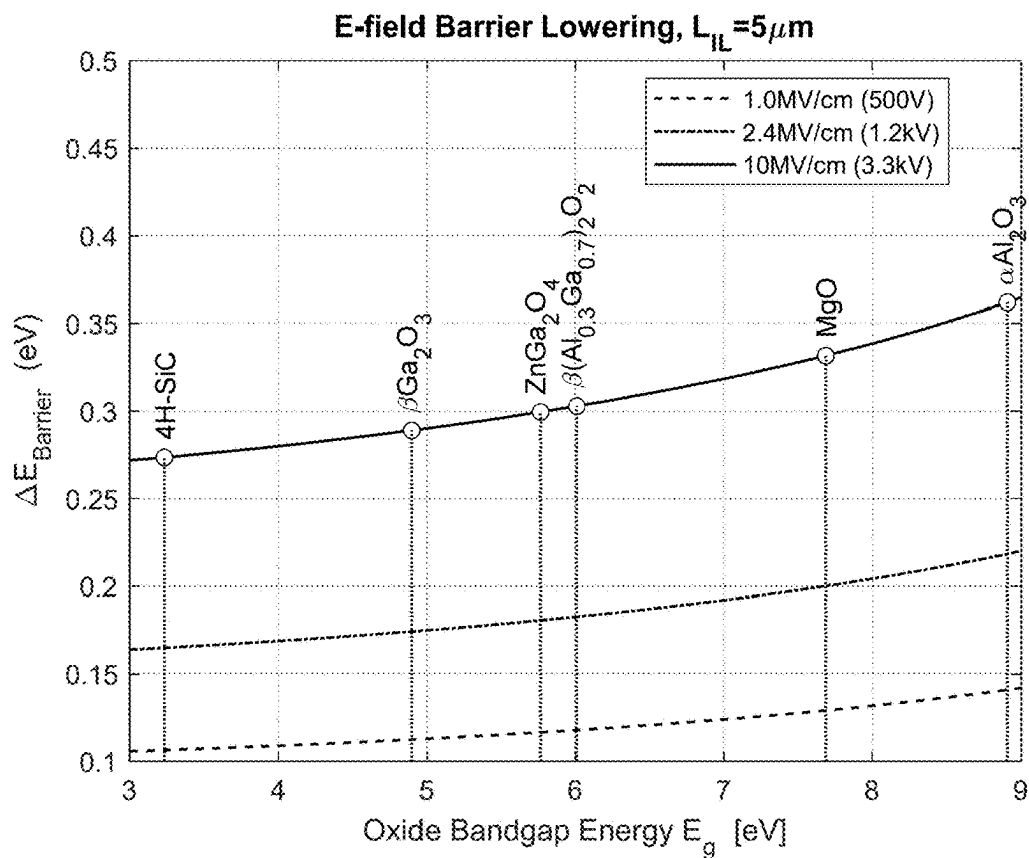


FIG. 29B

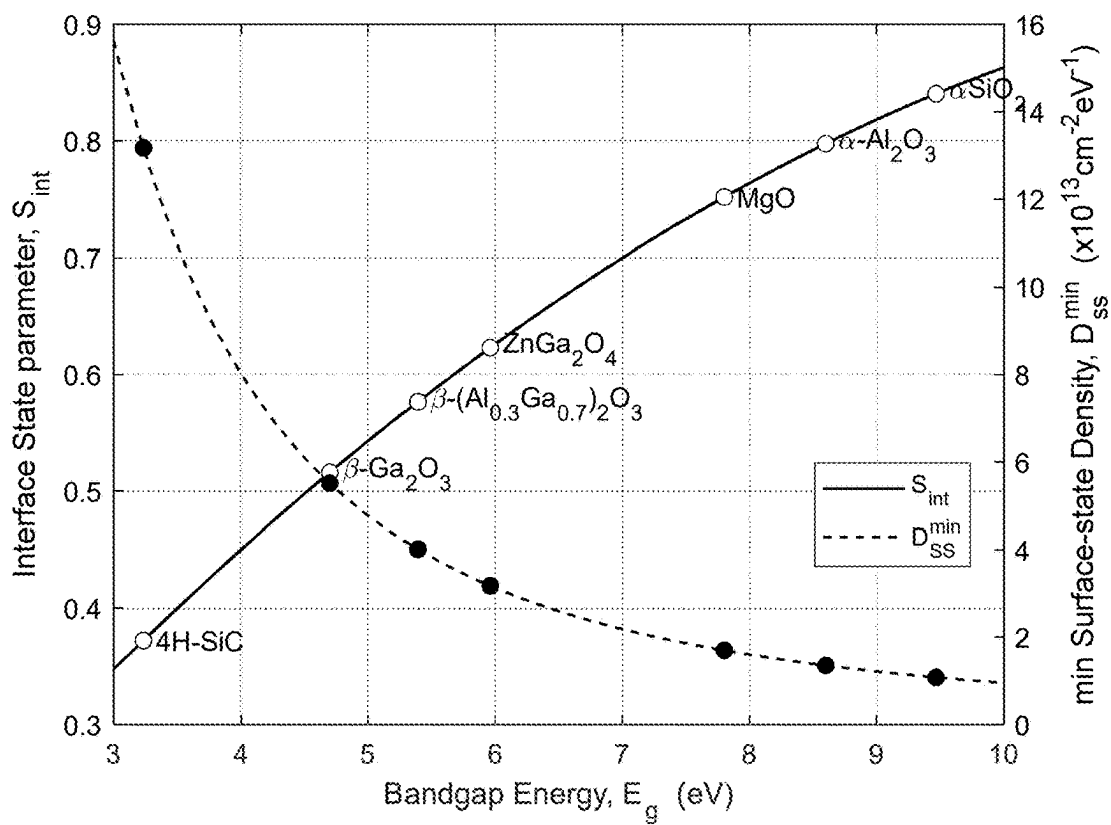


FIG. 29C

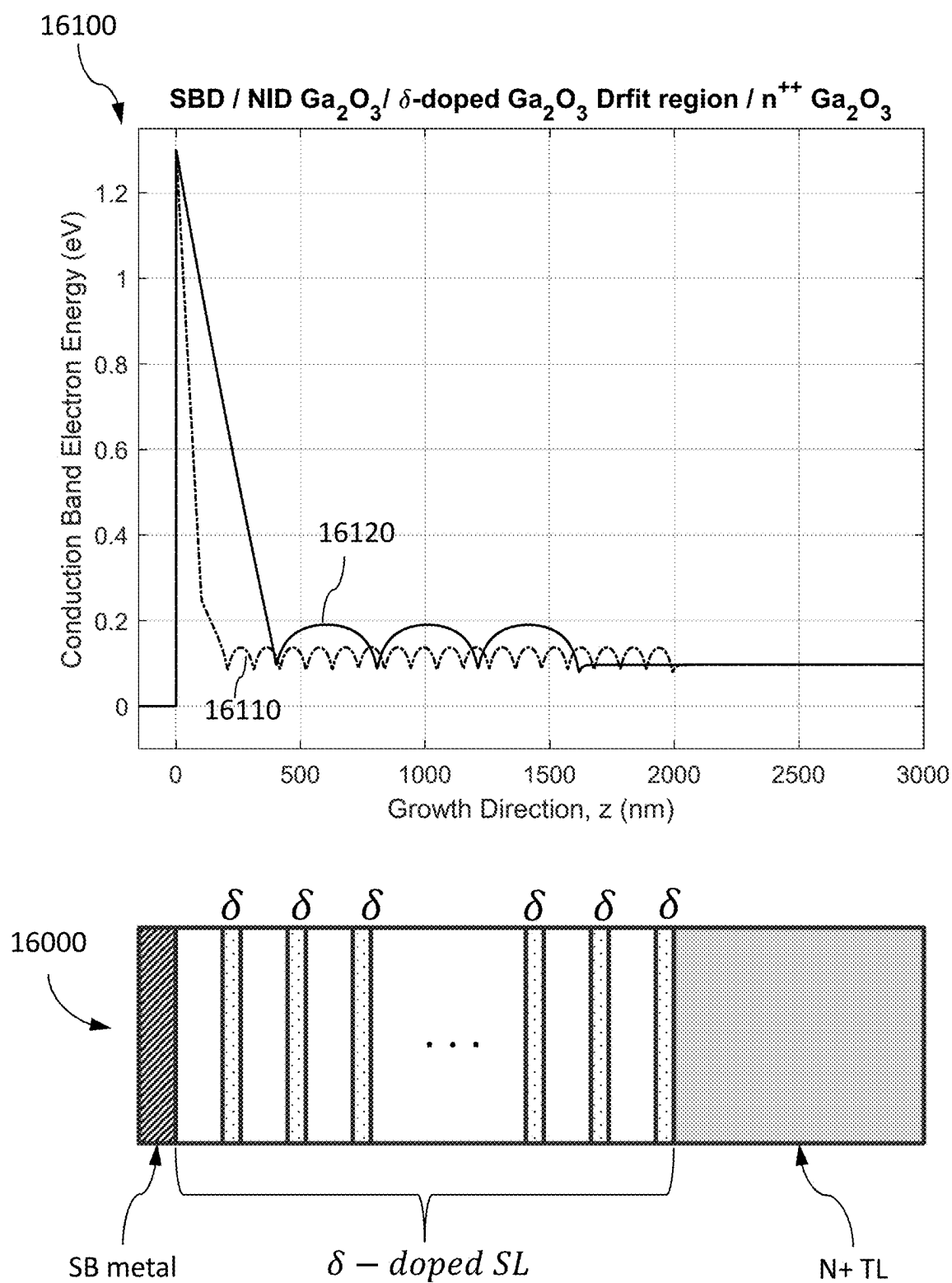


FIG. 30

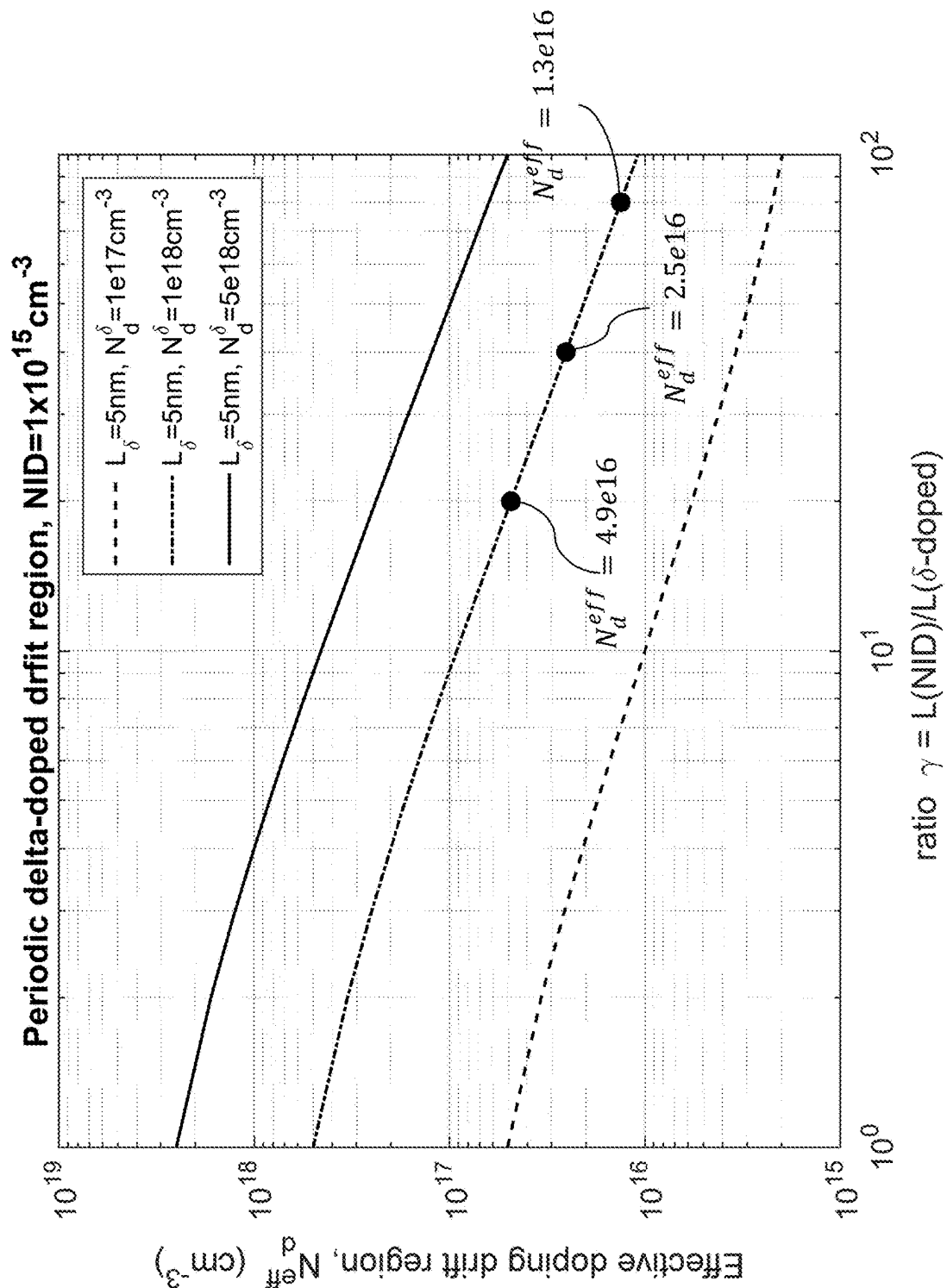


FIG. 31

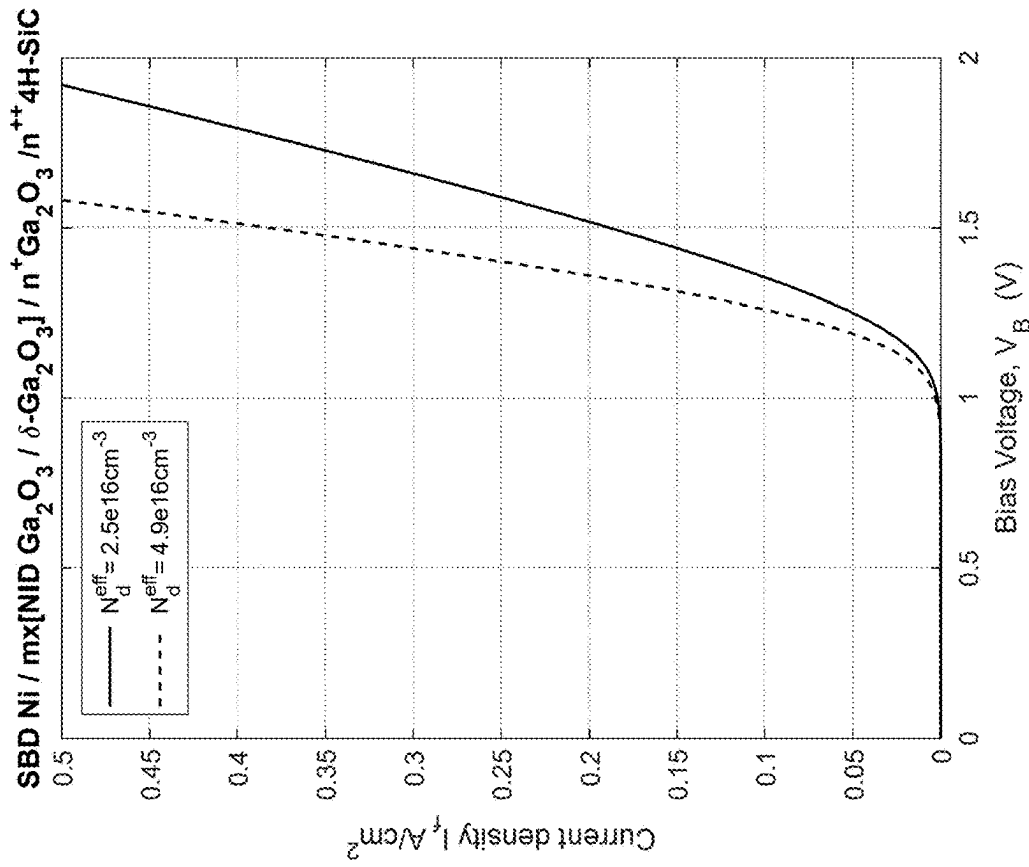


FIG. 32A

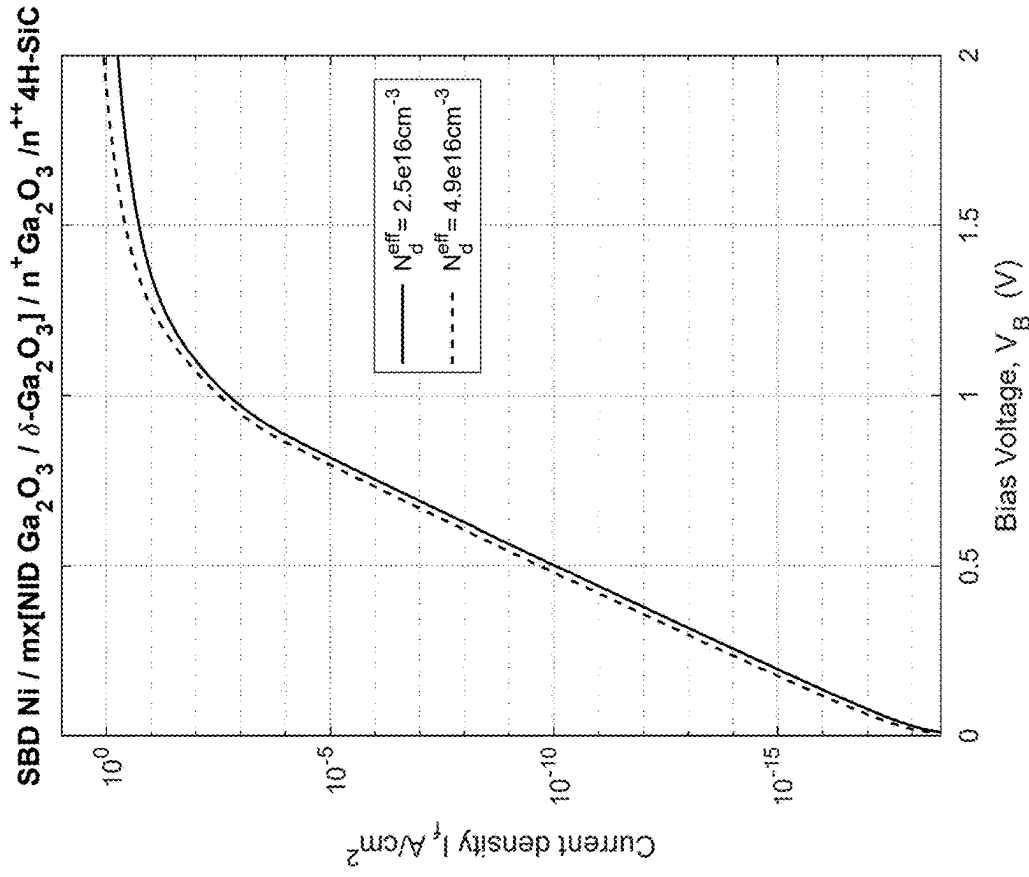


FIG. 32B

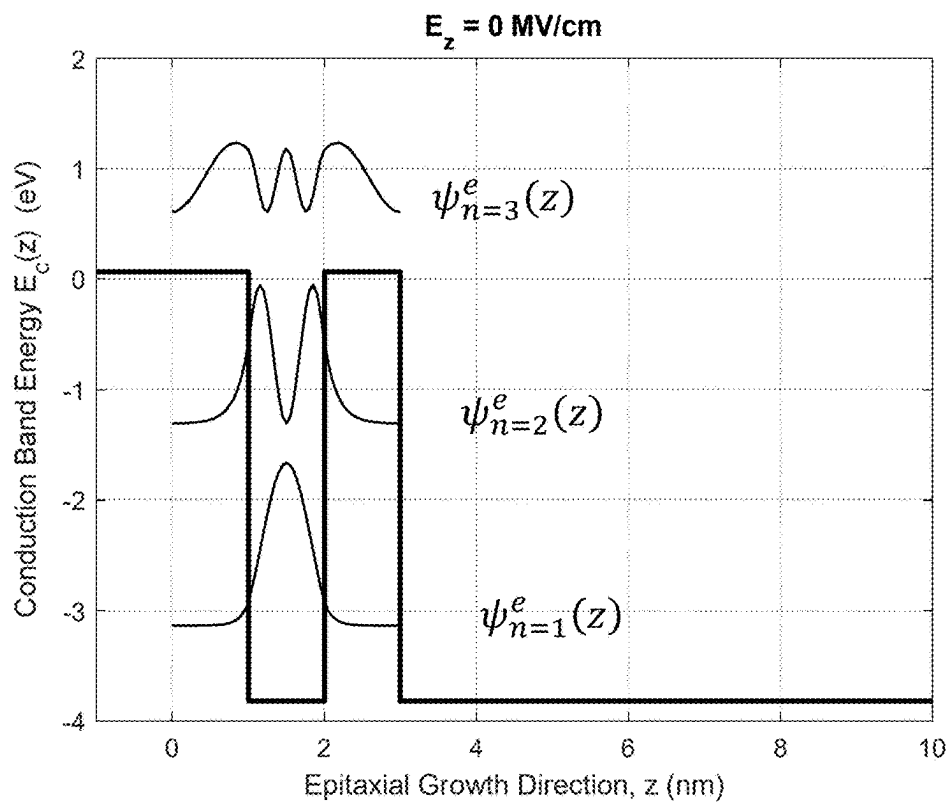


FIG. 33A

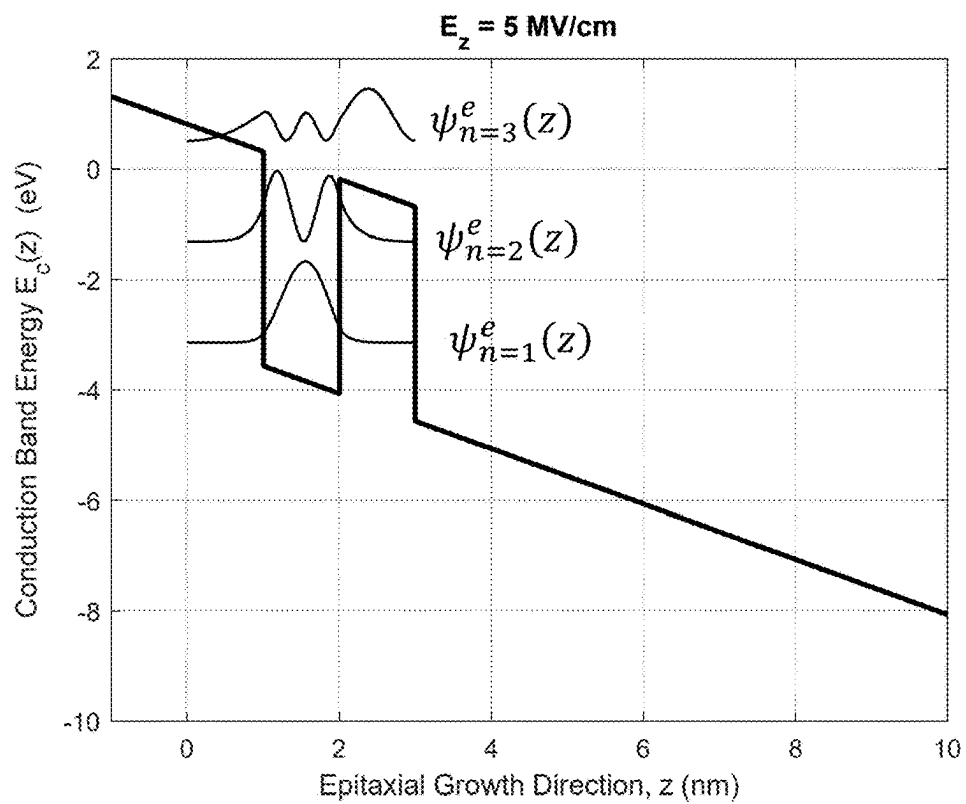


FIG. 33B

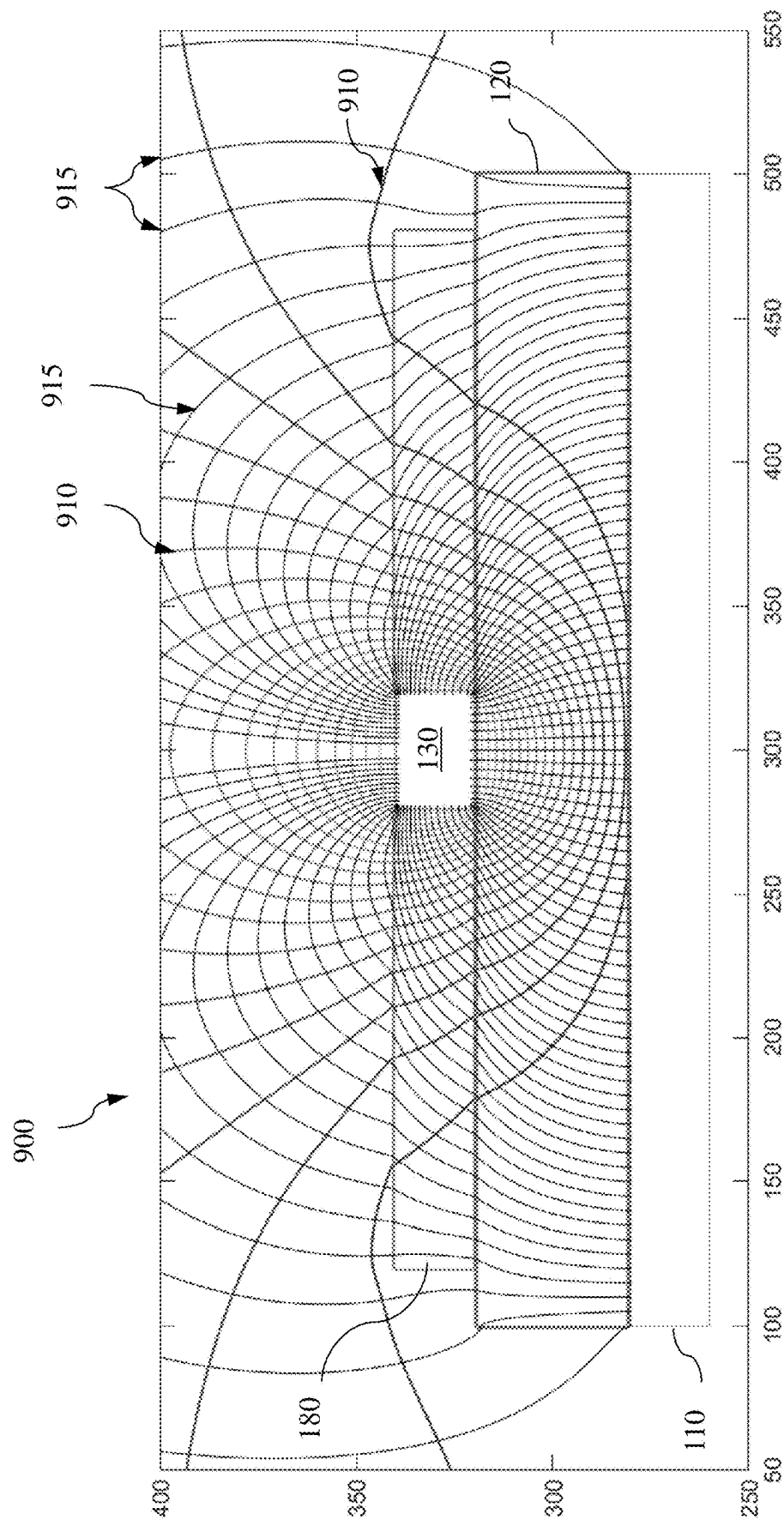


FIG. 34

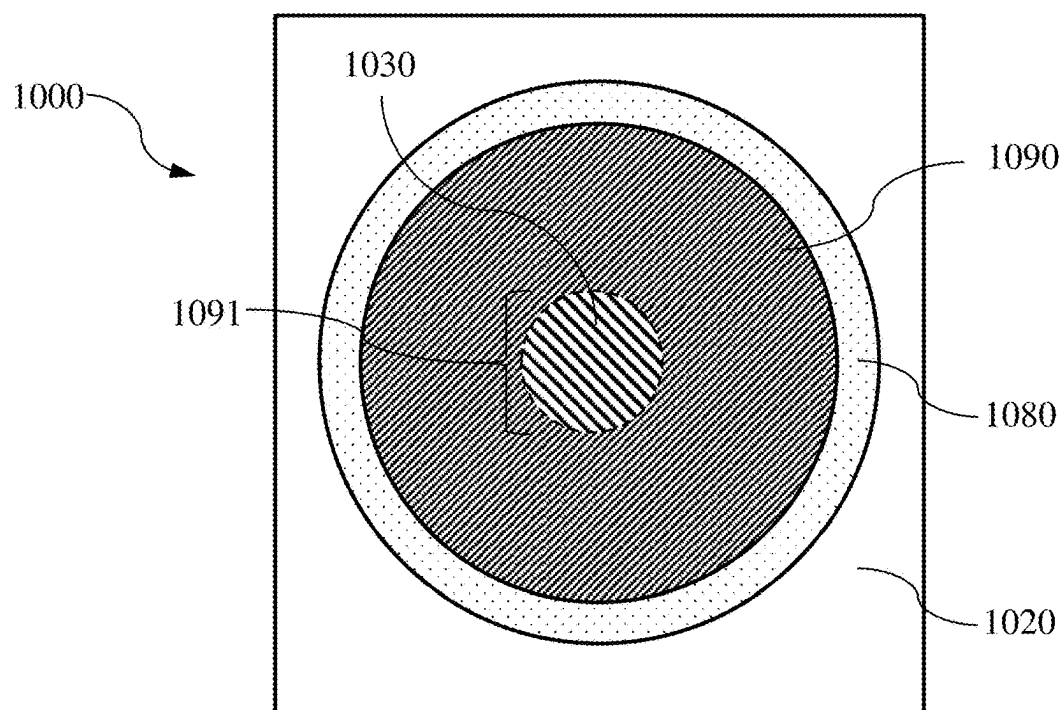


FIG. 35A

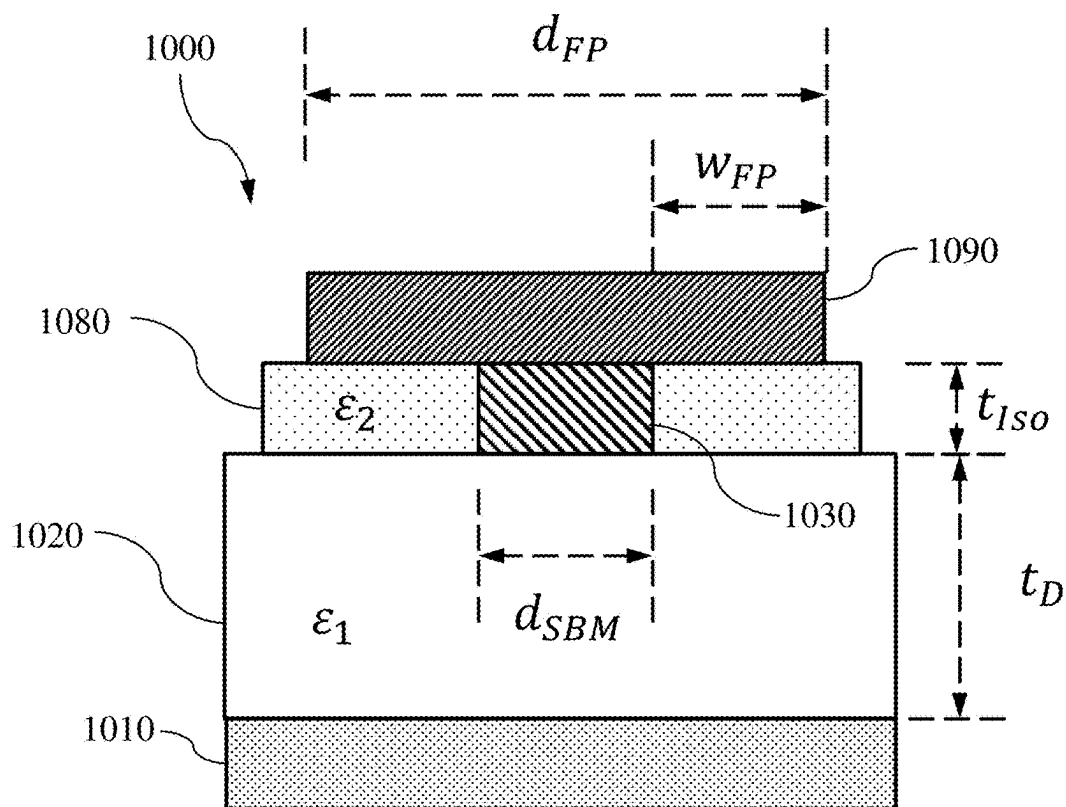


FIG. 35B

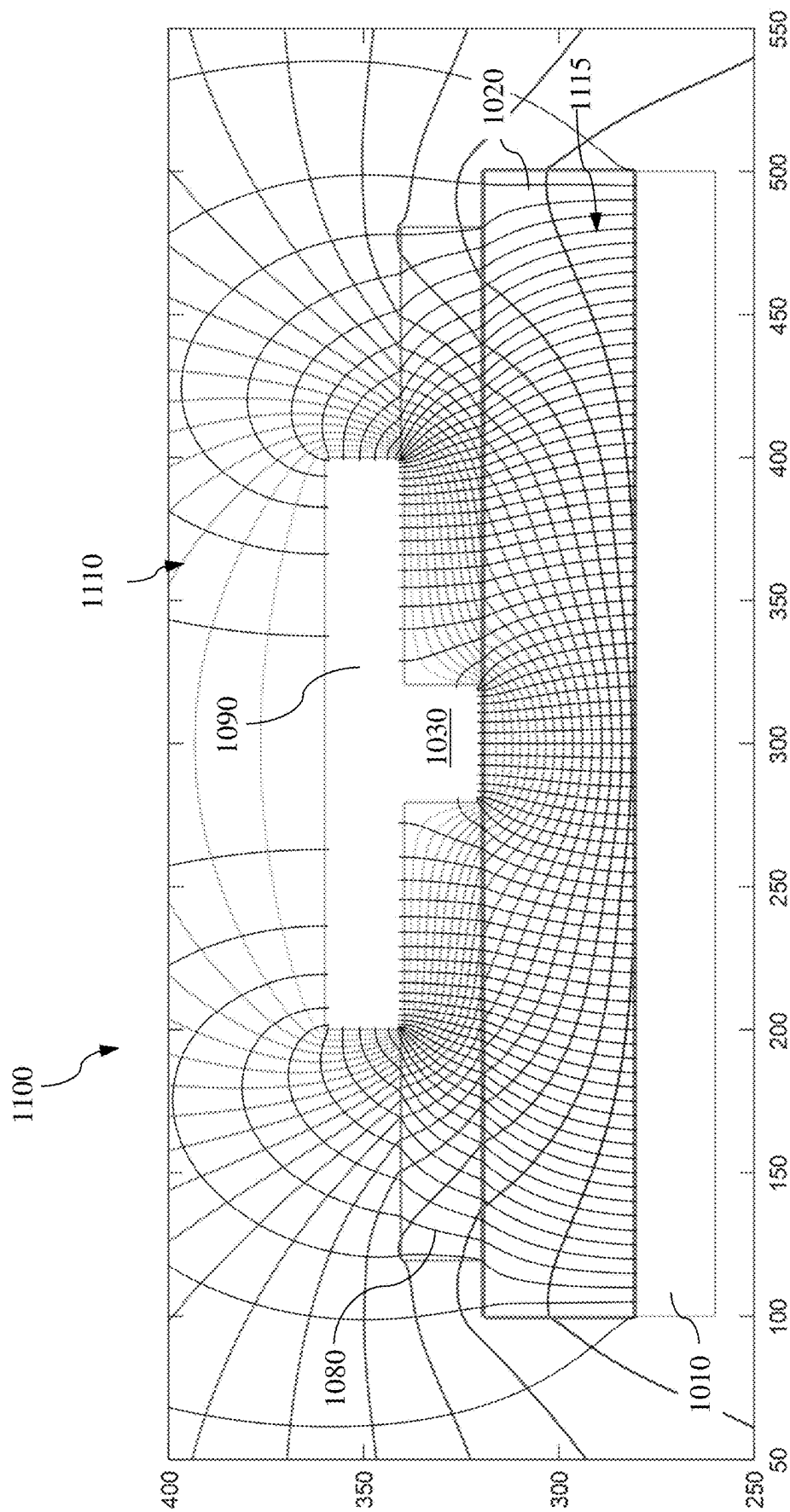


FIG. 36

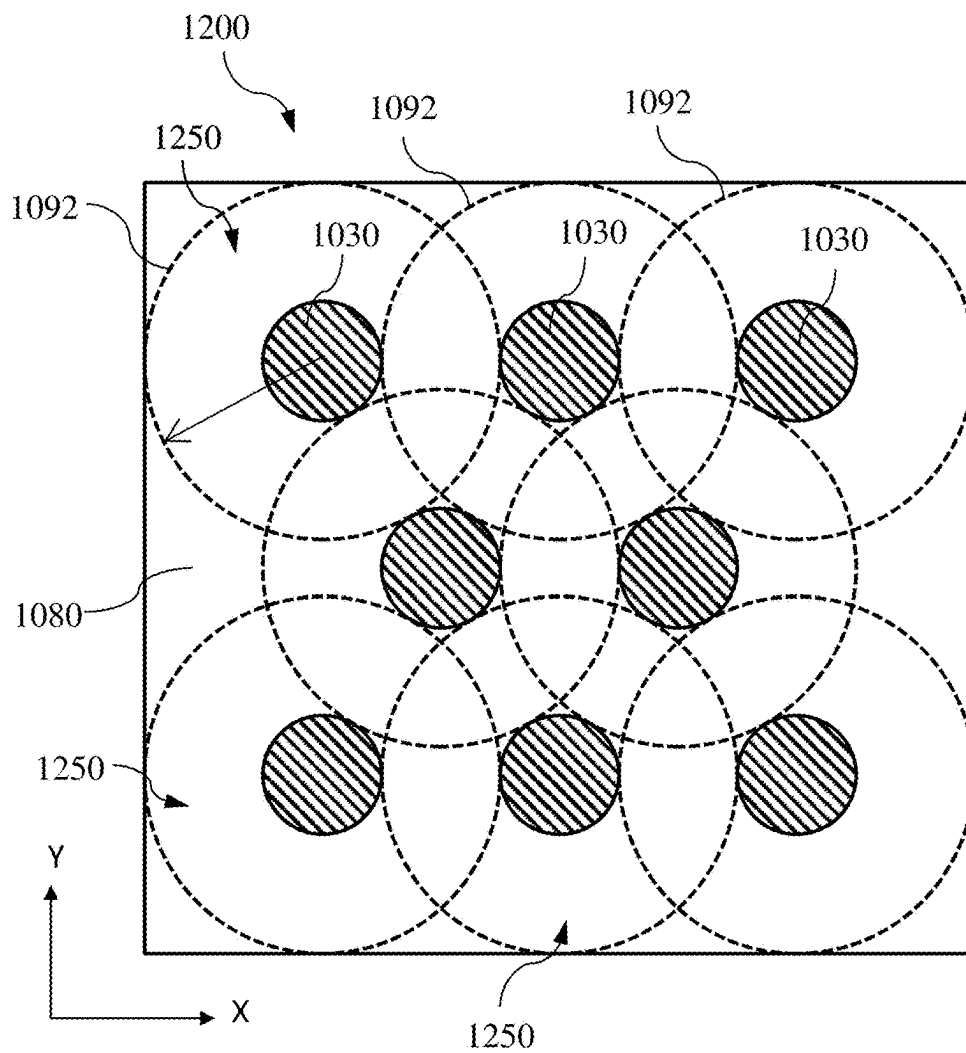


FIG. 37A

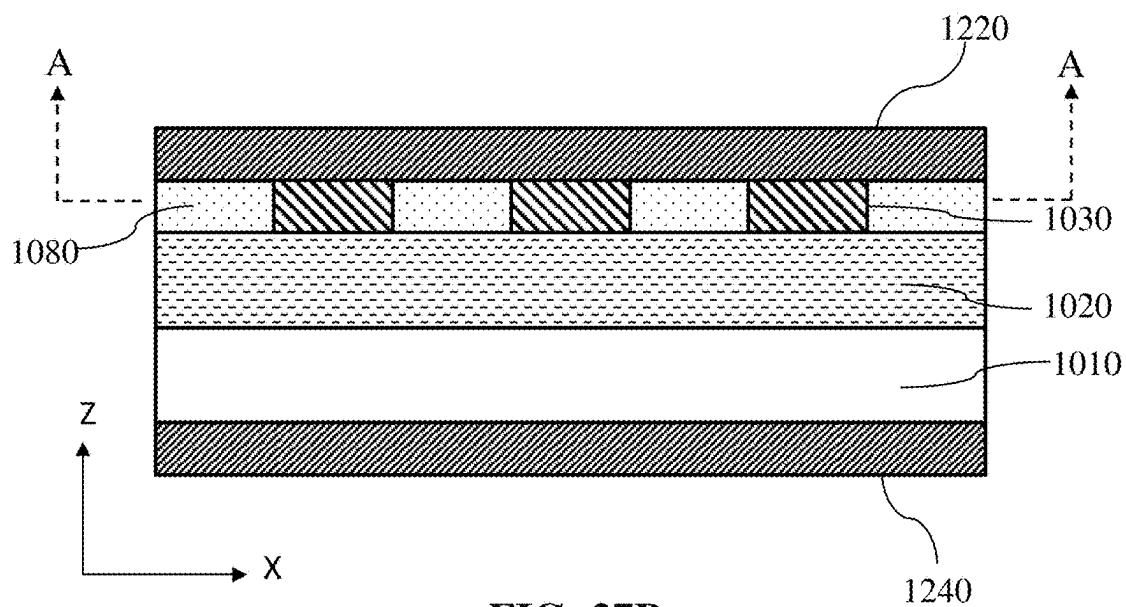


FIG. 37B

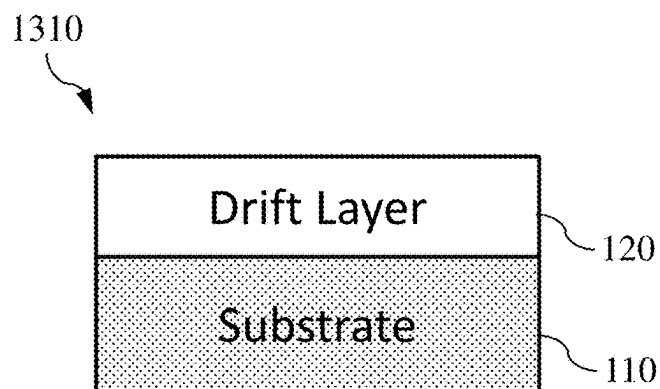


FIG. 38A

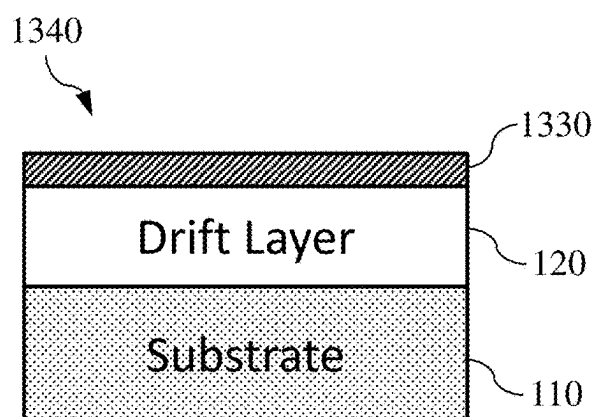


FIG. 38B

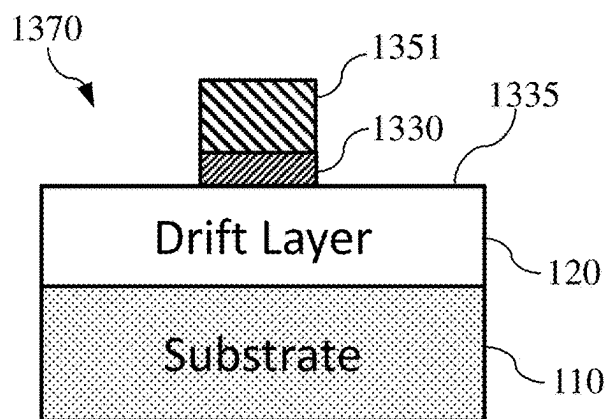


FIG. 38C

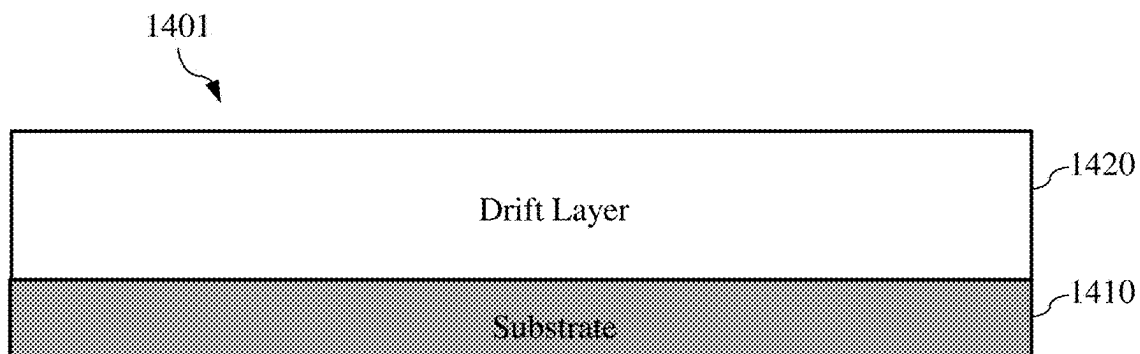


FIG. 39A

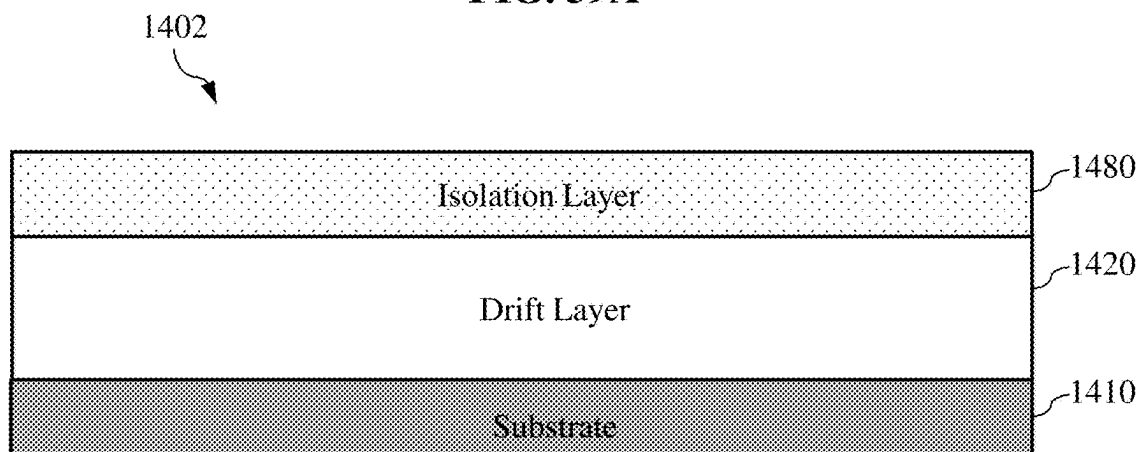


FIG. 39B

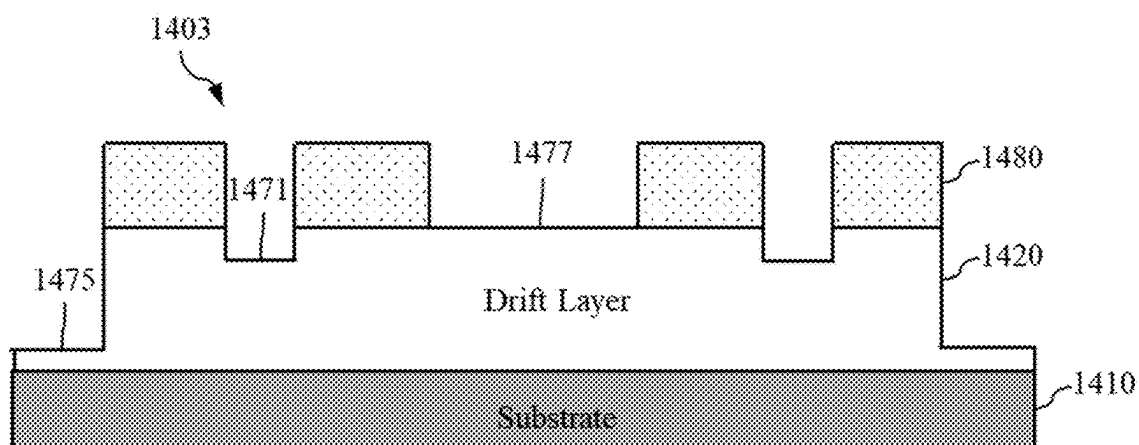


FIG. 39C

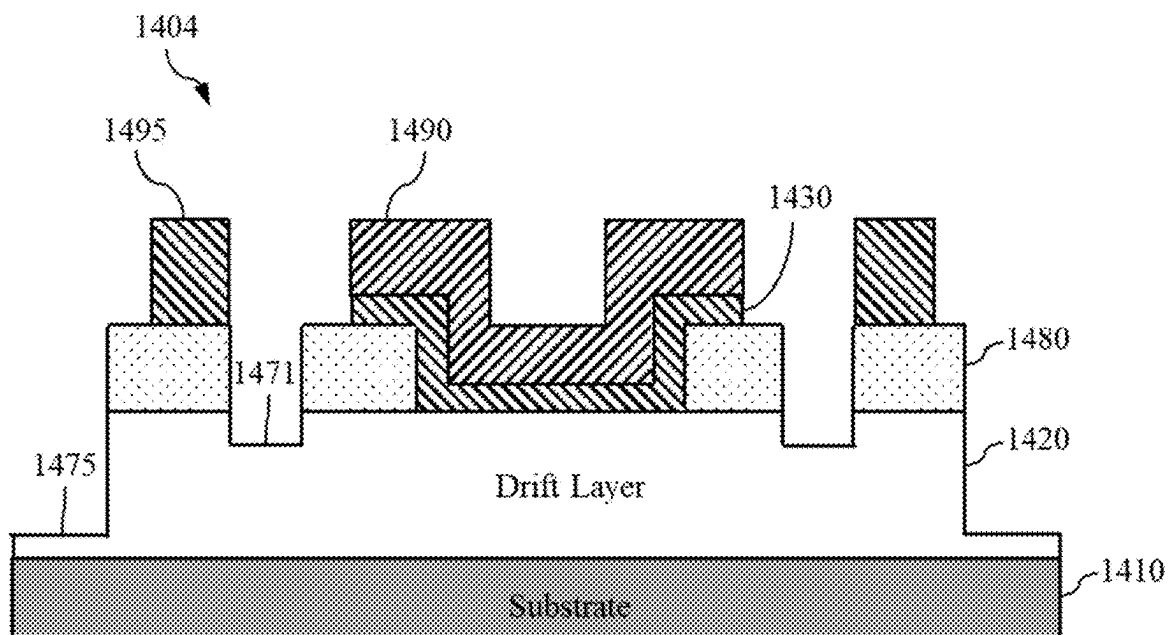


FIG. 39D

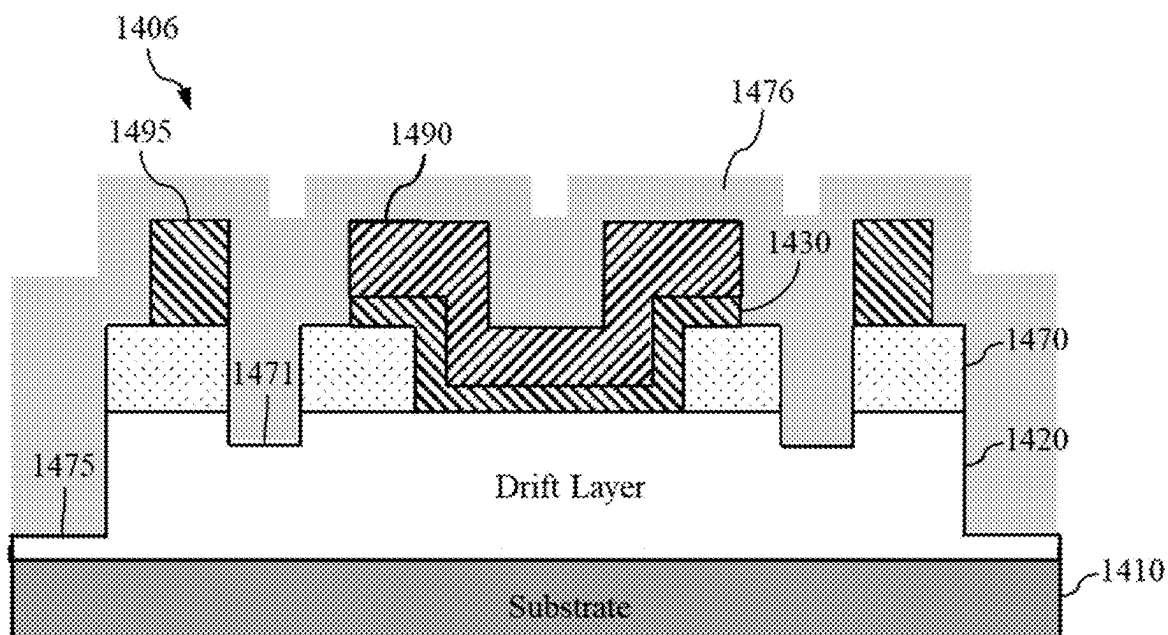


FIG. 39E

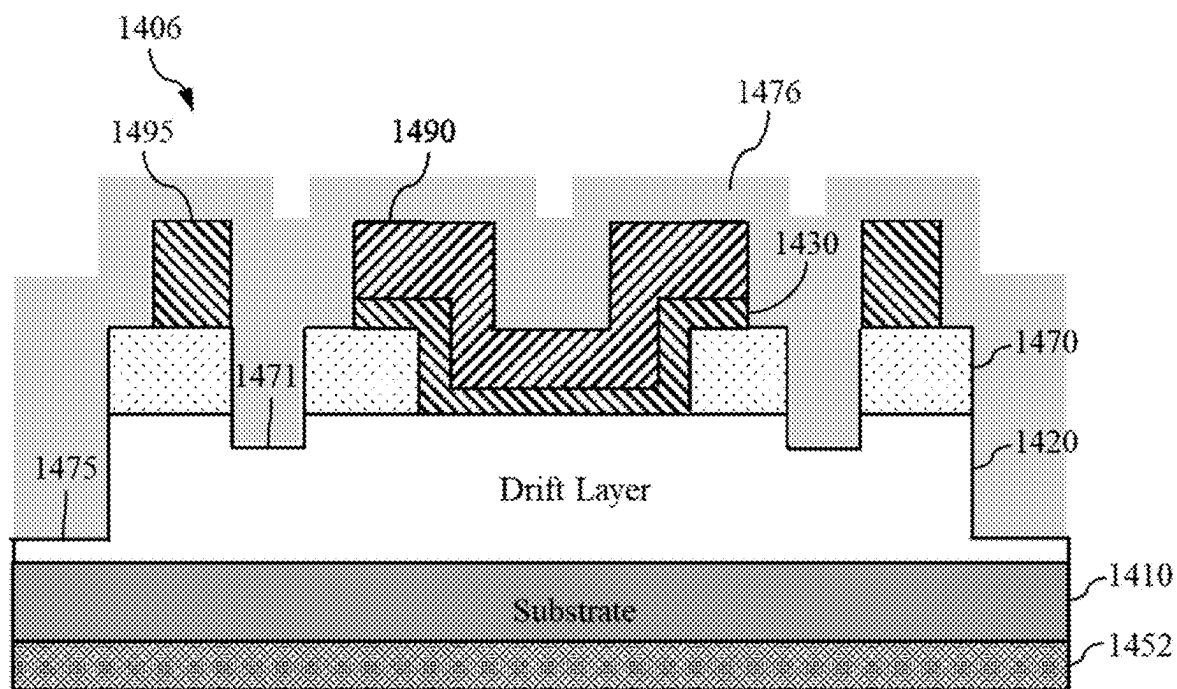


FIG. 39F

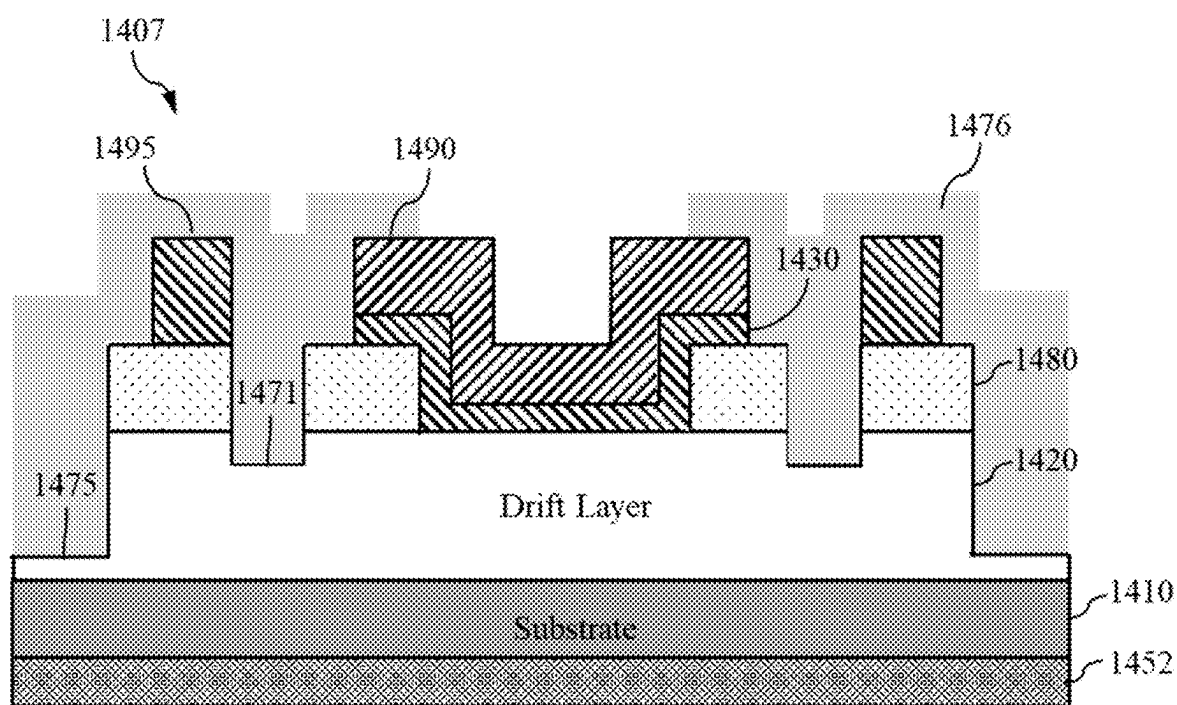


FIG. 39G

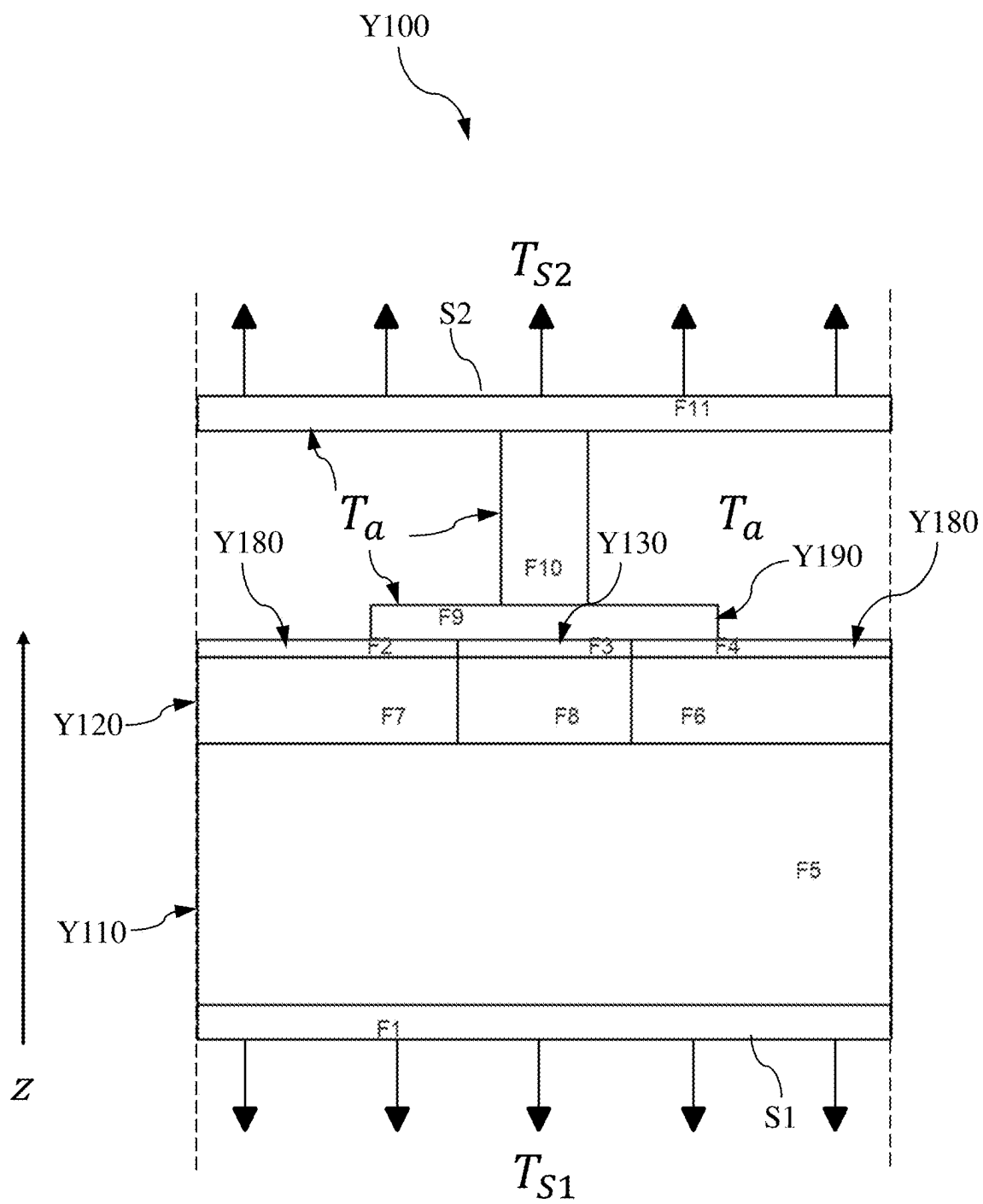


FIG. 40

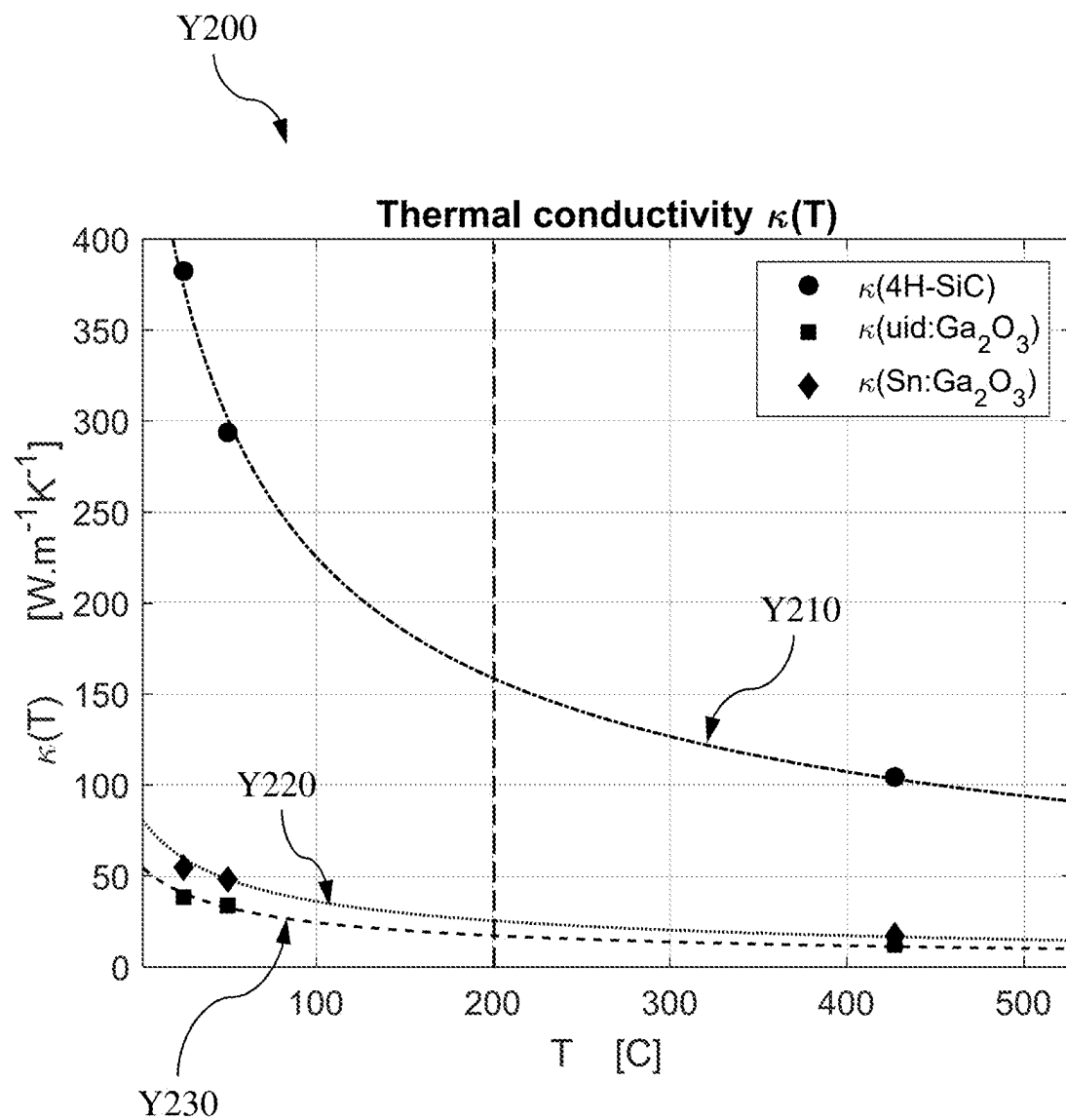


FIG. 41

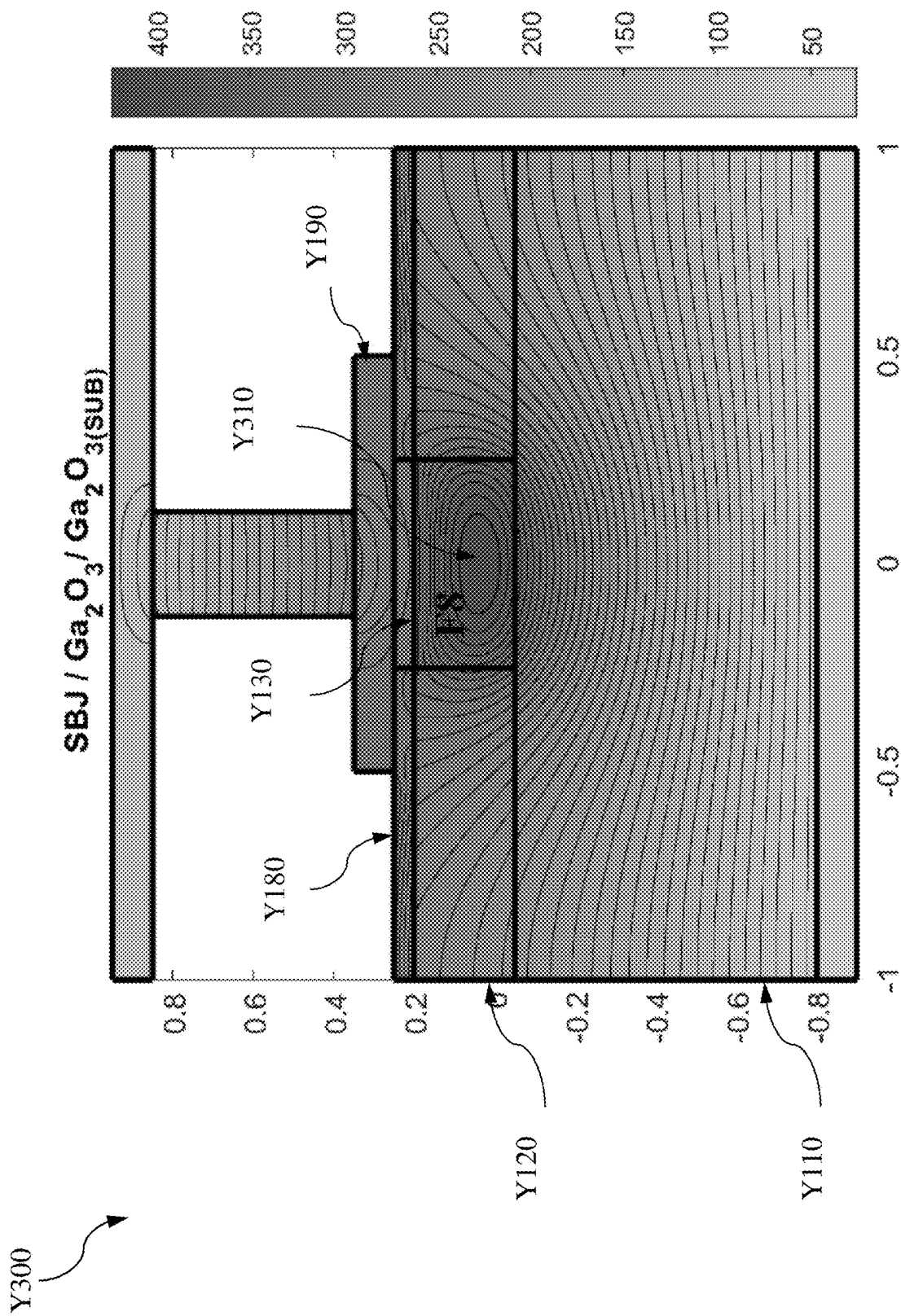


FIG. 42

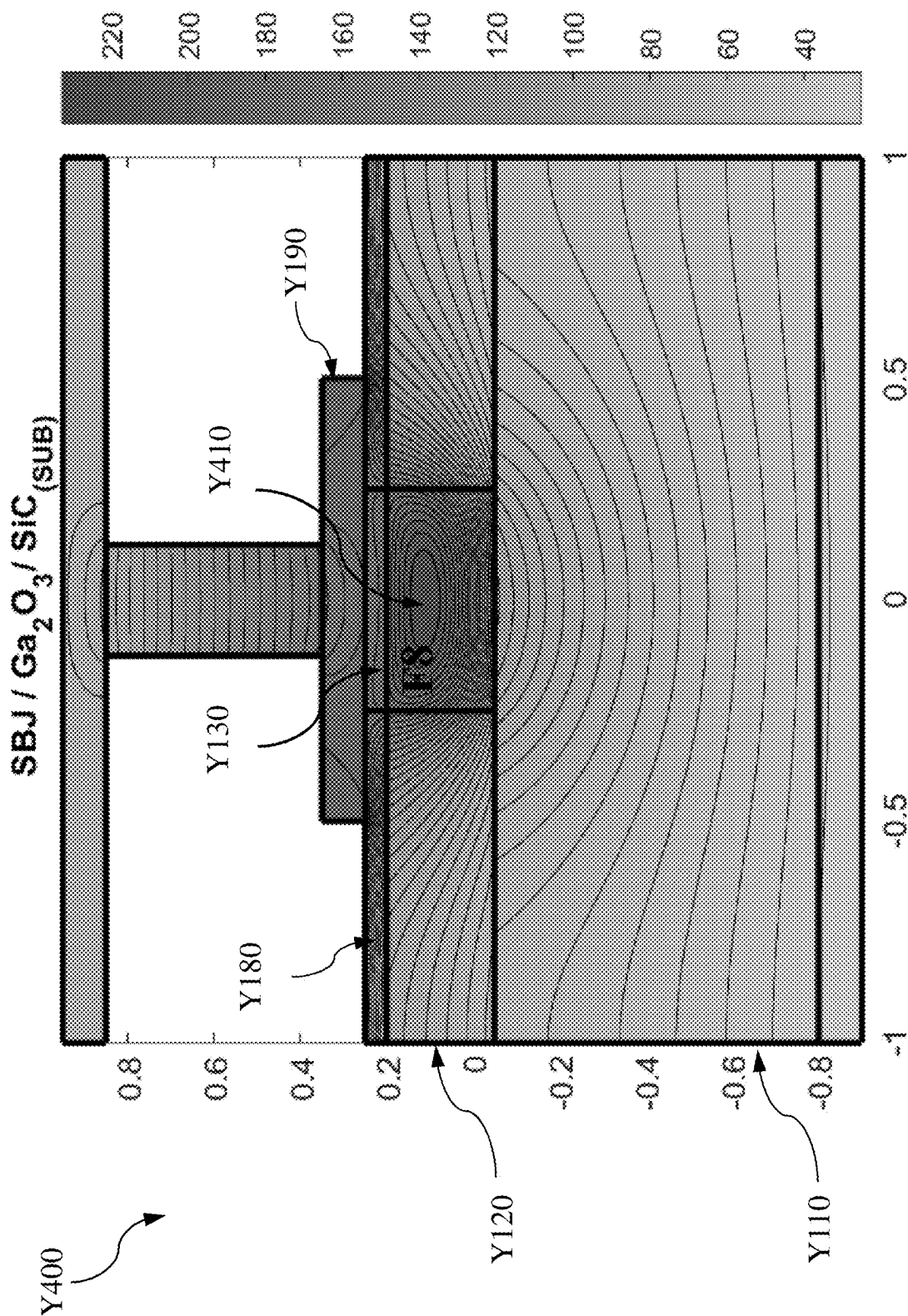


FIG. 43

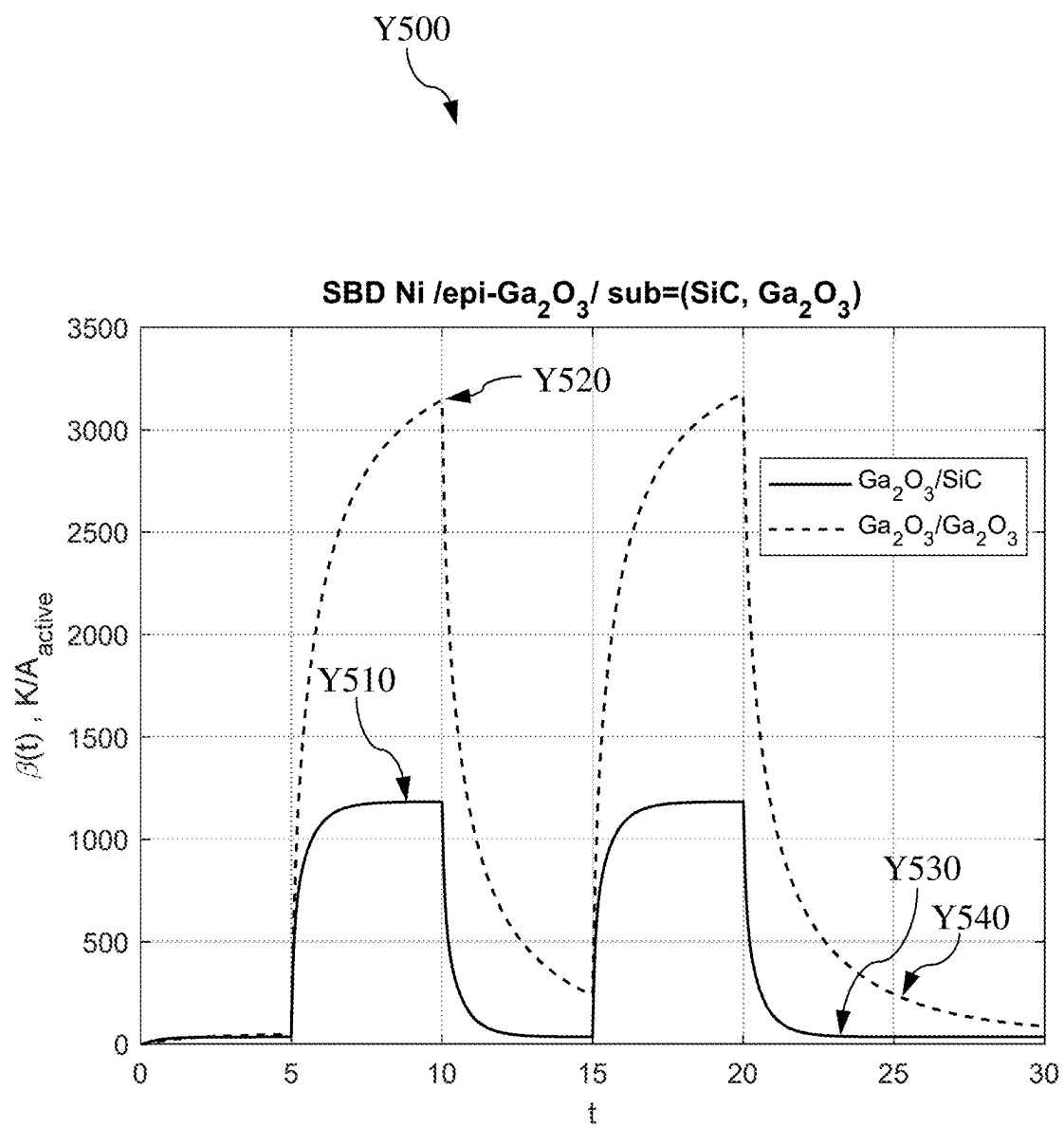


FIG. 44

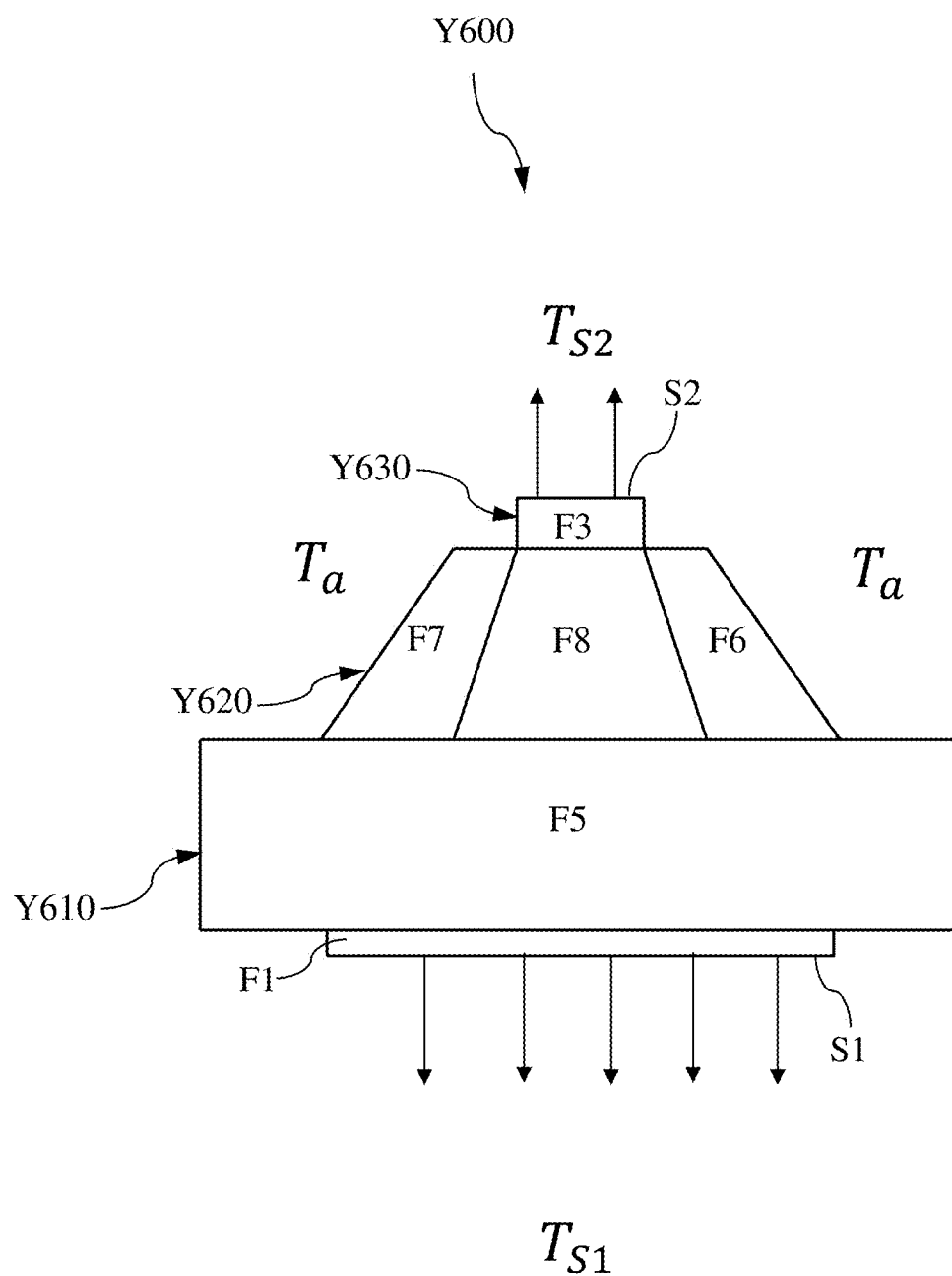


FIG. 45

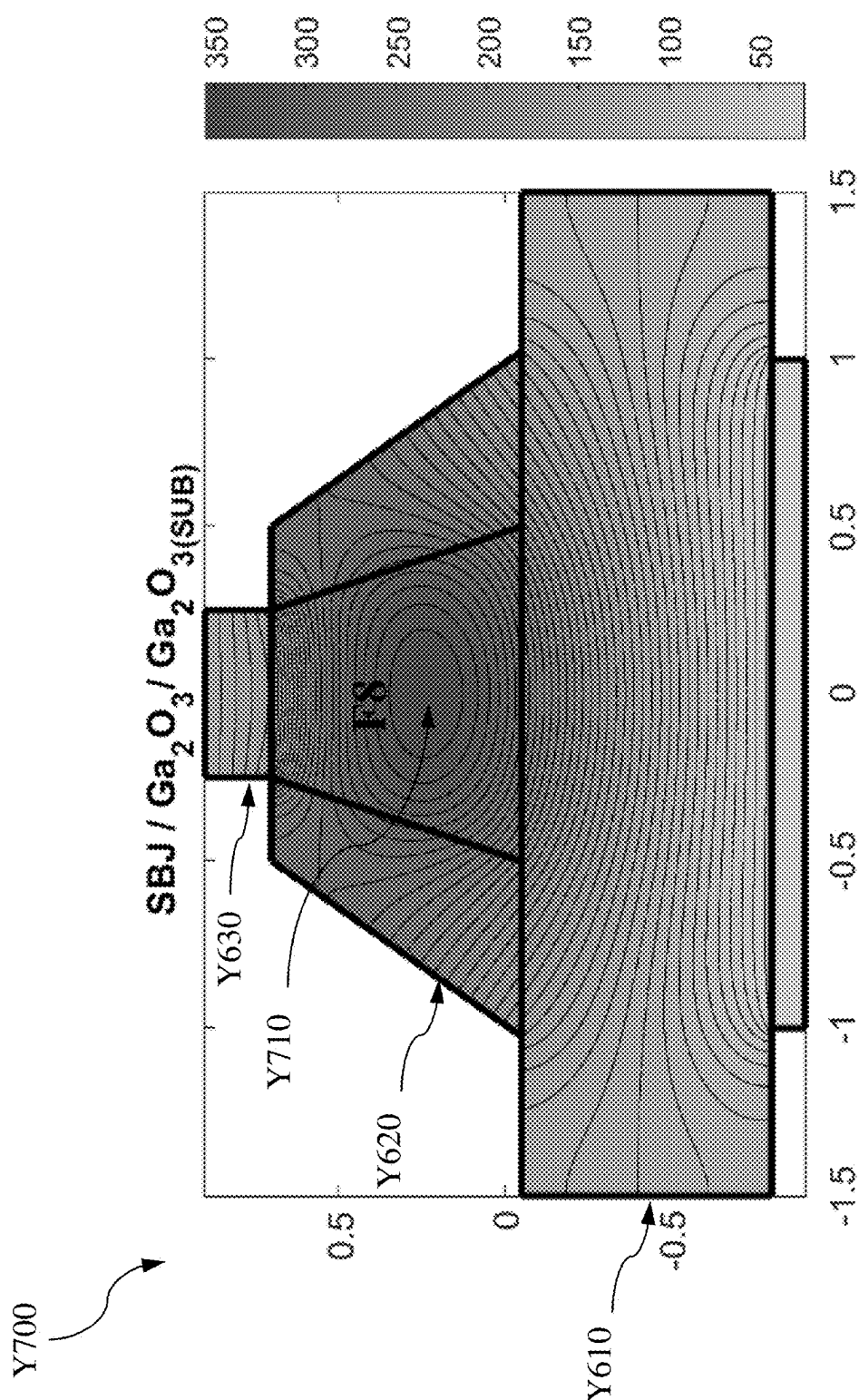


FIG. 46

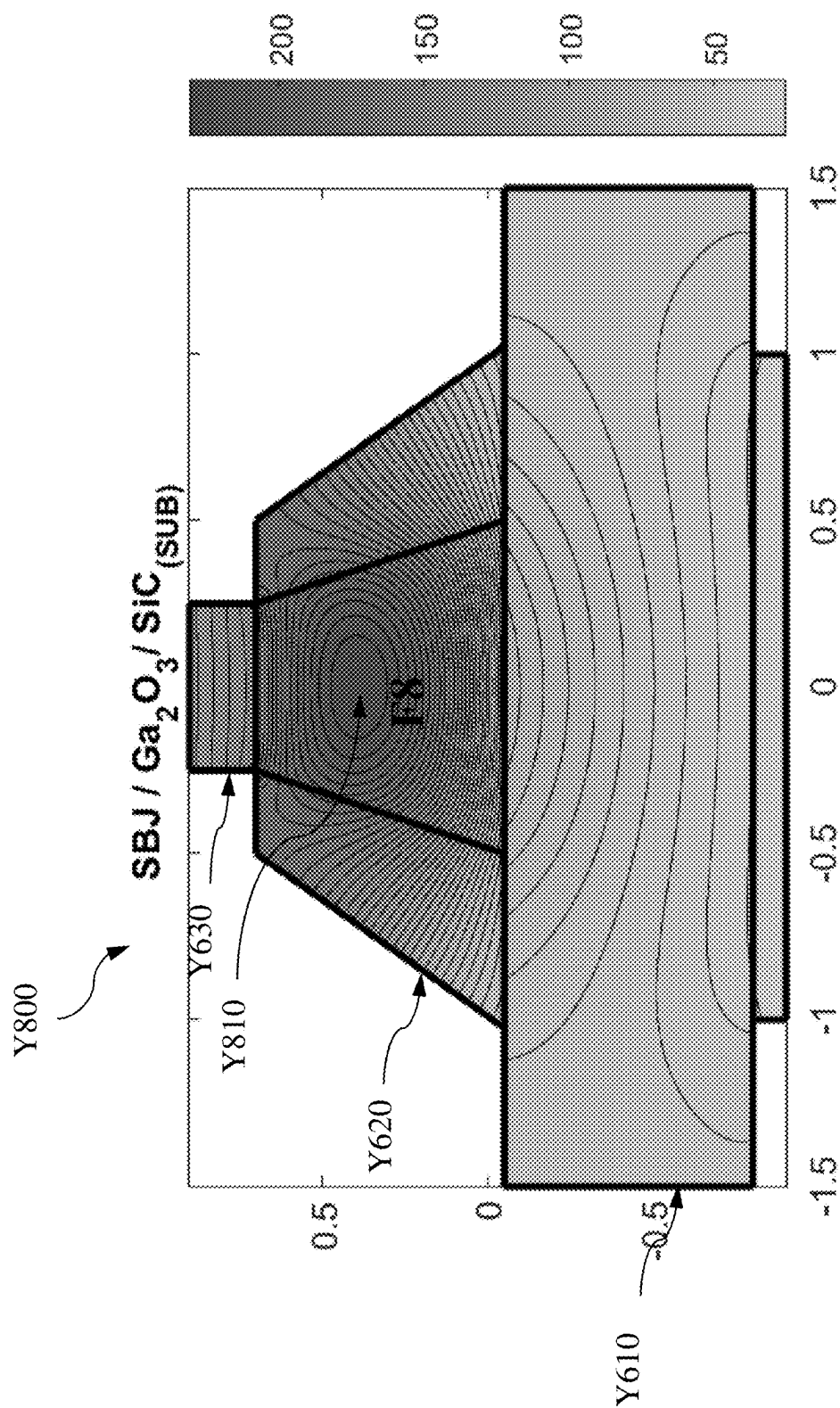


FIG. 47

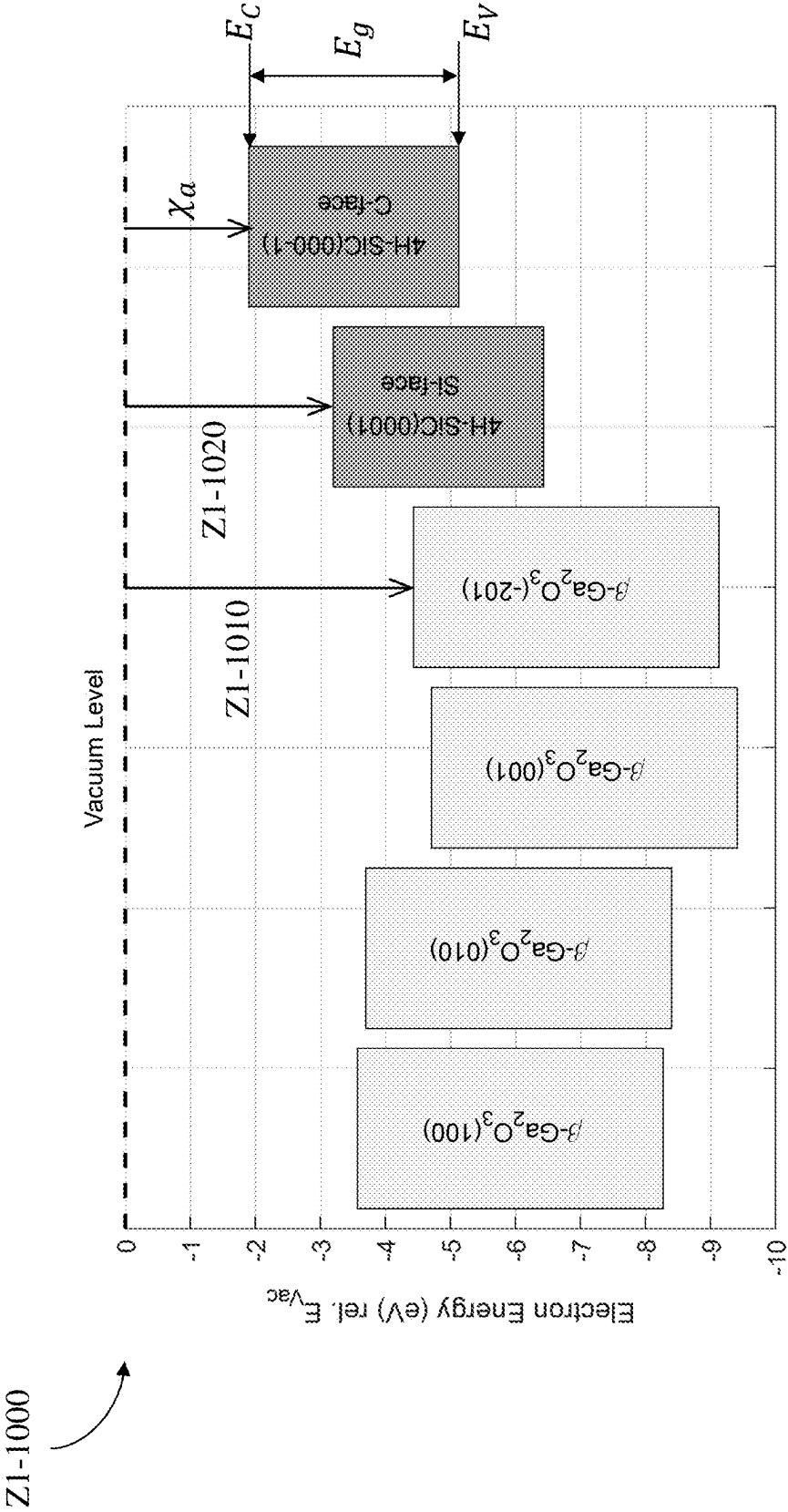


FIG. 48

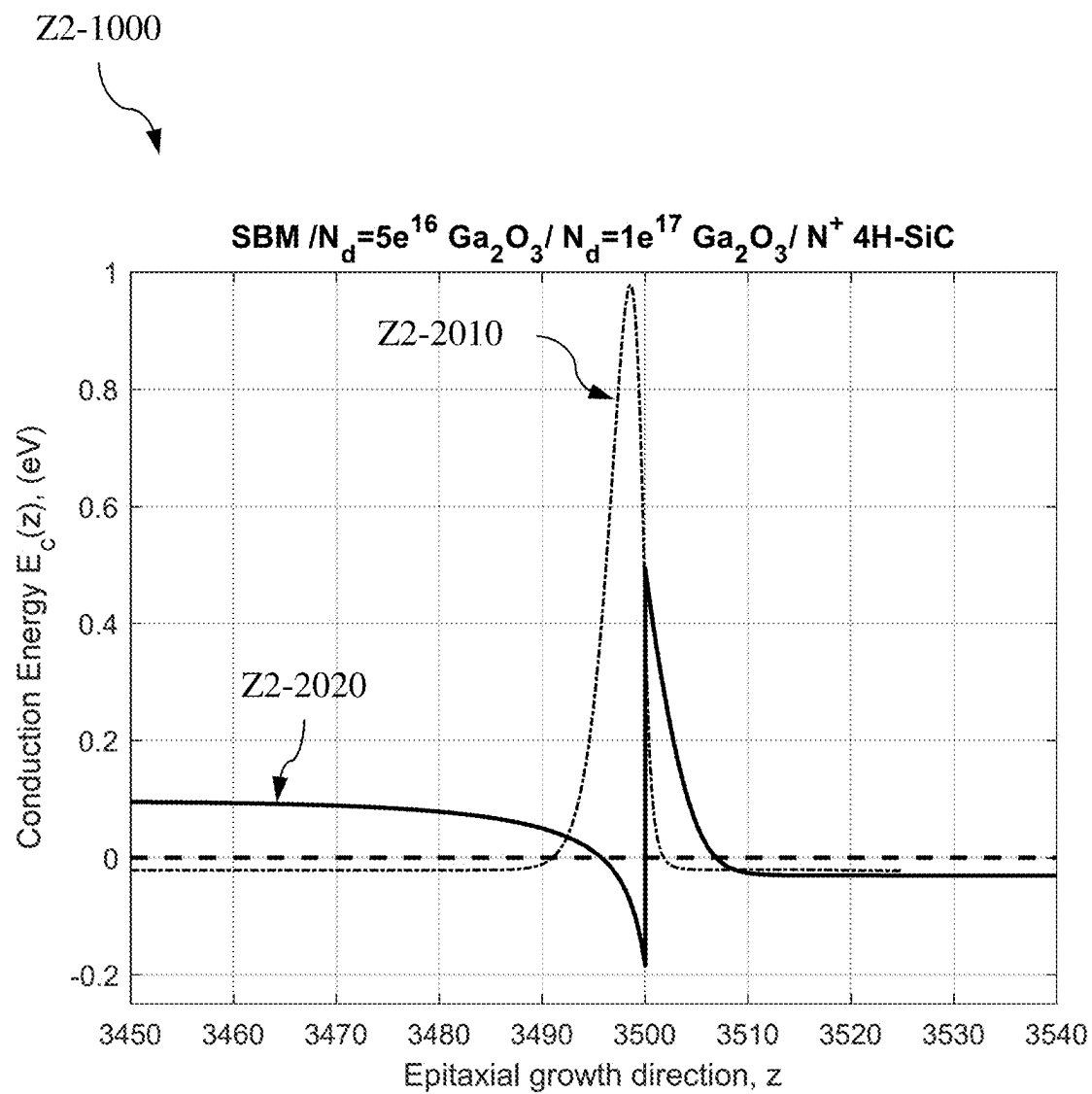


FIG. 49

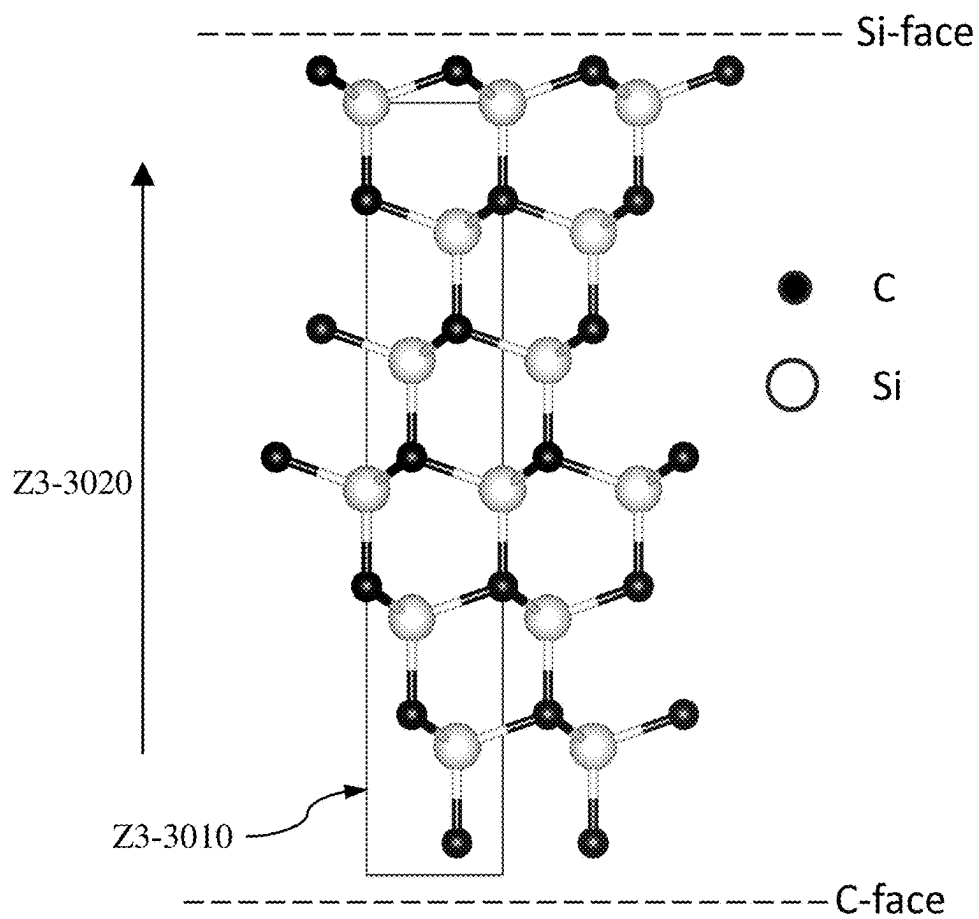


FIG. 50

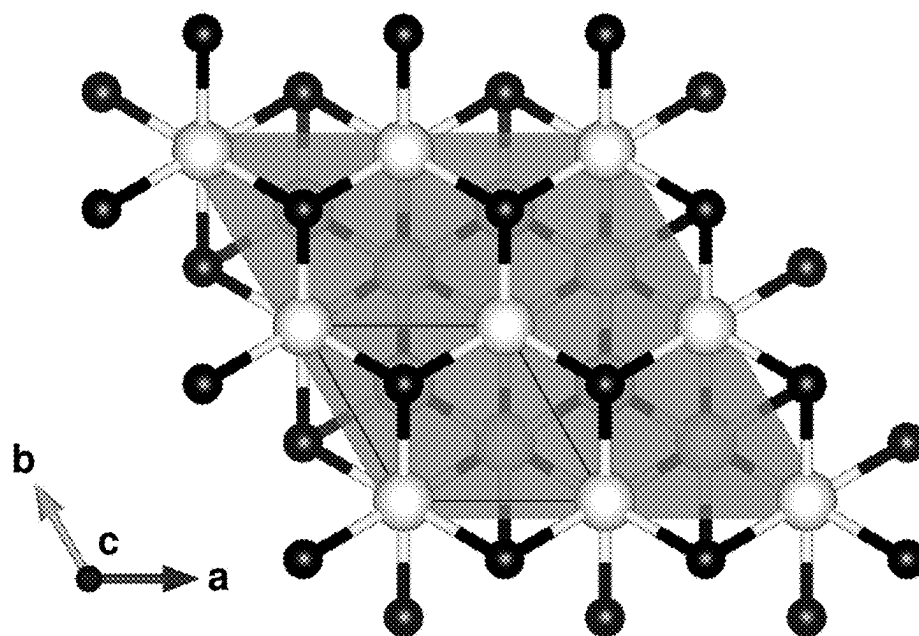


FIG. 51

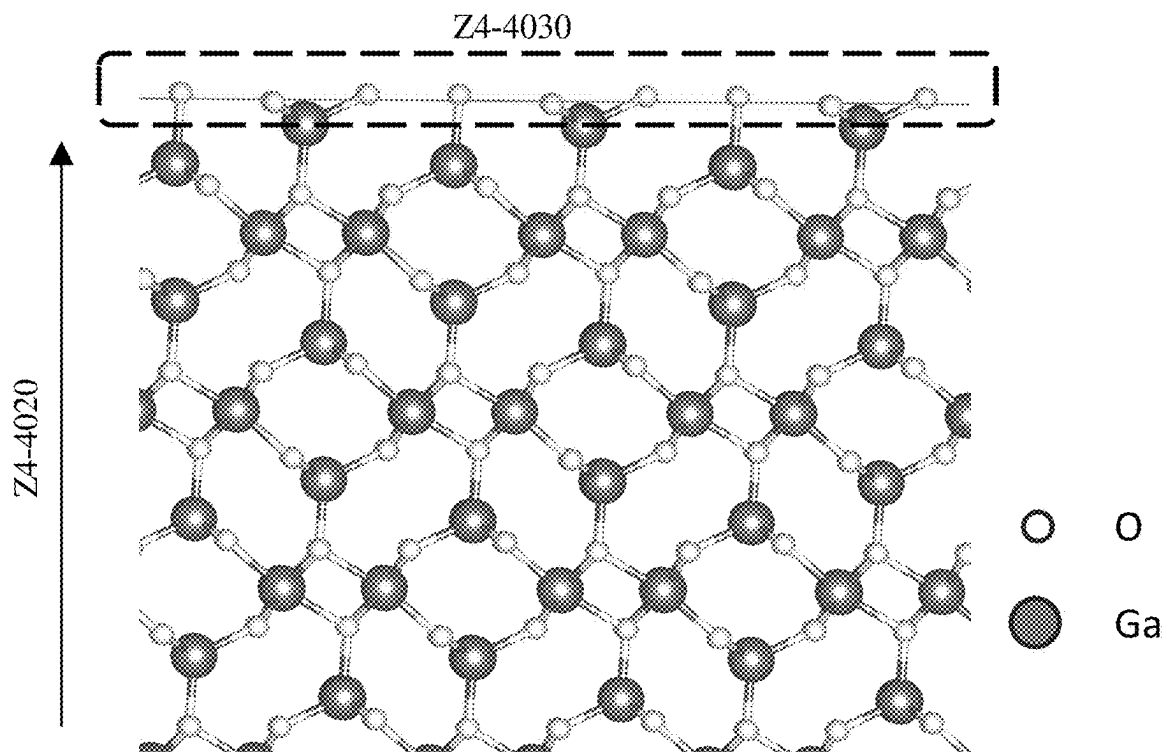


FIG. 52

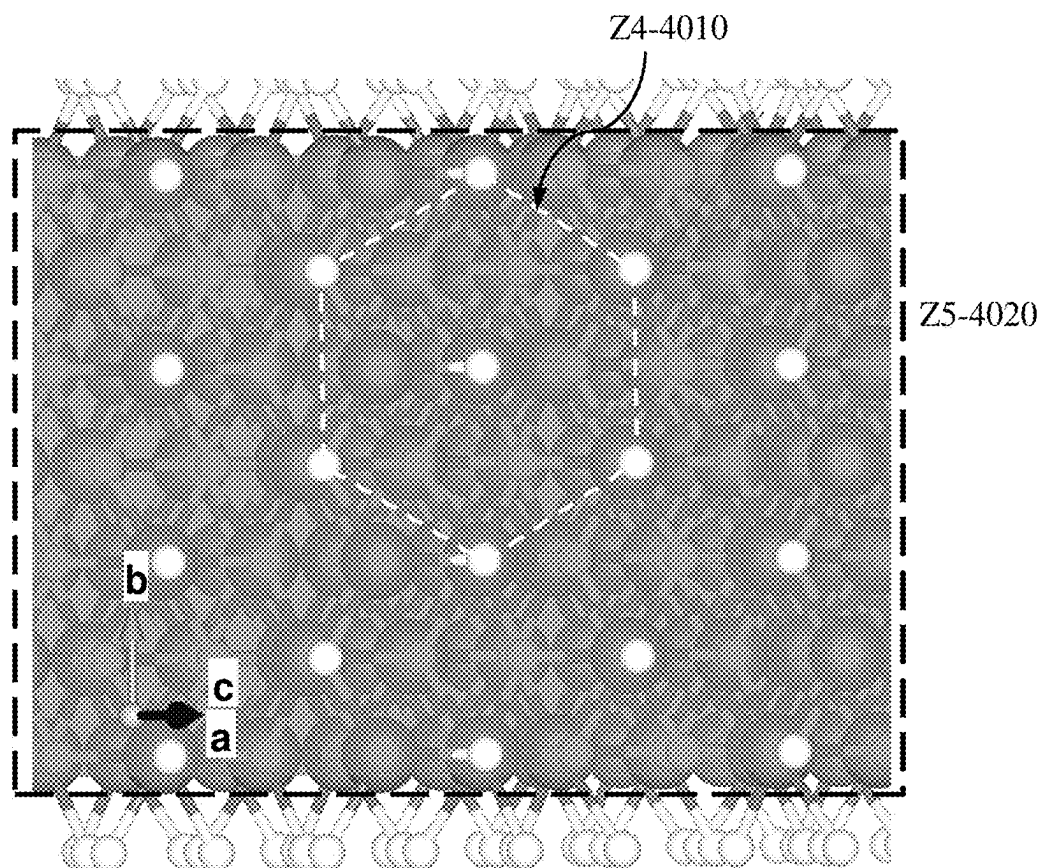


FIG. 53

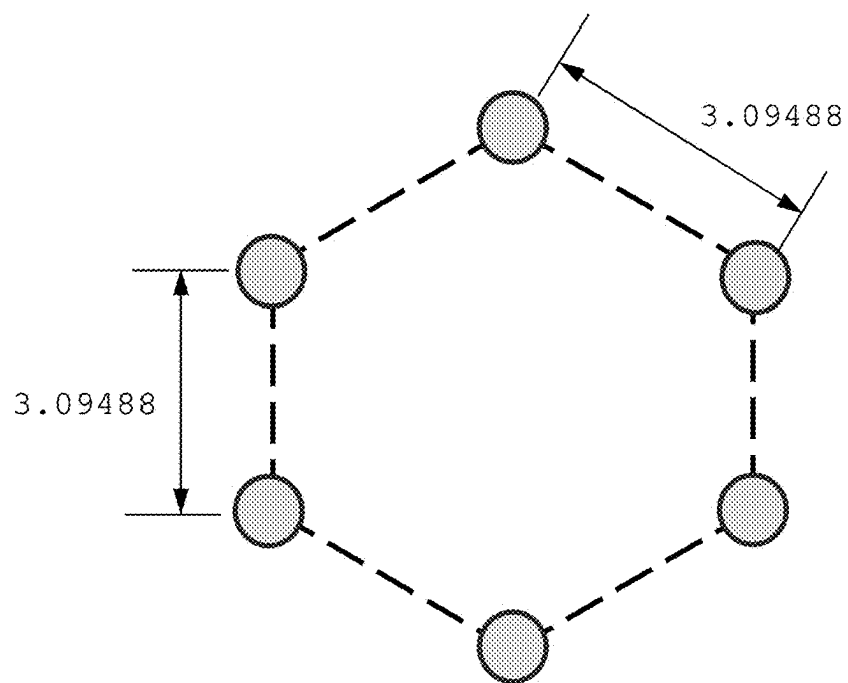


FIG. 54

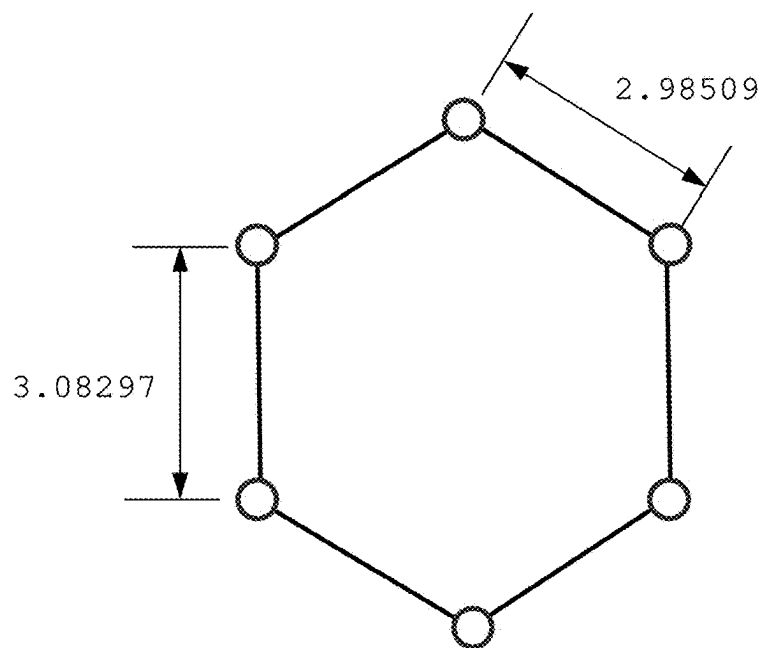


FIG. 55

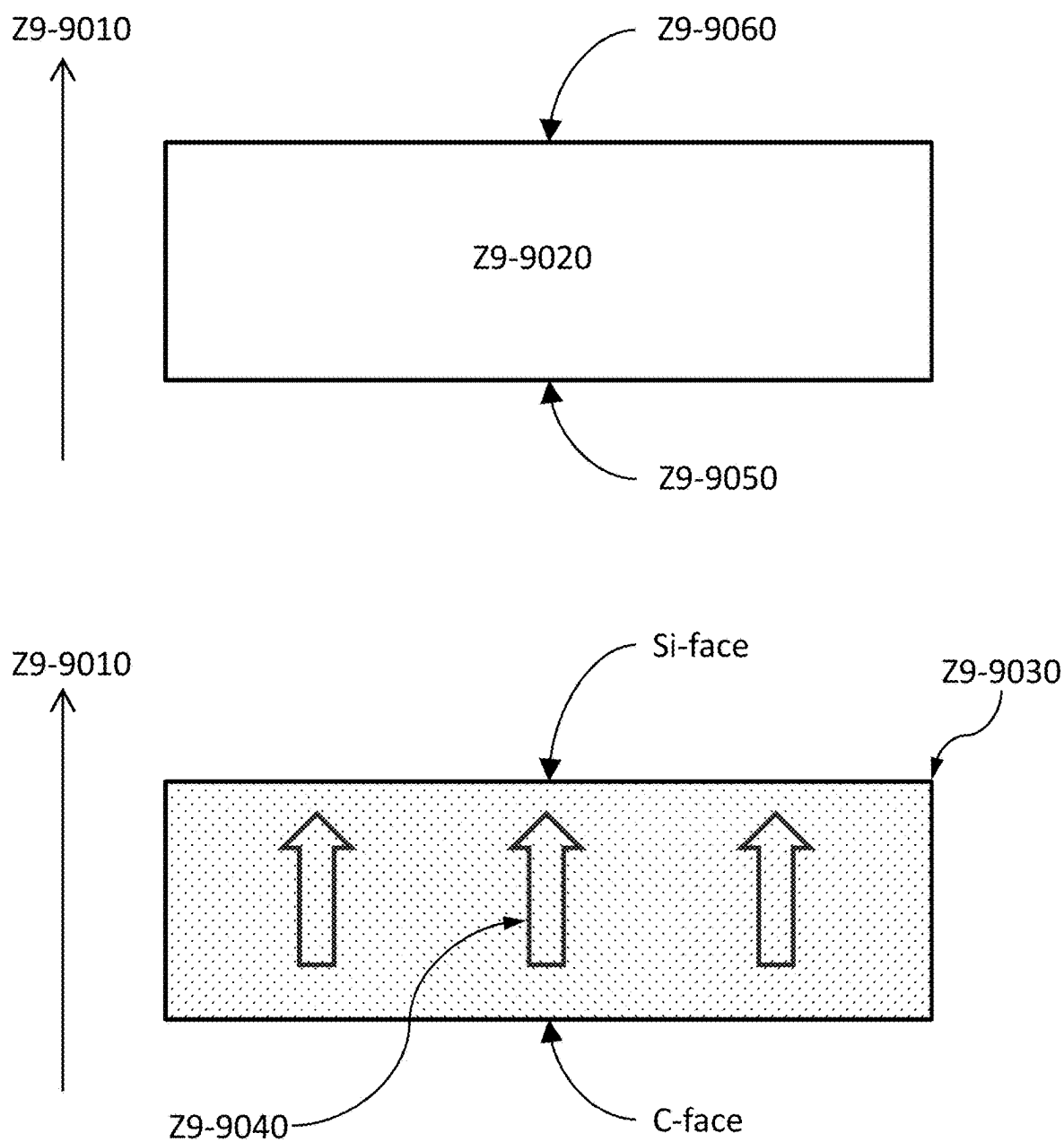


FIG. 56

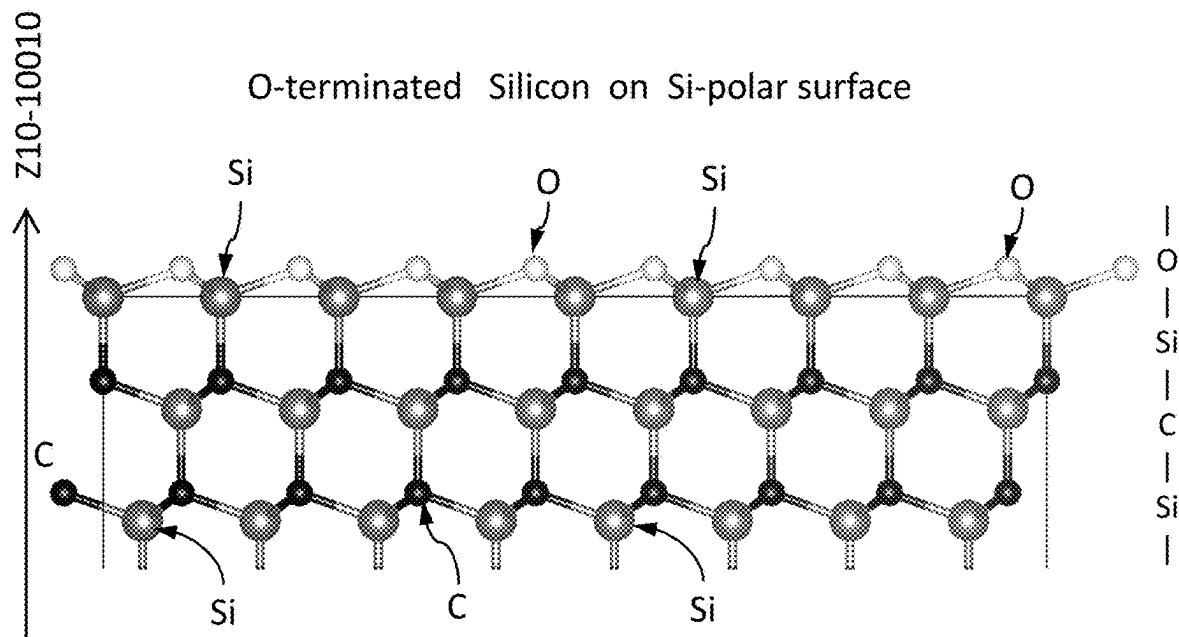


FIG. 57A

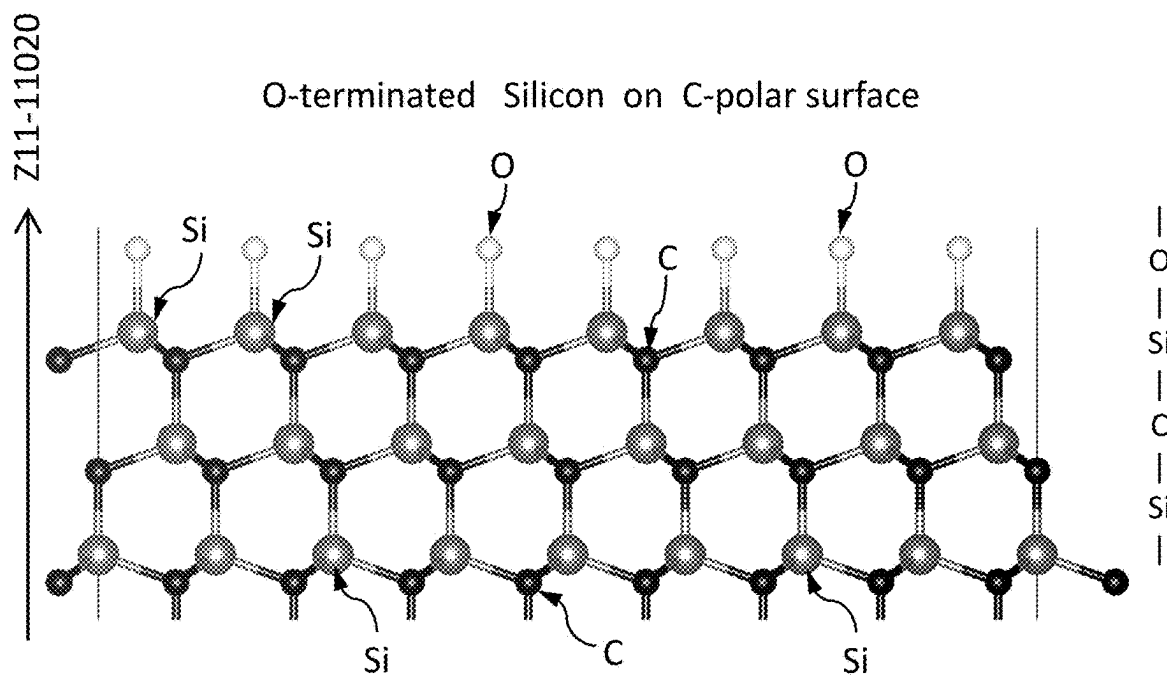


FIG. 57B

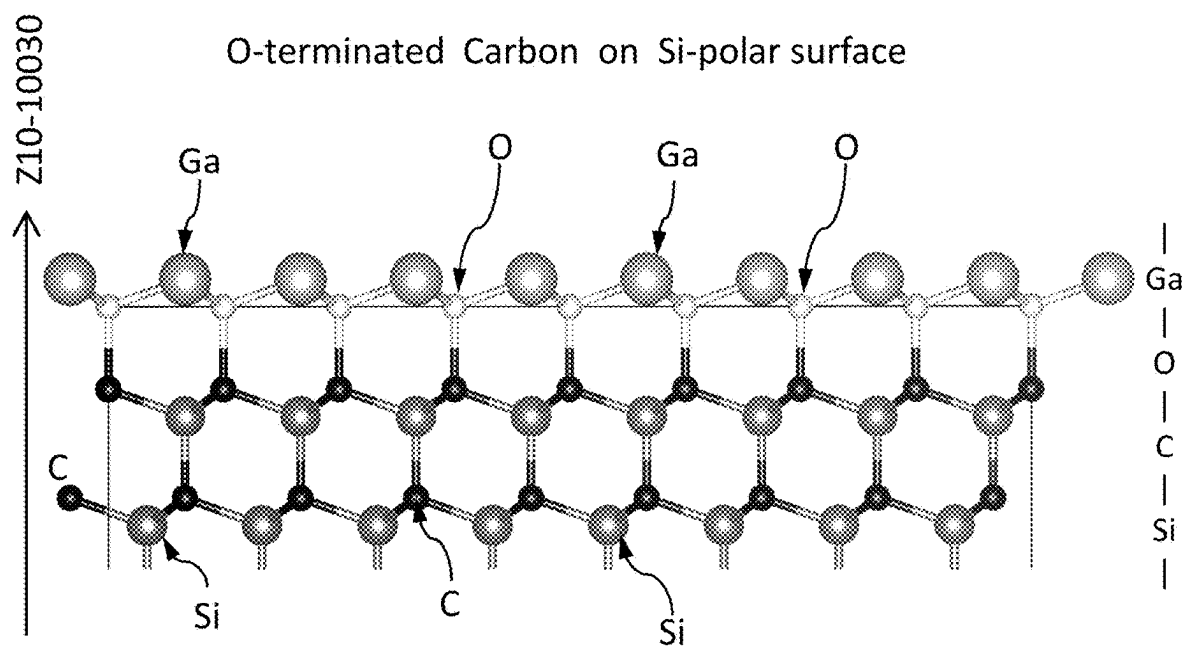


FIG. 57C

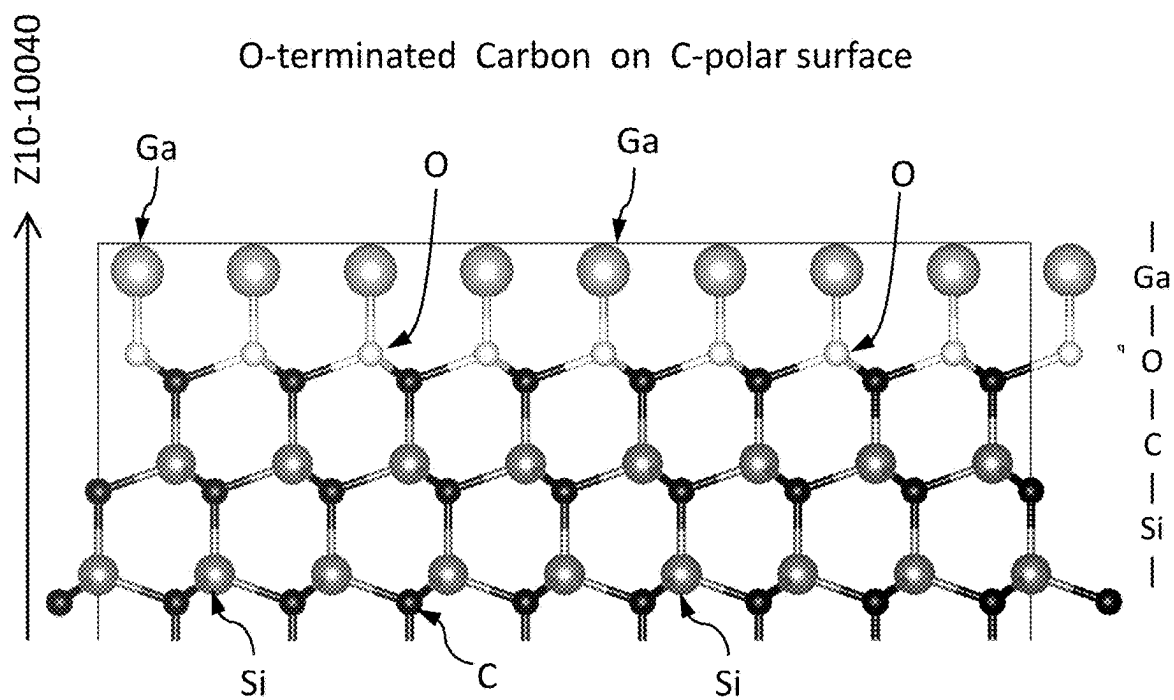


FIG. 57D

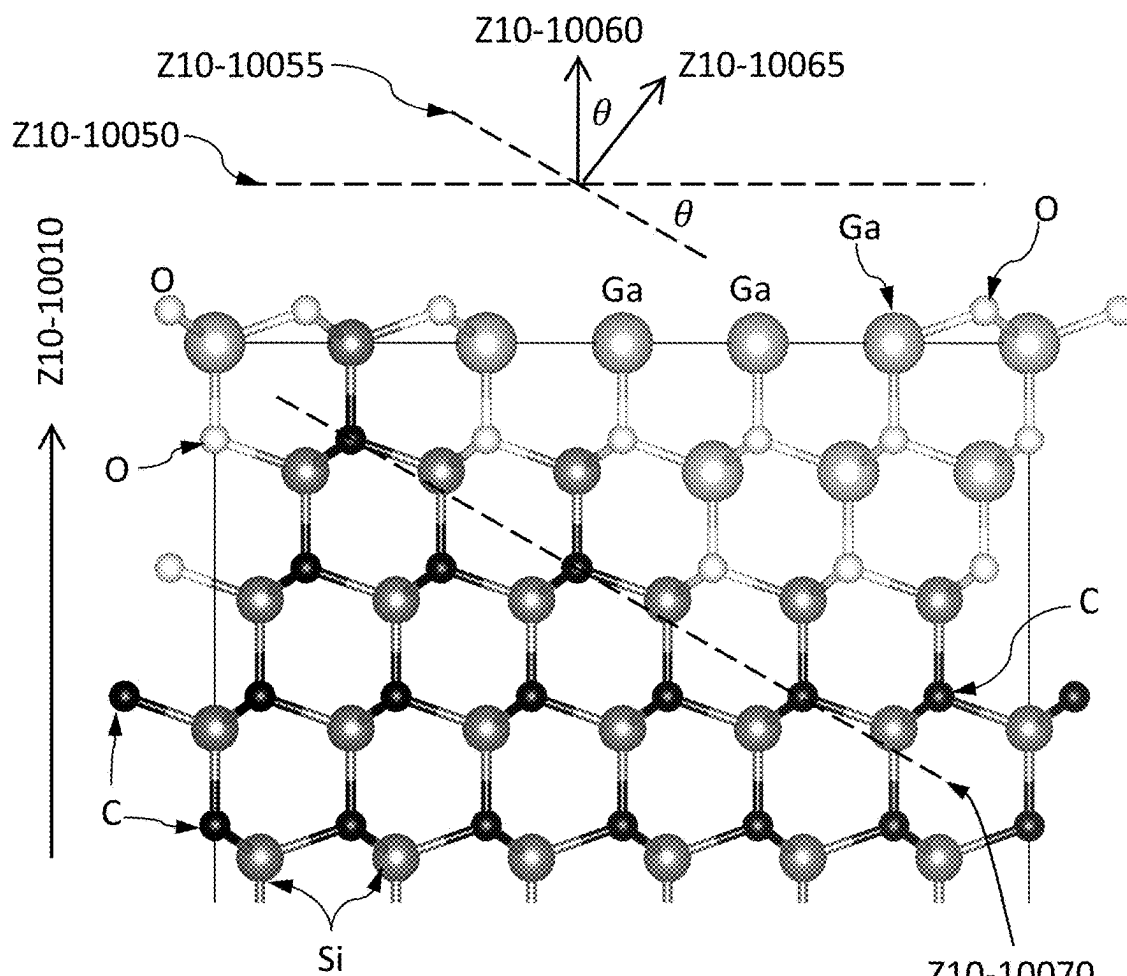


FIG. 57E

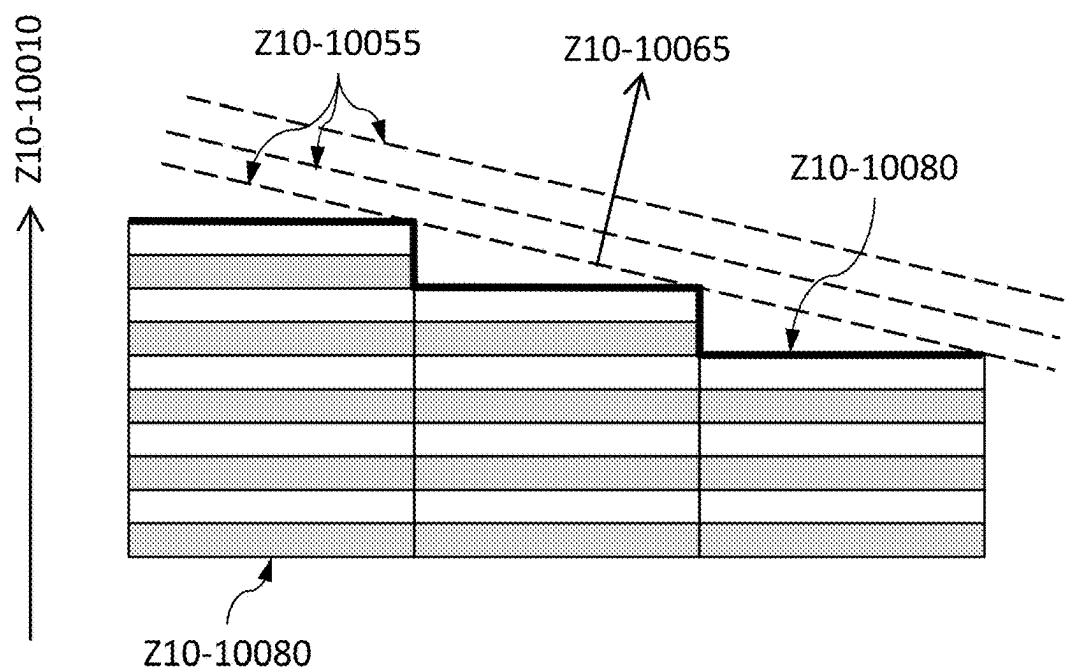


FIG. 57F

Z10G-1000 ↘

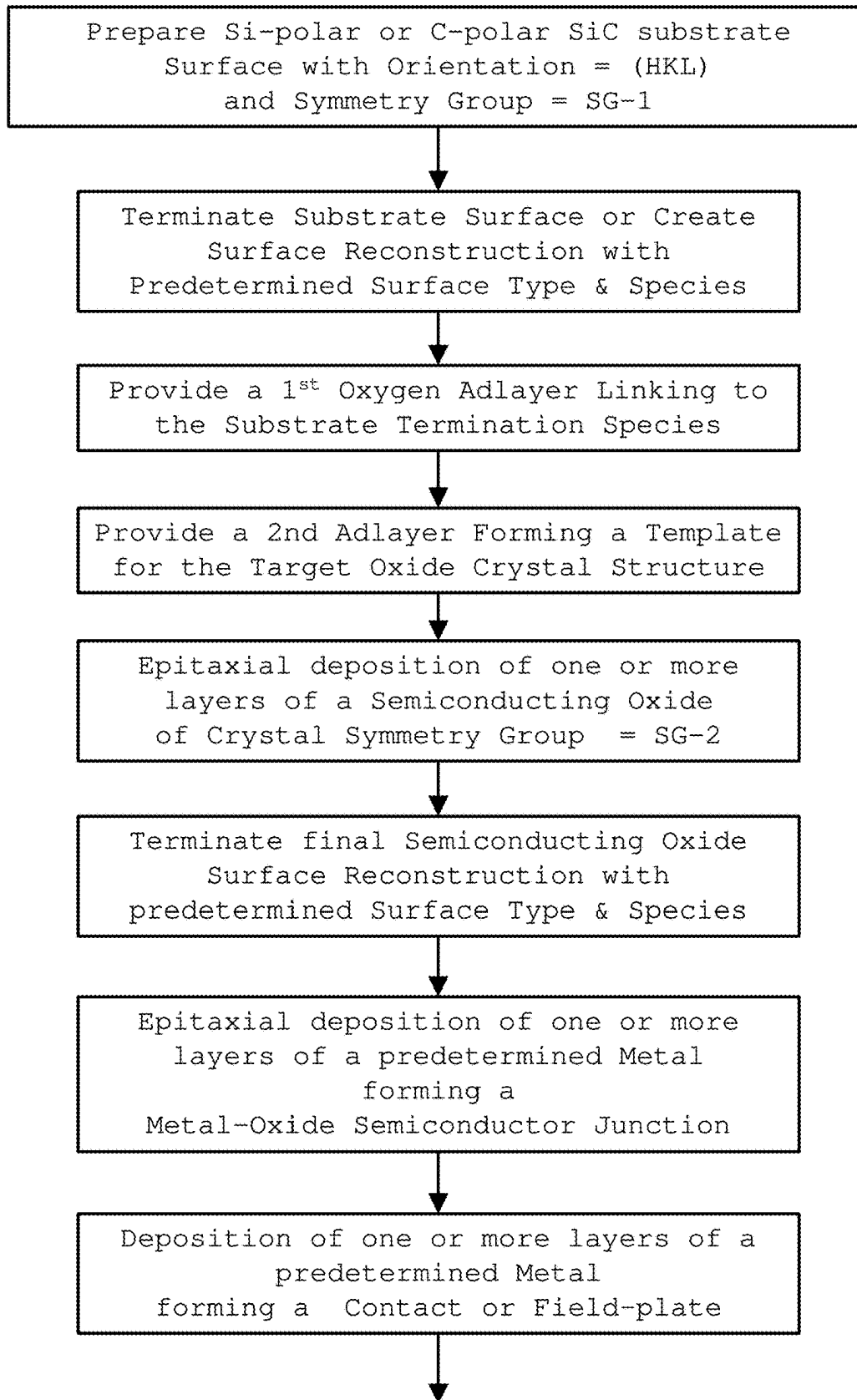


FIG. 57G

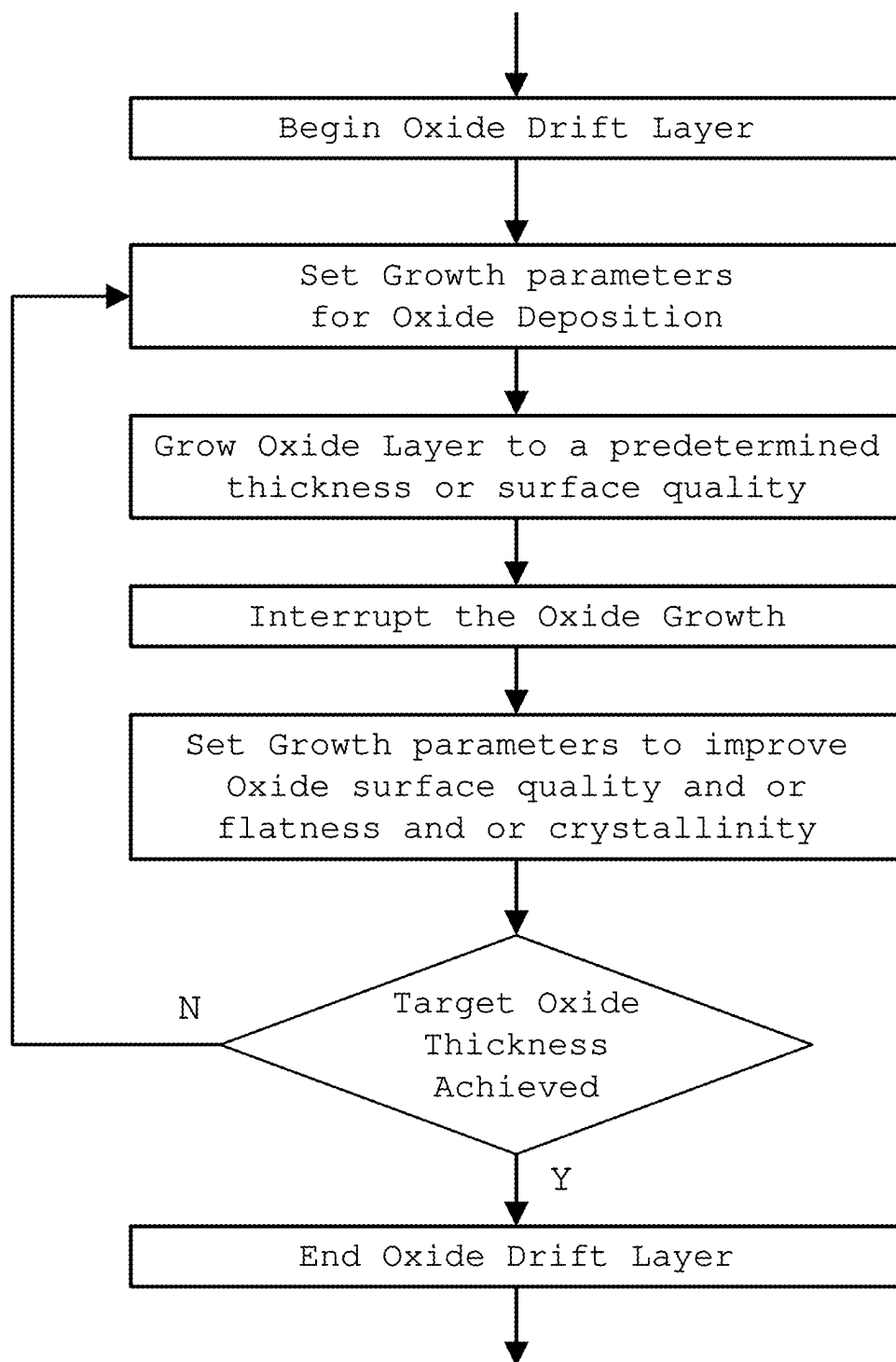


FIG. 57H

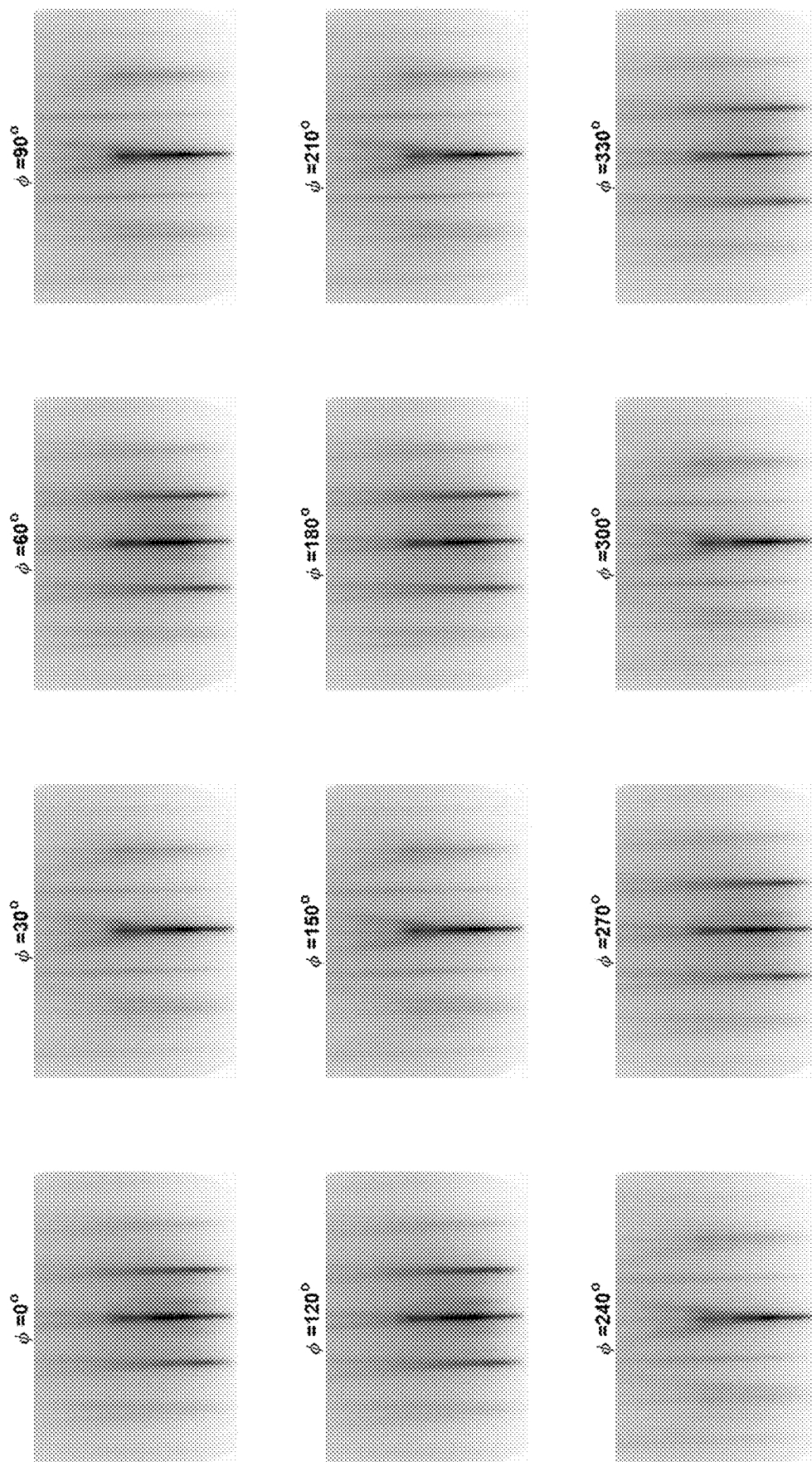


FIG. 58

SiC 4deg micut

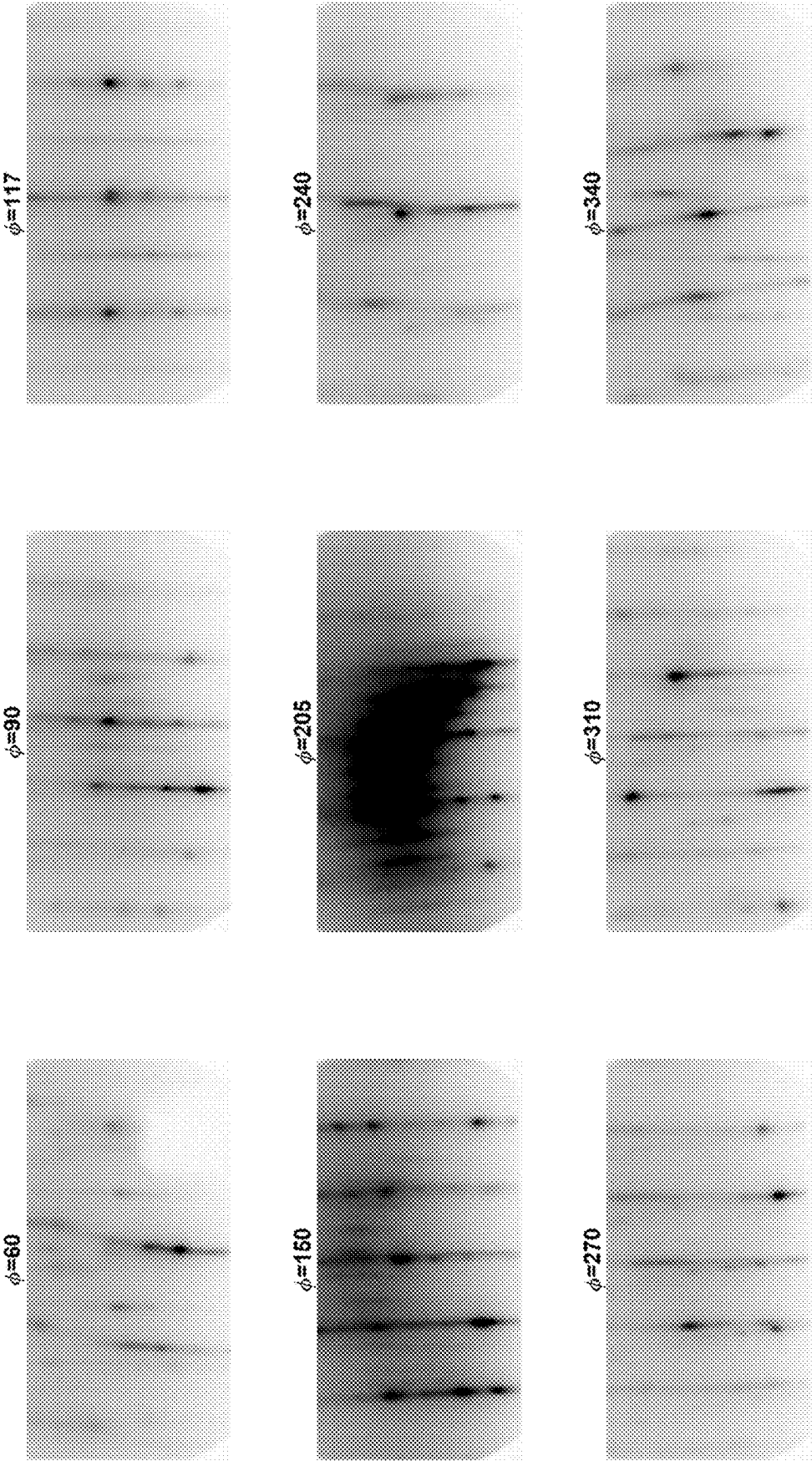


FIG. 59

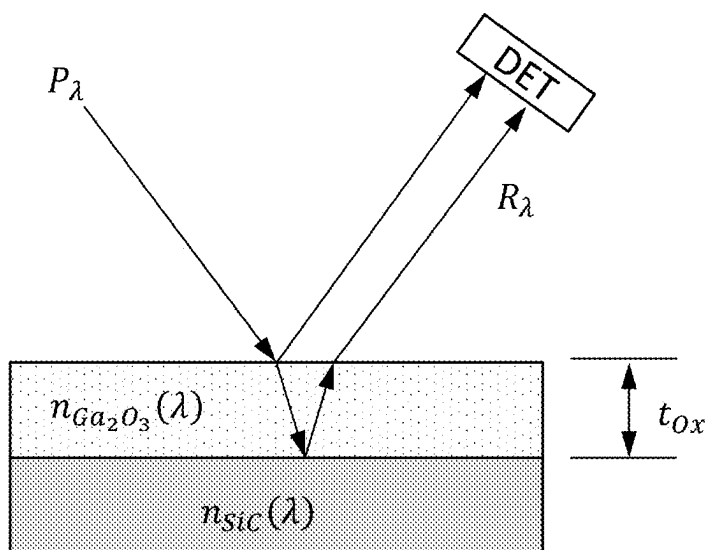


FIG. 60A

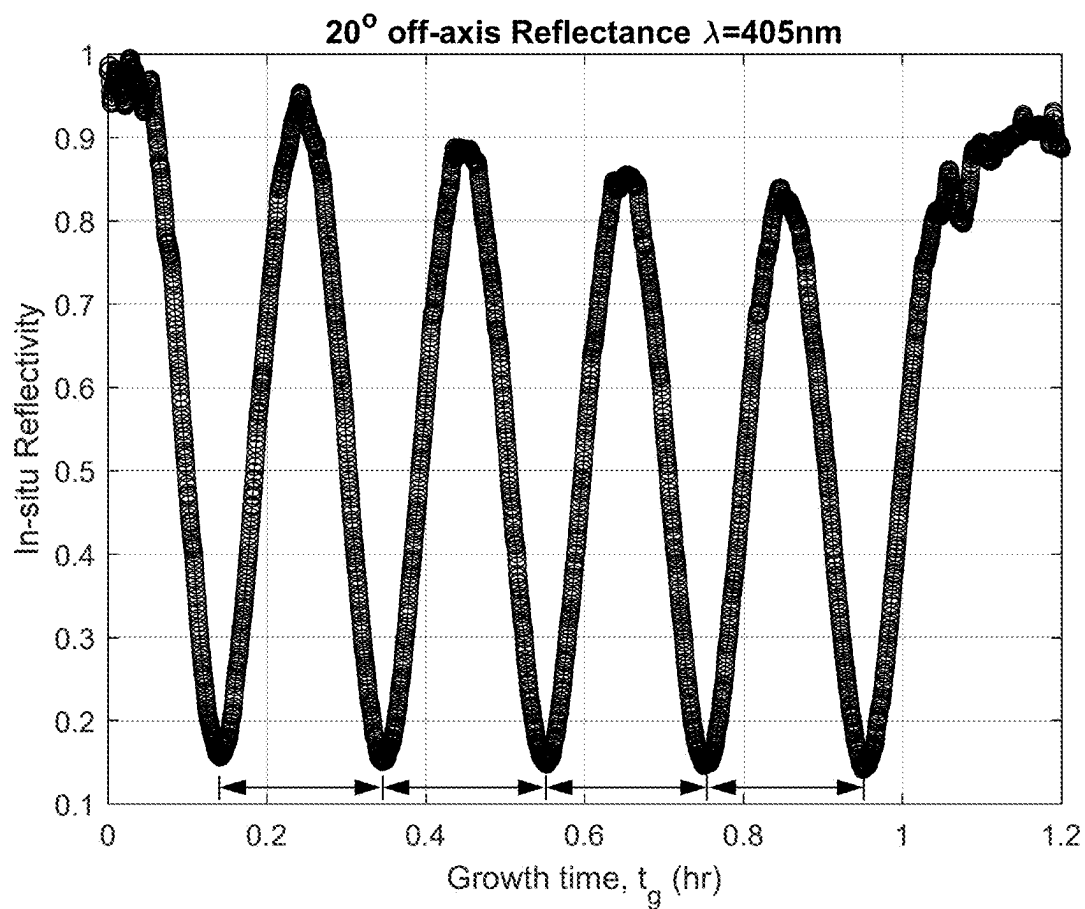


FIG. 60B

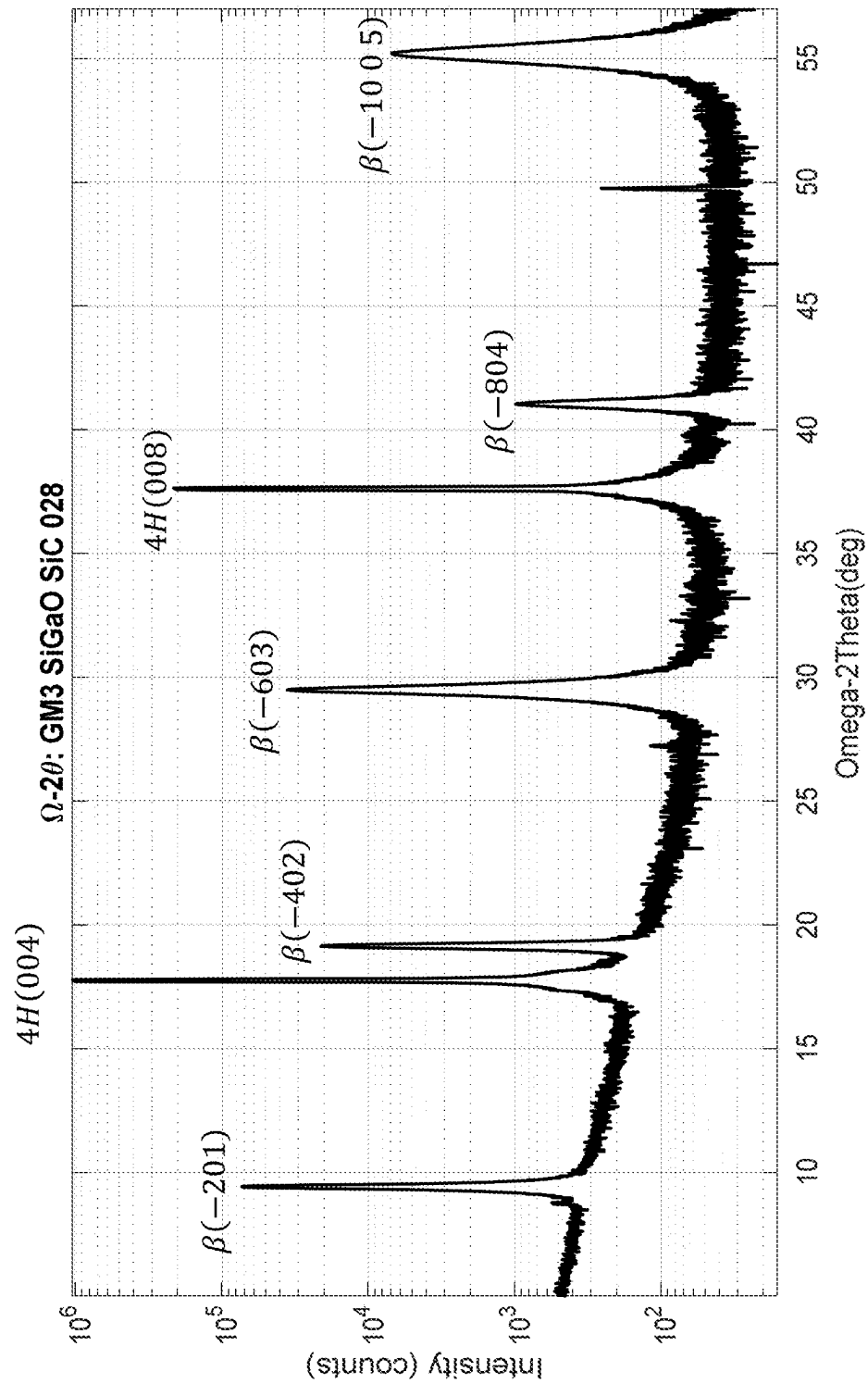


FIG. 61A

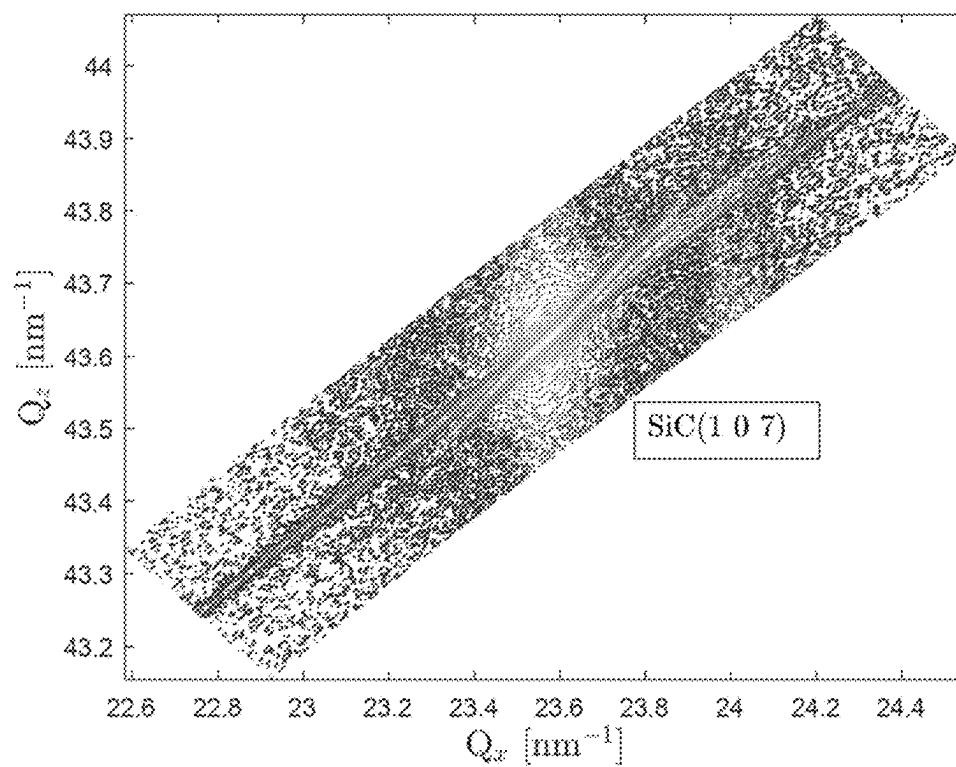


FIG. 61B

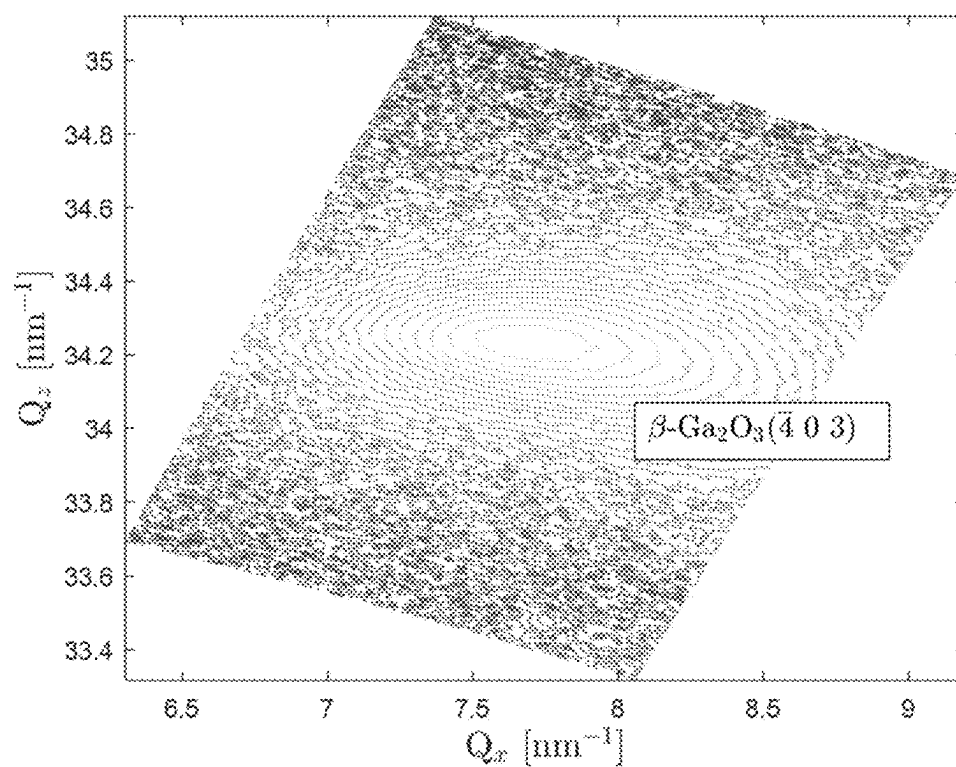


FIG. 61C

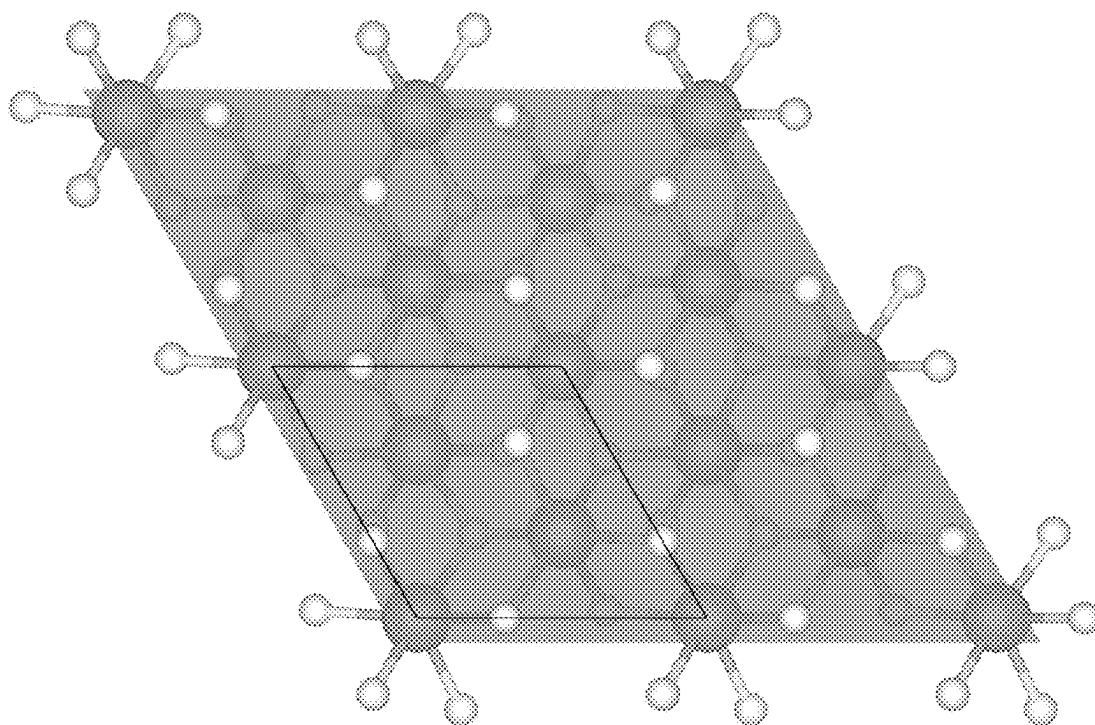
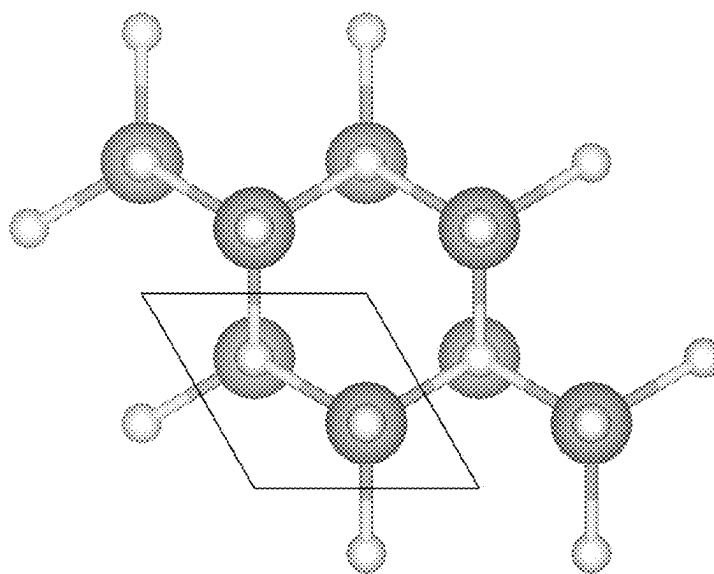


FIG. 62



P63mc WZ

FIG. 63

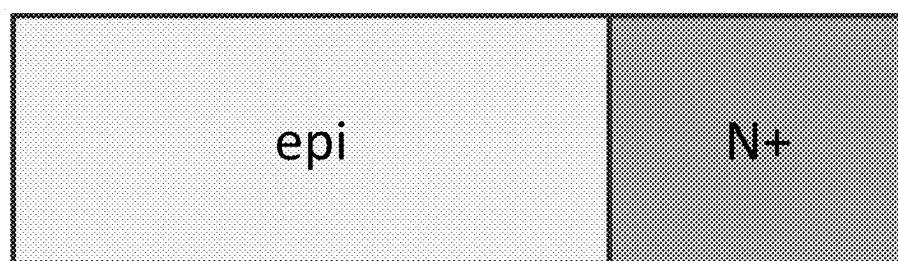
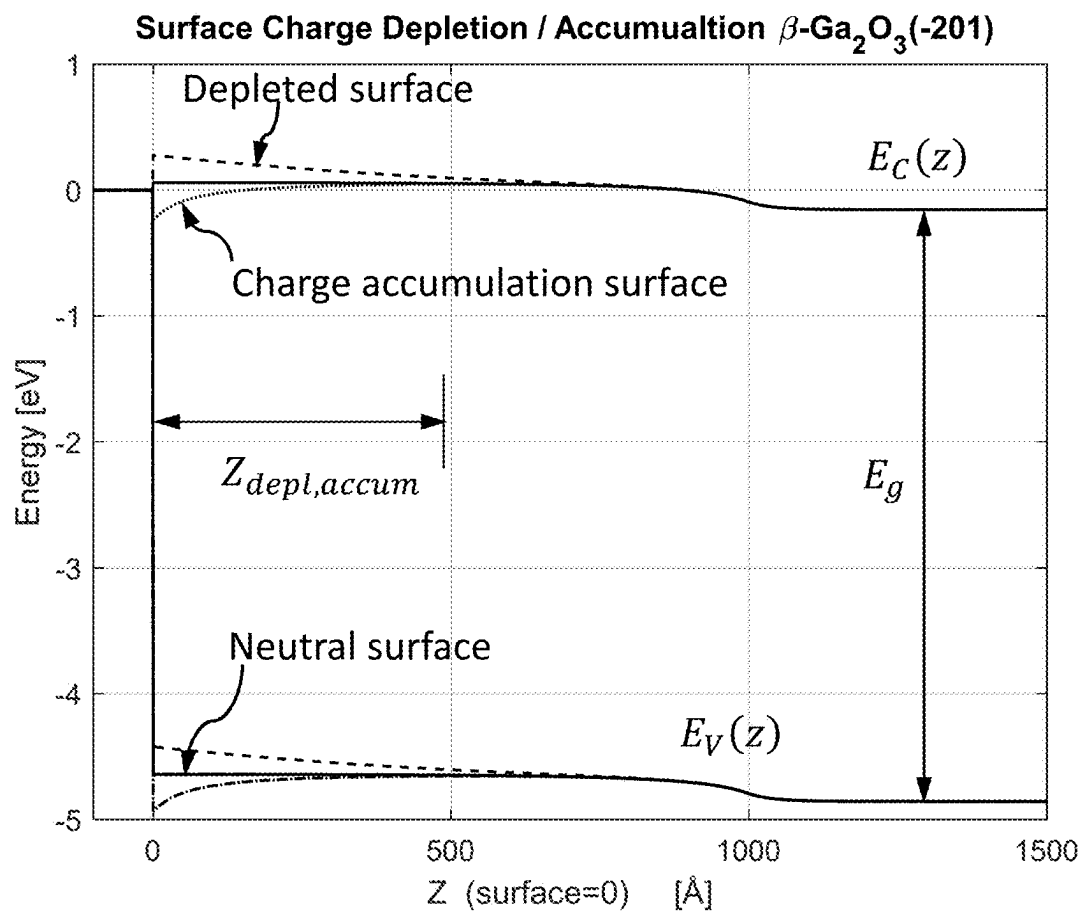


FIG. 65

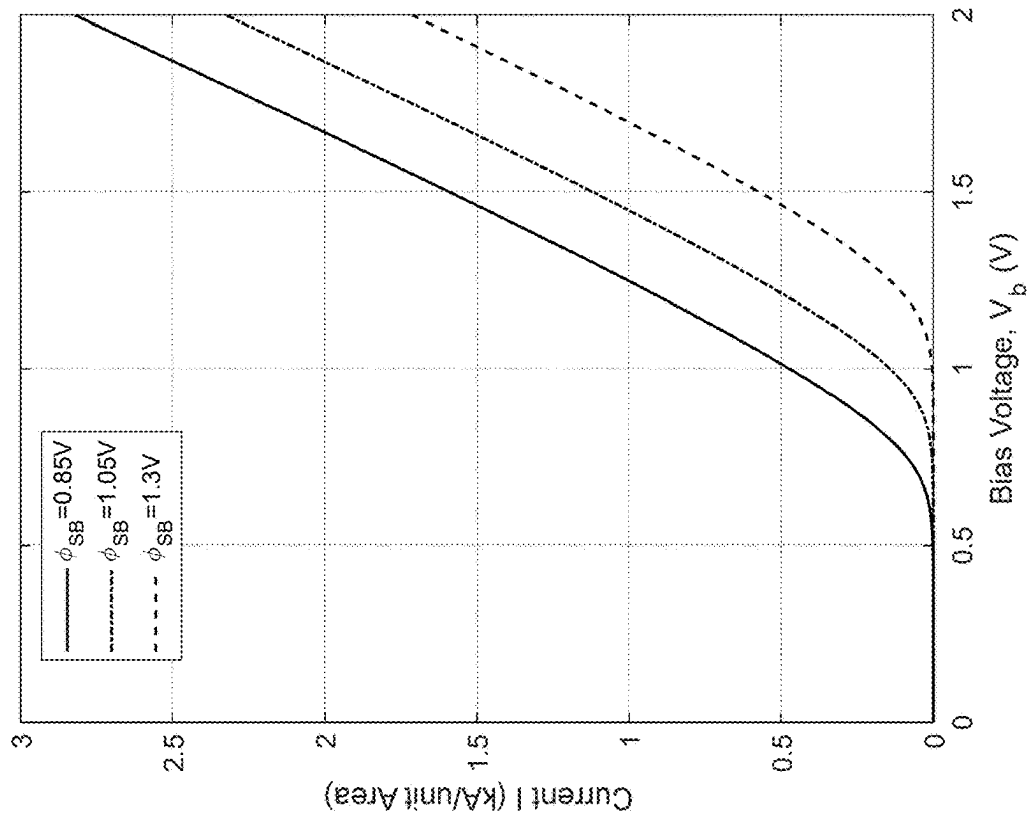


FIG. 66A

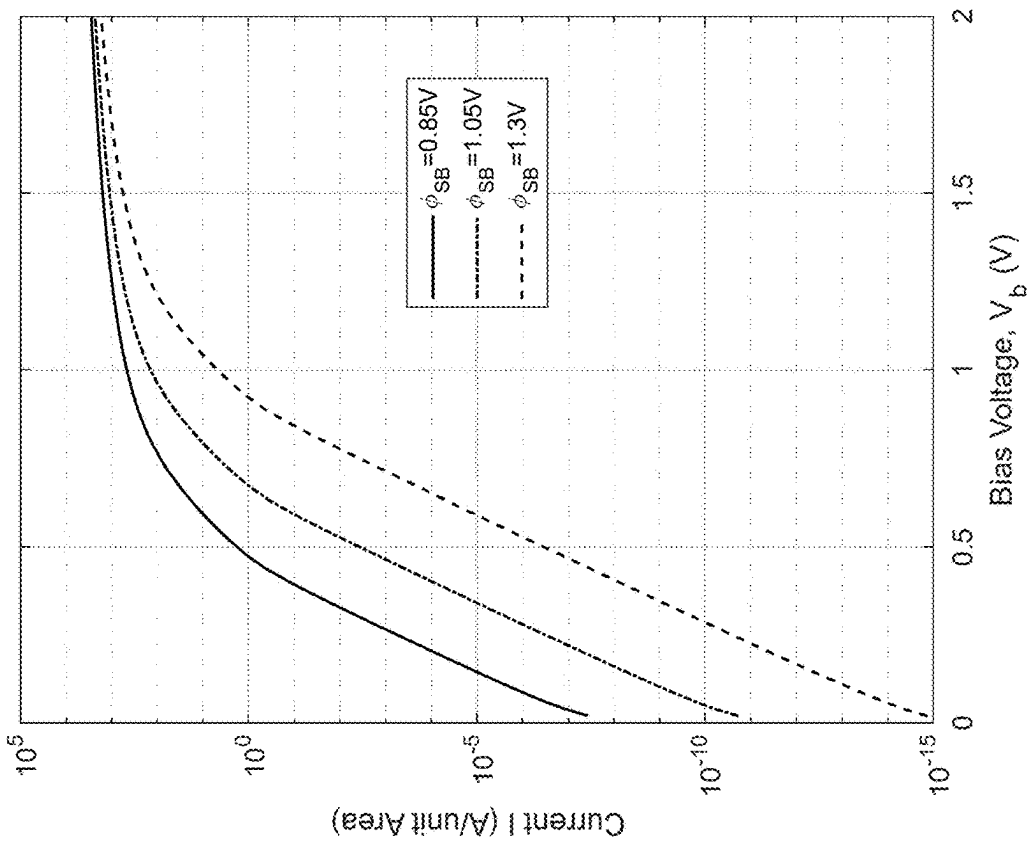


FIG. 66B

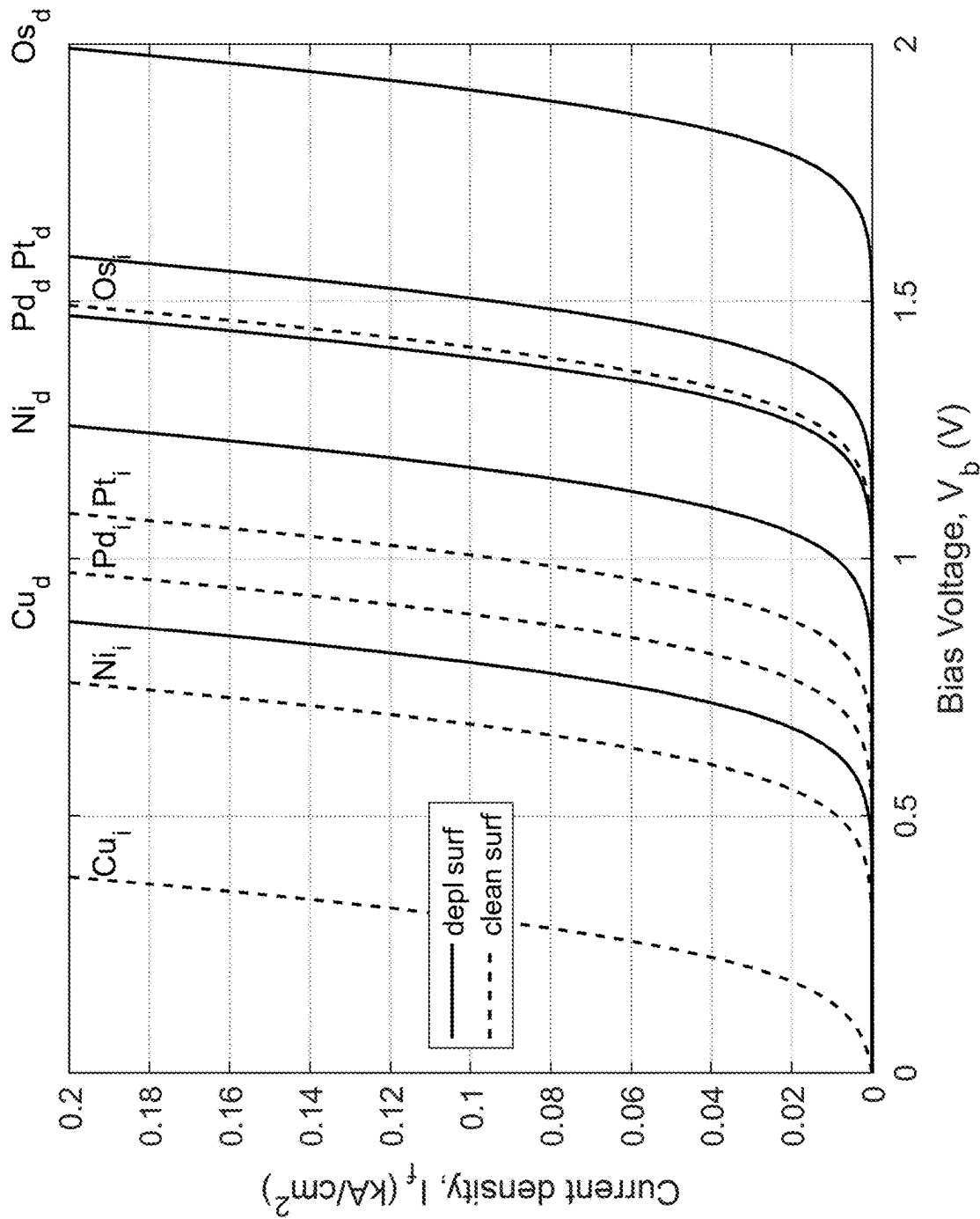


FIG. 67

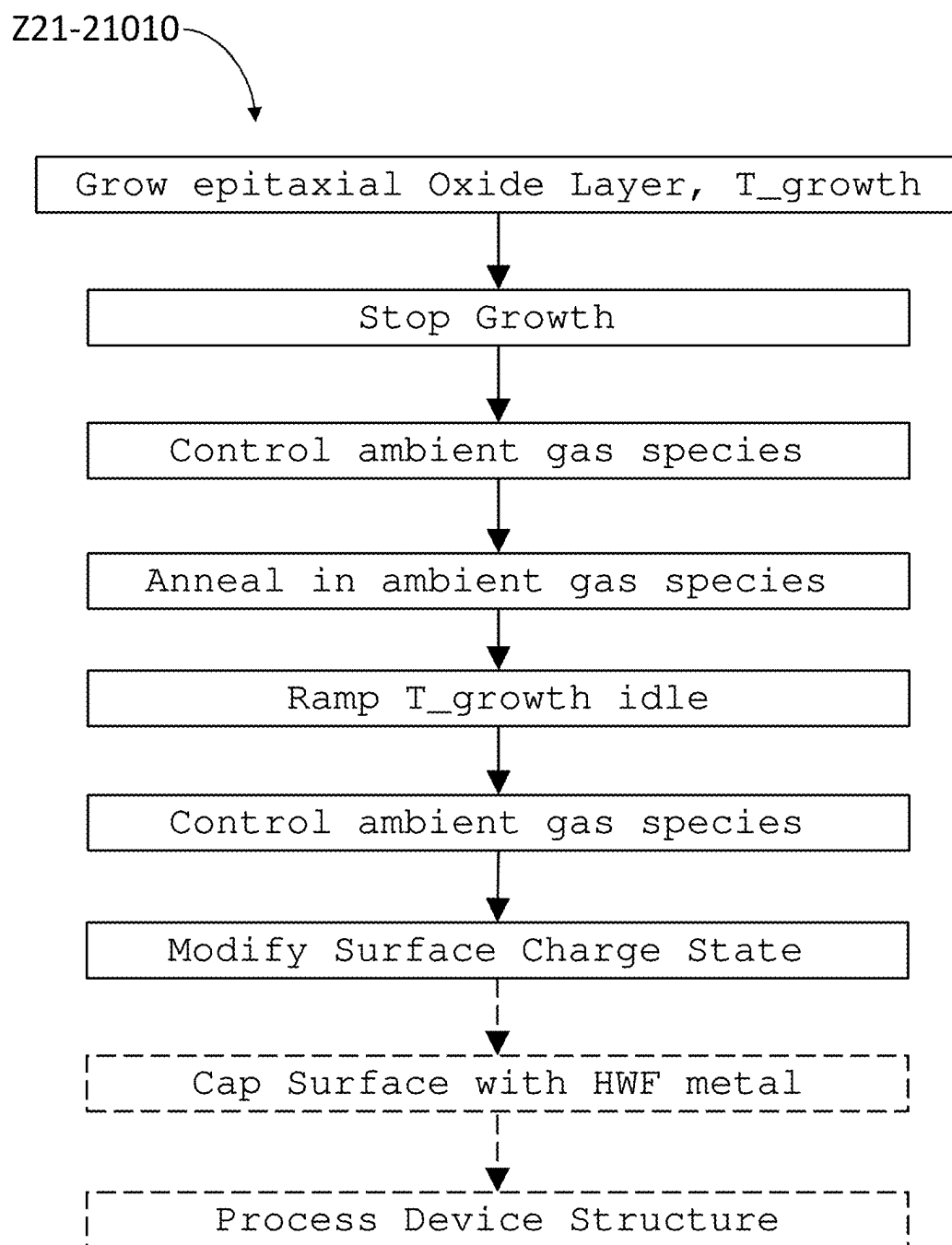


FIG. 68A

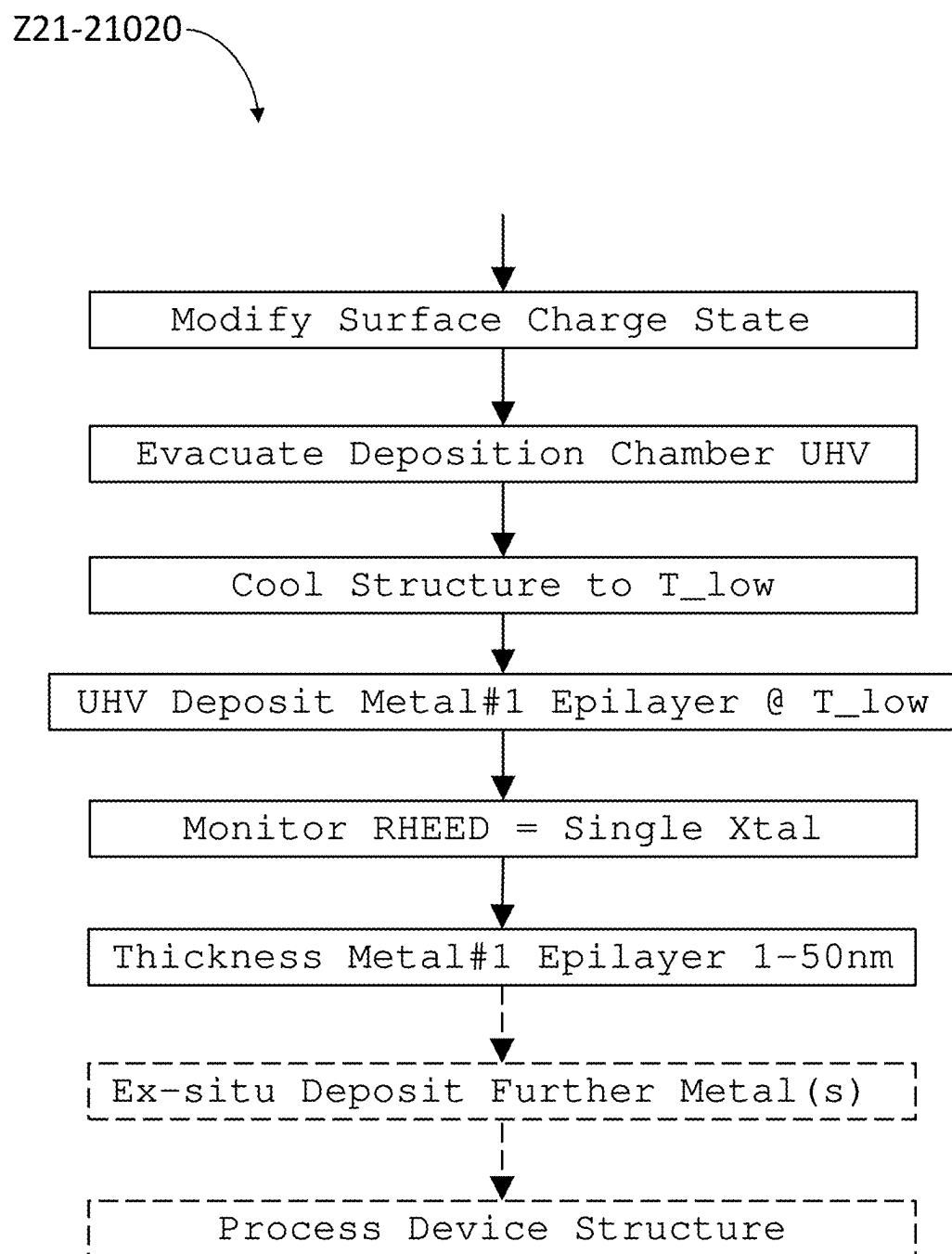


FIG. 68B

Epitaxial Ni on $\beta\text{Ga}_2\text{O}_3(-201)$ O-term surface

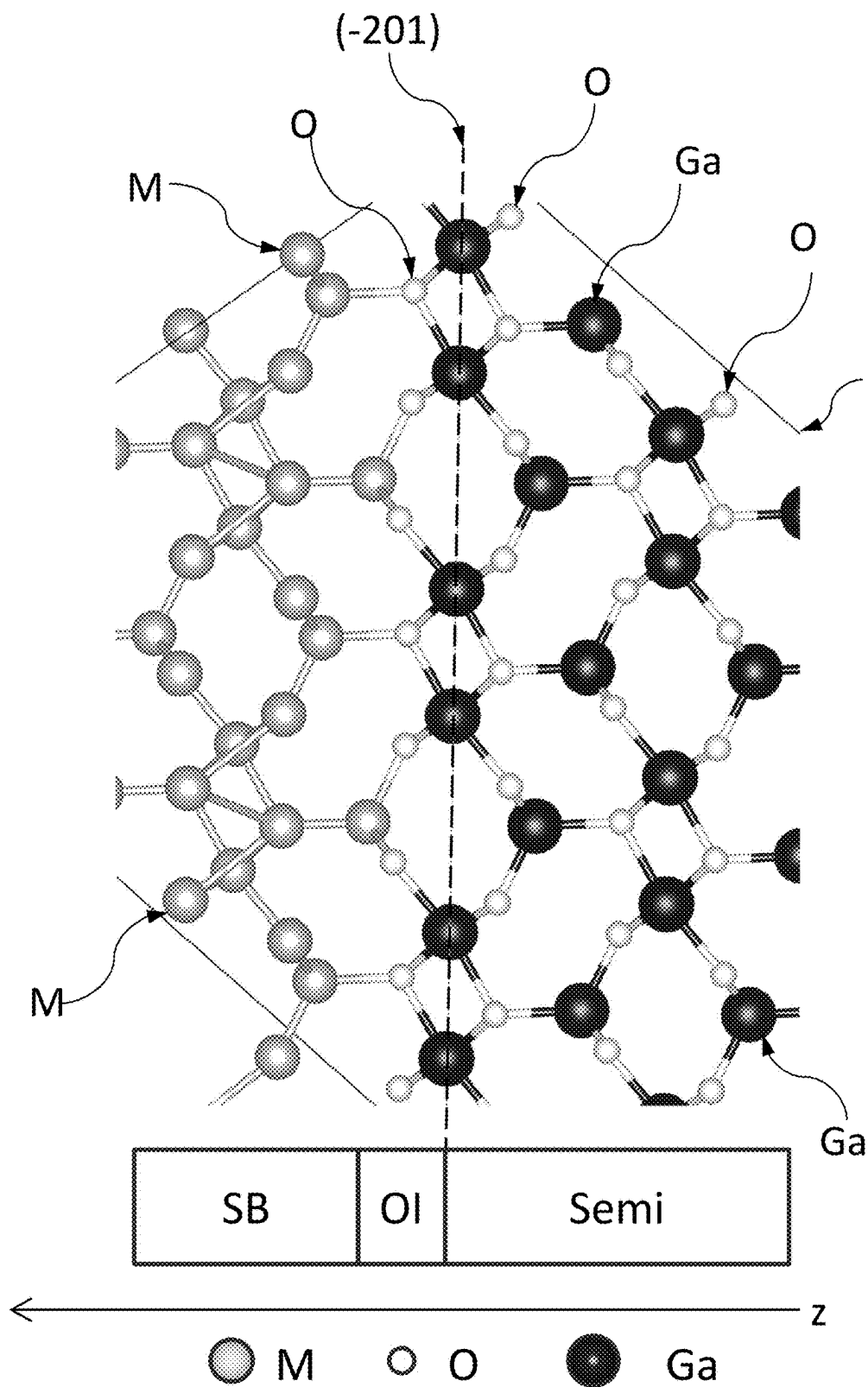


FIG. 69

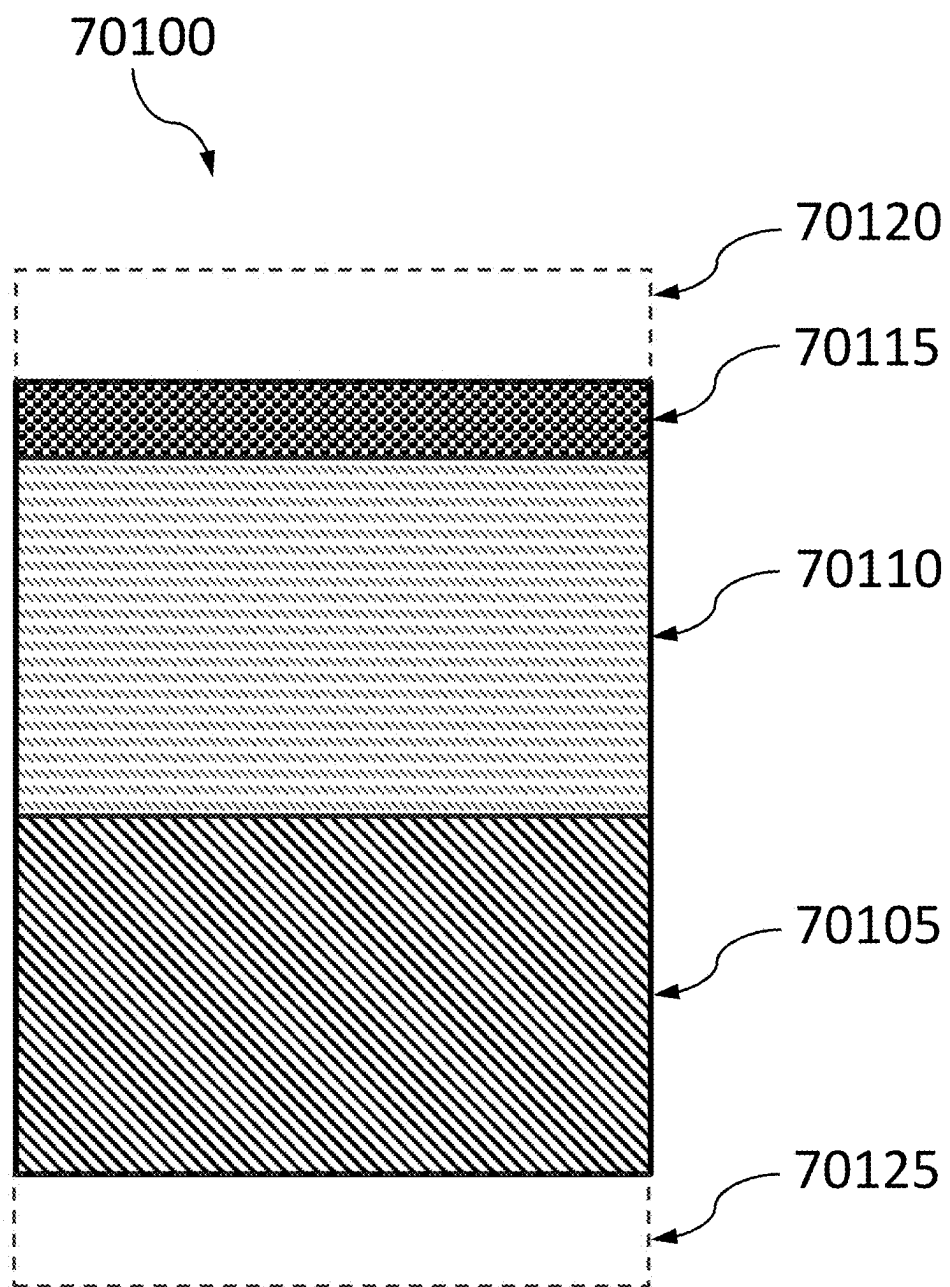


FIG. 70A

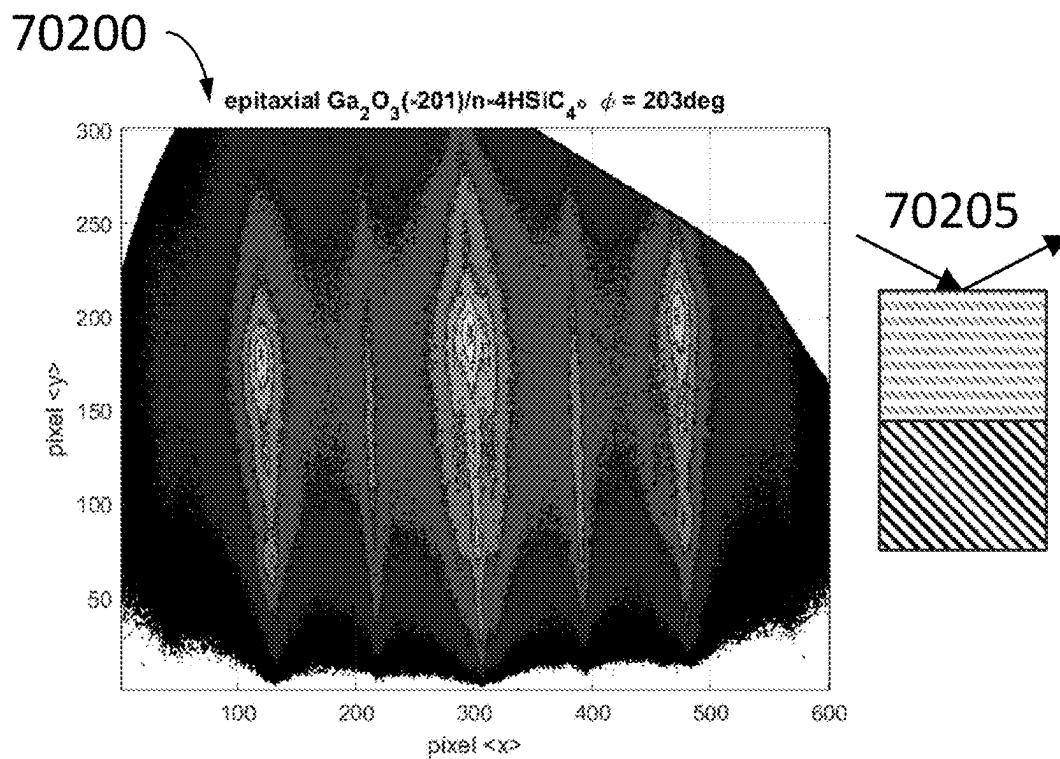


FIG. 70B

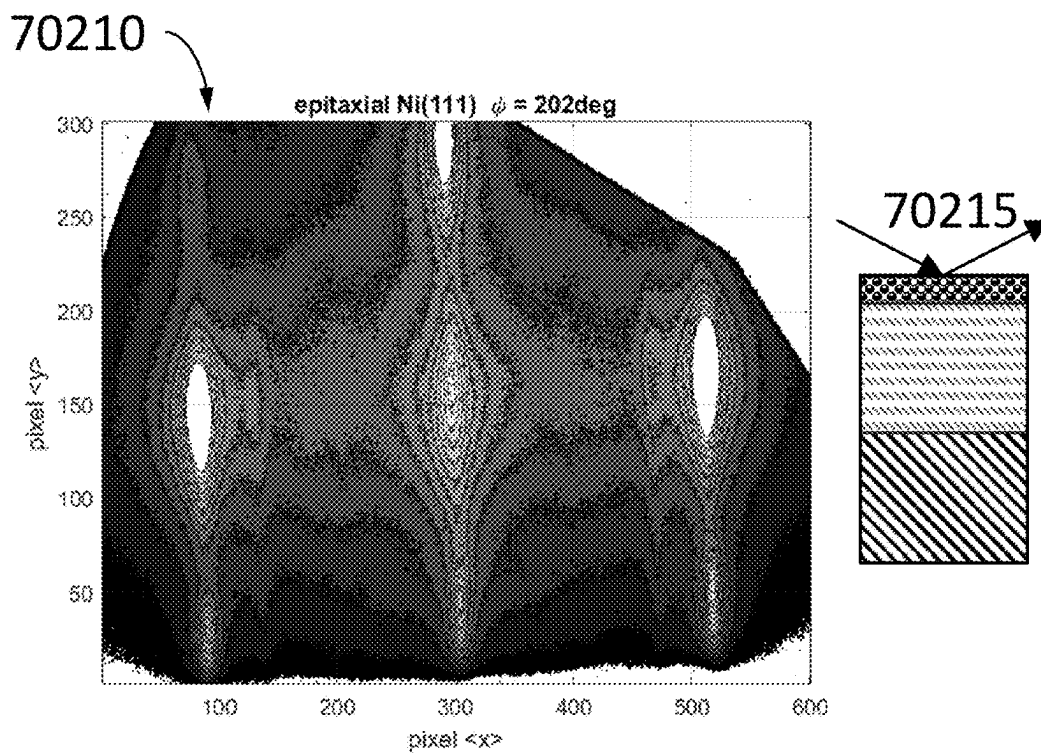


FIG. 70C

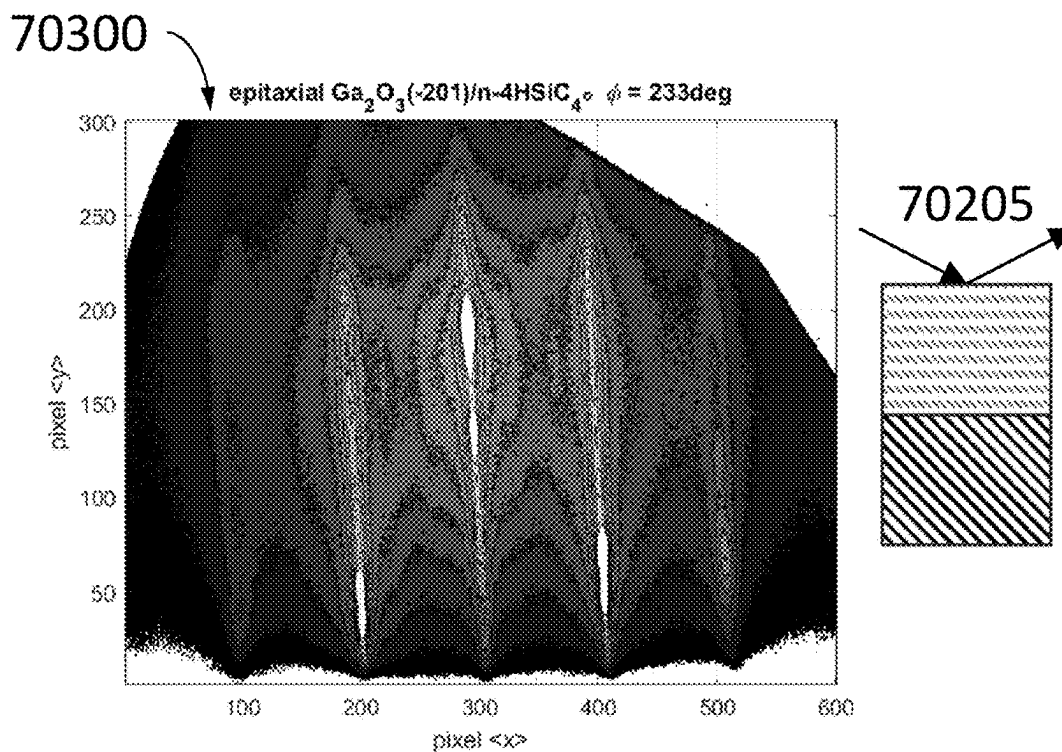


FIG. 70D

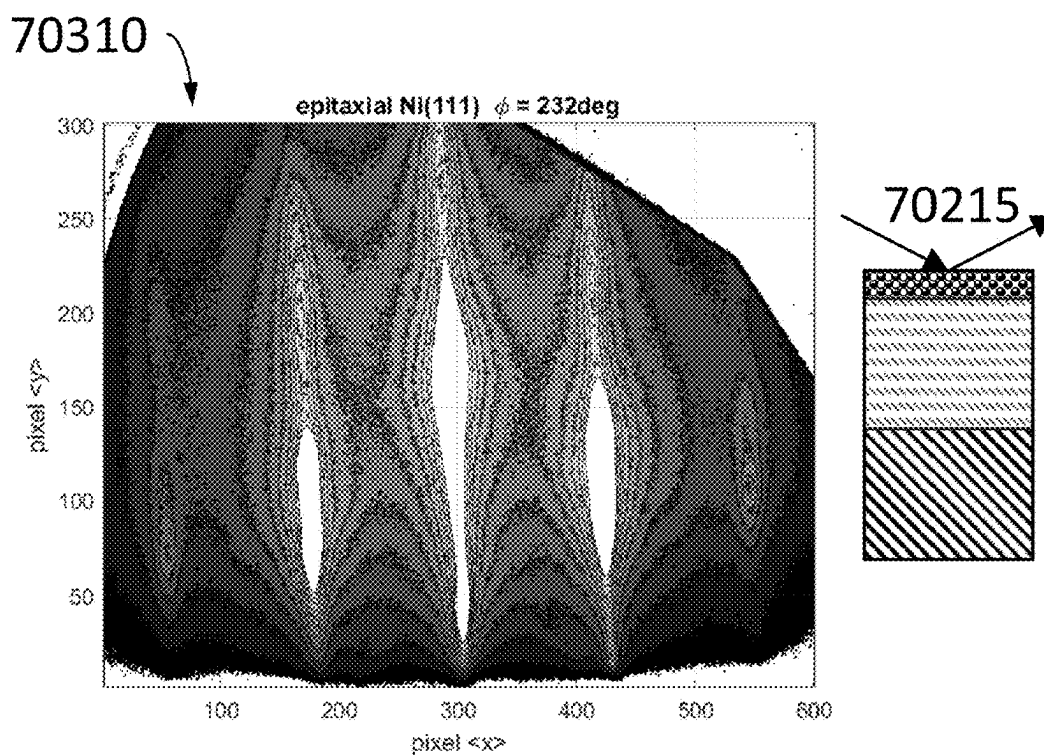


FIG. 70E

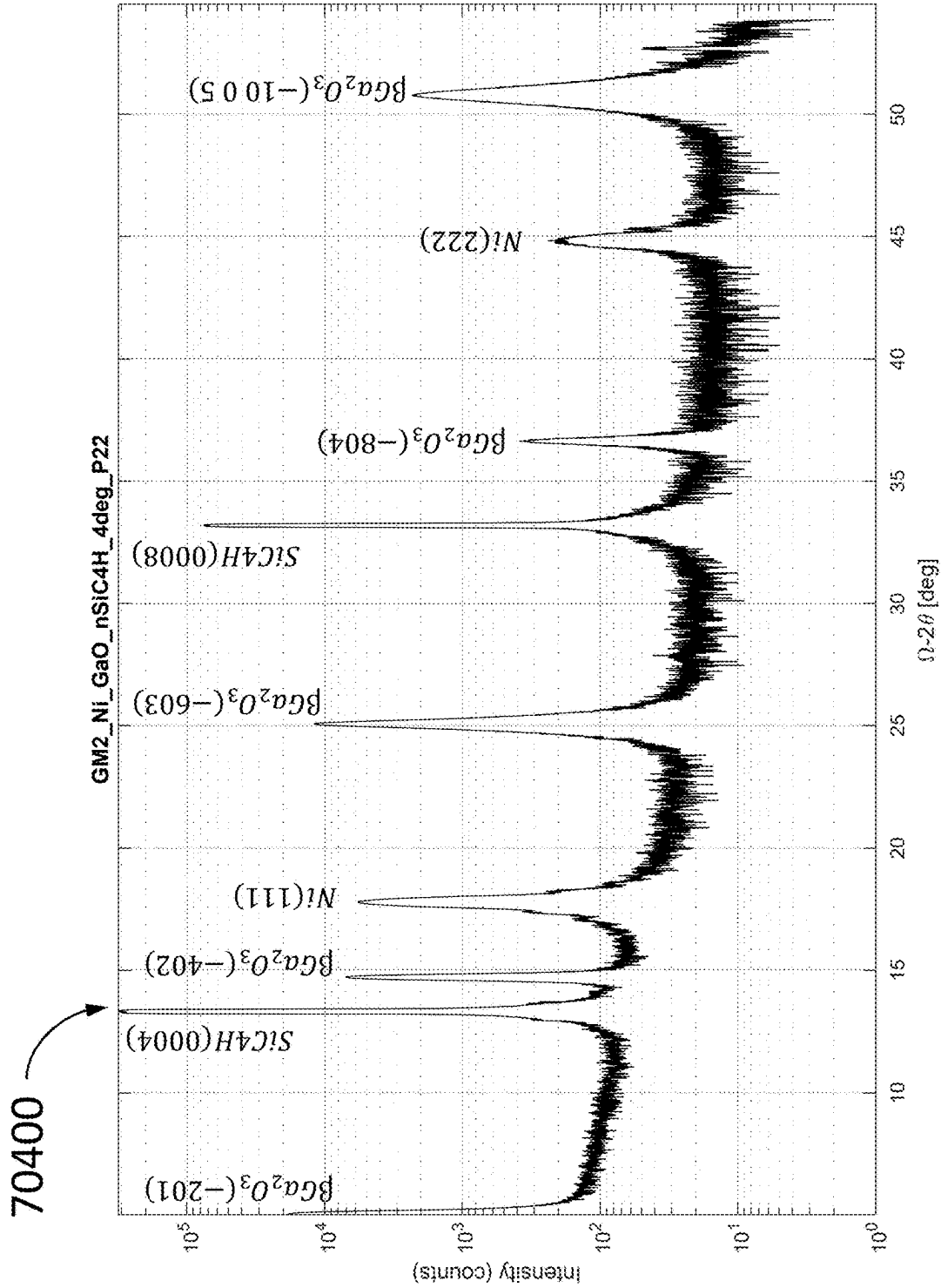


FIG. 70F

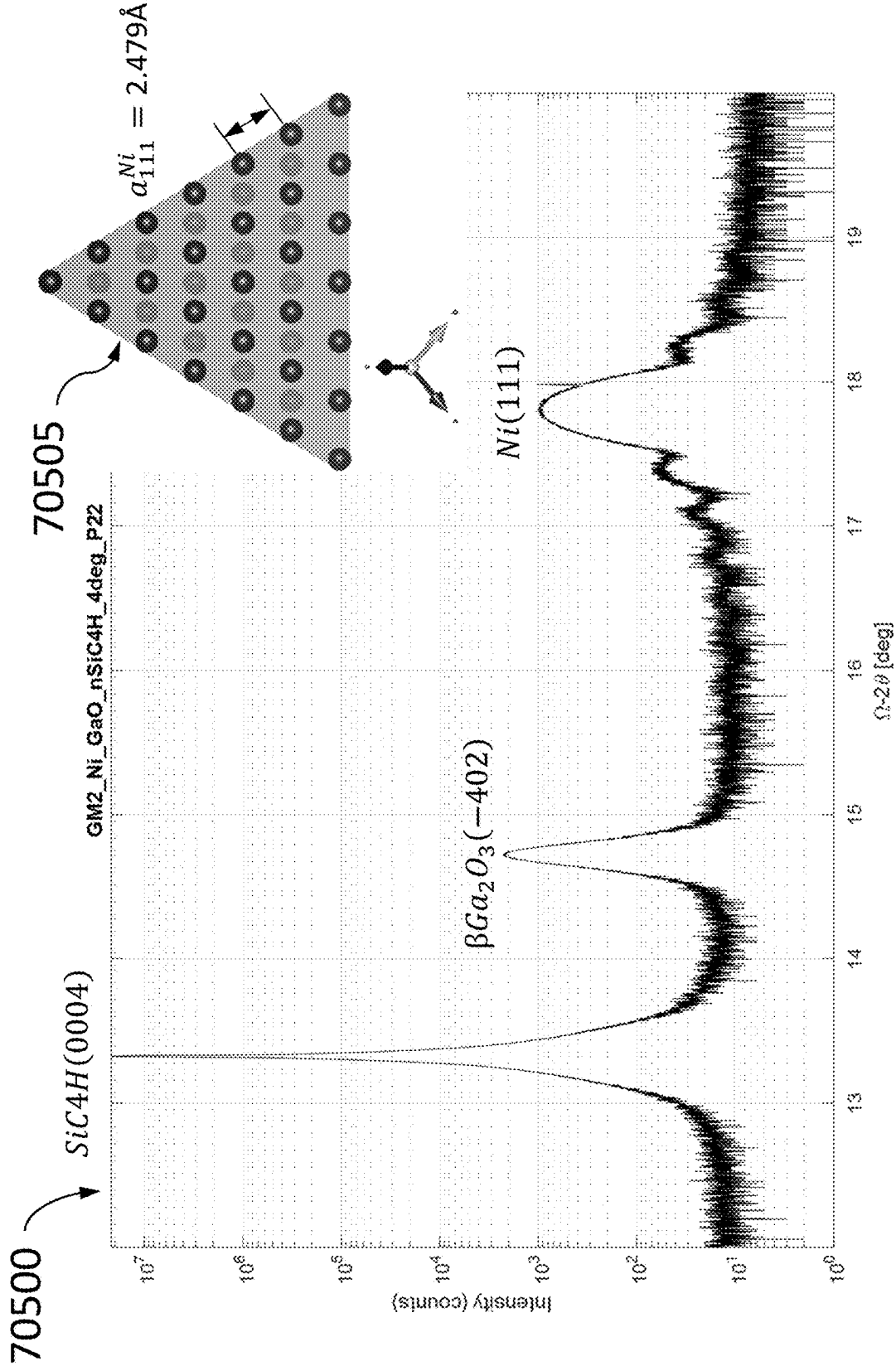


FIG. 70G

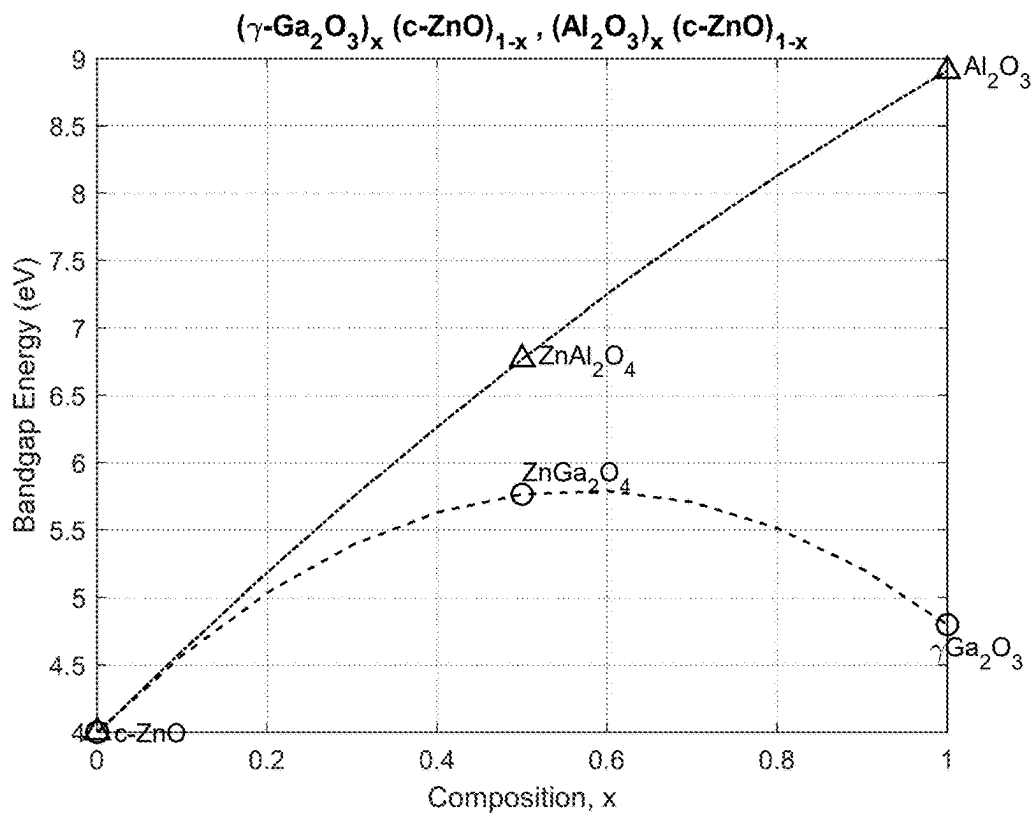


FIG. 71A

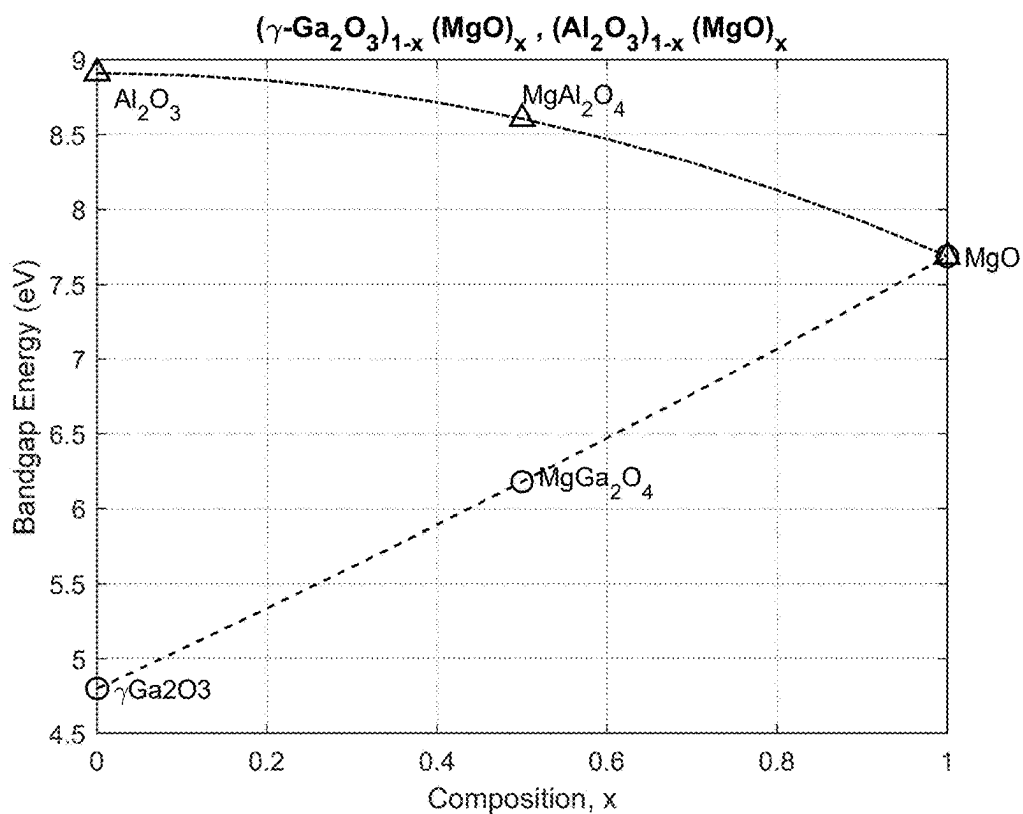


FIG. 71B

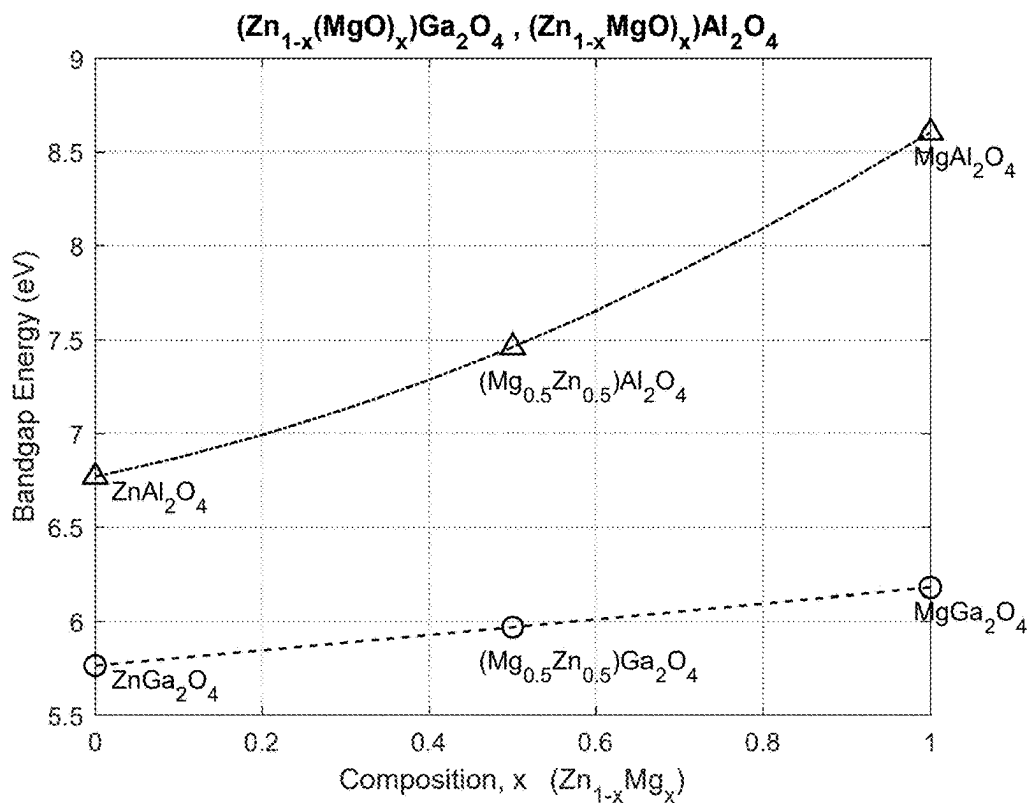


FIG. 72

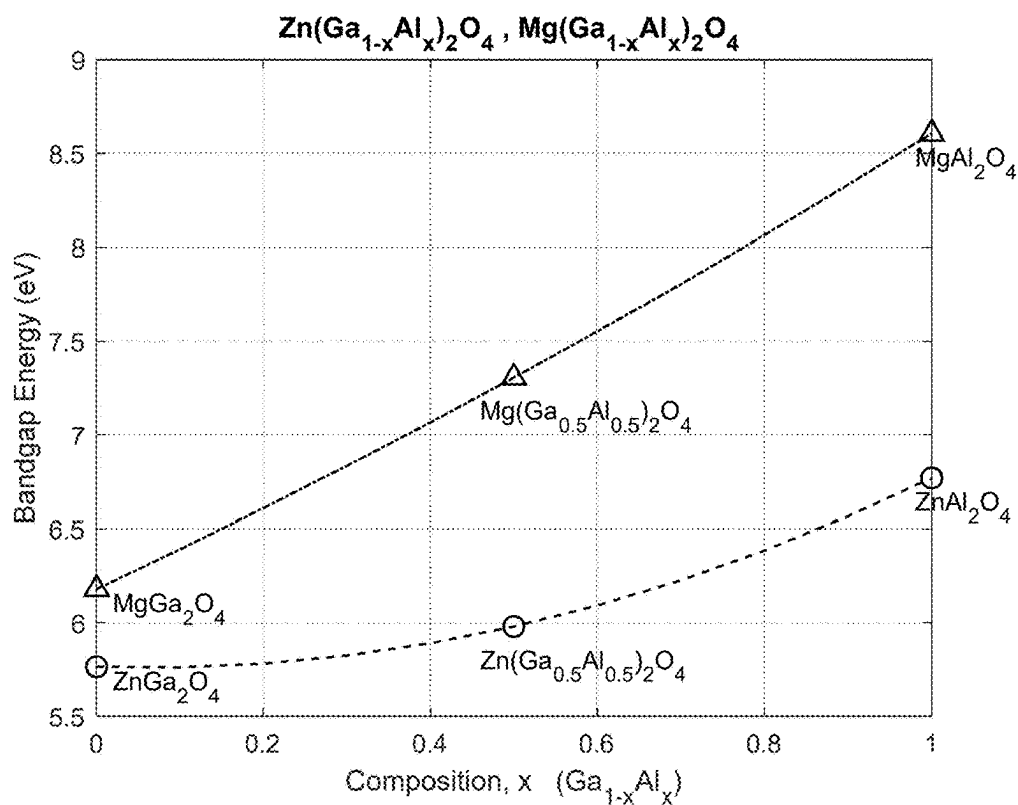


FIG. 73

75-28010

A	Ga_2O_3	$4H - SiC$	K
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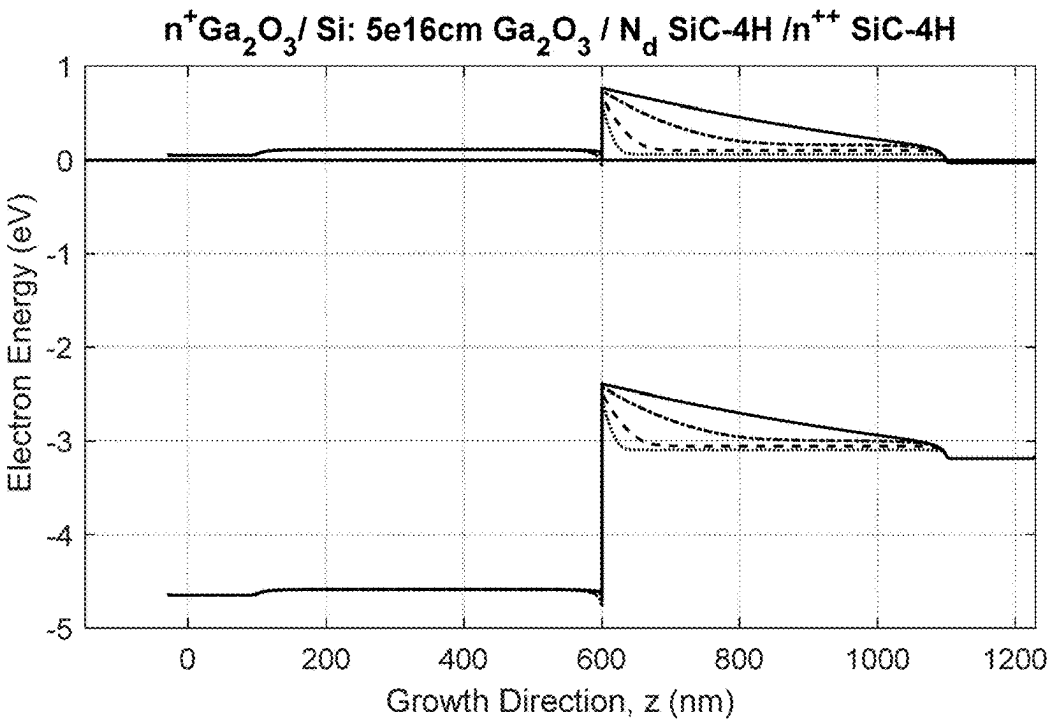


FIG. 74

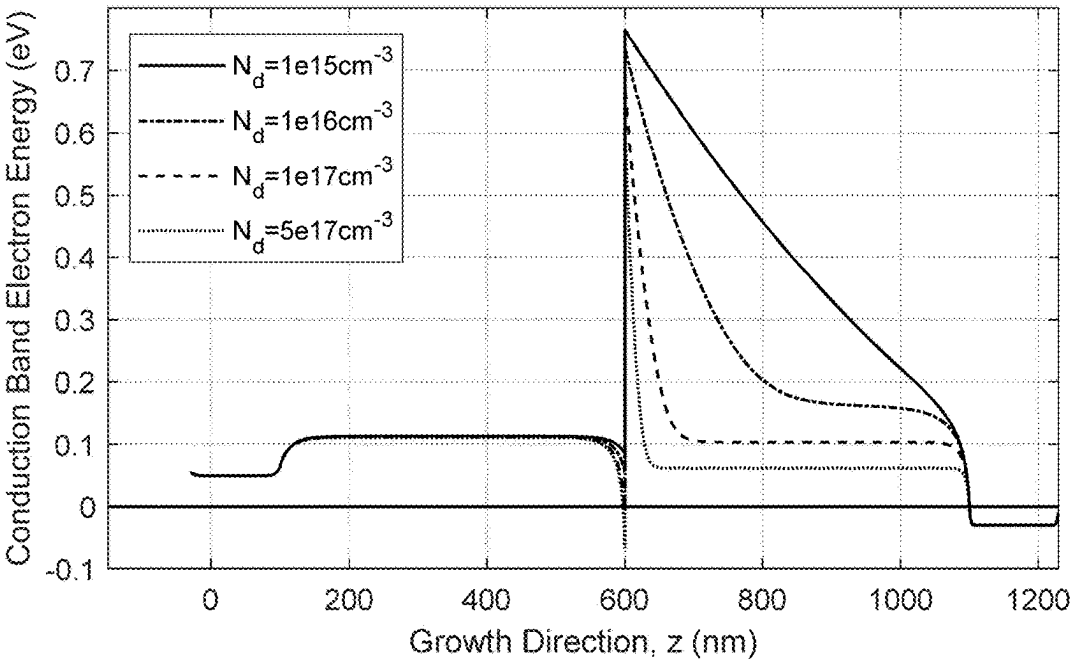


FIG. 75

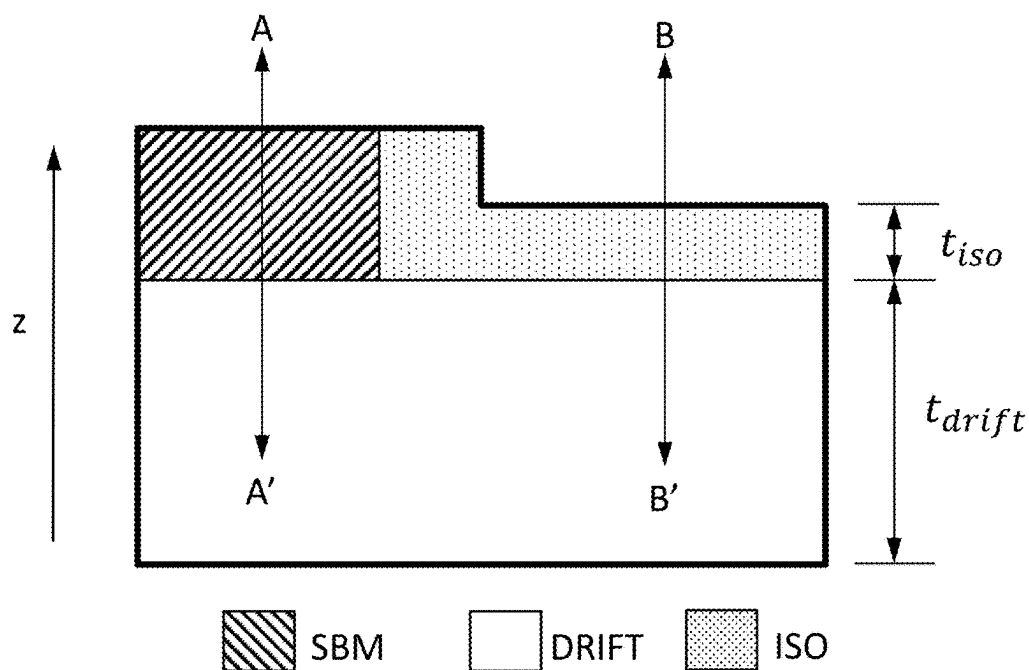


FIG. 76

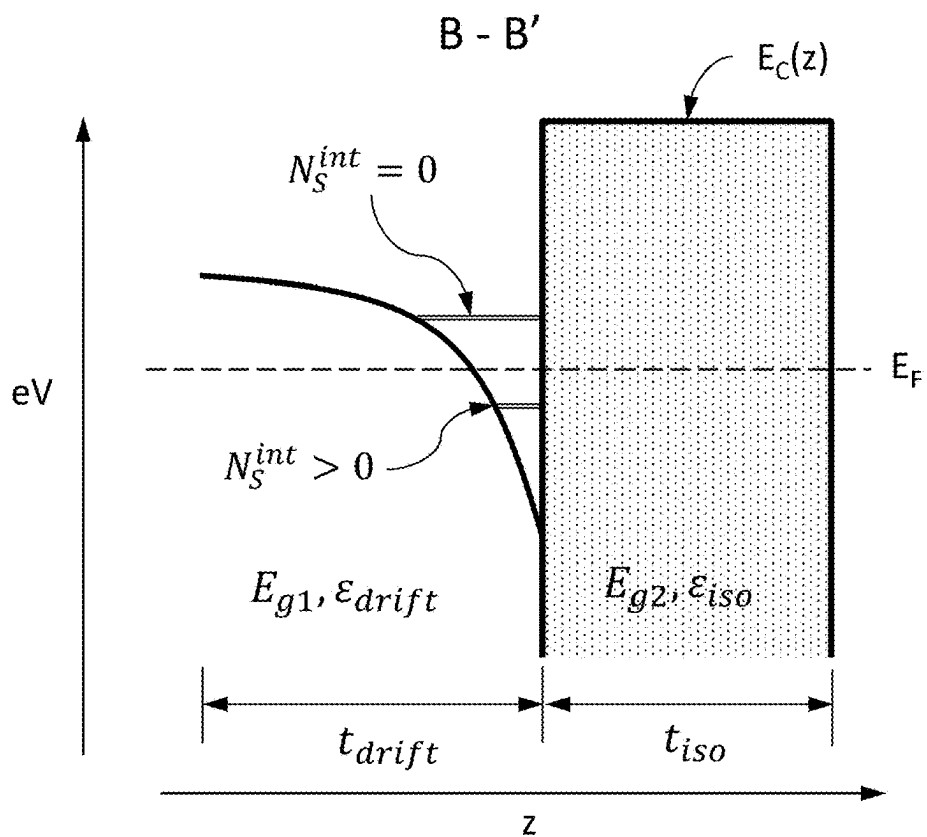


FIG. 77

DIELECTRIC CONSTANT UWBG OXIDES

material	Eg	eps_static	eps_inf
b-Ga2O3(100)	4.9	3.969	10.2
b-Ga2O3(010)	4.9	4.062	10.21
b-Ga2O3(001)	4.9	4.06	13.12
a-Ga2O3_x	5.1195	4.62	12.9816
a-Ga2O3_y	5.1195	4.62	12.9816
a-Ga2O3_z	5.1195	4.465	18.6645
AlGaO3_x	5.8799	3.703	14.23
AlGaO3_y	5.8799	3.73	15.5
AlGaO3_z	5.8799	4	15.78
MgO	7.6865	3.208	10.77
MgAl2O4	8.6054	3.079	8.476
MgF2_x	11.9118	2.079	4.474
MgF2_y	11.9118	2.04	5.418
MgF2_z	11.9118	2.04	5.418
a-Al2O3_x	8.9082	3.232	9.092
a-Al2O3_y	8.9082	3.232	9.092
a-Al2O3_z	8.9082	3.191	10.93
t-Al2O3_x	7.24	3.158	8.641
t-Al2O3_y	7.24	3.182	9.206
t-Al2O3_z	7.24	3.152	9.091
4H-SiC_x	3.3603	6.988	10.28
4H-SiC_y	3.3603	6.988	10.28
4H-SiC_z	3.3603	7.22	10.77
b-SiO2_x	9.8	2.184	3.828
b-SiO2_y	9.8	2.184	3.828
b-SiO2_z	9.8	2.187	3.991
a-SiO2_x	10	2.243	4.103
a-SiO2_y	10	2.243	4.103
a-SiO2_z	10	2.23	3.905
LiGaO2_x	6.2705	3.339	7.168
LiGaO2_y	6.2705	3.207	6.553
LiGaO2_z	6.2705	3.332	7.681
LiAlO2_x	8.5273	2.753	6.712
LiAlO2_y	8.5273	2.753	6.712
LiAlO2_z	8.5273	2.69	5.142
ZnGeO3_x	4.7671	4.044	10.66
ZnGeO3_y	4.7671	4.044	10.66
ZnGeO3_z	4.7671	4.044	10.66
MgGeO3_x	5.8052	3.374	9.45
MgGeO3_y	5.8052	3.331	8.435
MgGeO3_z	5.8052	3.46	8.54
NiO	1.96	5.074	14.41

FIG. 78A

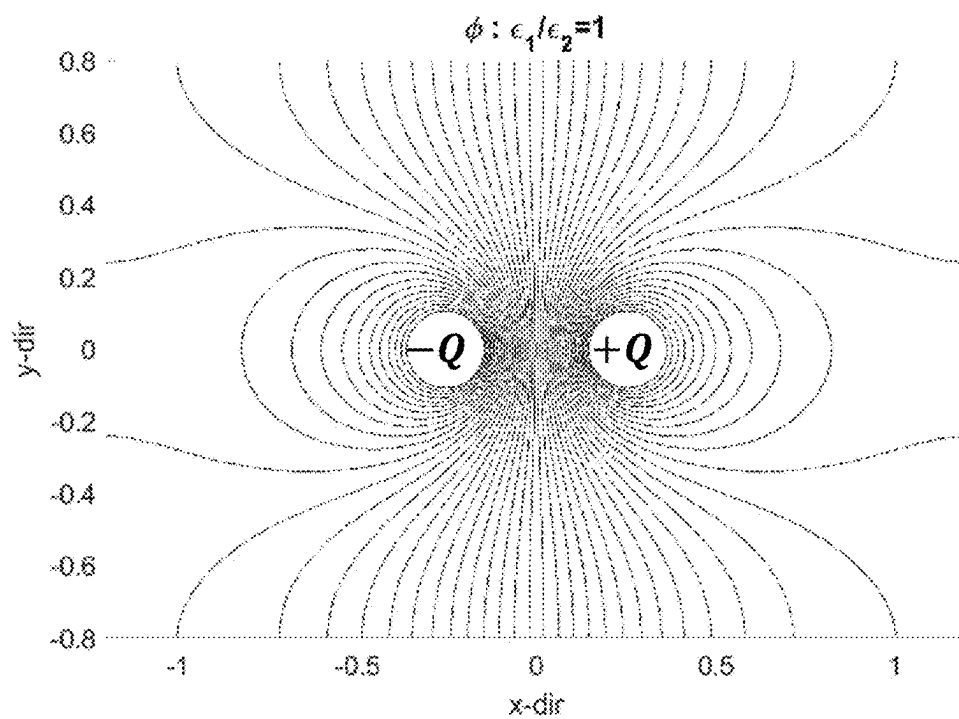


FIG. 78B

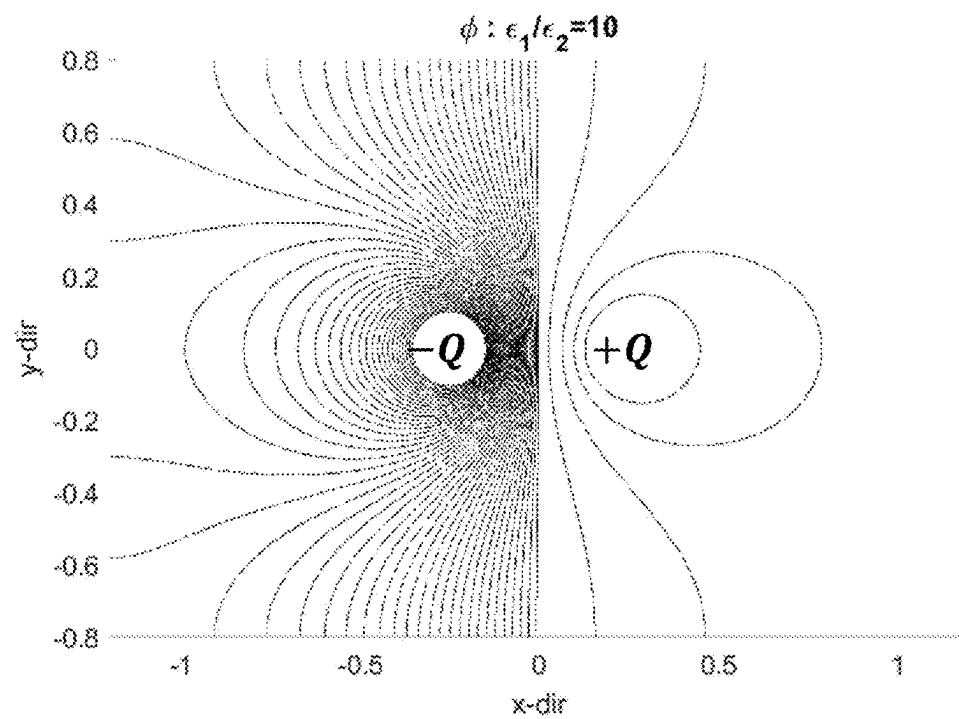


FIG. 78C

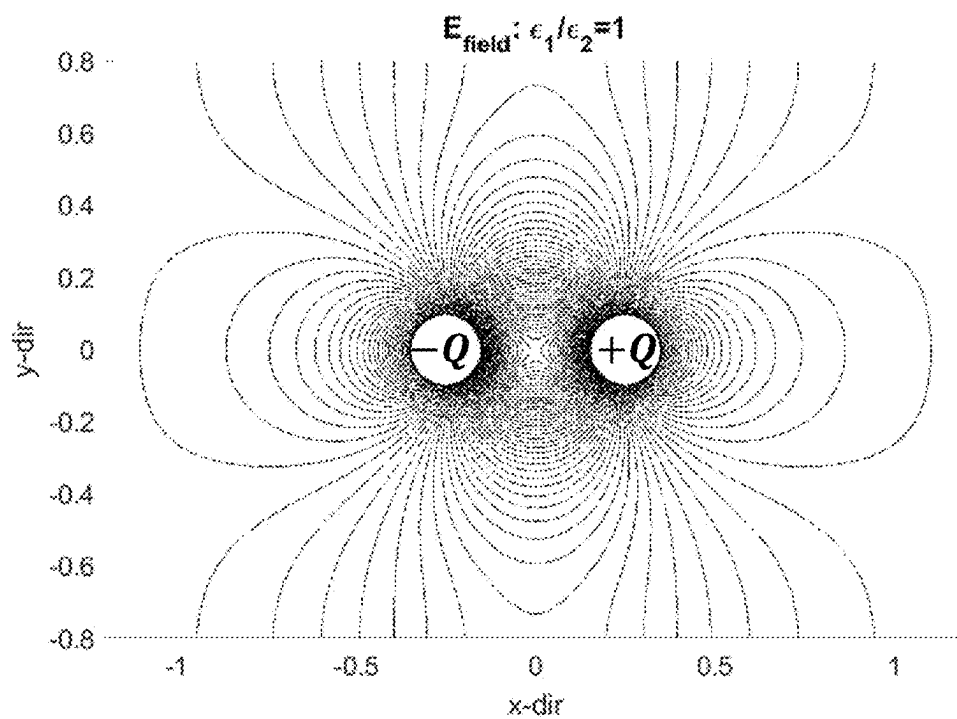


FIG. 78D

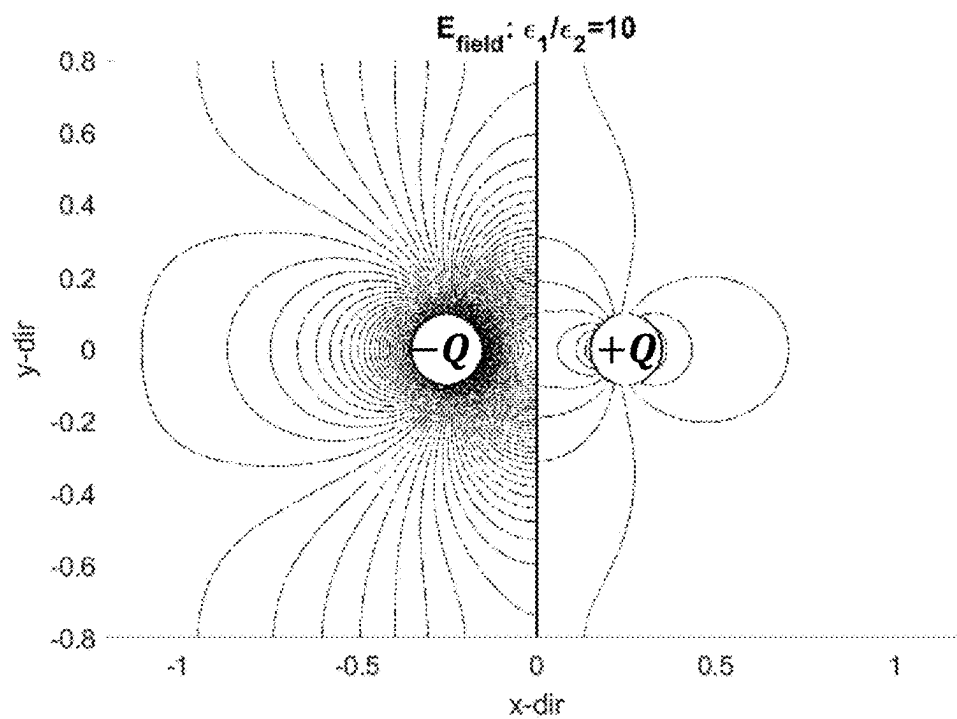


FIG. 78E

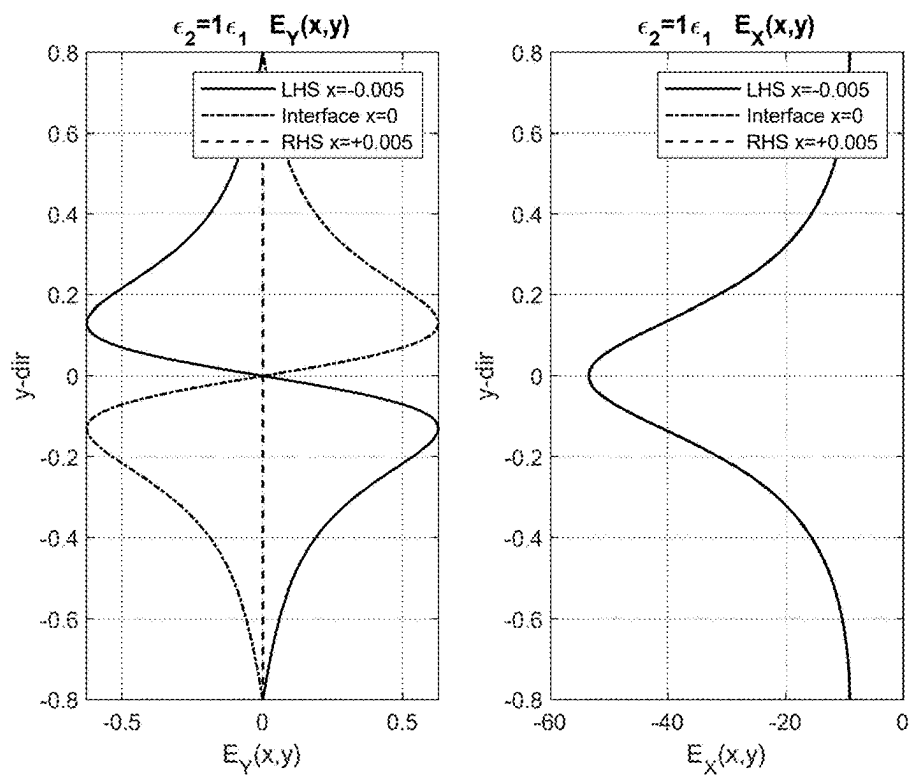


FIG. 78F

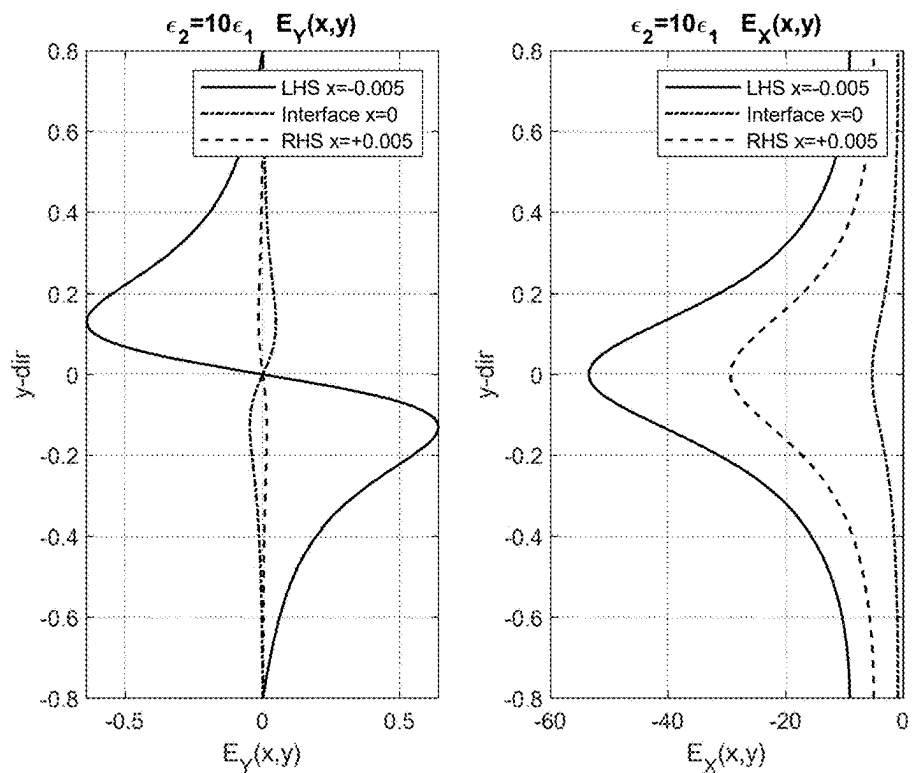
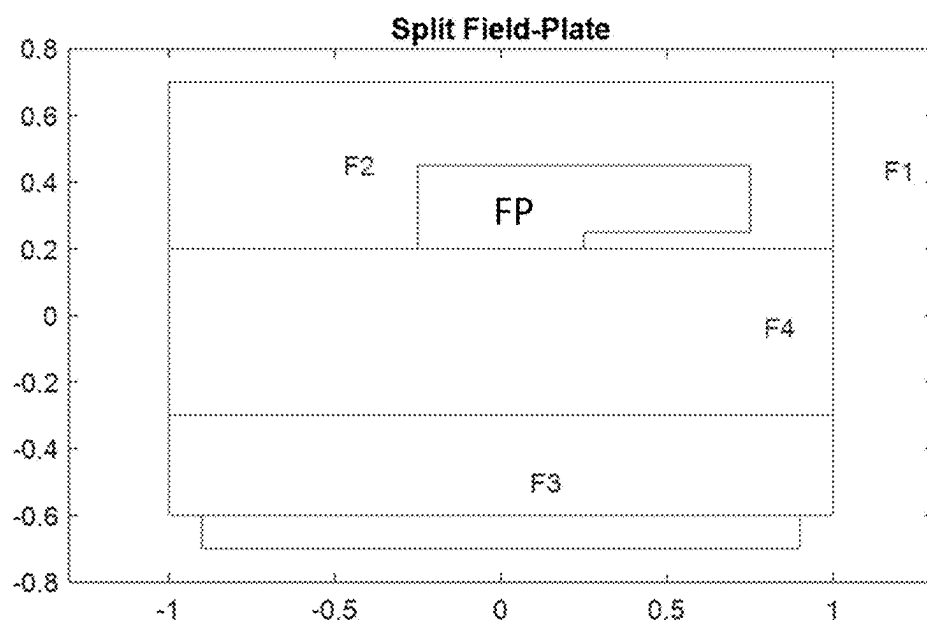
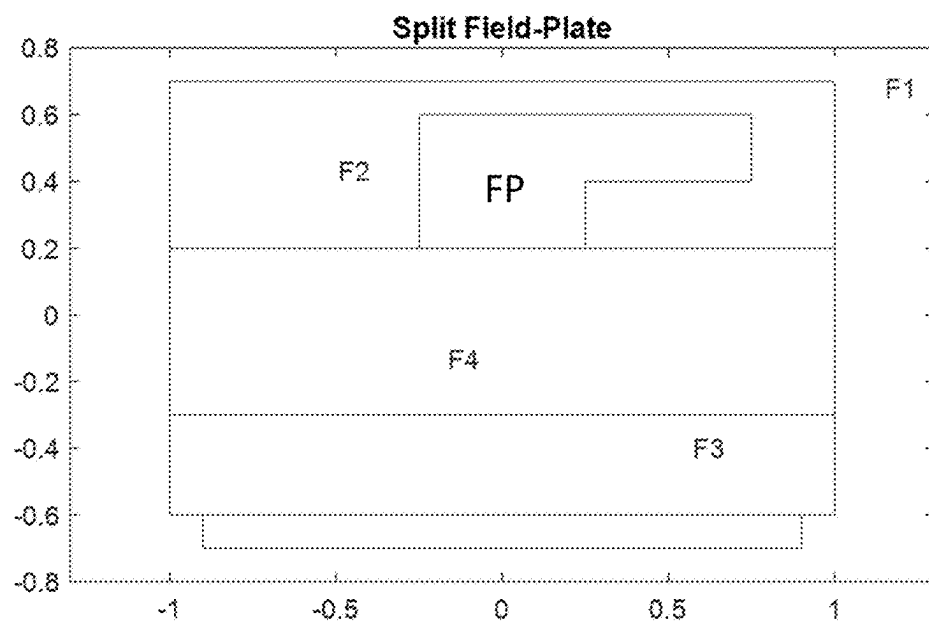


FIG. 78G



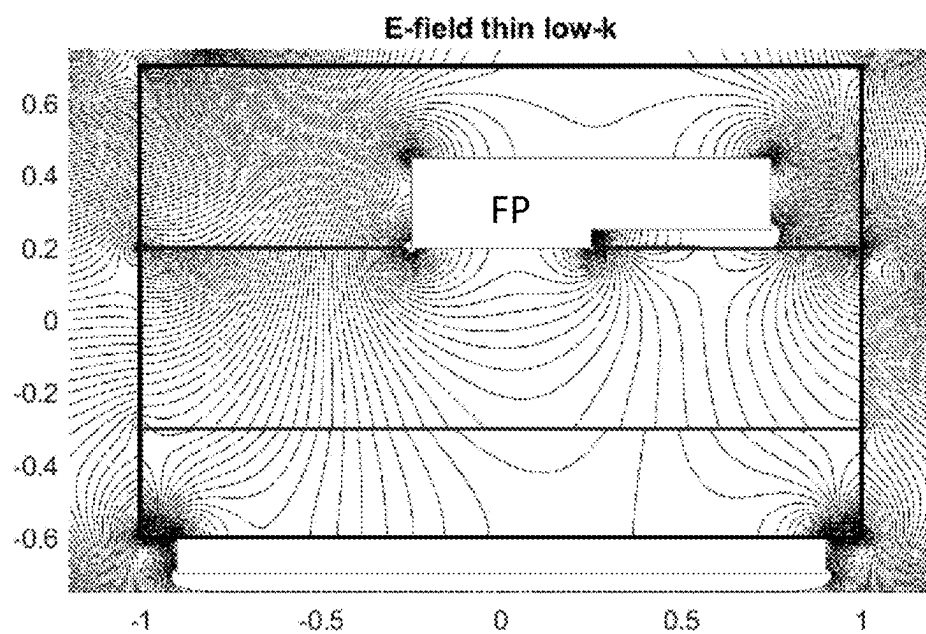


FIG. 81

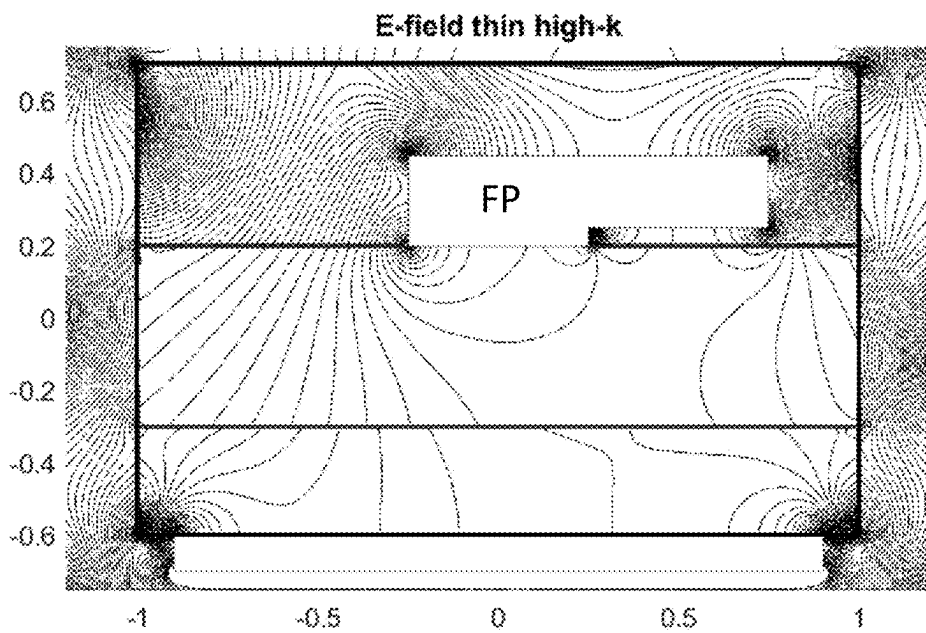


FIG. 82

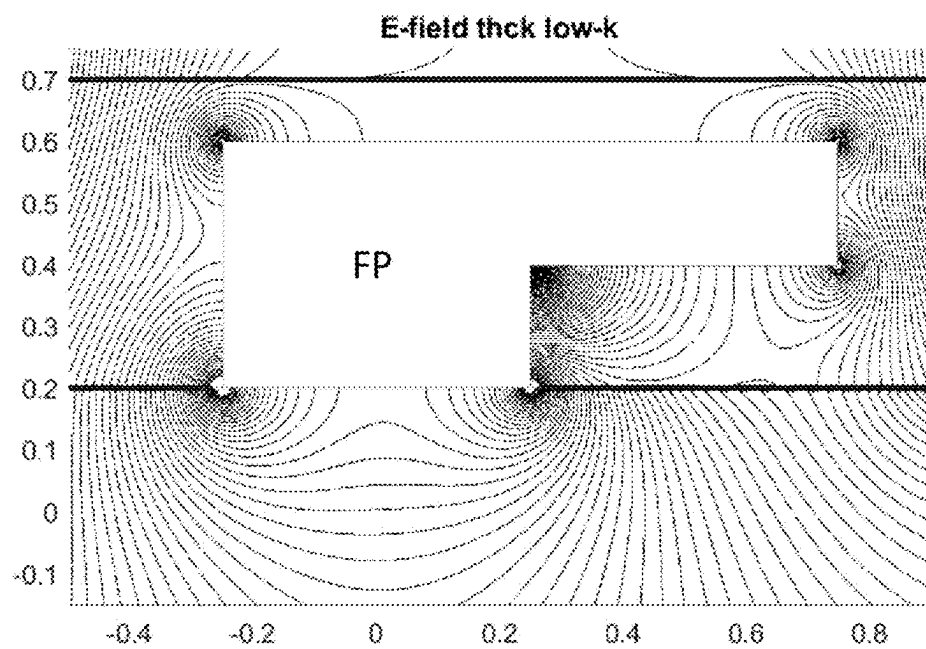


FIG. 83

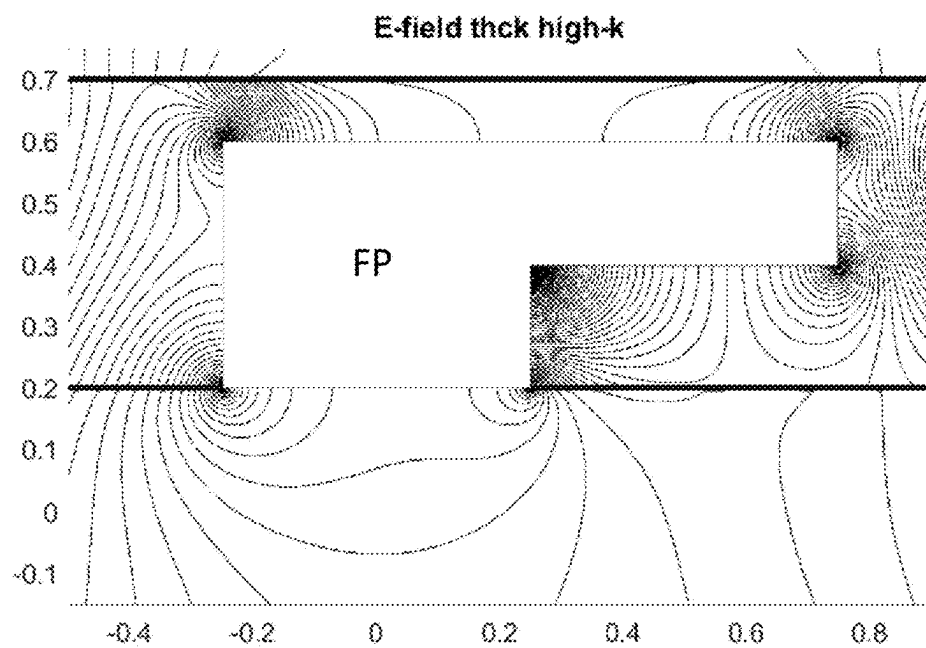


FIG. 84

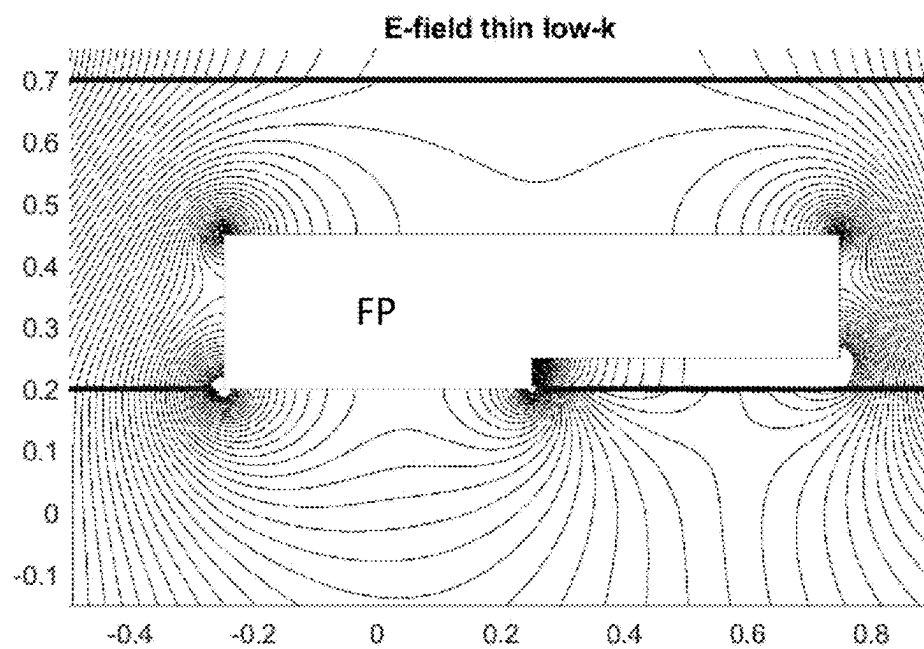


FIG. 85

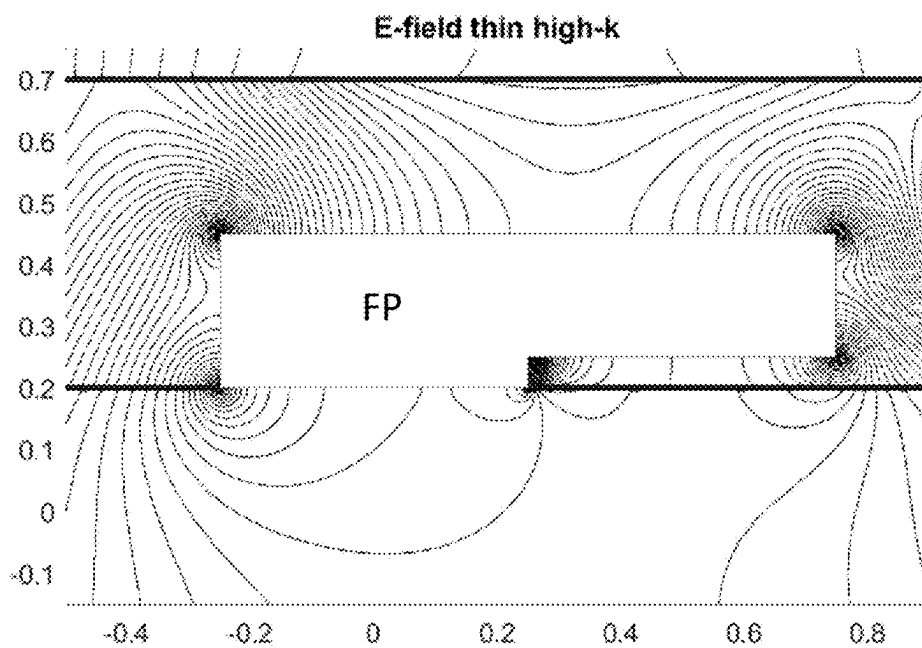


FIG. 86

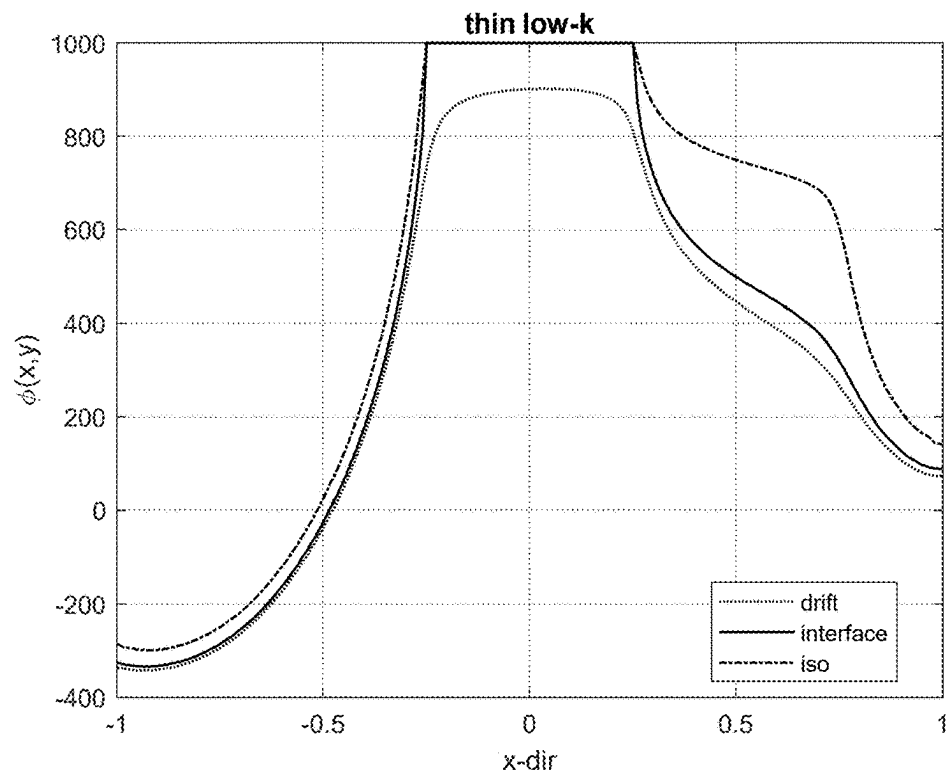


FIG. 87

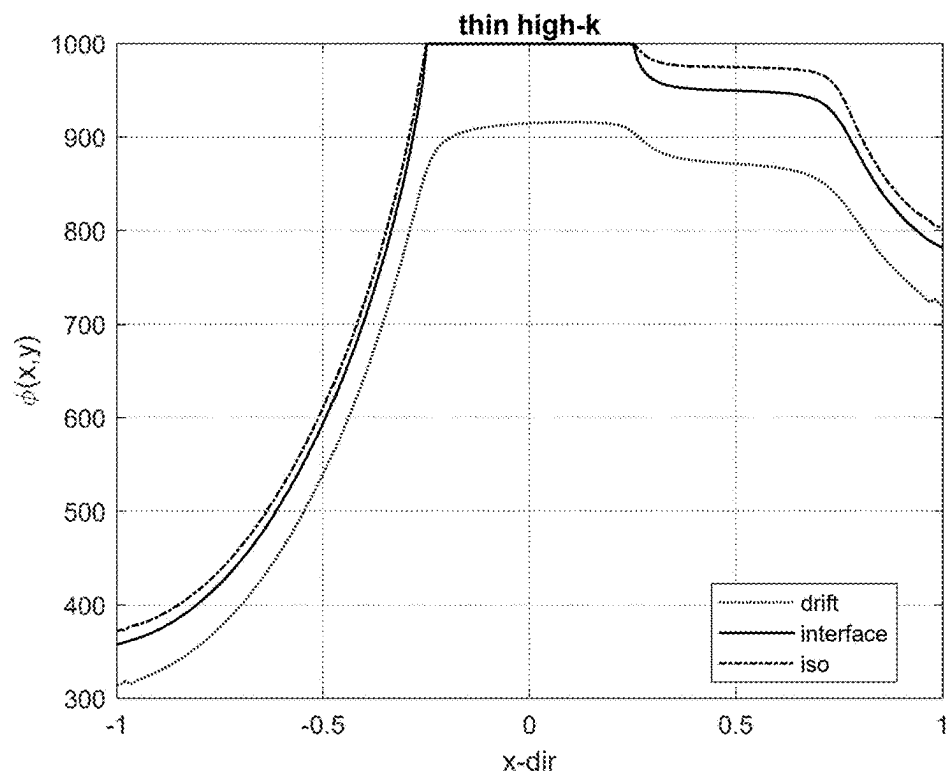


FIG. 88

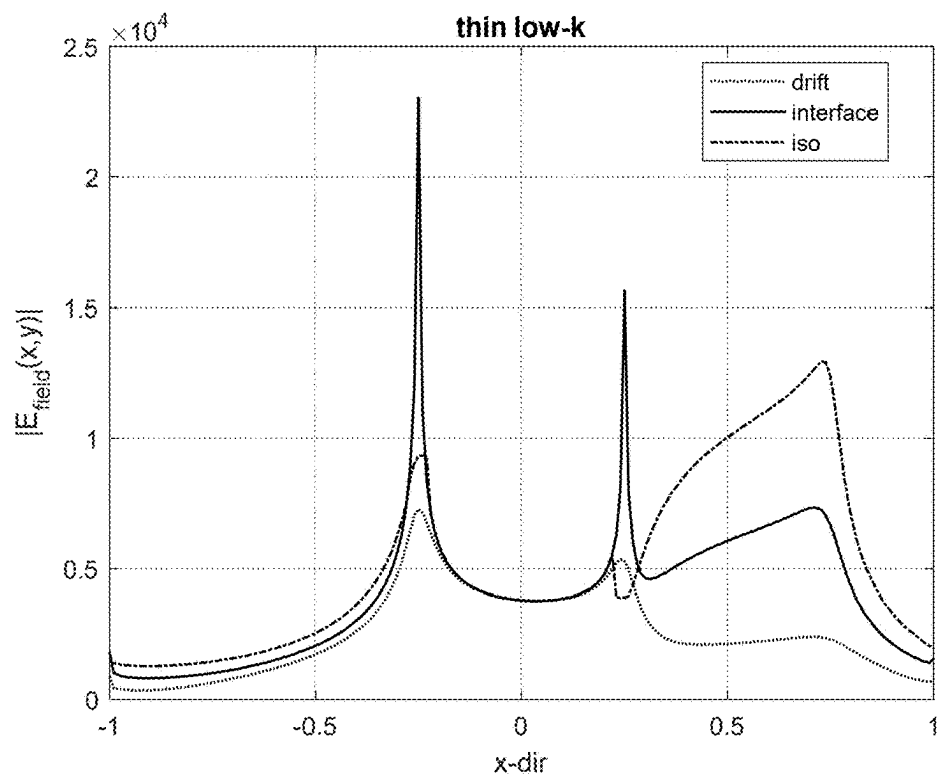


FIG. 89

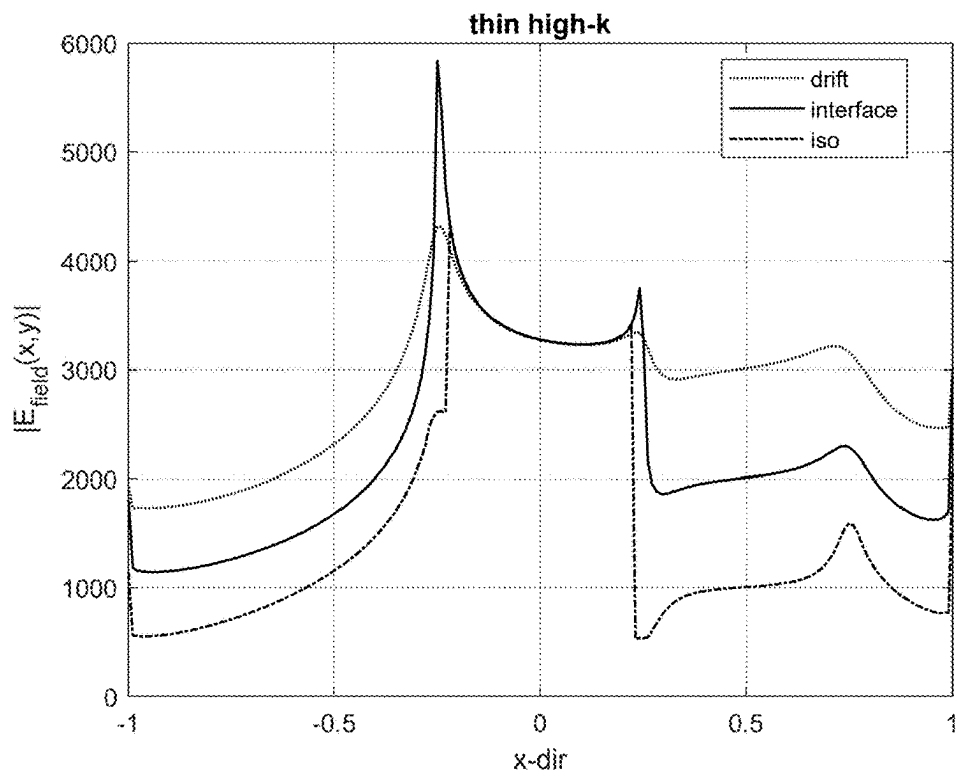


FIG. 90

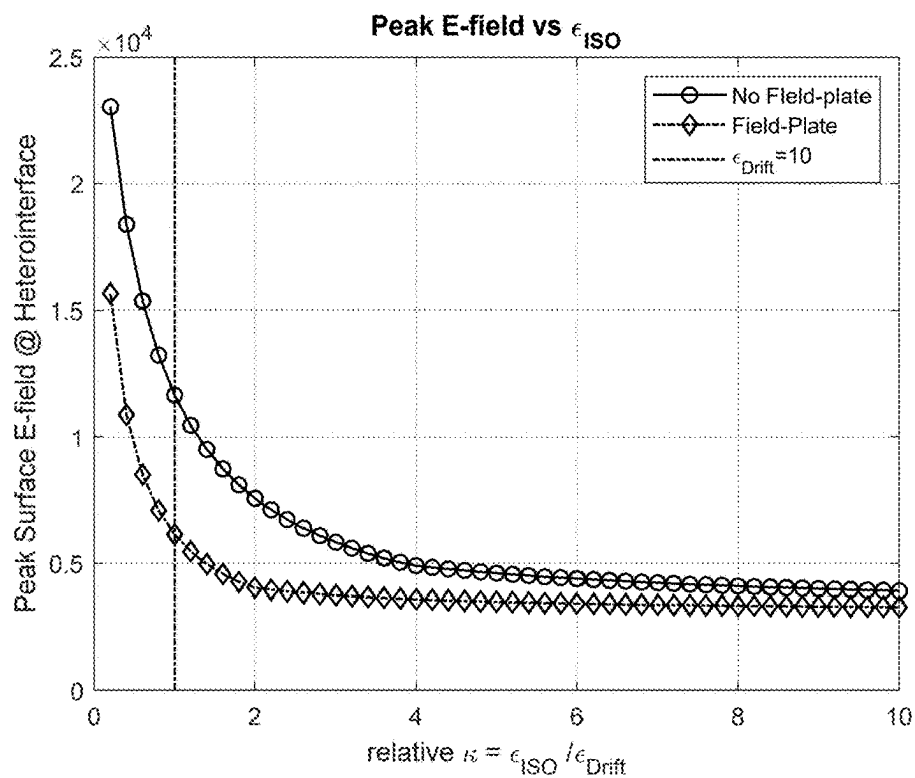


FIG. 91

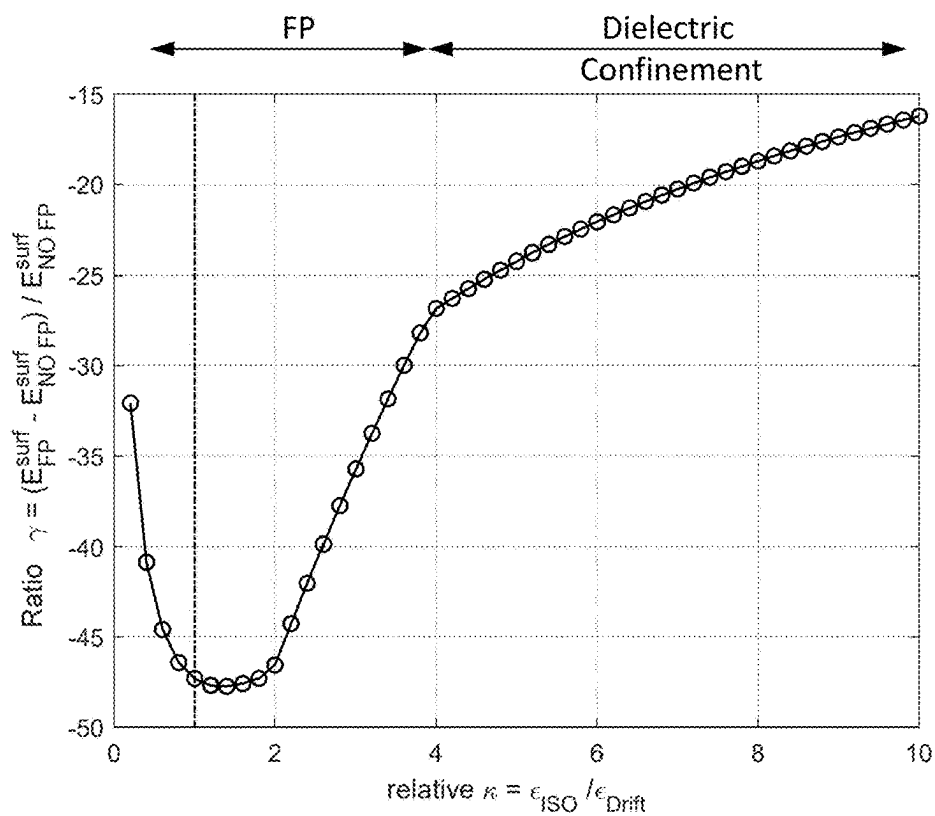


FIG. 92

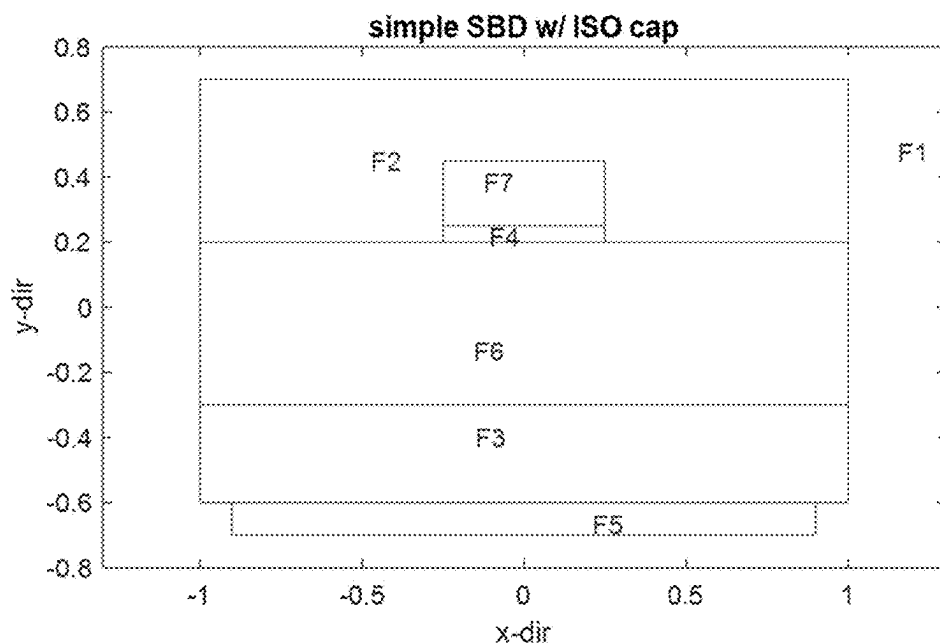


FIG. 93

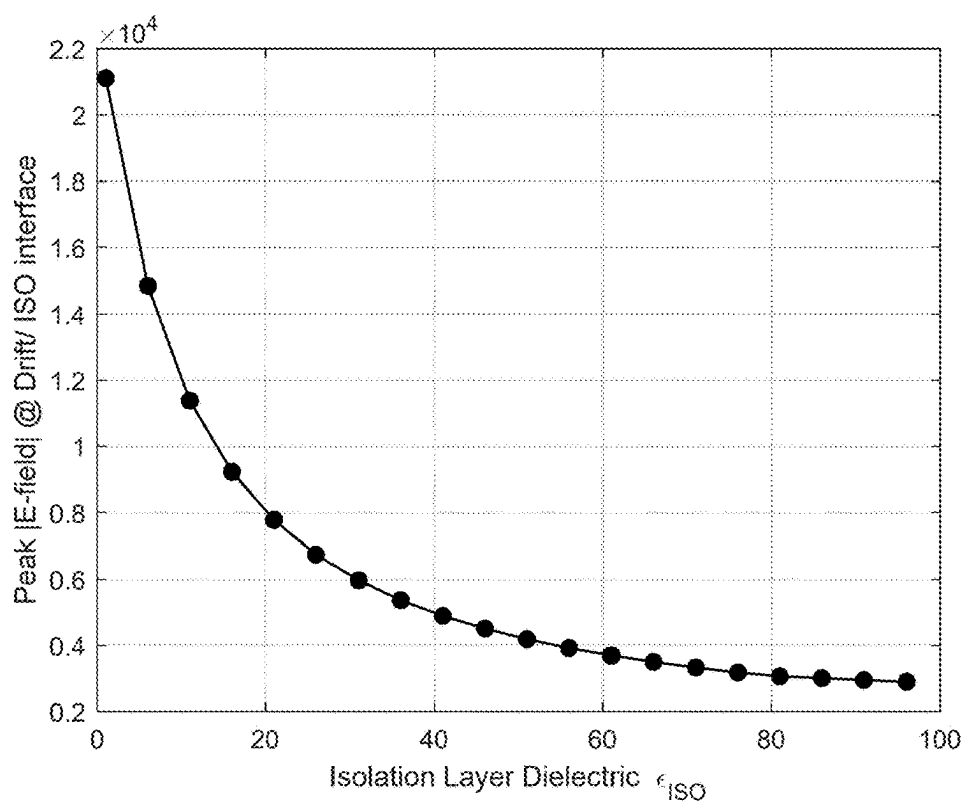


FIG. 94

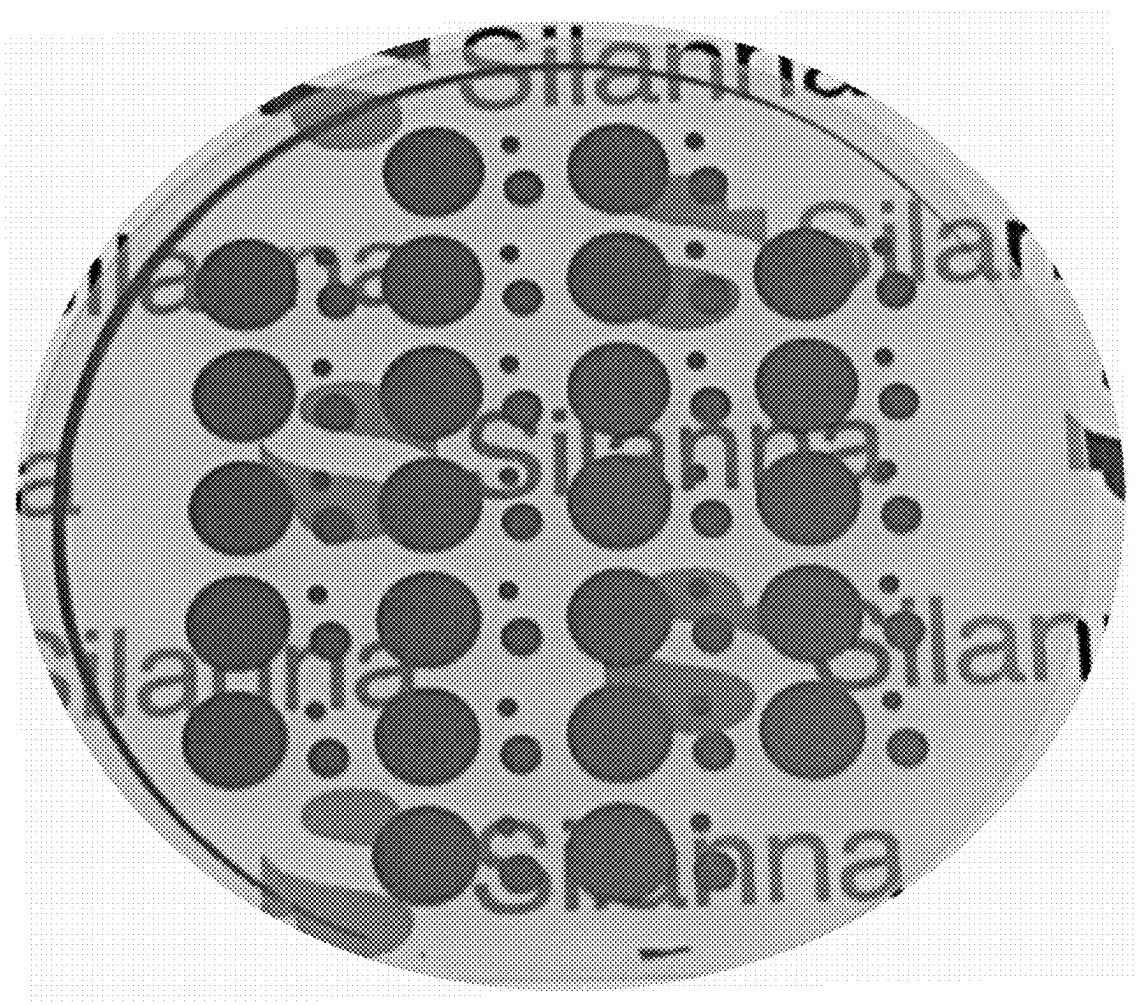


FIG. 95

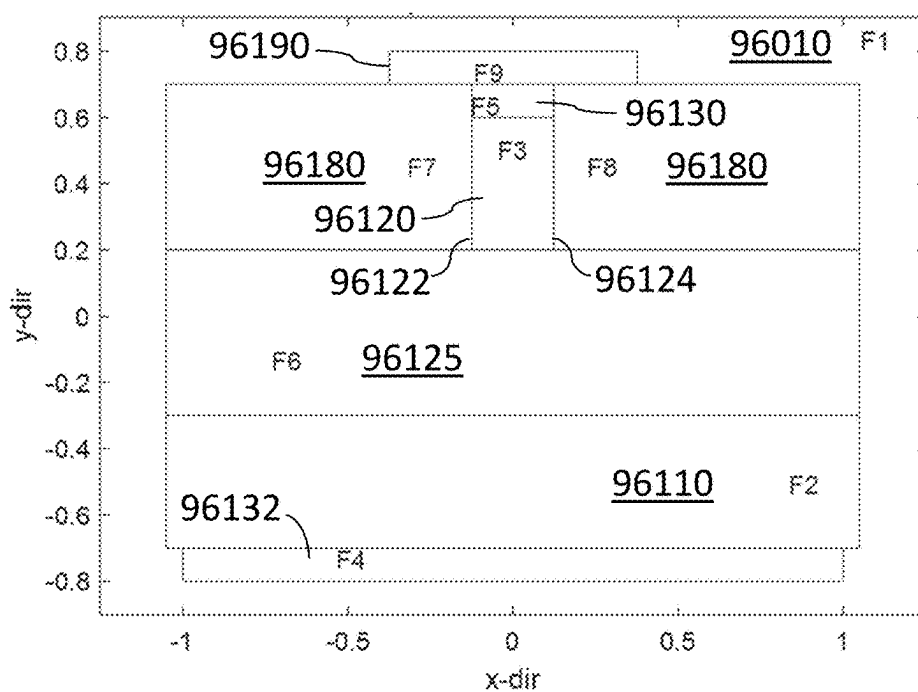


FIG. 96A

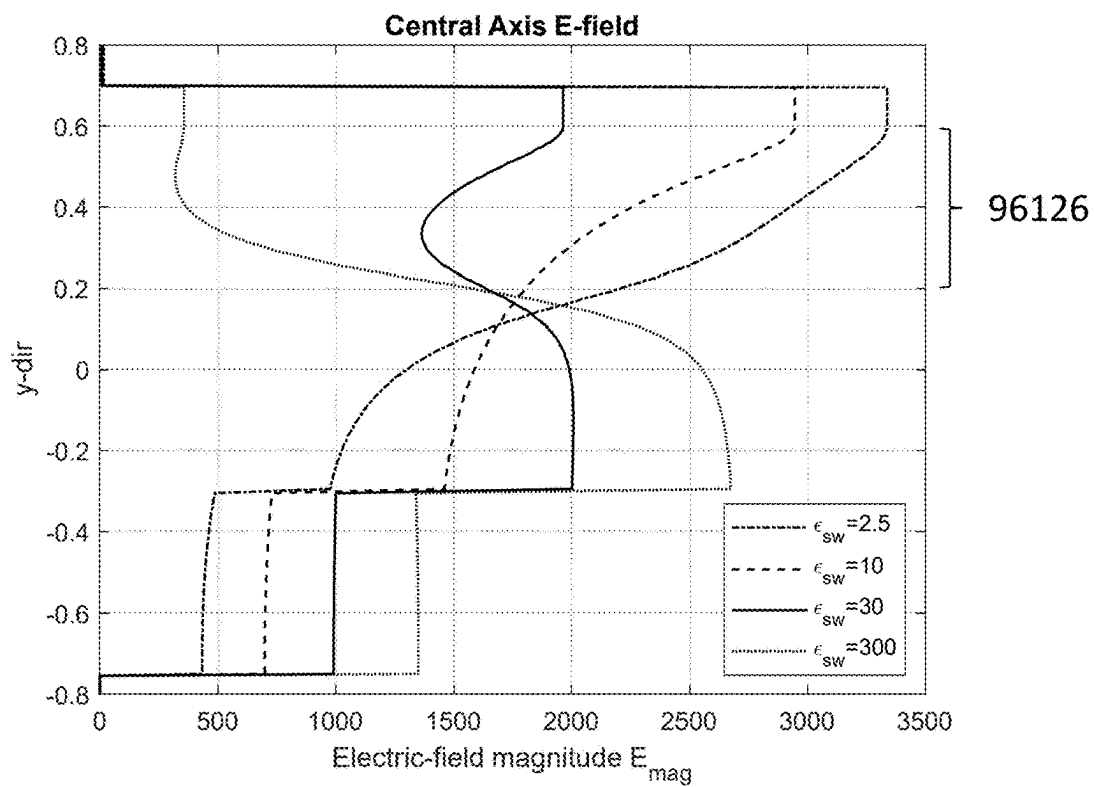


FIG. 96B

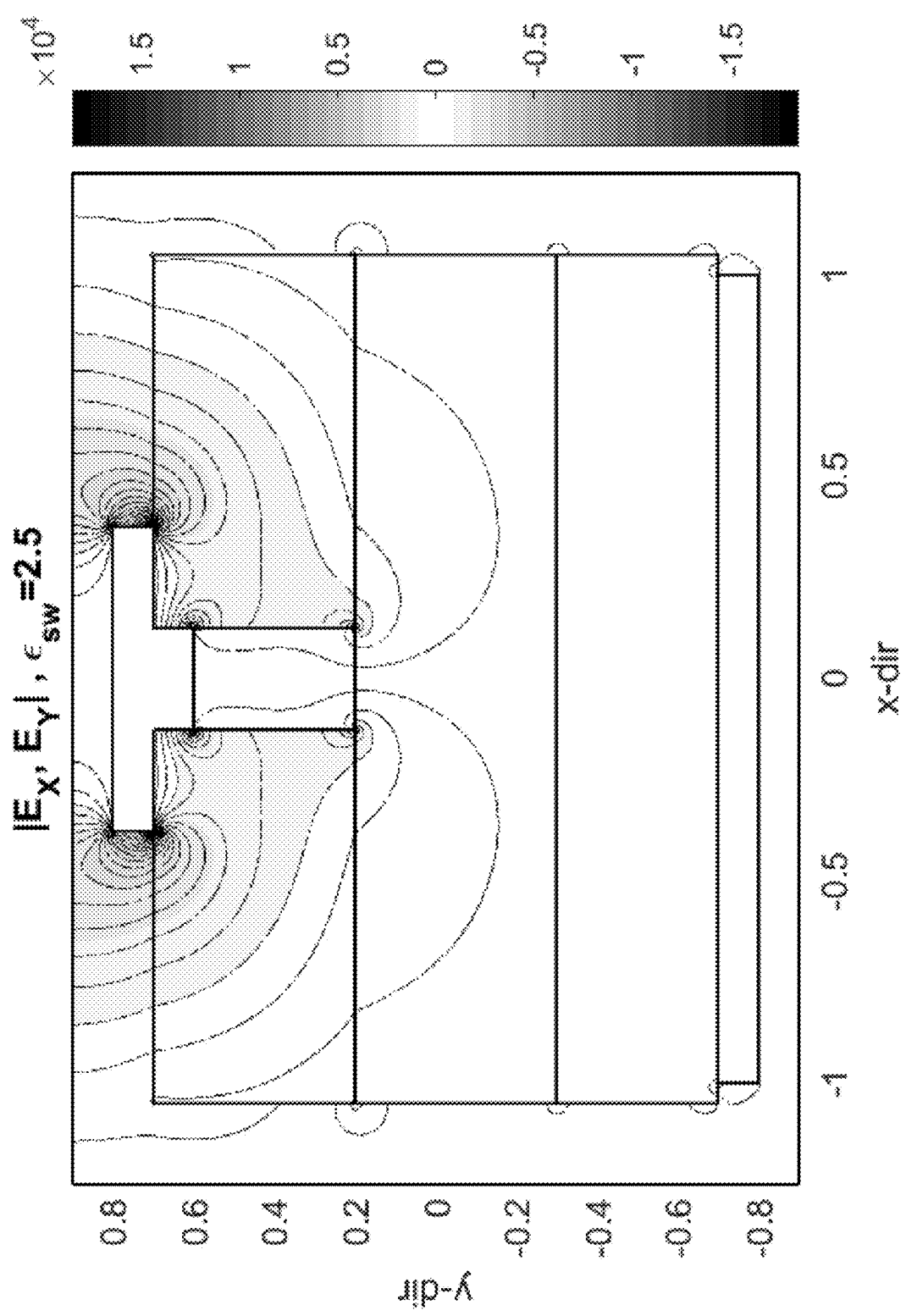


FIG. 96C

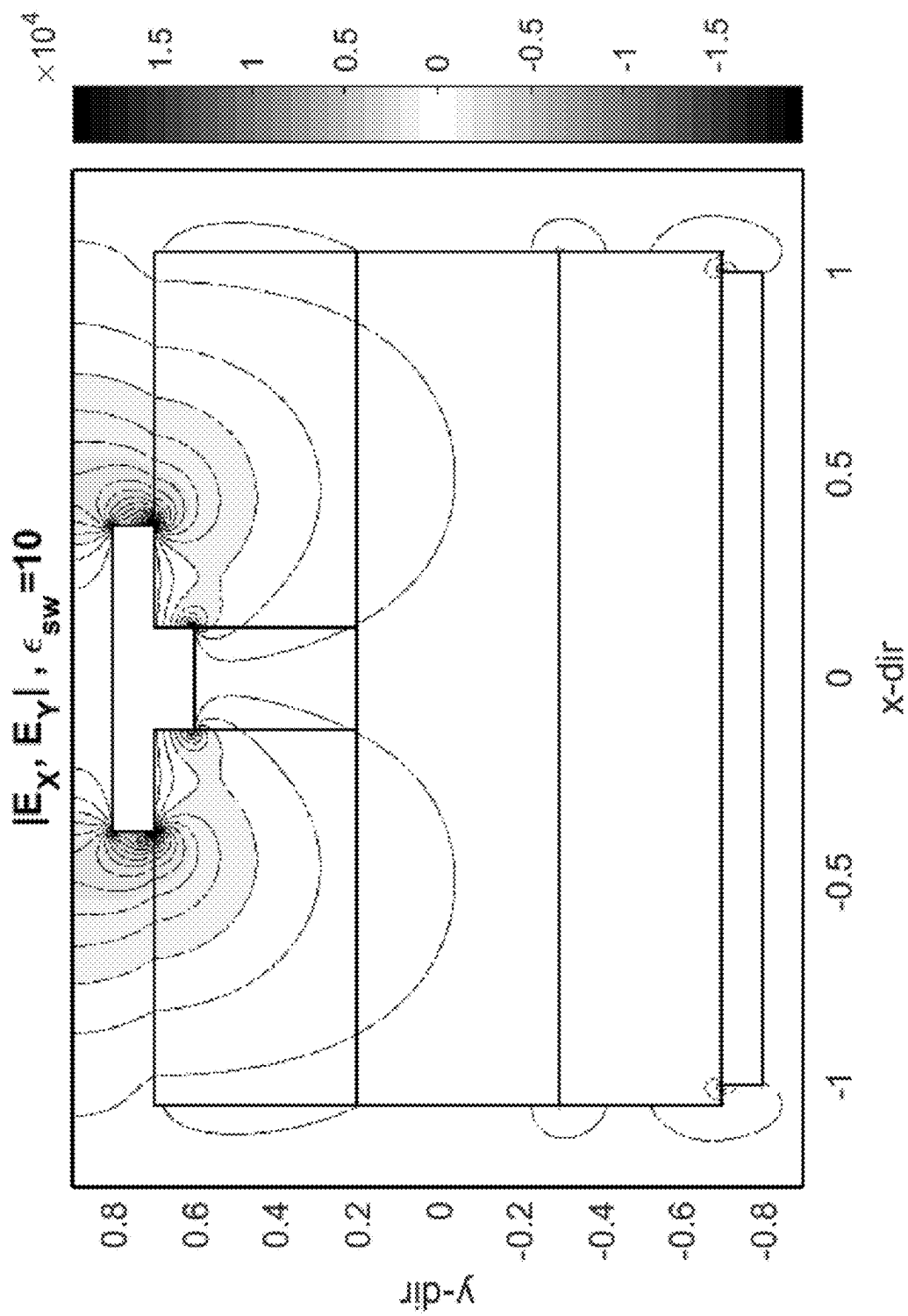


FIG. 96D

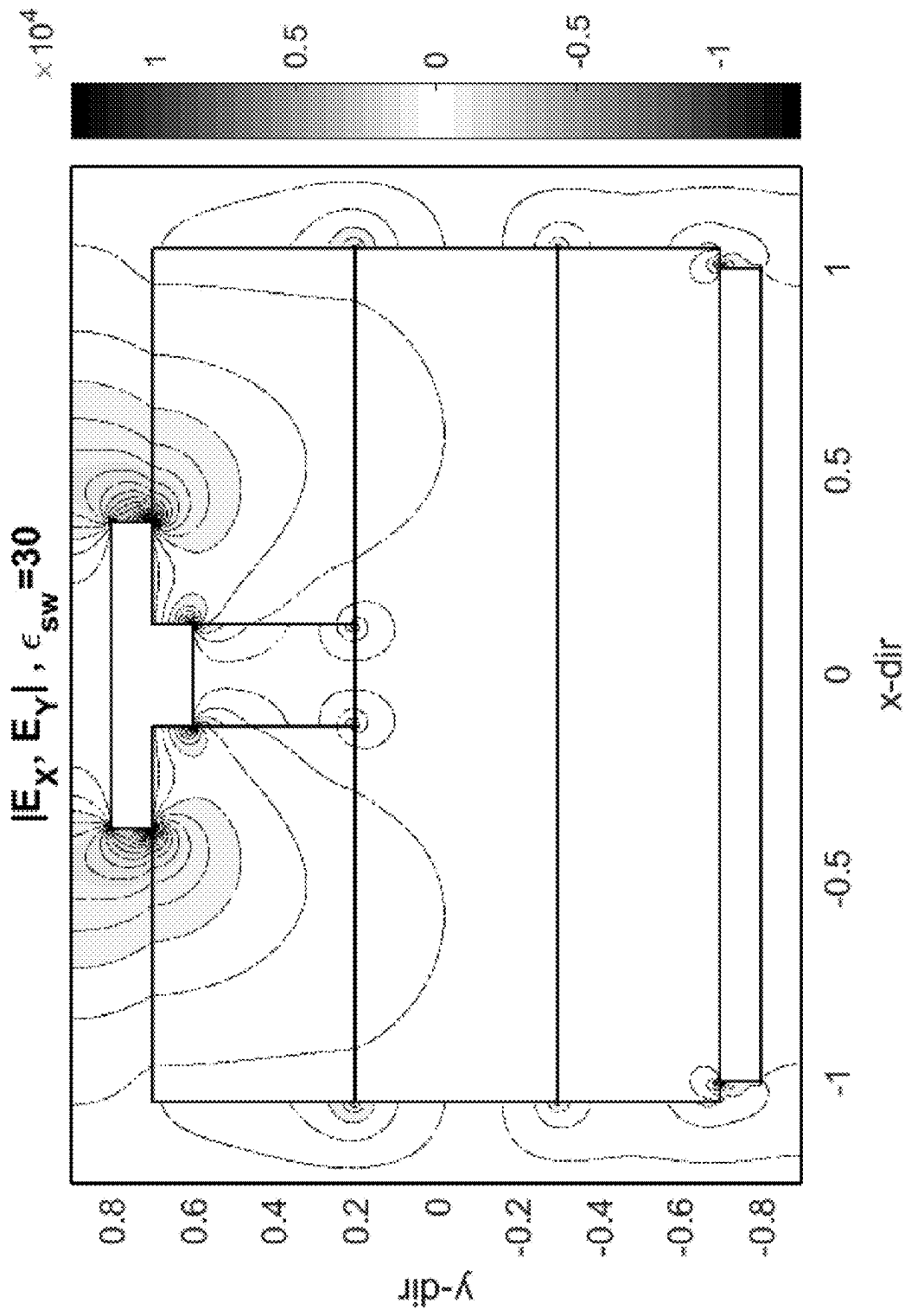


FIG. 96E

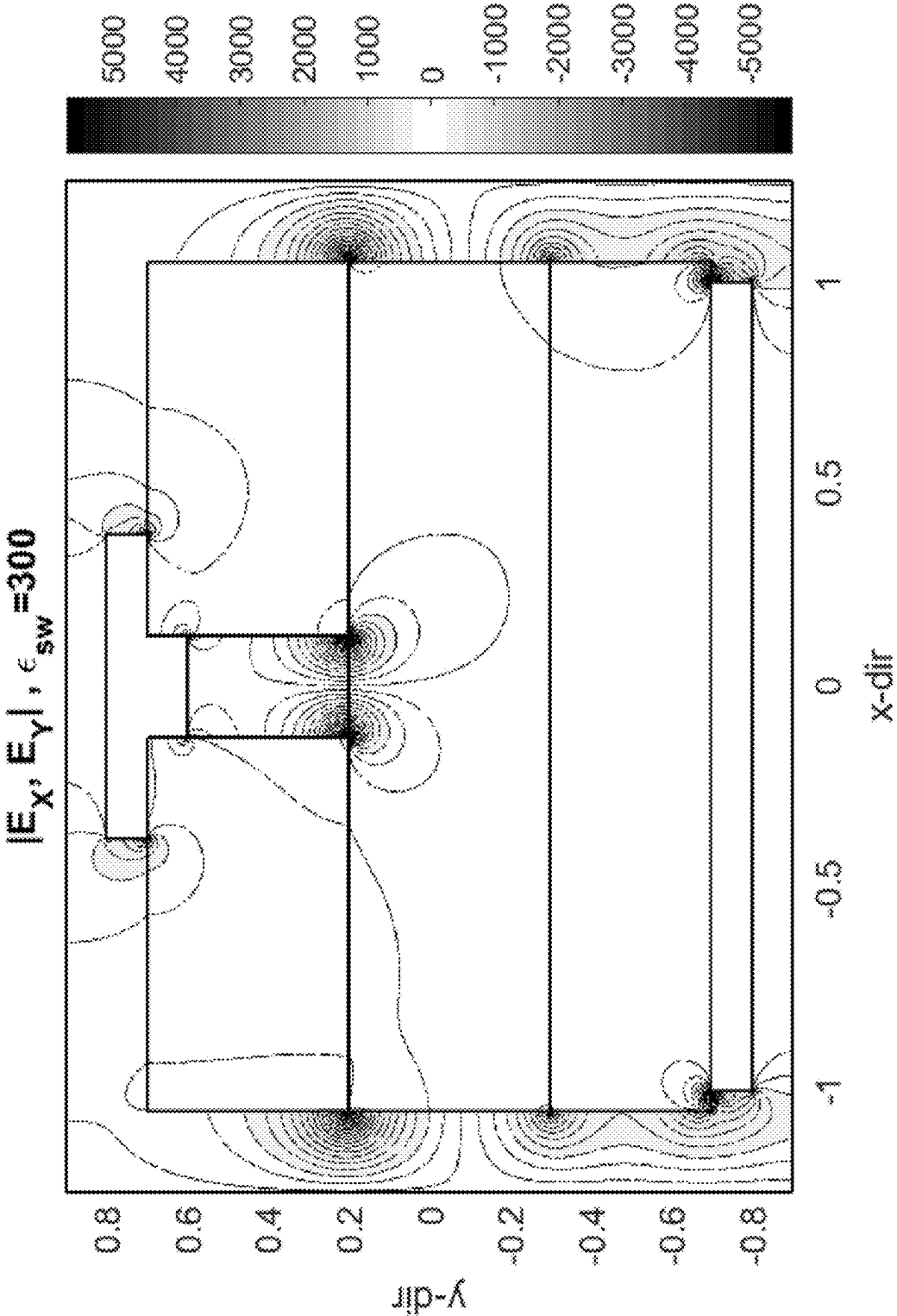


FIG. 96F

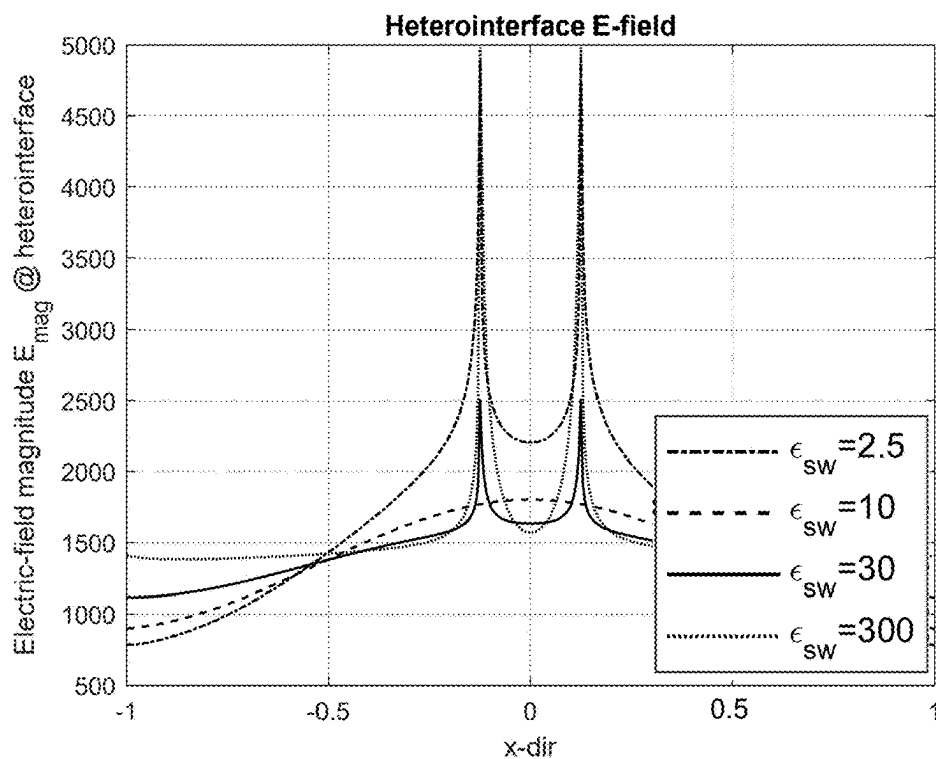


FIG. 96G

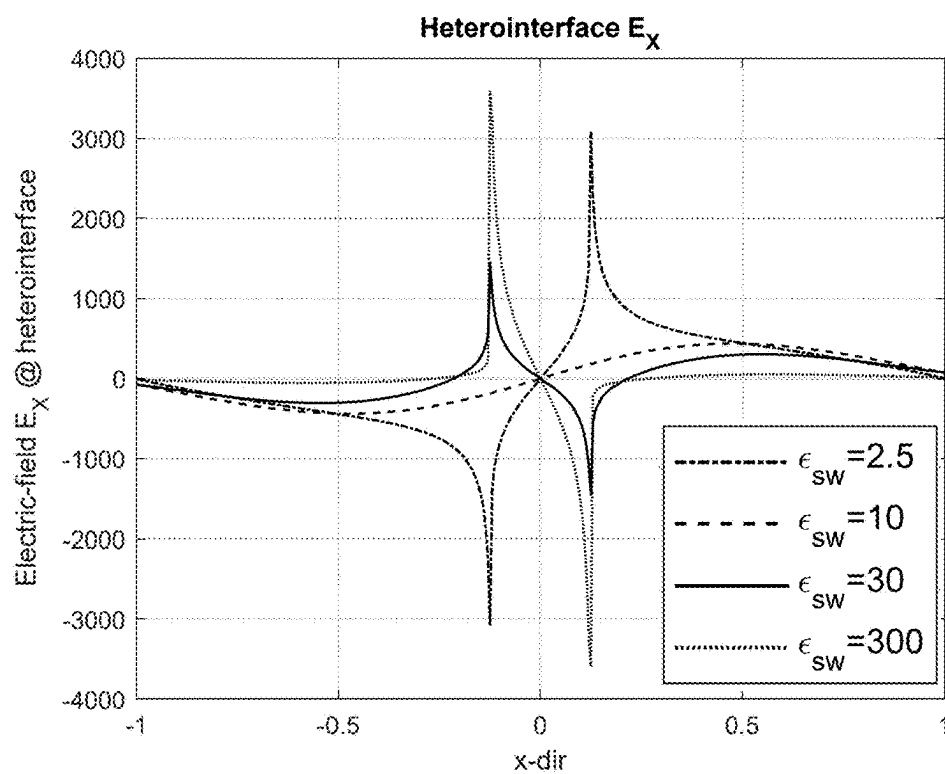


FIG. 96H

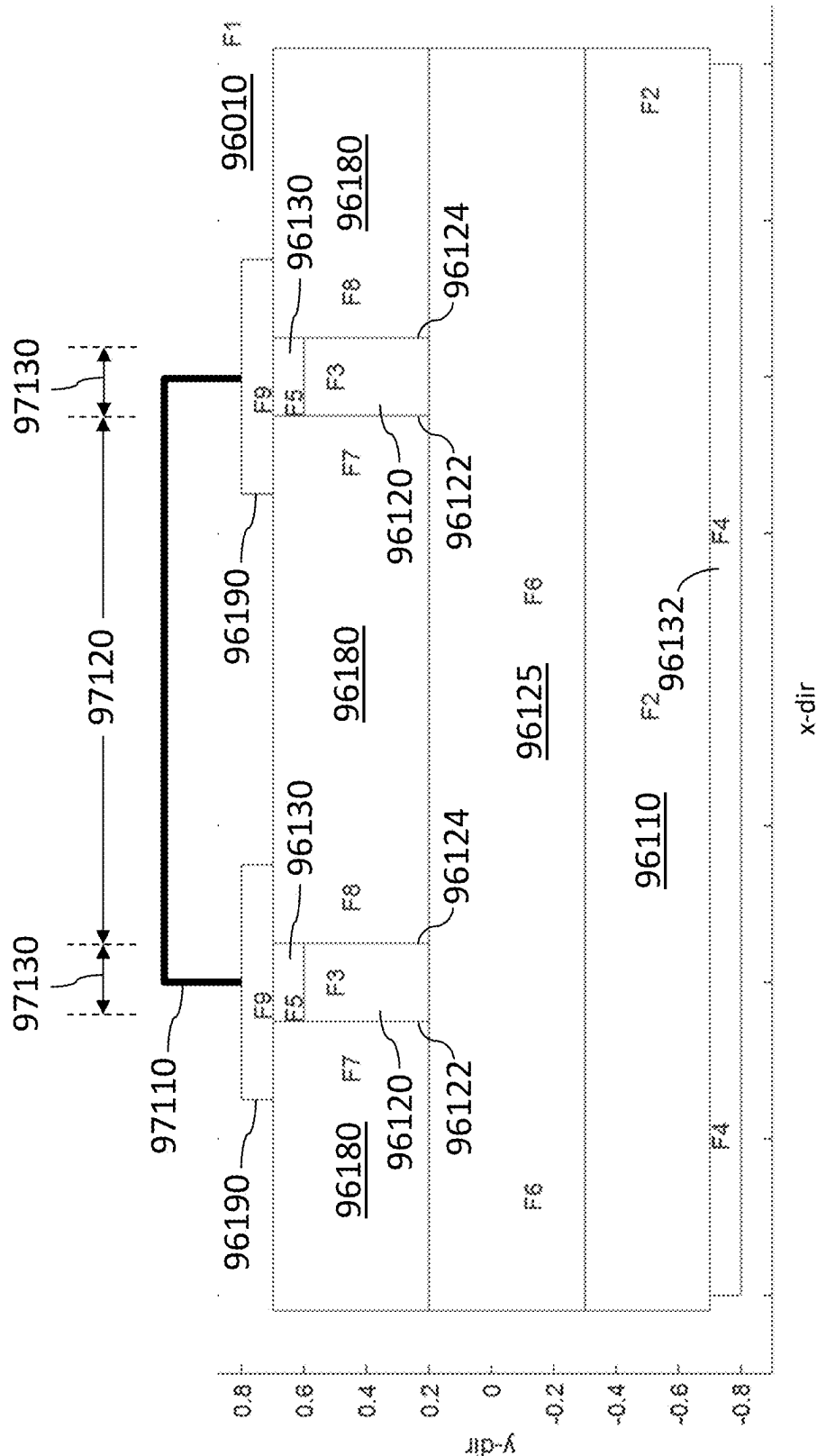


FIG. 97

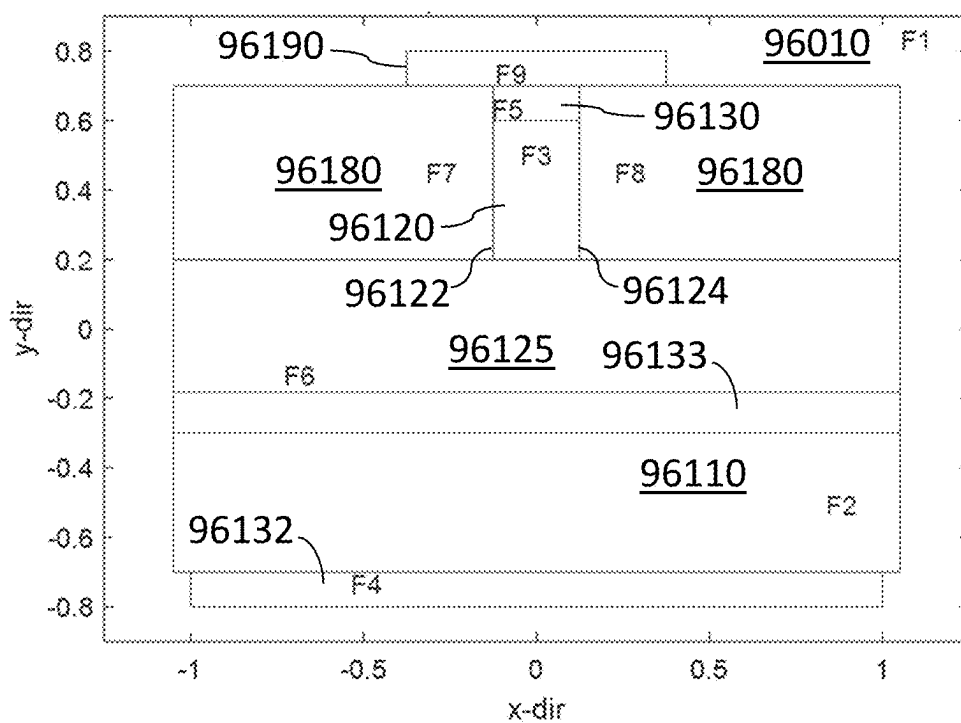


FIG. 98A

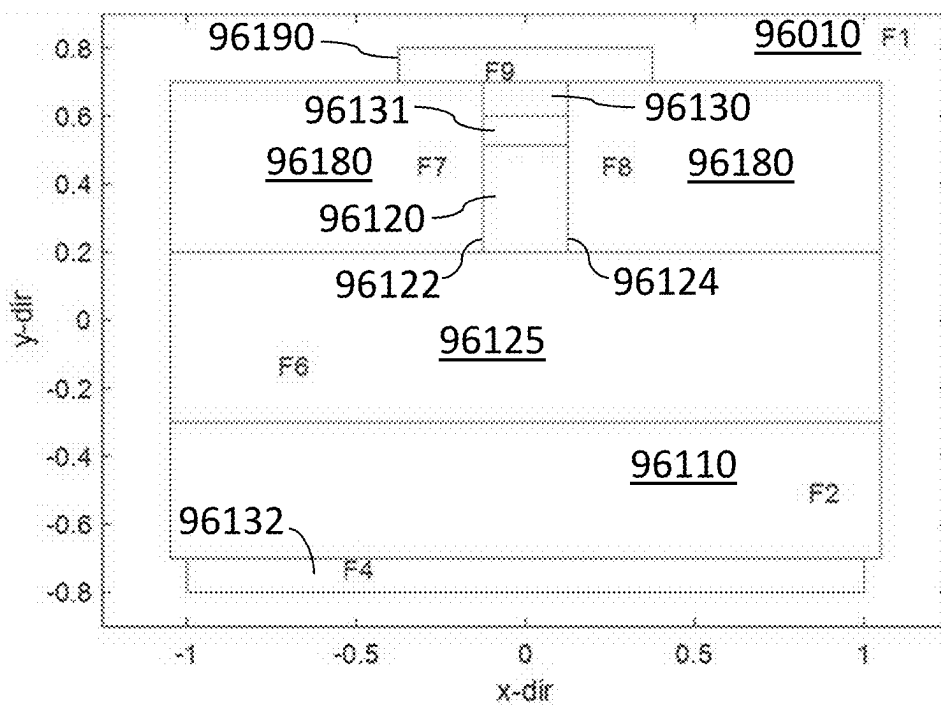


FIG. 98B

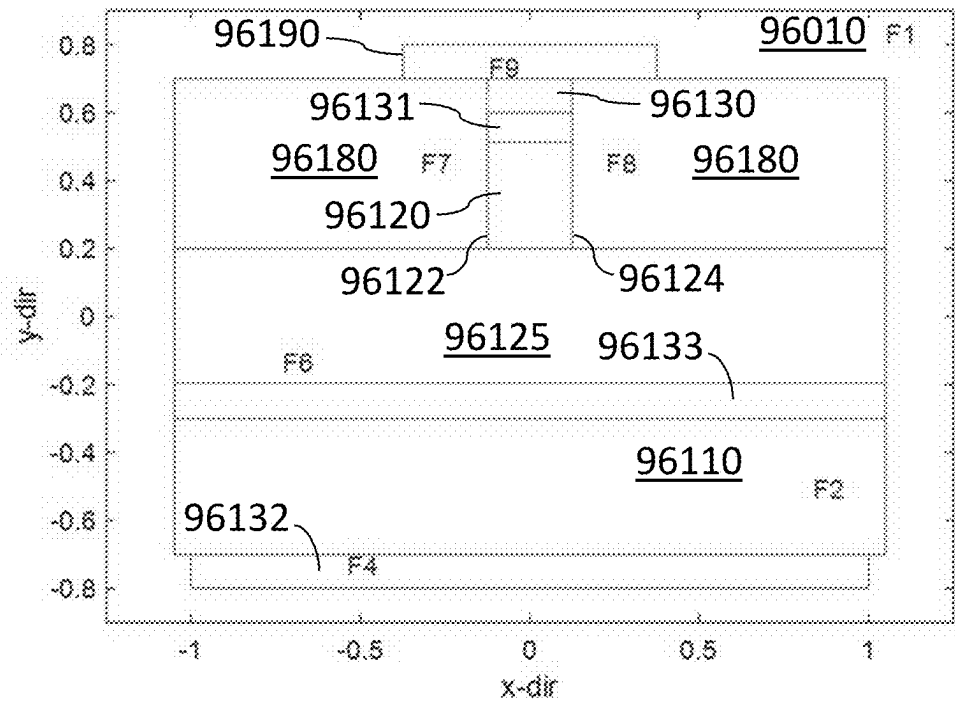


FIG. 98C

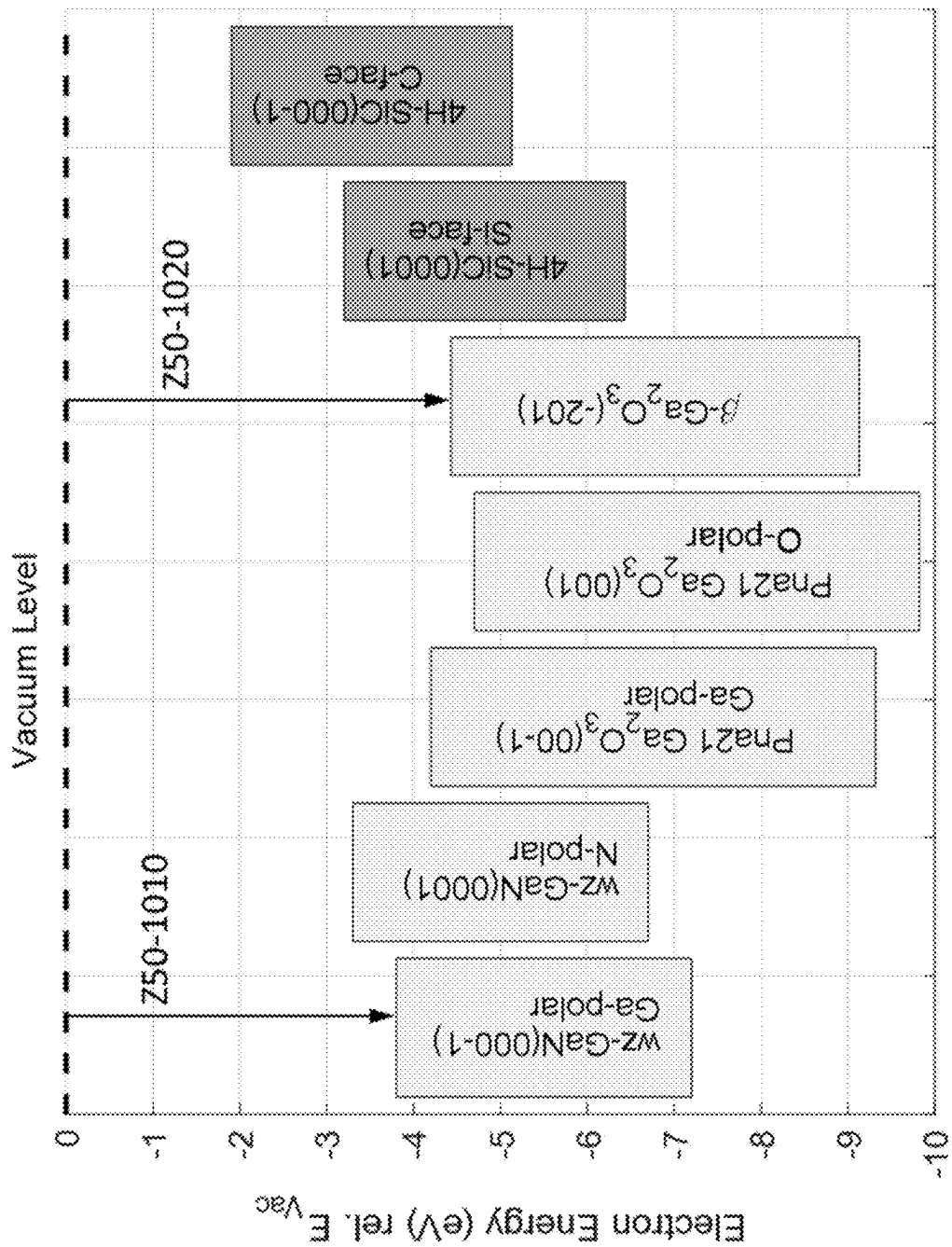


FIG. 99

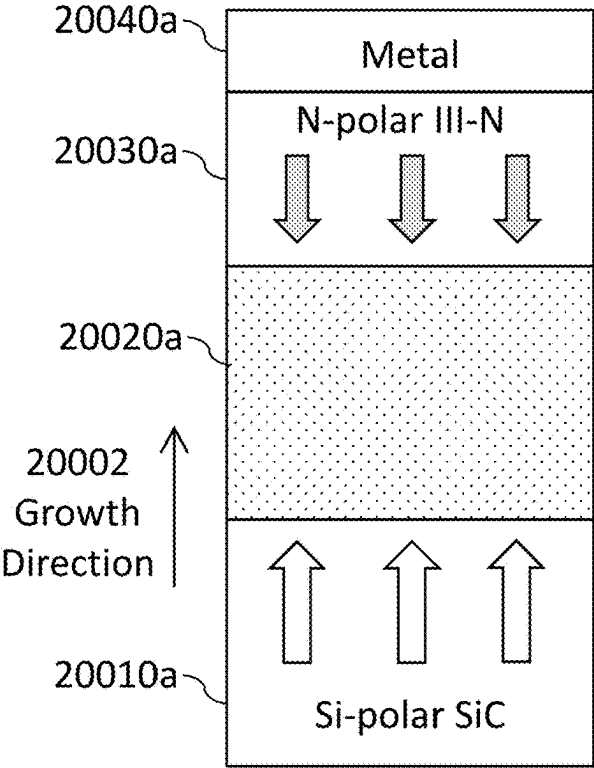


FIG. 100A

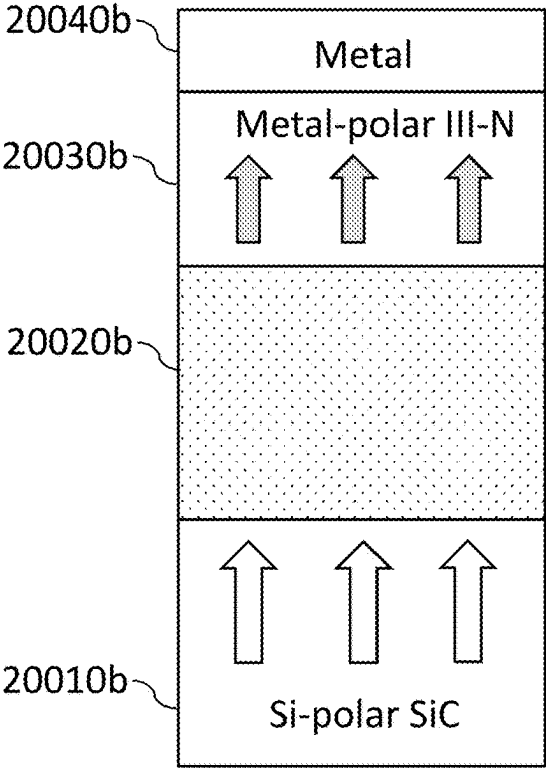


FIG. 100B

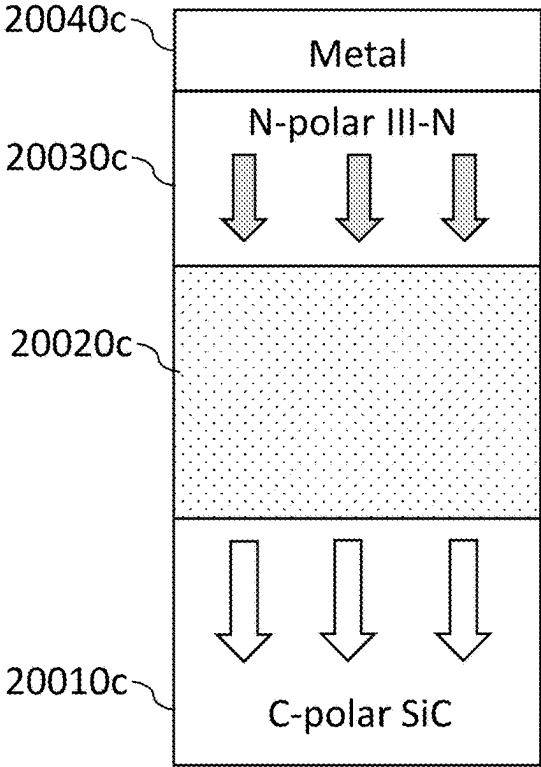


FIG. 100C

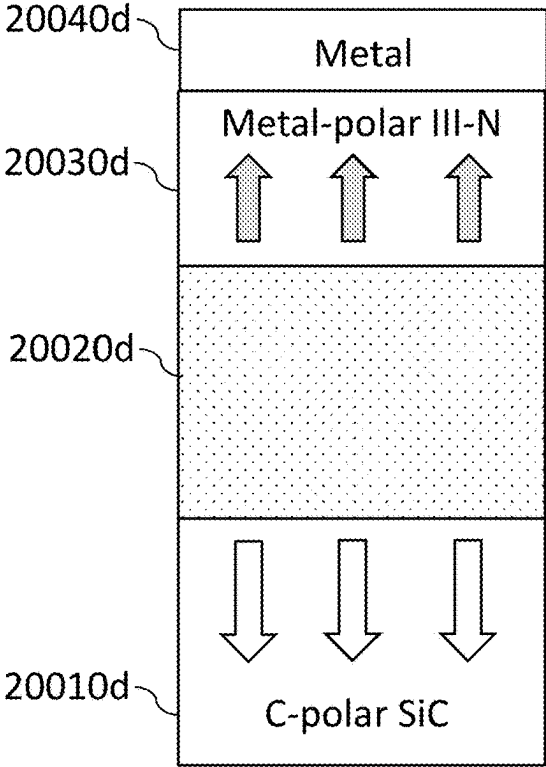


FIG. 100D

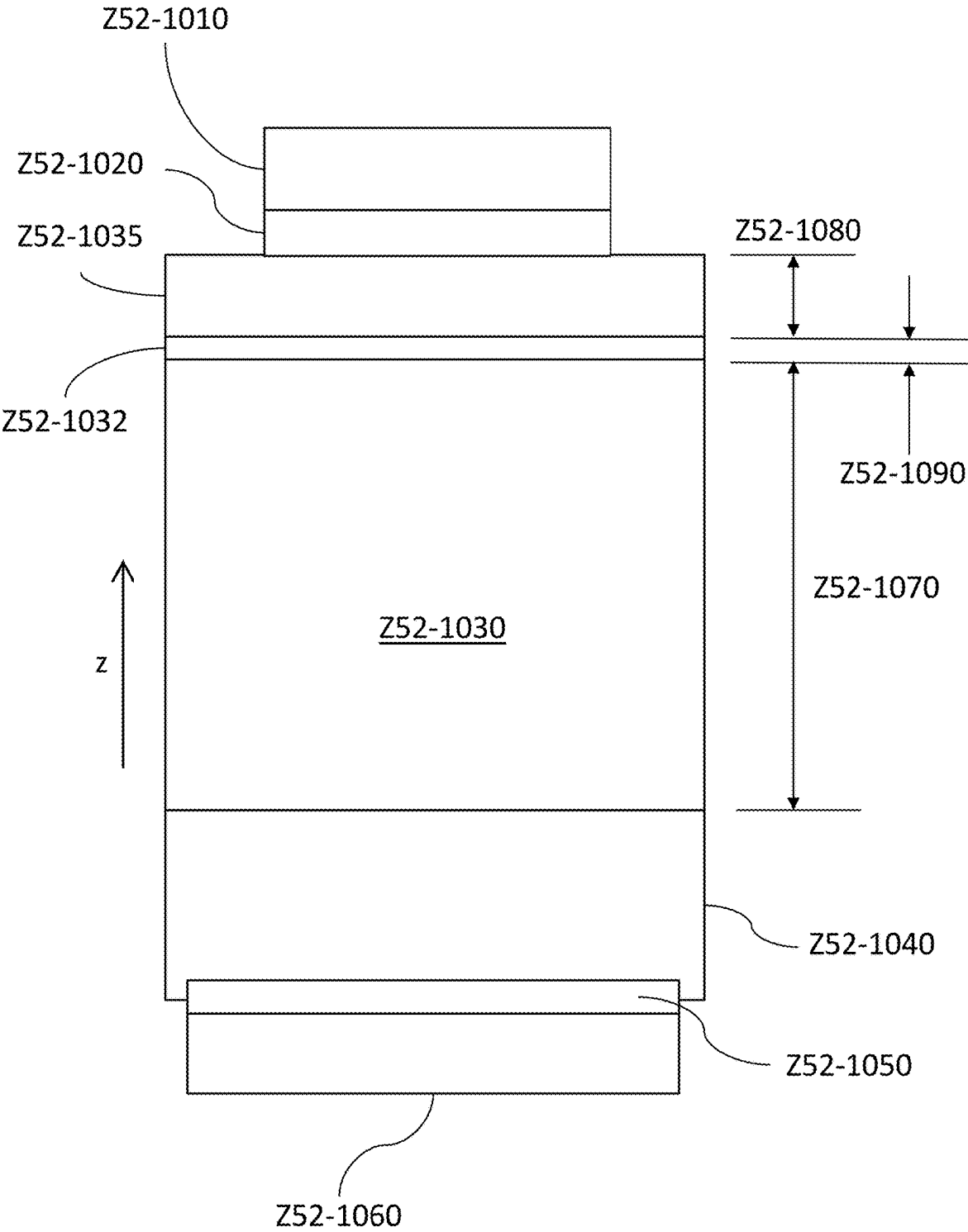


FIG. 101A

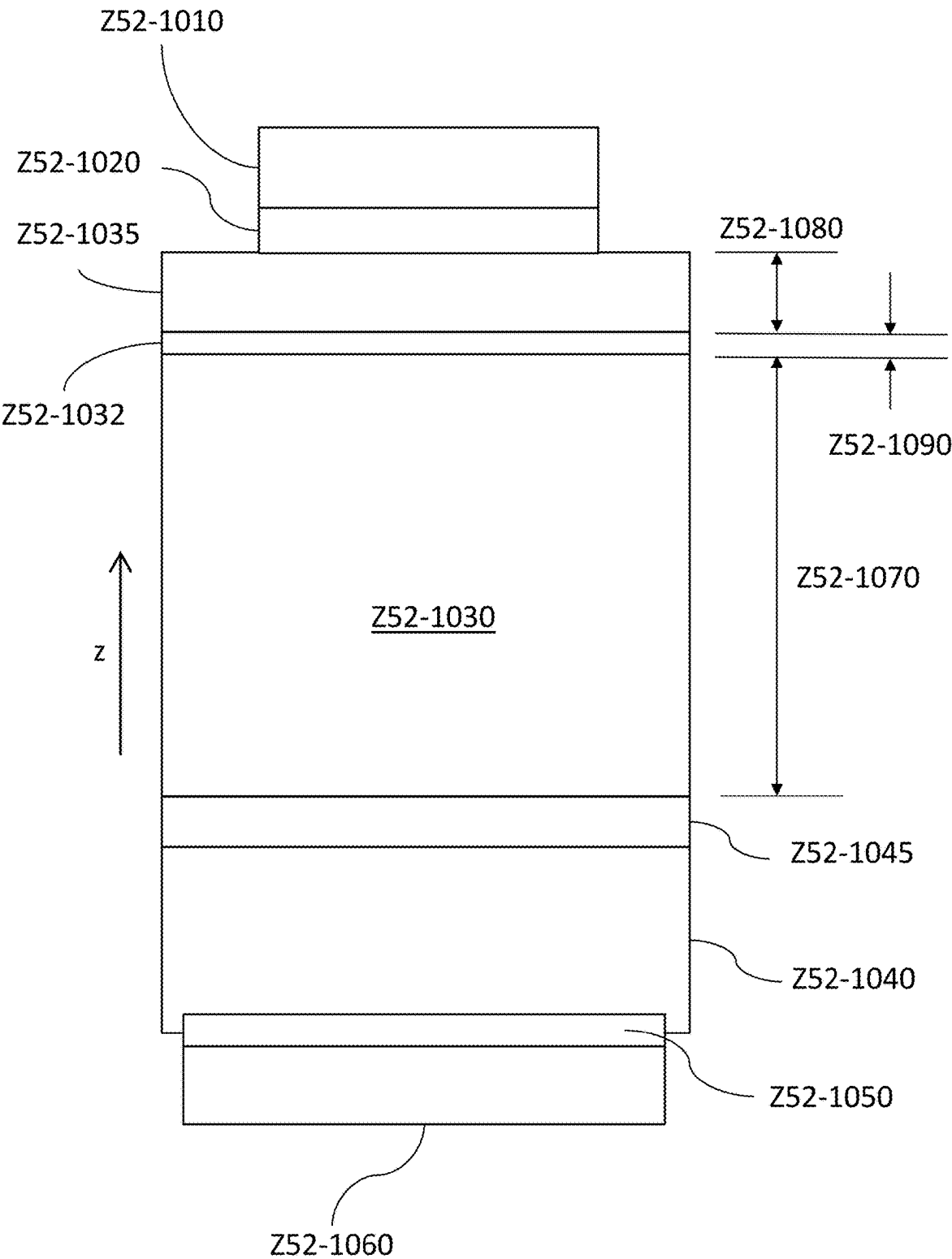


FIG. 101B

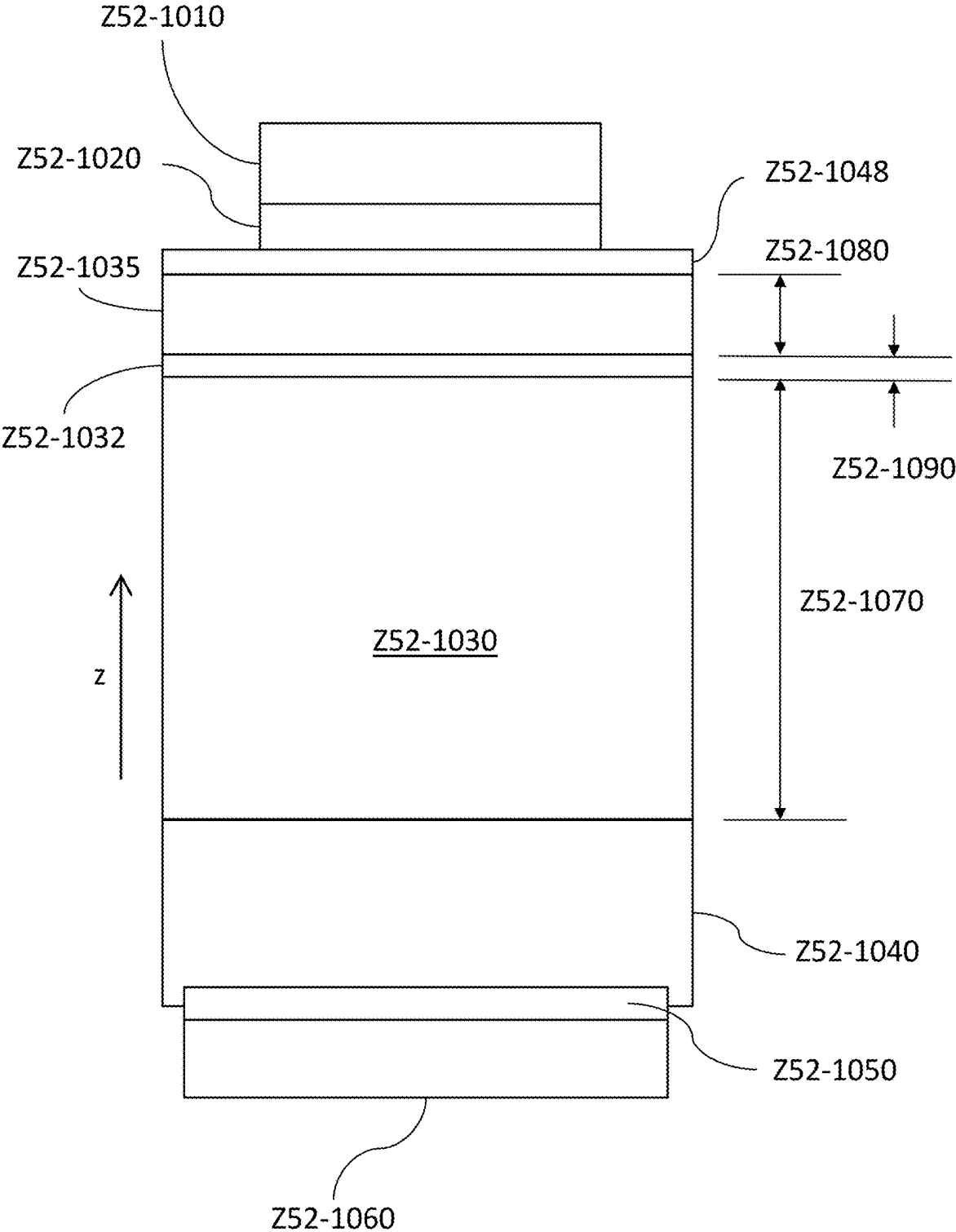


FIG. 101C

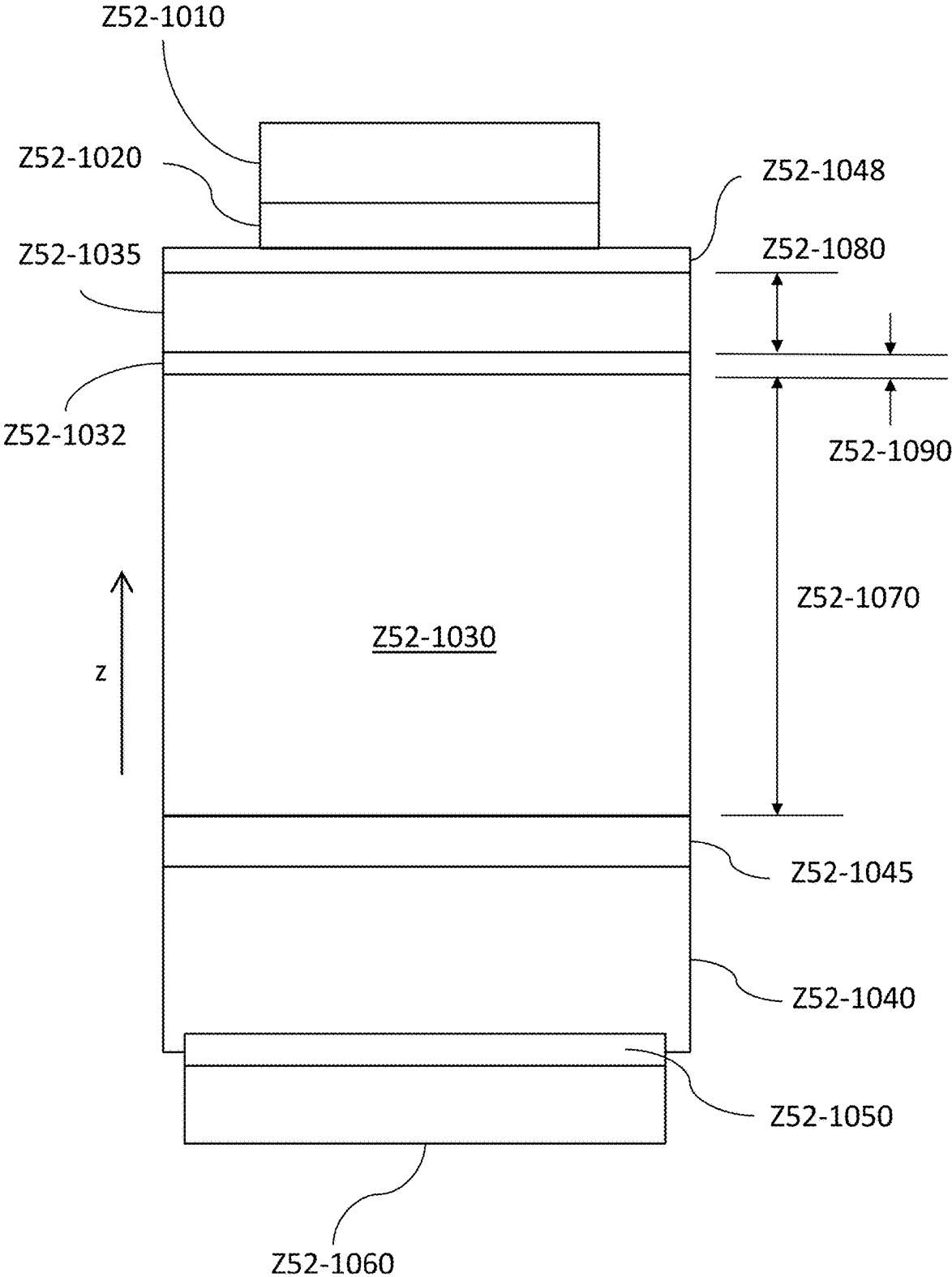


FIG. 101D

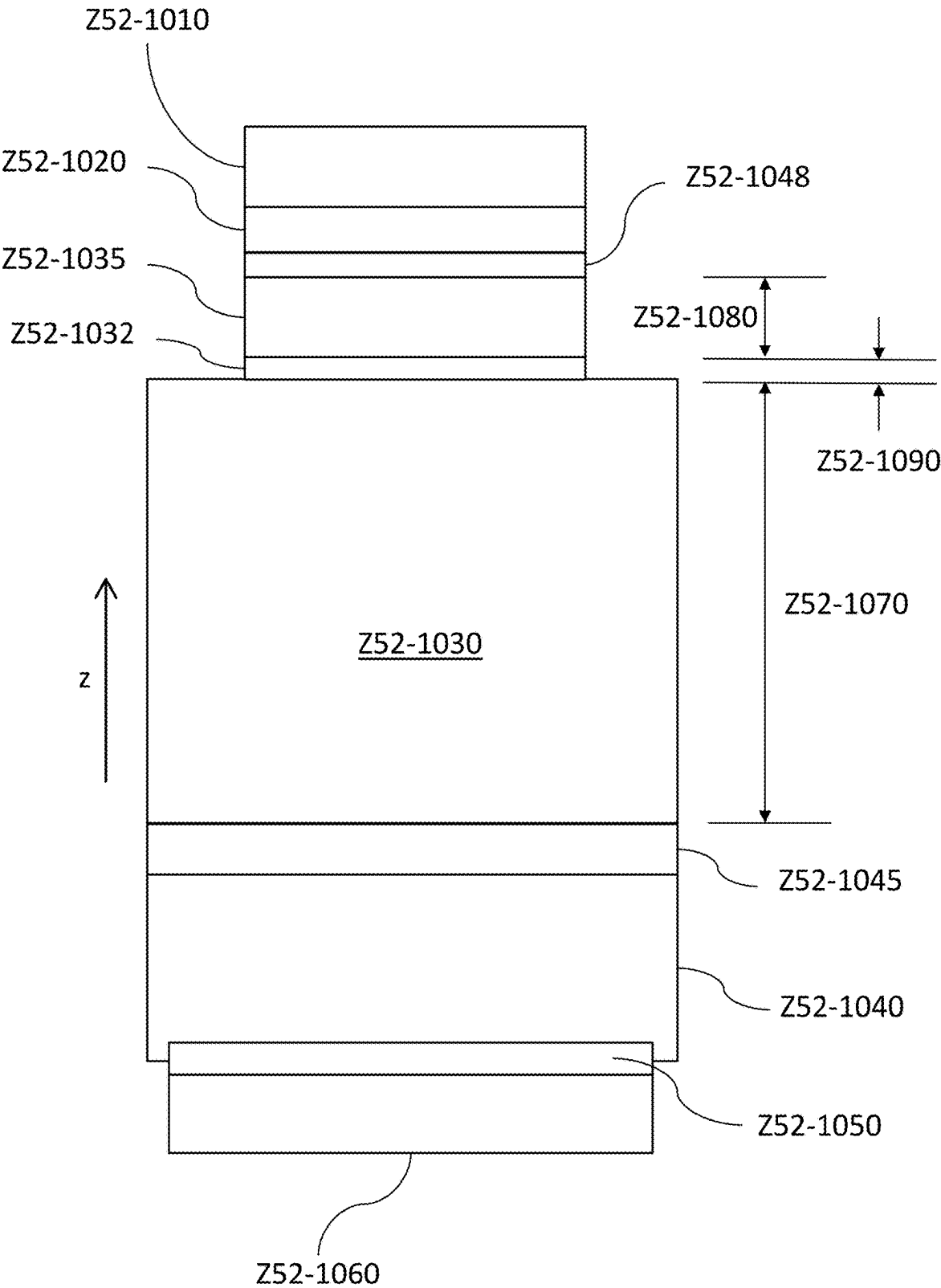


FIG. 101E

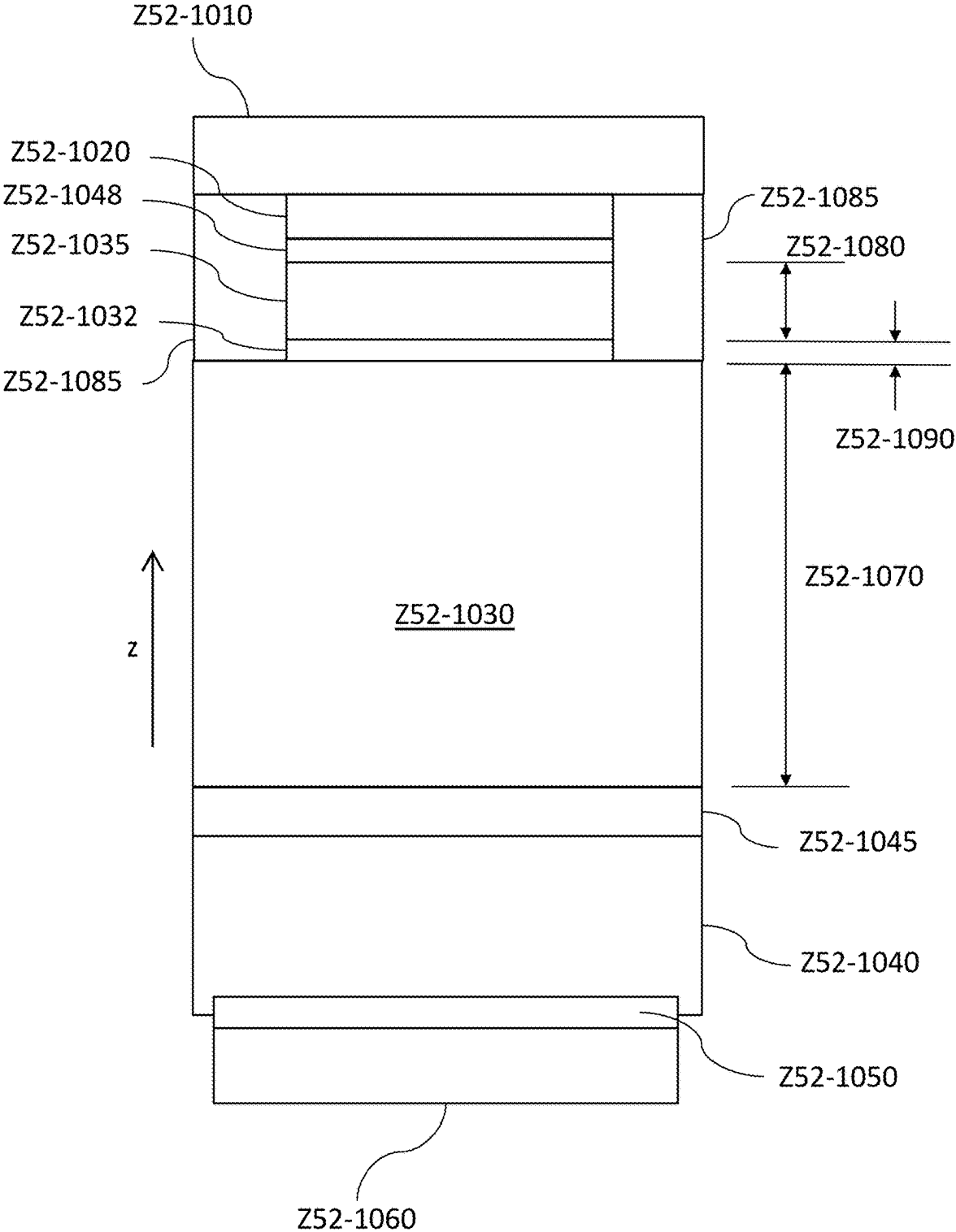


FIG. 101F

Metal E	Ohmic or Schottky							
Nitride	n	n	n	n	p	p	p	p
Oxide	n	p	n	p	n	p	n	p
Substrate	n	n	p	p	n	n	p	p
Metal S	Ohmic							

FIG. 102A

Metal E	Ohmic or Schottky															
IL	n	p	n	n	p	n	p	n	p	n	p	n	p	n	p	p
Nitride	n	n	n	n	n	n	n	p	p	p	p	p	p	p	p	p
Oxide	n	n	p	n	n	p	p	n	n	p	p	n	n	p	p	p
Substrate	n	n	n	p	p	p	p	n	n	n	n	p	p	p	p	p
Metal S	Ohmic															

FIG. 102B

Metal E	Ohmic or Schottky																			
IL	n	p	n	n	p	n	p	n	p	p	p	n	n	n	p	n	p	n	p	p
Nitride	n	n	n	n	n	p	p	p	p	p	p	n	n	n	n	n	n	p	p	p
Oxide	n	n	p	n	n	n	n	p	p	n	p	n	p	n	n	p	p	p	p	p
TL	n	n	n	n	n	n	n	n	n	n	n	p	p	p	p	p	p	p	p	p
Substrate	n	n	n	p	p	n	n	n	n	p	p	n	n	p	p	p	p	n	n	p
Metal S	Ohmic																			

FIG. 102C

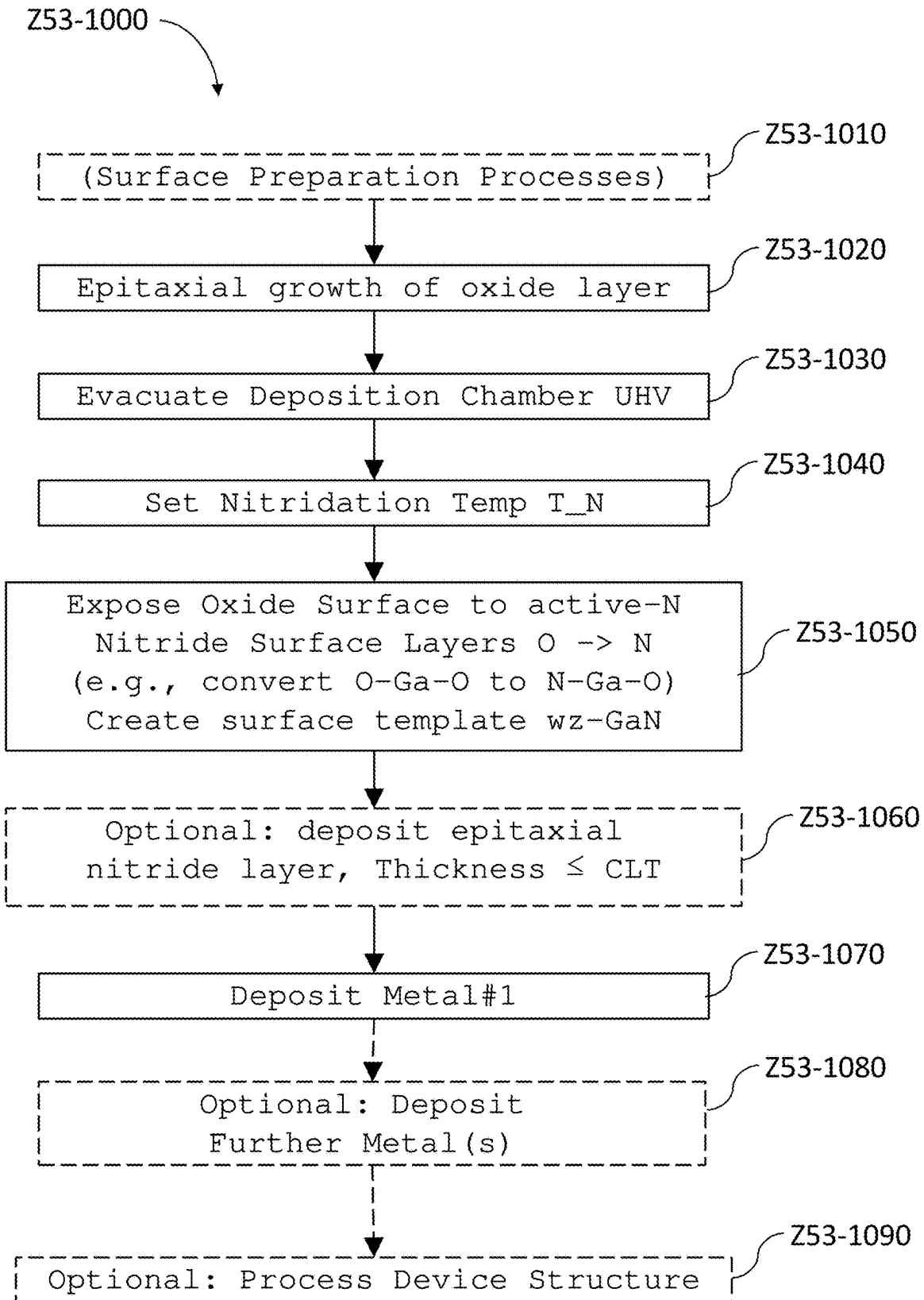


FIG. 103

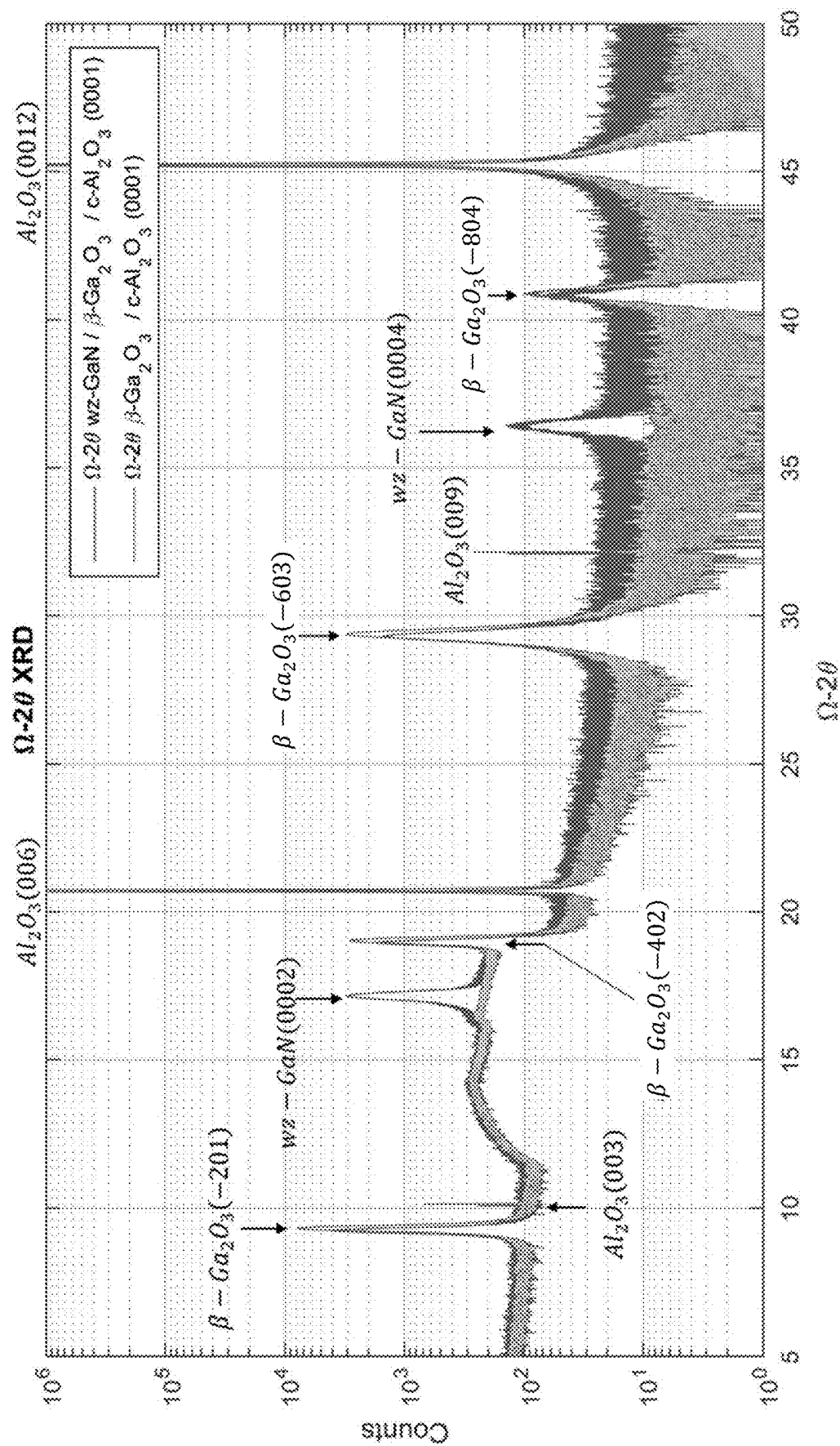


FIG. 104

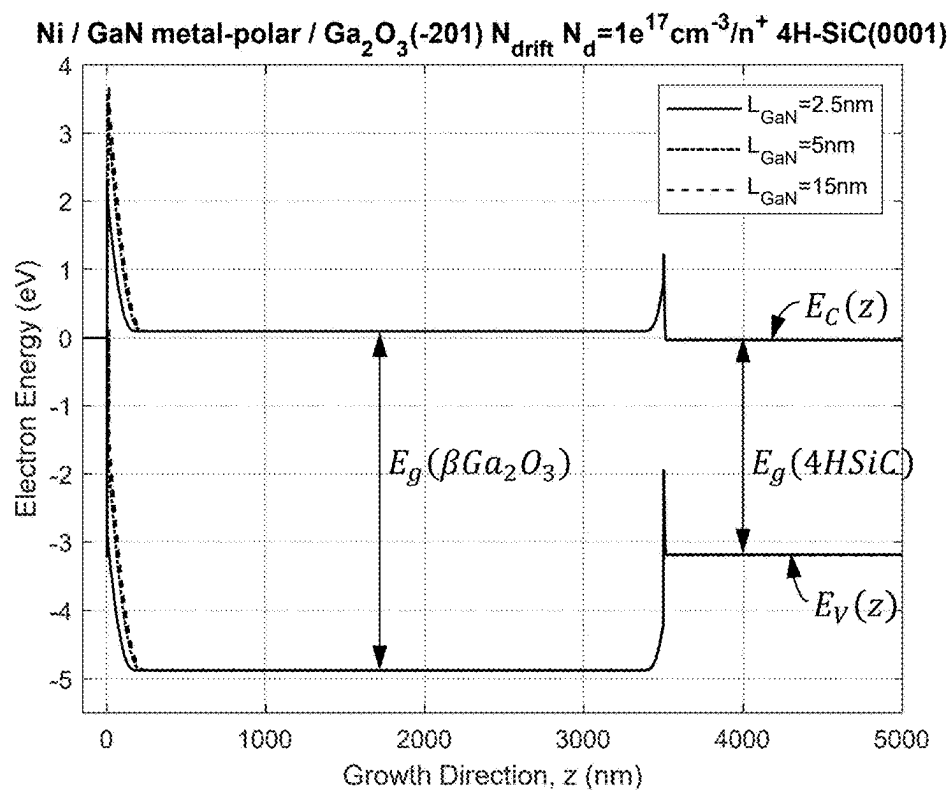


FIG. 105A

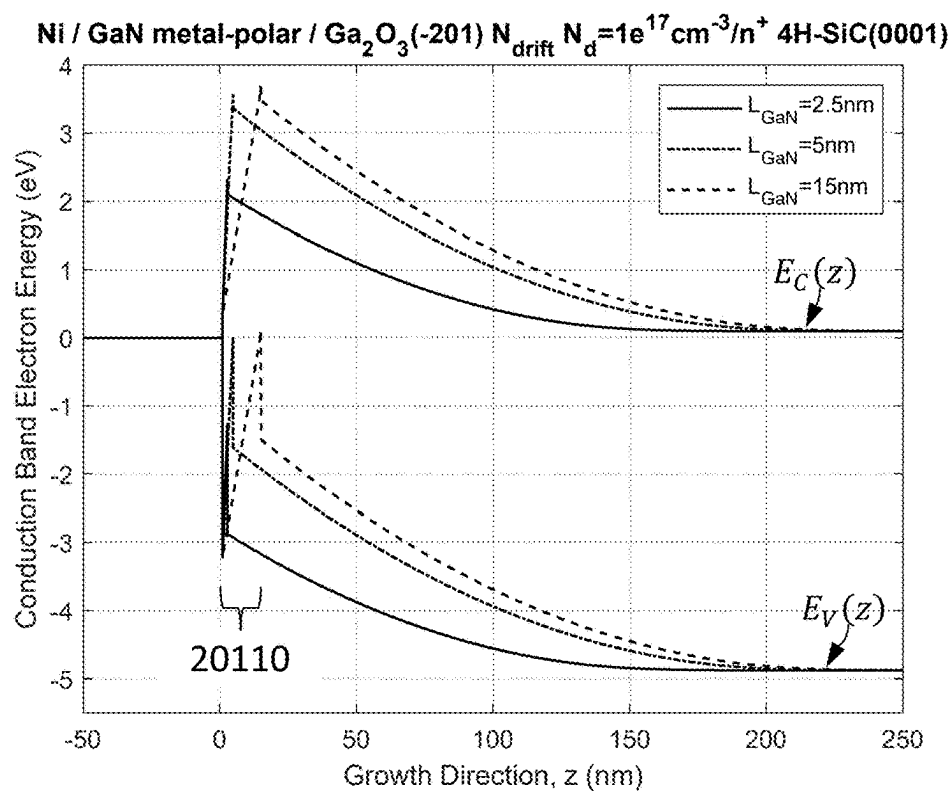


FIG. 105B

2D-Hole Gas may form P+GaN / N-Ga₂O₃ Junction

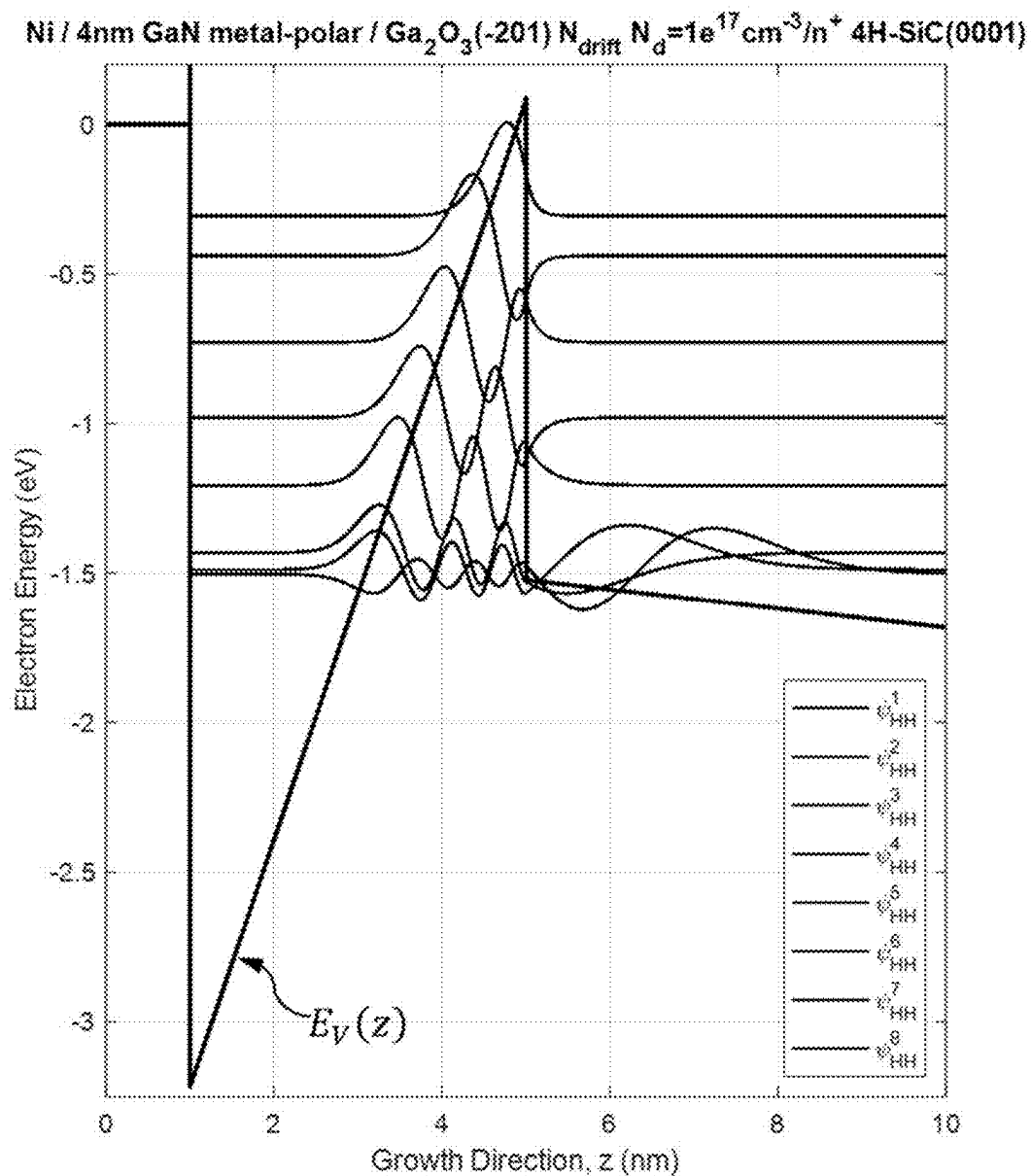


FIG. 105C

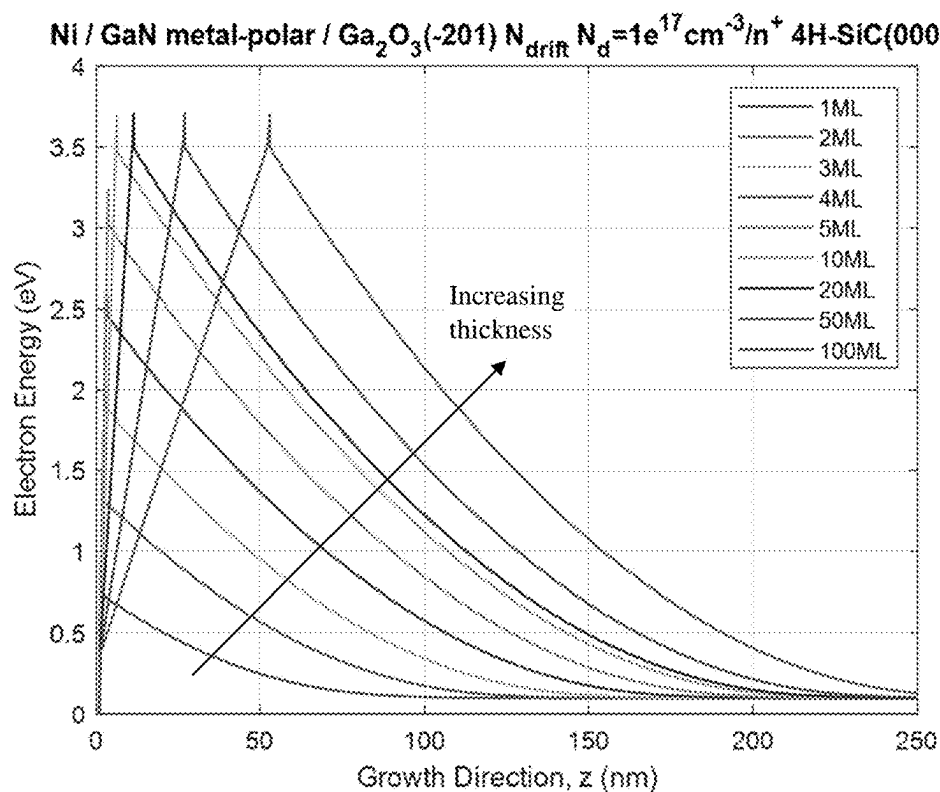


FIG. 105D

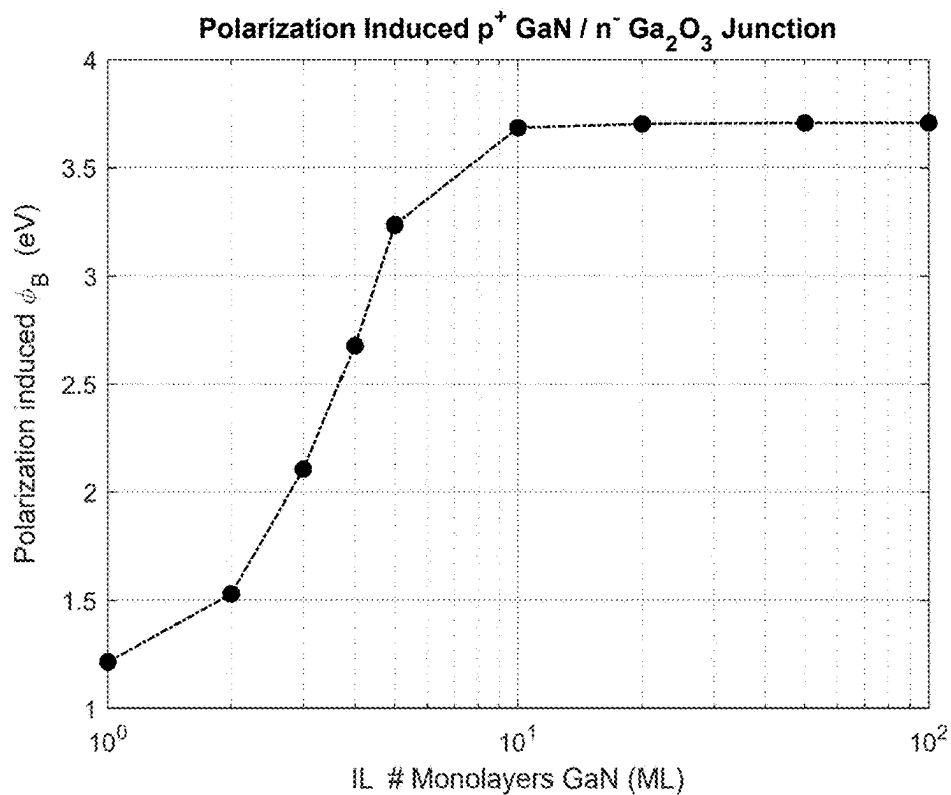


FIG. 105E

Epitaxial Oxide Deposition

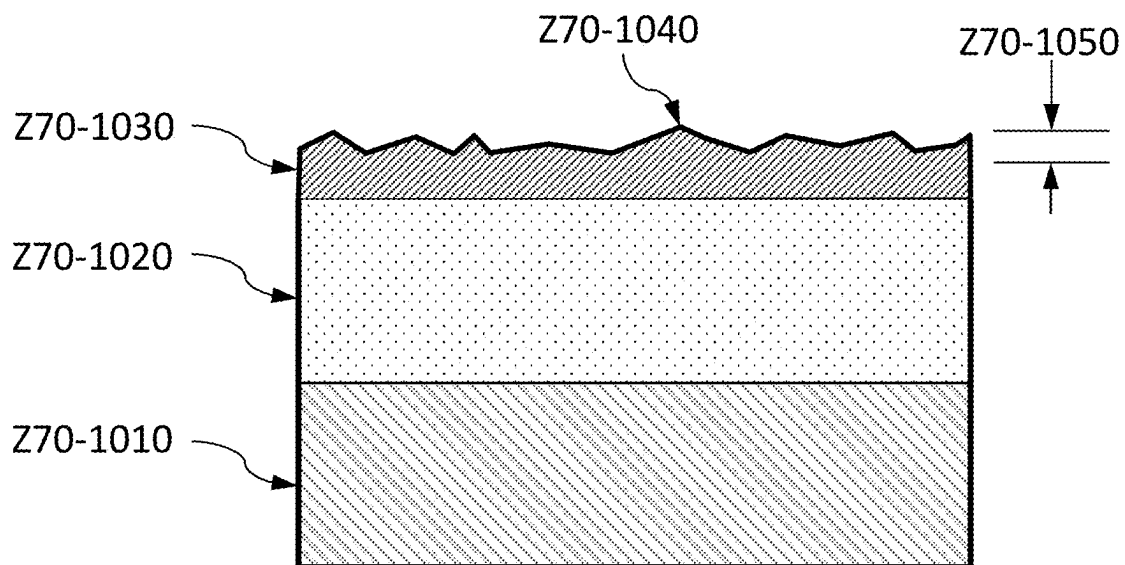


FIG. 106A

Post epi-growth planarization of surface

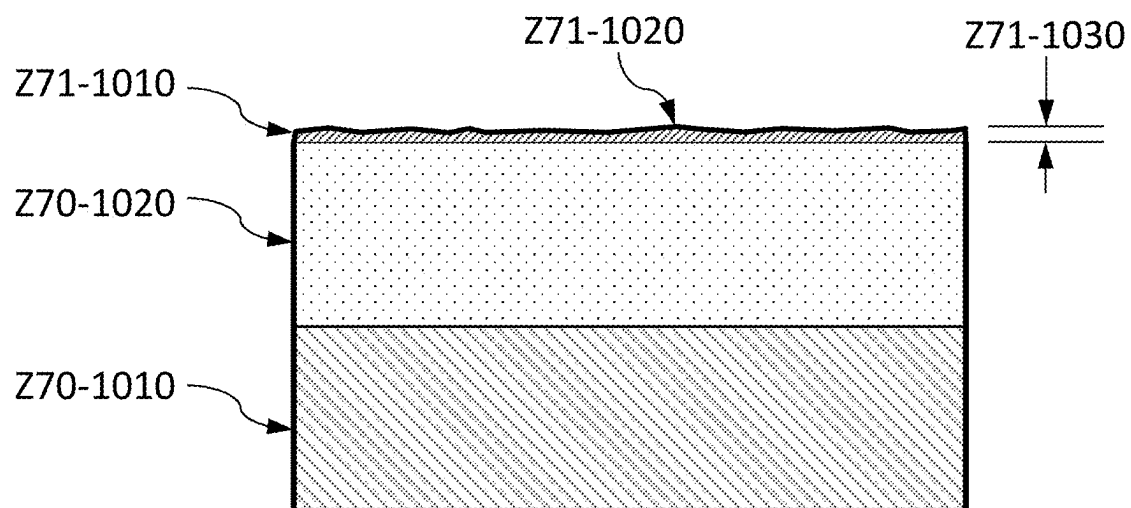


FIG. 106B

SEMICONDUCTOR DEVICE

RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/830,089 filed on Sep. 10, 2024 and entitled “SEMICONDUCTOR DEVICE”, which is a continuation-in-part of U.S. application Ser. No. 18/628,178 filed on Apr. 5, 2024 and entitled “SEMICONDUCTOR DEVICE”, which is a continuation-in-part of U.S. application Ser. No. 18/394,688 filed on Dec. 22, 2023, which issued as U.S. Pat. No. 12,074,195 on Aug. 27, 2024, and entitled “SEMICONDUCTOR DEVICE”; which claims the benefit of U.S. Provisional Patent Application No. 63/584,661 filed on Sep. 22, 2023, and entitled “SEMICONDUCTOR DEVICE”; all of which are hereby incorporated by reference for all purposes.

BACKGROUND

[0002] High power semiconductor devices operating at high voltages (e.g., greater than 1.3 kV) have been made using different types of semiconductor materials. For example, Schottky diodes using silicon carbide (SiC) materials have replaced Silicon (Si) for high power applications, such as in power conversion and automotive electronics. Recently, beta-phase Gallium Oxide (β -Ga₂O₃) has been offered as a standalone technology and used to form high power Schottky diodes as an alternative to comparable SiC—with potentially improved performance and lower cost. SiC is a wide bandgap (WBG) semiconductor compared to Si. Improved high power performance can be engineered by utilizing the fundamental advantages afforded by virtue of the WBG. In the same manner, an even larger bandgap semiconductor may potentially further improve device performance beyond SiC. Gallium Oxide (Ga₂O₃) semiconductors in various polytypes exhibit bandgaps (E_g) from 4.5 eV to 5.0 eV, compared to the bandgap of SiC (about 3.2 eV) and for Si (about 1.1 eV). This increased bandgap can be used advantageously to improve the diode ON-state resistance (i.e., reduce loss) and increase blocking voltage capability (i.e., improve isolation).

[0003] Despite intense development of high voltage β -Ga₂O₃ diodes and transistors, there still exist fundamental limitations toward the widespread acceptance for high power applications.

SUMMARY

[0004] In some embodiments, the techniques described herein relate to a multilayered semiconductor diode device including: a substrate including silicon carbide (SiC); an epitaxial transition layer including a first semiconductor oxide material or SiC, wherein the epitaxial transition layer is on the substrate; an epitaxial drift layer including a second semiconductor oxide material, wherein the epitaxial drift layer is on the epitaxial transition layer; and a metal layer above the epitaxial drift layer, wherein the metal layer and the epitaxial drift layer form a Schottky barrier junction.

[0005] In some embodiments, a method of forming a multilayered semiconductor diode device includes: providing a substrate including silicon carbide (SiC); forming, on the substrate, an epitaxial transition layer including a first semiconductor oxide material or SiC; forming, on the epitaxial transition layer, an epitaxial drift layer including a second semiconductor oxide material; and forming a metal

layer above the epitaxial drift layer, wherein the metal layer and the epitaxial drift layer form a Schottky barrier junction.

[0006] In some embodiments, the techniques described herein relate to a multilayered semiconductor diode device including: a substrate including silicon carbide (SiC); an epitaxial drift layer including a first semiconductor oxide material, wherein the epitaxial drift layer is on the substrate; an epitaxial channel layer including a second semiconductor oxide material, wherein the epitaxial channel layer is on the epitaxial drift layer; a Schottky metal layer above the epitaxial channel layer, wherein the Schottky metal layer and the epitaxial channel layer form a Schottky barrier junction, and wherein the epitaxial channel layer and the Schottky metal layer are formed into a first mesa structure; and a sidewall layer including a dielectric material, wherein the sidewall layer is on the epitaxial drift layer and contacts a wall of the first mesa structure.

[0007] In some embodiments, the techniques described herein relate to a method of forming a multilayered semiconductor diode device including: providing a substrate including silicon carbide (SiC); forming, on the substrate, an epitaxial drift layer including a first semiconductor oxide material; forming, on the epitaxial drift layer, an epitaxial channel layer including a first semiconductor oxide material; forming a metal layer on the epitaxial drift layer, wherein the metal layer and the epitaxial drift layer form a Schottky barrier junction; etching the epitaxial channel layer and the metal layer to form a mesa structure; forming a sidewall layer including a dielectric material, wherein the sidewall layer is on the epitaxial drift layer and contacts a wall of the mesa structure.

[0008] In some embodiments, the techniques described herein relate to a multilayered semiconductor diode device including: a substrate including silicon carbide (SiC); an epitaxial drift layer including a first semiconductor oxide material, wherein the epitaxial drift layer is above the substrate with respect to a growth direction; a polar nitride layer including a polar semiconductor nitride material, wherein the polar nitride layer is above the epitaxial drift layer with respect to the growth direction; and a metal layer above the polar nitride layer with respect to the growth direction.

[0009] In some embodiments, the techniques described herein relate to a method of forming a multilayered semiconductor diode device including: providing a substrate including silicon carbide (SiC); forming, on the substrate, an epitaxial drift layer including a first semiconductor oxide material; forming, on the epitaxial drift layer, an epitaxial polar nitride layer including a polar semiconductor nitride material; and forming a metal layer on the epitaxial polar nitride layer, wherein the metal layer and the epitaxial polar nitride layer form a Schottky barrier junction.

[0010] In some embodiments, the techniques described herein relate to a multilayered semiconductor diode device including: a substrate including silicon carbide (SiC); an epitaxial drift layer including a first semiconductor oxide material, wherein the epitaxial drift layer is above the substrate with respect to a growth direction, and wherein the epitaxial drift layer is doped n-type; a template layer including a nitride template material, wherein the template layer is in contact with the epitaxial drift layer; a polar nitride layer including a polar semiconductor nitride material, wherein the polar nitride layer is in contact with the template layer, and wherein a polarization axis of the polar semiconductor

nitride material is aligned with the growth direction such that the polar nitride layer includes p-type charge, wherein the polar nitride layer and the epitaxial drift layer form a p/n junction; and a metal layer above the polar nitride layer.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Embodiments of the present disclosure will be discussed with reference to the accompanying drawings.

[0012] FIG. 1 is a figurative sectional view of a vertical multilayered semiconductor diode device in accordance with an illustrative embodiment.

[0013] FIG. 2A is a simplified band diagram of a vertical multilayered semiconductor diode device, in accordance with some embodiments.

[0014] FIG. 2B is a plot of the trends for the critical electric-field that can be supported versus a semiconductor bandgap energy.

[0015] FIG. 2C is a plot of the specific ON-resistance versus the breakdown voltage for Silicon, Silicon-Carbide and Gallium Oxide semiconductors.

[0016] FIGS. 3A and 3B are figurative views of the substrate and drift layer of a vertical multilayered semiconductor diode device such as shown in FIG. 1 and associated plots of example variations in doping concentration (i.e., doping density) as a function of vertical height, in accordance with some embodiments.

[0017] FIG. 4 is a figurative sectional view of a vertical multilayered semiconductor diode device in accordance with another illustrative embodiment.

[0018] FIG. 5 is a figurative view of the substrate, transition layer and drift layer of a vertical multilayered semiconductor diode device such as shown in FIG. 4 and an associated plot of an example variation in doping concentration as a function of vertical height, in accordance with some embodiments.

[0019] FIG. 6 is a figurative view of the substrate, transition layer and drift layer of a vertical multilayered semiconductor diode device such as shown in FIG. 4 and an associated plot of another example variation in doping concentration as a function of vertical height, in accordance with some embodiments.

[0020] FIG. 7 is a figurative view of the substrate, transition layer and drift layer of a vertical multilayered semiconductor diode device such as shown in FIG. 4 and an associated plot of another example variation in doping concentration as a function of vertical height, as well as an associated plot of an example variation in composition as a function of vertical height, in accordance with some embodiments.

[0021] FIG. 8A is a figurative sectional view of a vertical multilayered semiconductor diode device in accordance with another illustrative embodiment.

[0022] FIG. 8B is a figurative sectional view of a vertical multilayered semiconductor diode device in accordance with another illustrative embodiment.

[0023] FIGS. 8C-8F show tables depicting combinations of n-type and p-type layers with Ohmic contacts to a SiC substrate, and Ohmic or Schottky contacts to an epitaxially grown semiconductor layer, in accordance with some embodiments.

[0024] FIG. 9A is an illustrative process flow for the design of a vertical multilayered semiconductor diode device in accordance with some embodiments.

[0025] FIG. 9B is the spatial variation of the conduction and valence band edge energy diagram of a vertical multilayered semiconductor device directed along a growth direction in accordance with an illustrative embodiment.

[0026] FIG. 9C is the spatial variation of the conduction band edge energy diagram of a vertical multilayered semiconductor device directed along a growth direction in accordance with an illustrative embodiment.

[0027] FIG. 10 is a plot of the variation of breakdown voltage versus drift region donor doping concentration for β -Ga₂O₃ for several film thicknesses, in accordance with some embodiments.

[0028] FIG. 11A is a plot of metal work function energies for a selection of metals and the corresponding Schottky barrier energy with respect to (-201) oriented surface of β -Ga₂O₃, in accordance with some embodiments.

[0029] FIG. 11B is a forward bias semilog-linear current-voltage for a Schottky barrier diode in accordance with some embodiments.

[0030] FIG. 11C is a linear-linear forward bias current-voltage for a Schottky barrier diode in accordance with some embodiments.

[0031] FIG. 12 is a plot of the spatial variation of the conduction band energy for the Schottky barrier formed between nickel and a (-201) oriented surface of β -Ga₂O₃ as a function of the drift donor carrier concentration in accordance with some embodiments.

[0032] FIG. 13 is a plot of the spatial variation of the conduction band energy for the Schottky barrier formed between nickel and a MgO intermediate layer and a (-201) oriented surface of β -Ga₂O₃ as a function of the drift donor carrier concentration in accordance with some embodiments.

[0033] FIG. 14 is a schematic cross-section of a vertical multilayered semiconductor structure forming a Schottky barrier diode comprising alpha-phase Ga₂O₃ and alpha-phase (Al_xGa_{1-x})₂O₃ epitaxially formed on A-plane, R-plane and M-plane SiC substrate in accordance with some embodiments.

[0034] FIG. 15 is a schematic cross-section of a vertical multilayered semiconductor structure forming a Schottky barrier diode comprising beta-phase Ga₂O₃ and beta-phase (Al_xGa_{1-x})₂O₃ epitaxially formed on C-plane SiC substrate in accordance with some embodiments.

[0035] FIG. 16 is a plot of the spatial variation of the conduction band energy for the Schottky barrier formed between nickel and a 5 nm thick MgO intermediate layer and a (-201) oriented surface of β -Ga₂O₃ as a function of the drift donor carrier concentration in accordance with some embodiments.

[0036] FIG. 17 is a plot of the spatial variation of the conduction band energy for the Schottky barrier formed between nickel and a 15 nm thick MgO intermediate layer and a (-201) oriented surface of β -Ga₂O₃ as a function of the drift donor carrier concentration in accordance with some embodiments.

[0037] FIG. 18 is a schematic current-voltage plot of a vertical multilayered semiconductor Schottky barrier diode representing features associated with the reverse and forward bias operation, in accordance with some embodiments.

[0038] FIG. 19A is a plot of the quantum mechanical tunnelling probability (coefficient) for electron transport in the conduction band calculated for Schottky barrier of a

[Ni/ β -Ga₂O₃] multilayered structure as a function of the incident electron energy, in accordance with some embodiments.

[0039] FIG. 19B is a plot of the quantum mechanical tunnelling probability (coefficient) for electron transport in the conduction band calculated for Schottky barrier of a [Ni/MgO/ β -Ga₂O₃] multilayered structure as a function of the incident electron energy, in accordance with some embodiments.

[0040] FIG. 19C is a plot of the spatial variation in the conduction and valence band edges along a growth direction for the multilayered diode structure comprising [Ni/ZnGa₂O₄ intermediate layer/ β -Ga₂O₃ drift-layer/4H-SiC substrate], in accordance with some embodiments.

[0041] FIG. 19D is a plot of the spatial variation in the conduction and valence band edges along a growth direction for the multilayered diode structure comprising [Ni/ZnGa₂O₄ intermediate layer/ β -Ga₂O₃ drift-layer/4H-SiC substrate], shown in the vicinity of the depletion region formed by the Schottky barrier, in accordance with some embodiments.

[0042] FIGS. 19E and 19F are simulated I-V plots of the multilayered semiconductor Schottky barrier diode in forward bias conditions, in accordance with some embodiments.

[0043] FIG. 20 schematically shows the distinct functional regions of a vertical multilayered semiconductor device comprising a Schottky metal, an intermediate region, a drift region, a transition region, a substrate and a rear contact (not shown), where the intermediate and transition regions are formed as a plurality of semiconductor layers, in accordance with some embodiments.

[0044] FIGS. 21A and 21B show plots of the spatial variation in the conduction and valence band edges along a growth direction for the multilayered diode structure comprising [Ni/superlattice intermediate layer/ β -Ga₂O₃ drift-layer/4H-SiC substrate], wherein the finite number of periods of the superlattice intermediate layer comprises for example bilayer pairs of [ZnGa₂O₄/Ga₂O₃], in accordance with some embodiments.

[0045] FIG. 21C shows a schematic spatial band energy diagram of a metal-oxide-metal (MOM) heterostructure, wherein the effect of high electric-field band bending is depicted for a biased MOM structure, in accordance with some embodiments.

[0046] FIG. 22 is a plot of the spatial variation in the conduction and valence band edges along a growth direction for the multilayered diode structure comprising an example [Ni/ β -Ga₂O₃ drift-layer/4H-SiC substrate] as a function of applied reverse bias including the barrier lowering effect due to the surface electric field developed across the Schottky barrier potential, in accordance with some embodiments.

[0047] FIG. 23 is a plot of the spatial variation in the conduction band edges along a growth direction for the multilayered diode structure comprising an example [Ni/ β -Ga₂O₃ drift-layer/4H-SiC substrate] as a function of applied reverse bias including the barrier lowering effect due to the surface electric field developed across the Schottky barrier potential, where the surface electric field is varied from 0 MV/cm to 1 MV/cm showing the Schottky barrier height and width variation, in accordance with some embodiments.

[0048] FIG. 24 shows a plot of the calculated effective barrier width and potential energy height of the Schottky

heterojunction as a function of applied reverse bias and therefore the surface electric field, in accordance with some embodiments.

[0049] FIG. 25 shows the calculated quantum mechanical tunnelling coefficient versus the incident electron energy traversing the Schottky barrier formed by the [Ni/ β -Ga₂O₃/4H-SiC] structure as a function of surface electric field, in accordance with some embodiments.

[0050] FIG. 26 shows the calculated quantum mechanical tunnelling coefficient versus the incident electron energy traversing the Schottky barrier formed by the [Ni/5 nm MgO/ β -Ga₂O₃/4H-SiC] structure as a function of surface electric field, in accordance with some embodiments.

[0051] FIG. 27 plots the calculated component contributions of the thermionic emission current and quantum mechanical tunneling currents versus applied surface electric field as a function of several substrate temperatures, where the device structure comprises a vertical multilayer semiconductor structure [Ni/ β -Ga₂O₃/4H-SiC], in accordance with some embodiments.

[0052] FIG. 28 plots the calculated component contributions of the thermionic emission current and quantum mechanical tunneling currents versus applied surface electric field as a function of several substrate temperatures, where the device structure comprises a vertical multilayer semiconductor structure [Ni/5 nm MgO/ β -Ga₂O₃/4H-SiC], in accordance with some embodiments.

[0053] FIG. 29A shows a plot of static and optical (high frequency) dielectric constants calculated for a selection of semiconductors and oxide materials, where a trend for dielectric constant versus material bandgap energy is disclosed, in accordance with some embodiments.

[0054] FIG. 29B is a plot of the electric-field induced conduction band edge barrier lowering energy versus material bandgap energy using the optical dielectric constants calculated in FIG. 29A, where three example electric-field strengths are plotted for 5 μ m slabs, in accordance with some embodiments.

[0055] FIG. 29C plots the empirical trends of interface state parameter and minimum surface state density that varies as a function of a material bandgap energy, in accordance with some embodiments.

[0056] FIG. 30 is a plot of the spatial variation of the conduction band edge along a growth direction (upper diagram) for superlattice delta-doped drift region using an example Ga₂O₃ host layer coupled to a Ni Schottky barrier and deposited on a SiC substrate, where the lower diagram depicts the periodic doped layers punctuating not intentionally doped regions, in accordance with some embodiments.

[0057] FIG. 31 shows a plot of the effective donor doping concentration of delta-doped superlattice drift region versus the thickness ratio γ , in accordance with some embodiments.

[0058] FIGS. 32A and 32B are plots of the forward bias I-V curves for two configurations of vertical multilayered Schottky barrier diode comprising a delta-doped SL Ga₂O₃ drift region, in accordance with some embodiments.

[0059] FIGS. 33A and 33B plot the spatial variation of the conduction band edge along a growth direction for a double barrier intermediate region comprising [ZnGa₂O₄/Ga₂O₃/ZnGa₂O₄] formed between a Ni metal and a Ga₂O₃ drift region, and the quantized electron energy states are shown for zero and high surface electric field applied to the structure, in accordance with some embodiments.

[0060] FIG. 34 plots the electrostatic potential and electric field lines calculated for an example [metal/Ga₂O₃/SiC] vertical structure with the topmost Ni contact surrounded by an isolation layer dielectric material, in accordance with some embodiments.

[0061] FIG. 35A shows a top view of an example of a diode formed on a Ga₂O₃/SiC multilayer structure using a circular Schottky barrier metal and circular field-plate metal contacts, in accordance with some embodiments.

[0062] FIG. 35B shows a cross-sectional view of an example diode formed on a Ga₂O₃/SiC multilayer structure using a circular Schottky barrier metal and circular field-plate metal contacts, where the top field-plate contact is separated from the surface of the Ga₂O₃ drift layer using an isolation layer, in accordance with some embodiments.

[0063] FIG. 36 plots the electrostatic potential and electric field lines calculated for an example [metal/Ga₂O₃/SiC] vertical structure with the topmost Ni contact surrounded by an isolation layer dielectric material and a further field-plate positioned upon the isolation layer and in electrical contact with the Schottky barrier contact, in accordance with some embodiments.

[0064] FIG. 37A schematically shows a top view of a plurality of close packed circular Schottky barrier metal contacts separated by a fixed radius, in accordance with some embodiments.

[0065] FIG. 37B is the cross-sectional view of the structure in FIG. 37A showing the relation between the Schottky barrier metal, field-plate contact, isolation layer, drift layer, substrate and rear ohmic contact, in accordance with some embodiments.

[0066] FIGS. 38A, 38B and 38C depict schematically the process flow for the formation of a top Schottky metal mesa region formed on an epitaxial drift layer formed on a substrate, in accordance with some embodiments.

[0067] FIGS. 39A-39G schematically depict a fabrication process flow for a vertical conduction multilayered semiconductor diode structure in accordance with an illustrative embodiment.

[0068] FIG. 40 is a two-dimensional thermal model of the vertical multilayered semiconductor diode used to solve conduction-dominant heat transfer with convection and radiation occurring at boundaries, in accordance with some embodiments.

[0069] FIG. 41 is a plot of the temperature dependent thermal conductivity for single crystal SiC and Ga₂O₃ materials, in accordance with some embodiments.

[0070] FIG. 42 is a steady-state heat distribution contour plot of the 2D diode model configured with a Ga₂O₃ drift layer and electrically conductive and low thermal conductance Ga₂O₃ substrate. A steady-state current induces ohmic heating in region F8, and the metal contacts are connected to thermal reservoirs.

[0071] FIG. 43 is a steady-state heat distribution contour plot of the 2D diode model configured with a Ga₂O₃ drift layer and electrically conductive and high thermal conductance SiC substrate, where a steady-state current induces ohmic heating in region F8, and the metal contacts are connected to thermal reservoirs, in accordance with some embodiments.

[0072] FIG. 44 is a plot of the transient heat transfer and dissipation rate in the multilayered semiconductor diode for

forward bias current pulses, where the Ga₂O₃ and SiC substrate configurations are compared, in accordance with some embodiments.

[0073] FIG. 45 is a two-dimensional thermal model of the vertical multilayered semiconductor diode configured as a mesa drift region and used to solve conduction-dominant heat transfer with convection and radiation occurring at boundaries, in accordance with some embodiments.

[0074] FIG. 46 is a steady-state heat distribution contour plot of the 2D diode model configured with a mesa Ga₂O₃ drift layer and electrically conductive and low thermal conductance Ga₂O₃ substrate, where a steady-state current induces ohmic heating in region F8, and the metal contacts are connected to thermal reservoirs, in accordance with some embodiments.

[0075] FIG. 47 is a steady-state heat distribution contour plot of the 2D diode model configured with a mesa Ga₂O₃ drift layer and electrically conductive and high thermal conductance SiC substrate, where a steady-state current induces ohmic heating in region F8, and the metal contacts are connected to thermal reservoirs, in accordance with some embodiments.

[0076] FIG. 48 plots the absolute energy alignment relative to vacuum energy of the conduction and valence bands for the surface orientation dependence of β -Ga₂O₃ and the Si-polar and C-polar 4H-SiC C-plane surfaces, in accordance with some embodiments.

[0077] FIG. 49 is a plot of the spatial conduction band energy diagram in the vicinity of the β -Ga₂O₃/Si-polar 4H-SiC heterojunction, where the lowest energy quantum confined electron wavefunction is positioned below the Fermi energy level, in accordance with some embodiments.

[0078] FIG. 50 shows a cross-section of the atomic crystal structure of 4H-SiC polytype along the [0001] direction, in accordance with some embodiments.

[0079] FIG. 51 shows the atomic crystal structure arrangement of 4H-SiC polytype viewed on the C-plane hexagonal surface, in accordance with some embodiments.

[0080] FIG. 52 is a cross-sectional view of the monoclinic β -Ga₂O₃ crystal structure along the [-201] direction, in accordance with some embodiments.

[0081] FIG. 53 shows the oxygen sub-lattice arrangement on the (-201) β -Ga₂O₃ surface, in accordance with some embodiments.

[0082] FIG. 54 shows the Si or C hexagonal sublattice for the 4H-SiC(0001) surface, in accordance with some embodiments.

[0083] FIG. 55 shows the oxygen hexagonal sublattice for the (-201) β -Ga₂O₃ surface, in accordance with some embodiments.

[0084] FIG. 56 depicts the surface orientations and polarity of two slabs comprising (-201)-oriented 3-Ga₂O₃ and 4H-SiC as Si-polar or C-polar, in accordance with some embodiments.

[0085] FIG. 57A shows the atomic arrangement a bulk 4H-SiC crystal having Si-polar surface and the SiC surface with preferentially terminated Si atoms with a first oxygen adatom layer, in accordance with some embodiments.

[0086] FIG. 57B shows the atomic arrangement a bulk 4H-SiC crystal having C-polar surface and the SiC surface with preferentially terminated Si atoms with a first oxygen adatom layer, in accordance with some embodiments.

[0087] FIG. 57C shows the atomic arrangement a bulk 4H-SiC crystal having Si-polar surface and the SiC surface

with preferentially terminated C atoms with a first oxygen adatom layer and second Ga adlayer, in accordance with some embodiments.

[0088] FIG. 57D shows the atomic arrangement a bulk 4H-SiC crystal having C-polar surface and the SiC surface with preferentially terminated C atoms with a first oxygen adatom layer and second Ga adlayer, in accordance with some embodiments.

[0089] FIG. 57E shows a possible atomic arrangement at an intentionally miscut 4H-SiC(0001) surface and the subsequent epitaxial Ga—O bond formation as a template for further Ga₂O₃ film growth, in accordance with some embodiments.

[0090] FIG. 57F depicts a cross-section of the 4H-SiC(0001) crystal miscut to create a plurality of steps and ledges on the SiC surface, creating a film formation surface that is tilted with respect to the 4H-SiC crystal axis, in accordance with some embodiments.

[0091] FIG. 57G is an example process flow for the heterogenous deposition of an oxide semiconductor upon a preselected surface orientation of a SiC substrate, where a process step of manipulating the substrate and/or oxide semiconductor surface species and state in preparation for oxide epitaxy and/or metal deposition sequence for the device structure is shown, in accordance with some embodiments.

[0092] FIG. 57H is an example process flow for the heterogenous deposition of an oxide semiconductor using the method of growth interruptions for improving the film quality, in accordance with some embodiments.

[0093] FIG. 58 is a complete series of reflection high energy electron diffraction (RHEED) images with azimuthal rotation around the rotation axis of the epitaxial growth, where the long sharp RHEED streaks indicate atomically flat and high crystal quality of a 2 μm of Si-doped β-Ga₂O₃ (−201) oriented film deposited upon a 0 deg miscut n+4H-SiC(0001) substrate, in accordance with some embodiments.

[0094] FIG. 59 is a series of RHEED images with azimuthal rotation around the rotation axis of the epitaxial growth. The long sharp RHEED streaks indicate atomically flat and high crystal quality of a 2 μm of Si-doped β-Ga₂O₃ (−201) oriented film deposited upon a 4 deg miscut n+4H-SiC(0001) substrate, where the RHEED dots represent transmission of the electron beam through the step edges, in accordance with some embodiments.

[0095] FIGS. 60A and 60B depict in-situ laser reflectance used for epitaxial deposition of 3-Ga₂O₃ (−201) oriented film on an n+4H-SiC(0001) substrate, where the periodic thickness oscillations shown in the experimental data in FIG. 60B are a result of constant Ga₂O₃ growth rate initially deposited on a bare n+SiC substrate, in accordance with some embodiments.

[0096] FIG. 61A is a plot of the X-ray diffraction symmetric double crystal rocking curve of a single crystal β-Ga₂O₃ (−201) oriented film on n+4H-SiC(0001)_0 deg miscut, where the sharp β-Ga₂O₃ peaks indicate beta-phase pure growth conditions, in accordance with some embodiments.

[0097] FIG. 61B is a contour plot of a reciprocal space map of a 3-Ga₂O₃ (−201) oriented film on n+4H-SiC(0001)_0 deg miscut, where a reciprocal space map (RSM) of the asymmetric Bragg SiC(107) peak is shown, in accordance with some embodiments.

[0098] FIG. 61C is a contour plot of a reciprocal space map of a 3-Ga₂O₃ (−201) oriented film on n+4H-SiC(0001)_0 deg miscut, where an RSM of the asymmetric Bragg β-Ga₂O₃ (−403) peak is shown, in accordance with some embodiments.

[0099] FIG. 62 is a top view of the atomic arrangement of an alpha-phase Ga₂O₃ crystal showing the (0001) plane and the hexagonal oxygen sublattice, in accordance with some embodiments.

[0100] FIG. 63 is a top view of the atomic arrangement of a wurtzitic-phase Ga₂O₃ crystal showing the (0001) plane and the hexagonal oxygen sublattice, in accordance with some embodiments.

[0101] FIG. 64 is a schematic representation of a bulk region and surface region of an oxide epilayer with hydrogenic impurities disposed within the structure, in accordance with some embodiments.

[0102] FIG. 65 is a plot of the spatial energy band diagram of a Ga₂O₃ multilayered structure showing surface depletion and accumulation effects at the surface, in accordance with some embodiments.

[0103] FIGS. 66A and 66B are plots of the forward bias I-V curves calculated for a Schottky barrier diode comprising Ni/β-Ga₂O₃/4H-SiC with three cases of surface modification of the β-Ga₂O₃ semiconductor and the effect on the turn-on voltage of the diode, in accordance with some embodiments.

[0104] FIG. 67 is a plot of the forward bias I-V curves calculated for a series of Schottky barrier diodes comprising [high work function metals/β-Ga₂O₃/4H-SiC] with two cases of surface modification (clean and depleted) of the β-Ga₂O₃ semiconductor and the effect on the turn-on voltage of the diode, in accordance with some embodiments.

[0105] FIG. 68A is an example process flow sequence for the fabrication of an oxide semiconductor region followed by a surface modification which is then capped advantageously by a high work function metal, in accordance with some embodiments.

[0106] FIG. 68B is an example process flow sequence for the fabrication of an in-situ deposition of a first metal layer on a modified surface of an epitaxial oxide layer, followed by further processing steps, in accordance with some embodiments.

[0107] FIG. 69 shows possible atomic arrangements in the vicinity of a metal-oxide heterojunction, where a surface portion of a single crystal β-Ga₂O₃ region with an epitaxial coordination of an epitaxial metal region is shown with an interfacial oxygen termination layer, in accordance with some embodiments.

[0108] FIG. 70A depicts schematically a multilayered epitaxial structure comprising a SiC substrate, an epitaxial semiconducting oxide layer, and an epitaxial metal layer, in accordance with some embodiments.

[0109] FIGS. 70B-70E show RHEED images of a high symmetry surface diffractogram of the structure of FIG. 70A, in accordance with some embodiments.

[0110] FIGS. 70F-70G respectively show a wide angle x-ray diffraction plot and a high-resolution x-ray diffraction plot for the completed multilayered structure of FIG. 70A, in accordance with some embodiments.

[0111] FIG. 71A plots the variation of bandgap energy versus composition for binary and ternary compositions of the form (Ga₂O₃)_x(ZnO)_{1-x} and (Al₂O₃)_x(ZnO)_{1-x} which are

suitable for integration with the vertical multilayered semiconductor devices disclosed herein, in accordance with some embodiments.

[0112] FIG. 71B plots the variation of bandgap energy versus composition for binary and ternary compositions of the form $(\text{Ga}_2\text{O}_3)_x(\text{MgO})_{1-x}$ and $(\text{Al}_2\text{O}_3)_x(\text{MgO})_{1-x}$ which are suitable for integration with the vertical multilayered semiconductor devices disclosed herein, in accordance with some embodiments.

[0113] FIG. 72 plots the variation of bandgap energy versus composition for ternary and quaternary compositions of the form $(\text{Zn}_x\text{MgO}_{1-x})\text{Ga}_2\text{O}_4$ and $(\text{Zn}_x\text{MgO}_{1-x})\text{Al}_2\text{O}_4$ which are suitable for integration with the vertical multilayered semiconductor devices disclosed herein, in accordance with some embodiments.

[0114] FIG. 73 plots the variation of bandgap energy versus composition for ternary and quaternary compositions of the form $(\text{Al}_x\text{GaO}_{1-x})_2\text{ZnO}_4$ and $(\text{Al}_x\text{GaO}_{1-x})_2\text{MgO}_4$ which are suitable for integration with the vertical multilayered semiconductor devices disclosed herein, in accordance with some embodiments.

[0115] FIGS. 74 and 75 are spatial energy band diagrams for a [metal/n+Ga₂O₃/4H-SiC/metal] multilayered structure showing the effect of donor doping concentration within the 4H-SiC region, in accordance with some embodiments.

[0116] FIG. 76 is cross-section of a portion of a device comprising the region of an epitaxial oxide, Schottky metal contact and a dielectric isolation region, in accordance with some embodiments.

[0117] FIG. 77 is a plot of the spatial energy band diagram of the heterojunction formed between the oxide semiconductor drift and the dissimilar oxide or material region forming the isolation region, in accordance with some embodiments.

[0118] FIG. 78A is a table of calculated dielectric constants for a series of example oxide and fluoride materials that are compatible with the vertical multilayered device structures disclosed herein, in accordance with some embodiments.

[0119] FIGS. 78B and 78C are models for electrostatic interaction between equal and opposite charges positioned separately within two different dielectric constant materials, in accordance with some embodiments. FIG. 78B shows the symmetric equipotential lines distributed between the +Q and -Q charges for the case of equal dielectric constant material slabs. FIG. 78C shows the asymmetric equipotential lines distributed between the +Q and -Q charges for the case of dissimilar dielectric constant material slabs.

[0120] FIGS. 78D and 78E are models for electrostatic interaction between equal and opposite charges positioned separately within two materials, in accordance with some embodiments. FIG. 78D shows the symmetric distribution of the magnitude of the electric-field as lines distributed between the +Q and -Q charges for the case of equal dielectric constant material slabs. FIG. 78E shows the asymmetric distribution of the magnitude of the electric-field as lines distributed between the +Q and -Q charges for the case of dissimilar dielectric constant material slabs. The concept of dielectric screening is demonstrated for the present disclosure utilizing UWBG materials with dissimilar dielectric constants.

[0121] FIGS. 78F and 78G are plots of the parallel E_y - and perpendicular E_x -electric field components in the vicinity of

the heterointerface for the electrostatic model structures in FIGS. 78D and 78E, respectively, in accordance with some embodiments.

[0122] FIGS. 79 and 80 are two-dimensional electrostatic models of an example multilayered semiconductor diode structure comprising a lower electrode, a substrate, a drift layer and a field-plate electrode embedded within an isolation region dielectric, in accordance with some embodiments. The field-plate electrode is drawn asymmetrically about the vertical center line of the structure to simulate simultaneously a structure without a field-plate on the left-hand side, and a structure with a field-plate on the right-hand side. FIG. 79 represents a field-plate separated from the drift region with a thick isolation layer dielectric. FIG. 80 represents a field-plate separated from the drift region with a thin isolation layer dielectric.

[0123] FIGS. 81 and 82 are electrostatic simulations of a split field-plate design having a thin dielectric spacing and the resulting spatial variation of the electric-field strength, where the two cases of a low-k dielectric (FIG. 81) and high-k dielectric (FIG. 82) isolation layer are compared, in accordance with some embodiments.

[0124] FIGS. 83 and 84 are electrostatic simulations of a split field-plate design having a thick dielectric spacing and the resulting spatial variation of the electric-field strength in the vicinity of the interface between the drift region and isolation region, where the two cases of a low-k dielectric (FIG. 83) and high-k dielectric (FIG. 84) isolation layer are compared, in accordance with some embodiments.

[0125] FIGS. 85 and 86 are electrostatic simulations of a split field-plate design having a thin dielectric spacing and the resulting spatial variation of the electric-field strength in the vicinity of the interface between the drift region and isolation region, where the two cases of a low-k dielectric (FIG. 85) and high-k dielectric (FIG. 86) isolation layer are compared, in accordance with some embodiments.

[0126] FIGS. 87 and 88 are plots of the magnitude of the electric field generated within the structures at the heterointerface between the drift layer and isolation layer and laterally extending along the x-direction. The device structure has a thin field-plate/isolation layer/drift layer spacing, in accordance with some embodiments. FIG. 87 is for the case of a low-k dielectric and FIG. 88 is for a high-k dielectric isolation layer.

[0127] FIGS. 89 and 90 are plots of the spatial variation of the potential energy generated within the structure for three horizontal planes, namely, at the heterointerface, slightly above and below the heterointerface between the drift layer and isolation layer. The device structure is a thin field-plate/isolation layer/drift layer spacing, in accordance with some embodiments. FIG. 89 is for the case of a low-k dielectric and FIG. 90 is for a high-k dielectric isolation layer.

[0128] FIG. 91 is a plot of simulations investigating the effect of the isolation layer dielectric constant relative to the drift region dielectric constant applied to the case of a thin field-plate separation, in accordance with some embodiments. The top electrode structure comprises a right-hand side (rhs) with a field-plate, and the left-hand side (lhs) has no field-plate. The maximum electric field at the heterointerface between the drift layer and the top electrode (or isolation layer) is calculated for the rhs and lhs portions of the electrode as a function of the isolation layer dielectric constant.

[0129] FIG. 92 is a plot resulting from the calculations in FIG. 91, showing the ratio γ of maximum electric field for the field-plate relative to no field-plate, in accordance with some embodiments.

[0130] FIG. 93 shows the two-dimensional electrostatic model used for simulating the effect of the isolation layer dielectric constant for the case of a simple top metal mesa contact, in accordance with some embodiments.

[0131] FIG. 94 is plot of the maximum surface electric-field at the heterointerface between the isolation layer (or top metal) and the drift layer surface as a function of the isolation layer dielectric constant, where the high-k dielectric isolation layer electric field screening effect is demonstrated, in accordance with some embodiments.

[0132] FIG. 95 is a photograph of a fabricated wafer comprising a plurality of vertical device structures showing a Ni Schottky barrier metal pattern utilized for an example design, in accordance with some embodiments. The rear contact was not processed at this stage, showing the transparent nature of the materials.

[0133] FIG. 96A shows an alternative device design referred to as a “T-Channel” design herein, in accordance with some embodiments.

[0134] FIGS. 96B-96H show modeled (or calculated) spatial electric fields within devices having structures like that shown in FIG. 96A, in accordance with some embodiments.

[0135] FIG. 97 shows an example device configuration, wherein the device configuration shown in FIG. 96A is repeated to form an array of devices on a common SiC substrate, in accordance with some embodiments.

[0136] FIGS. 98A, 98B and 98C show examples of devices with a “T-Channel” design as described with respect to FIG. 96A, and additionally with a transition layer, an intermediate layer, or both, in accordance with some embodiments.

[0137] FIG. 99 is an absolute energy scale comparison of the conduction and valence edge energy alignments for: N-polar and Ga-polar GaN surfaces; Ga-polar and O-polar Pna21 Ga_2O_3 surfaces; the monoclinic $\beta\text{-Ga}_2\text{O}_3$ (-201) surface; and the SI-polar and C-polar 4H-SiC surfaces, in accordance with some embodiments.

[0138] FIGS. 100A-100D show possible combinations of mixed polarity multilayered semiconductor structures comprising $[\text{GaN}/\text{Ga}_2\text{O}_3/4\text{H-SiC}]$, in accordance with some embodiments.

[0139] FIGS. 101A-101F show examples of device structures including a vertical multilayered semiconducting structures with epitaxial oxide and epitaxial nitride layers, in accordance with some embodiments.

[0140] FIGS. 102A-102C show tables depicting combinations of n-type and p-type layers with Ohmic contacts to a SiC substrate, and Ohmic or Schottky contacts to an epitaxially grown semiconductor layer, in accordance with some embodiments.

[0141] FIG. 103 is an example fabrication process flow for the creation of an epitaxial Ga_2O_3 layer formed on a SiC substrate, followed by a process to create a GaN template layer and optional GaN epilayer growth, in accordance with some embodiments. The structure is then processed to form the Schottky metal electrode(s), in accordance with some embodiments.

[0142] FIG. 104 is a plot of the double crystal x-ray diffraction showing the symmetric Q-20 scan for an epitaxial multilayered structure forming Ga-polar wurtzite GaN on

the final surface of a $\beta\text{-Ga}_2\text{O}_3$ (-201) epilayer, formed on SiC or Sapphire C-plane substrate, in accordance with some embodiments.

[0143] FIGS. 105A and 105B are plots of the spatial energy band diagram for a vertical multilayer semiconductor device comprising a: $[\text{Ni}/\text{Ga-polar GaN}/\beta\text{-Ga}_2\text{O}_3(-201) \text{N}_{\text{drift}}/\text{n+4H-SiC}]$, wherein polarization induced band bending is due to the thickness of the Ga-polar GaN top layer and is simulated for three GaN layer thicknesses, in accordance with some embodiments.

[0144] FIG. 105C is a plot of the spatial valence band energy profile in the vicinity of the $[\text{Ni}/\text{Ga-polar GaN}/\beta\text{-Ga}_2\text{O}_3]$ region showing the lowest 8 energy levels quantum confined heavy-hole wavefunction states due to the GaN at the heterointerface, in accordance with some embodiments.

[0145] FIG. 105D is a plot of a series of simulations of the spatial conduction band energy in the vicinity of the $[\text{Ni}/\text{Ga-polar GaN}/\beta\text{-Ga}_2\text{O}_3]$ region showing the variation in electron barrier height as a function of the GaN layer thickness, in accordance with some embodiments.

[0146] FIG. 105E is a plot of the calculated maximum conduction energy barrier height as a function of the GaN layer thickness for the device structure in FIGS. 55-58, in accordance with some embodiments.

[0147] FIGS. 106A and 106B are schematic diagrams of a cross-section of a vertical multilayer semiconductor stack. FIG. 106A shows a topmost portion on an oxide layer exhibiting surface roughness. FIG. 106B shows a topmost portion on an oxide layer that has been planarized after epitaxial growth to exhibit a smaller surface roughness, in accordance with some embodiments.

[0148] In the following description, like reference characters designate like or corresponding parts throughout the figures.

DETAILED DESCRIPTION

[0149] This disclosure describes semiconductor diode devices including at least one type of epitaxial oxide material deposited upon a substrate including silicon carbide (SiC), where the epitaxial oxide material forms an epitaxial oxide layer (or drift layer), and a metal layer (e.g., an epitaxial metal layer) on the epitaxial oxide layer. In some embodiments, the materials of the metal layer and the epitaxial oxide layer are chosen to form a Schottky junction. The epitaxial oxide material advantageously comprises a wide bandgap (e.g., from about 2.5 eV to about 10 eV, or from about 3 eV to about 10 eV, or from about 4.5 eV to about 7 eV).

[0150] In some embodiments, the semiconductor devices described herein include layers in addition to the epitaxial oxide layer (or drift layer) on the SiC substrate. For example, an epitaxial transition layer can be formed between the SiC substrate and the epitaxial oxide layer. In some cases, the transition layer can provide an intermediate level of doping between the substrate and epitaxial oxide layer and/or manage interfacial strain, and can thereby impact the electronic and structural properties of the semiconductor diode device, as described further herein. In another example, an intermediate layer can be formed between the epitaxial oxide layer and the metal layer. The intermediate layer can alter the Schottky barrier properties and can thereby impact the electronic properties of the semiconductor diode device such as the turn-on voltage and/or the reverse breakdown voltage, as described further herein. In some cases, a structure can

include both a transition layer and an intermediate layer. The intermediate layer and the transition layer can each be a single layer, or can include multiple layers of materials such as in a superlattice layer or a chirp layer.

[0151] Furthermore, in some cases, the semiconductor diode devices described herein can include an isolation layer surrounding the metal layer. The isolation layer can be advantageous to mitigate surface currents and manage the electrostatic vector fields of the devices described herein. The isolation layer can include a material with a higher bandgap than the materials of the epitaxial oxide layer (or drift layer), and can be crystalline (e.g., epitaxial materials) or can be amorphous, or polycrystalline materials. The dielectric constant of the materials of the isolation layer can be lower or higher than that of the epitaxial oxide layer (or drift layer). In some cases, the isolation layer is an oxide. In some cases, the isolation layer is an epitaxial layer formed on the epitaxial oxide layer (or drift layer).

[0152] The materials, structures and methods described herein can be used in two-terminal, three-terminal, or other multi-terminal semiconductor devices, such as Schottky barrier (SB) diodes, p/n junction diodes, p-i-n diodes, transistors, switches, high-power switches, RF-switches, varactors, light emitting devices, semiconductor devices with quantum wells or superlattices, semiconductor devices with compositional (or doping density) gradients, and others.

[0153] The semiconductor devices described herein (e.g., vertical semiconductor diode devices) enable high dielectric breakdown voltages in the off-state or blocking state (i.e., high breakdown voltages may be greater than 50 V, greater than 100 V, or from 50 V to 10,000 V). Fundamentally, electrical power P is the product of voltage (V) and current (I). It follows that for a given operating power, a higher voltage can be utilized with a commensurate decrease in current. Generally, current scales as the cross-sectional area of the device or conductor, therefore higher operating voltages enables smaller devices with lower ohmic heating losses ($I^2 R_{on}$). Reducing the ON-resistance (R_{on}) further reduces losses.

[0154] Increasing stringent requirements for high power semiconductor devices operating at greater than 1.3 kV address improvements in the simultaneous metric of Size-Weight and Power (SWaP). Compact high power density SWaP power conversion configurations present a problem for conventional technologies. Semiconductor devices comprising Ga_2O_3 have been proposed, but this material suffers from poor thermal conductivity and the resultant generation of heat often must be dissipated using complicated thermo-mechanical arrangements. On the other hand, currently available all-SiC devices offer lower breakdown voltages compared with that of comparable devices comprising Ga_2O_3 for the same thickness and doping concentration (or conductivity). The terms “doping concentration” and “doping density” are used interchangeably herein.

[0155] In the present disclosure, a solution to the above problem and avenue toward dramatic improvements includes a hybrid approach involving a silicon carbide (SiC) substrate which has good thermal and electrical conductivity in combination with layers of a semiconducting device formed of epitaxially deposited oxide materials (e.g., Ga_2O_3) which have an increased breakdown voltage, V_{br} ,

for a given vertical thickness and doping concentration (or conductivity) as compared to devices with SiC epitaxial layers on SiC substrates. A further advantage of coupling an oxide semiconducting active region to SiC is the improved flexibility of device configurations possible (i.e., heterojunction engineering).

[0156] The structures and methods described herein can also potentially enable reduced manufacturing cost.

[0157] In some examples of semiconductor diode devices including an epitaxial oxide layer on a silicon carbide (SiC) substrate, the SiC substrate can be made from crystal polytypes of 4H-SiC, 2H-SiC, 6H-SiC, or 3C-SiC. The orientation of the SiC substrate can be C-plane SiC (0001), A-plane SiC (11-20), R-plane SiC (10-12) or M-plane SiC (1-100). Single crystal SiC has a thermal conductivity of about 300-500 W/(m-K), which is significantly higher than other semiconductor substrate materials such as Ga_2O_3 (about 20-30 W/(m-K)) or Si (about 100-150 W/(m-K)). The high thermal conductivity of SiC is advantageous for high-power electronic device structures epitaxially formed on SiC substrates, since it improves heat dissipation through the substrate.

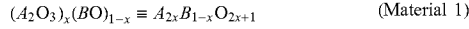
[0158] In some examples of semiconductor diode devices on a silicon carbide (SiC) substrate, the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer can include one or more of: Ga_2O_3 (e.g., α -phase or β -phase); $(Al_xGa_{1-x})_2O_3$ (e.g., α -phase or β -phase) where $0 \leq x \leq 1$; $Mg_xGa_{2(1-x)}O_{3-2x}$ where $0 \leq x \leq 1$; $Mg_xAl_{2(1-x)}O_{3-2x}$ where $0 \leq x \leq 1$; $(Mg_xZn_{1-x})_2(AlyGa_{1-y})_{2(1-z)}O_{3-2z}$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $Zn_xGa_yO_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $Zn_xAl_yO_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $Mg_xGa_yO_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $Mg_xAl_yO_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $Ni_xGa_yO_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $Ni_xAl_yO_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $(Ni_xMg_{1-x})_yGa_{2(1-y)}O_{3-2y}$ where $0 \leq x \leq 1$ and $0 \leq y \leq 1$; or $(Mg_xNi_{1-x})_y(AlyGa_{1-y})_{2(1-z)}O_{3-2z}$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$, or $(Ni_xMg_{1-x})_y(AlyGa_{1-y})_{2(1-z)}O_4$ where $0 \leq (x, y, z) \leq 1$, or $(Zn_pMg_{1-p})_z(AlyGa_{1-y})_{2(1-z)}O_{3-2z}$ where $0 \leq (p, x, y, z) \leq 1$.

[0159] The oxygen content of the epitaxial oxide materials described herein (e.g., the aforementioned epitaxial oxide materials) can vary, in some embodiments. For example, different crystal symmetries of the aforementioned epitaxial oxide materials can have compositions with oxygen contents that vary compared to the compositions listed (e.g., have smaller or larger oxygen fractions). In some cases, the ratio of metal to oxygen atoms can be smaller or larger than those in the compounds described herein due to defects (e.g., oxygen or metal vacancies and/or interstitial atoms) in the materials.

[0160] The epitaxial oxide materials described herein (e.g., the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer) can be doped p-type or n-type. For example, an epitaxial oxide such as $(Al_xGa_{1-x})_2O_3$ can be doped n-type using an extrinsic dopant such as Si, Ge, or Sn. In various examples, dopants may be spatially confined (e.g., to one or more doped layers within a given region). In one example, a dopant may be confined to two or more spatially separated doped layers in a region (e.g., two spatially separated doped layers in the transition layer) or in other examples the doped layer may span the region or a portion of the region.

[0161] In some embodiments, any of the epitaxial layers of the structures and devices described herein (e.g., the

epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer) can include:



[0162] In Material 1, $0 \leq x \leq 1$, $0 < x \leq 1$, $0 \leq x < 1$, or $0 < x < 1$. These ranges are distinct from each other, for example, since some embodiments of Material 1 (as described by the formula above) include binary materials, and others include binary or ternary materials. In other cases herein, the subscripts of formulas for materials can include or exclude zero and/or one to distinguish between binary, ternary, quaternary, or other materials. The atomic species of Material 1 can be selected from: trivalent $A = \{\text{Al, Ga, RE, Bi, B', In}\}$; and bivalent $B = \{\text{Ni, Mg, Zn}\}$, where O=Oxygen, Al=Aluminum, Ga=Gallium, RE=at least one Rare-earth (e.g., Lanthanide), Bi=Bismuth, B'=Boron, In=Indium, Zn=Zinc, Mg=Magnesium, and Ni=Nickel.

[0163] For example, when $x = 1/2$ in the above formula for Material 1, an epitaxial layer of a structure or device described herein can include $A_2B_1O_4$. Some examples of such materials are $Al_2Mg_1O_4$, $Ga_2Mg_1O_4$, $Al_2Zn_1O_4$, $Ga_2Zn_1O_4$, $Al_2Ni_1O_4$, $Ga_2Ni_1O_4$, and $RE_2Zn_1O_4$.

[0164] In another example, when $x = 1/3$ in the above formula for Material 1, an epitaxial layer of a structure or device described herein can include $A_2B_2O_5$. Some examples of such materials are $Al_2Mg_2O_5$, $Ga_2Mg_2O_5$, $Al_2Zn_2O_5$, $Ga_2Zn_2O_5$, $Al_2Ni_2O_5$, and $Ga_2Ni_2O_5$.

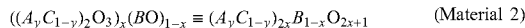
[0165] In another example, when $x = 1/4$ in the above formula for Material 1, an epitaxial layer of a structure or device described herein can include $A_2B_3O_6$. Some examples of such materials are $Al_2Mg_3O_6$, $Ga_2Mg_3O_6$, $Al_2Zn_3O_6$, $Ga_2Zn_3O_6$, $Al_2Ni_3O_6$, and $Ga_2Ni_3O_6$.

[0166] In another example, when $x = 1/5$ in the above formula for Material 1, an epitaxial layer of a structure or device described herein can include $A_2B_4O_7$. Some examples of such materials are $Al_2Mg_4O_7$, $Ga_2Mg_4O_7$, $Al_2Zn_4O_7$, $Ga_2Zn_4O_7$, $Al_2Ni_4O_7$, and $Ga_2Ni_4O_7$.

[0167] In another example, when $x = 3/4$ in the above formula, an epitaxial layer of a structure or device described herein can include $A_6B_1O_{10}$. Some examples of such materials are $Al_6Mg_1O_{10}$, $Ga_6Mg_1O_{10}$, $Al_6Zn_1O_{10}$, $Ga_6Zn_1O_{10}$, $Al_6Ni_1O_{10}$, $Ga_6Ni_1O_{10}$, and $RE_6Zn_1O_{10}$.

[0168] In other examples, x in the above formula for Material 1 can be any of the values $0 \leq x \leq 1$, $0 < x \leq 1$, $0 \leq x < 1$, and $0 < x < 1$. For example, more dilute composition cases are also possible, such as when $0 \leq x \leq 0.1$, or $0.9 \leq x \leq 1.0$.

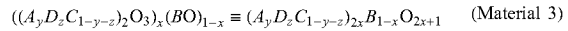
[0169] In some embodiments, any of the epitaxial layers of the structures and devices described herein (e.g., the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer) can include:



[0170] In Material 2, $0 \leq x \leq 1$, $0 < x \leq 1$, $0 \leq x < 1$, or $0 < x < 1$; and $0 \leq y \leq 1$, $0 < y \leq 1$, $0 \leq y < 1$, or $0 < y < 1$. The atomic species of Material 2 can be selected from: trivalent A and C selected from two of $\{\text{Al, Ga, RE, Bi, B', In}\}$; and bivalent $B = \{\text{Ni, Mg, Zn}\}$, where O=Oxygen.

[0171] For example, when $x = 1/2$ and $y = 1/10$ in the above formula for Material 2, an epitaxial layer of a structure or device described herein can include $(A_{0.1}C_{0.9})_2B_1O_4$. Some examples of such materials are $(Al_{0.1}Ga_{0.9})_2Mg_1O_4$, $(Al_{0.1}Ga_{0.9})_2Ni_1O_4$, $(Bi_{0.1}Ga_{0.9})_2Mg_1O_4$, $(RE_{0.1}Ga_{0.9})_2Zn_1O_4$, and $(RE_{0.1}Al_{0.9})_2Zn_1O_4$.

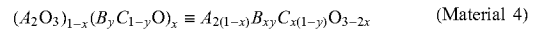
[0172] In some embodiments, any of the epitaxial layers of the structures and devices described herein (e.g., the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer) can include:



[0173] In Material 3, $0 \leq x \leq 1$, $0 < x \leq 1$, $0 \leq x < 1$, or $0 < x < 1$; $0 \leq y \leq 1$, $0 < y \leq 1$, $0 \leq y < 1$, or $0 < y < 1$; and $0 \leq z \leq 1$, $0 < z \leq 1$, $0 \leq z < 1$, or $0 < z < 1$. The above formula for Material 3 is also constrained such that the sum $y+z \leq 1$. The atomic species of Material 3 can be selected from: trivalent A, D and C selected from three of $\{\text{Al, Ga, RE, Bi, B', In}\}$; and bivalent $B = \{\text{Ni, Mg, Zn}\}$, where O=Oxygen.

[0174] For example, when $x = 1/2$, $y = 18/20$, and $z = 1/20$ in the above formula for Material 3, an epitaxial layer of a structure or device described herein can include $(A_{0.9}D_{0.05}C_{0.05})_2B_1O_4$. Some examples of such materials are $(Ga_{0.9}Al_{0.05}In_{0.05})_2Mg_1O_4$, $(Ga_{0.9}Al_{0.05}In_{0.05})_2Ni_1O_4$, $(B'_{0.05}Bi_{0.05}Ga_{0.9})_2Mg_1O_4$, $(RE_{0.05}In_{0.05}Ga_{0.9})_2Zn_1O_4$, and $(RE_{0.05}Ga_{0.9}Al_{0.05})_2Zn_1O_4$.

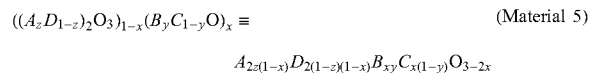
[0175] In some embodiments, any of the epitaxial layers of the structures and devices described herein (e.g., the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer) can include:



[0176] In Material 4, $0 \leq x \leq 1$, $0 < x \leq 1$, $0 \leq x < 1$ or $0 < x < 1$; and $0 \leq y \leq 1$, $0 < y \leq 1$, $0 \leq y < 1$, or $0 < y < 1$. The atomic species of Material 4 can be selected from: trivalent $A = \{\text{Al, Ga, RE, Bi, B', In}\}$; and bivalent B and C selected from at least two of $\{\text{Ni, Mg, Zn}\}$, where O=Oxygen.

[0177] For example, when $x = 1/2$ and $y = 1/4$ in the above formula for Material 4, an epitaxial layer of a structure or device described herein can include $A_2B_{0.25}C_{0.75}O_4$. Some examples of such materials are $Ga_2Mg_{0.25}Zn_{0.75}O_4$, $Al_2Mg_{0.25}Zn_{0.75}O_4$, $Ga_2Mg_{0.25}Ni_{0.75}O_4$, and $Ga_2Zn_{0.25}Mg_{0.75}O_4$.

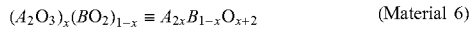
[0178] In some embodiments, any of the epitaxial layers of the structures and devices described herein (e.g., the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer) can include:



[0179] In Material 5, $0 \leq x \leq 1$, $0 < x \leq 1$, $0 \leq x < 1$, or $0 < x < 1$; $0 \leq y \leq 1$, $0 < y \leq 1$, $0 \leq y < 1$, or $0 < y < 1$; and $0 \leq z \leq 1$, $0 < z \leq 1$, $0 \leq z < 1$, or $0 < z < 1$. The atomic species of Material 5 can be selected from: trivalent A or D selected from at least two of $\{\text{Al, Ga, RE, Bi, B', In}\}$; and bivalent B and C selected from at least two of $\{\text{Ni, Mg, Zn}\}$, where O=Oxygen.

[0180] For example, when $x=1/2$, $y=1/2$, and $z=1/2$ in the above formula for Material 5, an epitaxial layer of a structure or device described herein can include $A_1D_1B_{0.5}C_{0.5}O_4$. Some examples of such materials are $Al_1Ga_1Ni_{0.5}Mg_{0.5}O_4$, $In_1Ga_1Zn_{0.5}Mg_{0.5}O_4$, and $RE_1Al_1Zn_{0.5}Ni_{0.5}O_4$.

[0181] In some embodiments, any of the epitaxial layers of the structures and devices described herein (e.g., the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer) can include:

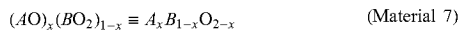


[0182] In Material 6, $0 \leq x \leq 1$, $0 < x \leq 1$, $0 \leq x < 1$, or $0 < x < 1$. The atomic species of Material 6 can be selected from: trivalent $A=\{Al, Ga, RE, Bi, B', In\}$; and bivalent $B=\{Ge, Si, Sn\}$, where $O=Oxygen$, $Al=Aluminum$, $Ga=Gallium$, $RE=at$ least one Rare-earth (e.g., Lanthanide), $Bi=Bismuth$, $B'=Boron$, $Ge=Germanium$, $Si=Silicon$, and $Sn=Tin$.

[0183] For example, when $x=1/2$ in the above formula for Material 6, an epitaxial layer of a structure or device described herein can include $A_2B_1O_5$. Some examples of such materials are $Al_2Si_1O_5$, $Ga_2Ge_1O_5$, $Al_2Ge_1O_5$, $RE_2Sn_1O_5$, $B'_2Ge_1O_5$, and $In_2Ge_1O_5$.

[0184] For example, when $x=3/4$ in the above formula for Material 6, an epitaxial layer of a structure or device described herein can include $A_6B_1O_{11}$. Some examples of such materials are $Al_6Ge_1O_{11}$, $RE_6Ge_1O_{11}$, and $B'_6Si_1O_{11}$.

[0185] In some embodiments, any of the epitaxial layers of the structures and devices described herein (e.g., the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer) can include:



[0186] In Material 7, $0 \leq x \leq 1$, $0 < x \leq 1$, $0 \leq x < 1$, or $0 < x < 1$. The atomic species of Material 7 can be selected from: trivalent $A=\{Mg, Ni, Zn\}$; and bivalent $B=\{Ge, Si, Sn\}$, where $O=Oxygen$, $Mg=Magnesium$, $Ni=Nickel$, $Zn=Zinc$, $Ge=Germanium$, $Si=Silicon$, and $Sn=Tin$.

[0187] For example, when $x=1/2$ in the above formula for Material 7, an epitaxial layer of a structure or device described herein can include $A_1B_1O_3$. Some examples of such materials are $Zn_1Si_1O_3$, $Zn_1Ge_1O_3$, $Ni_1Ge_1O_3$, $Mg_1Sn_1O_3$, $Mg_1Zn_1O_3$, and $Ni_1Sn_1O_3$.

[0188] For example, when $x=2/3$ in the above formula for Material 7, an epitaxial layer of a structure or device described herein can include $A_2B_1O_4$. Some examples of such materials are $Zn_2Si_1O_4$, $Mg_2Ge_1O_4$, $Ni_2Ge_1O_4$, $Mg_2Sn_1O_4$, $Mg_2Si_1O_4$, and $Ni_2Si_1O_4$.

[0189] It is to be understood that, for any of the epitaxial layers of the structures and devices described herein (e.g., the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer), the composition fractions (e.g., x , y and/or z) of any of the formulas for materials described herein (e.g., Materials 1-7 described above) may vary within the layer from the exact amount. For example, the composition fraction may vary due to crystalline lattice defects (e.g., vacancies, antisite defects, and/or interstitial species) and random fluctuations within physical compositions. Therefore, the composition fractions of the

formulas for materials described herein can be considered average compositions of the layer in some cases.

[0190] It is further to be understood that any of the epitaxial layers of the structures and devices described herein (e.g., the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer) can include stoichiometric compositions of the materials described herein (e.g., Materials 1-7 above), such as, Ga_2O_3 , or non-stoichiometric compositions such as, $Ga_{2-p}O_3$, where $0 < p < 2$ represents an oxygen-deficient or gallium-rich composition. Another example of a non-stoichiometric composition is $Ga_{2-q}O_3$, where $0 < q < 1$ represents compositions that are oxygen-rich and/or gallium-deficient. Any of the oxide compositions described herein (e.g., Materials 1-7) can be stoichiometric or non-stoichiometric in the present structures and devices.

[0191] Additionally, any of the epitaxial layers of the structures and devices described herein (e.g., the epitaxial oxide layer (or drift layer), the transition layer, the intermediate layer, and/or the isolation layer) can include a dilute alloy composition of any of the oxide compositions described herein (e.g., Materials 1-7). A dilute alloy composition is one in which a dilute component of the material is present in a concentration of less than 0.1. For example, one of Materials 1-7 can include a value of x , y and/or z that is less than 0.1, or less than 0.05, or less than 0.01. In some cases, a dopant species (e.g., Si , Ge or Sn) can be incorporated into a material described herein (e.g., into one of Materials 1-7 formed in an epitaxial layer) with a composition less than 0.1, or less than 0.05, or less than 0.01.

[0192] The epitaxial oxide layer can also have a crystal symmetry and orientation to enable epitaxial layer formation on the SiC substrate. In some examples, the epitaxial oxide material is chosen to have a lattice mismatch that is less than 10%, or less than 8%, or less than 6%, or less than 5%, or less than 2% with the SiC substrate. The thickness of the epitaxial oxide layer can be less than a critical layer thickness of the material (below which the epitaxial oxide layer may be elastically strained and is substantially coherent with the substrate, and above which the epitaxial oxide layer is relaxed or partially relaxed). For example, the thickness of the epitaxial oxide layer can be from 1 nm to 100 microns, or from 100 nm to 10 microns, or from 1 nm to 1 micron, or from 5 nm to 500 nm.

[0193] In some examples, the semiconductor diode devices disclosed herein comprise a short-period superlattice (SPSL). The epitaxial oxide layer can comprise an SPSL, and/or other layers of the device (e.g., a transition layer, or intermediate layer) can comprise an SPSL. In some cases, the SPSL contains alternating sub-layers of wider and narrower bandgap materials (barrier layers and well layers, respectively). The thickness of the barrier or well sub-layers in an SPSL can be from less than 1 monolayer (ML) to 5 MLs, or from less than 1 ML to 10 MLs, or from less than 1 ML to 20 MLs. Barrier and/or well sub-layers with a thickness of less than 1 ML or 1 unit cell along a growth direction may be discontinuous laterally (i.e., in a direction parallel with a major surface of the layer).

[0194] In some examples, the semiconductor diode devices disclosed herein comprise a chirp layer. The epitaxial oxide layer can comprise a chirp layer, and/or other layers of the device (e.g., a transition layer, or intermediate layer) can comprise a chirp layer. A chirp layer is similar to an SPSL in that it also contains alternating sub-layers of

wider and narrower bandgap materials (barrier layers and well layers, respectively), however, chirp layers have sub-layer properties that change throughout the chirp layer. For example, a chirp layer can contain alternating sub-layers of $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ with two different compositions (i.e., two different values of x) to form barrier layers and well sub-layers, where the barrier sub-layers and/or the well sub-layers change thickness through the chirp layer. In another example, a chirp layer can contain alternating sub-layers of $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ to form barrier sub-layers and well sub-layers, where the barrier sub-layers and/or the well sub-layers change composition through the chirp layer. In another example, a chirp layer can contain alternating sub-layers of $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ to form barrier sub-layers and well sub-layers, where the barrier sub-layers and/or the well sub-layers change thickness and/or composition through the chirp layer.

[0195] In yet another example, a bulk-like graded bandgap may be incorporated within the device that has a bandgap energy $E_g(z)$ varying along a growth direction, z . The variation can be linear or non-linear, monotonic or non-monotonic, along a growth direction. For example, a graded layer of $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ may be formed such that the alloy composition x is a function of the growth direction z , such that, $x(z)$.

[0196] A digital alloy formed using a SPSL or a bulk-like alloy is possible, and further variation of the effective material properties can be imparted along a growth direction.

[0197] Methods of forming the semiconductor diode devices disclosed herein, including an epitaxial oxide layer on a silicon carbide (SiC) substrate and a metal layer (such as an epitaxial metal layer) on the epitaxial oxide layer, are also described herein. Such methods can include forming highly crystalline and epitaxial oxide layers on SiC substrates using an epitaxial layer growth technique, such as molecular beam epitaxy (MBE), gas source molecular beam epitaxy (GS-MBE), plasma-source MBE (P-MBE), metal-organic chemical vapor deposition (MOCVD), atomic layer deposition (ALD), vacuum-based laser ablation sources, vapor phase epitaxy (VPE), gas source, sputter source, electron beam evaporation source, and plasma-based deposition methods. Other epitaxial layer growth techniques are also applicable eventuating in a high crystal quality oxide film formed on the surface of a substrate.

[0198] In general embodiments, ex-situ metal layers can be formed (or deposited) on the epitaxial oxide layer. In another embodiment, in-situ metal layers may be directly formed on a pristine final surface of an epitaxial oxide. Yet further, an in-situ metal layer may be formed on an intentionally prepared final epitaxial oxide surface. Yet even further, a metal layer may be formed on an intentionally prepared final epitaxial oxide surface wherein the process does not expose the structure or surface to a contaminating environment. Metal layers, such as elemental Ni, Cu, Al, Ti, Pd, Pt, Ir, Er, Gd, Mo, W, Os and alloys thereof, can be deposited using physical vapor deposition techniques, such as MBE, sputtering, pulsed laser deposition, or thermal evaporation of a metal, to form non-epitaxial metal layers. Alternatively, or additionally, epitaxial metal layers can be formed on the epitaxial oxide layers using an epitaxial layer growth technique (e.g., using MBE, MOCVD, or another epitaxial layer growth technique) to form a substantially crystalline first metal layer in direct contact with the epi-

taxial oxide layer to form a metal-oxide heterojunction. Further metal or conductive layers may be formed on the final surface of the first metal crystalline metal layer forming the metal-oxide heterojunction. In some cases, multiple epitaxial and/or non-epitaxial metal layers can be deposited. For example, epitaxial and/or non-epitaxial metal layers can be deposited in different areas of the device to perform different functions. A metal layer can also contain multiple epitaxial and/or non-epitaxial metal layers in the same region of the device, for example, where a first layer is a sticking layer, and subsequent layers adhere to the sticking layer. In some cases, a metal layer can contain a first epitaxial layer to form a low defect interface with a semiconductor layer below it, and a non-epitaxial metal layer can be deposited on top of the epitaxial metal layer.

[0199] Metal-induced-gap states (MIGS) are known in the prior-art for metal contacts to semiconductor surfaces. Simplistically, the MIGS arise from mismatch of dissimilar crystal structures comprising the metal and oxide semiconductor interface, resulting in disadvantageous bonding arrangements at the metal-oxide heterointerface. Passivation of the aforementioned interfacial dangling bonds may be used to reduce the effect of MIGS on the Schottky barrier lowering effect.

[0200] In some embodiments, the semiconductor devices disclosed herein include direct and connective arrangement of bonds across the metal-oxide heterointerface wherein a substantially epitaxial (i.e., crystalline) first metal layer is formed on an anion- or cation-terminated oxide surface.

[0201] In some cases, an isolation layer can be formed between the epitaxial oxide layer and the metal layer (e.g., an epitaxial metal layer) using an epitaxial layer growth technique (e.g., using MBE, MOCVD, or another epitaxial layer growth technique). In some cases, a passivation layer can be formed on the device. The passivation layer can include, for example, aluminum oxide Al_2O_3 , magnesium oxide MgO , zinc-magnesium-gallium-oxide $(\text{Zn}_x\text{Mg}_{1-x})\text{Ga}_2\text{O}_4$, aluminum-gallium oxide $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$, nickel-zinc-magnesium-gallium-aluminum oxide $(\text{Ni}_x\text{Zn}_y\text{Mg}_{1-x-y}) (\text{Ga}_p\text{Al}_{1-p})\text{O}_q$, magnesium-germanium-oxide $\text{Mg}_x\text{Ge}_y\text{O}_z$ or silicon dioxide SiO_2 , wherein subscripts x , y , z , p , q represent atomic fractions representing various crystalline, polycrystalline or amorphous forms.

[0202] In some cases, one or more layers (e.g., the epitaxial oxide layer, the epitaxial metal layer or metal layer, the isolation layer, and/or the passivation layer) can be patterned to form the devices, such as using photolithography and etching techniques (e.g., using photoresist, plasma etching, and/or metal lift-off processes). In other methods a damascene metal patterning can be utilized, wherein the oxide is used as the trench and subsequently chemical mechanical polished if required.

[0203] FIG. 1 shows a figurative sectional view of a vertical multilayered semiconductor diode device **100** according to an illustrative embodiment.

[0204] In this example, semiconductor device **100** comprises a substrate **110** comprising silicon carbide (SiC) and having a width, w_{sub} , and thickness, t_{sub} . In an example, the substrate **110** is formed from the 4H-SiC polytype of SiC. In another example, the substrate **110** is formed from 2H-SiC or 6H-SiC or 3C-SiC.

[0205] In some cases, substrate **110** is a bulk material, such as single crystal SiC. In some cases, substrate **110** can include an optional bulk substrate layer **112** and an optional

surface layer 114. Substrates 110 including both the bulk substrate layer 112 and the surface layer 114 are referred to as “composite substrates” herein. For example, bulk substrate layer 112 can be polycrystalline SiC, and surface layer 114 can be a single crystal layer of SiC. In such structures, surface layer 114 can be a thin layer (e.g., from 100 nm to 5 microns, or from 100 nm to 1 micron) that is a template for epitaxial growth, and bulk substrate layer 112 can be a material with high thermal conductivity and/or selected electrical conductivity (e.g., electrically conductive or insulating). For example, surface layer 114 can comprise one or more single crystal layers of a 4H-SiC polytype of SiC, or it can be 2H-SiC, 6H-SiC, or 3C-SiC. In other examples, surface layer 114 can include one or more single crystal oxide materials (e.g., Ga_2O_3 , Al_2O_3 , or $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ where $0 \leq x \leq 1$). For example, surface layer 114 can include any of the oxide materials described herein, for example, any of the compositions of Materials 1-7.

[0206] Such a composite substrate can be formed, for example, by removing a layer 114 from a single crystal substrate (e.g., a single crystal SiC substrate, or an oxide substrate) and coupling or bonding it to bulk substrate layer 112 (e.g., a bulk layer of polycrystalline SiC, or another material with a high thermal conductivity). In such cases, a layer of a single crystal substrate (e.g., SiC, Ga_2O_3 or Al_2O_3) can be removed and coupled or bonded to bulk substrate layer 112. In other examples, an epitaxially grown layer (e.g., any of the compositions of Materials 1-7, or epitaxially grown SiC) can be removed from the substrate upon which it was grown and coupled or bonded to bulk substrate layer 112. In some cases, bulk substrate layer 112 can be formed of a material with a high thermal conductivity, and optionally a high electrical conductivity, such as polycrystalline SiC, a metal (e.g., Ni, Al, or Cu) or a ceramic material (e.g., AlN or BN).

[0207] A method for transferring a single crystal surface layer 114 from a donor substrate to the bulk substrate layer 112 can include ion implantation (e.g., using H^+ or He^+ or Si^+ or C ions) to form a weak or defective crystal layer beneath the surface layer 114 (e.g., a surface layer 114 of the donor substrate, or a surface layer 114 that is an epitaxial layer on the donor substrate). The surface through which the ions were implanted can then be bonded to the bulk substrate layer 112. Subsequently, heat and/or pressure can be applied to form an optional bonding layer 116, as described further below. A force can then be applied between the donor substrate and the bulk substrate layer 112, and the structure will separate at the ion implanted layer (if it is the weakest layer), thereby leaving the surface layer 114 bonded to the bulk substrate layer 112 while the majority of the donor wafer is removed. In some cases, the surface of the transferred surface layer 114 can be subsequently smoothed, the composite substrate can be annealed, and/or other processes can be performed to prepare the composite substrate for epitaxial growth.

[0208] In other cases, the bulk substrate layer 112 can be single crystal SiC and surface layer 114 can be a surface region or near surface region that has been modified, or thermally treated, or chemically treated, for example to reconstruct the surface for epitaxial growth, or to convert it to graphene or to an oxide upon which epitaxial oxide layers can be grown. Composite substrates can be advantageous, since the bulk substrate layer 112 can be made from a material with fewer constraints than the surface layer 114. In

order to be used as a template for epitaxial growth of oxide layers, surface layer 114 can be a single crystal material with a compatible lattice constant and crystal structure, and have a sufficiently high materials quality (e.g., have a low concentration of threading dislocations). The bulk substrate layer 112 of a composite substrate does not have to be single crystal, and can therefore be made from an electrically and thermally conductive material onto which the surface layer 114 can be bonded. In this disclosure, substrates for the various embodiments may utilize the composite substrate described in FIG. 1.

[0209] In some cases, a thin bonding layer 116 forms between bulk substrate layer 112 and surface layer 114. For example, bonding layer 116 can include metallic or semi-metallic materials (e.g., metallic or semi-metallic Si, C, or Si_xC_y compounds). Bonding layer 116 can be thermally and electrically conductive, in some cases. For example, a metallic or semi-metallic bonding layer can be formed as a result of surface activation and thermal bonding between the surface layer 114 (e.g., single crystal SiC) and the bulk substrate layer 112 (e.g., polycrystalline SiC).

[0210] In some cases, the bulk substrate layer 112 may be thinned following device fabrication (e.g., by a mechanical process, an etching process, or chemical mechanical polishing) to further reduce the effective electrical and/or thermal resistance of the bulk substrate layer 112. This may be advantageous as the cost of layer 112 can be substantially lower than the cost of the single crystal high quality surface layer 114.

[0211] The vertical direction of the multilayered semiconductor device 100 is defined to be approximately perpendicular to a top surface 111 of substrate 110 as indicated by the arrow with the dimension “z” increasing in the vertical upwards direction. Formed above the SiC substrate 110, and in this example on the top surface 111 of substrate 110, is an epitaxial drift layer 120 comprising a semiconductor oxide material having a layer thickness t_D and itself having a top surface 121. Formed above the epitaxial drift layer 120, and in this case on the top surface 121 of epitaxial drift layer 120, is a metal layer 130. In some embodiments, the material properties of the drift layer 120 and metal layer 130 are selected to form a Schottky potential barrier at an interface region 140 between the metal layer 130 and epitaxial drift layer 120. In other embodiments, the material properties of the metal layer 130 are selected to form an ohmic contact to the semiconductor device 100. Metal layer 130 can be either an epitaxial metal layer, or a metal layer that is not epitaxial with the drift layer below it. In some cases, the metal layer 130 is an epitaxial metal layer that forms a low defect interface with the epitaxial oxide layer 120 (compared to an interface formed between the epitaxial oxide layer 120 and a non-epitaxial metal layer), to form the Schottky barrier junction. A second metal layer 132 forms an ohmic (or approximately ohmic) electrical contact with substrate 110, at a back surface opposite the top surface 121.

[0212] The thicknesses of the layers are not shown to scale in FIG. 1, or in any figures showing a figurative sectional view of a vertical multilayered semiconductor diode device herein. For example, the substrate thickness t_{sub} can be much thicker than the drift layer 120 thickness t_D . In some cases, the substrate thickness t_{sub} can be from 1 mm to 3 mm, or from 100 microns to 1 mm, or from 10 microns to 500 microns. The drift layer 120 thickness t_D can be from 10 nm to 100 microns, or from 1 micron to 100 microns.

[02113] As depicted in FIG. 1, device 100 may include an ohmic anode contact region 151 (“A”) and an ohmic cathode contact region 152 (“C”) which may be formed on or in the metal layer 130 and on or in the metal layer 132 respectively. In an example, the ohmic anode contact region 151 comprises the metal layer 130. In an example, the ohmic cathode contact region 152 comprises the metal layer 132.

[02114] In an example, the SiC substrate 110 may be doped n-type or p-type, imparting a conductivity type governed by charge carriers of electrons or holes, respectively. In cases where substrate 110 is a composite substrate, the bulk substrate layer 112 and/or the surface layer 114 may be doped n-type or p-type, imparting a conductivity type governed by charge carriers of electrons or holes, respectively.

[02115] In some cases, epitaxial drift layer 120 comprises a semiconductor oxide material. Epitaxial drift layer 120 can include any of the oxide materials described herein, for example, any of the compositions of Materials 1-7.

[02116] In various examples, the epitaxial drift layer 120 is a semiconductor oxide material in the form of Ga_2O_3 or $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ where $0 \leq x \leq 1$.

[02117] In another form, the epitaxial drift layer 120 is a semiconductor oxide material in the form of a ternary oxide including, but not limited to $\text{Mg}_x\text{Ga}_{2(1-x)}\text{O}_{3-2x}$ where $0 \leq x \leq 1$ or $\text{Mg}_x\text{Al}_{2(1-x)}\text{O}_{3-2x}$ where $0 \leq x \leq 1$.

[02118] In another form, the epitaxial drift layer 120 is a semiconductor oxide material in the form of $(\text{Mg}_x\text{Zn}_{1-x})_z(\text{Al}_y\text{Ga}_{1-y})_{2(1-z)}\text{O}_{3-2z}$, where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z < 1$.

[02119] In another form, the epitaxial drift layer 120 is a semiconductor oxide material in the form of $(\text{Mg}_x\text{Zn}_{1-x})_z(\text{Al}_y\text{Ga}_{1-y})_{2(1-z)}\text{O}_{3-2z}$, where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z < 1$.

[0220] In another form, the epitaxial drift layer 120 is a semiconductor oxide material in the form of $(\text{Mg}_x\text{Zn}_{1-x})_z(\text{Al}_y\text{Ga}_{1-y})_{2(1-z)}\text{O}_{3-2z}$, where $0 < x < 1$, $0 \leq y \leq 1$, and $0 \leq z < 1$.

[0221] In some examples epitaxial drift layer 120 can include one or more of: Ga_2O_3 (e.g., α -phase or β -phase); $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ (e.g., α -phase or β -phase) where $0 \leq x \leq 1$; $\text{Mg}_x\text{Ga}_{2(1-x)}\text{O}_{3-2x}$ where $0 \leq x \leq 1$; $\text{Mg}_x\text{Al}_{2(1-x)}\text{O}_{3-2x}$ where $0 \leq x \leq 1$; $(\text{Mg}_x\text{Zn}_{1-x})_z(\text{Al}_y\text{Ga}_{1-y})_{2(1-z)}\text{O}_{3-2z}$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Zn}_x\text{Ga}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Zn}_x\text{Al}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Mg}_x\text{Ga}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Mg}_x\text{Al}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Ni}_x\text{Ga}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Ni}_x\text{Al}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $(\text{Ni}_x\text{Mg}_{1-x})_y\text{Ga}_{2(1-y)}\text{O}_{3-2y}$ where $0 \leq x \leq 1$ and $0 \leq y \leq 1$; or $(\text{Mg}_x\text{Ni}_{1-x})_z(\text{Al}_y\text{Ga}_{1-y})_{2(1-z)}\text{O}_{3-2z}$, where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$, or $(\text{Ni}_x\text{Mg}_y\text{Zn}_{1-x-y})_z(\text{Al}_y\text{Ga}_{1-y})_{2(1-z)}\text{O}_4$ where $0 \leq (x, y, z) \leq 1$, or $(\text{Zn}_p\text{Mg}_x\text{Ni}_{1-x-p})_z(\text{Al}_y\text{Ga}_{1-y})_{2(1-z)}\text{O}_{3-2z}$, where $0 \leq (p, x, y, z) \leq 1$.

[0222] In other examples, the epitaxial drift layer 120 is doped n-type or p-type. In some cases, one or more impurity species can be added to the epitaxial drift layer 120 to extrinsically dope the layer. In some cases, epitaxial drift layer 120 can be a polar material that is doped via polarization doping.

[0223] In another example, the epitaxial drift layer 120 is formed from $\beta\text{-Ga}_2\text{O}_3$ (~ 201) and the substrate 110 is formed from C-plane SiC (0001). In cases where substrate 110 is a composite substrate, the surface layer 114 can comprise C-plane SiC (0001) and the bulk substrate layer 112 can comprise polycrystalline SiC.

[0224] In yet another example, the epitaxial drift layer 120 is formed from $\beta\text{-Ga}_2\text{O}_3$ (~ 201) and the substrate 110 is

formed from C-plane SiC (0001) wherein the oxide is formed on a Silicon-polar (Si-polar) SiC surface.

[0225] Another example has the epitaxial drift layer 120 that is formed from $\beta\text{-Ga}_2\text{O}_3$ (~ 201) and the substrate 110 is formed from C-plane SiC (0001) wherein the oxide is formed on a Carbon-polar (C-polar) SiC surface.

[0226] In an example, the epitaxial drift layer 120 is formed in whole or part from alpha-phase $\alpha\text{-Ga}_2\text{O}_3$ having crystal symmetry group R3m and the substrate 110 is formed from 4H-SiC. The epitaxial oxide $\alpha\text{-Ga}_2\text{O}_3$ formed primarily as an A-plane oriented epitaxial film on an A-plane oriented 4H-SiC (11-20), R-plane 4H-SiC (10-12) or M-plane 4H-SiC (10-10) surface. That is, A-plane $\alpha\text{-Ga}_2\text{O}_3$ oriented epitaxial films are possible on A-, R- and M-plane 4H-SiC surfaces. The SiC surface can be selected from a silicon-polar or carbon-polar orientation. The polarity of the surface directly affects the heterojunction formed with an epitaxial oxide which can be used as a form of polarization control at the interface and can also facilitate polarization doping. The epitaxial oxide of the epitaxial drift layer 120 may be non-polar, semi-polar or polar depending upon the oxide crystal symmetry group and film orientation with respect to the substrate surface.

[0227] In an example, the epitaxial drift layer 120 is formed from $\beta\text{-(Al}_x\text{Ga}_{1-x})_2\text{O}_3$ with $0 \leq x \leq 1$. In another example, the epitaxial drift layer 120 is formed from $\beta\text{-(Al}_x\text{Ga}_{1-x})_2\text{O}_3$ with $0 \leq x \leq 0.3$, and the substrate 110 is formed from C-plane SiC (0001). The SiC substrates described herein can be silicon polar, or carbon polar. In the case of a composite substrate, the surface layer 114 can be silicon polar or carbon polar single crystal SiC while the bulk substrate layer 112 can be another material. This is a potential advantage of composite substrates, since the surface layer 114 can be a material that can be used as a template for epitaxial growth of an oxide layer, while bulk substrate layer 112 can be made from a material with fewer constraints.

[0228] Furthermore, SiC substrates may be prepared with an intentional miscut in the surface, exposing a plurality of atomic steps and terraces. For example, a miscut C-plane 4H-SiC(0001) oriented surface can be prepared to form a miscut angle between 0 degrees and 6 degrees directed toward the A-plane(11-20). A 4-degree miscut is a standard type of substrate used in the prior-art of growing epitaxial SiC film on the n-doped 4H-SiC substrate. In some examples of the present disclosure, insight is provided in which a 4 deg miscut 4H-SiC can be advantageous for improving the crystallographic quality of epitaxial oxides.

[0229] In some cases, the SiC substrate is not miscut, meaning that a surface orientation plane of the substrate is approximately parallel to a crystal plane of the SiC of the substrate. For example, a substrate that is not miscut can have an angle between a surface orientation plane of the substrate and a crystal plane of the SiC of the substrate between 0 degrees and 1 degree. For example, a substrate that is not miscut can have a C-plane 4H-SiC(0001) oriented surface with a miscut angle that is from 0 to 1 degrees directed towards the A-plane(11-20).

[0230] In an example, the epitaxial drift layer 120 is formed from $\alpha\text{-(Al}_x\text{Ga}_{1-x})_2\text{O}_3$ with $0 \leq x \leq 1$ and the substrate 110 is formed from A-plane SiC (1-100) or R-plane SiC (2100) or M-plane SiC (1-100). The surface orientation of SiC substrates with these orientations may also be miscut by about 0 to 6 degrees to improve the crystal property of the

epitaxial oxide. In other cases, the SiC substrate can be not miscut (i.e., have a miscut angle from 0 to 1 degrees).

[0231] In some examples, the composition of the epitaxial drift layer **120** varies in the vertical direction. Similarly, the doping concentration may also vary in the vertical direction. In various examples, the drift layer **120** may have a combination of: (i) constant doping concentration and constant composition in the vertical direction; (ii) graded doping concentration and constant composition in the vertical direction; (iii) constant doping concentration and graded composition in the vertical direction; or (iv) graded doping concentration and graded composition in the vertical direction.

[0232] In some embodiments, the epitaxial drift layer **120** may be formed as a superlattice comprising a plurality of different or dissimilar oxide materials. In some embodiments, the drift layer **120** may be formed as a chirp layer comprising multiple oxide materials, where a chirp layer is similar to a superlattice with layers that change thickness through the drift layer **120**. The superlattice or chirp layer can contain wider and narrower bandgap materials forming potential energy barrier and well layers, respectively. In some cases, the superlattice or chirp layer of the drift layer **120** may have, in the vertical direction, a combination of: (i) constant doping concentration and constant composition; (ii) graded doping concentration and constant composition; (iii) constant doping concentration and graded composition; or (iv) graded doping concentration and graded composition. In cases, where the doping and/or composition are graded, the change in doping and/or composition may be formed by varying the barrier and/or well layers through the drift layer **120**. For example, the thicknesses of the barrier and/or well layers can be varied to form a graded composition through the drift layer **120**. In another example, the doping concentration of the barrier and/or well layers can be varied to form a graded doping profile through the drift layer **120**. In some cases, the doping concentration, thickness, and/or composition can be varied within one or more barrier and/or well layers of a superlattice or chirp layer of the drift layer **120** to form the graded doping concentration and/or graded composition.

[0233] In some examples, one or more layers formed from a non-semiconductor oxide material or a non-SiC material may be formed between the substrate **110** and the drift layer **120** while still maintaining the configuration that the epitaxial drift layer **120** is formed above the substrate **110**. For example, a single crystal metallic layer, a semi-metallic layer, or an insulating layer can be formed between the substrate **110** and the drift layer **120**. Such a layer can be useful for improving the contact resistance or providing a tunnel barrier within a low resistance (or ohmic) contact between the substrate **110** and the drift layer **120**.

[0234] FIG. 2A shows a simplified electronic energy band diagram **200** of a vertical multilayered semiconductor diode device where the direction of increasing “z” corresponds to moving vertically upwards as shown in FIG. 1. In this example, band diagram **200** depicts the Schottky barrier junction (SBJ) region **210**, the drift region **220**, the n-doped substrate **225** and an ohmic junction region **230** corresponding to an ohmic contact (OC) region for diode device **100** where the drift layer is an n-doped semiconductor oxide material. A Schottky metal (SM) and Ohmic metal (OM) are selected to achieve a high potential energy barrier ϕ_{SB} and low energy barrier ϕ_C , respectively. SM and OM may be dissimilar metals or the same. The energy of the valence

band, $E_v(z)$ and conduction band, $E_c(z)$ as a function of vertical distance z are shown with the band gap energy E_g also indicated.

[0235] As can be seen by inspection, at the interface between the epitaxial metal layer **130** and the epitaxial drift layer **120** at equilibrium, the Fermi energy level, E_F , matches across the interface and a Schottky potential barrier ϕ_{SB} is formed with a depletion zone extending downwardly into the epitaxial drift layer a distance of z_{depl} .

[0236] FIG. 2B shows the empirically derived relationship between the critical electric field E_{crit} that can be supported by various semiconductors possessing an electronic energy bandgap E_g . Avalanche breakdown process due to energetic carrier-induced impact ionization governs the maximum sustainable electric field that can be supported by the host semiconductor (i.e., the critical or breakdown voltage across a finite thickness of a semiconductor is defined as V_{br}). The maximum electric-field at breakdown (also called the critical electric field, E_{crit}) is known to increase with wider semiconductor bandgap E_g . FIG. 2B shows selected data and fitted relationships for not-intentionally-doped (NID) semiconductors ranging from small (“NBG”, e.g., InAs, InP, GaAs, InGaP), wide (“WBG”, e.g., GaN, ZnO, SiC) and ultrawide bandgaps (“UWBG”, e.g., diamond, Ga_2O_3 , AlN, ZnGaO). The dashed line **250** is a fit to the data, resulting in a power law as shown. The solid line **255** is a preferential fit to WBG and UWBG data, resulting in a power law as shown. This dependence (solid line **255**) provides the major impetus for the adoption of wide bandgap (WBG) and, in the present disclosure, ultrawide bandgap (UWBG) semiconductors for high power electronics.

[0237] FIG. 2C plots the important relationship between figures of merit describing a given semiconductor. The specific ON-resistance per unit thickness of material R_{ON}^{sp} governs efficiency or loss of diode when in the conducting state. Plotting R_{ON}^{sp} vs V_{br} shows the fundamental trends for a given semiconductor technology. The plot shows that both WBG 4H-SiC and UWBG β - Ga_2O_3 provide significant advantages over bulk Si, with 3 to 4 orders of magnitude reduction ON-resistance for a given maximum operating voltage. In particular, the devices of the present disclosure utilize the benefits of UWBG Ga_2O_3 to further extend the improvements possible for high power devices designs over WBG SiC. The shaded bands represent the variation in possible material properties such as crystallinity, defects, carrier mobility and the like. FIG. 2C can be used as a guide for designing high power diode operation and shows the trade-off between low loss and high voltage operation.

[0238] FIG. 3A shows a figurative view **300** of the substrate **110** and drift layer **120** of a vertical multilayered semiconductor diode device such as shown in FIG. 1, and an associated plot **301** of an example variation in doping concentration, $N_{A,D}$, (A and D referring to donor and acceptor sites respectively) as a function of vertical height z . For example, a highly doped n-type (e.g., n^+ doped) substrate **110** would have a large N_D concentration (e.g., from 10^{18} cm^{-3} to 10^{20} cm^{-3} , or from 10^{19} cm^{-3} to 10^{20} cm^{-3}), and a highly doped p-type (e.g., p^+ doped) substrate **110** would have a large N_A concentration (e.g., from 10^8 cm^{-3} to 10^{20} cm^{-3} , or from 10^{19} cm^{-3} to 10^{20} cm^{-3}). In this example, there is a discrete change in the drift layer doping concentration **320** and the substrate doping concentration **310** that occurs at the interface **111** between substrate layer **110** and drift layer **120**. In this example, the drift layer **120** has a

smaller dopant concentration than the substrate **110**. For example, a doped n-type drift layer **120** could have an N_D concentration from 10^{15} cm^{-3} to 10^{18} cm^{-3} , and a doped p-type drift layer **120** could have an N_A concentration from 10^{15} cm^{-3} to 10^{18} cm^{-3} . In some embodiments, the substrate **110** and the drift layer **120** are both doped n-type, or are both doped p-type.

[0239] FIG. 3B shows a figurative view **302** of the substrate **110** and drift layer **120** of a vertical multilayered semiconductor diode device such as shown in FIG. 1, and an associated plot **303** of an example variation in doping concentration, $N_{A,D}$, (A and D referring to donor and acceptor sites respectively) as a function of vertical height z . In this example, a p-type SiC substrate **110** has an N_A concentration **312** (e.g., from about 10^{17} cm^{-3} to about 10^{20} cm^{-3} , or about from 10^{19} cm^{-3} to about 10^{21} cm^{-3}), and an n-type drift layer doping concentration **320** has an N_D concentration (e.g., from about 10^{15} cm^{-3} to about 10^{21} cm^{-3} , or from about 10^{17} cm^{-3} to about 10^{21} cm^{-3} , or from about 10^{15} cm^{-3} to about 10^{18} cm^{-3}). In such embodiments, a p/n junction forms between the SiC substrate **110** and the drift layer **120**. For example, drift layer **120** can comprise epitaxial Ga_2O_3 . In such examples, the metal contact **130** in FIG. 1 can be a metal that forms an Ohmic contact with the drift layer **120**. When the SiC substrate **110** is doped p-type, then the metal **130** can be a metal selected from Ti, Al, Ni, Au, Pt, Mo, and/or Ta, as well as alloys and layered stacks thereof. The thickness of the drift layer **120** can be from about 100 nm to 100 microns, or from about 500 nm to about 5 microns thick, or from about 1 microns to about 15 microns, or from about 3 microns to about 5 microns. The SiC substrate **110** can be Si-polar or C-polar, and can optionally have a miscut oriented surface in some cases to modify oxide material quality. In some examples, the miscut angle can range from 0.5 degrees to 10 degrees, or 1 degree to 5 degrees, or 1 degree to 4 degrees. The substrate miscut may be selected to provide a step and terraced surface to assist oxide epitaxy.

[0240] FIG. 4 shows a figurative sectional view of a vertical multilayered semiconductor diode device **400** according to an illustrative embodiment. Device **400** in this example is similar to device **100** in FIG. 1, however, device **400** further comprises a transition layer **160** disposed between substrate **110** and drift layer **120**. In various examples, intermediate transition layer **160** may be formed of SiC or a semiconductor oxide material or combination of both these materials. The transition layer **160** has a layer thickness t_T .

[0241] In some cases, transition layer **160** comprises a semiconductor oxide material. Transition layer **160** can include any of the oxide materials described herein, for example, any of the compositions of Materials 1-7. Transition layer **160** can contain a single semiconductor oxide composition, or more than one sub-layer with different semiconductor oxide composition (e.g., the sub-layers can form a bilayer, a multilayered structure, a superlattice, or a chirp layer), any of which can be chosen from Materials 1-7. In some cases, transition layer **160** comprises a graded composition (e.g., a linear, a step-wise, or a chirped composition gradient) comprising one or more of Materials 1-7. In some cases, transition layer **160** comprises one or more of Materials 1-7 that is doped with a single dopant concentration, or with a dopant concentration that varies in depth in the transition layer **160**.

[0242] In some examples, transition layer **160** can include one or more of: Ga_2O_3 (e.g., α -phase or β -phase); $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ (e.g., α -phase or β -phase) where $0 \leq x \leq 1$; $\text{Mg}_x\text{Ga}_{2(1-x)}\text{O}_{3-2x}$ where $0 \leq x \leq 1$; $\text{Mg}_x\text{Al}_{2(1-x)}\text{O}_{3-2x}$ where $0 \leq x \leq 1$; $(\text{Mg}_x\text{Zn}_{1-x})_2(\text{Al}_y\text{Ga}_{1-y})_{2(1-z)}\text{O}_{3-2z}$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Zn}_x\text{Ga}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Zn}_x\text{Al}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Mg}_x\text{Ga}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Mg}_x\text{Al}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Ni}_x\text{Ga}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Ni}_x\text{Al}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $(\text{Ni}_x\text{Mg}_{1-x})_2\text{Ga}_2(1-y)\text{O}_{3-2y}$ where $0 \leq x \leq 1$ and $0 \leq y \leq 1$; or $(\text{Mg}_x\text{Ni}_{1-x})_2(\text{Al}_y\text{Ga}_{1-y})_{2(1-z)}\text{O}_{3-2z}$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$, or $(\text{Ni}_x\text{Mg}_x\text{Zn}_{1-x-z})(\text{Al}_y\text{Ga}_{1-y})_2\text{O}_4$ where $0 \leq (x, y, z) \leq 1$, or $(\text{Zn}_p\text{Mg}_x\text{Ni}_{1-x-p})_2(\text{Al}_y\text{Ga}_{1-y})_{2(1-z)}\text{O}_{3-2z}$ where $0 \leq (p, x, y, z) \leq 1$.

[0243] FIG. 5 shows a figurative view **500** of the substrate **110**, transition layer **160** and drift layer **120** of a vertical multilayered semiconductor diode device such as shown in FIG. 4 and an associated plot of an example variation in doping concentration, $N_{A,D}$, as a function of vertical height z .

[0244] In this example, the transition layer doping concentration **560** may have a doping concentration between that of the substrate doping concentration **510** and the drift layer doping concentration **520** with a discrete change in doping concentration between the substrate **110** and transition layer **160**, and between the transition layer **160** and the drift layer **120**. In an example, the substrate **110** may be n^+ doped (i.e., having a doping concentration of from 10^{19} to 10^{21} cm^{-3}), the Transition Layer **160** may be n^- Doped (i.e., Having a Doping concentration of approximately 10^{18} cm^{-3} , or from 10^{17} cm^{-3} to 10^{18} cm^{-3}) and the drift layer **120** may be n^{++} doped (i.e., having a doping concentration from 10^{15} cm^{-3} to 10^{17} cm^{-3}) as a result providing a transition between a n^+ doped region to a n^{++} doped region moving vertically upwards from the substrate **110** to the drift layer **120**. Similarly, a p-type device can be formed wherein the substrate **110** is p^+ doped, the transition layer **160** is p^- doped, and the drift layer **120** is p^{++} doped, where the layers have the doping concentrations described above of acceptors rather than donors. In other cases, the substrate **110** can be n^+ or p^+ doped, the drift layer **120** can be n^- or p^- doped, respectively (to form the Schottky barrier junction), and the transition layer **160** can have a doping concentration between that of the substrate **110** and the drift layer **120**. For example, the transition layer **160** may be formed from SiC and the drift layer **120** may be formed of a semiconductor oxide material. In another example, the transition layer **160** (TL) may be formed of a semiconductor oxide material, and the drift layer **120** may be formed of the same semiconductor oxide material or a different oxide material as the TL. In an example, the transition layer **160** may be formed of a material chosen to provide a structural matching region between the substrate **110** and the drift layer **120**. For example, a transition layer **160**, comprising a semiconductor oxide material of the form of $\text{Mg}_x\text{Zn}_y\text{Ni}_z(\text{Al}_p\text{Ga}_{1-p})\text{O}_q$ can be formed between an SiC substrate **110** and a drift layer **120**, where the drift layer **120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$. Another example of a semiconductor oxide material is of the form of a silicon oxycarbide $\text{Si}_x\text{C}_y\text{O}_z$, which can be comprised in the transition layer **160** and can be formed between an SiC substrate **110** and a drift layer **120**, where the drift layer **120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$. Yet another

example of a semiconductor oxide material is of the form of a silicon oxy-nitride $\text{Si}_x\text{N}_y\text{O}_z$ which can be comprised in the transition layer **160** and can be formed between an SiC substrate **110** and a drift layer **120**, where the drift layer **120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$. Another example of a transition layer **160** is one comprising a material of the form of a silicon-germanium-carbide $\text{Si}_x\text{Ge}_y\text{C}_z$ such that the transition layer **160** can be used to manage the interfacial strain formed between an SiC substrate **110** and a drift layer **120**, where the drift layer **120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$.

[0245] In some cases of structures **400** and **500**, the transition layer **160** and the drift layer **120** comprise n-type Ga_2O_3 (e.g., single crystal epitaxially deposited Ga_2O_3) and the transition layer doping concentration **560** is at least an order of magnitude higher than the drift layer doping concentration **520**. The doping concentration within transition layer **160** can be constant (as shown in FIG. **5**) or can be spatially varying in the growth direction (as shown in FIGS. **6** and **7**, and discussed further below). The doping concentration of the transition layer **160** can be lower than, or equal to, that of the SiC substrate **110**. The Ga_2O_3 transition layer **160** can have a thickness from about 10 nm to about 1 micron. The thickness of the Ga_2O_3 drift layer can be from about 1 micron to 100 microns, or from about 2 microns to about 15 microns, or from about 3 microns to about 10 microns. The SiC substrate **110** can be Si-polar or C-polar, and can optionally have a miscut oriented surface in some cases.

[0246] The surface of an SiC substrate or layer can be intentionally terminated with one or more species. For example, an intentional surface termination of an SiC surface can comprise a substantially Silicon- (Si) or Carbon- (C) or Oxygen- (O) or Nitrogen- (N) terminated species. Termination (e.g., uniform termination) with a single species may be used advantageously for seeding the epitaxial deposition of semiconductor oxide (or metal oxide film), such that, the interfacial bonds arrange with low lattice mismatch (or with reduced mismatch, or with an acceptable amount of mismatch) compared to not terminating the surface with a single species. For example, in vacuo, the SiC surface may be prepared such that a Si-terminated surface is configured. The Si-terminated surface can then be preferentially reacted with active oxygen species to form a —C—Si—O— bond sequence. Conversely, a C-terminated surface can be configured such that reaction with active oxygen produces Si—C—O— sequence at the immediate interface. The oxygen can then readily bond with Ga (or another cation atom) to subsequently form a Ga_2O_3 single crystal epilayer.

[0247] Other possible interfacial layers can also be used advantageously. For example, an interfacial surface comprising silicon oxy-carbide $\text{Si}_x\text{C}_y\text{O}_z$ comprising the transition layer can also be formed between a SiC substrate **110** and a drift layer **120**, where the drift layer **120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$.

[0248] Alternately, an interfacial surface comprising silicon-carbon-nitride $\text{Si}_x\text{C}_y\text{N}_z$, which can be comprised in the transition layer can also be formed between a SiC substrate **110** and a drift layer **120**, where the drift layer **120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$.

[0249] An interfacial surface comprising germanium-nitride Ge_xN_y , which can be comprised in the transition layer **160** can also be formed between a SiC substrate **110** and a

drift layer **120**, where the drift layer **120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$.

[0250] Another example of a transition layer oxide is of the form of magnesium oxide (MgO), which can be comprised in transition layer **160**, and can be formed between an SiC substrate **110** and a drift layer **120**, where the drift layer **120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$. For 4H-SiC(0001) surface a MgO(111) oriented transition layer may be of only a few unit cells in thickness.

[0251] Yet another example of a transition layer oxide is of the form of zinc gallium oxide (e.g., spinel crystal symmetry ZnGa_2O_4 , for example with a Fd3m space group), which can be comprised in transition layer **160**, and can be formed between an SiC substrate **110** and a drift layer **120**, where the drift layer **120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$. For 4H-SiC(0001) surface a ZnGa_2O_4 (111) oriented transition layer may be of one or only a few unit cells in thickness or more.

[0252] In other examples, one or more layers formed from a non-semiconductor oxide material or a non-SiC material may be formed between the substrate **110** and the transition layer **160** and/or between the transition layer **160** and the drift layer **120**. For example, a single crystal metallic layer, a semi-metallic layer, or an insulating layer can be formed between the substrate **110** and the transition layer **160**. Such a layer can be useful for improving the contact resistance or providing a tunnel barrier to form a low resistance (or ohmic) contact between the substrate **110** and the transition layer **160**.

[0253] FIG. **6** shows a figurative view **600** of the substrate **110**, transition layer **160** and drift layer **120** of a vertical multilayered semiconductor diode device such as shown in FIG. **4** and an associated plot **601** of another example variation in doping concentration, $N_{A,D}$, as a function of vertical height z . FIG. **6** shows an example wherein the transition layer **160** has a varying doping density (i.e., doping concentration). The transition layer **160** can have a variable doping density that varies in a vertical direction that is perpendicular to a top surface of the substrate. In some cases, the variable doping density of the transition layer **160** includes a doping density (e.g., an average doping density of the layer) that is between a doping density of the substrate and a doping density of the drift layer.

[0254] In this example, the transition layer doping concentration **660** has a monotonic gradient in the z direction, starting at a doping concentration below (i.e., less than) that of the substrate **110**, and ending at a doping concentration approximately equal to that of the drift layer **120**. In some cases, the transition layer doping concentration **660** has a doping concentration that monotonically varies starting at a doping concentration that is either approximately equal to that of the substrate or is between the substrate doping concentration **610** and the drift layer doping concentration **620**, and ending at a doping concentration that is either approximately equal to that of the drift layer or is between the substrate doping concentration **610** and the drift layer doping concentration **620**. In an example, the transition layer doping concentration **660** may match the substrate doping concentration **610** at the interface between the substrate **110** and the transition layer **160** and then continuously change to match the drift layer doping concentration **620** at the interface between the transition layer **160** and the drift layer **120**. In an example, the change in the transition layer doping concentration **660** may be substantially linear as a function

of vertical height z (e.g., such as shown in FIG. 6). In another example, the transition layer concentration 660 may change continuously (i.e., without significant step changes, and monotonically, or non-linearly, for example) as a function of vertical height z . In another example, the transition layer concentration 660 may change in a stepwise manner as a function of vertical height z . Transition layer 160 in FIG. 6 can have any of the attributes of transition layer 160 as described above (e.g., with respect to transition layer 160 in FIG. 4).

[0255] In an example, the initial transition layer doping concentration 660 at the interface between the substrate 110 and transition layer 160 may be different from the substrate doping concentration 610 (e.g., such as shown in FIG. 6). Similarly, and in another example, the final transition layer doping concentration 660 (moving upwards in the z direction) at the interface between the transition layer 160 and the drift layer 120 may be different from the drift layer doping concentration 620.

[0256] Referring back to FIG. 1, discrete changes in composition such as would occur with a SiC substrate 110 and an oxide based drift layer 120 (e.g., Ga_2O_3) may cause defects at the interface surface 111 which can affect the carrier transport across the heterointerface between the substrate 110 and the drift layer 120 (e.g., the conduction and valence band discontinuities providing potential energy confinement or barriers and/or electrically active defects and/or crystallographic defects). In some cases, threading dislocations can also form in the drift layer 120 due to differences between the materials properties (e.g., lattice constants, crystal symmetries and/or orientations) of the substrate 110 and drift layer 120, which can cause defects at the interface 121 between the drift layer and a layer above (e.g., the metal layer 130). Defects at the interface 121 can affect the carrier transport across the heterointerface between the drift layer 120 and a layer above (e.g., the metal layer 130), which can, for example, affect the Schottky barrier diode properties.

[0257] FIG. 7 shows a figurative view 700 of the substrate 110, transition layer 160 and drift layer 120 of a vertical multilayered semiconductor diode device such as shown in FIG. 4. FIG. 7 also shows an associated plot 701 of another example variation in doping concentration, $N_{A,D}$, as a function of vertical height z , as well as an associated plot 702 of an example variation band gap energy E_G such as would occur by varying the composition as a function of vertical height z . In this example, the SiC substrate would have a first band gap energy 770 of approximately 3.2 eV. As can be seen in this example, the band gap energy 780 E_G of transition layer 160 increases substantially linearly as a function of vertical height from a value close to the first band gap energy 770 of the substrate 110, to a value that is close to the band gap energy 790 E_G of the drift layer 120. In other examples, the changing composition of the transition layer 160 may be selected to cause a stepwise increase in band gap energy 790 E_G from the band gap energy of the substrate 110 to the drift layer 120.

[0258] The transition layer doping concentration 760 in this example is similar to transition layer doping concentration 660 in FIG. 6, and can have all of the same characteristics as described above. FIG. 7 shows an example where the transition layer doping concentration 760 starts at a doping concentration that is approximately equal to that of the substrate 110 (substrate doping concentration 710), and

ends at a doping concentration that is approximately equal to that of the drift layer 120 (drift layer doping concentration 720).

[0259] In some cases of structures 400, 600 and 700, the transition layer 160 and the drift layer 120 comprise n-type Ga_2O_3 (e.g., single crystal epitaxially deposited Ga_2O_3). In some such cases, a doping concentration 660 or 760 of the transition layer 160 is at least an order of magnitude higher than the drift layer doping concentration 620 or 720. The doping concentration 660 or 760 within transition layer 160 can be constant (as shown in FIG. 5) or can be spatially varying in the growth direction (as shown in FIGS. 6 and 7). In cases where the doping concentration varies spatially in transition layer 160, at least one of the values of the doping concentration 660 or 760 within the transition layer 160 is at least an order of magnitude higher than the drift layer doping concentration 620 or 720. For example, transition layer 160 can comprise a gradient in doping concentration 660 or 760, and a doping concentration equal to or similar to the drift layer doping concentration 620 or 720 at the interface between the transition layer 160 and the drift layer 120. In such cases, the gradient in doping concentration 660 or 760 can increase between the drift layer 120 and the SiC substrate 110 to at least an order of magnitude greater than the doping concentration 720 of the drift layer 120. In other examples, the doping concentration of the transition layer 160 can be higher than that of the drift layer 120, and there can be a discrete change in doping density between the transition layer 160 and the drift layer 120. In such cases, there can also be a spatially varying doping concentration within the transition layer 160 such that some or all of the doping concentrations within the transition layer 160 are at least an order of magnitude greater than that of the drift layer 120. The doping concentration of the transition layer 160 can be lower than, or equal to, that of the SiC substrate 110. The Ga_2O_3 transition layer 160 can have a thickness from about 10 nm to about 1 micron. The thickness of the Ga_2O_3 drift layer 120 can be from about 1 micron to 100 microns, or from about 2 microns to about 15 microns, or from about 3 microns to about 10 microns. The SiC substrate 110 can be Si-polar or C-polar, and can optionally have a miscut oriented surface in some cases.

[0260] FIG. 8A is a figurative sectional view of a vertical multilayered semiconductor device 800 in accordance with an illustrative embodiment comprising a substrate 110, a drift layer 120, an intermediate layer 170, and a metal layer 130. In this example, device 800 is similar to device 100 of FIG. 1 but further comprises an intermediate layer 170 formed of a semiconductor oxide disposed between drift layer 120 and metal layer 130. The intermediate layer 170 has a layer thickness t_i . In this example, intermediate layer 170 is formed directly on the top surface 121 of drift layer 120, and metal layer 130 is formed on a top surface 171 of the intermediate layer 170.

[0261] In some cases, intermediate layer 170 can comprise a high carrier concentration to form an Ohmic contact between the drift layer 120 and the metal layer 130. For example, intermediate layer 170 can comprise a high n-type doping concentration (e.g., N_D about 10^{19} cm^{-3} or 10^{20} cm^{-3}) when drift layer 120 is n-type, or a high p-type doping concentration (e.g., N_A about 10^{19} cm^{-3} or 10^{20} cm^{-3}) when drift layer 120 is p-type. In such cases, the doping concentration of the drift layer can be lower (e.g., from about 10^{15} cm^{-3} to about 10^{18} cm^{-3} , or lower than 10^{15} cm^{-3}) and the

doping concentration of the intermediate layer 170 can be at least an order of magnitude higher than that of the drift layer 120 to form an Ohmic contact with the metal layer 130. Such Ohmic contacts can be useful, for example, when there is a p/n junction between other layers of the structure. For example, a p/n junction can be formed using a p-type SiC substrate 110 and an n-type drift layer 120 (e.g., comprising epitaxial Ga_2O_3), or between an n-type drift-layer and a p-type epitaxial layer, as described further herein. The thickness of the drift layer 120 can be from about 1 micron to 100 microns, or from 3 about microns to about 15 microns, or from about 3 microns to about 5 microns. The SiC substrate 110 can be Si-polar or C-polar, and can optionally have a miscut oriented surface in some cases.

[0262] FIG. 8B shows a figurative sectional view of a vertical multilayered semiconductor diode device 802 according to an illustrative embodiment. In this example, device 802 is similar to device 800 but further comprises a transition layer 160 disposed between substrate 110 and drift layer 120, similar to device 400 in FIG. 4.

[0263] A further advantage of the epitaxial oxide—SiC hybrid structures described herein is the additional doping methods available to form doping regions within the host oxide region (e.g., planar doping regions, selective area doping regions, or doping regions through the depth or thickness of the layers). For example, SiC can be effectively doped p-type or n-type using in-situ epilayer growth or post epilayer growth ion implantation followed by high temperature recrystallization annealing. Diffusion doping in SiC is not an effective method for post growth doping and/or selective area doping because of the high SiC bond energy that prevents advantageous activated impurity species doping. In contrast, post growth diffusion doping can be used to dope oxides (e.g. Ga_2O_3), and can be highly effective. As an example, spin on glass dopants can be selectively applied to a surface of Ga_2O_3 as a blanket layer or for selected areas and diffusion driven processes may be employed to transport the impurity species into the oxide material. It is a property of oxides that the dopant species can be activated once driven into the oxide with relatively low thermal budget, for example using rapid thermal annealing techniques. For example, diffusion driven dopants can alter the conductivity type of Ga_2O_3 from low n-type doping used in a drift layer to a semi-insulating conductivity using doping species such as Mg, H, Fe, Zn. In another example, Si or Ge may be used to increase the n-type doping of Ga_2O_3 , in some cases, by at least an order of magnitude. In another example, spin on glasses can be used to provide a surface reservoir of dopant impurities that can be driven into the relevant oxide region. The spin on dopant can be patterned to provide selective area doping. The diffusion time and temperature can be used to control the lateral and vertical extent of the doped region. Spin on glass materials may include SiO_2 , GeO_2 , Fe_2O_3 , B_2O_3 , MgO , P_2O_5 or hydrogen containing materials. Therefore, not only can ion implantation or in-situ epilayer doping be used, but also post growth diffusion driven processes to dope epitaxial oxide layers n-type or p-type.

[0264] In an example, intermediate layer 170 is formed of a different semiconductor oxide having a wider band gap E_G than that of the semiconductor oxide material that the drift layer 120 is formed from. In an example, oxide material for the intermediate layer 170 is chosen to improve diode performance characteristics such as the turn-on voltage and/or the reverse breakdown voltage. In some cases, the

intermediate layer 170 changes the Schottky potential barrier ϕ_s , which in turn changes the diode performance characteristics such as the turn-on voltage and/or the reverse breakdown voltage. In some cases, intermediate layer 170 can be formed of a graded semiconductor oxide composition with a correspondingly varying band gap E_G . In another example, the intermediate layer has a multilayer, superlattice, or chirp layer configuration comprising two or more different semiconductor oxide materials, for example, selected to improve reverse breakdown voltage performance (e.g., by changing the Schottky potential barrier ϕ_s). In some cases, it can be advantageous for the bandgap of the intermediate layer 170 to be greater than that of the drift layer 120. For example, drift layer 120 can comprise Ga_2O_3 , and intermediate layer 170 can comprise $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ where $0 < x \leq 1$, or where $0 < x \leq 0.3$.

[0265] In another example, intermediate layer 170 is formed of a different semiconductor oxide having a narrower or smaller band gap E_G than that of the semiconductor oxide material that the drift layer 120 is formed from. In an example, the intermediate layer 170 is chosen to improve diode performance characteristics such as the turn-on voltage and/or the reverse breakdown voltage. For example, a narrower bandgap material may advantageously present opposite conductivity type charge carriers thereby forming an improved depletion region between drift layer 120 and intermediate layer 170. For example, Nickel oxide (NiO), copper oxide (Cu_2O), lithium oxide (Li_2O), or complex oxides such as spinel crystal structures NiGa_2O_4 , ZnGa_2O_4 , MgGa_2O_4 can provide p-type character which can be used advantageously within a heterojunction formed with an n-type $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ UWBG drift layer 120. Furthermore, ternary or spinel oxides of the form of, for example, NiGa_2O_4 , Cu_2GaO_4 and LiGaO_2 may also be utilized in intermediate layer 170. As a further example, an intermediate layer comprising multiple layers of p-type NiO/NiGaO formed on Ga_2O_3 may also be adopted. For example, NiGaO can be formed explicitly or by intermixing by high temperature annealing between NiO and Ga_2O_3 . In some cases, intermediate layer 170 has a smaller band gap than the drift layer 120, and, due to band alignment, there is a conduction band offset at the interface between the intermediate layer 170 and the drift layer 120. In such a case, the intermediate layer 170 can form a tunnel barrier for electrons between the drift layer 120 and the Schottky metal layer 130, even though intermediate layer 170 has a smaller band gap than the drift layer 120.

[0266] The structures in FIGS. 8A and 8B can include n-type or p-type 4H-SiC substrate 110, n-type or p-type epitaxial oxide drift layer 120 (e.g., n-type Ga_2O_3 or p-type NiO), an optional n-type or p-type intermediate layer 170 (e.g., n-type Ga_2O_3 or p-type NiO), and a metal contact layer 130 that can form an Ohmic contact or Schottky barrier with the intermediate layer 170. Additionally, an optional n-type or p-type transition layer 160 (e.g., SiC, or Ga_2O_3) can be included, as shown in FIG. 8B. The polar 4H-SiC substrate can be doped n-type or p-type using intrinsic and/or extrinsic dopants and/or can have an n-type or p-type induced charge due to the polarity. The epitaxial oxide drift layer 120 and the intermediate layer 170 can also be polar in some cases (e.g., using Pna21 Ga_2O_3), and in such cases, can be doped using intrinsic and/or extrinsic dopants and/or can have an n-type or p-type induced charge due to the polarity. In other cases, epitaxial oxide drift layer 120 and the intermediate

layer 170 can be non-polar and can be doped n-type or p-type using intrinsic and/or extrinsic dopants.

[0267] FIGS. 8C-8F show tables depicting combinations of n-type and p-type layers with Ohmic contacts (“Metal S”) to a SiC substrate, and Ohmic or Schottky contacts (“Metal E”) to an epitaxially grown semiconductor layer, in accordance with some embodiments. The structures shown in FIGS. 1, 4, 8A and 8B can have layers with the doping types shown in the respective tables in FIGS. 8C-8F. In some cases, the n-type or p-type doping of the SiC (e.g., 4H-SiC) substrate 110 (“Substrate” in FIGS. 8C-8F) can be from $1 \times 10^{15} \text{ cm}^{-3}$ to $1 \times 10^{21} \text{ cm}^{-3}$, or from $1 \times 10^{17} \text{ cm}^{-3}$ to $1 \times 10^{20} \text{ cm}^{-3}$, or less than $1 \times 10^{15} \text{ cm}^{-3}$, or greater than $1 \times 10^{21} \text{ cm}^{-3}$. The n-type or p-type doping of the epitaxial oxide drift layer 120 (“Oxide” in FIGS. 8C-8F) (e.g., n-type Ga_2O_3 or p-type NiO), intermediate layer 170 (“IL” in FIGS. 8C-8F) (e.g., n-type Ga_2O_3 or p-type NiO), and transition layer 160 (“TL” in FIGS. 8C-8F) can each be from $1 \times 10^{15} \text{ cm}^{-3}$ to $1 \times 10^{21} \text{ cm}^{-3}$, or from $1 \times 10^{16} \text{ cm}^{-3}$ to $1 \times 10^{20} \text{ cm}^{-3}$, or from $1 \times 10^{15} \text{ cm}^{-3}$ to $1 \times 10^{18} \text{ cm}^{-3}$, or less than $1 \times 10^{15} \text{ cm}^{-3}$, or greater than $1 \times 10^{20} \text{ cm}^{-3}$. An “i” (intrinsic or not intentionally doped) or “n--” or “p--” layer can have a doping density of about $1 \times 10^{15} \text{ cm}^{-3}$, or less than about $1 \times 10^{15} \text{ cm}^{-3}$. An “n--” or “p--” layer can have a doping density from about $1 \times 10^{15} \text{ cm}^{-3}$ to about $1 \times 10^{16} \text{ cm}^{-3}$. An “n” or “p” layer can have a doping density from about $1 \times 10^{16} \text{ cm}^{-3}$ to about $1 \times 10^{18} \text{ cm}^{-3}$. An “n+” or “p+” layer can have a doping density greater than about $1 \times 10^{18} \text{ cm}^{-3}$, or greater than about $1 \times 10^{19} \text{ cm}^{-3}$, or greater than about $1 \times 10^{20} \text{ cm}^{-3}$. It will be appreciated that these ranges can overlap by up to an order of magnitude. In general, increasing band gap energy of a host material for the same impurity incorporation for doping may result in substantially lower activated doping density. Therefore, substrate 110, epitaxial oxide drift layer 120 (e.g., n-type Ga_2O_3 or p-type NiO), intermediate layer 170 (e.g., n-type Ga_2O_3 or p-type NiO), and transition layer 160 can each be n-type (e.g., n-, n+, etc.), p-type (p-, p+, etc.), or i-type (e.g., intrinsic, or not intentionally doped, which can also be n-- or p--) layers in structures and devices.

[0268] The Ohmic contact (“Metal S” in FIGS. 8C-8F) to the SiC substrate is optional and may be omitted in some cases. For example, the layers of the structures can be patterned and selectively etched such that two or more contacts are made to two or more epitaxial layers of the structures instead of to the backside of the substrate.

[0269] For example, FIG. 8E includes cases where the [polar 4H-SiC 110/epitaxial oxide 120/intermediate layer 170/metal 130] structure is an n-n-n-Schottky metal structure, or a p-p-p-Schottky metal structure. In such cases, the Schottky metal contact 130 (“Metal E” in FIGS. 8C-8F) would form a Schottky barrier with the n-type or p-type epitaxial oxide drift layer 120 and the intermediate layer 170. The resulting structures can be used to form diodes, transistors, or other devices. For example, a metal-oxide-semiconductor field-effect transistor (MOSFET) can be formed using ion implantation to form doped regions under source and drain contacts, and the oxide layer of these structures can be used as a channel region, where the channel region is separated from a gate contact by a gate oxide. The epitaxial intermediate layer 170 can form the gate oxide, or in other cases, a non-epitaxial (e.g., sputtered) gate oxide can be deposited. Additionally, similar structures can be formed using the [polar 4H-SiC 110/transition layer 160/epitaxial oxide 120/metal 130] structures shown in FIG.

8D, which include an n-n-n-Schottky metal structure, or a p-p-p-Schottky metal structure.

[0270] In another example, FIG. 8E includes cases where the [polar 4H-SiC 110/epitaxial oxide 120/intermediate layer 170/metal 130] structure is an n-n-p-Ohmic contact structure, an n-n-p-Schottky metal structure, a p-p-n-Ohmic contact structure, or a p-p-n-Schottky metal structure. In such cases, a p/n junction is formed between the epitaxial oxide layer and the intermediate layer. Additionally, in cases including a Schottky metal, a Schottky barrier would form with the n-type or p-type intermediate layer. This Schottky barrier can serve to increase or decrease the built-in potential within the device, and/or form a well for a 2D electron gas or a 2D hole gas, as described further below. The resulting structures can be used to form diodes, transistors, or other devices. For example, a metal-oxide-semiconductor field-effect transistor (MOSFET) can be formed using ion implantation to form doped regions under source and drain contacts. In another example of a transistor, the intermediate layer can be patterned and selectively etched to form doped regions under source and drain contacts, and a gate oxide can be formed between the epitaxial oxide drift layer 120 and a gate contact. In other cases, an LED can be formed using such structures, wherein the epitaxial oxide and/or the polar III-nitride layer emit light. Additionally, similar structures can be formed using the [polar 4H-SiC 110/transition layer 160/epitaxial oxide 120/metal 130] structures shown in FIG. 8D, which include an n-n-p-Ohmic contact structure, an n-n-p-Schottky metal structure, a p-p-n-Ohmic contact structure, or a p-p-n-Schottky metal structure.

[0271] In another example, FIGS. 8E and 8F include cases where the [polar 4H-SiC 110/epitaxial oxide 120/intermediate layer 170/metal 130] structure can be a p-n-n-Ohmic contact structure, a p-n-n-Schottky metal structure, an n-p-p-Ohmic contact structure, or an n-p-p-Schottky metal structure. In such cases, a p/n junction is formed between the polar 4H-SiC substrate and the epitaxial oxide layer. Additionally, an optional n-type or p-type transition layer 160 (e.g., SiC, or Ga_2O_3) can be included between the polar 4H-SiC substrate and the epitaxial oxide layer, which can move the p/n junction interface to the interface between the polar 4H-SiC substrate and the transition layer, or to the interface between the transition layer and the epitaxial oxide layer. Additionally, in cases including a Schottky metal, a Schottky barrier can form with the n-type or p-type intermediate layer. This Schottky barrier can serve as a second built-in potential within the device, and if the potentials are in opposite directions (e.g., and form back-to-back potentials) the structure could be used as a transistor or switch in some cases (e.g., in a metal-insulator-semiconductor (MIS) transistor architecture). The resulting structures can be used to form diodes, transistors, or other devices. For example, a metal-oxide-semiconductor field-effect transistor (MOSFET) can be formed using ion implantation to form doped regions under source and drain contacts. In other cases, an LED can be formed using such structures, wherein the epitaxial oxide and/or the polar III-nitride layer emit light. Additionally, similar structures can be formed using the [polar 4H-SiC 110/transition layer 160/epitaxial oxide 120/metal 130] structures shown in FIG. 8D, which include a p-n-n-Ohmic contact structure, a p-n-n-Schottky metal structure, an n-p-p-Ohmic contact structure, or an n-p-p-Schottky metal structure.

[0272] In another example, FIGS. 8E and 8F include cases where the [polar 4H-SiC 110/epitaxial oxide 120/intermediate layer 170/metal 130] structure can be a p-n-p-Ohmic contact structure, a p-n-p-Schottky metal structure, an n-p-n-Ohmic contact structure, or an n-p-n-Schottky metal structure. In such cases, a p/n junction is formed between the polar 4H-SiC substrate and the epitaxial oxide layer and a p/n junction is formed (in the reverse direction) between the epitaxial oxide layer and the intermediate layer. Additionally, an optional n-type or p-type transition layer 160 (e.g., SiC, or Ga₂O₃) can be included between the polar 4H-SiC substrate and the epitaxial oxide layer, which can move the p/n junction interface to the interface between the polar 4H-SiC substrate and the transition layer, or to the interface between the transition layer and the epitaxial oxide layer. Additionally, in cases including a Schottky metal, a Schottky barrier would form with the n-type or p-type intermediate layer. This Schottky barrier can serve to increase or decrease the built-in potential within the device, and/or form a well for a 2D electron gas or a 2D hole gas, as described further below. The resulting structures can be used to form diodes, transistors, or other devices. For example, a bipolar junction transistor (BJT) can be formed by patterning and selectively etching the structure to form separate contacts to the p-type and n-type layers of the n-p-n or p-n-p structures. In such devices, the metal contact adjacent to the oxide or intermediate layer can form either the source or drain contact of the BJT, and an additional source or drain contact can be formed. In other cases, a metal-oxide-semiconductor field-effect transistor (MOSFET) can be formed, using patterning and selectively etching of the intermediate layer and/or ion implantation to form doped regions under source and drain contacts. In other cases, an LED can be formed using such structures, wherein the epitaxial oxide and/or the polar III-nitride layer emit light. Additionally, similar structures can be formed using the [polar 4H-SiC 110/transition layer 160/epitaxial oxide 120/metal 130] structures shown in FIG. 8D, which include a p-n-p-Ohmic contact structure, a p-n-p-Schottky metal structure, an n-p-n-Ohmic contact structure, or an n-p-n-Schottky metal structure.

[0273] Yet a further possible utility of intermediate layer 170 is the advantageous reaction of Schottky metal 130 with selective region 141 which may form an alloy. The selectively alloyed region 141 may be formed via rapid thermal anneal or diffusion process.

[0274] Any of drift layer 120, transition layer 160, and/or intermediate layer 170 (e.g., of devices 100, 400, 800, or 802 in FIGS. 1, 4, 8A and 8B respectively) can include single epitaxial oxide layers, or can include multiple layers of epitaxial oxide materials (e.g., bilayers, tri-layers, superlattices, and/or chirp layers). Bilayers, tri-layers and other multiple layer structures of layers 120, 160 and/or 170 can include different compositions and/or different doping densities. The total number of layers within any of drift layer 120, transition layer 160, and/or intermediate layer 170 can be from 1 to 100, or from 1 to 20, or from 1 to 10, in different examples.

[0275] Any of drift layer 120, transition layer 160, and/or intermediate layer 170 (e.g., of devices 100, 400, 800, or 802 in FIGS. 1, 4, 8A and 8B respectively) can include a superlattice and/or a chirp layer including epitaxial oxide semiconductor materials. Superlattices, such as short-period superlattices, contain alternating layers (i.e., barrier layers and well layers) forming a unit cell, and the unit cell is

repeated to form the superlattice. A chirp layer is similar to a superlattice in that it also contains alternating layers, however, the layer thicknesses and/or compositions of the layers change monotonically along a thickness (or growth direction) a chirp layer. The thickness of each of the barrier layers, well layers, and/or unit cells within a superlattice or chirp layer can be from less than 1 monolayer (ML) to 100 MLs, or from less than 1 ML to 20 MLs, or from less than 1 ML to 10 MLs, or from less than 1 ML to 5 MLs, or from 1 ML to 20 MLs, or 1 ML to 10 MLs, or 1 ML to 5 MLs. A layer with a thickness less than 1 ML may be discontinuous laterally (i.e., in a direction parallel with the surface of the substrate upon which the layer is grown).

[0276] Superlattices can be used to form digital alloys whereby barrier layers and well layers of the superlattice have materials properties such that the superlattice has overall materials properties (i.e., composite materials properties) that are different from the materials properties of the barrier layers and that are different from the materials properties of the well layers. Superlattices can also be used to modify the effects of the electrons and/or holes within a device, for example, by causing a barrier to electron and/or hole flow across the superlattice.

[0277] Chirp layers can also be used to form digital alloys and impact the electrons and/or holes within a device as described herein. Additionally, chirp layers introduce a changing materials parameter, such as a changing average lattice constant, a changing average bandgap, and/or a changing average doping density, within the chirp layer. For example, chirp layers can be used to form electron blocking layers (EBLs). Chirp layers can also be used to change the lattice constant within a structure, for example to form a drift layer on an SiC substrate, where the drift layer has a different lattice constant than SiC.

[0278] FIG. 9A discloses a design flow 9900 for achieving target performance for UWBG power device based on the desired R_{ON} and V_{br} specification. In practice, a device simulator utilizing Poisson solver and quantum mechanical models for the specific semiconductors is used to model the performance. This modeling may include block 9910 (select Schottky Barrier "SB" diode target R_{ON} and V_{br}), block 9912 (select bandgap of drift layer), block 9914 (select drift layer conductivity/doping), and block 9916 (select drift layer thickness for V_{br}). Flow 9900 may include a decision point 9920 to check if the target R_{ON} is acceptable before proceeding further. If R_{ON} is not acceptable, the flow returns to block 9914 to reselect drift layer parameters. If R_{ON} is acceptable, the flow 9900 continues with block 9930 (optionally select substrate/drift transition layer), block 9932 (select Schottky barrier configuration), block 9934 (select substrate ohmic configuration), and block 9936 (simulate design). If the design is within the specification at block 9940, then the device is fabricated in block 9950. If the design does not meet specifications, the flow returns to block 9912 to reselect parameters for the various layers of the device. Simulation tools for 1D, 2D and 3D may be utilized in conjunction with finite element electrostatic solvers to detail the electric field distribution within the device structure.

[0279] FIG. 9B is a detailed simulated energy band diagram 2000 of the conduction band and valence band edges of the vertical multilayered semiconductor device illustrated in FIG. 4, showing the following regions comprising (moving from left to right): a Schottky barrier metal 2010 (in this

example Ni), forming a rectifying junction with a drift layer **2012** and transition layer **2014** (in this example β -Ga₂O₃ (-201)), and a substrate **2016** (in this example Si-polar 4H-SiC(0001)). The band diagram was constructed from detailed knowledge of the bandstructure of each material and the absolute electron affinity, enabling accurate heterojunction offsets to be determined. The substrate is an n-type doped 4H-SiC, such that ($E_g^{sub}=3.23$ eV, $N_d^{sub}=1\times10^{19}$ cm⁻³), the transition layer an n-type β -Ga₂O₃ ($E_g^{TL}=4.7$ eV, $N_d^{TL}=1\times10^{17}$ cm⁻³), and a drift region ($E_g^{drift}=4.7$ eV, $1\times10^{16}\leq N_d^{drift}\leq1\times10^{17}$ cm⁻³). The idealized case of a Nickel Schottky metal having a barrier height of $\phi_{SB}=1.3$ eV @ $N_d^{drift}=5\times10^{16}$ cm⁻³ is shown, and the Fermi energy is selected as the zero energy reference. The diode is a substantially unipolar device, wherein the majority carriers are electrons. The holes in the valence band can be neglected to first order. Furthermore, the detailed band structure of β -Ga₂O₃ reveals substantially heavier hole masses as compared to 4H-SiC. The mobility of holes in the valence band is also limited by virtue of the flat valence band energy dispersion in crystal momentum space.

[0280] Referring to band diagram **2001** of FIG. 9C, the effective n-type doping of the drift layer is found to have a strong influence on the depletion width of Schottky potential penetrating into the drift region. Band diagram **2001** is a close-up view of energy band values (y-axis) of 0 eV to 1.4 eV from energy band diagram **2000**. The width of the depletion potential Z_{depl} influences the reverse bias leakage behavior, whereas ϕ_{SB} determines the thermionic emission process in forward bias. As would be expected, as the doping concentration increases the size or extent of the depletion region will reduce. Not shown in the band diagram **2001** is the ohmic contact layer (e.g., ohmic cathode contact region **152** in FIG. 4) that would be located beneath the substrate. Energy band diagram **2001** also illustrates an ideal Schottky barrier potential ϕ_{SB} resulting from the metal-semiconductor junction. Not shown are Schottky barrier energy lowering effects due to image forces and the like—which while important, do not detract from the main conclusion discussed above.

[0281] In this example of FIGS. 9B-9C, the doping concentration of the SiC substrate is highly n-type (i.e., n+), e.g., from about 10^{19} to about 10^{21} cm⁻³ and is therefore highly electrically conductive. The backside of the substrate is substantially C-polar if the epilayer is deposited on the Si-polar surface. Ohmic contact to the C-polar surface of SiC substrate can be formed using Ti, Ti/Al or Nickel. Nickel-silicides can be formed by high temperature annealing wherein a portion of the deposited Ni and Si from the SiC surface region react to form Ni_xSi_{1-x}. In some cases, high temperature processing may also carbonize the SiC surface which is disadvantageous for creating Ohmic contacts. In an example, a thin Si-based layer (~1-3 nm) can be deposited on the SiC surface followed by the deposition of Ni. This prevents Si loss from the SiC surface and enhances the Ni_xSi_{1-x} formation. That is, a n+SiC/Si/Ni stack can be thermally treated post deposition to form n+SiC/Ni_xSi_{1-x}/Ni. Alternatively, the Ni and or Si can be deposited in-situ at elevated temperature. In order to increase the Schottky barrier potential ϕ_{SB} beyond that possible with a simple metal-semiconductor junction, it is further possible to insert an intermediate layer (e.g., intermediate layer **170** in FIG. 8A or 8B), as discussed further herein.

[0282] FIG. 10 is a plot **3000** showing the breakdown voltage V_{br} of β -Ga₂O₃ as a function of n-type doping concentration for selected thicknesses of $L_{drift}=0.3, 1, 3, 10, 30$ μ m. For a given slab thickness the breakdown voltage decreases with increasing doping density, with inflection points **3005, 3010, 3015, 3020** and **3025** indicating the optimal choice of (L_{drift}, V_{br}, N_d). For example, selecting $L_{drift}=3$ μ m an optimal configuration is $N_d=5\times10^{16}$ cm⁻³ capable of $V_{br}=1.2$ kV.

[0283] In general terms, the ON-resistance of the vertical multilayered semiconductor device in forward bias will be determined primarily by N_d whereas the breakdown voltage V_{br} will be determined by N_d and the thickness of the drift region in reverse bias.

[0284] FIG. 11A is a plot **4000** of the experimental work functions for various high work function metals (shown on the x-axis) favorable for forming Schottky barrier junctions to a ultrawide bandgap semiconductor oxide material, as determined in the present disclosure. A selection of metal work function energies are plotted on the left-hand y-axis, where Cu has the lowest value and Osmium the highest. Defining the vacuum energy E_{vac} as zero, then the work function energy WF of a metal is understood as having values $WF<0$. Relative to E_{vac} the electron affinity X_A of the conduction band edge for a given semiconductor can be used to compare the metal-semiconductor absolute energy alignment. The ideal Schottky barrier height (ϕ_{SB}) for various metals in FIG. 11A for the case of metal/ β -Ga₂O₃ (-201) oriented clean surface is plotted on the right-hand y-axis as open circle data points. Selenium (Se) and Osmium (Os) provide the highest ϕ_{SB} ~1.4 eV. In practice, all of the metals listed in this figure can be used to form Schottky barrier diodes, in particular Ni, Pt and Pd are relatively stable.

[0285] A unique feature described in the present disclosure is the β -Ga₂O₃ (-201) oriented surface. As discussed later, the surface can be advantageously modified, such that, a surface depletion charge can be engineered to produce improved Schottky barrier diodes. By depleting a small depth of the n-doped β -Ga₂O₃ (-201) surface, a large increase in relative ϕ_{SB} can be achieved for all the metals as shown by the diamond data points. By careful inspection of FIG. 11A it can be seen that Cu-metal/depleted β -Ga₂O₃ (-201) may achieve a similar ϕ_{SB} as Ir-metal/clean β -Ga₂O₃ (-201). Clean and depleted surfaces are discussed later in this disclosure. Any of the metals in FIG. 11A can be used to form an oxide layer between the Schottky metal contact layer and the drift layer, as described further herein. For example, the Schottky metal can include platinum, the drift layer can include Ga₂O₃ (or a different epitaxial oxide, as described herein) and an intermediate layer including PtO₂ can be formed between the Schottky metal contact layer and the drift layer. A metal oxide (e.g., PtO₂, or a metal oxide containing any of the metals in FIG. 11A) for an intermediate layer can be deposited as an oxide (e.g., using MBE, ALD, MOCVD, or another epitaxial or non-epitaxial growth method), or as a metal that is subsequently oxidized to form the metal oxide.

[0286] FIGS. 11B and 11C plot the forward bias current-voltage (I-V) behavior of a diode formed as shown in FIGS. 9B and 9C using a Schottky barrier metal (SBM)=Ni, deposited on a depleted surface of β -Ga₂O₃ (-201) n-type drift layer having various doping concentrations N_{drift} and thickness of 2 μ m. The structure is deposited upon a Si-polar 4H-SiC n-type substrate. Referring to FIG. 11B in which the

room temperature $T=25$ C, the semilog I-V plot shows the subthreshold current increasing with forward bias voltage V_F until turn-on at $\sim V_F=1$ V. Thermionic emission (TE) transport of electrons across Φ_{SB} dominates the forward conduction current density J_{TE} with respect to the applied forward bias voltage V_F and follows the equation:

$$J_{TE} = A^* T^2 \exp\left(\frac{q\Phi_{SB}}{k_B T}\right) \left[\exp\left(\frac{qV_F}{nk_B T}\right) - 1 \right] = J_{sat} \left[\exp\left(\frac{qV_F}{nk_B T}\right) - 1 \right] \quad (\text{Eqn 1})$$

where the Richardson's constant can be calculated explicitly as $A^*=4\pi m^* k_B^2 q/h^3$, m^* is the electron effective mass, n ideality constant, q electron charge and k_B Boltzmann constant. It follows that a strong temperature dependence is expected for TE dominated conduction. For the example of FIGS. 11B-11C involving Ni/clean β -Ga₂O₃, the drift layer electron effective mass is $m^*=0.30 m_0$ and results in $A^*=36$ A/cm². In practice, the ideality factor n can be extracted from fabricated diodes and fitted to the subthreshold slope 4005 in FIG. 11B using Equation 1 above. Ideality factors ranging from $1 \leq n \leq 3$ are possible, such as $1 \leq n \leq 2$ in some examples, and are dependent upon N_{drift} .

[0287] FIG. 11C shows the linear I-V plot for the example of FIG. 11B, wherein the ON-resistance decreases with increasing n-type doping N_{drift} as shown by the differential resistance after the turn-on voltage 4010 (dI/dV)⁻¹.

[0288] FIG. 12 is a simulated energy band diagram 5000 of the conduction band edge of the vertical multilayered semiconductor device having the configuration 5050 that comprises a substrate layer of SiC (0001) (i.e., SiC with a (0001) orientation), a TL of n+ β -Ga₂O₃ (-201), a drift layer formed from β -Ga₂O₃ (-201) (i.e., 3-phase Ga₂O₃, with a (-201) orientation) and a metal layer formed of Ni (111) (i.e., Ni with a (111) orientation). In this example, the Schottky potential barrier has a value of approximately 1.15 eV and the drift layer doping is varied in the range of $1 \times 10^{15} \leq N_{drift} \leq 5 \times 10^{17}$ cm⁻³ as shown by the five curves plotted in energy band diagram 5000. The depletion layer width is strongly controlled by N_{drift} as shown, resulting in a quantum mechanically thin barrier for $N_{drift}=5 \times 10^{17}$ cm⁻³ (curve 5010) wherein the carrier transport will be a combination of TE and quantum mechanical tunneling. That is, the total current density across the barrier J_{tot} will have two components, a temperature and peak energy barrier dependent thermionic current J_{TE} and a quantum mechanical tunneling component J_{tun} , such that $J_{tot}=J_{TE}+J_{tun}$. For lower values of $N_{drift} \rightarrow 1 \times 10^{15}$ cm⁻³ the width of the depletion barrier reduces $J_{tun} \rightarrow 0$. Conversely, quantum tunneling through the potential barrier becomes particularly important for designing high voltage reverse bias operation.

[0289] FIG. 13 is a simulated energy band diagram 6000 of the conduction band edge of the vertical multilayered semiconductor device having configuration 6050 which is similar to that depicted in FIG. 12 except that it includes an additional intermediate layer (IL) having a thickness of 100 Angstroms and formed of single crystal MgO (111) (i.e., MgO with a (111) orientation). The simulation shows that the inclusion of the intermediate layer in this example increased the Schottky potential barrier to a value of approximately 3.1 eV. Of particular interest is MgO which exhibits a small electron affinity when compared to Ga₂O₃ and SiC. As expected, when the drift layer doping is varied between $1 \times 10^{15} \leq N_{drift} \leq 5 \times 10^{17}$ cm⁻³ the depletion layer

width is strongly controlled, however unlike the device of FIG. 12, the IL presents a fixed quantum mechanical tunnel barrier for electron transport. The increased barrier potential of the device in FIG. 13 further can be used to tune the turn-on voltage (or threshold voltage) of the diode. For example, selecting an IL oxide having a large bandgap and small electron affinity creates a large SB potential and thus can be used to engineer the turn-on voltage for the diode. This is particularly advantageous for creating high voltage diodes that are less susceptible to low voltage noise. Large barrier potentials are also advantageous for decreasing the reverse leakage currents at high reverse electric bias and high electric fields.

[0290] Intermediate layer 170 can comprise an epitaxial semiconductor oxide material formed on the epitaxial drift layer. In some cases, intermediate layer 170 comprises an epitaxial oxide material, and the structure further includes an epitaxial metal layer (e.g., metal layer 130 in FIG. 8B) formed on the epitaxial intermediate layer 170. Intermediate layer 170 can include any of the oxide materials described herein, for example, any of the compositions of Materials 1-7. Intermediate layer 170 can contain a single semiconductor oxide composition, or more than one sub-layer with different semiconductor oxide composition (e.g., the sub-layers can form a bilayer, a multilayered structure, a superlattice, or a chirp layer), any of which can be chosen from Materials 1-7. In some cases, intermediate layer 170 comprises a graded composition (e.g., a linear, a step-wise, or a chirped composition gradient) comprising one or more of Materials 1-7. In some cases, intermediate layer 170 comprises one or more of Materials 1-7 that is doped with a single dopant concentration, or with a dopant concentration that varies in depth in intermediate layer 170.

[0291] In some examples, the intermediate layer (e.g., intermediate layer 170 in FIG. 8A or 8B) may be formed from or comprise a material chosen from one or more of the following materials:

- [0292] epitaxial single crystal, polycrystalline or amorphous Al₂O₃
- [0293] epitaxial single crystal, polycrystalline or amorphous NiO
- [0294] epitaxial β -(Al_xGa_{1-x})₂O₃ with $0 \leq x \leq 0.3$ (see FIG. 8)
- [0295] epitaxial α -(Al_xGa_{1-x})₂O₃ with $0 \leq x \leq 1$ (see FIG. 7)
- [0296] epitaxial (Mg_xZn_{1-x})_z(Al_yGa_{1-y})_{2(1-z)}O_{3-2z}, where $0 \leq x, y, z \leq 1$
- [0297] epitaxial Zn_xGa_yO_z where $0 \leq x, y, z \leq 1$
- [0298] epitaxial Zn_xAl_yO_z where $0 \leq x, y, z \leq 1$
- [0299] epitaxial Mg_xGa_yO_z where $0 \leq x, y, z \leq 1$
- [0300] epitaxial Mg_xAl_yO_z where $0 \leq x, y, z \leq 1$
- [0301] epitaxial Ni_xGa_yO_z where $0 \leq x, y, z \leq 1$
- [0302] epitaxial Ni_xAl_yO_z where $0 \leq x, y, z \leq 1$
- [0303] epitaxial (Ni_xMg_{1-x})_yGa_{2(1-y)}O_{3-2y} where $0 < x < 1$ and $0 < y < 1$
- [0304] epitaxial (Mg_xNi_{1-x})_z(Al_yGa_{1-y})_{2(1-z)}O_{3-2z}, where $0 \leq x, y, z \leq 1$.

[0305] In some examples, the drift layer may have the same compositional makeup as the intermediate layer but with a different doping concentration. In some examples, the drift layer may have a different compositional makeup as the intermediate layer and have a different doping concentration.

[0306] Other crystal face orientations of SiC (e.g., in particular 4H-SiC) may also be used advantageously for the epitaxial growth of other oxide compositions and polytopes for the devices, structures and methods described herein. For example, A-plane, R-plane and M-plane 4H-SiC substrates can be prepared from bulk grown single crystal material by preferential dicing along specific crystal planes followed by chemical mechanical polishing (CMP) to form an epitaxy ready surface. Furthermore, some examples may include epitaxially depositing alpha-phase Ga₂O₃ films on the aforementioned vicinal SiC substrate surfaces. The ability to form α -Ga₂O₃ provides at least two improvements applied to the present disclosure. Firstly, at room temperature, α -Ga₂O₃ exhibits a larger fundamental bandgap energy than β -Ga₂O₃ ($E_g(\alpha\text{Ga}_2\text{O}_3)=5.4\text{ eV} > E_g(\beta\text{Ga}_2\text{O}_3)=4.7\text{ eV}$) which can advantageously increase the critical electric field possible and thus improve diode performance. Second, α -Ga₂O₃ grown on the A-, R- and M-planes enables a full alloy range to be realized when alloyed with Al, such that, $\alpha(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ for all alloy ranges $0 \leq x \leq 1$ are possible, including $E_g(\alpha\text{Al}_2\text{O}_3)=8.6\text{ eV}$. Such a large bandgap engineering range for creating heterojunctions dramatically improves the performance of diodes (and other devices, such as p/n junction diodes, transistors, switches, varactors, or light emitting devices) that are possible, compared to conventional devices.

[0307] FIGS. 14 and 15 are examples of other vertical multilayered semiconductor device configurations including an intermediate layer in accordance with the present disclosure. FIG. 14 shows an example of a structure 7000 with an SiC substrate oriented in the A-, R-, or M-plane, an α -Ga₂O₃ drift layer, an $\alpha(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ intermediate layer with $0 \leq x \leq 1$, and a metal layer that forms a Schottky barrier. A graded alloy of $\alpha(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ possessing a growth direction dependent alloy content $x(z)$ is also possible for the intermediate layer. FIG. 15 shows an example of a structure 8000 with an SiC substrate oriented in the C-plane, a β -Ga₂O₃ drift layer, a $\beta(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ intermediate layer with $0 \leq x \leq 0.3$, and a metal layer that forms a Schottky barrier. In practice, the highest Al content possible to preserve single crystal beta-phase is $x \sim 0.3$ or in some cases $x=0.4$, which represents the largest bandgap that can be implemented as a heterojunction to $x=0$, i.e., $\beta\text{-Ga}_2\text{O}_3/\beta(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$.

[0308] FIGS. 16 and 17 are simulated energy band diagrams of two example vertical multilayered semiconductor device configurations having intermediate layer (IL) thicknesses of approximately 50 Å and 150 Å, respectively. Both devices comprise a 4H-SiC substrate, a Ga₂O₃ drift layer, an MgO intermediate layer, and a Ni metal layer. The drift layer doping is varied in the range of $1 \times 10^{15} \leq N_{\text{drift}} \leq 5 \times 10^{17} \text{ cm}^{-3}$ as shown by the five curves plotted in energy band diagrams 9000 and 10000. In some examples, the thickness of the intermediate layer may be configured to control the quantum mechanical tunnelling current, for example, to control the quantum mechanical tunnelling current in reverse bias near the reverse breakdown voltage. In some examples, the thickness of the intermediate layer may be configured to mitigate the Schottky barrier height reduction arising from metal induced gap states at the metal and intermediate layer interface. In these examples of FIGS. 16 and 17, the IL is selected as an UWBG MgO, showing the band bending of the conduction band edge with various drift region doping for the two examples of MgO thicknesses (50 Å in FIGS. 16

and 150 Å in FIG. 17). FIG. 16 represents a thin quantum mechanical tunneling barrier, while the thicker barrier of FIG. 17 reduces or inhibits quantum mechanical tunneling. This design option is advantageous for managing the reverse leakage at high voltages and electric fields.

[0309] FIG. 18 shows a simplified example of a representative current-voltage characteristic curve 11000 for a vertical multilayered semiconductor device in accordance with the present disclosure. In reverse bias, the leakage current region before the voltage breakdown (V_{br}) will itself comprise three regions:

[0310] thermionic emission region (“Thermionic Emission Region”) occurring at low to medium electric fields where the current will arise from charge carriers having enough energy to overcome the potential barrier;

[0311] combined quantum tunnelling and thermionic emission region (“Combined Tunnelling & Thermionic Region”) occurring at medium electric fields where the current will arise from a combination of charge carriers having enough energy to overcome the potential barrier (i.e., thermionic emission) and charge carriers tunnelling through the potential barrier; and

[0312] quantum tunnelling region (“QM Tunnelling Region”) occurring at high electric fields where the current will arise from charge carriers tunnelling through the potential barrier.

[0313] In conventional devices the practical reverse bias breakdown voltage (V_{br}) is not solely determined by the breakdown strength of the drift region but also by the device structure, specific material properties utilized and possible alternate current leakage paths when subjected to extreme electric fields.

[0314] For example, a simple metal semiconductor Schottky barrier presents only a limited quantum mechanical barrier height and an asymmetric width potential. In some cases, tunnelling currents may cause devices with Schottky barrier junctions at extreme high electric fields to exhibit lower reverse breakdown voltage than expected due to barrier lowering effects (BLE).

[0315] The electric field values of the breakdown voltage, and the values of the low, medium and high electric fields in FIG. 18 can also vary due to the materials used and device structures. For example, the low electric fields can be less than 100 V/cm, or less than 1000 V/cm, the medium electric fields can be between 100 V/cm and 1000 V/cm, or between 1000 V/cm and 10,000 V/cm, and the high electric fields can be greater than 1000 V/cm, or greater than 10,000 V/cm.

[0316] The reverse-bias current comprises a superposition of the thermionic J_{TE} and quantum mechanical tunneling J_{nm} current components, such that:

$$J_{nm} = \frac{A^* T}{k_B} \int_{E_c^{min}}^{E_c^{max}} T(E) \times \ln \left[1 + \exp \left(\frac{E - E_f}{k_B T} \right) \right] dE \quad (\text{Eqtn 2})$$

where $T(E)$ is the quantum mechanical (QM) transmission probability as a function of incident electron energy relative to the potential barrier, and E_f is the Fermi energy reference to the conduction band energy. The integration limits $E_c^{min, max}$ span the range over incident quasi-ballistic electron injection into the structure.

[0317] FIGS. 19A and 19B are plots 12000 and 12100, respectively, of the QM tunnelling current transmission

coefficient/probability $T(E)$ as a function of injected electron energy relative to the Fermi level ($E_F=0$). The insets **12001** and **12101** show the respective energy barriers in the conduction band edges in these examples, and an electron tunneling through each of the barriers. The curves are calculated using a finite-element-method for piecewise approximation of the spatial potential coupled to a complex transfer matrix for each subregion, taking into account the real and imaginary parts of the reflected and transmitted wavefunctions. FIG. **19A** corresponds to $T(E)$ **12010** for the specific band structure shown in FIG. **9C** for the case of $N_{drift}=5 \times 10^{16} \text{ cm}^{-3}$, showing significant SB penetration below the potential barrier maximum (indicated by the arrow **12015**), and approaching unity above the barrier. Note $T(E)<1$ even for incident energy above the potential maximum. The triangular potential presents an asymmetric barrier which suppresses QM interference between reflected and transmitted electron wavefunctions. Arrow **12015** corresponds to the height of the Schottky barrier in this example.

[**0318**] Introducing a wider bandgap IL between the Schottky metal and Ga_2O_3 drift layer can be used to effectively increase the potential barrier of the rectifying contact. In FIG. **19B**, the IL is selected from MgO ($E_g=7.8 \text{ eV}$) having an electron affinity of $X_A(\text{MgO})=-1.0 \text{ eV}$ providing a maximum potential barrier **12115** of 3.0 eV . The drift layer doping is found to have small effect on the $T(E)$ features, whereas the MgO thickness has dramatic influence on the oscillatory behavior observed in FIG. **19B**. The ‘soft turn-on’ $T(E)$ curve **12110** is for the thin MgO layer $L_{IL}(\text{MgO})=5 \text{ nm}$, whereas the sharp ‘turn-on’ and narrow resonances $T(E)$ curve **12120** is for the case of thicker $L_{IL}(\text{MgO})=10 \text{ nm}$. Referring again to Eqn (2) the $T(E)$ can be used to calculate the reverse leakage current with respect to the surface electric field. It is found the reverse leakage current reduces significantly with the introduction of an UWBG IL and is therefore advantageous for application to high voltage diodes, for example to increase the forward-reverse (ON-OFF) ratio.

[**0319**] Yet a further example of an IL comprising a $\text{ZnGa}_2\text{O}_4(111)$ oriented layer is shown in FIGS. **19C** and **19D** inserted into the device of: Ni metal/IL= ZnGa_2O_4 NID $10 \text{ nm}/\text{Ge}: \beta\text{-Ga}_2\text{O}_3$ $5 \times 10^{16} \text{ cm}^{-3}/\text{TL}=\text{Ge}: \beta\text{-Ga}_2\text{O}_3$ $1 \times 10^{17} \text{ cm}^{-3}/1 \times 10^{19} \text{ cm}^{-3}$ 4H-SiC Si-polar substrate. FIG. **19C** shows the band diagram **12200** along a growth direction, z , and a close up diagram **12300** of the conduction band near the Schottky barrier (at $z=0 \text{ nm}$) in FIG. **19D**. The IL effectively increases the Schottky barrier potential which directly influences the diode performance as shown in FIGS. **19E** and **19F**. The semilog forward bias I-V curve of plot **12400** in FIG. **19E** shows the shift of the subthreshold slope to higher voltage for the metal-intermediate-substrate structure (MIS) (IL= $\text{Zn}_2\text{Ga}_2\text{O}_4$) when compared to the Ni/ $\beta\text{-Ga}_2\text{O}_3$ diodes (M-S curves) for the case of a clean and depleted $\beta\text{-Ga}_2\text{O}_3$ (~ 201) surface. The plot **12500** of FIG. **19F** shows the linear I-V forward bias comparison of the 3 cases from FIG. **19E**. Introducing the IL also has the effect of increasing the impedance of the device when in the conducting mode.

[**0320**] The inclusion of an intermediate layer in various embodiments provides a number of advantages including:

[**0321**] Improving the turn-on voltage to higher values if the bandgap of the intermediate layer is greater than that of the drift region;

[**0322**] Reducing the prevalence of defects and/or metal induced gap states (MIGS) between the metal and drift layers which reduce the height of the Schottky potential barrier from its theoretically expected value. While there may be MIGS between the intermediate layer and the metal layer, the effect on the barrier height is reduced since this interface is separated from the semiconductor/metal Schottky junction interface;

[**0323**] Increasing the reverse breakdown voltage due to both thermionic emission and quantum tunnelling; and

[**0324**] The wider bandgap intermediate layer spatially separates the metal induced gap states (MIGS) from the active semiconductor oxide material of the surface of the drift region which assists in maintaining Schottky barrier potential and reducing reverse leakage currents.

[**0325**] Devices and methods of the present disclosure utilize a unique insight identifying that the high field region performance may be improved under reverse bias by configuring the intermediate layer to present further limitations to the tunnelling current, as a result further improving the reverse breakdown voltage of a device.

[**0326**] In any of the examples described in the present disclosure, the intermediate layer can include a repeating characteristic. The repeating characteristic can be a repeating compositional variation (e.g., as in a superlattice structure), or a repeating doping concentration variation (e.g., in the concentration of extrinsic dopant species, or defects or vacancies within a crystalline layer).

[**0327**] FIG. **20** is a figurative sectional view of a vertical multilayered semiconductor device **14000** where a repeating characteristic of the intermediate layer (IL) **1450** or a transition layer (TL) **1460** is configured by forming the IL and or TL as a superlattice or multilayered epitaxial stack. The substrate **1400** is SiC such as the 4H polytype. The substrate can have a specific vicinal surface orientation, namely, C-, A-, R- or M-plane as known in the literature. The vicinal plane may be on-axis or intentionally miscut toward a particular crystal plane. For example, the miscut angle for C-plane may range from 0 to 6 degrees. The transition layer **1460** may be a semiconducting oxide or non-oxide or a combination. For example, the transition layer **1460** may comprise Silicon-Germanium-Carbide ($\text{Si}_x\text{Ge}_y\text{C}_{1-x-y}$) enabling in-plane lattice constant and strain tuning for the drift region composition. Other optional TL and IL compositions may include a polar or non-polar semiconductor. For example, nitrogen-polar or metal-polar Aluminum-nitride (AlN) may be formed.

[**0328**] In the example of FIG. **20**, the superlattice within at least a portion of the IL (or TL) is formed from two repeating sublayers **14010** and **14020** (or sublayers **14030** and **14040** for the TL) that vary in thickness traversing through the intermediate layer. In another example, the superlattice may form a digital alloy. In another example, the superlattice may be formed from three or more repeating sublayers of either constant or varying thickness. In various examples, the superlattice may be periodic or aperiodic.

[**0329**] The use of A-, R- and M-plane 4H-SiC enables the epitaxial and selective growth of alpha-phase $\alpha\text{-(Al}_x\text{Ga}_{1-x})_2\text{O}_3$ with space group R3c. Furthermore, the epitaxial growth of $\alpha\text{-(Al}_x\text{Ga}_{1-x})_2\text{O}_3$ is possible over the complete alloy range $0 \leq x \leq 1$. The TL or IL may therefore comprise a superlattice of $\alpha\text{-(Al}_x\text{Ga}_{1-x})_2\text{O}_3/\alpha\text{-(Al}_y\text{Ga}_{1-y})_2\text{O}_3$ $0 \leq x \leq 1$ and $0 \leq y \leq 1$, such that $x \neq y$. This enables the bandgap to be tuned from 4.7 eV ($x=0$) to 8.6 eV ($x=1$) with approximately 4 eV of

conduction band discontinuity to be imposed in electron confinement and transport designs. The lattice constant mismatch between the $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3/\alpha-(\text{Al}_y\text{Ga}_{1-y})_2\text{O}_3$ creates biaxial strain at the heterointerface(s) which can be managed by appropriate layer thickness selection.

[0330] FIG. 21A is a simplified energy band diagram 15000 of a vertical multilayered semiconductor device with a structure similar to structure 14000 in FIG. 20 and comprising an intermediate layer formed of a finite period superlattice comprising $\text{ZnGa}_2\text{O}_4(111)$ and $\text{B-Ga}_2\text{O}_3(-201)$ sublayers. Other numbers of layers, and/or other combinations of materials for the superlattice are also possible. The full conduction and valence energy band diagram 15000 shows the stepped potential barrier (ϕ_s) arising from the superlattice structure of the intermediate layer and a Ga_2O_3 drift layer. FIG. 21B shows a close-up energy band diagram 15020 in the vicinity of the Metal/SL IL/Drift layer with an effective $\phi_s=2.25$ eV and $N_{\text{drift}}=5 \times 10^{16} \text{ cm}^{-3}$. In one example, the superlattice structure is configured so that the equivalent potential barrier is increased, resulting in the quantum mechanical tunnelling transmission coefficient being reduced and thermionic emission onset tuned to a higher turn-on voltage. The increased effective barrier potential energy can also improve the high temperature performance of the device. The leakage current associated with quantum tunnelling through the potential barrier in reverse bias can also be reduced using such structures to increase the effective barrier potential energy, as discussed previously.

[0331] In practice, even though Schottky barriers and UWBG intermediate layer can be realized in the devices, structures, and methods described herein, there exists a phenomenological potential barrier lowering effect (BLE) under the influence of high electric fields. Notably, BLE is correlated with both the bandgap energy and dielectric constant of the specific materials used. FIG. 21C schematically shows the energy diagram 15030 of a semiconductor slab (oxide 15032) with thickness L_{IL} , comprising a bandgap E_g (distance between dashed lines), further sandwiched between two metals 15031 and 15033, thus forming a zero bias ($V=0$) potential barrier at the heterojunction ϕ_B . The Metal-Oxide-Metal (MOM) structure upon application of a bias generates an electric field $\mathcal{E}_z=V/L_{\text{IL}}$ that alters the band diagram as shown, wherein the rectangular barrier ($V=0$) becomes a distorted parallelogram 15035 (e.g., with some band curvature) ($V<0$).

[0332] An electron injected from the left-hand side metal electrode into the oxide layer induces an equal but opposite positive image charge in the metal. If the electron is injected a distance $z>0$ into the oxide, the Coulomb image force F_{im} generated between the charges separated by a distance $2z$ is:

$$F_{\text{im}}(z) = \frac{-q^2}{4\pi\epsilon_{\text{opt}}(2z)^2}$$

which in turn generates an electrostatic potential $V_{\text{im}}(z)$ given by:

$$V_{\text{im}}(z) = - \int_{-\infty}^z F_{\text{im}}(z) dz = \frac{-q^2}{16\pi\epsilon_{\text{opt}} z}$$

Clearly the MOM oxide barrier spatial potential under bias is composed of the: (i) rectangular barrier potential; (ii) the linear electric field; and (iii) the image potential, such that:

$$V_{\text{Oxide}}(z) = \phi_B - q\mathcal{E}_z z - \frac{q^2}{16\pi\epsilon_{\text{opt}} z}$$

[0333] FIG. 22 shows the image force band bending of the Schottky barrier for an example comprising an ideal $\text{Ni}/1 \times 10^{16} \text{ cm}^{-3} \text{ Ga}_2\text{O}_3/\text{n-Ga}_2\text{O}_3/\text{n}^+4\text{H-SiC}$ device. FIG. 23 shows closeup of the band bending near the peak of the potential. For UWBG oxides the sensitivity analysis shows the BLE is relatively small in comparison to the available ϕ_B .

[0334] The BLE is responsible for reducing the peak barrier potential ϕ_B by an amount $\Delta E_{\text{barrier}}$ which can be quantitatively expressed in terms of the high frequency (optical) dielectric constant and the applied field.

[0335] The condition $dV_{\text{oxide}}/dz=0$ yields the maximum and is located at

$$z_m = \left(\frac{q}{16\pi\epsilon_{\text{opt}}\mathcal{E}_z} \right)^{-1}$$

away from the heterointerface, such that,

$$\Delta E_{\text{barrier}} = \sqrt{\frac{q^3\mathcal{E}_z}{4\pi\epsilon_{\text{opt}}}}$$

Inclusion of the BLE into the metal Schottky barrier device example is shown in FIG. 22. This example case is for a Schottky barrier metal (SBM) having $\phi_B=1.3$ eV on a 1250 nm drift region comprising $N_{\text{drift}}=1 \times 10^{16} \text{ cm}^{-3} \beta\text{-Ga}_2\text{O}_3(-201)$. A transition layer of $\text{n}+\beta\text{-Ga}_2\text{O}_3(-201) 1 \times 10^{17} \text{ cm}^{-3}$ is deposited on a nitrogen-doped $\text{n}+1 \times 10^{18} \text{ cm}^{-3} 4\text{H-SiC}$ substrate. The zero bias (0V) and reverse bias (4V and 8V) band diagrams are plotted showing the change in the depletion width. FIG. 23 shows a close-up view of the conduction band of FIG. 22 in the vicinity of the Schottky barrier peak for a series of applied reverse bias voltages and resulting surface electric-fields. The band lowering effect and band bending are shown.

[0336] FIG. 24 plots the extracted values of the effective barrier height lowering energy $\Delta E_{\text{Barrier}}$ (curve 15110 and right-hand y-axis) and the effective barrier width W_{barrier} (curve 15120 and left-hand y-axis) as a function of the surface electric-field E_{F_z} . For $E_{F_z}=0$ MV/cm the built-in electric field is ~ 52 kV/cm due to the depletion field.

[0337] Referring again to Eqtn. 2, the tunneling current is also dependent on the temperature as well as the transmission coefficient of electrons through the barrier as a function of the surface electric-field (refer to FIG. 23).

[0338] FIGS. 25 and 26 calculate the quantum mechanical transmission coefficient for two device configurations of: (i) $\text{Ni SBM}/5 \times 10^{16} \text{ cm}^{-3} \text{ Ga}_2\text{O}_3/\text{n}+4\text{H-SiC}$ in FIG. 25; and (ii) $\text{Ni SBM}/5 \text{ nm MgO IL}/5 \times 10^{16} \text{ cm}^{-3} \text{ Ga}_2\text{O}_3/\text{n}+4\text{H-SiC}$ in FIG. 26, the device of FIG. 26 incorporating an intermediate layer of UWBG 5 nm MgO. The transmission coefficient $T(E_e, E_{F_z})$ is dependent on the incident electron energy E_e relative to the Fermi energy (E_{Fermi} is taken as zero), and the potential

profile is dependent on the surface electric-field E_{Fz} . The tunneling current integral is performed over E_e ranging from 0 to the maximum value of the potential energy shown in FIG. 23 yielding the curves in FIGS. 27 and 28.

[0339] FIGS. 27 and 28 plot the calculated contributions of the thermionic current (J_{therm}) and the tunneling current (J_{tun}) to the total current (J_{tot}) for each of the two device structures of FIGS. 25 and 26. Three example operating temperatures are shown, viz., $T_s=300K$, $400K$ and $500K$, indicative of the operating temperatures discussed above for the thermal performance of the Oxide/SiC hybrid device.

[0340] Close inspection of the curves detailed in FIG. 27 demonstrates that at low surface electric fields $E_z \leq 0.5$ MV/cm the reverse leakage current of the Schottky barrier diode is primarily determined by the thermionic current component. Above this critical surface electric field $E_z > 0.5$ MV/cm the temperature dependent tunneling current dominates and indeed swamps the contribution from the thermionic emission. This indicates the design of the heterojunction Schottky barrier is important for managing and limiting the reverse leakage current for high surface electric field regime.

[0341] FIG. 28 shows the analogous plots of the calculated reverse leakage current for the case of the diode structure intentionally incorporating a UWBG 5 nm MgO intermediate layer between the Schottky metal and the Ga_2O_3 drift region. Inspection of the curves in FIG. 28 shows a lowering of the critical surface electric field where thermionic emission and tunneling components dominate, $E_z \leq 0.25$ MV/cm. However, the absolute reverse leakage currents for both thermionic and tunneling components are dramatically reduced compared to the case of FIG. 27 where there is no IL present. Therefore, the present disclosure shows that an UWBG intermediate layer can be advantageous for improving the reverse leakage performance of Schottky barrier diodes as disclosed herein.

[0342] Ab-initio theoretical calculations and experimental data yield a trend of the static dielectric constant (low frequency) ϵ_r and (high frequency) optical dielectric constant ϵ_r^{opt} with respect to the bandgap E_g of non-oxide and oxide materials. Density Functional Theory was used to simulate the energy band structure and dielectric dispersion of a variety of crystal structures. A selection of oxide materials is plotted in FIG. 29A showing the dielectric constants in general decrease (reduce) with increasing band-gap energy E_g . The static ϵ_{Static} and optical $\epsilon_{Optical}$ dielectric constants exhibit similar trends but have differing bowing, as shown.

[0343] FIG. 29B shows the expected variation of $\Delta E_{barrier}$ with respect to E_g for three cases of applied electric field across a slab of thickness of 5 μm with selected oxide semiconductors. Barrier lowering is observed to be higher for large band gaps due in part to dynamic screening which decreases with increasing band gap. Further, the high-frequency (optical) dielectric constant should be used to model screening associated with barrier lowering since the time associated with an electron transiting through the barrier is very short. The high surface electric fields can be accommodated using an intermediate layer as described for application to high voltage rectifiers.

[0344] FIG. 29C discloses further trends available for advantageous incorporation of the intermediate layer oxide materials into the diode structure. A figure of merit for the interface quality S_{int} (interface state parameter) of a given

surface state can be estimated also by relation to the dielectric constant trend. Defining for clarity

$$S_{int} = \frac{1}{1 + 0.1(\epsilon_r^{opt} - 1)^2}$$

where $0 \leq S_{int} \leq 1$, and $S_{int}=1$ represents an ideal surface having a surface state density=0, and $S_{int}=0$ represents infinite surface state density. An empirical fit can also be estimated for the induced gap states resulting in a minimum surface state density D_{SS}^{min} and is also related to the dielectric constant by:

$$D_{SS}^{min} = \frac{C_{dipole}}{q} \left(\frac{1}{S_{int}} - 1 \right) \text{ where } C_{dipole} = \frac{\epsilon_r^{opt}}{\delta},$$

such that, a typical dipole is across at least one unit cell of the material, $\delta \sim 0.43$ nm for MgO.

[0345] Referring again to both FIGS. 19C and 21A, the heterostructure design incorporating an intermediate layer (e.g., MgO, or other material(s)) between the metal and Ga_2O_3 enables a reduction in surface state density while maintaining a high barrier potential required for forward bias rectification and low reverse leakage current.

[0346] In another example, a repeating characteristic of the intermediate layer or drift layer is configured by introducing a repeating spatial modulation of the doping concentration as a function of growth direction ("z"). For example, the intermediate layer or drift layer can have a single material composition, and the conduction band potential can be modified using a repeating spatial modulation of doping concentration. In one example, the spatial extent of the doping region is very small and has a very high doping concentration. In another example, the doping region extends over one or more unit cells of the host composition but less than about 10 nm. In another example, the doping region comprises an extremely high doping concentration, of the order of $N_{\delta}^{3D} = 10^{18} - 10^{20} \text{ cm}^{-3}$. In yet another example, the doping region is called a delta-doped region wherein the thickness over which is doped is about 10-20 nm or less. In some cases, the doping region thickness is of the order of 1 or more monolayers of the host material composition. For very thin doping region thickness an equivalent two-dimensional doping density N_{δ}^{2D} can be defined related to the bulk-like N_{δ}^{3D} , such that,

$$N_{\delta}^{3D} = (N_{\delta}^{2D})^{3/2}$$

That is, $N_{\delta}^{3D} = 10^{18} \text{ cm}^{-3}$ is equivalent to an areal density $N_{\delta}^{2D} = 10^{12} \text{ cm}^{-2}$.

[0347] In one example, the repeating spatial modulation of doping concentration may have a homoepitaxial superlattice type structure similar to superlattices described above except that instead of the sublayers having different compositions, in this example the sublayers will be of constant composition but have different doping concentration.

[0348] In another example, a repeating characteristic of the intermediate layer includes both repeating spatial modu-

lation of the doping concentration and repeating changes in the composition of the intermediate layer.

[0349] An example of a homoepitaxial delta-doped superlattice (δ -SL) forming the drift region of an Schottky barrier diode **16000** is shown in FIG. 30. Nickel metal forms a SB (“SB metal”) to a delta-doped SL drift region (“ δ -doped SL”) formed by using either Sn, Ge or Si shallow dopants into a host β -Ga₂O₃ composition (i.e., Ga₂O₃ is used throughout the drift region). Shown in plot **16100** is a sheet density doping $N_d^{2D}=10^{12}$ cm⁻² for two different periods of a periodic [δ -doped Ga₂O₃/NID Ga₂O₃] superlattice (δ -SL) extending over 2 μ m forming the drift layer of the overall structure, Ni/ δ -doped SL drift/n+ TL/n+ 4H-SiC. The high work function Ni metal is in direct contact with a first NID Ga₂O₃ layer of the δ -SL thereby maximizing the SB potential. Two δ -SL periods are selected, by way of example, and plotted are the conduction band diagrams as a function of growth direction. Plot **16100** shows curves representing a first δ -SL **16110** comprising 19 $\times$ periods of [N_d^{δ} cm⁻³ 5 nm, NID 100 nm] and second δ -SL **16120** comprising 4 $\times$ periods of [N_d^{δ} cm⁻³ 5 nm, NID 400 nm]. The SL doping region ratio is defined as:

$$\gamma = L(\text{NID})/L(\delta - \text{doped})$$

such that, the first and second δ -SL have $\gamma_1=20$ and $\gamma_2=80$, respectively.

[0350] The band structure again behaves as a Schottky barrier diode, with the δ -SL having an effective bulk doping density of $N_d^{eff}=4.9\text{e}16$ cm⁻³ and $1.3\text{e}16$ cm⁻³ for the first and second δ -SL, respectively.

[0351] An advantage to utilizing the periodic δ -doping method for the drift region in order to achieve an effective bulk-like n-type doping level compared to continuous co-doped growth of Ga₂O₃ is that higher epitaxial material quality is possible. In general, higher levels of n-type impurity doping in the range of $1\text{e}17$ cm⁻³, and above, result in relatively reduced crystalline quality during co-doping epitaxial growth. Furthermore, active oxygen species may interact with the Si, Ge or Sn doping source in the growth reactor, forming potentially undesired byproduct. Periodic and short doping bursts during epitaxial growth enable confinement of the dopant in a small layer, followed by NID material growth which tends to repair any crystalline distortion due to the incorporation of the impurity atom within the Ga₂O₃ crystal lattice. This method of periodic doping further improves the crystal quality of the effective doped drift region for application to thick drift regions of 5 microns and higher.

[0352] FIG. 31 shows a possible design space for achieving a target N_d^{eff} Musing a δ -SL. Example configurations using a length of the individual delta doped layer $L(\delta\text{-doped})=5$ nm at three different 2D sheet doping densities, namely, N_d^{2D} (N_d^{3D})= $2.2\text{e}11$ cm⁻² ($1\text{e}17$ cm⁻³), $1.0\text{e}12$ cm⁻² ($1\text{e}18$ cm⁻³), and $2.9\text{e}12$ cm⁻² ($5\text{e}18$ cm⁻³), and sheet doping densities in between the curves can be interpolated.

[0353] FIGS. 32A and 32B present the forward bias I-V curves for the Schottky barrier diode of FIG. 30 incorporating two examples of a δ -SL drift region having $N_d^{eff}=4.9\text{e}16$ cm⁻³ and $2.5\text{e}16$ cm⁻³, where FIG. 32A uses a semilog scale and FIG. 32B uses a linear scale. The subthreshold

slopes are comparable, whereas the differential resistance is higher for $N_d^{eff}=2.5\text{e}16$ cm⁻³, as expected.

[0354] In another example, the intermediate layer is formed comprising at least one quantum well (QW) sublayer sandwiched between wider bandgap layers, the QW formed by confinement of electrons due to the potential well thus formed. The QW has at least one quantized discrete energy levels within the narrow band gap QW layer. In one example, the quantum well sublayer is configured so that a resonant tunnelling current can occur for only a specific bias configuration, i.e., the quantum well sublayer is configured to have a bias dependent tunnelling current. In one example, this can be used for a voltage dependent switch for creating pulsed current or a bistable switch. Such a device can be a resonant tunneling diode having two electrical terminals and exhibiting a non-linear I-V response. The non-linear I-V response can have a property of negative differential resistance, which can be used advantageously in a switching or oscillator configuration of an electrical circuit.

[0355] In another example, this two-terminal switch can be used as part of a power converter for controlling a pulse width and duration of a current.

[0356] An example of a vertical conduction resonant tunneling diode is described below. A double barrier intermediate layer incorporating a single QW is formed using the heterostructure of β -(Al_{0.2}Ga_{0.8})₂O₃/ β -Ga₂O₃/ β -(Al_{0.2}Ga_{0.8})₂O₃ which is further deposited upon an n-type β -Ga₂O₃ drift layer, further deposited upon a SiC conductive substrate. The thickness of the β -(Al_{0.2}Ga_{0.8})₂O₃ barrier is sufficiently thin to enable tunnelling of electrons, for example $L_{barrier}=1\text{-}10$ nm or equal to a unit multiple (e.g., 1, or 2, or 3, or 5, or from 1 to 10) of the (Al_{0.2}Ga_{0.8})₂O₃ crystal unit cell along the growth direction. The thickness of the β -Ga₂O₃ layer comprising the QW can be selectively n-type doped or NID, and have a thickness of the order of 1 monolayer to about 10 monolayers, wherein 1 monolayer is defined as half the unit crystal dimension along the epitaxial growth direction. Due to the relatively large electron effective mass in both β -Ga₂O₃ ($m_e^*=0.29\text{-}0.31$) and 4H-SiC ($m_e^*=0.25\text{-}0.29$) (which is essentially isotropic with respect to crystal momentum), quantum confinement occurs within even shallow potential wells in physically thin QW layers. As an example, a double heterostructure comprising a thin QW: α -(Al_xGa_{1-x})₂O₃/QW α -Ga₂O₃/ α -(Al_xGa_{1-x})₂O₃/drift α -Ga₂O₃ is shown in FIGS. 33A and 33B for the cases of two electric field strengths applied to the structure along z, namely,

$$\mathcal{E}_z = 0 \frac{\text{MV}}{\text{cm}} (\text{FIG. 33A}) \text{ and } \mathcal{E}_z = 5 \frac{\text{MV}}{\text{cm}} (\text{FIG. 33B}).$$

The QW thickness is selected as 1 nm to achieve only two quantum confined states within the potential well, with barrier composition $x=1$ to provide the highest barrier potential. Wider QW thickness supports an increasing number of quantum confined states. The lowest energy quantum confined electron spatial wavefunctions profiles $\psi_n(z)$ are plotted with respect to their quantization energy level within the QW. An above barrier state is also plotted to represent a propagating state. In order to achieve significant barrier penetration of the electron wavefunction, the barrier thickness can be of the order of 1 nm or less. Such resonant

tunneling structures require precise epitaxial growth to accurately control the thickness of the layers and enable the resonance.

[0357] An advantage of the devices and structures described herein is that the thermal conductivity of SiC is circa two orders of magnitude greater than Ga_2O_3 , which advantageously enables the heat generated in the oxide drift layer to be rapidly conducted away from the device through the SiC substrate. Self-heating effects in conventional fully Ga_2O_3 power devices utilizing thick $\beta\text{-Ga}_2\text{O}_3$ substrates presents one of the most challenging issues for practical use in high power density technology. In contrast to a fully SiC device using a thick SiC substrate, a Ga_2O_3 substrate in a conventional device exhibits a low thermal conductivity thereby requiring other forms of heat management, such as, substrate thinning and integration with diamond heat spreaders, and the like. The present disclosure describes devices, structures and methods to simultaneously improve the electrical performance of device utilizing UWBG oxide materials, by integrating them with extremely high thermal conductance and low electrical resistance SiC substrates. That is, the simultaneous solution of the electrical and thermal performance of the device is enabled. This is a major improvement over thermally insulating and electrically conductive $\beta\text{-Ga}_2\text{O}_3$ substrates in terms of the overall performance of the device.

[0358] Yet a further advantage of the hybrid integration of semiconducting oxide-SiC materials in forming a Schottky barrier diode in the present disclosure is the improved electrostatic field distribution within the device.

[0359] FIG. 34 shows a plot 900 of the simulated electric field lines 915 (and associated electric potential lines 910) of a vertical multilayered semiconductor device similar in configuration to device 100 illustrated in FIG. 1 but including an isolation layer 180 surrounding the metal layer 130 that functions to mitigate surface currents and manage the electrostatic vector field. The electric field lines 915 and equipotential lines 910 shown in FIG. 34 are indicative of what would be expected for a potential difference between the conductive substrate 110 and metal layer 130 in the range of about 100 V to about 10,000 V.

[0360] In this example, a Ni-metal patterned contact (metal layer 130) is positioned on a $\beta\text{-Ga}_2\text{O}_3$ drift layer 120, which is disposed upon a conductive 4H-SiC substrate 110. The isolation layer 180 is selected from a lower dielectric constant composition, for example Al_2O_3 . The device is surrounded by vacuum or air. The contact (metal layer 130) is forced to positive potential (+V) and the conductive substrate 110 is forced to a negative potential (-V).

[0361] As can be seen from inspection, the field lines 915 are directed from the surface boundaries of the rectangular contact toward the conductive substrate surface 110 forming equipotential lines between metal layer 130 and substrate 110. The plot 900 shows high concentration of electric fields in regions of isolation layer 180 and drift layer 120 near the corners of the metal layer 130. As would be appreciated, in certain circumstances the magnitude of the electric field $|E_{x,y,z}|$ may become so concentrated that it exceeds the breakdown electric field of the host material (e.g., as shown in FIG. 2B). In practice, sharp corners at conductor features concentrate electric fields and can be made rounded if possible. Fabrication methods for metal patterning, such as lift-off or damascene methods, typically cannot create perfectly rounded features in vertical cross-section. How-

ever, the in-plane (top surface) patterning of contacts can be managed via lithographic expression to form circular pads or rounded corners and the like, which can also help to reduce the electric field densities.

[0362] It should be noted that isolation layer 180 has a material with a bandgap substantially larger than the oxide active drift layer. Otherwise, isolation layer 180 will not be effectively insulating. Low electrical conductivity also dictates the choice of suitable materials for isolation layer 180. Compatibility with UWBG active drift region 120 is another factor for isolation layer 180 materials. In some cases, isolation layer 180 simultaneously possesses a low dielectric constant such that the electric field strength can be reduced within the host isolation layer 180. Referring again to the example selection of materials in FIG. 29A showing the trend of dielectric constant with bandgap energy, there is a range of suitable materials that are compatible with a Ga_2O_3 active drift layer and can be used for isolation layer 180 materials.

[0363] In some examples isolation layer 180 can include one or more of lower dielectric constant and higher bandgap materials than the drift layer semiconductor. If drift layer comprises Ga_2O_3 , then isolation layer compositions of MgO and Al_2O_3 are possible. While SiO_2 has even better isolation layer properties, the practical methods for deposition (e.g. PECVD) produce hydrogenated H:SiO_2 films which have amorphous and substantially different properties to single crystal quartz ($\alpha\text{-SiO}_2$). Atomic layer deposition of Al_2O_3 exhibits one of the most favorable methods for producing high quality amorphous and conformal coatings while still exhibiting larger effective bandgap to Ga_2O_3 and low dielectric constant. It is also possible to use ALD deposited amorphous $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ for $x>0$, or other materials as described herein.

[0364] Referring again to the mesa metal contact device, FIG. 34 shows the electric field surrounding the contact 130 is relatively unbounded. An improved high voltage contact arrangement can be implemented to manage the lateral extent of electric field while keeping the contact area of 130 to the drift region fixed.

[0365] FIGS. 35A and 35B show top and figurative sectional views, respectively, of a vertical multilayered semiconductor diode device 1000 according to an illustrative embodiment incorporating a lateral electric field-plate arrangement to modify the vector electric field of the device.

[0366] In this example, device 1000 comprises a substrate 1010 (FIG. 35B), epitaxial drift layer 1020 and metal layer 1030 as has been previously described (e.g., in device 100 in FIG. 1). In this example, an isolation layer 1080 is formed on top of epitaxial drift layer 1020 and surrounds metal layer 1030. Formed on a top surface of metal layer 1030 and on a top surface of isolation layer 1080 is a field plate arrangement in the form of a field plate layer 1090 formed of metal and having a central aperture 1091 sized and shaped to expose metal layer 1030. In this example, both the metal layer 1030 and the field plate layer 1090 have a circular configuration, and isolation layer 1080 has a ring or annular configuration surrounding (or wrapping around) metal layer 1030. The field plate layer has an outer diameter d_{FP} and an overhang of w_{FP} over the isolation layer 1080.

[0367] In this example, the metal layer 1030, isolation layer 1080 and field plate layer 1090 each have a complementary circular or annular configuration and this assists in minimizing edges where more intense electric fields may

otherwise form. In other examples, the Schottky barrier metal of metal layer **1080** may have a n-polygon (e.g., rectangular, rounded corner elongated stripes, hexagonal, etc.) configuration and the other layers may have complementary configurations as required. Shown in FIG. **35B** is the diameter of the Schottky barrier metal layer **1030**, d_{SBM} , the diameter and annulus width of the field plate layer **1090**, d_{FP} and W_{FP} , in addition to the thickness of the drift layer **1020** and isolation layer **1070**, t_D and t_{iso} respectively.

[**0368**] In an example, drift layer **1020** will have a dielectric constant of ϵ_1 and band gap energy E_{g1} , and isolation layer **1080** will have a dielectric constant of ϵ_2 and band gap energy E_{g2} . In an example, the materials of the drift layer **1020** and isolation layer **1080** are chosen such that $\epsilon_2 \ll \epsilon_1$ and $E_{g2} \gg E_{g1}$.

[**0369**] A design principle that can be used to design the isolation layers and field-plates described herein is now discussed. Consider a plane parallel-metal plate capacitor separated by a thickness t , with a dielectric positioned between the finite lateral extent metal plates of area A . The capacitance C_p is defined as: $C_p = \epsilon_r \epsilon_0 A / t$. To minimize the effective parasitic capacitance C_p of the field-plate overhang w_{FP} , it is possible to select a low- k (i.e. low ϵ_r) material and/or increase the isolation layer thickness t_{iso} . Furthermore, a low- k dielectric material can be compared to Ga_2O_3 by using the relation:

$$\frac{C_p(\text{low}-k)}{C_p(Ga_2O_3)} = \frac{\epsilon_r^2(\text{low}-k)}{\epsilon_r^2(Ga_2O_3)} \times \frac{t_{Ga_2O_3}}{t_{low-k}}$$

If a $t_{Ga_2O_3} = 100$ nm thickness is used, then $(C_p/A) = 89$ fF/ μm^2 . To achieve a relative reduction in overhang capacitance density to a target value of

$$\frac{C_p(\text{low}-k)}{A} = 10 \text{ fF}/\mu m^2$$

device can use Al_2O_3 with a thickness of $t_{low-k} = 640$ nm.

[**0370**] In an example, the field plate layer **1090** and SB metal contact **1030** may be formed from the same metal, for example, a high work function metal or alloy of Cu, Te, Be, Rh, Co, C, Ni, Au, Ir, Pd, Pt, Se, Os and combinations thereof (refer to FIG. **11A**). In another example, the metal layer **1030** may be formed from high work function metals including, but not limited to Cu, Te, Be, Rh, Co, C, Ni, Au, Ir, Pd, Pt, Se, Os and combinations thereof, whereas field-plate layer **1090** metal may be selected from non-high work function metals, such as, Al, Ti, Mo, W and others. High melting point metals can be advantageous, for example for use in high temperature operation (e.g., in excess of 300° C.).

[**0371**] Ohmic contacts to the substrate may also comprise high work function metals to enable co-processing of front and backside of the device structure. For example, Ni can be used as a SB metal for the drift layer surface whereas Ni may act as an Ohmic contact to the highly doped SiC substrate. Furthermore, silicides may be used to form low resistance Ohmic contacts to the substrate. Silicon may be scavenged from the SiC substrate by thermal anneal processing of contact metal to form a silicide. For example, Ni may form Ni_xSi_{1-x} and similarly Ti may form Ti_xSi_{1-x} . Furthermore,

multiple metal layers may form an Ohmic contact to the substrate, such as $SiC/Ti_xSi_{1-x}/Ti/Al$.

[**0372**] Some examples of field-plate metal layer **1090**/isolation layer **1080**/Schottky barrier metal **1030**/wideband gap oxide drift layer **1020**/substrate **1010**/Ohmic contact layer structures are:

[**0373**] $Al/Al_2O_3/Al/Ni/Ga_2O_3/4H-SiC/Ni/Al$

[**0374**] $Cu/Al_2O_3/Al/Ni/Ga_2O_3/4H-SiC/Ni/Cu$

[**0375**] $Cu/Al_2O_3/Al/Ni/NiO/Ga_2O_3/4H-SiC/Ni/Cu$

[**0376**] $Cu/Al_2O_3/Al/Ni/Ni_xCu_yO/Ga_2O_3/4H-SiC/Ni/Cu$

[**0377**] $Al/Al_2O_3/Al/Ni/Pd/Ga_2O_3/4H-SiC/Ni/Cu$.

[**0378**] In one embodiment, isolation layer **1080** is formed from a wider band gap material than drift layer **1020**. In an example, isolation layer **1080** is formed from Al_2O_3 . In another example, isolation layer **1080** is formed from SiO_2 .

[**0379**] In other examples, the isolation layer can be a wide bandgap oxide that has p-type conductivity type. For example, oxides for the isolation layer may include Nickel-Oxide, Nickel-Copper-Oxide, Copper-Oxide, Tellurium-Oxide, or Osmium-Oxide. The isolation layer may also comprise a portion of a metal oxide formed from the SB metal.

[**0380**] In other examples the IL may also be of p-type character which may or may not possess a larger bandgap than the drift layer.

[**0381**] FIG. **36** shows a plot **1100** of the simulated electric field distribution **1110** of a vertical multilayered semiconductor device similar in configuration to the device **1000** illustrated in FIGS. **35A** and **35B** incorporating a field-plate arrangement as has been previously described (e.g., with respect to FIGS. **35A** and **35B**).

[**0382**] As can be seen from inspection, the field plate layer **1090**, including the outer diameter (e.g., d_{FP} in FIG. **35B**) and the overhang (e.g., w_{FP} in FIG. **35B**) over the isolation layer **1080**, functions to reduce the concentration of electric-field lines proximal to the vertical boundary or corners of the metal layer **1030** that is in contact with the drift layer surface. The equipotential lines **1110** between the portion of the contact metal **1030** interface with the drift layer **1020** are significantly flattened in comparison to the case with no field-plate layer **1090**. The concentration of electric-field lines **1115** is also significantly reduced near the contact region (metal layer **1030**) which is in direct contact with the drift region **1020**. The low- k isolation layer **1080** also acts to spread out the equipotential lines laterally, thereby reducing the maximal electric field strength beneath the field plate layer **1090**. Furthermore, the conducting channel that is formed between the metal layer **1030** in contact with the drift region and laterally disposed conductive substrate **1010** exhibits a substantially uniform electric field. Not to be limited by theory, the reduction of the electric field density in regions adjacent to metal layer **1030** (especially at a corner region near the interface between drift layer **1020** and metal layer **1030**, such as interface region **140** shown in FIG. **1**) can improve the device performance, for example by reducing the peak electric field within the device to well below the critical breakdown field of the host drift material.

[**0383**] For a fixed vertical device configuration, the current-carrying capacity of the diode structure in FIG. **1** can be increased by scaling the contact area **140** to larger areas, or by multiplexing a number of parallel connected diodes. A trade-off in epitaxial area consumed, maximum vertical

current capacity and thermal effects is possible by partitioning vertical current across a number of parallel interconnected diodes.

[0384] FIGS. 37A and 37B respectively show horizontal and vertical cross-sectional views of a composite diode device 1200 having a plurality of unit cells 1250 similar to the vertical multilayered semiconductor diode device 1000 illustrated in FIGS. 35A and 35B. The top view cross-section in FIG. 37A is taken through section A-A, representing a plane parallel with the substrate surface that intersects circular metal layers 1030 and the isolation layer 1080. The vertical cross-sectional view in FIG. 37B relates the features of an optional continuous top contact metal 1220 interconnecting the plurality of Schottky barrier metal contacts 1030, in which the top contact metal 1220 serves as a top field-plate 1090. Drift layer 1020 is disposed upon substrate 1010 as well as an Ohmic contact 1240. Metal layer 1030 comprises several circular metal regions. Isolation layer 1080 extends across composite diode device 1200 and surrounds each of the circular metal regions of metal layer 1030. A drift layer (e.g., drift layer 1020 in FIG. 35B) can be between the metal layers 1030 and a substrate (e.g., substrate 1010 in FIG. 35B), and can extend across composite diode device 1200. A field plate layer 1220 (e.g., 1090 in FIG. 35B) can be formed above metal layer 1030, and can also extend across composite diode device 1200. The unit cells 1250 each have a circumference 1092, which can be related to the outer diameter and overhang dimensions of a field plate needed for each unit cell to reduce the electric field concentrations at each of the circular regions of metal layer 1030. Therefore, in this example, the dimension of the field plate layer 1090/1220 are a limiting factor as to the number of unit cells that may be arranged within a given surface area.

[0385] In various examples, each unit cell may have a width that varies between 1 micron and 100 microns, or from 100 microns up to 1 cm. In an example, each unit cell will share a common substrate (1010) and drift layer 1020. In these examples, a highly resistive edge termination region may be formed between each unit cell. In an example, the termination region is formed as a physical trench or channel extending downwardly into the drift layer 120 and forming a boundary between each of the unit cells 1250. In another example, the termination region may be formed by ion implantation where a resistive trench or channel region is formed in the oxide material of the drift layer 1020.

[0386] In various embodiments, the metal layer (e.g., metal layer 130 in FIG. 1, or metal layer 1030 in FIGS. 35A and 35B) is deposited using a metal deposition process (such as, sputtering, thermal evaporation, electron beam deposition or by electro-deposition) following the formation of the epitaxial drift layer (e.g., drift layer 120 in FIG. 1, or drift layer 1020 in FIGS. 35A and 35B) and in any further epitaxial layers formed above the drift layer. That is, the metal deposition process is ex-situ to the epitaxial oxide deposition. As would be appreciated, this requires a process change from an epitaxial oxide layer deposition process to a metal deposition process. The topmost surface of the final deposited epitaxial oxide may be protected by limiting the exposure to a contaminating environment prior to metal deposition. This can be achieved by configuring separate deposition chambers with a vacuum wafer transfer mechanism. In other methods, the metal deposition can be performed in the same process chamber as the oxide deposition once disadvantageous species are removed from the cham-

ber. In some cases, the metal can be epitaxially deposited in the epitaxial oxide deposition chamber, as described further herein.

[0387] FIGS. 38A-38C depict a process flow for forming a vertical multilayered semiconductor device including an epitaxial drift layer of an oxide material with a metal layer forming a Schottky barrier junction, in accordance with some embodiments. The metal layer 1330 can be an epitaxial metal layer (e.g., formed using MBE, ALD or MOCVD), or a non-epitaxial metal layer (e.g., formed using sputtering, pulsed laser deposition, or thermal evaporation of a metal). Each process block (or unit operation) of the process results in an intermediate product shown in FIGS. 38A-38C.

[0388] At a stage/block of the process, the epitaxial drift layer 120 is formed on substrate 110, as depicted by intermediate product 1310 in FIG. 38A.

[0389] At another block of the process, an (epitaxial) metal layer 1330 is formed on drift layer 120, as depicted by intermediate product 1340 in FIG. 38B. The lattice constant and/or crystal structure mismatch between the specific metal and the final surface reconstruction of the oxide surface plays an important role in the metal-oxide interface quality. Methods are disclosed later in this disclosure that form advantageous metal-oxide interfaces. In other examples, the (epitaxial) metal layer 1330 may be formed on any intervening or intermediate epitaxially deposited layer formed on drift layer 120. In one example, the thickness of the (epitaxial) metal layer 1330 is between 1 Angstrom and 100 Angstroms. The epitaxial metal or first deposited metal layer may be relatively thin, and another metal layer may be deposited subsequently to increase the total layer thickness. The second metal deposition step may comprise a dissimilar metal to the first metal composition, or the same metal.

[0390] At another block of the process, metal layer 1330 may be patterned as depicted by intermediate product 1370 in FIG. 38C. Intermediate product 1370 may be produced via photolithographic patterning of a mask and subsequent selective material removal via physical etch process. The etch process may be at least one of a dry reactive gas plasma process or wet etching process or other. The device may also be further processed as required to form a vertical multilayered semiconductor diode device in accordance with the present disclosure. For example, in some cases a patterned contact layer 1351 is included on top of the epitaxial metal layer. In some cases, a metal layer that is not epitaxial can also be used to form the Schottky barrier junction with the epitaxially deposited drift layer 120. In some cases, a transition layer (e.g., as shown in structure 400 in FIG. 4, and described above) can be deposited between the substrate 110 and the drift layer 120. In some cases, an intermediate layer (e.g., as shown in structure 800 in FIG. 8, and described above) can be deposited between the drift layer 120 and the (epitaxial) metal layer 1330. In some cases, some of the layers are patterned (e.g., using wet etching, plasma etching, and/or photolithography) to form the structures described herein. Optionally an oxide surface exposed by metal patterning 1335 may be treated to reduce surface damage, reduce surface leakage and passivate the surface.

[0391] Any of the device structures described herein (e.g., structures 100 in FIG. 1, 300 in FIGS. 3A and 3B, 400 in FIG. 4, 600 in FIG. 6, 800 in FIG. 8, 1000 in FIGS. 35A and 35B, and 1200 in FIG. 37A) can be made using the process above.

[0392] FIGS. 39A-39G depict a process flow for forming a vertical multilayered semiconductor diode device according to another illustrative embodiment, including an epitaxial drift layer of an oxide material with a metal layer forming a Schottky barrier junction. The metal layer 1430 in this process can also be an epitaxial metal layer (e.g., formed using MBE or MOCVD), or a non-epitaxial metal layer (e.g., formed using sputtering, pulsed laser deposition, or thermal evaporation of a metal). Each process block (or unit operation) of the process results in an intermediate product shown in FIGS. 39A-39G.

[0393] At a block of the process, an epitaxial drift layer 1420 comprising a semiconductor oxide material is formed on a SiC substrate 1410, as depicted by intermediate product 1401 shown in FIG. 39A. In some cases, SiC substrate 1410 is a composite substrate comprising a bulk substrate layer and a surface layer (e.g., bulk substrate layer 112 and surface layer 114 of substrate 110 in FIG. 1), where the surface layer comprises single crystal SiC.

[0394] At another block of the process, an isolation layer 1480 is formed on drift layer 1420, as depicted by intermediate product 1402 shown in FIG. 39B. In an example, isolation layer 1480 is formed from a suitable oxide material, such as Al_2O_3 , NiGa_2O_4 , AlCu_2O_4 , GaCu_2O_4 , MgO , ZnGa_2O_4 , or SiO_2 . In an example, isolation layer 1480 is formed on drift layer 1420 by a conformal deposition process such as atomic layer deposition (ALD). In another example, isolation layer 1480 is an epitaxial layer formed using an epitaxial deposition process (e.g., MBE or MOCVD). In an example, the isolation layer 1480 is formed to have a layer thickness from 100 nm to 300 nm.

[0395] At another block of the process, opposed intra-device field termination regions 1471 and opposed inter-device isolation regions 1475 are formed by physical removal of portions of the isolation layer 1470 and drift layer 1420 respectively, as depicted by intermediate product 1403 shown in FIG. 39C. Alternatively, physical alteration of selective area and depth can be attained via ion-implantation process. Furthermore, a central contact area region 1477 is formed by removing the isolation layer 1480 to expose drift layer 1420. The opposed intra-device field termination regions 1471 and opposed inter-device isolation regions 1475 can be formed by patterning and etching the isolation layer 1480 and the drift layer 1420 (e.g., using wet etching, plasma etching, and/or photolithography). In one example, the portions are removed by applying a photoresist followed by etching.

[0396] At another block of the process, a top metal layer is deposited which in this example will form the metal layer 1430 and the field plate layer 1490, as depicted by intermediate product 1404 shown in FIG. 39D. In some cases, metal layer 1430 is thinner than field plate layer 1490. For example, metal layer 1430 can be deposited using an epitaxial technique and be thinner than field plate layer 1490, which can be deposited using a method with a faster growth rate (e.g., sputtering, or thermal evaporation). In this example, a guard ring 1495 is formed on the isolation layer 1480 on the outer side of field termination region 1471. The guard ring 1495 can provide a reference voltage and/or mitigate parasitic currents and voltages. In various examples, the top metal layers (e.g., 1430, 1490 and/or 1495) may be formed from Ti/Al, Al, Ni, Mo, Pd, or combinations thereof. In some cases, the device does not include an intermediate layer (e.g., as shown in structure 100

in FIG. 1, or structure 400 in FIG. 4), and the guard ring 1495 can be deposited on the epitaxial oxide layer. Portions of the top metal layers (e.g., 1430, 1490 and/or 1495) corresponding to the field termination regions 1471 and isolation regions 1475 are also removed, for example using a patterning and plasma etch process. In another example, patterning and metal lift-off (MLO) processes are used.

[0397] At another block of the process, a top passivation layer 1476 is formed over the entire device, as depicted by intermediate product 1405 shown in FIG. 39E. In an example, the top passivation layer 1476 is formed from a suitable oxide material such as Al_2O_3 or SiO_2 .

[0398] At another block of the process, a bottom contact layer 1452 is formed on the backside of substrate 1410, as depicted by intermediate product 1406 shown in FIG. 39F. Bottom contact layer 1452 can make an ohmic contact with the substrate 1410, and can be deposited using any of the metal deposition techniques described herein.

[0399] At another block of the process, the metal layer 1430, which in this example also forms the top contact, is exposed by a further etching process that removes a portion of the passivation layer 1476, as depicted by intermediate product 1407 shown in FIG. 39G.

[0400] FIG. 40 is an example of a figurative sectional view of a vertical multilayered semiconductor diode device configuration Y100 adopted to simulate thermal heating effects. In the thermal models in this example, semiconductor diode device configuration Y100 is assumed to be continuous to the left and right (i.e., semi-infinite) with no lateral boundary conditions.

[0401] Moving generally upwards (in the positive z-direction) from the bottom of FIG. 40, semiconductor device configuration Y100 comprises a horizontally extending back side drain metal ohmic contact layer F1 (cathode) whose lower surface (i.e., Si) is held at temperature T_{S1} . The next layer comprises substrate Y110 (F5). In various examples as set out below, thermal modelling is performed comparing different substrate materials such as highly doped (n- or p-type) SiC or Ga_2O_3 . To first order, the doping level and type does not significantly alter the host material thermal conductivity.

[0402] Formed above the substrate Y110 is an epitaxial drift layer Y120 (F6+F7+F8) comprising a semiconductor oxide material which in this example is modelled as three regions including a central region F8 and opposed border regions F6 and F7. This division of epitaxial drift layer Y120 into three regions allows the temperature performance characteristics of the central region F8 to be separately characterized (e.g., see FIG. 44 below). The central region F8 is where the most ohmic heating occurs due to a large forward conducting channel. Regions F7 and F6 represent semiconductor regions which are subject to less or no vertical current flow.

[0403] Located above the central region F8 of the epitaxial drift layer Y120 is a Schottky barrier metal layer Y130 (in region F3) that forms a Schottky barrier junction between the metal layer Y130 and epitaxial drift layer Y120 central region F8.

[0404] Located above epitaxial drift layer border regions F6 and F7 are respective isolation layer Y180 border regions F2 and F4 formed from a wider band gap material compared to the semiconductor oxide material forming the epitaxial drift layer Y120. In general, isolation layer regions F2 and F4 may comprise lower thermal conductivity composition

than the drift layer Y120 (F7, F8 and F6). Extending over Schottky barrier metal layer Y130 and partway over isolation layer Y180 regions F2 and F4 is field plate layer Y190 (in region F9) formed from a low, medium or high work function metal. Centrally disposed metal interconnect F10 extends upwardly from the field plate layer Y190 to an overlying, horizontally extending, top contact bus metal layer F11 (anode). In one example, field plate layer Y190, metal interconnect F10 and top contact metal layer F11 may be formed of Al. In another example, the electrical conductivity types can be changed to the opposite conductivity type in which case the top bus metal layer F11 will function as a cathode. As would be appreciated, the thermal conductivity performance will be equivalent.

[0405] In this example, the exposed bottom surface of top contact bus metal layer F11, the exposed side surfaces of interconnect F10, the exposed side and top surfaces of field plate layer Y190 as well as the exposed top surfaces of isolation layer Y180 border regions F2 and F4 are all assumed to be at the ambient air temperature T_a (as depicted in FIG. 40). In this manner, these exposed surfaces are modelled to allow convective transfer of heat generated by the device configuration Y100 to the “air.” This will be seen as a worst-case scenario as in other examples, the “air” region will be a thermally insulating dielectric material (e.g., a polymer planarization layer) also minimally transferring heat from device configuration Y100 (e.g., interconnect F10 implemented as a wire bond connected to contact bus metal layer F11 in the form of a bond pad). Additionally, the lower surface S1 of metal contact region F1 is assumed to be held at temperature T_{S1} and the upper surface of metal contact bus region F11 is assumed to be held at temperature T_{S2} .

[0406] FIG. 41 shows a plot Y200 of the thermal conductivity of various semiconducting materials as a function of temperature that can be used for the temperature modelling of vertical multilayered semiconductor device configuration Y100 as will be described below. The plot shows the thermal conductivity curves of semiconducting materials including those for potential substrate materials 4H-SiC Y210 and n-doped (Sn) Ga_2O_3 Y220, as well as for unintentionally doped Ga_2O_3 Y230 which may in one example be employed to form the epitaxial drift layer Y120 of device configuration Y100 as shown in FIG. 40.

[0407] As can be seen by inspection, the thermal conductivity of all the depicted materials reduces with increasing temperature. At an operating temperature of 200° C. the thermal conductivity of SiC is at least five times greater than that of the depicted Ga_2O_3 -based materials. This aspect represents one of the major advantages of the hybrid approach disclosed herein.

[0408] In an example, the thermal modelling comprises simulating a voltage being initially applied to an approximately 1 μm central region at the top contact layer F11 with the drain contact layer F1 held at ground potential. In this example, the simulated voltage applied corresponds to an initial current of 65 Amps being conducted by model configuration Y100 between the top contact layer F11 and the drain contact layer F1. In addition, the temperature T_{S1} of surface S1 of configuration model Y100 is set to 25° C. and the temperature T_{S2} of surface S2 is set to 75° C. The model is then allowed to evolve to an equilibrium state with a steady state forward current being conducted by semiconductor device configuration Y100.

[0409] The model is calculated using finite element method for piecewise construction of the spatial distribution of the material properties in a high-density tensor mesh. Heat transfer throughout the 2D model structure is applied and solved using the parabolic partial differential equation:

$$Q = \rho C_h \frac{\partial T}{\partial t} - \nabla \cdot (\kappa \nabla T)$$

where Q is the heat transfer, T is the temperature, C_h the specific heat, κ is the thermal conductivity, and ρ is the material density. Each parameter is a function of time and spatial position of each element (t, x, y).

[0410] The simulation workflow includes the steps or blocks of: (i) create a 2D physical model for steady-state or transient simulation; (ii) apply a spatial mesh algorithm to discretize the model; (iii) assign matrix elements $C_h(t=0, x, y)$, $\rho(t=0, x, y)$, $\kappa(t=0, x, y)$; (iv) specify internal heat sources Q within the geometry; (v) specify temperatures on the boundaries or heat fluxes through the boundaries, $T(t=0, x, y)$; (v) specify the ambient temperature and the convective heat transfer coefficients, and radiative heat flux; (vi) specify emissivity F, and Stefan-Boltzmann constant σ for each material of the structure. Once the model is fully specified, the simulation heat flow algorithm is initialized with an initial temperature or initial guess.

[0411] Referring now to FIG. 42, there is shown a temperature contour shade plot Y300 showing the resultant steady-state temperature distribution following thermal modelling of the semiconductor device configuration Y100 comprised of in this example a highly doped Ga_2O_3 substrate Y110 (e.g., material Y220 of FIG. 41) and a Ga_2O_3 epitaxial drift layer Y120 (e.g., material Y230 of FIG. 41). Schottky barrier metal layer Y130 is selected to be a high work function metal, such as Ni, and the Ga_2O_3 -based epitaxial drift layer Y120 is n⁻ doped. In other examples, the Schottky barrier metal layer Y130 may be selected from other high work function metals such as Pt, Pd, Cu or Os. In an alternative configuration, the Ga_2O_3 epitaxial oxide layer is p⁻ doped and the Schottky barrier metal layer Y130 may be selected from a low work function metal such as Al or Ti. The thermal conductivity of most metals selected do not significantly alter the conclusions herein. An internal heat source within the structure Y300 is generated by resistive heating effect due to vertical current (e.g., constant current) flowing within region F8 of the drift region Y120.

[0412] As can be seen from FIG. 42, the maximum temperature region Y310 is within the central region F8 of drift layer Y120 and is approximately 400° C., with a temperature gradient extending laterally and into the contact and field-plate Y190. The top contact and isolation layer are coupled to open surfaces for air convection. Even so, the top contact Y190 is raised to approximately 300° C., which implies refractory metals may be required for stability (or reliability, or durability). In general, metal junction temperatures exceeding 250° C. are problematic for high power and high reliability applications. For this example model, it is concluded the materials comprising the Schottky barrier metal layer Y130, isolation layer Y180, and field-plate layer Y190 would be required to operate at temperatures in the range of 250° C. to 300° C. which would be expected to adversely impact reliability.

[0413] The forward biased diode operation is modelled as the central epitaxial drift layer Y130 region F8 as a current “filament” conducting where the heating power will follow the IR^2 relationship, where I is the current flowing through the region and R is the effective resistance of the region governed by the doping concentration.

[0414] Referring to FIG. 42 again, the steady-state temperature of the interface between the Ga_2O_3 drift region and substrate junction is also shown to be $\sim 350^\circ C$. with a gradient extending toward the rear contact held at $25^\circ C$. The low thermal conductivity of the Ga_2O_3 substrate causes this self-heating effect of the drift region.

[0415] In contrast to the structure of FIG. 42, a high thermal conductivity substrate is now substituted in FIG. 43. There is shown a temperature contour shade plot Y400 showing the resultant temperature distribution following thermal modelling of the semiconductor device configuration Y100 modelled in FIG. 42 except that a SiC substrate (e.g., see material Y210 of FIG. 41) is substituted for the Ga_2O_3 substrate of the structure adopted in FIG. 42.

[0416] In contrast to the modeling results shown in FIG. 42, the results in FIG. 43 show that the maximum temperature Y410 for central region F8 has a maximum temperature is in the range of $200^\circ C$. to approximately $220^\circ C$. which is substantially lower than the maximum temperature of the structure with the Ga_2O_3 substrate modeled in FIG. 42. Note, the same thermal heat source is applied for both FIG. 42 and FIG. 43. Additionally, the materials comprising the Schottky barrier metal layer Y130, isolation layer Y180, and field-plate layer Y190 operate at temperatures of approximately $175^\circ C$., which is more favorable toward conventional thermal dissipation methods. The SiC substrate having a high thermal conductivity provides efficient heat-sinking of the drift region and provides lateral thermal management throughout the structure. The lower temperature enabled by the improved thermal management herein also relaxes constraints on the types of metals that can be used, and advantageously impacts device reliability.

[0417] Many material parameters are dependent on temperature. Thermal conductivity, heat capacity and bandgap energy, in general, decrease with increasing temperature. Thermally activated carrier generation also increases the number of free-carriers and reduces the electrical resistance in semiconductors. Conversely, in metals, the electrical resistance typically increases with increasing temperature. The self-heating effect disclosed in FIG. 42 may, therefore, lead to thermal “runaway” and premature breakdown of the device when the thermal dissipation is limited.

[0418] Transient thermal response of the model reveals further advantages and disadvantages of the oxide drift region being coupled to high and low thermal conducting substrates.

[0419] Referring to FIG. 44, there is shown a transient response of the device models Y300 and Y400 for two current pulse cycles generated in the current filament regions F8. The plot Y500 of the integrated thermal intensity $\beta(t)$ for region F8 of the drift layer Y120 as a function of time (in microseconds) is provided for the two device configurations shown in FIG. 42 (i.e., with the Ga_2O_3 substrate) and FIG. 43 (i.e., with the SiC substrate) assuming a pulsed current input into region F8 having a 5 μs on-period and a 5 μs off-period (with a 50% duty cycle). By inspection, the maximum thermal intensity Y510 of the SiC substrate-based device is approximately a factor of three below the maxi-

um thermal intensity Y520 of the Ga_2O_3 substrate-based device, even though both devices have the same Ga_2O_3 drift layer and top contact structure. Furthermore, the thermal intensity of the SiC substrate-based device is able to recover to zero Y530 before the next 5 μs second pulse commences, which contrasts with the recovery Y540 of the Ga_2O_3 substrate-based device.

[0420] As would be appreciated, the increased time required for the Ga_2O_3 substrate-based diode device to dissipate heat will increase the minimum dwell time required for any switching application as compared to the SiC substrate-based diode device. Alternatively, if such a minimum dwell time (or off-period duration) is not sufficiently long, then the Ga_2O_3 substrate-based device could suffer from thermal “runaway.”

[0421] FIG. 45 shows a figurative sectional view of a vertical multilayered semiconductor diode device configuration Y600 adopted to simulate the thermal performance of a mesa structure semiconductor device. In this example, the epitaxial drift layer Y620 (F6+F7+F8) of device model configuration Y600 has been formed into a mesa structure with angled walls that extend upwardly from the substrate Y610 (F5), and the regions outside of the mesa are assumed to be ambient air temperature T_a (as shown in FIG. 45).

[0422] Again, moving generally upwards from the bottom of FIG. 45, device model configuration Y600 comprises a horizontally extending back side drain metal ohmic contact layer F1 (cathode) whose lower surface (i.e., Si) is held at temperature T_{S1} . The next layer comprises substrate Y610 (F5). Formed above the substrate F5 is the epitaxial drift layer Y620 comprising a semiconductor oxide material and which again is modelled as three regions including a central region F8 and opposed border regions F6 and F7 to allow characterization of central region F8. Located above the central region F8 of the epitaxial drift layer Y620 is a Schottky barrier metal layer Y630 (F3) that forms a Schottky barrier junction between the metal layer Y630 (F3) and epitaxial drift layer Y620 central region F8. In this example, metal layer Y630 also forms the top contact layer, and its upper surface is assumed to be held at temperature T_{S2} (e.g., the upper surface is coupled to a thermal heat sink at temperature T_{S2}).

[0423] In this example, the thermal modelling comprises simulating a voltage being initially applied to an approximately 1 μm central region at the Schottky barrier metal layer Y630 with the drain contact layer F1 held at ground potential. In this example, the simulated voltage applied corresponds to an initial current of 65 Amps being conducted by model configuration Y600 between the Schottky barrier metal layer Y630 and the drain contact layer F1. In addition, the temperature T_{S1} of surface S1 of configuration model Y600 is set to $25^\circ C$. and the temperature T_{S2} of surface S2 is set to $75^\circ C$. The model is then allowed to evolve to an equilibrium state with a steady state forward current being conducted by device configuration Y600.

[0424] Referring now to FIG. 46, there is shown a steady-state temperature contour shade plot Y700 showing the resultant temperature distribution following thermal modelling of configuration Y600 comprised of in this example a highly doped Ga_2O_3 substrate (e.g., see material Y220 of FIG. 41) and a Ga_2O_3 epitaxial drift layer (e.g., see material Y230 of FIG. 41). Schottky barrier metal layer Y630 is selected to be a high work function metal such as Ni, and Ga_2O_3 epitaxial oxide layer is n doped. In other examples,

the Schottky barrier metal layer **Y630** may be selected from other high work function metals such as Pt, Pd or Os. In an alternative configuration, the Ga_2O_3 epitaxial oxide layer is p^- doped and the Schottky barrier metal layer **Y630** may be selected from a low work function metal such as Al.

[0425] As can be seen from the modeling results shown in FIG. 46, which are similar to those in FIG. 42, the maximum temperature **Y710** for central region **F8** is approximately 350°C . which again far exceeds a desired operating temperature of 200°C . to 220°C . for a semiconductor diode device. Additionally, extended regions of the epitaxial drift layer **Y620** would be required to operate at temperatures in the range of 250°C . to 300°C . which would be expected to impact reliability.

[0426] Referring now to FIG. 47, there is shown a steady-state temperature contour shade plot **Y800** showing the resultant temperature distribution following thermal modeling of the semiconductor device configuration **Y600** modelled in FIG. 46 except now a SiC substrate (e.g., see material **Y210** of FIG. 41) is substituted for the Ga_2O_3 substrate. By contrast to FIG. 46, the maximum temperature **Y810** for central region **F8** has a maximum temperature in the range of 200°C . to 220°C . which is substantially lower than the maximum temperature of the Ga_2O_3 substrate-based device shown in FIG. 46. Additionally, the epitaxial drift layer **Y620** considered in total has an overall operating temperatures of approximately 150°C . which is well within the conventional operating temperatures for these types of diode devices.

[0427] The thermal models described herein illustrate advantageous hybrid integration of a oxide drift layer on a high thermal conductivity SiC substrate to dramatically improve steady-state and transient device performance, which can be applied to high power switching.

[0428] Further detailed aspects of the hybrid metal oxide semiconductor and SiC substrate device architecture and manufacturing methods are now discussed.

[0429] FIG. 48 shows a plot **Z1-1000** of the crystal surface orientation electron-affinity (χ_A) on an absolute energy scale relative to vacuum for vicinal surfaces labelled by the miller indices (hkl) of $\beta\text{-Ga}_2\text{O}_3$ and the (000 $\pm$ 1) surface of 4H-SiC (i.e., Si-polar and C-polar surface). The bandgap of each material is represented by a shaded box with the topmost level representing the conduction band edge and the lower boundary of each box representing the valence band edge. The bandgap and (χ_A) in FIG. 48 were calculated using Density Functional Theory (DFT) for specific material crystal slabs.

[0430] In particular, the Si-polar 4H-SiC(0001) surface is advantageous for the epitaxial deposition of $\beta\text{-Ga}_2\text{O}_3$ (-201) oriented films. Electron affinity of $\beta\text{-Ga}_2\text{O}_3$ (-201) is shown as **Z1-1010** and 4H-SiC(0001) is **Z1-1020**, assuming not intentionally doped materials.

[0431] The interface between an epitaxially grown $\beta\text{-Ga}_2\text{O}_3$ (-201) layer and a 4H-SiC(0001) layer results in the energy band lineup as shown in FIG. 49. The conduction band edge **Z2-2020** is shown along a growth direction of the device in FIG. 9C at the heterointerface junction. The drift region is n-type ($5 \times 10^{16}\text{ cm}^{-3}$) and the n-type 4H-SiC region is $1 \times 10^{19}\text{ cm}^{-3}$ and results in a conduction band discontinuity as shown, with a triangular potential well induced on the Ga_2O_3 side. The device configuration of FIG. 9C further shows the first quantized energy level in the potential well is below the Fermi energy, thereby inducing a two-dimen-

sional electron gas (2DEG) at the heterojunction. The lowest energy quantized electron wavefunction, which is an example of a 2DEG localized at the heterojunction is shown as **Z2-2010**. The presence of the 2DEG is used advantageously in the present disclosure to form an electrical contact between the two regions and effectively pins the conduction band locally.

[0432] The devices, structures and methods described herein can include the crystallographic matching of two dissimilar crystal space groups (SG) comprising Ga_2O_3 (SG=C2m) and 4H-SiC (SG=P63mc).

[0433] FIGS. 50 and 51 show detailed atomic arrangements of the Si and C atoms comprising a single crystal 4H-SiC polytope, oriented along the C-direction **Z3-3020** and in the C-plane, respectively. FIG. 50 shows the single crystal unit cell **Z3-3010** comprising the wurtzite-like hexagonal atomic arrangement. The polar orientation refers to the directed Si—C bond along the [0001] or [000-1] axis, as is evident with the Si-face and C-face which are shown. The C-plane (i.e. (0001) or (000-1)) oriented surface is shown in FIG. 51 which has in-plane Si—Si and C—C lattice spacing $a(\text{Si—Si})=3.095\text{ \AA}$. The stacking sequence along the Si-polar direction is: (A-B-C-B-A-B-C-B- . . .), having an equal number of cubic and hexagonal bonds.

[0434] Epitaxial bonding of a single crystal Ga_2O_3 composition is possible to the dissimilar SiC surface, however, Ga_2O_3 may form in a variety of polytopes exhibiting distinct symmetry groups selected from monoclinic (C2m), rhombohedral (R3c or R3m), orthorhombic (Pna21), cubic (Fm3m), and even hexagonal (P63mc). Ideally a single homogeneous polytope is desired for the epilayer to achieve the highest electronic performance for the present application. Mixed polytope epilayer compositions are possible but may produce lower quality materials and devices. The stability of a given polytope epitaxially formed on the SiC surface is governed by several factors: (i) in-plane lattice mismatch between dissimilar crystal symmetry space groups of the epilayer and substrate for a given crystal oriented plane and epilayer; (ii) the surface energy of the SiC surface; (iii) the first bonding atomic species of the SiC surface to a first complementary atomic specie of Ga_2O_3 (i.e. Ga or O); (v) the relative ratio of incident fluxes Ga and active-O atoms chemisorbed and physisorbed at a given surface deposition temperature, as well as other factors.

[0435] The devices, structures and methods described herein can include the beta-phase monoclinic form of Ga_2O_3 which can be stabilized to a wide growth window by selectively depositing on a prepared C-plane surface of 4H-SiC. Furthermore, the $\beta\text{-Ga}_2\text{O}_3$ film epitaxially forms in a layer-by-layer growth mode oriented in the $[-201]$ crystal direction **Z4-4020**, as shown in FIG. 52. Monoclinic Ga_2O_3 has 50% of the Ga—O bonds tetrahedrally coordinated and the other 50% of the Ga—O bonds are octahedrally coordinated. The coordination of an O-terminated SiC surface is discussed later with implications to the stacking sequence of Ga_2O_3 epitaxy. Of interest is the stacking sequence of the Ga and O atoms along the epitaxial growth direction **Z4-4020**. Atomic planes comprising: a buckled Ga—Ga layer, followed by a substantially planar two-dimensional O-lattice, a substantially planar two-dimensional Ga-lattice, followed again by a substantially planar two-dimensional O-lattice—the sequence repeating along the growth direction. During co-deposition of Ga and O species the self-assembly of the crystalline structure occurs. An optional final surface on the

(−201) plane is shown as an oxygen terminated surface **Z4-4030**. This can be achieved post epitaxial deposition by exposing the surface to active-O species (or exclusively active-O species). The surface reconstruction of an O-terminated plane is shown in FIG. 53, showing the hexagonal arrangement of O-atoms **Z4-4010**.

[0436] FIG. 54 shows a further simplified lattice model for assessing the lattice mismatch of a first Si-terminated or C-terminated surface for an on-axis C-plane-oriented 4H-SiC in FIG. 54 having planar hexagonal inter-atomic distances $a(\text{Si-Si})=a(\text{C-C})=3.095 \text{ \AA}$.

[0437] FIG. 55 shows the oxygen hexagonal lattice for an oxygen terminated (−201) oriented $\beta\text{-Ga}_2\text{O}_3$ surface. The O-lattice is a deformed hexagon with $2.985 \leq a(\text{O-O}) \leq 3.083 \text{ \AA}$. This enables an estimation of the in-plane lattice constant mismatch (LM) to range from:

$$-0.38 \leq \Delta a = \frac{(a_{\text{O-O}} - a_{\text{Si-Si}})}{a_{\text{Si-Si}}} \leq -3.6\%$$

In practice, $\text{LM} < 5\%$ is desirable for lower defect density epilayers, indicating that the O-lattice should be able to accommodate the in-plane compressive strain elastically.

[0438] FIG. 56 depicts schematically the epitaxial orientations possible for producing the Oxide-SiC article. The C-plane cut 4H-SiC substrate **Z9-9030** can be oriented to present a Si-face or C-face for epitaxial deposition. The intrinsic polarity **Z9-9040** of 4H-SiC influences the surface electron affinity (as shown in FIG. 48) and therefore the heterojunction band alignment with an epitaxial oxide. If the 4H-SiC is oriented with surface normal **Z9-9010**, this defines the epitaxial growth direction **Z9-9010**. It is also possible the epilayer may grow/assemble with a tilt such that the surface normal may not completely coincide with the epilayer crystal plane. For example, the semiconducting oxide **Z9-9020** can be $\beta\text{-Ga}_2\text{O}_3$. The first surface of the epilayer **Z9-9050** can be predetermined by the growth sequence initiating the epitaxy process. Furthermore, the final epilayer surface **Z9-9060** may also be predetermined by virtue of the growth process.

[0439] Further analysis of the 4H-SiC C-plane oriented surface structures are shown in FIGS. 57A, 57B, 57C and 57D.

[0440] FIG. 57A shows a cross-sectional view of the atomic arrangements for a prepared C-plane 4H-SiC crystal which is Si-polar **Z10-10010**, and presenting a final surface that has a topmost plane of Si-atoms that are preferentially bonded to oxygen (O) atoms. This can be achieved by ultrahigh vacuum anneal up to approximately 1000°C . Optionally, exposure of a low flux of Si atoms impinging on the surface can also ensure a final Si surface state once the native silicon-oxy-carbide has been desorbed. Exposure of the Si-terminated surface with active species of oxygen can then self-limit the surface with a O-terminated surface. The stacking sequence would then be from the surface downwards: —O—Si—C—Si—C—.

[0441] FIG. 57B shows a cross-sectional view of the atomic arrangements for a prepared C-plane 4H-SiC crystal which is C-polar **Z10-10020**, and presenting a final surface that has a topmost plane Si-atoms that are preferentially bonded to oxygen (O) atoms. This can be achieved by ultrahigh vacuum anneal up to approximately 1000°C . Optionally, exposure of a low flux of Si atoms impinging on

the surface can also ensure a final Si surface state once the native silicon-oxy-carbide has been desorbed. Exposure of the Si-terminated surface with active species of oxygen can then self-limit the surface with a O-terminated surface. The stacking sequence would then be from the surface downwards: —O—Si—C—Si—C—.

[0442] Comparison of FIGS. 57A and 57B shows the oxygen bonding between the topmost O—Si atoms having distinct bonding symmetry which dictates the preferred next adatom layer of Ga arrangement (refer the Ga and O arrangement in FIG. 52).

[0443] Oxygen can bond to either Si or C with bond strengths:

$$E_{\text{bond}}(\text{Si-O}) = 452 \text{ kJ/mol}, E_{\text{bond}}(\text{Si=O}) = 590 \text{ kJ/mol},$$

$$E_{\text{bond}}(\text{C-O}) = 360 \text{ kJ/mol}, \text{ and } E_{\text{bond}}(\text{C=O}) = 715 \text{ kJ/mol}$$

[0444] FIGS. 57C and 57D show an alternate first oxygen bonding arrangement to a carbon topmost surface configured on a Si-polar or C-polar 4H-SiC surface, respectively.

[0445] FIG. 57C shows a cross-sectional view of the atomic arrangements for a prepared C-plane 4H-SiC crystal which is Si-polar **Z10-10030**, and presenting a final surface that has a topmost plane C-atoms that are preferentially bonded to oxygen (O) atoms. This can be achieved by ultrahigh vacuum anneal in excess of approximately 1000°C . Optionally, exposure of a low flux of C atoms impinging on the surface can also ensure a final C surface state once the native silicon-oxy-carbide has been desorbed. Exposure of the C-terminated surface with active species of oxygen can then self-limit the surface with a O-terminated surface. The stacking sequence would then be from the surface downwards: —O—C—Si—C—. Further crystal growth is shown with an additional Ga adlayer shown with tetragonal bonding arrangement forming sequence —Ga—O—C—Si—C—.

[0446] FIG. 57D shows a cross-sectional view of the atomic arrangements for a prepared C-plane 4H-SiC crystal which is C-polar **Z10-10040**, and presenting a final surface that has a topmost plane C-atoms that are preferentially bonded to oxygen (O) atoms. This can be achieved by ultrahigh vacuum anneal in excess of approximately 1000°C . Optionally, exposure of a low flux of C atoms impinging on the surface can also ensure a final C surface state once the native silicon-oxy-carbide has been desorbed. Exposure of the C-terminated surface with active species of oxygen can then self-limit the surface with a O-terminated surface. The stacking sequence would then be from the surface downwards: —O—C—Si—C—. Further crystal growth is shown with an additional Ga adlayer shown with tetragonal or octahedral bonding arrangement forming sequence —Ga—O—C—Si—C—.

[0447] The SiC surface preparation is an important step for the realization of high-quality single crystal $\beta\text{-Ga}_2\text{O}_3$ (−201) oriented epilayers, with methods and structures disclosed herein.

[0448] FIGS. 57E and 57F depict influence of a C-plane 4H-SiC surface that is intentionally misoriented from the on-axis surface (i.e., miscut at an angle >0 from the on-axis surface) on the epilayer growth of Ga_2O_3 .

[0449] FIG. 57E shows the cross-sectional view of the Si-polar 4H-SiC crystal along the [0001] direction **Z10-**

10010 with a miscut surface **Z10-10070** inclined an angle θ from the on-axis c-plane surface **Z10-10050**. **Z10-10065** is a miscut direction normal to the substrate surface, while direction **Z10-10060** is the c-axis of the SiC. The angle θ (or miscut angle) in FIG. 57E is not drawn to scale. In some cases, θ is from 2 to 6 degrees, or about 4 degrees. Below the plane of miscut surface **Z10-10070** comprises the SiC crystal and above comprises the epitaxial Ga_2O_3 growth. The epilayer can either grow along the **Z10-10060** direction with stacking faults or grow inclined with tilt substantially directed with epilayer crystal planes **Z10-10055**. FIG. 57F schematically depicts the epilayer crystal planes growing with tilted planes **Z10-10055** on the 4H-SiC crystal **Z10-10080** having a step-terrace surface **Z10-10080**.

[0450] FIG. 57G further discloses an example process flow **Z10G-1000** for the fabrication of the structure template surfaces disclosed in FIGS. 57A, 57B, 57C and 57D that can be further incorporated into structures and devices disclosed herein.

[0451] The integration of two dissimilar crystal symmetry groups (SGs) begins by the epitaxial deposition of an oxide having a symmetry group SG-2 (e.g., monoclinic Ga_2O_3) onto the substrate having a symmetry group SG-1 (e.g., hexagonal 4H-SiC). Advantageous alignment of SG-2 with SG-1 can occur via judicious choice of substrate crystal plane in order to match to a possible epilayer layer plane of the oxide SG-2. For example, the Si-polar face of C-plane 4H-SiC can be terminated with a Si sub-lattice surface. The Si sub-lattice can be terminated with an oxygen adlayer thereby forming a linking layer with a Ga-adatom layer.

[0452] The oxide epilayer (e.g., $\beta\text{-Ga}_2\text{O}_3$) final surface can also be terminated surface type and with a species conducive for subsequent formation of a high quality first metal layer. The first metal layer can be of the order of 1-10 nm in thickness or more, and can be followed by another second metal layer.

[0453] FIG. 57H is another example of a fabrication process flow for the epitaxial deposition of thick oxide films deposited on SiC substrates. Growth conditions necessary for the self-assembly of Ga and O into a specific polytope of Ga_2O_3 (e.g., monoclinic) can be achieved by providing a growth temperature and a predetermined incident flux and ratio of co-deposited Ga and active-oxygen species. Stacking faults and other crystalline misalignments and coordination are possible and may accumulate as a function of film thickness. An example method to achieve thick epitaxial films in excess of some limiting co-deposition thickness provides for a growth interruption wherein the film growth is suspended and the surface quality is recovered. For example, epitaxial growth of $\beta\text{-Ga}_2\text{O}_3$ (-201) can be interrupted after approximately ~100 nm and annealed in an oxygen environment by suspending the Ga-flux. The surface improvement may be evidenced by in-situ diagnostics. The epitaxial growth can be recommenced again for predetermined time or figure of merit indicative of film quality. The cycle can be repeated a plurality of times until a desired thickness is achieved. In some examples, the growth interruption introduces non-growth species, such as, nitrogen, atomic nitrogen, a surfactant (e.g., Bismuth or Zinc or Magnesium or Germanium), hydrogen or atomic hydrogen, and others.

[0454] FIG. 58 shows experimental data from an epitaxial $\beta\text{-Ga}_2\text{O}_3$ (-201) film deposited directly onto a prepared on-axis n-type 4H-SiC(0001) oriented substrate surface.

Atomic beam epitaxy was used in an ultrahigh vacuum deposition chamber, utilizing a pure Ga metal and an activated oxygen source from high power quartz bulb inductively coupled RF-plasma source. In some example growths O_2 and N_2 and N_2O were co-fed into one or more RF plasma sources. Epitaxy systems and methods are described for example in U.S. Pat. Nos. 10,964,537, 11,282,704, 11,342,484, and 11,670,508, which are hereby incorporated by reference.

[0455] The 4H-SiC substrate for the example of FIG. 58 was directly heated with a high temperature or mid infrared emissivity SiC heater. The wafer was rotated during growth and positioned with respect to the sources so that a highly uniform film could be deposited over a 4" diameter wafer. The substrate growth temperature was between 500-1000° C. depending on the SiC substrate resistivity. N-type 4H-SiC (0001) substrates were grown at about 500-700° C. without a backside coating due to the direct emissivity match between the heater and substrate. Semi-insulating 4H-SiC substrates were grown at about 700-950° C. without a backside coating due to the direct emissivity match between the heater and substrate. It was found in some example epitaxial growths that the highest quality single crystal films resulted from higher substrate deposition temperatures.

[0456] Reflection high energy electron diffraction (RHEED) is an essential in-situ diagnostic for direct and real time assessment of the epilayer film quality and crystal structure. FIG. 58 shows the RHEED images at high symmetry points as a function of azimuthal rotation of the substrate about the surface normal for an approximately 2 micron thick $\beta\text{-Ga}_2\text{O}_3$ (-201) epilayer grown upon n-type 4H-SiC(0001) Si-polar on-axis. The excellent single crystal quality is evidenced by the long and sharp RHEED diffraction streaks indicating an atomically flat film. The RHEED streaks were monitored and maintained throughout the film growth. The (-201)-oriented film has a 60-degree periodicity as expected and is completely devoid of any polycrystalline arcs or pyramidal surface features as spots.

[0457] FIG. 59 shows RHEED images of a thick $\beta\text{-Ga}_2\text{O}_3$ (-201) epilayer grown upon n-type 4H-SiC(0001) Si-polar substrate with an intentional 4-degree miscut from the on-axis surface toward the A-plane. Sharp and intense streaks are also shown and the streaks are bent due to the miscut of the substrate. Spots are superimposed on the streaks indicating the incident RHEED electron beam being partially transmitted through the terrace steps. It is determined in the present disclosure that growth on miscut SiC surfaces may provide superior Ga_2O_3 film quality with respect to rotation domains that are possible for the (-201) reconstruction.

[0458] FIGS. 60A and 60B show yet further evidence of the high crystal and optical quality of the epilayer Ga_2O_3 grown on SiC substrate, in the form of in-situ oblique laser reflectance (equipment set-up shown in FIG. 60A) at a wavelength of $\lambda=405$ nm. The high initial reflectance (plot in FIG. 60B) from the bare SiC surface decreases upon deposition of the Ga_2O_3 film with optical thickness oscillations appearing every $\lambda/2n$ where $n(\text{Ga}_2\text{O}_3)=1.9$. Growth rates (GR) between 0.1 and 4 Angstroms per second are possible. Higher growth rates are also possible with scaled elemental sources and chamber characteristics.

[0459] FIG. 61A shows the wide-angle double crystal diffraction Ω -2 θ symmetric scan of a Ga_2O_3 epilayer on an

SiC substrate. Sharp and strong β -Ga₂O₃ n(-201), n=1, 2, 3, 4 and 5 Bragg peaks are shown with no evidence of parasitic polytopes.

[0460] FIGS. 61B and 61C disclose the asymmetric x-ray diffraction reciprocal lattice maps in the vicinity of two Bragg peaks centered on the 4H-SiC(107) and β -Ga₂O₃(-403) for a hetero-epitaxial structure formed as a 2 μ m thick β -Ga₂O₃ (-201)/4H-SiC(0001) 0 deg miscut. The RSM maps show the oxide film is single crystal and of high quality and shows that dissimilar crystal symmetry space groups can be conformed.

[0461] It is also possible in another example for the Ga₂O₃ epilayer to have other pure polytope when deposited on alternate SiC surface orientations. For example, FIG. 62 depicts the crystal structure of the alpha-phase Ga₂O₃ (SG=R3c) which is similar to sapphire and may be formed on A-R- or M-plane SiC (2H-, 4H- or 6H-). FIG. 63 also shows the possible wurtzite crystal polytope of Ga₂O₃ having a SG=P63mc that may be formed on some SiC substrate orientations. Other polytopes are also possible such as the SG=Pna21 or a cubic (defective SG=Fm3m). Of particular interest are the Pna21 and P63 ms space groups due to the intrinsic property that they are polar crystals. It is determined in the present disclosure that forming Ga-polar or O-polar Ga₂O₃ films (e.g., with Pna21 and P63 ms space groups) on Si-polar or C-polar SiC enables new electronic devices to be constructed potentially utilizing spontaneous and piezoelectric polarization induced carrier doping.

[0462] FIG. 64 schematically depicts a novel aspect of the present disclosure directed toward pre-determining the electronic character of a final surface of an oxide epitaxial film. It was found that the β -Ga₂O₃ (-201) surface has an electron affinity that is sensitive to the deposition condition, post deposition condition, thermal annealing, annealing in oxygen, and exposure to other environments. It was further found that hydrogen can play an important role in controlling the final state of the surface electron affinity. The bulk region of a β -Ga₂O₃ film or substrate is depicted as Z17-17010 which may have previously incorporated hydrogenic (or hydrogen containing) impurities Z17-17020 distributed throughout the bulk or within a thickness Z17-17030. For example, bulk grown Ga₂O₃ substrates exhibit high levels of hydrogenic impurities. These hydrogenic impurities may be liberated from surface region Z17-17040 by thermal or chemical treatment Z17-17050, wherein hydrogenic (or hydrogen containing) particle Z17-17045 diffuses or migrates to the surface Z17-17055 and is removed from the structure Z17-17060. A topmost surface portion of thickness Z17-17035 can therefore be dissimilar to a bulk region Z17-17030.

[0463] Conversely, a surface region Z17-17040 can be chemical or thermally treated with a process Z17-17050 that introduces hydrogenic species into the surface region Z17-17040 to a depth of Z17-17035. In one example, the high temperature epitaxial growth of oxide layer is terminated, and growth temperature ramped to ambient over a time (t) while still being subjected to a large oxygen partial pressure. The oxygen annealing during substrate temperature ramp-down post-growth of the epilayer may selectively deplete the surface region Z17-17040 of impurities or induce an effect to deplete the surface of a charge carrier. Such techniques can be useful, for example, if the epilayer is co-doped with Si, Ge or Sn during epitaxy then the final surface region Z17-17040 may be significantly lower doping

than the bulk region Z17-17010. Such a depleted surface, if left unmodified, would directly impact the Schottky barrier creation, as discussed herein.

[0464] FIG. 65 shows the effect on a β -Ga₂O₃ (-201) oriented epilayer surface deposited as an n-type drift layer on top of a highly doped n+ layer or substrate. The bandgap E_g is shown separating the conduction $E_c(z)$ and valence $E_v(z)$ band edges. For this example, Si-doped Ga₂O₃ with impurity donor doping is used for the epilayer ("epi") $N_{drift}=5 \times 10^{16} \text{ cm}^{-3}$.

[0465] Intentionally exposing the post epitaxial growth surface to hydrogen and atomic hydrogen can result in a neutral surface, however at elevated substrate temperatures the hydrogen exposure can create a surface electron accumulation layer that penetrates into the epilayer to a depth Z_{accum} ("Z_{depl,accum}"). Conversely, a post grown epitaxial layer surface when exposed to oxygen at elevated temperatures creates a depleted surface. The surface band bending for the possible scenarios is shown in FIG. 65 as 'Depleted surface' or as a 'Charge accumulation surface'.

[0466] FIGS. 65, 66A and 66B show the effect of surface modification to the β -Ga₂O₃ (-201) drift layer for a device similar to that shown in FIG. 9B using a Ni metal contact. A neutral drift layer surface will result in SB potential of $\phi_{SB}=1.05 \text{ eV}$, a depleted surface will have a high barrier at the heterointerface $\phi_{SB}=1.3 \text{ eV}$, whereas the surface charge accumulated case will result in $\phi_{SB}=0.85 \text{ eV}$. The results of the forward bias I-V are shown in FIGS. 66A (semilog) and 66B (linear) plots.

[0467] The plots show that the surface state of the post growth drift layer directly influences the turn-on voltage and subthreshold slope of the diodes.

[0468] Comparing 'clean' and 'depleted' surfaces of a β -Ga₂O₃ (-201) drift layer with a range of high work function metals for the device structure similar to that shown in FIG. 9B (but with different contact metal) is shown in FIG. 67. A clean surface is treated with hydrogen at elevated temperatures, whereas the depleted surface is annealed in oxygen.

[0469] FIG. 67 shows the large variation in the possible forward I-V curves for the various high work function metals when deposited upon the two types of surfaces (depleted or clean). This helps explain the variation in experimental results for various devices and processing methods.

[0470] FIGS. 68A and 68B show example process flows for the manipulation of the surface charge state of a final surface of an epitaxial oxide layer. An epitaxial oxide layer is grown in a highly oxidizing environment by design to form the metal oxide composition and at a high growth temperature T_{growth} . Post epilayer growth the metal source is abruptly removed from growth region, shown as the step 'Stop growth'. The oxygen and active oxygen species in the growth chamber may be in the range of 1e-6 to 1e-4 torr and takes a finite time to evacuate from the chamber such that the epilayer surface is not substantially interacting. The substrate may be cooled at a ramp rate to a cooler temperature or a hotter temperature. In one example the growth temperature is reduced to a lower idle value while the ambient gas environment is substantially oxygen.

[0471] In this example, the epilayer surface may be modified to exhibit a depleted surface charge density. In another example, once the epilayer growth is stopped then oxidizing gas is removed and replaced with an alternate gas species or

none. An alternate gas species may be nitrogen or argon such that the surface does not substantially interact with the gas while ramping to idle. In another example, the epilayer post growth is held at a high temperature or lower temperature while replacing the chamber gas with a non-oxidizing hydrogenic species, such as hydrogen, forming gas or hydrogen plasma, or methane, as well as others. In other examples, the surface may be subjected to hydrogen and oxygen or hydrogen peroxide or a hydrocarbon. During the ramp to idle the hydrogenic specie may be maintained or pumped out.

[0472] After the surface charge state is achieved, the epilayer article may be transferred to another chamber under vacuum or remain in the growth chamber for a subsequent in-situ direct metal deposition. Metal deposition may be a high work function metal that is deposited directed onto the charge state modified surface of the epilayer. In one example a Nickel effusion source is used to deposit pure Ni in UHV environment directly onto the $\beta\text{-Ga}_2\text{O}_3$ (-201) surface. In an example, the Ni may be epitaxial and form a Ni(111) oriented first metal layer directly on the hexagonal $\beta\text{-Ga}_2\text{O}_3$ (-201) surface. The electronic surface charge of the $\beta\text{-Ga}_2\text{O}_3$ (-201) can be preserved with the in situ Ni(111) cap. Subsequent ex-situ processing may deposit further metal(s) to increase the thickness of the contact layer for mechanical strength and/or increased conductivity.

[0473] In some cases, a Ni-oxide layer can be formed directly on the oxygen terminated $\beta\text{-Ga}_2\text{O}_3$ (-201) surface followed by a Ni metal cap.

[0474] FIG. 69 describes the epitaxial Ni metal contact layer (SB) deposition directly onto the semiconducting $\beta\text{-Ga}_2\text{O}_3$ (-201) surface ("Semi"). A monolayer of oxygen (OI) can be used on the O-terminated surface to form linking bonds to the Nickel (M) metal contact (SB). In some cases, a thick intentional deposition of NiO can be formed followed by the Ni metal cap. Structures similar to the structure shown in FIG. 69 are also possible. In some cases, any of the metal materials used for the Schottky metal described herein can form an oxide at the interface between the drift layer and the Schottky metal layer. For example, any of the metal materials in FIG. 11A can form an oxide containing the metal species and oxygen (e.g., PtO_2) at the interface between a drift layer (e.g., including Ga_2O_3 or other epitaxial oxide material) and a layer of the metal (the Schottky contact). In some cases, the oxide of the metal forming the Schottky barrier can form the intermediate layer, as described herein. In some cases, the oxide of the metal forming the Schottky barrier can be thicker than the example shown in FIG. 69. For example, it can have a thickness in the range of the intermediate layers described herein.

[0475] The devices, structures and methods described herein can include in-situ single crystal Ni(111) oriented films directly deposited on the $\beta\text{-Ga}_2\text{O}_3$ (-201) surface at low temperatures and high vacuum $<1\text{e-}8$ torr.

[0476] There are also many other possible configurations for implementing intermediate layers (IL) or drift layers comprising UWBG oxide compositions.

[0477] FIGS. 70A-70G show an experimental example of an epitaxial structure comprising an epitaxial $\beta\text{-Ga}_2\text{O}_3$ (-201) layer formed on a SiC(0001) substrate, and an epitaxial Ni (111) layer. The Ni(111) layer in this example was 20 nm thick. The epitaxial structure in this example was formed using MBE using the techniques described herein,

for example as described in methods Z10G-10000 in FIG. 57G, or Z21-21010 and Z21-21020 in FIGS. 68A-68B.

[0478] FIG. 70A depicts schematically a multilayered epitaxial structure 70100 comprising a SiC substrate 70105, an epitaxial semiconducting oxide layer 70110, an epitaxial metal layer 70115, an optional further contact layer 70120 to the metal layer 70115, and an optional contact layer 70125 to the substrate 70105.

[0479] FIG. 70B shows a RHEED image 70200 of a high symmetry surface in-plane azimuth ($\phi=203^\circ$) diffractogram of structure 70205, which includes a single crystal epitaxial $\beta\text{-Ga}_2\text{O}_3$ (-201) oriented layer at the top surface, which was formed on a 4-degree miscut n-type Si-polar 4H polytope SiC(0001) substrate. The single crystal high quality oxide film is evidenced by the elongated and thin RHEED streaks in FIG. 70B.

[0480] FIG. 70C shows a RHEED image 70210 of the same high symmetry surface in-plane azimuth ($\phi=203^\circ$) diffractogram of structure 70215, which includes a single crystal epitaxial metal Ni(111) oriented layer at the top surface, which was deposited directly upon the single crystal epitaxial $\beta\text{-Ga}_2\text{O}_3$ (-201) oriented surface of structure 70205. The single crystal high quality metal film is evidenced by the elongated and thin RHEED streaks in FIG. 70C.

[0481] FIG. 70D shows a RHEED image 70300 of a high symmetry surface in-plane azimuth ($\phi=233^\circ$) diffractogram of the same structure 70205 of FIG. 70B, which includes the single crystal epitaxial $\beta\text{-Ga}_2\text{O}_3$ (-201) oriented surface deposited directly upon a 4-degree miscut n-type Si-polar 4H polytope SiC(0001) substrate. The single crystal high quality oxide film is again evidenced by the elongated and thin RHEED streaks as shown.

[0482] FIG. 70E shows a RHEED image 70310 of the same high symmetry surface in-plane azimuth ($\phi=233^\circ$) diffractogram of the structure 70215, which includes a single crystal epitaxial metal Ni(111) oriented surface deposited directly upon the single crystal epitaxial $\beta\text{-Ga}_2\text{O}_3$ (-201) oriented surface of structure 70205. The single crystal high quality metal film is again evidenced by the elongated and thin RHEED streaks as shown.

[0483] The RHEED results shown in FIGS. 70C and 70E confirm that the periodicity of the surface azimuth RHEED diffractograms occurs every $\Delta\phi=60^\circ$, which is indicative of a Ni(111) oriented film comprising a face-centered cubic crystal.

[0484] FIGS. 70F and 70G respectively show a wide angle x-ray diffraction plot 70400 and a high-resolution x-ray diffraction plot 70500 for the completed multilayered structure 70215 comprising epitaxial Ni(111), which was formed on an epitaxial $\beta\text{-Ga}_2\text{O}_3$ (-201) layer, which was formed on an Si-polar 4H polytope SiC(0001) substrate. Inset 70505 shows an atomic structure of a Ni(111) oriented surface. The high resolution x-ray diffraction plot 70500 in FIG. 70G includes thickness fringes for the Ni(111) peak, which shows the high quality of the interfaces and the low amount of surface roughness of the structure 70215.

[0485] These RHEED and x-ray diffraction results in FIGS. 70B-70G clearly show phase pure (or single phase) semiconducting oxide and Ni-metal Bragg diffraction patterns, indicating a coherent epitaxial relationship between the deposited layers.

[0486] FIGS. 71A, 71B, 72 and 73 disclose a range of suitable UWBG semiconducting oxide compositions that are

possible and advantageous for increasing the performance of semiconductor devices, such as high electrical power devices and diodes.

[0487] In general, any of the materials described herein (e.g., any of the compositions of Materials 1-7, $(\text{Mg}_x\text{Zn}_{1-x})(\text{Al}_y\text{Ga}_{1-y})_2\text{O}_4$ where $0 \leq (x, y) \leq 1$ or $(\text{Mg}_x\text{Ni}_{1-x})_z(\text{Al}_y\text{Ga}_{1-y})_2(1-z)\text{O}_{3-2z}$, where $0 \leq (x, y, z) \leq 1$, or $(\text{Ni}_z\text{Mg}_x\text{Zn}_{1-x-z})(\text{Al}_y\text{Ga}_{1-y})_2\text{O}_4$ where $0 \leq (x, y, z) \leq 1$, or $(\text{Zn}_p\text{Mg}_x\text{Ni}_{1-x-p})_z(\text{Al}_y\text{Ga}_{1-y})_2(1-z)\text{O}_{3-2z}$, where $0 \leq (p, x, y, z) \leq 1$) can be used for the drift layers, intermediate layers, and other layers of the devices, structures and methods described herein.

[0488] For example, FIG. 71A discloses the bandgap range possible for compositions of the form of Zinc-Gallium-Oxide and Zinc-Aluminum-Oxide. The spinel structures ZnGa_2O_4 and ZnAl_2O_4 exhibit SG=Fd3m and deposit as (111)-oriented films on $\beta\text{-Ga}_2\text{O}_3$ (-201) or C-plane 4H-SiC.

[0489] In another example, FIG. 71B discloses the bandgap range possible for compositions of the form of Magnesium-Gallium-Oxide and Magnesium-Aluminum-Oxide.

[0490] The spinel structures MgGa_2O_4 and MgAl_2O_4 exhibit SG=Fd3m and deposit as (111)-oriented films on $\beta\text{-Ga}_2\text{O}_3$ (-201) or C-plane 4H-SiC.

[0491] In other examples, FIGS. 72 and 73 show quaternary composition of oxide materials that can be used for the drift layers, intermediate layers, and other layers of the devices, structures and methods described herein.

[0492] FIG. 73 shows the possible bandgap variation of magnesium-zinc-gallium-oxide and magnesium-zinc-aluminum-oxides.

[0493] FIG. 74 shows the possible bandgap variation of magnesium-aluminum-gallium-oxide and zinc-aluminum-gallium-oxides.

[0494] In some cases, the drift layers, intermediate layers, and other layers of the devices, structures and methods described herein can include the spinel forms (SG=Fd3m) of $(\text{Mg}_x\text{Zn}_{1-x})(\text{Al}_y\text{Ga}_{1-y})_2\text{O}_4$ where $0 \leq (x, y) \leq 1$ forming complex oxides can be realized as intrinsic p-type or n-type conductivity types (without impurity doping) thereby alleviating the need for impurity doping. In another form, the spinels can be deposited as (111)-oriented films on $\beta\text{-Ga}_2\text{O}_3$ (-201) or C-plane 4H-SiC.

[0495] In an example, $(\text{Ni}_z\text{Mg}_x\text{Zn}_{1-x-z})(\text{Al}_y\text{Ga}_{1-y})_2\text{O}_4$ where $0 \leq (x, y, z) \leq 1$, or $(\text{Mg}_x\text{Ni}_{1-x})_z(\text{Al}_y\text{Ga}_{1-y})_2(1-z)\text{O}_{3-2z}$, where $0 \leq (x, y, z) \leq 1$ forming complex oxides (e.g., with the spinel forms (SG=Fd3m)) can be used as intrinsic, p-type, or n-type conductivity types (without impurity doping) in the structures and devices described herein, thereby alleviating the need for impurity doping. In another form, the spinels can be deposited as (111)-oriented films on $\beta\text{-Ga}_2\text{O}_3$ (-201) or C-plane 4H-SiC in the structures and devices described herein.

[0496] Other SG forms of Ga_2O_3 (e.g., α -phase or β -phase); $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ (e.g., α -phase or β -phase) where $0 \leq x \leq 1$; $\text{Mg}_x\text{Ga}_{2(1-x)}\text{O}_{3-2x}$ where $0 \leq x \leq 1$; $\text{Mg}_x\text{Al}_{2(1-x)}\text{O}_{3-2x}$ where $0 \leq x \leq 1$; $(\text{Mg}_x\text{Zn}_{1-x})_z(\text{Al}_y\text{Ga}_{1-y})_2(1-z)\text{O}_{3-2z}$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Zn}_x\text{Ga}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Zn}_x\text{Al}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Mg}_x\text{Ga}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Mg}_x\text{Al}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $\text{Ni}_x\text{Al}_y\text{O}_z$ where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$; $(\text{Ni}_z\text{Mg}_{1-x})_y\text{Ga}_{2(1-y)}\text{O}_{3-2y}$ where $0 \leq x \leq 1$ and $0 \leq y \leq 1$; or $(\text{Mg}_x\text{Ni}_{1-x})_z(\text{Al}_y\text{Ga}_{1-y})_2(1-z)\text{O}_{3-2z}$, where $0 \leq x \leq 1$, $0 \leq y \leq 1$, and $0 \leq z \leq 1$, or $(\text{Ni}_z\text{Mg}_x\text{Zn}_{1-x-z})(\text{Al}_y\text{Ga}_{1-y})_2\text{O}_4$ where $0 \leq (x, y,$

$z) \leq 1$, or $(\text{Zn}_p\text{Mg}_x\text{Ni}_{1-x-p})_z(\text{Al}_y\text{Ga}_{1-y})_2(1-z)\text{O}_{3-2z}$, where $0 \leq (p, x, y, z) \leq 1$, can also be used in the materials, structures and devices described herein.

[0497] Yet another aspect of the heterojunction — Ga_2O_3 (-201)/4H-SiC(0001) is disclosed, which relates to the doping density of the SiC.

[0498] Consider the diode formed as shown in FIGS. 74 and 75 as structure 75-28010, showing an anode contact (A) and — Ga_2O_3 (-201) epilayer deposited on Si-polar on-axis 4H-SiC(0001) layer or substrate with cathode contact (K). Consider further the oxide is n-type and doped to a concentration of $1 \times 10^{17} \text{ cm}^{-3}$, whereas the n-type doping concentration of the 4H-SiC layer is varied $1 \times 10^{15} \leq N_d^{\text{SiC}} \leq 5 \times 10^{17} \text{ cm}^{-3}$.

[0499] FIG. 74 shows the full conduction and valence band structure as a function of growth direction. The conduction band heterointerface is expanded in FIG. 75 showing the depletion effect at the junction extending into the SiC region. In this case, low N_d^{SiC} generates a large potential barrier which can significantly impact device performance, for example in the diode structure in FIG. 9C.

[0500] Therefore, it can be advantageous to control the epitaxy process utilized to fabricate the structure to also control the doping density of the SiC (substrate or layer). For example, a highly conductive 4H-SiC substrate can be prepared for epitaxial oxide growth. While the SiC substrate is heated to the required growth temperature in vacuo, the surface of the SiC substrate may alter the surface charge density and desorb potential impurity dopants. This may significantly reduce the 4H-SiC surface donor density and therefore create a significantly wider depletion region extending into the SiC material. The result may be a higher ON-resistance for the diode and increased electrical losses.

[0501] In some cases, the SiC substrate is heated to the required growth temperature in an environment that avoids altering the surface charge density. For example, the SiC substrate may be heated in an inert environment, or under an Si flux, to prevent Si desorption.

[0502] It may not be fully appreciated by workers in the field that heterojunctions formed at the interface between two dissimilar Oxide materials may also form complex interactions. The Metal-[Oxide Semiconductor] junction may form a Schottky Barrier-type contact or Ohmic-type contact, as discussed above. Heterojunctions between two Oxides having dissimilar bandgap energies, composition, electron affinity and furthermore dissimilar dielectric constants directly affects the properties in the vicinity of the heterojunction.

[0503] FIG. 76 shows the simplified cross-sectional physical structure of a drift oxide semiconductor with a portion of the drift oxide surface deposited with a metal along the A-A' direction parallel to the growth direction, z. Similarly, the isolation layer discussed above serving as the dielectric insulator between a field-plate and the oxide drift layer (refer to FIGS. 79-86) is also formed along a portion of the oxide drift surface along the direction B-B', as shown. The thickness of the isolation layer (ISO) t_{iso} and the thickness of the drift layer t_{drift} are indicated. Assuming the isolation layer further comprises an oxide material composition that possesses a wider bandgap energy than the drift layer oxide composition, such that, $E_{g1} < E_{g2}$, then the equivalent energy band diagram along the B—B' direction is shown in FIG. 77.

[0504] FIG. 77 is a plot of the spatial energy band diagram of the heterojunction formed between the oxide semiconductor drift and the dissimilar oxide or material region

forming the isolation region. The left-hand side of the spatial conduction band diagram $E_C(z)$ corresponds to the drift layer and the right-hand side corresponds to the isolation layer, with the equilibrium Fermi energy E_F position between two possible 1st electron quantized states in the inverted potential well located at the heterointerface.

[0505] If the E_F is positioned above the 1st quantized electron state, then heterointerface two-dimensional electron gas may exist with a surface charge $N_s^{int} > 0$, forming an interfacial negatively charged sheet. This charge sheet may be continuous or discontinuous depending on the surface roughness at the interface, however, this represents a surface leakage path that is disadvantageous for vertical conduction device operation. This may be utilized in a lateral transport device.

[0506] Conversely, if E_F is positioned below the 1st quantized electron state then the heterointerface will not accumulate a surface charge $N_s = 0$, and therefore may act as a suitable blocking barrier.

[0507] Yet a further aspect specific to interactions between abrupt interfaces between dissimilar oxide material compositions is the large variation in dielectric constant that is possible and the resulting large dielectric constant discontinuity $\Delta\epsilon = \epsilon_{drift} - \epsilon_{iso}$ at the heterointerface. Dielectric confinement (deconfinement) effects are possible to increase (decrease) the Coulomb interaction between charge carriers located at or in the vicinity of the discontinuity.

[0508] For example, if the drift layer comprises a Ga_2O_3 , and the isolation layer a low-k dielectric constant material then $\Delta\epsilon = \epsilon_{drift} - \epsilon_{iso} > 0$. Alternately, in another example, if the drift layer comprises a Ga_2O_3 and the isolation layer a high-k dielectric constant material then $\Delta\epsilon = \epsilon_{drift} - \epsilon_{iso} \leq 0$.

[0509] FIG. 78A tabulates the calculated static and optical dielectric constants for some example UWBG oxide materials suitable for incorporation into the present disclosure. For some materials there exist crystallographic direction dependent properties.

[0510] The concept of dielectric confinement and screening is treated herein by specific reference to a boundary between two slabs of oxide materials having dissimilar dielectric constants.

[0511] FIGS. 78B and 78C plot the electrostatic equipotential lines between two equal and opposite charge regions separated by a distance of 0.5 units in the x-direction and hosted in differing dielectric constants slabs of materials. FIG. 78B provides the case for equal dielectric constants between both the left-hand side (ϵ_1) and right-hand side (ϵ_2) slabs. The symmetric equipotential lines are shown. However, for the case where the ratio of $\epsilon_2/\epsilon_1 = 10$ is shown in FIG. 78C the equipotential lines are highly asymmetric across the hetero-dielectric interface with the potential screened within the higher dielectric constant slab.

[0512] From FIGS. 78B and 78C are calculated the electric-field strengths shown in the contour plots of FIGS. 78D and 78E, respectively. The electric field screening effect of the high dielectric region is shown. Quantitative values of the electric-field vector are calculated in the vicinity of the hetero-dielectric interface and are shown in FIGS. 78F and 78G.

[0513] FIG. 78F shows the spatial variation of the x- and y-components of the electric-field E_x (E_y). If both halves of the structure have equal dielectric constants, s.t., $\epsilon_2/\epsilon_1 = 1$, then the symmetry in the E_x and E_y components along the interface are shown for three cases at the interface $x=0$ and

a small amount on either side ($x = \pm 0.005$ units). As expected, the symmetry of the curves reflects the equal and opposite charges.

[0514] FIG. 78G shows the simulation for the case of $\epsilon_2/\epsilon_1 = 10$, in which the electric field within the high dielectric region is substantially screened as shown. This phenomenon can be utilized in the present disclosure for engineering the maximum electric-fields created in device structures due to fabrication geometry-induced effects. For example, sharp edges that are typically formed at the periphery of metal contacts by photolithographic processing, lift-off or etching can create disadvantageous fringing fields. It is therefore desirable to manage these fringing fields at sharp metal features because electric fields can be generated in excess of the oxide electric field breakdown strength.

[0515] FIGS. 79 and 80 show two-dimensional electrostatic models of two types of vertical multilayered semiconductor devices used for the formation of Schottky barrier diodes.

[0516] Shown are two similar structures differentiated by the thickness between the field-plate (FP) overhang region and the drift region final surface (F4). The SiC substrate F3 is electrically conductive, whereas the isolation region F2 is given the properties of having a bandgap energy in excess of the drift region but having a dielectric constant that can be selected to be different to the drift layer. The region F1 is taken as air and the drift region F4 is Ga_2O_3 having a low charge density. The gap between the FP and drift region is 0.4 units in the vertical direction for the model of FIG. 79 and 0.05 units in the vertical direction for the model of FIG. 80. Furthermore, the Schottky metal contact is drawn as a split field-plate wherein the left-hand side does not have a field-plate overhang. This enables the same simulation to compare designs with and without a field-plate. The contribution to the overall device capacitance of the FP will be larger for the small gap configuration which is important for high-speed switching application.

[0517] FIGS. 81 and 82 show the magnitude of the calculated spatial electric fields as contours for the thin FP biased at +1000V and the rear contact at -1000V. The dielectric constants of all the regions are taken into account with FIG. 81 having $\epsilon_{iso}/\epsilon_{drift} = 0.2$ and FIG. 82 having $\epsilon_{iso}/\epsilon_{drift} \leq 3$.

[0518] FIGS. 83 and 84 show a 'thick' FP closeup of the electrostatic electric field contours for the case of low-k and high-k, respectively.

[0519] FIGS. 85 and 86 show a 'thin' FP closeup of the electrostatic electric field contours for the case of low-k and high-k, respectively.

[0520] Quantitative values for the potential energy in the vicinity of the drift layer horizontal interface in contact with the metal contact and isolation layer were calculated. Plotted in FIGS. 87 and 88 are the cases of a thin FP having an isolation material of low-k and high-k, respectively. The potential is at a maximum at the metal interface and slightly lower in the vicinity of the contact but positioned slightly below the interface. The lateral extent of the FP shows the potential is maintained for the case of high-k isolation material, whereas the potential drops off rapidly for a low-k isolation material.

[0521] The spatial magnitude of the electric field in the vicinity of the interface is plotted in FIGS. 89 and 90 for the cases of a thin FP having an isolation material of low-k and high-k, respectively. The large peaks in the electric field

located at the metal contact corners are evident as well as the effect of the lateral FP. The results show that the high-k isolation layer is effective in lowering the peak electric fields at the metal contact critical points.

[0522] Further analysis of the split FP structure enables direct comparison of the maximum electric-field generated at the interface between the drift layer and the metal contact.

[0523] FIGS. 91 and 92 are plots of the spatial variation of the magnitude of the electric field generated within the structure for three horizontal planes, namely, at the hetero-interface, slightly above and below the heterointerface between the drift layer and isolation layer. The device structure is a thin field-plate/isolation layer/drift layer spacing.

[0524] FIG. 91 plots the maximum electric field within the structure at the drift layer and metal contact interface region for the case of 'no field-plate' and 'with field-plate', as a function of the dielectric constant used in the isolation material. The FP is shown to reduce the maximal electric field for all cases of the isolation layer dielectric constant, either lower or higher than the drift layer material. There are diminishing returns with the FP beyond $\epsilon_{ISO}/\epsilon_{Drift} > 4$, wherein a simple metal-mesa contact along with a conformal high-k dielectric can in an alternative example provide superior performance. It is also possible to deposit a high-k isolation layer and then etch a region for defining a Schottky metal contact. FIG. 92 further investigates the ratio γ of maximum electric field with and without using a FP by defining:

$$\gamma = \frac{E_{FP}^{surf} - E_{no FP}^{surf}}{E_{no FP}^{surf}}$$

[0525] Maximum benefit of the FP is witnessed in FIG. 92 at the minimum of Ratio γ when the ratio of isolation layer to drift layer dielectric constant is in the range of about $0.5 \leq \kappa \leq 2.0$. For this range, an ~45% reduction in maximum electric field is possible. For high-k isolation materials, i.e., $\kappa > 3.0$, the utility of the FP is diminished as dielectric confinement and screening dominates and a simple metal mesa contact is efficient.

[0526] In various embodiments, the results of the above exploration of isolation layer materials and the effect of the FP may be taken into account when optimizing a diode design in accordance with the present disclosure.

[0527] FIG. 93 depicts the two-dimensional electrostatic model used for exploring a simple metal mesa Schottky contact embedded within an isolation materials of varying dielectric constants.

[0528] The resulting calculation for the maximal surface electric field strength at the drift and the metal/isolation layer interface is plotted in FIG. 94. An order of magnitude reduction in peak surface electric field can result by utilizing isolation layer materials selected from high dielectric constants. High dielectric materials that have the property of bandgaps greater than the drift region and dielectric constants well in excess of Ga_2O_3 can be found in rare-earth oxides, titanium oxide and the like. Of interest are rare-earth oxides which can be alloyed with aluminum to increase the effective bandgap while maintaining high dielectric constant.

[0529] Of note is nickel-oxide which exhibits a large dielectric constant by low bandgap energy.

[0530] FIG. 95 shows a photograph of a partially completed Schottky barrier diode formed on a Ga_2O_3 epilayer deposited on a 4H-SiC substrate. The Schottky barrier metal is Ni, and a second metal of Al was used to thicken the contact. The rear ohmic contact was not processed, showing the transparency of the substrate.

[0531] In some embodiments, the structures and devices described herein include a "T-Channel" design, which can further improve the operating characteristics of the epitaxial oxide/Schottky metal devices described herein.

[0532] As discussed above, the dielectric constants of the materials incorporated into the device structure govern the electrostatic behavior of the device under high (or intense) electric fields. Furthermore, as described herein, the isolation layer surrounding the Schottky metal contact can be used to engineer the surface electric field at the oxide drift layer and metal interface. For example, high dielectric constant isolation layer materials can be used to minimize surface electric fields at the critical points of the Schottky metal contact.

[0533] FIG. 96A shows an alternative device design referred to as a "T-Channel" design herein. A rear contact 96132 (F4) is made to a SiC substrate 96110 (F2) and an oxide semiconductor drift layer 96125 (F6) are shown. In the "T-Channel" design, an oxide semiconductor channel region 96120 (F3) with a Schottky metal layer 96130 (F5) is formed (e.g., by selective area etching) into a mesa structure. In some cases, oxide semiconductor drift layer 96125 has a higher doping density than oxide semiconductor channel region 96120. A high doping density of the oxide semiconductor drift layer 96125 can be advantageous, for example, to reduce the ON-resistance of the device. The doping density of the oxide semiconductor channel region 96120 can be tuned to create a Schottky junction with tunable characteristics (e.g., to have a desired depletion width). Therefore, the doping density of the oxide semiconductor channel region 96120 can be either higher or lower than that of the drift layer 96125, depending on the desired Schottky junction properties. Sidewall layer 96180 (F7 and F8) is formed around the mesa structure. Sidewall layer 96180 (F7 and F8) can surround the mesa structure. Sidewall layer 96180 (F7 and F8) can be in physical contact with one or more walls 96122 and 96124 of the mesa structure. The walls 96122 and 96124 of the mesa structure include interfaces of the oxide semiconductor channel region 96120 and the Schottky metal layer 96130. Field plate layer 96190 (F9) can be on a top surface of Schottky metal layer 96130, and extend laterally (i.e., in the x-direction) beyond the Schottky metal layer 96130 (and extend laterally beyond the mesa structure) such that it also is on a top surface of the sidewall layer 96180 (as shown in FIG. 96A). Schottky metal layer 96130 can comprise any suitable metal described herein, for example, any material of FIG. 11A (i.e., Cu, Te, Be, Rh, Co, C, N, Ni, Au, Ir, Pd, Pt, Se, or Os), or alloys of the metals in FIG. 11A. Additionally, in some cases, substrate 96110 is a composite substrate comprising a surface layer and a bulk substrate layer (e.g., bulk substrate layer 112 and surface layer 114 of substrate 110 in FIG. 1), where the surface layer comprises single crystal SiC.

[0534] For example, the mesa structure in FIG. 96A, including the oxide semiconductor channel region 96120 and the Schottky metal layer 96130, can have a circular

cross-section, and the sidewall layer **96180** can surround the mesa structure in an annular shape (or wrap around the mesa structure). In such an example, wall **96122** and **96124** (or sidewalls) would be the same wall surrounding the cylindrical shaped mesa structure. In another example, the mesa structure can be rectangular shaped, and the sidewall layer **96180** can be shaped like a rectangular annulus surrounding the mesa structure. In such an example, the sidewall layer **96180** can be in physical contact with walls **96122** and **96124** as well as two other walls (not shown) of the rectangular prism shaped mesa structure. In some cases, sidewall layer **96180** can have a dielectric constant (k) greater than or equal to that of the oxide semiconductor drift layer **96125** and the oxide semiconductor channel region **96120**. For example, sidewall layer **96180** can have a dielectric constant (k) greater than 10, or greater than 100, or from about 10 to about 1200. For example, the drift layer can comprise Ga_2O_3 , and the sidewall layer **96180** can comprise TiO_2 (dielectric constant about 100), barium titanate BaTiO_3 and associated polymorphs (dielectric constant about 1200), or diamond (dielectric constant about 16.5).

[0535] In another example, the walls **96122** and **96124** (or sidewalls) can be tapered inwardly or outwardly from the top of the drift layer **96125**. In another example, the taper may be curved so as to adopt a generally convex or concave profile. In another example, the taper smoothly varies towards the slope or tangent of the interface defined by the oxide semiconductor channel region **96120** and drift layer **96125**. In yet another example, the taper smoothly varies towards the slope or tangent of the interface defined by the oxide semiconductor channel region **96120** and contact layer Schottky metal layer **96130**.

[0536] In other examples, the shape of the oxide semiconductor channel region **96120** may be configured to extend longitudinally in the “z-direction” perpendicular to the view shown in FIG. 96A to enable the cross-sectional area of the channel to accommodate a predetermined current density.

[0537] Region **96010** (F1) in FIG. 96A represents air with a permittivity of air. The sidewall layer **96180** can have a dielectric constant greater than, equal to, or less than that of the oxide drift and channel regions. For example, oxide semiconductor regions **96125** and **96120** can be Ga_2O_3 and the sidewall dielectric can have a lower, equal, or higher dielectric constant than that of Ga_2O_3 . In some embodiments, the sidewall layer **96180** dielectric material has a bandgap energy in excess of the drift and channel oxide materials so that it behaves as an insulator. In other cases, the sidewall layer **96180** dielectric material has a bandgap energy equal to or less than those of the drift and channel materials, and the sidewall layer **96180** dielectric material is semi-insulating or insulating.

[0538] In some cases, the oxide semiconductor channel region **96120** has a different doping density and/or material composition compared to those of the oxide semiconductor drift layer **96125**. For example, the doping density of the oxide semiconductor channel region **96120** can be lower compared to that of the oxide semiconductor drift layer **96125**. In another example, the composition of the oxide semiconductor channel region **96120** can be different than that of the oxide semiconductor drift layer **96125**, such that the oxide semiconductor channel region **96120** has a lower, the same or higher bandgap compared to that of the oxide semiconductor drift layer **96125**.

[0539] The oxide semiconductor drift layer **96125** and the oxide semiconductor channel region **96120** may be formed of the same semiconductor oxide material or different oxide materials. In some cases, the oxide semiconductor drift layer **96125** is formed of an oxide semiconductor material with the same composition and crystal symmetry as those of an oxide semiconductor material forming oxide semiconductor channel region **96120**. In other cases, the oxide semiconductor drift layer **96125** is formed of an oxide semiconductor material with a different composition and/or crystal symmetry as those of an oxide semiconductor material forming oxide semiconductor channel region **96120**. In an example, the oxide semiconductor drift layer **96125** may be formed of a material chosen to provide a structural matching region between the substrate **96110** and the oxide semiconductor channel region **96120**. The oxide semiconductor drift layer **96125** and the oxide semiconductor channel region **96120** also may be formed of materials with the same or different bandgaps. For example, the oxide semiconductor drift layer **96125** can have a bandgap that is larger or smaller than that of the oxide semiconductor channel region **96120**. In some cases, the oxide semiconductor drift layer **96125** and the oxide semiconductor channel region **96120** have a conduction band offset that does not substantially block the movement of electrons from the oxide semiconductor drift layer **96125** to the oxide semiconductor channel region **96120** or vice-versa. In some cases, the oxide semiconductor drift layer **96125** and the oxide semiconductor channel region **96120** have a valence band offset that does not substantially block the movement of holes from the oxide semiconductor drift layer **96125** to the oxide semiconductor channel region **96120** or vice-versa. In general, the bandgap of any region (including the channel region) of a structure or device described in the present disclosure may be modified by a suitable change in composition of the material. For example, for a ternary material such as $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ the bandgap can be varied by changing the molar fraction x from Al_2O_3 to Ga_2O_3 . In practice, there can be limits on the composition range possible, while maintaining high quality (i.e., low defect) materials. In the example of $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$, $x < 0.35$ is the limit, and it can be difficult to obtain high quality single phase $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ on a Ga_2O_3 bulk layer with $x > 0.35$.

[0540] For example, a oxide semiconductor drift layer **96125**, comprising a semiconductor oxide material of the form of $\text{Mg}_x\text{Zn}_y\text{Ni}_z(\text{Al}_p\text{Ga}_{1-p})\text{O}_q$ (or another oxide semiconductor material described herein) can be formed between an SiC substrate **96110** and the oxide semiconductor channel region **96120**, where the oxide semiconductor channel region **96120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples. Another example of a semiconductor oxide material is of the form of a silicon oxy-carbide $\text{Si}_x\text{C}_y\text{O}_z$, which can be comprised in the oxide semiconductor drift layer **96125** and can be formed between an SiC substrate **96110** and the oxide semiconductor channel region **96120**, where the oxide semiconductor channel region **96120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples. Yet another example of a semiconductor oxide material is of the form of a silicon oxy-nitride $\text{Si}_x\text{N}_y\text{O}_z$ which can be comprised in the oxide semiconductor drift layer **96125** and can be formed between an SiC substrate **96110** and the oxide semiconductor channel region **96120**, where the oxide semiconductor channel region **96120** comprises Ga_2O_3 or compositionally graded

$\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples. Another example of an oxide semiconductor drift layer **96125** is one comprising a material of the form of a silicon-germanium-carbide $\text{Si}_x\text{Ge}_y\text{C}_z$ such that the oxide semiconductor drift layer **96125** can be used to manage the interfacial strain formed between an SiC substrate **96110** and the oxide semiconductor channel region **96120**, where the oxide semiconductor channel region **96120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples.

[0541] In some cases, the oxide semiconductor channel region **96120** may have a doping concentration between that of the substrate **96110** doping concentration and oxide semiconductor channel region **96120** with a discrete change in doping concentration between the substrate **96110** and the oxide semiconductor drift layer **96125**, and between the oxide semiconductor drift layer **96125** and the oxide semiconductor channel region **96120**. In an example, the substrate **96110** may be n^+ doped (i.e., having a doping concentration of from 10^{19} to 10^{20} cm^{-3}), the oxide semiconductor drift layer **96125** may be n^- doped (i.e., having a doping concentration of approximately 10^{18} cm^{-3} , or from 10^{17} cm^{-3} to 10^{18} cm^{-3}) and the oxide semiconductor channel region **96120** may be n^{++} doped (i.e., having a doping concentration from 10^{15} cm^{-3} to 10^{17} cm^{-3}), where this configuration provides transition between a n^+ doped region to a n^{++} doped region moving vertically upwards from the substrate **96110** to the oxide semiconductor channel region **96120**. Similarly, a p-type device can be formed wherein the substrate **96110** is p^+ doped, the oxide semiconductor drift layer **96125** is p^- doped, and the oxide semiconductor channel region **96120** is p^{--} doped, where the layers have the doping concentrations described above of acceptors rather than donors.

[0542] In some cases, a doping density of the oxide semiconductor drift layer **96125** has a monotonic gradient in the z direction. For example, the doping density of the oxide semiconductor drift layer **96125** can start (closer to the substrate **96110**) with a doping concentration near or below (i.e., less than) that of the substrate **96110**, and end (farther from the substrate **96110**) at a doping concentration approximately equal to that of the oxide semiconductor channel region **96120**. In some cases, the oxide semiconductor drift layer **96125** has a doping concentration that monotonically varies starting at a doping concentration that is either approximately equal to that of the substrate **96110**, or is between that of the substrate **96110** and an oxide semiconductor channel region **96120** doping concentration, and ending at a doping concentration that is either approximately equal to that of the oxide semiconductor channel region **96120** or is between the substrate **96110** doping concentration and oxide semiconductor channel region **96120** doping concentration. In an example, the oxide semiconductor drift layer **96125** doping concentration may match the substrate **96110** doping concentration at the interface between the oxide semiconductor drift layer **96125** and the layer below (e.g., substrate **96110**) and then continuously change to match the oxide semiconductor channel region **96120** doping concentration at the interface between the oxide semiconductor drift layer **96125** and the oxide semiconductor channel region **96120**. In an example, the change in the oxide semiconductor drift layer **96125** doping concentration may be substantially linear as a function of vertical height z . In some cases, the oxide semiconductor drift layer **96125** doping concentration may change continuously (i.e., without

significant step changes, and monotonically, or non-linearly, for example) as a function of vertical height z . In some cases, the oxide semiconductor drift layer **96125** doping concentration may change in a stepwise manner as a function of vertical height z .

[0543] In other examples, the doping density of the oxide semiconductor channel region **96120** may be higher or the same as that of the oxide semiconductor drift layer **96125**, and the composition of the oxide semiconductor channel region **96120** can be different than that of the oxide semiconductor drift layer **96125**, such that the oxide semiconductor channel region **96120** has a lower, the same or higher bandgap compared to that of the oxide semiconductor drift layer **96125**. In some embodiments, there is no substantial (or minimal) barriers to the flow of electrons between the rear contact **96132** and the oxide semiconductor channel region **96120**. In other words, in some cases electrons may freely flow from the rear contact **96132** to the oxide semiconductor channel region **96120**. Therefore, in some cases, the bandgap of the drift region **96125** is the same as or smaller than that of the oxide semiconductor channel region **96120**. In other cases, the bandgap of the drift region **96125** is larger than that of the oxide semiconductor channel region **96120**, and there is no substantial (or a minimal) barrier to the flow of electrons from the drift region **96125** to the oxide semiconductor channel region **96120**, due to a band alignment with no substantial (or a minimal) conduction band offset between the materials.

[0544] FIGS. **96B-96H** show modeled (or calculated) spatial electric fields within devices having structures like that shown in FIG. **96A** where the sidewall material **96180** has various dielectric constants with respect to the oxide semiconductor of the drift and channel layers. Specifically, FIGS. **96B-96H** show the results of applying a potential of -1000V to the lower electrode and $+1000\text{V}$ to the upper field-plate for sidewall layer **96180** material dielectric constant of $2.5 \leq \epsilon_{\text{SW}} \leq 300$. The modeled devices therefore cover a range of sidewall layer **96180** materials, ranging from a low- k dielectric to a high- k dielectric.

[0545] FIG. **96B** shows the calculated (i.e., modeled) electric field magnitude along a vertical centerline positioned at $x=0$ in the structure shown in FIG. **96A**. FIG. **96B** also shows the channel region **96126** (from $y=0.2$ to $y=0.6$) along the vertical centerline ($x=0$). As the sidewall layer **96180** dielectric constant ϵ_{SW} is varied from low- k to high- k relative to the Ga_2O_3 channel, the electric field shows some unique attributes. For the case of $\epsilon_{\text{SW}}=2.5$ (e.g., approximately the dielectric constant of Al_2O_3) the electric field is maximal at the Schottky metal interface with the channel. For the case of $\epsilon_{\text{SW}}=\epsilon_{\text{channel}}=10$, the electric field is still maximal at the Schottky metal interface with the channel. For the case of $\epsilon_{\text{SW}}=30$ the electric field has a minimum within the channel, and for the case of $\epsilon_{\text{SW}}=300$ the situation is inverted, and the electric field is minimum at the Schottky metal interface with the channel.

[0546] FIGS. **96C-96F** show the contour plots of the electric field for each of the cases plotted in FIG. **96B**. FIG. **96C** shows the case of $\epsilon_{\text{SW}}=2.5$, FIG. **96D** shows the case of $\epsilon_{\text{SW}}=10$, FIG. **96E** shows the case of $\epsilon_{\text{SW}}=30$, and FIG. **96F** shows the case of $\epsilon_{\text{SW}}=300$.

[0547] FIGS. **96G** and **96H** respectively plot the electric field magnitude and the x -component of the electric field at the hetero-dielectric interface (i.e., at $y=0.2$ in FIG. **96A**) for the values of dielectric constants in FIG. **96A**. Extracting the

electric field components at the hetero-dielectric interface further illustrates the effect of sidewall dielectric constant on the device behavior.

[0548] The electric field results shown in FIGS. 96C-96H therefore show that selecting the value of the sidewall dielectric constant can be used control the maximum electric field at the hetero-dielectric interface and alleviate some of the occurrences of high peak electric fields observed in some other devices described herein (e.g., without a “T-channel” design). Furthermore, moving (or remoting) the Schottky barrier contact away from the surface of the channel regions, using the oxide channel region of the “T-channel” design can be used to dramatically reduce the peak electric fields at the surface of the channel regions (i.e., at the hetero-dielectric interface) in epitaxial oxide/Schottky devices.

[0549] Additionally, the “T-channel” device design shown in FIG. 96A and described herein can also be combined with any of the other device layer configurations and geometries described herein. For example, a device with a transition layer, such as those shown in FIGS. 4-7, could also incorporate a “T-channel” design by adding a transition layer between SiC substrate 96110 and oxide semiconductor drift layer 96125 of FIG. 96A. In another example, an intermediate layer can be added between the oxide semiconductor channel region 96120 and the Schottky metal layer 96130 as described herein, for example in FIGS. 13-17 and the associate description. In another example, a guard ring, such as shown in FIGS. 39D-39G, can be added surrounding the mesa structure. As one skilled in the art would appreciate, the features of many of the devices described herein can be combined with the “T-channel” design, including by not limited to those shown in FIGS. 20, 28-30, 40, 45, 48, 78A, and 79-80.

[0550] FIG. 97 shows an example device configuration, wherein the device configuration shown in FIG. 96A is repeated “(i.e., a plurality of the devices placed side by side)” to form an array of devices on a common SiC substrate 96110. In some cases, the top contacts (Schottky metal layer 96130 and optionally, field plate layer 96190) can be interconnected using an electrical interconnect schematically shown by interconnect 97110. Interconnect 97110 can be a metal layer, metal traces, or wire bonded wires coupling the field plate layers 96190 to one another. For example, interconnect 97110 can be a blanket metal layer covering the field plate layers 96190. In other cases, interconnect 97110 can be a metal layer that is patterned to electrically couple the field plate layers 96190. In some cases, the SiC substrate 96110 and the rear contact 96132 are shared by multiple devices in the array of devices. In some cases, the oxide semiconductor drift layer 96125 is shared by multiple devices in the array of devices, and the devices are separated by individual mesa structures including the oxide semiconductor channel region 96120 and the Schottky metal layer 96130, as shown in FIG. 97.

[0551] FIG. 97 shows two devices on a common SiC substrate 96110, however, in other cases, from 2 to 1000 devices can be formed on a common SiC substrate 96110 (e.g., in a rectangular array, hexagonal array, or any other arrangement of multiple mesa structures). Such configurations including arrays of devices can be advantageous. For example, the current-carrying capacity can be increased by multiplexing a number of parallel connected devices on a common SiC substrate 96110. A trade-off (or optimization) of epitaxial area consumed, maximum vertical current

capacity, and/or thermal heating effects is possible by partitioning vertical current across a number of parallel inter-connected devices. In some cases, a distance 97120 (e.g., a minimum distance) between the devices, shown as a distance from a wall (or sidewall) of one mesa to a wall (or sidewall) of an adjacent mesa, is tuned (or optimized) to balance the epitaxial area consumed, maximum vertical current capacity, and/or thermal heating effects. In some cases, a width 97130 of the mesa (or oxide semiconductor channel region 96120) in the x-direction, and a width of the mesa (or oxide semiconductor channel region 96120) in the z-direction (out of the plane of the page of FIG. 97), are tuned (or optimized) to balance the epitaxial area consumed, maximum vertical current capacity, and/or thermal heating effects. In some cases, minimum distance 97120 between the devices, width 97130 of the mesa (or oxide semiconductor channel region 96120) in the x-direction, and/or a width of the mesa (or oxide semiconductor channel region 96120) in the z-direction (out of the plane of the page of FIG. 97), are tuned (or optimized) to balance the epitaxial area consumed, maximum vertical current capacity, and/or thermal heating effects.

[0552] FIGS. 98A, 98B and 98C show examples of devices with a “T-Channel” design as described with respect to FIG. 96A, and additionally with a transition layer 96133, an intermediate layer 96131, or both. These structures can be advantageous, since they can combine the advantages of the “T-Channel” device designs with advantages of the devices designs with transition layers and/or intermediate layers described herein. For example, the transition layer 96133 can impact the electronic and structural properties of the semiconductor diode device, while the intermediate layer 96131 can alter the Schottky barrier properties and impact the electronic properties of the semiconductor diode device such as the turn-on voltage and/or the reverse breakdown voltage. The intermediate layer 96131 and the transition layer 96133 can each be a single layer, or can include multiple layers of materials such as in a superlattice layer or a chirp layer.

[0553] FIG. 98A shows an example of a device with a “T-Channel” design as described with respect to FIG. 96A, and additionally with a transition layer 96133 between the substrate layer 96110 and the oxide semiconductor drift layer 96125. Transition layer 96133 is similar to transition layer 160 in FIGS. 4-7 and 8B, and can have the same properties as described with respect to transition layer 160 in FIGS. 4-7 and 8B. Transition layer 96133 can have properties of any of the transition layers described herein. For example, the transition layer 96133 can provide an intermediate level of doping between the substrate 96110 and oxide semiconductor drift layer 96125 and/or manage interfacial strain, and can thereby impact the electronic and structural properties of the semiconductor diode device, as described further herein.

[0554] Transition layer 96133 may be formed of a semiconductor oxide material, and the oxide semiconductor drift layer 96125 and the oxide semiconductor channel region 96120 may be formed of the same semiconductor oxide material or different oxide materials as the transition layer 96133. In an example, the transition layer 96133 may be formed of a material chosen to provide a structural matching region between the substrate 96110 and the oxide semiconductor drift layer 96125. For example, a transition layer 96133, comprising a semiconductor oxide material of the

form of $\text{Mg}_x\text{Zn}_y\text{Ni}_z(\text{Al}_p\text{Ga}_{1-p})\text{O}_q$ (or another oxide semiconductor material described herein) can be formed between an SiC substrate **96110** and the oxide semiconductor drift layer **96125**, where the oxide semiconductor drift layer **96125** and/or the oxide semiconductor channel region **96120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples. Another example of a semiconductor oxide material is of the form of a silicon oxy-carbide $\text{Si}_x\text{C}_y\text{O}_z$, which can be comprised in the transition layer **96133** and can be formed between an SiC substrate **96110** and the oxide semiconductor drift layer **96125**, where the oxide semiconductor drift layer **96125** and/or the oxide semiconductor channel region **96120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples. Yet another example of a semiconductor oxide material is of the form of a silicon oxy-nitride $\text{Si}_x\text{N}_y\text{O}_z$ which can be comprised in the transition layer **96133** and can be formed between an SiC substrate **96110** and the oxide semiconductor drift layer **96125**, where the oxide semiconductor drift layer **96125** and/or the oxide semiconductor channel region **96120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples. Another example of a transition layer **96133** is one comprising a material of the form of a silicon-germanium-carbide $\text{Si}_x\text{Ge}_y\text{C}_z$ such that the transition layer **96133** can be used to manage the interfacial strain formed between an SiC substrate **96110** and the oxide semiconductor drift layer **96125**, where the oxide semiconductor drift layer **96125** and/or the oxide semiconductor channel region **96120** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples.

[0555] In some cases, the transition layer **96133** may have a doping concentration between that of the substrate **96110** doping concentration and oxide semiconductor drift layer **96125** with a discrete change in doping concentration between the substrate **96110** and the transition layer **96133**, and between the transition layer **96133** and the oxide semiconductor drift layer **96125**. In an example, the substrate **96110** may be n^+ doped (i.e., having a doping concentration of from 10^{19} to 10^{20} cm^{-3}), the transition layer **96133** and oxide semiconductor drift layer **96125** may both be n doped (i.e., having a doping concentration of approximately 10^{18} cm^{-3} , or from 10^{17} cm^{-3} to 10^{18} cm^{-3}) and the oxide semiconductor channel region **96120** may be n^- doped (i.e., having a doping concentration from 10^{15} cm^{-3} to 10^{17} cm^{-3}), where this configuration provides a transition between a n^+ doped region to a n^- doped region moving vertically upwards from the substrate **96110** to the oxide semiconductor channel region **96120**. Similarly, a p -type device can be formed wherein the substrate **96110** is p^+ doped, the transition layer **96133** and the oxide semiconductor drift layer **96125** are both p^- doped, and the oxide semiconductor channel region **96120** is p^- doped, where the layers have the doping concentrations described above of acceptors rather than donors.

[0556] In some cases, the transition layer **96133** doping concentration has a monotonic gradient in the z direction, starting at a doping concentration below (i.e., less than) that of the substrate **96110**, and ending at a doping concentration approximately equal to that of the oxide semiconductor drift layer **96125**. In some cases, the transition layer **96133** doping concentration has a doping concentration that monotonically varies starting at a doping concentration that is either approximately equal to that of the substrate **96110** or is between the substrate **96110** doping concentration and the

oxide semiconductor drift layer **96125** doping concentration, and ending at a doping concentration that is either approximately equal to that of the oxide semiconductor drift layer **96125** or is between the substrate **96110** doping concentration and the oxide semiconductor drift layer **96125** doping concentration. In an example, the transition layer **96133** doping concentration may match the substrate **96110** doping concentration at the interface between the substrate **96110** and the transition layer **96133** and then continuously change to match the oxide semiconductor drift layer **96125** doping concentration at the interface between the transition layer **96133** and the oxide semiconductor drift layer **96125**. In an example, the change in the transition layer **96133** doping concentration may be substantially linear as a function of vertical height z . In some cases, the transition layer **96133** doping concentration may change continuously (i.e., without significant step changes, and monotonically, or non-linearly, for example) as a function of vertical height z . In some cases, the transition layer **96133** doping concentration may change in a stepwise manner as a function of vertical height z . Transition layer **96133** can have any of the attributes of transition layer **160** as described above (e.g., with respect to transition layer **160** in FIGS. 4-7).

[0557] FIG. 98B shows an example of a device with a “T-Channel” design as described with respect to FIG. 96A, and additionally with an intermediate layer **96131** between the Schottky metal layer **96130** and the oxide semiconductor channel region **96120**. Intermediate layer **96131** is similar to intermediate layer **170** in FIG. 8A, and can have the same properties as described with respect to intermediate layer **170** in FIG. 8A. Intermediate layer **96131** can have properties of any of the intermediate layers described herein. For example, the intermediate layer **96131** can alter the Schottky barrier properties and can thereby impact the electronic properties of the semiconductor diode device such as the turn-on voltage and/or the reverse breakdown voltage, as described further herein.

[0558] FIG. 98C shows an example of a device with a “T-Channel” design as described with respect to FIG. 96A, and additionally with both transition layer **96133** and intermediate layer **96131**.

[0559] In some embodiments, the epitaxial oxide structures and devices on SiC substrates described herein can further include a polar nitride layer between the epitaxial oxide layer and the Schottky metal layer. The addition of the nitride layer can be beneficial, for example, to increase a potential barrier formed between the epitaxial oxide layer and the electrical contact across the Schottky junction. For example, an epitaxial Group-III metal nitride layer (e.g., GaN, AlN, AlGaIn, or InAlGaIn) can be formed between an epitaxial oxide layer (e.g., Ga_2O_3 , or $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ where $0 \leq x \leq 1$) and a Schottky metal (e.g., Ni or a metal in FIG. 11A) and the polar nature of the nitride material can increase the potential barrier between the epitaxial oxide layer and a metal contact coupled to the Schottky metal.

[0560] In some embodiments, the epitaxial oxide structures and devices on SiC substrates described herein can further include a polar nitride layer between the epitaxial oxide layer and a metal layer, where the metal layer can form a Schottky junction or an Ohmic contact with the nitride. In some cases, the nitride material can have a p -type conductivity and form a p/n junction with the epitaxial oxide layer. In such cases, the metal layer can additionally form a Schottky junction or an Ohmic contact with the nitride layer.

[0561] For example, $\text{Al}_{0.3}\text{Ga}_{0.7}\text{N}$ can be epitaxially deposited on Ga_2O_3 , as described further below. In other cases, the drift layer can comprise any of Materials 1-7 and a nitride layer can be formed on the drift layer. For example, the drift layer can be formed using an epitaxial growth technique within an epitaxial growth chamber. After the epitaxial oxide drift layer is grown, a surface region of the drift layer can be converted into a nitride layer, for example, by exposing the surface to activated nitrogen species within the growth chamber as described further below. The material of the drift layer will thereby be converted to a nitride layer with the cation species of the drift layer (e.g., one of Materials 1-7) and wherein the oxygen has been replaced with nitrogen. For example, a thin region at the surface of an epitaxially grown layer of ZnMgO_3 (an example of Material 7) can be converted into ZnMgN_3 by exposing the surface of the oxide layer to a nitrogen plasma. The ratio of cations to oxygen atoms in the drift layer can be the same or different than the ratio of cations to nitrogen atoms in the nitride layer. Such a nitride layer can be epitaxial with the drift layer and can have a relatively low number of defects (e.g., threading dislocations). Such a thin (e.g., from about 1 nm to about 100 nm) converted nitride layer is an example of a template layer (as described further below). In some cases, a thicker (e.g., from about 50 nm to about 3 microns, or from about 100 nm to about 10 microns) nitride layer is subsequently grown on the nitride template layer.

[0562] In some embodiments, the addition of a nitride layer can beneficially form a p/n junction with the epitaxial oxide, and an Ohmic metal electrical contact can be coupled to the nitride layer. For example, a p/n junction can be formed using a n-type epitaxial oxide layer and a nitride layer with p-type conductivity. The p-type conductivity of the nitride layer can be induced by the polar nature of the nitride material or the interface between the nitride and the oxide materials. The nitride can additionally, or alternatively, include extrinsic dopants in some cases to make the nitride material p-type or n-type. For example, an epitaxial Group-III metal nitride layer (e.g., GaN, AlN, AlGaIn, or InAlGaIn) can be formed between an epitaxial oxide layer (e.g., Ga_2O_3 , or $(\text{Al}_x\text{Ga}_{1-x})_2\text{O}_3$ where $0 \leq x \leq 1$), and an Ohmic metal (e.g., Ti, Al, Ni, Au, Mo, and/or Ta) can form an Ohmic contact with the nitride. The polar nature of the nitride material can induce a strong p-type conductivity in the nitride, which can form a p/n junction with the n-type epitaxial oxide layer (e.g., Ga_2O_3).

[0563] FIG. 99 plots the bandgaps and electron affinities of semiconductor material surfaces on an absolute energy scale relative to vacuum energy ($E_{\text{vac}}=0$ eV), in accordance with some embodiments. The materials shown include C-plane oriented wz-GaN which has wurtzite crystal symmetry oriented Ga-polar (000-1) and N-polar (0001), Ga-polar and O-polar polytypes of Pna21 Ga_2O_3 , non-polar $\beta\text{-Ga}_2\text{O}_3$ (-201), and the C-plane surfaces of Si-polar and C-polar 4H-SiC materials. The electron affinity energy Z50-1010 of the conduction band edge for the Ga-polar wz-GaN surface, and the electron affinity energy Z50-1020 for the $\beta\text{-Ga}_2\text{O}_3$ (-201) surface are shown.

[0564] In some cases, a polar form of crystalline GaN can be formed on a $\beta\text{-Ga}_2\text{O}_3$ (-201) surface using the methods described herein. In some embodiments, the structures, devices, and methods described herein include vertical multilayer semiconductor structures with mixed polar and non-polar materials and mixed crystal symmetry groups. For

example, a structure, device, or method described herein can include a vertical multilayer semiconductor structure with a layer of an epitaxial oxide and a layer of a polar nitride material. In some cases, the structure can include a non-polar epitaxial oxide layer and a polar epitaxial nitride layer. In some cases, the structure can include a polar SiC substrate, a polar or non-polar epitaxial oxide layer, and a polar epitaxial nitride layer.

[0565] FIGS. 100A-100C depict some examples of possible multilayer stacks comprising epitaxial oxide drift layer 20020a-d (e.g., Ga_2O_3) sandwiched between two dissimilar but polar crystal layers comprising a 4H-SiC substrate 20010a-d and a wurtzitic group-III Nitride layer 20030a-d (e.g., GaN). A contact metal layer 20040a-d is in contact with the group-III Nitride layer 20030a-d, which can be an Ohmic contact or a Schottky barrier depending on the work function of the metal used and the polarity (n-type or p-type, as it impacts the work function) of the group-III Nitride layer 20030a-d. The [polar III-Nitride/Oxide/polar 4H-SiC] heterostructure of these figures can be configured to provide unique properties advantageous for the devices and applications of the present disclosure. The polarization direction influences how the layer will be doped within the structures. Depending on the orientation of the crystals of the SiC and the nitride, there can be charges induced at the polar/non-polar interfaces between the polar SiC and a non-polar epitaxial oxide drift layer (e.g., beta Ga_2O_3 oriented in the -201 direction), and between the polar nitride and the non-polar epitaxial oxide drift layer.

[0566] The epitaxial oxide drift layer 20020a-d can include non-polar oxide materials, or polar oxide materials, in some cases. For example, beta-phase Ga_2O_3 is a non-polar material whereas orthorhombic Pna21 (also known as epsilon-phase) Ga_2O_3 is polar. Therefore, the selection of the drift layer crystal phase may directly impact the operation of the device. As an example, a drift layer comprising epsilon Ga_2O_3 coupled to a polar GaN layer would form a polar-polar junction, which can induce charge in the epitaxial oxide drift layer 20020a-d making it either have a p-type or a n-type conductivity depending on the polarities of the epitaxial oxide drift layer 20020a-d and the wurtzitic group-III Nitride layer 20030a-d. Conversely, if a junction is formed between beta-phase Ga_2O_3 and polar GaN a non-polar to polar junction is formed.

[0567] The structures in FIGS. 100A-100D can include n-type or p-type 4H-SiC substrate 20010a-d, n-type or p-type epitaxial oxide drift layer 20020a-d (e.g., n-type Ga_2O_3 or p-type NiO), n-type or p-type wurtzitic group-III Nitride layer 20030a-d (e.g., GaN), and a metal contact layer 20040a-d that can form an Ohmic contact or Schottky barrier with the wurtzitic group-III Nitride layer 20030a-d. The polar 4H-SiC substrate 20010a-d and the wurtzitic group-III Nitride layer 20030a-d can each be doped n-type or p-type using intrinsic and/or extrinsic dopants and/or they can have an n-type or p-type induced charge due to their polarity. The epitaxial oxide drift layer 20020a-d can also be polar in some cases, and in such cases, can be doped using intrinsic and/or extrinsic dopants and/or can have an n-type or p-type induced charge due to the polarity. In other cases, the epitaxial oxide drift layer 20020a-d can be non-polar and can be doped n-type or p-type using intrinsic and/or extrinsic dopants.

[0568] The structures shown in FIGS. 100A-100D contain polar materials with different combinations of orientations

of the SiC and Group III-N polar crystals, such that their polarization axes are aligned with growth direction and face in the same or opposite directions. The polarization axis is indicated for the polar SiC substrate and the polar nitride layers by the arrows in FIGS. 100A-100D. The structure shown in FIG. 100A contains Si-polar SiC 20010a (with polarization axis indicated by the upward arrows), which in some cases induces electrons at the interface with the oxide increasing an n-type conductivity of the SiC substrate. In contrast, the structure shown in FIG. 100C contains C-polar SiC 20010c (with polarization axis indicated by the downward arrows), which in some cases induces holes at the interface with the oxide increasing a p-type conductivity of the SiC substrate. Furthermore, the structure shown in FIG. 100A contains an N-polar Group III nitride layer 20030a (with polarization axis indicated by the downward arrows), which in some cases induces electrons at the interface with the oxide increasing an n-type conductivity of the Group-III nitride layer. In contrast, the structure shown in FIG. 100B contains a metal-polar (e.g., Ga-polar in the case of GaN) Group III nitride layer 20030b (with polarization axis indicated by the upward arrows), which in some cases induces holes at the interface with the oxide increasing a p-type conductivity of the Group-III nitride layer. FIG. 100D contains a metal-polar Group III nitride layer 20030d, which in some cases induces holes at the interface with the oxide increasing a p-type conductivity of the Group-III nitride layer. FIG. 100D also contains C-polar SiC 20010d, which in some cases induces holes at the interface with the oxide increasing a p-type conductivity of the SiC substrate.

[0569] The structures in FIGS. 100A-100D can be formed by starting with a 4H-SiC substrate with a particular orientation, such that the polarization axis of the SiC substrate is aligned with the growth direction 20002. The polarization axes of the SiC substrate 20010a and SiC substrate 20010b are in the same direction as the growth direction 20002 (as shown in FIGS. 100A and 100B), while the polarization axes of the SiC substrate 20010c and SiC substrate 20010d are in the opposite direction as the growth direction 20002 (as shown in FIGS. 100C and 100D). The crystal orientation of the nitride layer 20030a-d can be determined by the growth conditions of the nitride layer, and the surface of the oxide layer 20020a-d below. The nitride layer 20030a-d can be a template layer, or can include a template layer (e.g., a nitride template layer) at the interface with the oxide layer 20020a-d below. For example, the surface of the oxide layer 20020a-d can be terminated with a metal-rich surface (e.g., Ga-rich) to form a metal-polar template, and optionally form a metal-polar nitride film on the template. Alternatively, the surface of the oxide layer 20020a-d can be terminated with an oxygen-rich surface (e.g., O-rich) to form a nitrogen-polar (N-polar) template, and optionally form an N-polar nitride film on the template. The growth conditions of the template layer can also impact the crystal orientation of the nitride layer. The polarization axes of the nitride layer 20030a and nitride layer 20030c are in the opposite direction as the growth direction 20002 (as shown in FIGS. 100A and 100C), while the polarization axes of the nitride layer 20030b and nitride layer 20030d are in the same direction as the growth direction 20002 (as shown in FIGS. 100B and 100D).

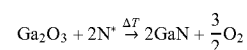
[0570] FIG. 101A shows an example of a device structure including a vertical multilayered semiconducting structure, in accordance with some embodiments. The structure in this

example includes: (i) n+ type 4H-SiC substrate Z52-1040 (e.g., Si-polar C-plane 4H-SiC); (ii) a rear contact Z52-1060; (iii) an optional interfacial ohmic region Z52-1050; (iv) an epitaxial oxide layer Z52-1030 (e.g., epitaxial β -Ga₂O₃ (-201)) which can be an active oxide layer; (v) a group-III Nitride layer Z52-1035 (e.g., epitaxially grown Ga-polar GaN); (vi) a metal contact Z52-1020 (forming a Schottky or Ohmic contact); (vii) a metal electrode Z52-1010. In some cases, metal contact Z52-1020 forms an Ohmic contact with group-III Nitride layer Z52-1035. For example, the group-III Nitride layer Z52-1035 can have a p-type conductivity and form a p/n junction with the epitaxial oxide layer Z52-1030. The epitaxial oxide layer Z52-1030 can be formed on the front side of the SiC substrate Z52-1040, where the growth direction z is shown by the arrow. The group-III nitride layer Z52-1035 can be formed on the active oxide layer, optionally with template layer Z52-1032 between them. The metal contact Z52-1020 (forming a Schottky or Ohmic contact) can be formed on the group-III nitride layer Z52-1035. The rear contact Z52-1060 can be formed on a backside of the substrate Z52-1040, optionally with interfacial ohmic region Z52-1050 between them. Additionally, in some cases, substrate Z52-1040 is a composite substrate comprising a surface layer and a bulk substrate layer (e.g., bulk substrate layer 112 and surface layer 114 of substrate 110 in FIG. 1), where the surface layer comprises single crystal SiC.

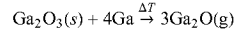
[0571] The thickness Z52-1070 of the epitaxial oxide layer is designed according to the embodiments disclosed herein. For example, the thickness Z52-1070 of the epitaxial oxide layer can be from 10 nm to 100 microns, or from 1 micron to 100 microns. The group-III Nitride layer Z52-1035 can be formed on an optional template layer Z52-1032 having thickness Z52-1090 and the epitaxial grown group-III Nitride layer Z52-1035 can have a thickness Z52-1080. In some cases, the thickness Z52-1090 of the template layer Z52-1032 can be about a few unit cells, or from about 1 nm to about 10 nm, or from about 1 nm to about 100 nm. In some cases, the thickness Z52-1080 of the group-III Nitride layer Z52-1035 can be from about 50 nm to about 1 micron, or from about 100 nm to about 5 microns, or from about 100 nm to about 10 microns.

[0572] In some cases, the group-III Nitride layer Z52-1035 can comprise one or more of GaN, AlN, AlGa_xN_{1-x} (e.g., Al_xGa_{1-x}N, where 0 ≤ x ≤ 1), InAlGa_xN_{1-x} (e.g., In_xAl_{1-x}Ga_{1-x-y}N_y, where 0 ≤ x ≤ 1, 0 ≤ y ≤ 1, and x + y ≤ 1), zinc nitride (e.g., Zn₃N₂), magnesium nitride (e.g., Mg₃N₂), zinc magnesium nitride (e.g., (Zn_xMg_{1-x})₃N₂, where 0 ≤ x ≤ 1), zinc germanium nitride (e.g., (Zn_xGe_{1-x})₃N₂, where 0 ≤ x ≤ 1), and magnesium germanium nitride (e.g., (Mg_xGe_{1-x})₃N₂, where 0 ≤ x ≤ 1).

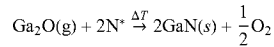
[0573] In some cases, the formation of the group-III Nitride layer Z52-1035 can start via a transformation of a portion (with thickness Z52-1090) of the top surface of the epitaxial oxide (e.g., Ga₂O₃) layer Z52-1030. In such cases, the template layer Z52-1032 can be formed by transforming a portion of the epitaxial oxide (e.g., Ga₂O₃) layer Z52-1030. For example, exposure of a Ga₂O₃ surface to an active species of nitrogen, such as atomic nitrogen from a plasma (e.g., a nitrogen or an ammonia plasma), can transform at least a portion of the topmost Ga—O bonds into Ga—N bonds, such that,



A thicker GaN layer can also be formed via decomposition of Ga_2O_3 to form a suboxide such that:



Wherein the Ga_2O vapor can react with a beam of atomic nitrogen under Ga-rich conditions to produce the reaction at the surface:



Once a template of hexagonal GaN (e.g., wz-GaN) is formed, then further epitaxial growth can be performed using beams of Ga and N precursors to form a thicker GaN layer.

[0574] The SiC substrate **Z52-1040** can be Si-polar C-plane 4H-SiC, or another type of SiC substrate, as described herein.

[0575] The rear contact **Z52-1060** can be a low-resistance or an Ohmic contact to the SiC substrate **Z52-1040**, and may also comprise high work function metals to enable co-processing of the front and backside of the device structure. For example, Ni can be used as a SB metal for the drift layer surface whereas Ni may act as an Ohmic contact to the highly doped SiC substrate. The optional interfacial ohmic region **Z52-1050**, for example containing silicides, may be used to form low resistance Ohmic contacts to the substrate. Silicon may be scavenged from the SiC substrate by thermal anneal processing of contact metal to form a silicide. For example, Ni may form $\text{Ni}_x\text{Si}_{1-x}$ and similarly Ti may form $\text{Ti}_x\text{Si}_{1-x}$. Furthermore, multiple metal layers may form an Ohmic contact to the substrate, such as SiC/Ti_xSi_{1-x}/Ti/Al.

[0576] The epitaxial oxide layer **Z52-1030** can be epitaxial $\beta\text{-Ga}_2\text{O}_3$ (−201), or another epitaxial oxide material described herein. For example, epitaxial oxide layer **Z52-1030** can include any of the oxide materials described herein, for example, any of the compositions of Materials 1-7 described above.

[0577] The group-III Nitride layer **Z52-1035** can be metal-polar GaN, AlGaN, AlN, InAlGaN, or another Group-III metal nitride. Other polarities can be used in other cases, for example, as shown in FIGS. 100A-100D.

[0578] The metal contact **Z52-1020** (forming a Schottky or Ohmic contact) and the metal electrode **Z52-1010** (e.g., a field plate layer, in some cases) may be formed from the same metal or different metals. For example, metal contact **Z52-1020** can form a Schottky contact and can include a high work function metal or alloy of Cu, Te, Be, Rh, Co, C, Ni, Au, Ir, Pd, Pt, Se, Os and combinations thereof (refer to FIG. 11A). In some cases, the metal contact **Z52-1020** can form an Ohmic contact with the Group-III metal nitride **Z52-1035** and can include metals such as Ti, Al, Ni, Au, Mo, and/or Ta. In another example, the metal contact **Z52-1020** (forming a Schottky or Ohmic contact) may be formed from high work function metals including, but not limited to Cu, Te, Be, Rh, Co, C, Ni, Au, Ir, Pd, Pt, Se, Os and combinations thereof, whereas metal electrode **Z52-1010** may be selected from non-high work function metals, such as, Al, Ti, Mo, W and others. High melting point metals can be

advantageous, for example for use in high temperature operation (e.g., in excess of 300° C.).

[0579] FIGS. 101B, 101C, and 101D show examples of devices with an epitaxial oxide layer **Z52-1030** and an epitaxial nitride layer **Z52-1035** as described with respect to FIG. 101A, and additionally with a transition layer **Z52-1045**, an intermediate layer **Z52-1048**, or both. These structures can be advantageous, since they can combine the advantages of devices with an epitaxial oxide layer and an epitaxial nitride layer with advantages of the device designs with transition layers and/or intermediate layers described herein. For example, the transition layer **Z52-1045** can impact the electronic and structural properties of the semiconductor diode device, while the intermediate layer **Z52-1048** can alter the Schottky barrier properties and impact the electronic properties of the semiconductor diode device such as the turn-on voltage and/or the reverse breakdown voltage. The intermediate layer **Z52-1048** and the transition layer **Z52-1045** can each be a single layer, or can include multiple layers of materials such as in a superlattice layer (e.g., short-period superlattices) or a chirp layer.

[0580] FIG. 101B shows an example of a device with an epitaxial oxide layer **Z52-1030** and an epitaxial nitride layer **Z52-1035** as described with respect to FIG. 101A, and additionally with a transition layer **Z52-1045** between the substrate layer **Z52-1040** and the oxide semiconductor drift layer **Z52-1070**. Transition layer **Z52-1045** is similar to transition layer **160** in FIGS. 4-7 and 8B, and can have the same properties as described with respect to transition layer **160** in FIGS. 4-7 and 8B. Transition layer **Z52-1045** can have properties of any of the transition layers described herein. For example, the transition layer **Z52-1045** can provide an intermediate level of doping between the substrate **Z52-1040** and oxide semiconductor layer **Z52-1030** and/or manage interfacial strain, and can thereby impact the electronic and structural properties of the semiconductor diode device, as described further herein.

[0581] Transition layer **Z52-1045** may be formed of a semiconductor oxide material, and the oxide semiconductor layer **Z52-1030** may be formed of the same semiconductor oxide material or different oxide materials as the transition layer **Z52-1045**. In an example, the transition layer **Z52-1045** may be formed of a material chosen to provide a structural matching region between the substrate **Z52-1040** and the oxide semiconductor layer **Z52-1030**. For example, a transition layer **Z52-1045**, comprising a semiconductor oxide material of the form of $\text{Mg}_x\text{Zn}_y\text{Ni}_z(\text{Al}_p\text{Ga}_{1-p})\text{O}_q$ (or another oxide semiconductor material described herein) can be formed between an SiC substrate **Z52-1040** and the oxide semiconductor layer **Z52-1030** (including Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples). Another example of a semiconductor oxide material is of the form of a silicon oxy-carbide $\text{Si}_x\text{C}_y\text{O}_z$, which can be comprised in the transition layer **Z52-1045** and can be formed between an SiC substrate **Z52-1040** and the oxide semiconductor layer **Z52-1030**, where the oxide semiconductor layer **Z52-1030** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples. Yet another example of a semiconductor oxide material is of the form of a silicon oxy-nitride $\text{Si}_x\text{N}_y\text{O}_z$, which can be comprised in the transition layer **Z52-1045** and can be formed between an SiC substrate **Z52-1040** and the oxide semiconductor layer **Z52-1030**, where the oxide semiconductor layer **Z52-1030** comprises Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples. Another example

of a transition layer **Z52-1045** is one comprising a material of the form of a silicon-germanium-carbide $\text{Si}_x\text{Ge}_y\text{C}_z$ such that the transition layer **Z52-1045** can be used to manage the interfacial strain formed between an SiC substrate **Z52-1040** and the oxide semiconductor layer **Z52-1030** (comprising Ga_2O_3 or compositionally graded $\text{Al}_x\text{Ga}_{1-x}\text{O}_3$ in some examples).

[0582] In some cases, the transition layer **Z52-1045** may have a doping concentration between that of the substrate **Z52-1040** doping concentration and oxide semiconductor layer **Z52-1030** with a discrete change in doping concentration between the substrate **Z52-1040** and the transition layer **Z52-1045**, and between the transition layer and the oxide semiconductor layer. In an example, the substrate **Z52-1040** may be n^+ doped (i.e., having a doping concentration of from 10^{19} to 10^{20} cm^{-3}), the transition layer **Z52-1045** and oxide semiconductor layer **Z52-1030** may both be n^- doped (i.e., having a doping concentration of approximately 10^{18} cm^{-3} , or from 10^{17} cm^{-3} to 10^{18} cm^{-3}), and/or may be n^{--} doped (i.e., having a doping concentration from 10^{15} cm^{-3} to 10^{17} cm^{-3}). In some cases, the oxide semiconductor layer **Z52-1030** can include a transition between an n^+ doped region to an n^{--} doped region moving vertically upwards from the substrate **Z52-1040** to the oxide semiconductor layer **Z52-1030**. Similarly, a p-type device can be formed wherein the substrate **Z52-1040** is p^+ doped, the transition layer **Z52-1045** and the oxide semiconductor layer **Z52-1030** are both p^- and/or p^{--} doped, where the layers have the doping concentrations described above of acceptors rather than donors.

[0583] In some cases, the transition layer **Z52-1045** doping concentration has a monotonic gradient in the z direction, starting at a doping concentration below (i.e., less than) that of the substrate **Z52-1040**, and ending at a doping concentration approximately equal to that of the oxide semiconductor layer **Z52-1030**. In some cases, the transition layer **Z52-1045** doping concentration has a doping concentration that monotonically varies starting at a doping concentration that is either approximately equal to that of the substrate **Z52-1040** or is between the substrate doping concentration and the oxide semiconductor layer **Z52-1030** doping concentration, and ending at a doping concentration that is either approximately equal to that of the oxide semiconductor layer or is between the substrate doping concentration and the oxide semiconductor layer doping concentration. In an example, the transition layer **Z52-1045** doping concentration may match the substrate **Z52-1040** doping concentration at the interface between the substrate and the transition layer and then continuously change to match the oxide semiconductor layer **Z52-1030** doping concentration at the interface between the transition layer and the oxide semiconductor layer. In an example, the change in the transition layer **Z52-1045** doping concentration may be substantially linear as a function of vertical height z . In some cases, the transition layer **Z52-1045** doping concentration may change continuously (i.e., without significant step changes, and monotonically, or non-linearly, for example) as a function of vertical height z (where z is in the growth direction). In some cases, the transition layer **Z52-1045** doping concentration may change in a stepwise manner as a function of vertical height z . Transition layer **Z52-1045** can have any of the attributes of transition layer **160** as described above (e.g., with respect to transition layer **160** in FIGS. 4-7).

[0584] FIG. 101C shows an example of a device with an epitaxial oxide layer **Z52-1030** and an epitaxial nitride layer **Z52-1035** as described with respect to FIG. 101A, and additionally with an intermediate layer **Z52-1048** between the epitaxial nitride layer and the metal contact **Z52-1020** (forming a Schottky or Ohmic contact). Intermediate layer **Z52-1048** is similar to intermediate layer **170** in FIG. 8A, and can have the same properties as described with respect to intermediate layer **170** in FIG. 8A. Intermediate layer **Z52-1048** can have properties of any of the intermediate layers described herein. For example, the intermediate layer **Z52-1048** can alter the Schottky barrier properties and can thereby impact the electronic properties of the semiconductor diode device such as the turn-on voltage and/or the reverse breakdown voltage, as described further herein.

[0585] FIG. 101D shows an example of a device with an epitaxial oxide layer **Z52-1030** and an epitaxial nitride layer **Z52-1035** as described with respect to FIG. 101A, and additionally with both transition layer **Z52-1045** and intermediate layer **Z52-1048** as described with respect to FIGS. 101B and 101C.

[0586] FIGS. 101E and 101F show examples of devices with an epitaxial oxide layer **Z52-1030** and an epitaxial nitride layer **Z52-1035** as described with respect to FIG. 101A, and additionally with a “T-Channel” design as described with respect to FIG. 96A, and additionally with a transition layer and an intermediate layer, in accordance with some embodiments. The structures in FIGS. 101E and 101F include mesa structures containing the epitaxial nitride layer **Z52-1035**, and the metal contact **Z52-1020** (forming a Schottky or Ohmic contact), as well as optional template layer **Z52-1032**, and optional intermediate layer **Z52-1048**. FIG. 101F shows an example with an optional sidewall material **Z52-1085**.

[0587] Sidewall material **Z52-1085** can have the properties of any of the sidewall materials described herein, such as sidewall material **96180** in FIG. 96A.

[0588] FIGS. 102A-102C show tables depicting combinations of n-type and p-type layers with Ohmic contacts (“Metal S”) to a SiC substrate, and Ohmic or Schottky contacts (“Metal E”) to an epitaxially grown semiconductor layer, in accordance with some embodiments. The structures shown in FIGS. 101A-101F can have layers with the doping types shown in the respective tables in FIGS. 102A-102C. In some cases, the n-type or p-type doping of the 4H-SiC substrate **Z52-1040** (“Substrate” in FIGS. 102A-102C) can be from $1\text{e}15 \text{ cm}^{-3}$ to $1\text{e}21 \text{ cm}^{-3}$, or from $1\text{e}17 \text{ cm}^{-3}$ to $1\text{e}20 \text{ cm}^{-3}$, or less than $1\text{e}15 \text{ cm}^{-3}$, or greater than $1\text{e}21 \text{ cm}^{-3}$. The n-type or p-type doping of the epitaxial oxide drift layer **Z52-1030** (“Oxide” in FIGS. 102A-102C) (e.g., n-type Ga_2O_3 or p-type NiO), transition layer **Z52-1045** (“TL” in FIGS. 102A-102C), wurtzitic group-III Nitride layer **Z52-1035** (“Nitride” in FIGS. 102A-102C) (e.g., n-type or p-type GaN), and intermediate layer **Z52-1048** (“IL” in FIGS. 102A-102C) (e.g., n-type Ga_2O_3 or p-type NiO), can each be from $1\text{e}15 \text{ cm}^{-3}$ to $1\text{e}21 \text{ cm}^{-3}$, or from $1\text{e}16 \text{ cm}^{-3}$ to $1\text{e}20 \text{ cm}^{-3}$, or from $1\text{e}15 \text{ cm}^{-3}$ to $1\text{e}18 \text{ cm}^{-3}$, or less than $1\text{e}15 \text{ cm}^{-3}$, or greater than $1\text{e}20 \text{ cm}^{-3}$. Therefore, substrate **Z52-1040**, epitaxial oxide drift layer **Z52-1030**, transition layer **Z52-1045**, wurtzitic group-III Nitride layer **Z52-1035**, and intermediate layer **Z52-1048** can each be n-type (e.g., n^- , n , n^+ , etc.), p-type (p^- , p , p^+ , etc.), or i-type (e.g., intrinsic, or not intentionally doped, or n^{--} , or p^{--}) layers in structures and devices.

[0589] The Ohmic contact (“Metal S” in FIGS. 102A-102C) to the SiC substrate is optional and may be omitted in some cases. For example, the layers of the structures can be patterned and selectively etched such that two or more contacts are made to two or more epitaxial layers of the structures instead of to the backside of the substrate.

[0590] For example, a p-n-p device structure can be formed in cases where the polarization axis (shown by the arrows in FIGS. 100A-100D) is oriented such that the polar SiC and the polar nitride layer have p-type conductivity, and where the non-polar Ga₂O₃ drift layer is doped n-type. In another example, an n-p-n device structure can be formed in cases where the polarization axis (shown by the arrows in the figures) is oriented such that the polar SiC and the polar nitride layer have n-type conductivity, and where a non-polar epitaxial oxide drift layer is a p-type epitaxial oxide (e.g., NiO).

[0591] For example, FIG. 102A includes cases where the [polar 4H-SiC/epitaxial oxide/polar III-nitride/metal] structure can be an n-n-n-Schottky metal structure, or a p-p-p-Schottky metal structure. In such cases, the Schottky metal contact (“Metal E” in FIGS. 102A-102C) would form a Schottky barrier with the n-type or p-type polar III-nitride layer (“Nitride” in FIGS. 102A-102C). The resulting structures can be used to form diodes, transistors, or other devices. For example, a metal-oxide-semiconductor field-effect transistor (MOSFET) can be formed using ion implantation to form doped regions under source and drain contacts, and the oxide layer of these structures can be used as a channel region, where the channel region is separated from a gate contact by a gate oxide. In some cases, an epitaxial intermediate layer 170 can form the gate oxide, or in other cases, a non-epitaxial (e.g., sputtered) gate oxide can be deposited. Additionally, as shown in FIG. 102B, similar structures can be formed further including n-type or p-type intermediate layers (e.g., “IL” in FIG. 102B, and Z52-1045 in FIG. 101F), for example to form an n-n-n-n-Schottky metal structure, or a p-p-p-p-Schottky metal structure. Additionally, as shown in FIG. 102C, similar structures can be formed further including n-type or p-type transition layers (e.g., “TL” in FIG. 102C, and Z52-1045 in FIG. 101F), and intermediate layers (e.g., “IL” in FIG. 102B, and Z52-1045 in FIG. 101F), for example to form an n-n-n-n-n-Schottky metal structure, or a p-p-p-p-p-Schottky metal structure.

[0592] In another example, FIG. 102A includes cases where the [polar 4H-SiC/epitaxial oxide/polar III-nitride/metal] structure is an n-n-p-Ohmic contact structure, an n-n-p-Schottky metal structure, a p-p-n-Ohmic contact structure, or a p-p-n-Schottky metal structure. In such cases, a p/n junction is formed between the epitaxial oxide layer and the polar III-nitride layer. Additionally, in cases including a Schottky metal (“Metal E” in FIGS. 102A-102C), a Schottky barrier would form with the n-type or p-type polar III-nitride layer (“Nitride” in FIGS. 102A-102C). This Schottky barrier can serve to increase or decrease the built-in potential within the device, and/or form a well for a 2D electron gas or a 2D hole gas, as described further below. Additionally, as shown in FIG. 102B, similar structures can be formed further including n-type or p-type intermediate layers (e.g., “IL” in FIG. 102B, and Z52-1045 in FIG. 101F), for example to form an n-n-p-p-Schottky metal structure, or a p-p-n-n-Schottky metal structure. Additionally, as shown in FIG. 102C, similar structures can be formed further including n-type or p-type transition layers (e.g., “TL” in

FIG. 102C, and Z52-1045 in FIG. 101F), and n-type or p-type intermediate layers (e.g., “IL” in FIG. 102C, and Z52-1045 in FIG. 101F), for example to form an n-n-n-p-p-Schottky metal structure, or a p-p-p-n-n-Schottky metal structure. The resulting structures can be used to form diodes, transistors, or other devices. For example, a metal-oxide-semiconductor field-effect transistor (MOSFET) can be formed using ion implantation to form doped regions under source and drain contacts. In another example of a transistor, the nitride layer and/or intermediate layer can be patterned and selectively etched to form doped regions under source and drain contacts, and a gate oxide can be formed between the epitaxial oxide drift layer 120 and a gate contact. In other cases, an LED can be formed using such structures, wherein the epitaxial oxide and/or the polar III-nitride layer emit light.

[0593] In another example, FIGS. 102A and 102B include cases where the [polar 4H-SiC/epitaxial oxide/polar III-nitride/metal] structure is a p-n-n-Ohmic contact structure, a p-n-n-Schottky metal structure, an n-p-p-Ohmic contact structure, or an n-p-p-Schottky metal structure. In such cases, a p/n junction is formed between the polar 4H-SiC substrate and the epitaxial oxide layer. Additionally, an optional n-type or p-type transition layer (e.g., SiC, or Ga₂O₃) can be included between the polar 4H-SiC substrate and the epitaxial oxide layer which can move the p/n junction interface to the interface between the polar 4H-SiC substrate and the transition layer, or to the interface between the transition layer and the epitaxial oxide layer. The resulting structures can be used to form diodes, transistors, or other devices. For example, in cases including a Schottky metal, a Schottky barrier would form with the n-type or p-type polar III-nitride layer. This Schottky barrier can serve as a second built-in potential within the device, and if the potentials are in opposite directions (e.g., as in the case of back-to-back potentials) the structure could be used as a transistor or switch in some cases (e.g., in a metal-insulator-semiconductor (MIS) transistor architecture). In another example, a metal-oxide-semiconductor field-effect transistor (MOSFET) can be formed using ion implantation to form doped regions under source and drain contacts. In other cases, an LED can be formed using such structures, wherein the epitaxial oxide and/or the polar III-nitride layer emit light. Additionally, as shown in FIG. 102C, similar structures can be formed further including n-type or p-type transition layers (e.g., “TL” in FIG. 102C, and Z52-1045 in FIG. 101F), and n-type or p-type intermediate layers (e.g., “IL” in FIG. 102C, and Z52-1045 in FIG. 101F), for example to form a p-n-n-n-p-Ohmic contact structure, or an n-n-n-p-n-Ohmic contact structure. The resulting structures can be used to form BJT devices, for example, patterning and selectively etching the structure to form separate contacts to the p-type and n-type layers of the n-p-n or p-n-p types of structures.

[0594] In another example, FIGS. 102A and 102B include cases where the [polar 4H-SiC/epitaxial oxide/polar III-nitride/metal] structure can be a p-n-p-Ohmic contact structure, a p-n-p-Schottky metal structure, an n-p-n-Ohmic contact structure, or an n-p-n-Schottky metal structure. In such cases, a p/n junction is formed between the polar 4H-SiC substrate and the epitaxial oxide layer and a p/n junction is formed (in the reverse direction) between the epitaxial oxide layer and the polar III-nitride layer. Additionally, an optional n-type or p-type transition layer (e.g.,

SiC, or Ga₂O₃) can be included between the polar 4H-SiC substrate and the epitaxial oxide layer, which can move the p/n junction interface to the interface between the polar 4H-SiC substrate and the transition layer, or to the interface between the transition layer and the epitaxial oxide layer. Additionally, in cases including a Schottky metal, a Schottky barrier would form with the n-type or p-type polar III-nitride layer. This Schottky barrier can serve to increase or decrease the built-in potential within the device, and/or form a well for a 2D electron gas or a 2D hole gas, as described further below. Additionally, as shown in FIG. 102C, similar structures can be formed further including n-type or p-type transition layers (e.g., “TL” in FIG. 102C, and Z52-1045 in FIG. 101F), and n-type or p-type intermediate layers (e.g., “IL” in FIG. 102C, and Z52-1045 in FIG. 101F), for example to form a p-n-n-p-p-Ohmic contact structure, or an n-n-n-p-n-Ohmic contact structure. The resulting structures can be used to form diodes, transistors, or other devices. For example, a bipolar junction transistor (BJT) can be formed by patterning and selectively etching the structure to form separate contacts to the p-type and n-type layers of the n-p-n or p-n-p types of structures. In such devices, the metal contact adjacent to the oxide or intermediate layer can form either the source or drain contact of the BJT, and an additional source or drain contact can be formed. In other cases, a metal-oxide-semiconductor field-effect transistor (MOSFET) can be formed, using patterning and selectively etching of the intermediate layer and/or ion implantation to form doped regions under source and drain contacts. In other cases, an LED can be formed using such structures, wherein the epitaxial oxide and/or the polar III-nitride layer emit light.

[0595] FIG. 103 provides an example fabrication process flow Z53-1000 for the nitridation of a portion of a surface of a Ga₂O₃ epilayer in order to create a GaN template or nitrogen terminated surface with hexagonal surface reconstruction. The template can be a monolayer of Ga—N forming a partial GaN layer. Subsequent deposition can be used to create a GaN layer of a target thickness. The GaN layer thickness can be used to control device performance and/or generate polarization induced band bending.

[0596] In block Z53-1010 surface preparation processes are performed on a SiC substrate (e.g., substrate Z52-1040 in FIG. 101A). In some cases, a 4H-SiC substrate can be processed in preparation for an oxide epitaxial layer deposition. For example, the 4H-SiC substrate surface can be cleaned, a buffer layer can be deposited, and/or a transition layer (e.g., transition layer Z52-1045 in FIG. 101A) can be deposited, as described herein.

[0597] In block Z53-1020 an epitaxial oxide layer (e.g., β-Ga₂O₃ (−201) or oxide semiconductor layer Z52-1030 in FIG. 101A) is grown on the surface prepared in block Z53-1020. The epitaxial layer can be grown using a method described herein, for example, the method shown in FIGS. 57G-57H.

[0598] In block Z53-1030 a deposition chamber is evacuated using a vacuum pump coupled to the chamber, and in block Z53-1040 a nitridation substrate temperature is set using a heater positioned to heat the substrate.

[0599] In block Z53-1050 the oxide surface is exposed to active nitrogen species (“active-N”) such that the surface of the oxide is converted to a nitride (e.g., O—Ga—O bonds are converted to N—Ga—O bonds at the surface) to create a surface template layer of wz-GaN (i.e., GaN with a

wurtzite crystal symmetry). In some cases, the oxide surface is exposed to active nitrogen species at room temperature or at elevated temperature (e.g., from 25° C. to 1200° C., from 50° C. to 1200° C., or from 500° C. to 1000° C.) to form the template layer. For example, the template layer can be the template layer Z52-1032 in FIG. 101A.

[0600] In some cases, in block Z53-1020, a thin (e.g., from about 1 nm to about 100 nm, or from about 1 nm to about 10 nm) region at the gallium oxide surface is formed with an excess Ga content (i.e., is formed with a Ga:O ratio greater than 2:3). Then in block Z53-1050, the gallium oxide surface with excess Ga content is exposed to active nitrogen species (e.g., at elevated temperature) to convert it to gallium nitride.

[0601] In block Z53-1060, a nitride layer (e.g., GaN or Al_xGa_{1-x}N, or nitride layer Z52-1035 in FIG. 101A) is optionally epitaxially deposited to form a thicker nitride layer on the epitaxial oxide semiconductor layer. This nitride layer can be highly strained, and in such cases the nitride layer thickness can be kept below a critical layer thickness (CLT) to prevent a high concentration of defects (e.g., dislocations) from forming in the nitride layer.

[0602] In block Z53-1070 a metal layer (e.g., metal layer Z52-1020 in FIG. 101A) is deposited on the nitride layer. This metal layer can be a Schottky metal layer to form a Schottky barrier on the nitride layer, or can be an Ohmic contact metal for form an Ohmic contact with the nitride layer. The thickness of this metal layer can be from about 1 nm to about 50 nm in some cases. In block Z53-1080 an optional second metal layer can be deposited on the previous metal layer. For example, the first metal layer can include a Schottky metal and the second metal layer can be thicker (e.g., from 10 nm to 100 nm, or greater than 100 nm) and be any conductive metal. In some cases, an optional intermediate layer (e.g., intermediate layer Z52-1048 in FIG. 101A) can be deposited on the nitride layer before the metal layer is deposited in block Z53-1070. The metal layers in blocks Z53-1070 and Z53-1080 can be formed using an in-situ and/or ex-situ deposition process. For example, the metal layer in block Z53-1070 can be deposited epitaxially (e.g., while keeping the structure in vacuum), while the metal layer in block Z53-1080 can be deposited using thermal evaporation or sputtering in an ex-situ process (e.g., after exposing the structure to vacuum).

[0603] In block Z53-1090 the structure can be fabricated into a device as described herein. For example, a “T-channel” design, as described in FIG. 96A, can be formed by etching a mesa structure.

[0604] FIG. 104 shows example data from a vertical multilayer semiconductor structure including a C-plane on-axis Sapphire substrate, a β-Ga₂O₃ (−201) drift layer deposited on the substrate, and a layer of Ga-polar GaN (0001) deposited on the β-Ga₂O₃ (−201) drift layer. The structure in this example was formed using the process Z53-1000 shown in FIG. 103, wherein 100 nm of GaN was grown in optional step Z53-1060. FIG. 104 shows double crystal XRD scans, and the symmetric reflections of the materials. The peaks of the XRD data in FIG. 104 have low full width half maximum (FWHM) peak widths indicative of high-quality growth. For comparison, FIG. 104 also shows an XRD scan for β-Ga₂O₃ (−201) drift layer deposited on a C-plane on-axis Sapphire substrate. The XRD scan shows that the GaN phase has a wurtzite crystal symmetry and is Ga-polar or N-polar.

[0605] In some embodiments, vertical semiconductor structures can be formed including III-Nitride layers on epitaxial oxide layers, where the III-Nitride (e.g., GaN) layer has an N-polar or group-III-polar surface, such as, N-polar GaN. For example, N-polar GaN can be formed using N-rich GaN growth conditions at high deposition temperatures (e.g., from 700° C. to 1000° C.).

[0606] Integrating a polar GaN thin film onto an epitaxial oxide surface of an epitaxial stack (e.g., on a β -Ga₂O₃/4H-SiC device structure) can be beneficial to tune the diode properties, such as the built-in potential of a diode fabricated from the structure. The utility of integrating a polar GaN thin film onto the non-polar β -Ga₂O₃ surface of an epitaxial stack of a β -Ga₂O₃/4H-SiC device structure is now discussed.

[0607] FIGS. 105A and 105B show simulated engineered spatial bandstructure diagrams of a novel vertical device comprising: [Ni/Ga-polar GaN/ β -Ga₂O₃ n-type drift/n-type 4H-SiC(0001)]. The Schottky barrier potential of Ni with unintentionally doped GaN is about 0.66 eV. The calculated band diagrams of FIG. 105B show the Ga-polar GaN polarity induced band bending that occurs as a function of the GaN layer thickness. This example corresponds to the structure shown in FIG. 100B, wherein the polarization induced band bending in the GaN layer extends downwards and inverts the junction type as shown in FIGS. 105A and 105B. A β -Ga₂O₃ drift region n-type doping density of $1 \times 10^{17} \text{ cm}^{-3}$ was used in the simulations, which improves the ON-resistance of the diode compared to lower doping densities (e.g., $1 \times 10^{16} \text{ cm}^{-3}$). The Ga-polar GaN region 20110 is shown in FIG. 105B, which is zoomed in compared to FIG. 105A, and is 2.5 nm, 5 nm, or 15 nm thick (from approximately $z=0$ nm) in the different examples shown in the figures. The height and width of the polarization induced potential barrier in these examples is shown to increase with GaN thickness $2.5 \leq L_{\text{GaN}} \leq 15$ nm. The potential energy barrier formed at the GaN/Oxide interface resembles a Schottky barrier potential as described herein. This effect can be used to tailor the barrier height in excess of that possible with metal/Ga₂O₃ Schottky junctions (e.g., compared to barrier heights formed when using metals in FIG. 11A with Ga₂O₃). This can be advantageous, for example, to reduce the leakage current in a diode formed using such structures.

[0608] FIG. 105C shows the simulated structure in the vicinity of the GaN/Oxide valence band edge where GaN layer is 4 nm, and a triangular quantum well is induced within the GaN side of the heterojunction. The calculated lowest energy quantized heavy-hole wavefunctions $\psi_{\text{HH}}^n(z)$ and the associated quantized energy levels are plotted. The lowest HH-state $\psi_{\text{HH}}^{n=1}(z)$ lies below the Fermi energy and thus does not form a 2D-Hole gas (2DHG). Impurity doping the GaN layer with Mg to form p-type Mg:GaN can also be used in some cases to form a 2DHG.

[0609] It is also possible that an AlGaN alloy or AlN can be used as the polarization inducing material. For example, an Al(111) monolayer film may be deposited at low temperature on a β -Ga₂O₃ (-201) surface followed by nitridation to form a template of wurtzite AlN. The AlN layer may then be increased in thickness.

[0610] Along the c-axis of wurtzite GaN the lattice constant is 0.5185 nm (5.185 Angstroms), which can be used to calibrate the GaN thickness dependence for polarization induced band bending. FIG. 105D shows the calculated conduction barrier for electrons induced by the polar GaN

layer formed epitaxially on the drift β -Ga₂O₃ layer surface, calibrated in units of GaN monolayer thickness.

[0611] FIG. 105E shows a plot of the extracted peak polarization induced barrier height as a function of the Ga-polar GaN film thickness. Beyond ~10 ML of GaN the barrier height saturates to $\phi_B=3.7$ eV. Comparing the polarization induced barrier width with a typical depletion width of the Schottky barrier (e.g., as shown in FIG. 9B) shows that comparable depletion widths can be achieved. Therefore, the tunnelling transmission coefficient as a function of surface electric field can be similarly analyzed as discussed previously herein.

[0612] Another possible fabrication method is disclosed for the creation of thick epitaxial oxide semiconductors deposited upon SiC substrates. Depending upon the oxide crystal type and film deposition plane, there exists the possibility the oxide film will roughen its growth front surface as a function of film deposition thickness. Various growth modes can be utilized for the epitaxial growth of oxide films, viz., (i) Frank-van der Merwe mode; (ii) Volmer-Weber mode; and (iii) Stranski-Krastanov mode. The kinetics of epitaxial growth can further be via: (i) layer-by-layer wherein surface diffusion, nucleation and attachment mechanisms occur during growth; and (ii) step-flow growth wherein upon surface diffusion, adatoms stick to a step and the nucleation and growth proceeds via step flow.

[0613] In practice, lattice defects, stacking faults and dislocations can accumulate and disrupt an otherwise homogeneous growth mode.

[0614] Regardless of the cause, FIG. 106A shows schematically a cross-section of an oxide layer Z70-1020 (e.g., Ga₂O₃) epitaxially deposited on the SiC substrate Z70-1010. The oxide layer Z70-1020 may be formed up to a thickness with excellent material quality and acceptable surface roughness. Beyond a critical oxide film thickness the topmost portion Z70-1030 of the oxide epitaxial film can become rough, such that rough surface Z70-1040 has a peak to valley fluctuation Z70-1050 which can also expose undesired nucleation on some crystal planes. Additionally, in some cases, substrate Z70-1010 is a composite substrate comprising a surface layer and a bulk substrate layer (e.g., bulk substrate layer 112 and surface layer 114 of substrate 110 in FIG. 1), where the surface layer comprises single crystal SiC.

[0615] The rough surface Z70-1040 may be unacceptable for immediate device fabrication but the deposition process may utilize an advantageous growth method providing a high growth rate. It is possible to process the structure after epitaxial growth to achieve a surface which exhibits improved properties. For example, the structure of FIG. 106A can be processed using surface polishing methods, such as chemical mechanical polishing (CMP), cluster beam polishing, ion-beam polishing, selective oxidation and etching, chemical etching, as well as others.

[0616] FIG. 106B shows the resulting multilayer structure producing a smoother surface Z71-1020 compared to surface Z70-1040 of the as-deposited oxide film in FIG. 106A. The topmost portion Z71-1010 of the oxide epitaxial film is thinner than the topmost portion Z70-1030 of the oxide epitaxial film in FIG. 106A. For example, CMP can be utilized to surface polish the oxide layer (e.g., Ga₂O₃) to an atomic flatness, for example, having peak to valley fluctuation Z71-1030 from 1 nm to 10 nm, in some cases. Chemical etching may also be utilized to further activate the surface to

provide a beneficial surface state for creating a Schottky barrier. For example, post CMP, the surface can be exposed to chemical treatment, such as, hydrogen peroxide, HCl and other treatments.

[0617] The structure of FIG. 106B may further be processed to deposit yet further semiconductor materials.

[0618] The structure of FIG. 106B may further be processed to perform direct surface wafer-to-wafer bonding to further semiconductor materials.

EMBODIMENTS

[0619] Clause 1. A multilayered semiconductor diode device comprising: a substrate comprising silicon carbide (SiC); an epitaxial transition layer comprising a first semiconductor oxide material or SiC, wherein the epitaxial transition layer is on the substrate; an epitaxial drift layer comprising a second semiconductor oxide material, wherein the epitaxial drift layer is on the epitaxial transition layer; and a metal layer above the epitaxial drift layer, wherein the metal layer and the epitaxial drift layer form a Schottky barrier junction.

[0620] Clause 2. The multilayered semiconductor diode device of clause 1, wherein the metal layer is an epitaxial metal layer.

[0621] Clause 3. The multilayered semiconductor diode device of clause 1, wherein the epitaxial transition layer further comprises a lattice constant that varies in a vertical direction that is perpendicular to a top surface of the substrate.

[0622] Clause 4. The multilayered semiconductor diode device of clause 1, wherein the epitaxial transition layer further comprises a bandgap that varies in a vertical direction that is perpendicular to a top surface of the substrate.

[0623] Clause 5. The multilayered semiconductor diode device of clause 1, wherein the substrate comprises a first doping density, the epitaxial transition layer comprises a second doping density and the epitaxial drift layer comprises a third doping density, and wherein the first doping density is greater than the second doping density, and wherein the second doping density is greater than the third doping density.

[0624] Clause 6. The multilayered semiconductor diode device of clause 1, wherein the epitaxial transition layer further comprises a variable doping density that varies in a vertical direction that is perpendicular to a top surface of the substrate.

[0625] Clause 7. The multilayered semiconductor diode device of clause 1, further comprising an epitaxial intermediate layer between the epitaxial drift layer and the metal layer, wherein the epitaxial intermediate layer has a wider bandgap than the epitaxial drift layer, wherein the metal layer, the epitaxial intermediate layer, and the epitaxial drift layer form the Schottky barrier junction.

[0626] Clause 8. The multilayered semiconductor diode device of clause 1, further comprising an isolation layer on the epitaxial drift layer, and wherein the isolation layer has a dielectric constant greater than or equal to the epitaxial drift layer.

[0627] Clause 9. The multilayered semiconductor diode device of clause 1, wherein the first or the second semiconductor oxide material comprises $A_{2x}B_{1-x}O_{2x+1}$, wherein $0 \leq x \leq 1$, wherein A comprises Al, Ga, RE, Bi, B', or In, and wherein B comprises Zn, Mg, or Ni, where O is oxygen, Al

is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0628] Clause 10. The multilayered semiconductor diode device of clause 9, wherein x is 0.5 and the first or the second semiconductor oxide material comprises $A_2B_1O_4$.

[0629] Clause 11. The multilayered semiconductor diode device of clause 10, wherein the first or the second semiconductor oxide material comprises $Al_2Mg_1O_4$, $Ga_2Mg_1O_4$, $Al_2Zn_1O_4$, $Ga_2Zn_1O_4$, $Al_2Ni_1O_4$, $Ga_2Ni_1O_4$, or $RE_2Zn_1O_4$.

[0630] Clause 12. The multilayered semiconductor diode device of clause 1, wherein the first or the second semiconductor oxide material comprises $(A_yC_{1-y})_{2x}B_{1-x}O_{2x+1}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein A and C comprise two of: Al; Ga; RE; Bi; B'; and In; and wherein B comprises Zn, Mg, or Ni, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0631] Clause 13. The multilayered semiconductor diode device of clause 12, wherein x is 1, and $0 \leq y \leq 1$, and the first or the second semiconductor oxide material comprises $(A_yC_{1-y})_2O_3$.

[0632] Clause 14. The multilayered semiconductor diode device of clause 13, wherein A is Al and C is Ga.

[0633] Clause 15. The multilayered semiconductor diode device of clause 1, wherein the first or the second semiconductor oxide material comprises $(A_yD_zC_{1-y-z})_{2x}B_{1-x}O_{2x+1}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein $0 \leq z \leq 1$, wherein $y+z \leq 1$, wherein A, D and C comprise three of: Al; Ga; RE; Bi; B'; and In; and wherein B comprises Zn, Mg, or Ni, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0634] Clause 16. The multilayered semiconductor diode device of clause 15, wherein x is 0.5, wherein y is 0.9, wherein z is 0.05, and the first or the second semiconductor oxide material comprises $(A_{0.9}D_{0.05}C_{0.05})_2B_1O_4$.

[0635] Clause 17. The multilayered semiconductor diode device of clause 16, wherein the first or the second semiconductor oxide material comprises $(Ga_{0.9}Al_{0.05}In_{0.05})_2Mg_1O_4$, $(Ga_{0.9}Al_{0.05}In_{0.05})_2Ni_1O_4$, $(B'_{0.05}Bi_{0.05}Ga_{0.9})_2Mg_1O_4$, $(RE_{0.05}In_{0.05}Ga_{0.9})_2Zn_1O_4$, and $(RE_{0.05}Ga_{0.9}Al_{0.05})_2Zn_1O_4$.

[0636] Clause 18. The multilayered semiconductor diode device of clause 1, wherein the first or the second semiconductor oxide material comprises $A_{2(1-x)}B_{xy}C_{x(1-y)}O_{3-2x}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein A and C comprise two of: Al; Ga; RE; Bi; B'; and In; and wherein B comprises Zn, Mg, or Ni, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0637] Clause 19. The multilayered semiconductor diode device of clause 18, wherein x is 0.5, wherein y is 0.25, and the first or the second semiconductor oxide material comprises $A_2B_{0.25}C_{0.75}O_4$.

[0638] Clause 20. The multilayered semiconductor diode device of clause 19, wherein the first or the second semiconductor oxide material comprises $Ga_2Mg_{0.25}Zn_{0.75}O_4$, $Al_2Mg_{0.25}Zn_{0.75}O_4$, $Ga_2Mg_{0.25}Ni_{0.75}O_4$, and $Ga_2Zn_{0.25}Mg_{0.75}O_4$.

[0639] Clause 21. The multilayered semiconductor diode device of clause 1, wherein the first or the second semiconductor oxide material comprises $A_{2z(1-x)}D_{2(1-z)(1-x)}B_{xy}C_{x(1-y)}O_{3-2x}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein $0 \leq z \leq 1$,

wherein A and D comprise two of: Al; Ga; RE; Bi; B'; and In; and wherein B and C comprise two of: Zn; Mg; and Ni, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel

[0640] Clause 22. The multilayered semiconductor diode device of clause 21, wherein x is 0.5, wherein y is 0.5, wherein z is 0.5, and the first or the second semiconductor oxide material comprises $A_1D_1B_{0.5}C_{0.5}O_4$

[0641] Clause 23. The multilayered semiconductor diode device of clause 22, wherein the first or the second semiconductor oxide material comprises $Al_1Ga_1Ni_{0.5}Mg_{0.5}O_4$, $In_1Ga_1Zn_{0.5}Mg_{0.5}O_4$, and $RE_1Al_1Zn_{0.5}Ni_{0.5}O_4$.

[0642] Clause 24. The multilayered semiconductor diode device of clause 1, wherein the first or the second semiconductor oxide material comprises $A_{2x}B_{1-x}O_{x+2}$, wherein $0 \leq x \leq 1$, wherein A comprises Al, Ga, RE, Bi, B', or In, and wherein B comprises Ge, Si, or Sn, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Ge is germanium, Si is silicon, and Sn is tin

[0643] Clause 25. The multilayered semiconductor diode device of clause 24, wherein x is 0.5 and the first or the second semiconductor oxide material comprises $A_2B_1O_5$, or x is 0.75 and the first or the second semiconductor oxide material comprises $A_6B_1O_{11}$.

[0644] Clause 26. The multilayered semiconductor diode device of clause 25, wherein the first or the second semiconductor oxide material comprises $Al_2Si_1O_5$, $Ga_2Ge_1O_5$, $Al_2Ge_1O_5$, $RE_2Sn_1O_5$, $B'_2Ge_1O_5$, $In_2Ge_1O_5$, $Al_6Ge_1O_{11}$, $RE_6Ge_1O_{11}$, or $B'_6Si_1O_{11}$.

[0645] Clause 27. The multilayered semiconductor diode device of clause 1, wherein the first or the second semiconductor oxide material comprises $A_xB_{1-x}O_{2-x}$, wherein $0 \leq x \leq 1$, wherein A comprises Zn, Mg, or Ni, and wherein B comprises Ge, Si, or Sn, where O is oxygen, Zn is zinc, Mg is magnesium, Ni is nickel, Ge is germanium, Si is silicon, and Sn is tin.

[0646] Clause 28. The multilayered semiconductor diode device of clause 27, wherein x is $\frac{1}{2}$ and the first or the second semiconductor oxide material comprises $A_1B_1O_3$, or x is $\frac{2}{3}$ and the first or the second semiconductor oxide material comprises $A_2B_1O_4$.

[0647] Clause 29. The multilayered semiconductor diode device of clause 28, wherein the first or the second semiconductor oxide material comprises $Zn_1Si_1O_3$, $Zn_1Ge_1O_3$, $Ni_1Ge_1O_3$, $Mg_1Sn_1O_3$, $Mg_1Zn_1O_3$, $Ni_1Sn_1O_3$, $Zn_2Si_1O_4$, $Mg_2Ge_1O_4$, $Ni_2Ge_1O_4$, $Mg_2Sn_1O_4$, $Mg_2Si_1O_4$, or $Ni_2Si_1O_4$.

[0648] Clause 30. A method of forming a multilayered semiconductor diode device comprising: providing a substrate comprising silicon carbide (SiC); forming, on the substrate, an epitaxial transition layer comprising a first semiconductor oxide material or SiC; forming, on the epitaxial transition layer, an epitaxial drift layer comprising a second semiconductor oxide material; and forming a metal layer above the epitaxial drift layer, wherein the metal layer and the epitaxial drift layer form a Schottky barrier junction.

[0649] Clause 31. The method of clause 30, further comprising patterning the metal layer.

[0650] Clause 32. The method of clause 30, wherein the forming the metal layer comprises growing the metal layer on the epitaxial drift layer using an epitaxial growth technique.

[0651] Clause 33. The method of clause 30, wherein the substrate comprises a first doping density, the epitaxial transition layer comprises a second doping density, and the epitaxial drift layer comprises a third doping density, and wherein the first doping density is greater than the second doping density, and wherein the second doping density is greater than the third doping density.

[0652] Clause 34. The method of clause 30, wherein the epitaxial transition layer comprises a lattice constant that varies in a vertical direction that is perpendicular to a top surface of the substrate.

[0653] Clause 35. The method of clause 30, wherein the epitaxial transition layer comprises a bandgap that varies in a vertical direction that is perpendicular to a top surface of the substrate.

[0654] Clause 36. The method of clause 30, wherein the epitaxial transition layer comprises a doping density that varies in a vertical direction that is perpendicular to a top surface of the substrate.

[0655] Clause 37. The method of clause 30, further comprising forming an epitaxial intermediate layer between the epitaxial drift layer and the metal layer.

[0656] Clause 38. The method of clause 37, further comprising forming a guard ring on the epitaxial intermediate layer.

[0657] Clause 39. The method of clause 30, further comprising forming a second metal layer on the substrate.

[0658] Clause 40. The method of clause 30, further comprising forming passivation layer on the metal layer.

[0659] Clause 41. The method of clause 30, further comprising etching the epitaxial drift layer to form opposed intra-device field termination regions or opposed inter-device isolation regions.

[0660] Clause 42. The method of clause 30, wherein the first or the second semiconductor oxide material comprises $A_{2x}B_{1-x}O_{2x+1}$, wherein $0 \leq x \leq 1$, wherein A comprises Al, Ga, RE, Bi, B', or In, and wherein B comprises Zn, Mg, or Ni, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0661] Clause 43. The method of clause 42, wherein x is 0.5 and the first or the second semiconductor oxide material comprises $A_2B_1O_4$.

[0662] Clause 44. The method of clause 43, wherein the first or the second semiconductor oxide material comprises $Al_2Mg_1O_4$, $Ga_2Mg_1O_4$, $Al_2Zn_1O_4$, $Ga_2Zn_1O_4$, $Al_2Ni_1O_4$, $Ga_2Ni_1O_4$, or $RE_2Zn_1O_4$.

[0663] Clause 45. The method of clause 30, wherein the first or the second semiconductor oxide material comprises $(A_yC_{1-y})_{2x}B_{1-x}O_{2x+1}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein A and C comprise two of: Al; Ga; RE; Bi; B'; and In; and wherein B comprises Zn, Mg, or Ni, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0664] Clause 46. The method of clause 45, wherein x is 0.5, and y is 0.1, and the first or the second semiconductor oxide material comprises $(A_{0.1}C_{0.9})_2B_1O_4$.

[0665] Clause 47. The method of clause 46, wherein the first or the second semiconductor oxide material comprises $(Al_{0.1}Ga_{0.9})_2Mg_1O_4$, $(Al_{0.1}Ga_{0.9})_2Ni_1O_4$, $(Bi_{0.1}Ga_{0.9})_2Mg_1O_4$, $(RE_{0.1}Ga_{0.9})_2Zn_1O_4$, or $(RE_{0.1}Al_{0.9})_2Zn_1O_4$.

[0666] Clause 48. The method of clause 30, wherein the first or the second semiconductor oxide material comprises

$(A_xD_zC_{1-y-z})_{2x}B_{1-x}O_{2x+1}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein $0 \leq z \leq 1$, wherein $y+z \leq 1$, wherein A, D and C comprise three of: Al; Ga; RE; Bi; B'; and In; and wherein B comprises Zn, Mg, or Ni, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0667] Clause 49. The method of clause 48, wherein x is 0.5, wherein y is 0.9, wherein z is 0.05, and the first or the second semiconductor oxide material comprises $(A_{0.9}D_{0.05}C_{0.05})_2B_1O_4$.

[0668] Clause 50. The method of clause 49, wherein the first or the second semiconductor oxide material comprises $(Ga_{0.9}Al_{0.05}In_{0.05})_2Mg_1O_4$, $(Ga_{0.9}Al_{0.05}In_{0.05})_2Ni_1O_4$, $(B'_{0.05}Bi_{0.05}Ga_{0.9})_2Mg_1O_4$, $(RE_{0.05}In_{0.05}Ga_{0.9})_2Zn_1O_4$, and $(RE_{0.05}Ga_{0.9}Al_{0.05})_2Zn_1O_4$.

[0669] Clause 51. The method of clause 30, wherein the first or the second semiconductor oxide material comprises $A_{2(1-x)}B_{xy}C_{x(1-y)}O_{3-2x}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein A and C comprise two of: Al; Ga; RE; Bi; B'; and In; and wherein B comprises Zn, Mg, or Ni, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0670] Clause 52. The method of clause 51, wherein x is 0.5, wherein y is 0.25, and the first or the second semiconductor oxide material comprises $A_2B_{0.25}C_{0.75}O_4$.

[0671] Clause 53. The method of clause 52, wherein the first or the second semiconductor oxide material comprises $Ga_2Mg_{0.25}Zn_{0.75}O_4$, $Al_2Mg_{0.25}Zn_{0.75}O_4$, $Ga_2Mg_{0.25}Ni_{0.75}O_4$, and $Ga_2Zn_{0.25}Mg_{0.75}O_4$.

[0672] Clause 54. The method of clause 30, wherein the first or the second semiconductor oxide material comprises $A_{2z(1-x)}D_{2(1-z)(1-x)}B_{xy}C_{x(1-y)}O_{3-2x}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein $0 \leq z \leq 1$, wherein A and D comprise two of: Al; Ga; RE; Bi; B'; and In; and wherein B and C comprise two of: Zn; Mg; and Ni, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0673] Clause 55. The method of clause 54, wherein x is 0.5, wherein y is 0.5, wherein z is 0.5, and the first or the second semiconductor oxide material comprises $A_1D_1B_{0.5}C_{0.5}O_4$.

[0674] Clause 56. The method of clause 55, wherein the first or the second semiconductor oxide material comprises $Al_1Ga_1Ni_{0.5}Mg_{0.5}O_4$, $In_1Ga_1Zn_{0.5}Mg_{0.5}O_4$, and $RE_1Al_1Zn_{0.5}Ni_{0.5}O_4$.

[0675] Clause 57. The method of clause 30, wherein the first or the second semiconductor oxide material comprises $A_{2x}B_{1-x}O_{x+2}$, wherein $0 \leq x \leq 1$, wherein A comprises Al, Ga, RE, Bi, B', or In, and wherein B comprises Ge, Si, or Sn, where O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Ge is germanium, Si is silicon, and Sn is tin.

[0676] Clause 58. The method of clause 57, wherein x is 0.5 or 0.75, and the first or the second semiconductor oxide material respectively comprises $A_2B_1O_5$ or $A_6B_1O_{11}$.

[0677] Clause 59. The method of clause 58, wherein the first or the second semiconductor oxide material comprises $Al_2Si_1O_5$, $Ga_2Ge_1O_5$, $Al_2Ge_1O_5$, $RE_2Sn_1O_5$, $B'_2Ge_1O_5$, $In_2Ge_1O_5$, $Al_6Ge_1O_{11}$, $RE_6Ge_1O_{11}$, or $B'_6Si_1O_{11}$.

[0678] Clause 60. The method of clause 30, wherein the first or the second semiconductor oxide material comprises $A_xB_{1-x}O_{2-x}$, wherein $0 \leq x \leq 1$, wherein A comprises Zn, Mg,

or Ni, and wherein B comprises Ge, Si, or Sn, where O is oxygen, Zn is zinc, Mg is magnesium, Ni is nickel, Ge is germanium, Si is silicon, and Sn is tin.

[0679] Clause 61. The method of clause 60, wherein x is $\frac{1}{2}$ or $\frac{2}{3}$, and the first or the second semiconductor oxide material respectively comprises $A_1B_1O_3$ or $A_2B_1O_4$.

[0680] Clause 62. The method of clause 61, wherein the first or the second semiconductor oxide material comprises $Zn_1Si_1O_3$, $Zn_1Ge_1O_3$, $Ni_1Ge_1O_3$, $Mg_1Sn_1O_3$, $Mg_1Zn_1O_3$, $Ni_1Sn_1O_3$, $Zn_2Si_1O_4$, $Mg_2Ge_1O_4$, $Ni_2Ge_1O_4$, $Mg_2Sn_1O_4$, $Mg_2Si_1O_4$, or $Ni_2Si_1O_4$.

[0681] Clause 63. A multilayered semiconductor diode device comprising: a substrate comprising silicon carbide (SiC); an epitaxial drift layer comprising a first semiconductor oxide material, wherein the epitaxial drift layer is on the substrate; an epitaxial channel layer comprising a second semiconductor oxide material, wherein the epitaxial channel layer is on the epitaxial drift layer; a Schottky metal layer above the epitaxial channel layer, wherein the Schottky metal layer and the epitaxial channel layer form a Schottky barrier junction, and wherein the epitaxial channel layer and the Schottky metal layer are formed into a first mesa structure; and a sidewall layer comprising a dielectric material, wherein the sidewall layer is on the epitaxial drift layer and contacts a wall of the first mesa structure.

[0682] Clause 64. The multilayered semiconductor diode device of clause 63, wherein the Schottky metal layer is an epitaxial metal layer.

[0683] Clause 65. The multilayered semiconductor diode device of clause 63, wherein the dielectric material comprises a dielectric constant that is greater than a dielectric constant of the epitaxial channel layer.

[0684] Clause 66. The multilayered semiconductor diode device of clause 63, wherein the first semiconductor oxide material has a doping density between those of the substrate and the second semiconductor oxide material.

[0685] Clause 67. The multilayered semiconductor diode device of clause 63, wherein the first semiconductor oxide material and the second semiconductor oxide material are configured such that there is no substantial barrier to a flow of electrons from the first semiconductor oxide material to the second semiconductor oxide material.

[0686] Clause 68. The multilayered semiconductor diode device of clause 63, further comprising an epitaxial transition layer comprising a third semiconductor oxide material or SiC, wherein the epitaxial transition layer is between the substrate and the epitaxial drift layer.

[0687] Clause 69. The multilayered semiconductor diode device of clause 68, wherein the first semiconductor oxide material has substantially the same composition and crystal symmetry as the second semiconductor oxide material.

[0688] Clause 70. The multilayered semiconductor diode device of clause 68, wherein the epitaxial transition layer further comprises a lattice constant that is different than a lattice constant of the substrate.

[0689] Clause 71. The multilayered semiconductor diode device of clause 68, wherein the epitaxial transition layer further comprises a bandgap that is different than a bandgap of the substrate.

[0690] Clause 72. The multilayered semiconductor diode device of clause 68, wherein the substrate comprises a first doping density, the epitaxial transition layer comprises a second doping density and the epitaxial drift layer comprises a third doping density, and wherein the first doping density

is greater than the second doping density, and wherein the second doping density is greater than the third doping density.

[0691] Clause 73. The multilayered semiconductor diode device of clause 68, wherein the epitaxial transition layer further comprises a variable doping density that varies in a vertical direction that is perpendicular to a top surface of the substrate.

[0692] Clause 74. The multilayered semiconductor diode device of clause 63, further comprising an epitaxial intermediate layer between the epitaxial channel layer and the Schottky metal layer, wherein the epitaxial intermediate layer has a wider bandgap than the epitaxial channel layer, wherein the Schottky metal layer, the epitaxial intermediate layer, and the epitaxial channel layer form the Schottky barrier junction.

[0693] Clause 75. The multilayered semiconductor diode device of clause 63, further comprising an epitaxial intermediate layer between the epitaxial channel layer and the Schottky metal layer, wherein the epitaxial intermediate layer has a wider bandgap than the epitaxial channel layer, wherein the Schottky metal layer, the epitaxial intermediate layer, and the epitaxial channel layer form the Schottky barrier junction.

[0694] Clause 76. The multilayered semiconductor diode device of clause 63, wherein the first or the second semiconductor oxide material comprises $A_{2x}B_{1-x}O_{2x+1}$, wherein $0 \leq x \leq 1$, wherein A comprises Al, Ga, RE, Bi, B', or In, wherein B comprises Zn, Mg, or Ni, wherein O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0695] Clause 77. The multilayered semiconductor diode device of clause 76, wherein x is 0.5 and the first or the second semiconductor oxide material comprises $A_2B_1O_4$.

[0696] Clause 78. The multilayered semiconductor diode device of clause 77, wherein the first or the second semiconductor oxide material comprises $Al_2Mg_1O_4$, $Ga_2Mg_1O_4$, $Al_2Zn_1O_4$, $Ga_2Zn_1O_4$, $Al_2Ni_1O_4$, $Ga_2Ni_1O_4$, or $RE_2Zn_1O_4$.

[0697] Clause 79. The multilayered semiconductor diode device of clause 63, wherein the first or the second semiconductor oxide material comprises $(A_yC_{1-y})_2B_{1-x}O_{2x+1}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein A and C comprise two of: Al; Ga; RE; Bi; B'; and In; wherein B comprises Zn, Mg, or Ni, wherein O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0698] Clause 80. The multilayered semiconductor diode device of clause 79, wherein x is 1, and the first or the second semiconductor oxide material comprises $(A_yC_{1-y})_2O_3$.

[0699] Clause 81. The multilayered semiconductor diode device of clause 80, wherein A is Al and C is Ga.

[0700] Clause 82. The multilayered semiconductor diode device of clause 63, wherein the first or the second semiconductor oxide material comprises $(A_yD_zC_{1-y-z})_{2x}B_{1-x}O_{2x+1}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein $0 \leq z \leq 1$, wherein $y+z \leq 1$, wherein A, D and C comprise three of: Al; Ga; RE; Bi; B'; and In; wherein B comprises Zn, Mg, or Ni, wherein O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0701] Clause 83. The multilayered semiconductor diode device of clause 82, wherein x is 0.5, wherein y is 0.9,

wherein z is 0.05, and the first or the second semiconductor oxide material comprises $(A_{0.9}D_{0.05}C_{0.05})_2B_1O_4$.

[0702] Clause 84. The multilayered semiconductor diode device of clause 83, wherein the first or the second semiconductor oxide material comprises $(Ga_{0.9}Al_{0.05}In_{0.05})_2Mg_1O_4$, $(Ga_{0.9}Al_{0.05}In_{0.05})_2Ni_1O_4$, $(B'_{0.05}Bi_{0.05}Ga_{0.9})_2Mg_1O_4$, $(RE_{0.05}In_{0.05}Ga_{0.9})_2Zn_1O_4$, and $(RE_{0.05}Ga_{0.9}Al_{0.05})_2Zn_1O_4$.

[0703] Clause 85. The multilayered semiconductor diode device of clause 63, wherein the first or the second semiconductor oxide material comprises $A_{2(1-x)}B_{xy}C_{x(1-y)}O_{3-2x}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein A and C comprise two of: Al; Ga; RE; Bi; B'; and In; wherein B comprises Zn, Mg, or Ni, wherein O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0704] Clause 86. The multilayered semiconductor diode device of clause 85, wherein x is 0.5, wherein y is 0.25, and the first or the second semiconductor oxide material comprises $A_2B_{0.25}C_{0.75}O_4$.

[0705] Clause 87. The multilayered semiconductor diode device of clause 86, wherein the first or the second semiconductor oxide material comprises $Ga_2Mg_{0.25}Zn_{0.75}O_4$, $Al_2Mg_{0.25}Zn_{0.75}O_4$, $Ga_2Mg_{0.25}Ni_{0.75}O_4$, and $Ga_2Zn_{0.25}Mg_{0.75}O_4$.

[0706] Clause 88. The multilayered semiconductor diode device of clause 63, wherein the first or the second semiconductor oxide material comprises $A_{2z(1-x)}D_{2(1-z)(1-x)}B_{xy}C_x(1-y)O_{3-2x}$, wherein $0 \leq x \leq 1$, wherein $0 \leq y \leq 1$, wherein $0 \leq z \leq 1$, wherein A and D comprise two of: Al; Ga; RE; Bi; B'; and In; wherein B and C comprise two of: Zn; Mg; and Ni, wherein O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Zn is zinc, Mg is magnesium, and Ni is nickel.

[0707] Clause 89. The multilayered semiconductor diode device of clause 88, wherein x is 0.5, wherein y is 0.5, wherein z is 0.5, and the first or the second semiconductor oxide material comprises $A_1D_1B_{0.5}C_{0.5}O_4$.

[0708] Clause 90. The multilayered semiconductor diode device of clause 89, wherein the first or the second semiconductor oxide material comprises $Al_1Ga_1Ni_{0.5}Mg_{0.5}O_4$, $In_1Ga_1Zn_{0.5}Mg_{0.5}O_4$, and $RE_1Al_1Zn_{0.5}Ni_{0.5}O_4$.

[0709] Clause 91. The multilayered semiconductor diode device of clause 63, wherein the first or the second semiconductor oxide material comprises $A_2B_{1-x}O_{x+2}$, wherein $0 \leq x \leq 1$, wherein A comprises Al, Ga, RE, Bi, B', or In, wherein B comprises Ge, Si, or Sn, wherein O is oxygen, Al is aluminum, Ga is gallium, RE is a rare-earth, Bi is bismuth, B' is boron, In is indium, Ge is germanium, Si is silicon, and Sn is tin.

[0710] Clause 92. The multilayered semiconductor diode device of clause 91, wherein x is 0.5 and the first or the second semiconductor oxide material comprises $A_2B_1O_5$, or x is 0.75 and the first or the second semiconductor oxide material comprises $A_6B_1O_{11}$.

[0711] Clause 93. The multilayered semiconductor diode device of clause 92, wherein the first or the second semiconductor oxide material comprises $Al_2Si_1O_5$, $Ga_2Ge_1O_5$, $Al_2Ge_1O_5$, $RE_2Sn_1O_5$, $B'_2Ge_1O_5$, $In_2Ge_1O_5$, $Al_6Ge_1O_{11}$, $RE_6Ge_1O_{11}$, or $B'_6Si_1O_{11}$.

[0712] Clause 94. The multilayered semiconductor diode device of clause 63, wherein the first or the second semiconductor oxide material comprises $A_xB_{1-x}O_{2-x}$, wherein $0 \leq x \leq 1$, wherein A comprises Zn, Mg, or Ni, wherein B

comprises Ge, Si, or Sn, wherein O is oxygen, Zn is zinc, Mg is magnesium, Ni is nickel, Ge is germanium, Si is silicon, and Sn is tin.

[0713] Clause 95. The multilayered semiconductor diode device of clause 94, wherein x is $\frac{1}{2}$ and the first or the second semiconductor oxide material comprises $A_1B_1O_3$, or x is $\frac{2}{3}$ and the first or the second semiconductor oxide material comprises $A_2B_1O_4$.

[0714] Clause 96. The multilayered semiconductor diode device of clause 95, wherein the first or the second semiconductor oxide material comprises $Zn_1Si_1O_3$, $Zn_1Ge_1O_3$, $Ni_1Ge_1O_3$, $Mg_1Sn_1O_3$, $Mg_1Zn_1O_3$, $Ni_1Sn_1O_3$, $Zn_2Si_1O_4$, $Mg_2Ge_1O_4$, $Ni_2Ge_1O_4$, $Mg_2Sn_1O_4$, $Mg_2Si_1O_4$, or $Ni_2Si_1O_4$.

[0715] Clause 97. The multilayered semiconductor diode device of clause 63, further comprising a field plate on a top surface of the Schottky metal layer, wherein the field plate layer comprises a metal and extends laterally beyond the first mesa structure and onto a top surface of the sidewall layer.

[0716] Clause 98. The multilayered semiconductor diode device of clause 63, further comprising a second mesa structure and an interconnect, wherein: the first mesa structure comprises a first portion of the epitaxial channel layer and a first portion of the Schottky metal layer; the second mesa structure comprises a second portion of the epitaxial channel layer and a second portion of the Schottky metal layer; the sidewall layer further contacts a wall of the second mesa structure; and the first portion of the Schottky metal layer and the second portion of the Schottky metal layer are coupled with the interconnect.

[0717] Clause 99. The multilayered semiconductor diode device of clause 63, wherein the Schottky metal layer is selected from a metal of FIG. 11A.

[0718] Clause 100. A method of forming a multilayered semiconductor diode device comprising: providing a substrate comprising silicon carbide (SiC); forming, on the substrate, an epitaxial drift layer comprising a first semiconductor oxide material; forming, on the epitaxial drift layer, an epitaxial channel layer comprising a first semiconductor oxide material; forming a metal layer on the epitaxial drift layer, wherein the metal layer and the epitaxial drift layer form a Schottky barrier junction; etching the epitaxial channel layer and the metal layer to form a mesa structure; forming a sidewall layer comprising a dielectric material, wherein the sidewall layer is on the epitaxial drift layer and contacts a wall of the mesa structure.

[0719] Clause 101. A multilayered semiconductor diode device comprising: a substrate comprising silicon carbide (SiC); an epitaxial drift layer comprising a first semiconductor oxide material, wherein the epitaxial drift layer is above the substrate with respect to a growth direction; a polar nitride layer comprising a polar semiconductor nitride material, wherein the polar nitride layer is above the epitaxial drift layer with respect to the growth direction; and a metal layer above the polar nitride layer with respect to the growth direction.

[0720] Clause 102. The multilayered semiconductor diode device of clause 101, wherein the metal layer comprises a high work function metal such that a Schottky barrier is formed between the metal layer and the polar nitride layer.

[0721] Clause 103. The multilayered semiconductor diode device of clause 102, wherein the metal layer comprises a metal selected from a metal of FIG. 11A, or alloys thereof.

[0722] Clause 104. The multilayered semiconductor diode device of clause 102, wherein the metal layer is an epitaxial metal layer.

[0723] Clause 105. The multilayered semiconductor diode device of clause 101, wherein the substrate and the first semiconductor oxide material are configured such that there is no substantial barrier to a flow of electrons from the substrate to the first semiconductor oxide material.

[0724] Clause 106. The multilayered semiconductor diode device of clause 101, further comprising an epitaxial transition layer comprising a third semiconductor oxide material or SiC, wherein the epitaxial transition layer is between the substrate and the epitaxial drift layer.

[0725] Clause 107. The multilayered semiconductor diode device of clause 105, wherein the epitaxial transition layer further comprises a lattice constant that is different than a lattice constant of the substrate.

[0726] Clause 108. The multilayered semiconductor diode device of clause 105, wherein the epitaxial transition layer further comprises a bandgap that is different than a bandgap of the substrate.

[0727] Clause 109. The multilayered semiconductor diode device of clause 105, wherein the substrate comprises a first doping density, the epitaxial transition layer comprises a second doping density and the epitaxial drift layer comprises a third doping density, and wherein the first doping density is greater than the second doping density, and wherein the second doping density is greater than the third doping density.

[0728] Clause 110. The multilayered semiconductor diode device of clause 105, wherein the epitaxial transition layer further comprises a variable doping density that varies in a vertical direction that is perpendicular to a top surface of the substrate.

[0729] Clause 111. The multilayered semiconductor diode device of clause 101, wherein the epitaxial drift layer is doped n-type, wherein a polarization axis of the polar semiconductor nitride material is aligned with the growth direction such that the polar nitride layer comprises p-type charge, and wherein the polar nitride layer and the epitaxial drift layer form a p/n junction.

[0730] Clause 112. The multilayered semiconductor diode device of clause 101, wherein the first semiconductor oxide material comprises Ga_2O_3 , and wherein the polar semiconductor nitride material comprises $Al_{0.3}Ga_{0.7}N$.

[0731] Clause 113. The multilayered semiconductor diode device of clause 101, further comprising: an isolation layer on the polar nitride layer, and surrounding the Metal layer, wherein the isolation layer comprises a material with a higher dielectric constant than a dielectric constant of the first semiconductor oxide material; and a field plate on a top surface of the metal layer, wherein the field plate comprises a metal and extends laterally beyond the metal layer and onto a top surface of the isolation layer.

[0732] Clause 114. The multilayered semiconductor diode device of clause 101, further comprising: a first mesa structure comprising the polar nitride layer and the metal layer; and a sidewall layer comprising a dielectric material, wherein the sidewall layer is on the epitaxial drift layer and contacts a wall of the first mesa structure, wherein a dielectric constant of the dielectric material is higher than a dielectric constant of the first semiconductor oxide material.

[0733] Clause 115. The multilayered semiconductor diode device of clause 113, further comprising a second mesa

structure and an interconnect, wherein: the first mesa structure comprises a first portion of the polar nitride layer and a first portion of the metal layer; the second mesa structure comprises a second portion of the polar nitride layer and a second portion of the metal layer; the sidewall layer further contacts a wall of the second mesa structure; and the first portion of the metal layer and the second portion of the metal layer are coupled with the interconnect.

[0734] Clause 116. A method of forming a multilayered semiconductor diode device comprising: providing a substrate comprising silicon carbide (SiC); forming, on the substrate, an epitaxial drift layer comprising a first semiconductor oxide material; forming, on the epitaxial drift layer, an epitaxial polar nitride layer comprising a polar semiconductor nitride material; and forming a metal layer on the epitaxial polar nitride layer, wherein the metal layer and the epitaxial polar nitride layer form a Schottky barrier junction.

[0735] Clause 117. The method of clause 115, further comprising forming a template layer comprising a second nitride material, on a surface of the epitaxial drift layer before forming the epitaxial polar nitride layer on the epitaxial drift layer.

[0736] Clause 118. The method of clause 116, wherein the second nitride material of the template layer is formed by exposing the surface of the epitaxial drift layer to active nitrogen species at an elevated temperature.

[0737] Clause 119. A multilayered semiconductor diode device comprising: a substrate comprising silicon carbide (SiC); an epitaxial drift layer comprising a first semiconductor oxide material, wherein the epitaxial drift layer is above the substrate with respect to a growth direction, and wherein the epitaxial drift layer is doped n-type; a template layer comprising a nitride template material, wherein the template layer is in contact with the epitaxial drift layer; a polar nitride layer comprising a polar semiconductor nitride material, wherein the polar nitride layer is in contact with the template layer, and wherein a polarization axis of the polar semiconductor nitride material is aligned with the growth direction such that the polar nitride layer comprises p-type charge, wherein the polar nitride layer and the epitaxial drift layer form a p/n junction; and a metal layer above the polar nitride layer.

[0738] Clause 120. The multilayered semiconductor diode device of clause 118, wherein the template layer is from 1 nm to 100 nm thick, and wherein the polar nitride layer is from 100 nm to 5 microns thick.

[0739] Clause 121. The multilayered semiconductor diode device of clause 118, wherein the metal layer comprises a high work function metal such that a Schottky barrier is formed between the metal layer and the polar nitride layer.

[0740] In some cases, a single embodiment may, for succinctness and/or to assist in understanding the scope of the disclosure, combine multiple features. It is to be understood that in such a case, these multiple features may be provided separately (in separate embodiments), or in any other suitable combination. Alternatively, where separate features are described in separate embodiments, these separate features may be combined into a single embodiment unless otherwise stated or implied. This also applies to the claims which can be recombined in any combination. That is, a claim may be amended to include a feature defined in any other claim. Furthermore, a phrase referring to “at least one of” a list of items refers to any combination of those

items, including single members. As an example, “at least one of: a, b, or c” is intended to cover: a, b, c, a-b, a-c, b-c, and a-b-c.

[0741] Reference has been made in detail to embodiments of the disclosed invention, one or more examples of which have been illustrated in the accompanying figures. Each example has been provided by way of explanation of the present technology, not as a limitation of the present technology. In fact, while the specification has been described in detail with respect to specific embodiments of the invention, it will be appreciated that those skilled in the art, upon attaining an understanding of the foregoing, may readily conceive of alterations to, variations of, and equivalents to these embodiments. For instance, features illustrated or described as part of one embodiment may be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present subject matter covers all such modifications and variations within the scope of the appended claims and their equivalents. These and other modifications and variations to the present invention may be practiced by those of ordinary skill in the art, without departing from the scope of the present invention, which is more particularly set forth in the appended claims. Furthermore, those of ordinary skill in the art will appreciate that the foregoing description is by way of example only and is not intended to limit the invention.

What is claimed is:

1. A multilayered semiconductor diode device comprising:

- a substrate comprising silicon carbide (SiC);
- an epitaxial drift layer comprising a first semiconductor oxide material, wherein the epitaxial drift layer is above the substrate with respect to a growth direction;
- a polar nitride layer comprising a polar semiconductor nitride material, wherein the polar nitride layer is above the epitaxial drift layer with respect to the growth direction; and
- a metal layer above the polar nitride layer with respect to the growth direction.

2. The multilayered semiconductor diode device of claim 1, wherein the metal layer comprises a high work function metal such that a Schottky barrier is formed between the metal layer and the polar nitride layer.

3. The multilayered semiconductor diode device of claim 2, wherein the metal layer comprises a metal selected from a metal of FIG. 11A, or alloys thereof.

4. The multilayered semiconductor diode device of claim 2, wherein the metal layer is an epitaxial metal layer.

5. The multilayered semiconductor diode device of claim 1, wherein the substrate and the first semiconductor oxide material are configured such that there is no substantial barrier to a flow of electrons from the substrate to the first semiconductor oxide material.

6. The multilayered semiconductor diode device of claim 1, further comprising an epitaxial transition layer comprising a third semiconductor oxide material or SiC, wherein the epitaxial transition layer is between the substrate and the epitaxial drift layer.

7. The multilayered semiconductor diode device of claim 6, wherein the epitaxial transition layer further comprises a lattice constant that is different than a lattice constant of the substrate.

8. The multilayered semiconductor diode device of claim 6, wherein the epitaxial transition layer further comprises a bandgap that is different than a bandgap of the substrate.

9. The multilayered semiconductor diode device of claim 6, wherein the substrate comprises a first doping density, the epitaxial transition layer comprises a second doping density and the epitaxial drift layer comprises a third doping density, and wherein the first doping density is greater than the second doping density, and wherein the second doping density is greater than the third doping density.

10. The multilayered semiconductor diode device of claim 6, wherein the epitaxial transition layer further comprises a variable doping density that varies in a vertical direction that is perpendicular to a top surface of the substrate.

11. The multilayered semiconductor diode device of claim 1, wherein the epitaxial drift layer is doped n-type, wherein the polar nitride layer comprises p-type charge, and wherein the polar nitride layer and the epitaxial drift layer form a p/n junction.

12. The multilayered semiconductor diode device of claim 1, wherein the first semiconductor oxide material comprises Ga_2O_3 , and wherein the polar semiconductor nitride material comprises $\text{Al}_{0.3}\text{Ga}_{0.7}\text{N}$.

13. The multilayered semiconductor diode device of claim 1, further comprising:

an isolation layer on the polar nitride layer, and surrounding the Metal layer, wherein the isolation layer comprises a material with a higher dielectric constant than a dielectric constant of the first semiconductor oxide material; and

a field plate on a top surface of the metal layer, wherein the field plate comprises a metal and extends laterally beyond the metal layer and onto a top surface of the isolation layer.

14. The multilayered semiconductor diode device of claim 1, further comprising:

a first mesa structure comprising the polar nitride layer and the metal layer; and

a sidewall layer comprising a dielectric material, wherein the sidewall layer is on the epitaxial drift layer and contacts a wall of the first mesa structure, wherein a dielectric constant of the dielectric material is higher than a dielectric constant of the first semiconductor oxide material.

15. The multilayered semiconductor diode device of claim 14, further comprising a second mesa structure and an interconnect, wherein:

the first mesa structure comprises a first portion of the polar nitride layer and a first portion of the metal layer; the second mesa structure comprises a second portion of the polar nitride layer and a second portion of the metal layer;

the sidewall layer further contacts a wall of the second mesa structure; and

the first portion of the metal layer and the second portion of the metal layer are coupled with the interconnect.

16. A method of forming a multilayered semiconductor diode device comprising:

providing a substrate comprising silicon carbide (SiC);

forming, on the substrate, an epitaxial drift layer comprising a first semiconductor oxide material;

forming, on the epitaxial drift layer, an epitaxial polar nitride layer comprising a polar semiconductor nitride material; and

forming a metal layer on the epitaxial polar nitride layer, wherein the metal layer and the epitaxial polar nitride layer form a Schottky barrier junction.

17. The method of claim 16, further comprising forming a template layer comprising a second nitride material, on a surface of the epitaxial drift layer before forming the epitaxial polar nitride layer on the epitaxial drift layer.

18. The method of claim 17, wherein the second nitride material of the template layer is formed by exposing the surface of the epitaxial drift layer to active nitrogen species at an elevated temperature.

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